Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

August 9, 2026

The campaign in numbers (snapshot 2026-08-09; see §75)

200	374	1 281	87	69	3 015
current T0	registry	`fr` cycles	work sessions	campaign days	proof nodes

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1 Orientation and rigour boundary

The north star, and its status

The target of this project is the classical (stochastic) stability statement `op-classical`: there are universal constants $\eta_0, C > 0$, independent of the dimension, such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta \le \eta_0$ admits a stochastic idempotent E (`def-stochastic`) with

$$\|Q - E\|_{\infty\to\infty} \;\le\; C\sqrt{\eta}, \tag{1}$$

the exponent $\frac{1}{2}$ being sharp.

This is OPEN. Nothing in this document proves it, and nothing in this document should be read as evidence that it is close to proved. What follows is a rigorous *partial* record: the components of the currently live route that have cleared this repository's top in-repo rigour rung.

Two pictures, one bridge

Two equivalent formulations are in play throughout. In the *stochastic picture* the object is a genuinely positive row-stochastic Q that is only approximately idempotent, with defect η measured in the $\infty\to\infty$ map norm (`def-almost-idempotent`). In the *signed picture* the object is an exact signed affine retraction P — $P\mathbf{1} = \mathbf{1}$ and $P^2 = P$ *exactly* (`def-signed-idempotent`) — whose failure of positivity is quarantined in the single scalar δ, the negative mass (`def-negative-mass`). The two are interchangeable up to universal constants, and the licence to cross between them is `lem-classical-equiv` (§3); a bound must always name the picture it lives in and cite the bridge when it crosses.

What this report contains

Five blocks, in dependency order.

1. **The bridge** (§3). `lem-classical-equiv`, the signed$\leftrightarrow$stochastic equivalence, with explicit constants in both directions.

2. **Positive-retract hardening** (§4). `lem-prh` converts a *positive approximate retract* — a pair of positive unital maps A, M between finite commutative C^*-algebras whose composite MA is ε-close to the identity (`def-positive-approximate-retract`) — into an exact stochastic idempotent at distance $2\sqrt{2\varepsilon}$. It is the step that turns approximate retraction data into the exact object (1) asks for, at the sharp square-root rate.

3. **The COMP tier** (§§5–18, interleaved). This tier builds, over an *extended* ε-C^*-algebra (`def-extended-epsilon-cstar-algebra`), the approximate corner calculus of δ-projections: the compressed corners $\mathcal{S}_{P,Q}$, the compression maps $\mathrm{Co}_{P,Q}$ and the compressed product (`def-compressed-corner`). Its content is that all of this is *matrix-uniform*: every estimate holds at every amplification level n with one constant. The keystones are the amplification identity $1_{M_n} \otimes \mathrm{Co}_{P,Q} = \mathrm{Co}_{I_n\otimes P, I_n\otimes Q}$, the uniform rectangular-product estimate, and the statement that a nonvanishing corner $\mathcal{S}_P$ is itself an *extended* $O(e)$-C^*-algebra — the object the next tier needs.

4. **The H-CB tier** (§§12–30, interleaved). Over a closed H-CB datum (`def-hcb-datum`) this tier establishes the amplified estimates for the map $\mathrm{Ha}^{Q}_{P,R}$ of `def-ha-map`: the column-Hilbert comparison, the variational identity that defines it, its column action, exact adjointness, its

multiplicative defect, the diagonal unit and upper-norm estimates, and — under explicit level-one hypotheses — the diagonal lower modulus and the diagonal inverse bound, all uniform in n.

5. **The H-CB parent contract and EXT tier** (§31 and §§32–37). This tier develops corner arithmetic through a one-dimensional index, the dimension selection $(r, 1)$ of an EXT-CB datum, the four-corner merge, and the EXT parent that adds one dimension to a level-one datum.

The honesty boundary

Three rules govern what may appear here.

Only af-validated results are stated as lemmas. Every numbered lemma below is a registry result whose adversarial proof tree has root node `validated` with clean taint, produced by fresh `codex` provers and *separate* fresh per-node verifiers; the orchestrator never judged a proof. Each shard records the node count and names its workspace. Nothing else in the registry — the inherited `proved-mod-audit` transcriptions, the `conjecture` rows, the `numerical` evidence, the `heuristic` arguments — is restated here as a theorem; those keep their tags in `argument/` and are surveyed, without promotion, in §76.

Every lemma is exactly as strong as its registry contract. Each statement is a typeset transcription of the one-line `contract` of its `argument/` shard, reproduced byte-verbatim beneath it. Where a contract is deliberately conditional or one-sided — the two-sided compressed-unit bound only for *nonvanishing* projections, the level-one hypotheses of the inverse tier — that shape is load-bearing and is preserved. In particular a level-one conclusion is never glossed as a matrix-uniform one.

No definition is restated. Terms resolve to their canonical `definitions/` shards, cited by `def-`identifier. Where a proof tree consumed a byte-matched auxiliary registered inside its own `af` workspace — the ε-C^* norm axioms, the compressed-product display, the one-dimensional nonvanishing bridge — it is named as such, and identified as a workspace external rather than a registry definition.

What is deliberately absent

As of 2026-08-08, `op-classical` is `proved` with `af:  validated`: the H-CB tier feeds `conj-hcb` (§31), that feeds `conj-extcb` (§37), the MAIN assembly carrier `lem-thmainext-conditional`, the strengthened `lem-routef-k-ledger` with its three helper rows, and `lem-routef-f0-assembly` are all `proved` with `af:  validated`, and the user-ratified root rewire discharged the theorem through Route F (§76). Sharpness of the exponent is now carried at T0 by `cor-classical-sharpness` (§4). What this document does NOT claim is any rigour rung above af-validation (no Lean proof exists). Also absent: any numerical statement (those live in `runs/`, and are never rigorous), and the earlier signed-geometry material, which is off the live route and inventoried in §76.

2 The vocabulary

What this section is, and why it is generated

Every later section names its terms by `def-` identifier and never restates them. This section is where those identifiers resolve. It reproduces the project's Layer-0 definitions database — one shard per term under `definitions/` — as typeset definitions, in a deliberate reading order, each carrying its own provenance.

The material below is *generated*, and that is a load-bearing design decision rather than a convenience. The governing law of this repository is that each term is defined exactly once (`CLAUDE.md` L2): a second, hand-maintained statement of a definition inside the paper is drift, and drift here is fatal, because a lemma proved against one reading of δ or of exposedness and quoted against another is a wrong result that no gate would catch. So the canonical shard stays the single source of truth and the pages below are a deterministic projection of it, produced by `scripts/gen-report-defs.py` into `report/generated/defs/` in the same way that `argument/INDEX.md` and `argument/DAG.md` are produced from the registry shards. The generator's `--check` mode fails the local gate whenever the committed projection differs from a fresh render, so a definition edited in `definitions/` and not re-rendered here cannot be committed. Nothing in this section may be edited in place: corrections belong in the named shard, after which the projection is regenerated.

The statement is the definition; the source text is the second check

Each entry is a typeset `definition` environment, and that environment is the definition. Underneath it sits a small provenance block; underneath that, where the shard carries one, a SOURCE CHECK (BYTE-VERBATIM) block. The order is deliberate. A definition transcribed from the literature is only usable if it can be *read*, and a block of detokenized monospace is not a readable definition; but the byte-verbatim text is what the repository's `cited` criterion actually locks (`L1`), so it cannot simply be dropped. It is therefore kept and demoted: the small print at the end, against which the typeset statement above may be checked, rather than the statement itself.

Where a shard carries a harmonised **Statement** section, that section is what is typeset — it exists in the shard precisely to be the readable form, and it is where a transcription's known corrections are recorded. Where the shard's only content is the pinned source's own TEX, the generator typesets *that*, through an explicit macro-translation table. The table exists because a quoted source is written in that source's private macro dialect (\\`eps`, \\`calA`, \\`Co`, . . .), which this document's preamble does not define. Every row of the table maps a source macro to the expansion *the source itself gives it*, taken from that source's own preamble and recorded with the line number that defines it; the table is reproduced in full in `report/generated/defs/MANIFEST.md` for audit, together with its two declared glyph-only deviations. Translation changes spelling, never meaning. Each definition typeset this way says so in its provenance block, names the table version, and lists the macros that fired.

The honesty boundary is mechanical, not editorial. A translated fragment is validated before it is emitted — every surviving control sequence must be one this document defines, braces and math delimiters must balance, no alignment or script character may escape math mode — and a fragment that fails is *not* typeset. It is replaced on the page by a loud **NOT TYPESET** block naming the reason, its bytes are left to the source check below, and it is listed in `report/generated/defs/MANIFEST.md`. Partial success is allowed: what translates is typeset, what does not is flagged, and nothing in between is guessed at. As of this render there are no such flags. The byte-verbatim blocks themselves are set with TEX's detokenizer, which prints a control word followed by one space; apart from that spacing artefact, and from the line breaks the quote block inserts to keep long source lines inside the text block, what is printed is the source's own bytes.

Reading the kind and status of a definition

Each definition is followed by a small provenance block naming its *kind*, its *status*, its source, its locus, and — for literature transcriptions — the SHA256 prefix of the pinned source file. The vocabulary of that block is the repository's rigour ladder applied to definitions, and it is not the same ladder that grades results.

- `cited` — transcribed from the literature and byte-matched to a local source under `refs/` at the recorded locus and hash. The gate `scripts/check-defs.py` recomputes that hash. What this certifies is *provenance*: that the text below is what the source says. It certifies nothing about whether the source's claim is true, and several of the cited shards here carry explicit scope notes recording exactly which neighbouring assertions of the source are *not* part of the definition.
- `consensus` — a formulation this project adopted and agreed on, with no single external source to byte-match against.
- `original` — a notion introduced here, typically to name a configuration or a hypothesis package that the exploration kept re-inlining.

The status is orthogonal. `locked` means usable downstream: byte-verified for a `cited` shard, signed off for a `consensus` or `original` one. `draft` means the shard is pinned but not yet gated — fine as working vocabulary, and flagged whenever a validated result leans on it. A definition is never `proved`: it carries no truth claim, so there is nothing in it to prove. When a statement about a defined object needs to be true, it is a registry result with its own status, not a clause smuggled into the vocabulary.

The two pictures, and which one a term lives in

The single most reliable way to misread this project is to forget which picture a symbol belongs to. The stochastic picture holds genuinely positive row-stochastic maps that are only approximately idempotent (Definition 3, Definition 4). The signed picture holds exactly idempotent signed affine retractions whose failure of positivity is quarantined in a single scalar (Definition 2, Definition 1); everything built on row geometry — exposedness, the visible set, height, invisible mass — is stated there. The licence to cross between the pictures, with explicit constants, is `lem-classical-equiv` (§3), and it is the only such licence.

Two symbol collisions are deliberate in the sources and are preserved rather than silently renamed. The letter δ denotes the negative mass of a signed idempotent in the classical layer and the defect of a δ-projection in the approximate-C^* layer (Definition 21); these are different quantities, and `CONVENTIONS.md` is the registry that keeps them apart. Likewise ε is the ambient approximate-C^* parameter, never the retract defect of Definition 43. When a bound is quoted, the picture and the parameter must be named with it.

Which definitions appear here

This section renders the vocabulary of the *current proof strategy*, not the whole Layer-0 database. The strategy is the Route-F landing chain, and it is not re-described here: the generator imports `scripts/gen-report-dag.py` and reuses the very same subgraph computation that draws the atlas of §74, so the map and the vocabulary cannot drift apart. A definition is rendered when it is imported by a registry result in that subgraph, or by a registry result reproduced in this document, or when it is reached from one of those by following the `[[def-...]]` crosslinks that the rendered statements themselves use. Anything outside that closure is not printed, and a generated paragraph at the end of the section says how many such definitions there are and names them; they remain canonical in `definitions/` and in `definitions/INDEX.md`, they are simply not part of what this document needs. A crosslink pointing at one of them degrades to a bare identifier rather than to a link this document could not honour.

The three groups below are computed from that set, not curated. The first is the connected component of the crosslink graph that the classical signed and stochastic vocabulary forms: those definitions genuinely define each other, and they are ordered so that a term appears after everything its statement uses, with mutually recursive clusters (negative mass, signed idempotent, stochastic idempotent; exposedness and the visible set) kept together. The second group is the approximate-C^* vocabulary transcribed from the pinned source, which is the language of the tiers reproduced in §§5–37. The third is the project-internal packaging built over that language: the datum bundles that fix the hypotheses of a tier, and the notation the proofs of that tier share. A group with no member in scope would not be printed at all.

One honest asymmetry survives the scoping. Being *needed by* the strategy is a much weaker condition than being *proved on* it: much of the classical group is the vocabulary of the signed-geometry trunk, whose results are ancestors of the target but are recorded in `argument/` and whitelisted in `report/UNWIRED.md` rather than reproduced here. Those definitions belong in this section — the target `op-classical` of §1 is stated in their terms — but the reader should not read their presence as a claim that the surrounding argument is in this document. Each definition's pointer line says how many registry results import it and which of those, if any, appear here; a term with no reproduced consumer says so instead of leaving the reader to guess.

Citing a definition

Every definition carries `\label{def:`*slug*`}` and a matching `\hypertarget`, where *slug* is the shard identifier with its first hyphen replaced by a colon — `def-co-top` becomes `def:co-top`. That is the same identifier-to-label transform the provenance gate uses for results, so a lemma section may write `\ref{def:co-top}` and the knowledge-graph page may anchor at the same target. Inside the generated statements, a crosslink is a live link exactly when its target is in this document — a definition rendered in this section, or a registry result reproduced in a later one — and otherwise is printed as a bare identifier, because a link into material this document does not contain would be a promise it cannot keep. Each definition also names its canonical shard path, so the reader can go from the page to the source of truth in one step, and from there to `definitions/INDEX.md` and `argument/DAG.md`.

The classical picture: the signed and stochastic vocabulary

Definition 1 (negative mass (`def-negative-mass`)). For an exact signed idempotent $P \in \mathbb{R}^{n\times n}$ (or any signed matrix), the *negative mass* (equivalently *negativity*) is

$$\delta(P) := \max_{1\le i\le n} \sum_{j=1}^{n} \max\{-P_{ij}, 0\},$$

the largest total negative mass in any single row. The *row polytope* is

$$K := \operatorname{conv}\{p_1, \dots, p_n\},$$

the convex hull of the rows $p_i \in \mathbb{R}^n$. A stochastic idempotent is exactly a signed idempotent with $\delta(P) = 0$.

Kind. original (introduced by this project); `status: locked`. **Aliases.** negativity; row polytope; row polytope K.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (def:esi, δ(P)); row polytope from ../almost-idempotent-positive-maps/definitions/def-stochastic.md`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (negativity δ(P)) and ../almost-idempotent-positive-maps definitions/def-stochastic.md (row polytope K).
Used in this report by `lem-classical-equiv` (165 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-negative-mass.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. $\delta(P)$ quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Exact signed idempotent*); the row polytope K adopted from `../almost-idempotent-positive-maps/definitions/def-stochastic.md`. This scalar is the single knob that controls the geometry: the three derived scales $\tau = \sqrt{\delta}$, $\rho = 4\tau$, $\kappa = \tau/4$ live in `def-visible-set`. The δ-positivity of `def-near-positive-projection` is the same quantity: a unital idempotent R is δ-positive iff $\delta(R) \le \delta$.

Definition 2 (exact signed idempotent (`def-signed-idempotent`)). Let $n \in \mathbb{N}$. A matrix $P \in \mathbb{R}^{n\times n}$ is an *exact signed idempotent* if

$$P\mathbf{1} = \mathbf{1} \qquad \text{and} \qquad P^2 = P,$$

where $\mathbf{1} = (1, \dots, 1)^{\mathsf{T}}$. Equivalently, P is a linear self-map of $\mathbb{R}^n$ fixing the all-ones vector with $P^2 = P$ *exactly*, whose rows are signed measures of total mass 1 ($\sum_j P_{ij} = 1$ for each i). A row-stochastic idempotent — a stochastic idempotent — is exactly an exact signed idempotent whose negative mass $\delta(P) = 0$.

Row geometry. Writing $p_i \in \mathbb{R}^n$ for the i-th row with the ℓ^1 metric, each row satisfies $\sum_j p_{ij} = 1$ and $\sum_j |p_{ij}| \le 1 + 2\delta$; hence all rows lie in a common affine hyperplane and pairwise ℓ^1 distances are at most $2 + 4\delta$ (with $\delta = \delta(P)$).

Kind. original (introduced by this project); `status: locked`. **Aliases.** signed idempotent; signed affine retraction; signed stochastic retraction.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (Definition, exact signed idempotent)`; no source hash (not a literature transcription); record: adapted from ../almost-idempotent-positive-maps/agent-A/explorations/classical-portfolio/report/kernel-conjecture.tex §Setting (def:esi); exact-idempotence form adopted for this repo.
Used in this report by `lem-classical-equiv` (186 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-signed-idempotent.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Exact signed idempotent*, and the row-geometry remark). The exact-idempotence form is the working object: $P^2 = P$ holds exactly while non-positivity is quarantined into the scalar negative mass $\delta(P)$. This is the matrix/geometry framing of `def-near-positive-projection` (map/positivity framing). Its exposed row vertices, visible set, height and invisible mass are `def-exposed`, `def-visible-set`, `def-height`, `def-invisible-mass`.

Definition 3 (stochastic idempotent (`def-stochastic`))**.** A *unital positive map* of $\ell_n^\infty = \mathbb{R}^n$ is a *row-stochastic matrix* Q ($Q \geq 0$ entrywise, $Q\mathbf{1} = \mathbf{1}$) — an affine self-map of the probability simplex Δ_n. A *stochastic idempotent* E is a row-stochastic E with $E^2 = E$: an affine retraction of Δ_n onto a sub-polytope.

Dropping positivity but keeping *exact* idempotence gives the *signed* generalization — a signed affine retraction P with $P\mathbf{1} = \mathbf{1}$ and $P^2 = P$ exactly, whose rows are signed measures of total mass 1. Its failure of positivity is quantified by the negative mass $\delta(P)$; a stochastic idempotent is exactly a signed idempotent with $\delta(P) = 0$.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** row-stochastic idempotent; stochastic retraction; unital positive map on ℓ∞_n.
Provenance. `source: internal`; locus `adopted from ../almost-idempotent-positive-maps/definitions/def-stochastic.md`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-stochastic.md (B's classical formulation, report sec:classical def:stochastic).
Used in this report by `cor-classical-sharpness`, `lem-classical-equiv`, `lem-prh`, `lem-prh-sharpness`, `lem-prh-sharpness-row-coincidence`, `lem-routef-f0-assembly` and 9 further reproduced results (24 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-stochastic.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adopted from `../almost-idempotent-positive-maps/definitions/def-stochastic.md` (B's classical formulation, report sec:classical `def:stochastic`); re-tagged `consensus / source: internal` for this repo. The signed form is the working object because $P^2 = P$ holds *exactly* (non-positivity quarantined in δ, see `def-negative-mass`). This is the commutative specialization of `def-near-positive-projection`; the approximate-idempotence sibling is `def-almost-idempotent` (a genuinely positive Q with $\|Q^2 - Q\|$ small). Exposed row vertices and the visible set are `def-exposed` / `def-visible-set`.

Definition 4 (almost idempotent (`def-almost-idempotent`))**.** A row-stochastic matrix Q on ℓ_n^∞ is *almost idempotent* (with defect η) if

$$\|Q^2 - Q\|_{\infty\to\infty} \leq \eta, \qquad \eta \in [0, \tfrac{1}{4}),$$

where $\|\cdot\|_{\infty\to\infty}$ is the operator norm induced by ℓ_n^∞ — i.e. the maximum over rows of the ℓ^1-norm of the corresponding row of $Q^2 - Q$. The range $\eta < \frac{1}{4}$ is what makes the binomial series producing the exact idempotent converge.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** η-idempotent; eta-idempotent; almost-idempotent stochastic matrix.
Provenance. `source: internal`; locus `adopted from ../almost-idempotent-positive-maps/definitions/def-almost-idempotent.md (stochastic specialization)`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-almost-idempotent.md (operator-norm formulation, report rem:cb-norm); $\infty \to \infty$ specialization per kernel-conjecture.tex thm:chain(c).
Used in this report by `cor-classical-sharpness`, `lem-classical-equiv`, `lem-routef-ai-defect-linearization`, `lem-routef-f0-assembly`, `lem-routef-f0-defect-identity`, `lem-routef-factor-estimate-packet` and 4 further reproduced results (10 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-almost-idempotent.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adopted from `../almost-idempotent-positive-maps/definitions/def-almost-idempotent.md` (the operator-norm — *not* cb-norm — formulation, report rem:cb-norm), specialized to the $\|\cdot\|_{\infty\to\infty}$ map norm on stochastic matrices as in `classical-portfolio/report/kernel-conjecture.tex` Theorem (chain)(c). Measuring non-idempotence in the map operator norm (rather than the cb-norm) is why the bridge exponent is $\sqrt{\eta}$ rather than η. The classical stability question (`op-classical`) asks whether every such Q lies within $C\sqrt{\eta}$ (max-row-ℓ^1) of an exactly idempotent row-stochastic matrix; its parent object is `def-near-positive-projection`. Distinguish η (this defect) from the negative mass δ of the associated `def-signed-idempotent`, with $\delta = O(\eta)$.

Definition 5 (exposed vertex (`def-exposed`))**.** Fix an exact signed idempotent $P \in \mathbb{R}^{n\times n}$ and write $p_i \in \mathbb{R}^n$ for its i-th row, with the ℓ^1 metric $\|x - y\|_1 = \sum_j |x_j - y_j|$.

A row p_v is a *row vertex* of P if $p_v \notin \operatorname{conv}\{\, p_j : p_j \neq p_v \text{ as vectors of } \mathbb{R}^n \,\}$ (geometrically coincident duplicate rows count as a single point; a repeated point is still a vertex if it lies outside the hull of the *other* points).

An *admissible exposer* for a row vertex v is an affine function $h : \mathbb{R}^n \to \mathbb{R}$ with

$$h(p_v) = 0 \qquad \text{and} \qquad 0 \le h(p_j) \le 1 \quad \text{for every row } p_j.$$

The *exposedness margin* of v is

$$t^*(v) := \sup_{h \text{ admissible}} \min\{\, h(p_j) : \|p_j - p_v\|_1 \ge \rho \,\},$$

with the convention $t^*(v) = +\infty$ when no row is at ℓ^1-distance $\ge \rho$ from p_v. The vertex v is (ρ, κ)-*exposed* if $t^*(v) \ge \kappa$, and *hidden* otherwise. The scales $\rho = 4\tau$, $\kappa = \tau/4$ (with $\tau = \sqrt{\delta}$) and the resulting *visible set* $\mathcal{W}(P)$ of exposed vertices are recorded in `def-visible-set`.

Informally: an exposed vertex can be linearly separated, with margin $\ge \kappa$, from every row that is genuinely far ($\ge \rho$) from it; rows inside the ρ-ball are exempt.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** (ρ,κ)-exposed; (rho,kappa)-exposed; exposedness margin; admissible exposer; hidden row vertex; well-exposed vertex.
Provenance. `source: internal`; locus `adapted from classical-portfolio/report/kernel-conjecture.tex §Setting (def:vertex, def:exposed); ../almost-idempotent-positive-maps/definitions/def-exposed.md`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-exposed.md (B's exposed-circuit machinery, report sec:classical); reconciled with kernel-conjecture. tex def:exposed.
Used by 129 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-exposed.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adapted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (def:vertex, def:exposed) and `../almost-idempotent-positive-maps/definitions/def-exposed.md`. The latter states the equivalent *exposedness modulus* $e_v(\rho) = \sup_h \min_{\|p_i - v\|_1 \ge \rho} h(p_i)$ and calls a vertex *well-exposed* at $\sqrt{}$-scale if $e_v(\rho) \ge c\sqrt{\delta}$ for some $\rho = O(\sqrt{\delta})$, where δ is the negative mass. Central to the proved classical special cases (well-exposed $\Rightarrow$ simplex) and to the open global exposed-hull lemma (`op-exposed-hull`). A pointwise exposed-or-redundant dichotomy is provably *insufficient* (dense regular polygons) — the gap must be stated globally (see `def-height`).

Definition 6 (visible set (`def-visible-set`))**.** Given an exact signed idempotent P with $\delta = \delta(P)$ (the negative mass), the *three scales* are

$$\tau := \sqrt{\delta}, \qquad \rho := 4\tau, \qquad \kappa := \tau/4.$$

The *visible set* is

$$\mathcal{W} = \mathcal{W}(P) := \{\, v : \ p_v \text{ is a } (\rho,\kappa)\text{-exposed row vertex of } P \,\},$$

the set of row vertices that are (ρ, κ)-exposed at these scales; all other row vertices are *hidden*. At $\delta = 0$ the visible vertices are exactly the distinct rows of the recurrent (equal-input) blocks of a stochastic idempotent.

Kind. original (introduced by this project); `status: locked`. **Aliases.** visible set W; W(P); (ρ,κ)-exposed set; three scales.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (def:scales, def:exposed)`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (Definition: three scales; Definition: (ρ,κ)-exposedness and the visible set W).
Used by 137 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-visible-set.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adapted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *The three scales*; Definition: (ρ,κ)*-exposedness and the visible set* $\mathcal{W}$). The scales couple exposedness (`def-exposed`) to the negative mass (`def-negative-mass`). The convex hull $C_{\mathcal{W}} := \operatorname{conv}\{p_w : w \in \mathcal{W}\}$ is consumed by `def-height` (max ℓ^1-distance of a row from $C_{\mathcal{W}}$) and `def-invisible-mass` (positive mass a row places outside $C_{\mathcal{W}}$).

Definition 7 (height (`def-height`))**.** Let $C_{\mathcal{W}} := \operatorname{conv}\{p_w : w \in \mathcal{W}\}$, where $\mathcal{W} = \mathcal{W}(P)$ is the `def-visible-set`, and let $\operatorname{dist}_1(x, S) = \inf_{s \in \operatorname{conv} S} \|x - s\|_1$. If $\mathcal{W} \neq \emptyset$, the *height* of a row p_i is $\operatorname{dist}_1(p_i, C_{\mathcal{W}})$, and the *height of P* is

$$H = H(P) := \max_{1 \le i \le n} \operatorname{dist}_1\bigl(p_i,\ \operatorname{conv}\{p_w : w \in \mathcal{W}\}\bigr).$$

Because $\operatorname{dist}_1(\cdot, C_{\mathcal{W}})$ is convex and every row is a convex combination of the geometrically distinct row vertices, the maximum is attained at a row vertex; if $H > 0$, any maximizing vertex is necessarily hidden. Such a vertex is called a *hidden top vertex*.

Kind. original (introduced by this project); `status: locked`. **Aliases.** height H; H(P); hidden top vertex.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (Definition: Height)`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (Definition: Height, def:height).
Used by 112 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-height.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Height*). H measures how far the rows spill outside the hull of the visible (`def-visible-set`, `def-exposed`) vertices; the linear hull bound / HLC target is $H(P) \le C\sqrt{\delta}$ in the negative mass δ. The invisible mass `def-invisible-mass` of a hidden top vertex is the hypothesis quantity that governs whether H can be large.

Definition 8 (always-tight hull (`def-actor-hull`))**.** Fix an exact signed idempotent P and a hidden geometrically distinct row vertex u with $t^*(u) > 0$. Let $T(u)$, $O(u)$, $Z(u)$ be the always-tight ρ-far / upper-box / zero-face constraint families of the exposedness LP at u (`lem-always-tight-dual-support`). The **always-tight hulls** (or *actor hulls*) are the displacement convex bodies

$$K_T(u) := \operatorname{conv}\{p_f - p_u : f \in T(u)\}, \qquad K_O(u) := t^*(u) \cdot \operatorname{conv}\{p_i - p_u : i \in O(u)\}.$$

They are **disjoint** when $g := \operatorname{dist}_1(K_T(u), K_O(u)) > 0$ (the huddle / Branch-I predicate), and **intersect** otherwise (Branch II, the alpha-free optimal display of `lem-optimal-face-conic-reduction`). Boundary ownership: tangency / any common point counts as intersecting.

Kind. original (introduced by this project); `status: draft`. **Aliases.** actor hull; K_T; K_O; K_T(u); K_O(u); always-tight hulls; disjoint always-tight hulls; displacement hull.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-downhill-cotop-conic-mass.md and lem-always-tight-dual-support.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-always-tight-dual-support (the T/O families) + lem-downhill-cotop-conic-mass (the hulls K_T, K_O).
Used by 27 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-actor-hull.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "actor hull" and "always-tight hull(s)" name the same displacement bodies K_T, K_O built from the actors (far / upper-box rows) the dual witness charges. The disjoint-vs-intersect dichotomy on K_T, K_O is the root split $S1$ of the W54 huddle-charge tree. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-zero-face`, `def-dual-witness`, `def-co-top`.

Definition 9 (hiddenness dual witness (`def-dual-witness`))**.** Fix an exact signed idempotent P and a hidden row vertex v ($\rho = 4\tau$, $\kappa = \tau/4$, $\tau = \sqrt{\delta(P)}$), with $F_v = \{j : \|p_j - p_v\|_1 \ge \rho\}$ the (nonempty) ρ-far row set.

A **hiddenness dual witness** of v is a triple (λ, α, β) of nonnegative coefficients — λ_f ($f \in F_v$) with $\sum_f \lambda_f = 1$ (a probability on the far rows), and $\alpha_i, \beta_i \ge 0$ over all rows with $\sum_i \beta_i = t^*(v) < \kappa$ — satisfying the *witness balance*

$$\sum_{f \in F_v} \lambda_f\,(p_f - p_v) \;+\; \sum_i \alpha_i\,(p_i - p_v) \;=\; \sum_i \beta_i\,(p_i - p_v).$$

It is the LP dual of the exposedness program (see `def-exposed`); the *small-beta* property is $\sum_i \beta_i = t^*(v) < \kappa$. A witness is **reduced** when redundant centered-zero constraints are deleted; by `lem-always-tight-dual-support` a reduced optimal witness has $\mathrm{supp}(\lambda) \subseteq T$, $\mathrm{supp}(\beta) \subseteq O$, and $\mathrm{supp}(\alpha) \subseteq Z$ (the always-tight far / upper-box / zero-face families).

Kind. original (introduced by this project); `status: draft`. **Aliases.** dual witness; dual certificate; small-beta witness; reduced optimal witness; witness balance; (lambda, alpha, beta).
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-hiddenness-dual-witness.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-hiddenness-dual-witness (the LP-dual object) + lem-always-tight-dual-support (support localization).
Used by 1 registry result, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-dual-witness.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; from `lem-hiddenness-dual-witness` (existence via finite LP duality). The witness converts the scalar hiddenness fact $t^*(v) < \kappa$ into a geometric object: a convex combination of ρ-far rows reproducing p_v up to controlled signed slack. "dual certificate" is used interchangeably in the shards. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-zero-face`, `def-actor-hull`, `def-co-top`.

Definition 10 (zero-face row (`def-zero-face`))**.** Fix an exact signed idempotent P and a hidden (not (ρ,κ)-exposed) geometrically distinct row vertex u, with the exposedness LP at u (the program defining $t^*(u)$, see `def-exposed`) and an *optimal exposer* h^* (an admissible affine h attaining $t^*(u)$).

A row z is a **zero-face row** of u if $h^*(p_z) = 0$, i.e. z lies on the zero face $\{x : h^*(x) = 0\}$ of the optimal exposer. The **always-tight zero-face family** $Z(u)$ is the set of rows whose lower-box constraint $h \geq 0$ is tight on the *whole* primal optimal face of the LP (equivalently, $\mathrm{supp}(\alpha) \subseteq Z(u)$ for every reduced optimal hiddenness dual witness (λ, α, β), by `lem-always-tight-dual-support`). The **zero-face conic mass** is $\sum_{z \in Z(u)} a_z$ for the reduced optimal display's zero-face coefficients $a_z \geq 0$.

By `lem-zero-face-localization` every zero-face row is ρ-near u ($\|p_z - p_u\|_1 < 4\tau$, $\tau = \sqrt{\delta}$); if u is within 4τ of a hidden top v of height H then additionally $\|p_z - p_v\|_1 < 8\tau$ and z has depth $> H - 8\tau$.

Kind. original (introduced by this project); `status: draft`. **Aliases.** zero face; zero-face family; always-tight zero-face family; Z(u); h-zero row; zero-face conic mass.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-zero-face-localization.md and lem-always-tight-dual-support.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-zero-face-localization (rho-nearness) + lem-always-tight-dual-support (the alpha-carrier family Z).
Used by 2 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-zero-face.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the vocabulary in which the huddle-charge terminal node is posed (kill or bound the zero-face conic mass). Distinguish two uses in the shards, kept consistent here: the *pointwise* zero-face row ($h^*(p_z) = 0$ for one fixed optimal exposer) and the *always-tight family* $Z(u)$ (tight on the entire optimal face). $Z(u)$ is the always-tight version; statements that fix one optimal exposer h^* use the pointwise version, and the two agree on the relative interior optimal exposer. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-actor-hull` (the always-tight far/upper-box hulls K_T, K_O), `def-dual-witness`.

Definition 11 (co-top (`def-co-top`))**.** Fix an exact signed idempotent P with nonempty visible set $\mathcal{W}$ and a hidden top vertex v of height H; write $d_j = \mathrm{dist}_1(p_j, \mathrm{conv}\{p_w : w \in \mathcal{W}\})$, $\tau = \sqrt{\delta(P)}$.

A row f is **co-top** (relative to v, at slack c) if it sits nearly as deep as the top:

$$d_f \;>\; H - c\,\tau \qquad (c > 0 \text{ a small universal constant, typically } c \in \{4, 8\}).$$

The **co-top far web** of v is the set of rows that are simultaneously co-top **and** ρ-far from v ($\|p_f - p_v\|_1 \geq 4\tau$) — the **starved set** $\{j : \|p_j - p_v\|_1 \geq 4\tau,\ d_j > H - 8\tau\}$. By `lem-cotop-`

`witness-pinning` more than 13/16 of a hidden top's dual-witness λ-mass sits in this starved set (at $c = 4$, $\delta \leq 1/4$), and by `lem-downhill-cotop-conic-mass` (disjoint always-tight hulls) a definite share of the zero-face conic mass lands on nonclone, ρ-near, co-top rows.

Kind. original (introduced by this project); `status: draft`. **Aliases.** co-top row; co-top vertex; co-top web; co-top far web; co-top zero-face mass; starved set.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-cotop-witness-pinning.md and lem-downhill-cotop-conic-mass.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-cotop-witness-pinning + lem-downhill-cotop-conic-mass.
Used by 40 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-co-top.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "co-top" names the deep-far actors that a hidden top's dual witness is forced to charge yet its positive coefficient mass need not reach (the dual-required / primal-starved tension at the heart of the huddle charge and `conj-cotop-web-coupling`). 'status: draft' — A+B sign-off pending (Rule 7). Related: `def-dual-witness`, `def-actor-hull`, `def-near-cluster`.

Definition 12 (invisible mass (`def-invisible-mass`))**.** Let $C_{\mathcal{W}} := \mathrm{conv}\{p_w : w \in \mathcal{W}\}$ (`def-visible-set`) and let p_v be a row vertex. The *invisible (positive) mass* of v is

$$\widetilde{\sigma}_v := \sum_{j\,:\,\mathrm{dist}_1(p_j,\,C_{\mathcal{W}})>0} \max\{P_{vj},\, 0\},$$

the total positive coefficient mass that row v places on rows lying strictly outside $C_{\mathcal{W}}$ — *including* the index $j = v$ itself when $p_v \notin C_{\mathcal{W}}$ and $P_{vv} > 0$.

Halo robustness. Replacing the condition $\mathrm{dist}_1(p_j, C_{\mathcal{W}}) > 0$ by $\mathrm{dist}_1(p_j, C_{\mathcal{W}}) > \varepsilon$ changes $\widetilde{\sigma}_v$ only by mass sitting in the ε-halo of $C_{\mathcal{W}}$, which is absorbable as an ε-loss in the height; the $\varepsilon = 0$ form is used throughout.

Kind. original (introduced by this project); `status: locked`. **Aliases.** invisible positive mass; $\widetilde{\sigma}$_v; invisible mass $\widetilde{\sigma}$_v.
Provenance. `source: internal`; locus `quoted from classical-portfolio/report/kernel-conjecture.tex §Setting (Definition: Invisible mass` $\widetilde{\sigma}$`_v; Remark: Halo robustness)`; no source hash (not a literature transcription); record: adapted from kernel-conjecture.tex §Setting (Definition: Invisible mass $\widetilde{\sigma}$_v, def:sigt).
Used by 35 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-invisible-mass.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Quoted from `classical-portfolio/report/kernel-conjecture.tex` §Setting (Definition: *Invisible mass* $\widetilde{\sigma}_v$) with the *Halo robustness* remark. $\widetilde{\sigma}_v$ is the hypothesis quantity of the Kernel Conjecture: a hidden top vertex (`def-height`) with $\widetilde{\sigma}_v > \tau = \sqrt{\delta}$ is the missing branch, while the $\widetilde{\sigma}_v \leq \tau$ branch is the proved height cap. Uses `def-visible-set`, `def-negative-mass`.

Definition 13 (near-cluster (`def-near-cluster`))**.** Fix an exact signed idempotent P with nonempty visible set $\mathcal{W}$, a hidden top vertex v of height H, $\tau = \sqrt{\delta(P)}$, $\rho = 4\tau$, and a width constant $a \geq 4$. Write $d_j = \mathrm{dist}_1(p_j, \mathrm{conv}\{p_w : w \in \mathcal{W}\})$.

The **near-cluster** (or ρ-near deep cluster) of v is

$$C(v) \;:=\; \{\, j : \;\|p_j - p_v\|_1 < 4\tau \;\text{ and }\; d_j > a\,\tau \,\}$$

— rows simultaneously ρ-near v **and** deep (both conditions required). The **cluster mass** is $\sum_{j\in C(v)} \max(P_{vj}, 0)$; a top is **heavy** when this is $\geq 1 - \theta_0$. A **cluster vertex** is a geometrically distinct row vertex u carrying positive disintegrated weight from $C(v)$-rows (or, in the re-rooted assembly, $u := v$ itself).

Kind. original (introduced by this project); `status: draft`. **Aliases.** rho-near deep cluster; near-deep cluster; cluster C(v); C(v); huddle cluster; cluster mass.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/conj-near-cluster-absorption.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by conj-near-cluster-absorption (the summand set C(v)) and the W54 huddle-charge tree.
Used by 3 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-near-cluster.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the exact summand set of `conj-near-cluster-absorption` and the pinned target of the W54 huddle-charge decomposition tree. The width a is a calibration constant (any universal $a \geq 4$; $a = 16$ in the W54 assembly), not load-bearing downstream. NOTE ON DIVERGENCE: the bare word "cluster" is also used generically for an arbitrary row-index subset C (e.g. `lem-cluster-return-flow`'s return flow over any subset) — that generic set-theoretic use is common knowledge and is NOT this term; this shard defines only the huddle near-cluster $C(v)$. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-co-top`, `def-slab`.

Definition 14 (near-positive projection (`def-near-positive-projection`)). A *near-positive projection* is a linear map $R\colon \ell_n^\infty \to \ell_n^\infty$ that is an exact unital idempotent, $R^2 = R$ and $R(\mathbf{1}) = \mathbf{1}$, and is *δ-positive*: for every $0 \leq x$ (entrywise) with $\|x\|_\infty \leq 1$ one has $Rx \geq -\delta\,\mathbf{1}$, for a small parameter $\delta \in [0, \delta_0)$. As a matrix this is exactly an exact signed idempotent whose negative mass satisfies $\delta(R) \leq \delta$ (and $\|R\| \leq 1 + O(\delta)$).

The motivating operator-algebra example is the spectral idempotent $P = \theta(2\Phi - \mathbf{1})$ built from an `def-almost-idempotent` Φ, with $\delta = O(\eta)$; such a P need **not** be a genuinely positive map. *Near-positive-projection stability* asks whether every near-positive projection is within $C\sqrt{\delta}$ (operator norm) of a genuine *positive* unital idempotent — a stochastic idempotent E. The signed and stochastic-defect formulations are equivalent up to universal constants by `lem-classical-equiv` (af-validated), and the corresponding stochastic upper-bound theorem is `op-classical` (af-validated 2026-08-08). Sharpness of the $\sqrt{}$ exponent in the stochastic parameter η is the registry row `cor-classical-sharpness` (see its shard for status); it is not a certificate for sharpness in the signed parameter δ used by THIS definition. No signed-δ sharpness claim is currently established at any rigorous rung: the historical 3×3 family record (`ex-hume`) is `disproved` as literally stated, and its corrected distance-to-set statement remains an unproved candidate.

Kind. original (introduced by this project); `status: locked`. **Aliases.** δ-positive unital idempotent; delta-positive unital idempotent; near-idempotent stochastic map; near-positive idempotent.
Provenance. `source: internal`; locus `adopted from ../almost-idempotent-positive-maps/definitions/def-near-positive-projection.md (commutative shadow)`; no source hash (not a literature transcription); record: adopted from ../almost-idempotent-positive-maps definitions/def-near-positive-projection.md (B's factorization program, op:npps); commutative shadow agreed for this repo; sharpness/status remarks corrected 2026-08-08 (fresh hostile review: AUDIT-EXHUME-SHARPNESS-V3.md finding 1; user-ratified W139 package; defined object unchanged).
Used in this report by `op-classical` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-near-positive-projection.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Adopted from `../almost-idempotent-positive-maps/definitions/def-near-positive-projection.md` (B's factorization program, `op:npps`), taken here as the commutative shadow / general parent of the classical object. This is the map/positivity framing of the same object as `def-signed-idempotent` (matrix/geometry framing): R is δ-positive iff $\delta(R) \leq \delta$. The *stability* statement is not part of this definition; see `lem-classical-equiv`, `op-classical`, and `cor-classical-sharpness` for the separate formulations and their statuses.

Definition 15 (top support functional (`def-top-support-functional`)). Fix an exact signed idempotent P with nonempty visible set $\mathcal{W}$, write $C_{\mathcal{W}} = \operatorname{conv}\{p_w : w \in \mathcal{W}\}$, and let v be a hidden top vertex of height H.

A **top support functional** at v is an affine $\phi : \mathbb{R}^n \to \mathbb{R}$ with

$$\phi(p_v) = H, \qquad \phi \leq 0 \text{ on } C_{\mathcal{W}}, \qquad \phi \text{ is 1-Lipschitz for } \ell^1.$$

The set of all such is Φ_v; it is nonempty (`lem-top-deficit-price`), convex and compact, and by `lem-top-support-dual-face` equals $\{\phi_y(x) = y \cdot x - h_C(y) : y \in Y_v\}$ where $h_C(y) = \sup_{c \in C_{\mathcal{W}}} y \cdot c$ and $Y_v = \{y : \|y\|_\infty \le 1,\ y \cdot p_v - h_C(y) = H\}$. A finite convex average of top support functionals (an **averaged** $\bar{\phi}$) is again a top support functional (the three defining conditions are convex).

Kind. original (introduced by this project); `status: draft`. **Aliases.** top support functional phi; Phi_v; support functional at the top; averaged phi-bar.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-top-deficit-price.md and lem-top-support-dual-face.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-top-deficit-price (existence) + lem-top-support-dual-face (the dual face Phi_v).
Used by 39 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-top-support-functional.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the affine functional that realizes the height of the top and pins the top-deficit $z_j = H - \phi(p_j) \ge 0$. It is the charging channel of `lem-top-deficit-price`. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-top-deficit`, `def-height`, `def-visible-set`.

Definition 16 (selected-corner configuration (`def-selected-corner`)**).** Fix a finite exact signed idempotent P with $0 < \delta(P)$, $\tau = \sqrt{\delta(P)}$, $D = 2+4\delta(P)$, nonempty visible set $\mathcal{W}$, $C_{\mathcal{W}} = \mathrm{conv}\{p_w : w \in \mathcal{W}\}$, and a hidden top v of height $H > 16\tau$. A **selected-corner configuration** is such (P, v) together with:

- an affine ϕ with $\phi(p_v) = H$, $\phi \le 0$ on $C_{\mathcal{W}}$, and $|\phi(a) - \phi(b)| \le \|a - b\|_1$ (a top support functional), an admissible exposer h at v, and a **selected corner row** f with $\|p_f - p_v\|_1 \ge 4\tau$, $\mathrm{dist}_1(p_f, C_{\mathcal{W}}) > H - 4\tau$ (co-top and ρ-far) satisfying the corner inequality $2(H - \phi(p_f))/D + h(p_f) \le 12\tau/13$;
- a **disintegration kernel** $\xi_x(u)$ from row points to geometrically distinct row vertices, constant on clone fibers, with $p_x = \sum_u \xi_x(u) p_u$ and Dirac at vertex points.

Write $P^+_{fx} = \sum_{j : p_j = p_x} \max(P_{fj}, 0)$, $\Gamma_f(x, u) = P^+_{fx}\, \xi_x(u)$, $C_f = \{(x, u) : H - \phi(p_x) < 4\tau,\ h(p_x) < 4\tau,\ H - \phi(p_u) < 4\tau,\ h(p_u) < 4\tau\}$, $B_F = C_f \cap \{\|p_u - p_v\|_1 \ge 4\tau\}$, $B_N = C_f \cap \{\|p_u - p_v\|_1 < 4\tau\}$. For a block $B \in \{B_F, B_N\}$ the three **corner masses** are

$$M_X(B) = \Gamma_f\{(x, u) \in B : p_x \ne p_u\}, \quad M_I(B) = \Gamma_f\{(x, u) \in B : p_x = p_u,\ K_T(u) \cap K_O(u) \ne \emptyset\},$$

$$M_D(B) = \Gamma_f\{(x, u) \in B : p_x = p_u,\ K_T(u) \cap K_O(u) = \emptyset\},$$

with $K_T(u), K_O(u)$ the always-tight hulls at u (off-diagonal X = distinct $p_x \ne p_u$; intersection I = diagonal with intersecting hulls; deep D = diagonal with disjoint hulls).

Kind. original (introduced by this project); `status: draft`. **Aliases.** selected corner; corner configuration; disintegration kernel xi; Gamma_f; M_X; M_I; M_D; B_F; B_N; C_f.
Provenance. `source: internal`; locus `internal; first pinned (inlined verbatim) in argument/lemmas/lem-sl1a-three-cell-reduction.md and the three conj-sl1a-*-cell shards`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-sl1a-three-cell-reduction (the SL1a three-cell surface).
Used by 37 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-selected-corner.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; the single home of the SL1a "corner" machinery inlined verbatim in `lem-sl1a-three-cell-reduction` and the three cell conjectures. The three cells partition the selected corner's mass into the $X/I/D$ diagonal cells. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-actor-hull`, `def-co-top`, `def-top-support-functional`, `def-top-deficit`.

Definition 17 (top-deficit (`def-top-deficit`))**.** In the setting of `def-top-support-functional` — an exact signed idempotent P, nonempty visible set, hidden top v of height H, and a top support functional $\phi \in \Phi_v$ — the **top-deficit** of a row j under ϕ is

$$z_j \ := \ H - \phi(p_j) \ \geq \ 0,$$

nonnegative because $\phi(p_j) \leq \mathrm{dist}_1(p_j, C_{\mathcal{W}}) \leq H$ (and bounded above, $z_j \leq 2{+}4\delta$). The **top-deficit supremum** of a row f is $Z_v(f) := \sup_{\phi\in\Phi_v} \big(H - \phi(p_f)\big) = \sup_{y\in Y_v} y \cdot (p_v - p_f)$ (`lem-top-support-dual-face`). A **top-deficit slab** is a band of rows cut by a threshold on z, e.g. $\{j : z_j \geq L\}$.

Kind. original (introduced by this project); `status: draft`. **Aliases.** top deficit; z_j; deficit; top-deficit slab; Z_v(f); top-deficit supremum.
Provenance. `source: internal`; locus `internal; first pinned in argument/lemmas/lem-top-deficit-price.md and lem-top-support-dual-face.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by lem-top-deficit-price (z_j) + lem-top-support-dual-face (the supremum Z_v(f)).
Used by no registry result yet (vocabulary registered ahead of the argument).
Canonical shard. `definitions/def-top-deficit.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "deficit" alone always means this top-deficit in the huddle-charge shards. The charging identity (`lem-top-deficit-price`, from $P^2 = P$ and $P\mathbf{1} = \mathbf{1}$) is $\sum_j a_j^+ z_j = \sum_j a_j^- z_j \leq \nu_v(2 + 4\delta)$ with $a_j = P_{vj}$; its structural blind spot is that rows in the ρ-ball of v carry $z_j < 4\tau$. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-top-support-functional`, `def-slab`, `def-co-top`.

Definition 18 (slab (`def-slab`))**.** Fix an exact signed idempotent P, $\tau = \sqrt{\delta(P)}$, nonempty visible set $\mathcal{W}$. A **slab** is the set of rows on which a fixed affine functional lies in a prescribed threshold band. Two families recur, kept distinct here:

- **Low slab** (exposer-value): for an optimal exposer h^* at a vertex and a threshold s, the set $\{j : h^*(p_j) < s\}$. The **deep / very-low slab** takes $s = \kappa = \tau/4$ or $s = \tau/8$; combined with depth ($d_j > a\tau$) it is the summand set of `conj-low-slab-cap` and `conj-far-low-slab-cap`. The pincer `lem-cs-low-slab-pincer` controls its complementary shell $\{s \leq h^* < \kappa\}$.
- **Top slab** (depth / top-deficit): the near-top band $\{j : d_j > H - c\tau\}$ (equivalently small top-deficit $z_j < c'\tau$); the **far slab** additionally imposes ρ-farness. `lem-top-slab-companion` forces a ρ-far row in the top slab.

Kind. original (introduced by this project); `status: draft`. **Aliases.** low slab; low-slab; deep low slab; very-low slab; top slab; top-slab; far slab; slab leakage.
Provenance. `source: internal`; locus `internal; first pinned across argument/lemmas/conj-low-slab-cap.md, lem-cs-low-slab-pincer.md, lem-top-slab-companion.md`; no source hash (not a literature transcription); record: project-original; sign-off pending (Rule 7 lock). First pinned by conj-low-slab-cap (exposer-value slab) and lem-top-slab-companion (depth slab).
Used by 2 registry results, none of them reproduced in this report (exploration track; see `argument/INDEX.md` and `report/UNWIRED.md`).
Canonical shard. `definitions/def-slab.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Project-original; "slab" is overloaded in the shards between the exposer-value cut (h^*) and the depth cut (d / z) — this shard records BOTH canonical instances and which functional each uses, so downstream statements can name the functional explicitly and avoid drift. `status: draft` — A+B sign-off pending (Rule 7). Related: `def-top-deficit`, `def-near-cluster`, `def-exposed`.

The approximate C^*-algebra vocabulary (transcribed from the pinned source)

Definition 19 (level-one and amplified column-Hilbert corner (`def-column-hilbert-corner`))**.** Let P and Q be δ-projections in an ε-C^* algebra. If Q is one-dimensional, then $\mathcal{S}_{P,Q}$ is a Hilbert space with the inner product $\langle\cdot,\cdot\rangle$ defined by the equation

$$Y^\dagger \cdot X = \langle Y, X\rangle\, \widetilde{Q} \qquad (X, Y \in \mathcal{S}_{P,Q}),$$

where $Y^\dagger \cdot X = \mathrm{C}_Q(Y^\dagger X)$ and $\widetilde{Q} = \mathrm{C}_Q(Q)$.

One-dimensional projections and the corresponding Hilbert spaces. Lemma [`lem_PQ_Hilb`] says that if $P, Q \in \mathcal{A}$ are δ-projections and Q is one-dimensional, then the space $\mathcal{S}_{P,Q}$ is equipped with the Hermitian inner product given by [`1dQ_ip`]. The extended version of this space, $\mathbb{C}^n \otimes \mathcal{S}_{P,Q} = \mathbf{M}_{n,1} \otimes \mathcal{S}_{P,Q}$, satisfies essentially the same equation:

$$Y^\dagger \cdot X = \langle Y, X\rangle\, \widetilde{Q} \qquad (X, Y \in \mathbf{M}_{n,1} \otimes \mathcal{S}_{P,Q}),$$

where $\langle Y, X\rangle = \sum_l \langle [Y]_{l1}, [X]_{l1}\rangle$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** column-Hilbert corner; amplified column space.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1123-1128,1546-1550`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\CC`, `\Co`, `\Ma`, `\braket`, `\calA`, `\calS`, `\dag`, `\eps`, `\wt`; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source theorem-environment delimiters dropped; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-compcb-row-column-product`, `lem-hcb-column-hilbert-squared` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-column-hilbert-corner.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking. The known false unsquared display at source lines 1551-1555 is deliberately not part of this definition.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; the exclusion is required by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and is consistent with `VERDICT-W74F-H-STAGE1.md`.

SOURCE CHECK (BYTE-VERBATIM) — `KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1123-1128,1546-1550`

```
\begin {Lemma}\label {lem_PQ_Hilb}
Let $P$ and $Q$ be $\delta $-projections in an $\eps $-$C^*$ algebra. If $Q$ is one-dimensional, then $\calS
_{P,Q}$ is a Hilbert space with the inner product $\braket {\cdot }{\cdot }$ defined by the equation
\begin {equation}\label {1dQ_ip}
Y^\dag \cdot X=\braket {Y}{X}\,\wt {Q}\qquad (X,Y\in \calS _{P,Q}),
\end {equation}
where $Y^\dag \cdot X=\Co _{Q}(Y^{\dag }X)$ and $\wt {Q}=\Co _Q(Q)$.
```

```
\noindent \textbf {One-dimensional projections and the corresponding Hilbert spaces.} Lemma~\ref {lem_PQ_Hilb}
says that if $P,Q\in \calA $ are $\delta $-projections and $Q$ is one-dimensional, then the space $\calS
_{P,Q}$ is equipped with the Hermitian inner product given by \eqref {1dQ_ip}. The extended version of this
space, $\CC ^n\otimes \calS _{P,Q}=\Ma {n,1}\otimes \calS _{P,Q}$, satisfies essentially the same equation:
\begin {equation}\label {1dQ_ip_ext}
Y^\dag \cdot X=\braket {Y}{X}\,\wt {Q}\qquad (X,Y\in \Ma {n,1}\otimes \calS _{P,Q}),
\end {equation}
where $\braket {Y}{X}=\sum _{l}\braket {[Y]_{l1}}{[X]_{l1}}$.
```

Definition 20 (compressed corner (`def-compressed-corner`))**.** To each pair of projections (P, Q) in a C^* algebra $\mathcal{B}$, one assigns the vector space $\mathcal{S}_{P,Q} = \{PXQ : X \in \mathcal{B}\}$. In this section, we discuss an approximate version of this construction. Recall that a δ-projection in an ε-C^* algebra $\mathcal{A}$ is a Hermitian element P such that $\|P^2 - P\| \le \delta$. For each pair (P, Q) of δ-projections, there are two slightly different maps that could be called "compressions": $\mathrm{L}_P\mathrm{R}_Q\colon X \mapsto P(XQ)$ and $\mathrm{R}_Q\mathrm{L}_P\colon X \mapsto (PX)Q$. We will use their symmetric combination, $\frac{1}{2}(\mathrm{L}_P\mathrm{R}_Q + \mathrm{R}_Q\mathrm{L}_P)$. It is an $O(\delta+\varepsilon)$-idempotent element of the algebra of operators acting on $\mathcal{A}$, and therefore, can be approximated by an idempotent using Proposition [`prop_P`]. The *compression map* thus defined,

$$\mathrm{C}_{P,Q}\colon \mathcal{A}\to\mathcal{A},\qquad \mathrm{C}_{P,Q}=\theta(\mathrm{L}_P\mathrm{R}_Q+\mathrm{R}_Q\mathrm{L}_P-1),$$

satisfies the equations

$$\mathrm{C}_{P,Q}^2=\mathrm{C}_{P,Q},\qquad \mathrm{C}_{P,Q}(X)^\dagger=\mathrm{C}_{Q,P}(X^\dagger)\quad (X\in\mathcal{A}),$$

and both $\|\mathrm{L}_P\mathrm{R}_Q-\mathrm{C}_{P,Q}\|$ and $\|\mathrm{R}_Q\mathrm{L}_P-\mathrm{C}_{P,Q}\|$ are bounded by $O(\delta+\varepsilon)$. The image of this map, $\mathcal{S}_{P,Q}=\operatorname{Im}\mathrm{C}_{P,Q}=\operatorname{Ker}(1-\mathrm{C}_{P,Q})$, is a closed linear subspace of $\mathcal{A}$. There is a variant of this construction where only L_P or only R_Q is applied, and one writes $\mathrm{C}_{P,1},\mathcal{S}_{P,1}$ or $\mathrm{C}_{1,Q},\mathcal{S}_{1,Q}$, respectively. (This is the same as $\mathrm{C}_{P,I}$, etc. if the unit is exact.)

When $P=Q$, the abbreviations C_P, $\mathcal{S}_P$ are used.

For any triple (P,Q,R) of δ-projections, one defines the *compressed product* between elements of the corresponding subspaces:

$$(X,Y)\mapsto X\cdot Y:\ \mathcal{S}_{P,Q}\times\mathcal{S}_{Q,R}\to\mathcal{S}_{P,R},\qquad X\cdot Y=\mathrm{C}_{P,R}(XY).$$

It is close to the ambient product in $\mathcal{A}$, namely, $\|X\cdot Y-XY\|\le O(\delta+\varepsilon)\|X\|\|Y\|$. If P is a nonvanishing projection, the compressed product turns the Banach space $\mathcal{S}_P$ (which is closed under the involution $X\mapsto X^\dagger$) into an $O(\delta+\varepsilon)$-C^* algebra with unit $\widetilde{P}=\mathrm{C}_P(P)$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** compression map; compressed product; compressed unit.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1054-1066,1077-1082`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: \Co, \Img, \Ker, \La, \Ra, \calA, \calB, \calS, \dag, \eps, \ts, \wt; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-compcb-amplified-almost-containment`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-action`, `lem-compcb-compressed-unit-norm`, `lem-compcb-corner-algebra` and 25 further reproduced results (31 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-compressed-corner.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1054-1066,1077-1082

```
To each pair of projections $(P,Q)$ in a $C^*$ algebra $\calB $, one assigns the vector space $\calS
_{P,Q}=\{PXQ:\,X\in \calB \}$. In this section, we discuss an approximate version of this construction. Recall
that a $\delta $-projection in an $\eps $-$C^*$ algebra $\calA $ is a Hermitian element $P$ such that
$\|P^2-P\|\le \delta $. For each pair $(P,Q)$ of $\delta $-projections, there are two slightly different maps
that could be called ``compressions'': $\La _P\Ra _Q\colon X\mapsto P(XQ)$ and $\Ra _Q\La _P\colon X\mapsto
(PX)Q$. We will use their symmetric combination, $\frac {1}{2}(\La _P\Ra _Q+\Ra _Q\La _P)$. It is an $O(\delta
+\eps )$-idempotent element of the algebra of operators acting on $\calA $, and therefore, can be approximated
by an idempotent using Proposition~\ref {prop_P}. The \emph {compression map} thus defined,
\begin {equation}
\Co _{P,Q}\colon \calA \to \calA ,\qquad
\Co _{P,Q}=\theta (\La _P\Ra _Q+\Ra _Q\La _P-1),
\end {equation}
satisfies the equations
\begin {equation}
```

```
\Co _{P,Q}^2=\Co _{P,Q},\qquad \:
\Co _{P,Q}(X)^\dag =\Co _{Q,P}(X^\dag )\quad \: (X\in \calA ),
\end {equation}
and both $\|\La _P\Ra _Q-\Co _{P,Q}\|$ and $\|\Ra _Q\La _P-\Co _{P,Q}\|$ are bounded by $O(\delta +\eps )$.
The image of this map, $\calS _{P,Q}=\Img \Co _{P,Q}=\Ker (1-\Co _{P,Q})$, is a closed linear subspace of
$\calA $. There is a variant of this construction where only $\La _P$ or only $\Ra _Q$ is applied, and one
writes $\Co _{P,1},\calS _{P,1}$ or $\Co _{1,Q},\calS _{1,Q}$, respectively. (This is the same as $\Co
_{P,I}$, etc.\ if the unit is exact.)

When $P=Q$, the abbreviations $\Co _{P}$, $\calS _P$ are used.
```

```
For any triple $(P,Q,R)$ of $\delta $-projections, one defines the \emph {compressed product} between elements
of the corresponding subspaces:
\begin {equation}\label {compr_prod}
(X,Y)\mapsto X\cdot Y\,\colon \,
\calS _{P,Q}\times \calS _{Q,R}\to \calS _{P,R},\qquad X\cdot Y=\Co _{P,R}(XY).
\end {equation}
It is close to the ambient product in $\calA $, namely, $\|X\cdot Y-XY\|\le O(\delta +\eps )\|X\|\ts \|Y\|$.
If $P$ is a nonvanishing projection, the compressed product turns the Banach space $\calS _P$ (which is closed
under the involution $X\mapsto X^\dag $) into an $O(\delta +\eps )$-$C^*$ algebra with unit $\wt {P}=\Co
_{P}(P)$.\medskip
```

Definition 21 (δ-projection and nonvanishing alternative (`def-delta-projection`))**.** A *projection* in an ε-C^* algebra $\mathcal{A}$ is a Hermitian element P such that $P^2 = P$. Such elements are generally hard to come by, see Remark [**`rem_X2`**], so will content ourselves with *δ-projections* for sufficiently small δ. They are defined by the conditions

$$P^\dagger = P, \qquad \|P^2 - P\| \le \delta.$$

$$\|P\| \le O(\delta) \quad \text{or} \quad \big|\|P\| - 1\big| \le O(\delta + \varepsilon).$$

A δ-projection is called *nonvanishing* if the second alternative holds. If P is a δ-projection, then $I - P$ is a δ'-projection, where $\delta' = \delta$ if $\mathcal{A}$ has exact unit and $\delta' = \delta + O(\varepsilon)$ in general. A δ-projection P is called *nontrivial* if both P and $I - P$ are nonvanishing.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** delta-projection; nonvanishing delta-projection.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:917-920,926-929`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: \calA, \dag, \eps; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `conj-extcb`, `conj-hcb`, `lem-compcb-amplified-almost-containment`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-action` and 27 further reproduced results (33 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-delta-projection.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

SOURCE CHECK (BYTE-VERBATIM) — `KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:917-920,926-929`

```
A \emph {projection} in an $\eps $-$C^*$ algebra $\calA $ is a Hermitian element $P$ such that $P^2=P$. Such
elements are generally hard to come by, see Remark~\ref {rem_X2}, so will content ourselves with \emph
{$\delta $-projections} for sufficiently small $\delta $. They are defined by the conditions
\begin {equation}\label {delta_P}
```

```
P^\dag =P,\qquad \|P^2-P\|\le \delta .
\end {equation}
```

```
\begin {equation}\label {P_alternatives}
\|P\|\le O(\delta )\quad \text {or}\quad \bigl |\|P\|-1\bigr |\le O(\delta +\eps ).
\end {equation}
A $\delta $-projection is called \emph {nonvanishing} if the second alternative holds. If $P$ is a $\delta
$-projection, then $I-P$ is a $\delta '$-projection, where $\delta '=\delta $ if $\calA $ has exact unit and
$\delta '=\delta +O(\eps )$ in general. A $\delta $-projection $P$ is called \emph {nontrivial} if both $P$
and $I-P$ are nonvanishing.
```

Definition 22 (ε-C^*-algebra (`def-epsilon-cstar-algebra`))**.** An *ε-Banach algebra* is a Banach space $\mathcal{A}$ endowed with a bilinear multiplication map $\mathcal{A} \times \mathcal{A} \to \mathcal{A}$ such that

$$\|XY\| \le (1+\varepsilon)\|X\|\|Y\| \qquad (X, Y \in \mathcal{A}),$$
$$\|(XY)Z - X(YZ)\| \le \varepsilon\|X\|\|Y\|\|Z\| \qquad (X, Y, Z \in \mathcal{A}).$$

Such an algebra is called *ε'-commutative* if

$$\|XY - YX\| \le \varepsilon'\|X\|\|Y\| \qquad (X, Y \in \mathcal{A}).$$

A *$*\varepsilon$-Banach algebra* is a complex ε-Banach algebra with a conjugate linear involution $X \mapsto X^\dagger$ satisfying the equations

$$\|X^\dagger\| = \|X\|, \qquad (XY)^\dagger = Y^\dagger X^\dagger \qquad (X, Y \in \mathcal{A}).$$

An *ε-C^* algebra* is one with this additional property:

$$\|X^\dagger X\| \ge (1-\varepsilon)\|X\|^2 \qquad (X \in \mathcal{A}).$$

(A bound from the other side, $\|X^\dagger X\| \le (1+\varepsilon)\|X\|^2$, follows from [`ax_prodnorm`] and [`ax_*`].) We assume that all algebras are unital. The unit element $I \in \mathcal{A}$ should satisfy the approximate or exact conditions

$$\begin{array}{llll} \|XI - X\| \le \varepsilon\|X\|, & \|IX - X\| \le \varepsilon\|X\|, & \big|\|I\| - 1\big| \le \varepsilon & \text{(by default)}, \quad \text{or} \\ XI = X, & IX = X, & \|I\| = 1 & \text{(if specified)}. \end{array}$$

If the involution is defined, one also requires that $I^\dagger = I$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** epsilon-C^*-algebra; approximate C^*-algebra.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:407-440`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\calA`, `\dag`, `\eps`, `\ts`; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source theorem-environment delimiters dropped; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-extcb-corner-dimension-additivity`, `lem-extcb-one-dimensional-corner-dimension`, `lem-extcb-one-dimensional-product`, `lem-maincb-corner-nontriviality`, `lem-stage1-approximate-group-laws`, `lem-stage1-approximate-group-laws-transport` and 34 further reproduced results (40 registry results import it in total; see `argument/INDEX.md`).

Canonical shard. `definitions/def-epsilon-cstar-algebra.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft transcription; ratification is required before locking.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:407-440

```
\begin {Definition}
An \emph {$\eps $-Banach algebra} is a Banach space $\calA $ endowed with a bilinear multiplication map $\calA
\times \calA \to \calA $ such that
\begin {alignat}{2}
\label {ax_prodnorm}
\|XY\| &\le (1+\eps )\ts \|X\|\ts \|Y\|\qquad &&(X,Y\in \calA ),\\[2pt]
\label {ax_assoc}
\|(XY)Z-X(YZ)\| &\le \eps \ts \|X\|\ts \|Y\|\ts \|Z\|\qquad &&(X,Y,Z\in \calA ).
\end {alignat}
Such an algebra is called \emph {$\eps '$-commutative} if
\begin {equation}
\label {ax_comm}
\|XY-YX\|\le \eps '\ts \|X\|\ts \|Y\|\qquad (X,Y\in \calA ).
\end {equation}
A \emph {$*\eps $-Banach algebra} is a complex $\eps $-Banach algebra with a conjugate linear involution
$X\mapsto X^\dag $ satisfying the equations
\begin {equation}
\label {ax_*}
\|X^\dag \|=\|X\|,\qquad (XY)^\dag =Y^\dag X^\dag \qquad (X,Y\in \calA ).
\end {equation}
An \emph {$\eps $-$C^*$ algebra} is one with this additional property:
\begin {equation}
\label {ax_C*}
\|X^{\dag }X\|\ge (1-\eps )\ts \|X\|^{2}\qquad (X\in \calA ).
\end {equation}
(A bound from the other side, $\|X^{\dag }X\|\le (1+\eps )\ts \|X\|^{2}$, follows from \eqref {ax_prodnorm}
and \eqref {ax_*}.) We assume that all algebras are unital. The unit element $I\in \calA $ should satisfy the
approximate or exact conditions
\begin {alignat}{4}
\label {ax_eps_unit}
\|XI&-X\|\le \eps \ts \|X\|,\qquad &\|IX&-X\|\le \eps \ts \|X\|,\qquad
&\bigl |\|I\|&-1\bigr |\le \eps \qquad &&\text {(by default)},\quad \text {or}\\[2pt]
\label {ax_exact_unit}
XI&=X,\qquad &IX&=X,\qquad
&\|I\|&=1\qquad &&\text {(if specified)}.
\end {alignat}
If the involution is defined, one also requires that $I^\dag =I$.
\end {Definition}
```

Definition 23 (extended ε-C^*-algebra (`def-extended-epsilon-cstar-algebra`))**.** An *extended ε-C^* algebra* is a complete self-adjoint operator space $\mathcal{A}$ with a multiplication and a unit that make each space $\mathbf{M}_n \otimes \mathcal{A}$ into an ε-C^* algebra. An *extended δ-homomorphism* is a linear map $v\colon \mathcal{A}' \to \mathcal{A}''$ such that for each n, the map $1_{\mathbf{M}_n} \otimes v\colon \mathbf{M}_n \otimes \mathcal{A}' \to \mathbf{M}_n \otimes \mathcal{A}''$ is a δ-homomorphism.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** extended epsilon-C^*-algebra; extended approximate C^*-algebra.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1477-1479`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\Ma`, `\calA`, `\eps`. The byte-verbatim source is reproduced below as the second check.
Used in this report by `conj-extcb`, `conj-hcb`, `lem-compcb-amplified-almost-containment`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-action` and 49 further reproduced results (56 registry results import it in total; see `argument/INDEX.md`).

Canonical shard. `definitions/def-extended-epsilon-cstar-algebra.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notation. Registry contracts write the source's `\(\eps\)` as `\(\varepsilon\)` and `\(\Ma n\)` as `\(M_n\)`.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:1477-1479`

```
An \emph {extended $\eps $-$C^*$ algebra} is a complete self-adjoint operator space $\calA $ with a
multiplication and a unit that make each space $\Ma {n}\otimes \calA $ into an $\eps $-$C^*$ algebra. An \emph
{extended $\delta $-homomorphism} is a linear map $v\colon \calA '\to \calA ''$ such that for each $n$, the
map $1_{\Ma {n}}\otimes v\colon \Ma {n}\otimes \calA '\to \Ma {n}\otimes \calA ''$ is a $\delta
$-homomorphism.
```

Definition 24 (diagonal of a finite-dimensional C^*-algebra (`def-fd-cstar-diagonal`))**.** For arbitrary $\tilde{D} = \sum_j A_j \otimes B_j \in \mathcal{B} \otimes \mathcal{B}$ and $X \in \mathcal{B}$, we write

$$X\tilde{D} = \sum_j XA_j \otimes B_j, \qquad \tilde{D}X = \sum_j A_j \otimes B_jX, \qquad \pi(\tilde{D}) = \sum_j A_jB_j.$$

These operations extend from $\mathcal{B} \otimes \mathcal{B}$ to $\mathcal{B} \hat{\otimes} \mathcal{B}$. An element $D \in \mathcal{B} \hat{\otimes} \mathcal{B}$ is called a *diagonal* if

$$XD = DX \quad \text{for all } X \in \mathcal{B}, \qquad \pi(D) = I_{\mathcal{B}}.$$

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** finite-dimensional C^*-algebra diagonal; algebra diagonal.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1233-1242`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\calB`, `\hotimes`; display environments unnumbered (labels are stripped). The byte-verbatim source is reproduced below as the second check.
Used in this report by `cor-kitaev-diagonal-cpization`, `lem-extcb-exact-target-correction`, `lem-kitaev-diagonal-repair`, `lem-maincb-improvement-iteration`, `lem-maincb-improvement-one-step`, `lem-routef-delta-prime-closeness` and 1 further reproduced result (7 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-fd-cstar-diagonal.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Specialization. The registry uses this definition only when `\(\mathcal B\)` is a finite-dimensional `\(C^*\)`-algebra.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:1233-1242`

```
For arbitrary $\tilde {D}=\sum _{j}A_j\otimes B_j\in \calB \otimes \calB $ and $X\in \calB $, we write
\begin {equation}
X\tilde {D}=\sum _{j}XA_j\otimes B_j,\qquad
\tilde {D}X=\sum _{j}A_j\otimes B_jX,\qquad
\pi (\tilde {D})=\sum _{j}A_jB_j.
\end {equation}
These operations extend from $\calB \otimes \calB $ to $\calB \hotimes \calB $. An element $D\in \calB
\hotimes \calB $ is called a \emph {diagonal} if
\begin {equation}
XD=DX\quad \text {for all }\, X\in \calB ,\qquad \pi (D)=I_\calB .
\end {equation}
```

Definition 25 (H-space and left inversion (`def-h-space-left-inversion`))**.** An *H-space* is a topological space M with basepoint e and a continuous multiplication $\mu\colon M \times M \to M$ with $\mu(e,e) = e$ such that $x \mapsto \mu(x,e)$ and $x \mapsto \mu(e,x)$ are each basepoint-preserving homotopic to the identity. It is *associative* if the two triple-product maps are homotopic. An *inversion map* is a continuous basepoint-preserving $\sigma\colon M \to M$ with both $x \mapsto \mu(\sigma(x), x)$ and $x \mapsto \mu(x, \sigma(x))$ homotopic to the constant map e; when only the first homotopy is required, σ is a *left inversion map*.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** H-space; associative H-space; inversion map; left inversion map.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:895-912`; SHA256 prefix `e7eb512a2ec2438d`; record: byte-match lock per definitions/README.md (cited criterion); provisioned by DESIGN-FUDW-DECOMP-v4.1.md §4.1 (kind "cited / lock after byte-match").
Used in this report by `lem-stage1-bound-inversion-isolation`, `lem-stage1-bound-quotient-index-data`, `lem-stage1-bound-quotient-left-inversion`, `lem-stage1-bound-quotient-local-index`, `lem-stage1-exterior-cohomology`, `lem-stage1-extra-fixed-class` and 5 further reproduced results (11 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-h-space-left-inversion.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Definition only (per the DESIGN-FUDW-DECOMP-v4.1 §4.1 register row: "pinned TeX 895–912, definition only") — no claim about $\mathcal{U}$ or any Stage-1 object is carried by this shard. Source typo note: the source writes $\mu\colon M \to M$ for the multiplication map's type where $\mu\colon M \times M \to M$ is meant (its uses $\mu(x, y)$ are two-argument throughout); the harmonised statement records the corrected type. Elisions `[...]` in the verbatim block skip source sentences about the specific space $\mathcal{U}$ (claims, not definition content). Used by the Stage-1 topology rows `lem-topology-hopf-structure` and (via fixed-point data) the Lefschetz rows; see `def-lefschetz-fixed-point-data`.

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:895-912`

```
By definition, an \emph {H-space} is a topological space $M$ with a basepoint $e$ and a continuous
multiplication map $\mu \colon M\to M$ such that $\mu (e,e)=e$ and
\begin {equation}\label {H-unit}
\bigl (x\mapsto \mu (x,e)\bigr )
\,\sim \,\bigl (x\mapsto x\bigr )
\,\sim \,\bigl (x\mapsto \mu (e,x)\bigr ),
\end {equation}
where ‘‘$f\sim g$’’ means that $f$ and $g$ are homotopic through basepoint-preserving maps. [...] An H-space
$M$ is called \emph {associative} if
\begin {equation}
\bigl ((x,y,z)\mapsto \mu (\mu (x,y),z)\bigr )
\,\sim \,\bigl ((x,y,z)\mapsto \mu (x,\mu (y,z))\bigr ).
\end {equation}
[...] An \emph {inversion map} is a continuous basepoint-preserving map $\sigma \colon M\to M$ such that
\begin {equation}\label {homo_inverse}
\bigl (x\mapsto \mu (\sigma (x),x)\bigr )
\,\sim \,\bigl (x\mapsto e\bigr )
\,\sim \,\bigl (x\mapsto \mu (x,\sigma (x))\bigr ).
\end {equation}
[...] When only the existence of that homotopy is required, $\sigma $ is called a \emph {left inversion map}.
```

Definition 26 (Ha-map (`def-ha-map`))**.** Let Q be a one-dimensional δ-projection in an ε-C^* algebra $\mathcal{A}$. For any δ-projections $P, R \in \mathcal{A}$, we define a map $\mathrm{H}^{Q}_{P,R}\colon \mathcal{S}_{P,R} \to \mathbf{B}(\mathcal{S}_{R,Q}, \mathcal{S}_{P,Q})$. (Here $\mathbf{B}(\mathcal{S}_{R,Q}, \mathcal{S}_{P,Q})$ is the space of bounded linear maps from $\mathcal{S}_{R,Q}$ to $\mathcal{S}_{P,Q}$ with the norm induced by the Euclidean norm $\|X\|_{\mathrm{E}} = \sqrt{\langle X, X\rangle}$. Due to inequality [`1dQ_ip_norm`], the latter is equal to $\|X\|$ up to a $1 \pm O(\varepsilon + \delta)$ factor.) For each $Z \in \mathcal{S}_{P,R}$ and $X \in \mathcal{S}_{R,Q}$, the element $\mathrm{H}^{Q}_{P,R}(Z)(X) \in \mathcal{S}_{P,Q}$ is defined by the condition

$$(Y^\dagger \cdot Z) \cdot X + Y^\dagger \cdot (Z \cdot X) = 2 \left\langle Y, \mathrm{H}^{Q}_{P,R}(Z)(X) \right\rangle \widetilde{Q} \quad \text{for all } Y \in \mathcal{S}_{P,Q}.$$

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** Ha map; Kitaev Ha-map.

Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1146-1149`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\Bo`, `\Euc`, `\Ha`, `\bbraket`, `\braket`, `\calA`, `\calS`, `\dag`, `\eps`, `\wt`; display environments unnumbered (labels are stripped); source cross-reference keys shown in brackets; source-internal \labels dropped. The byte-verbatim source is reproduced below as the second check.
Used in this report by `conj-extcb`, `conj-hcb`, `lem-hcb1-column-action`, `lem-hcb1-variational-identity`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and 7 further reproduced results (13 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-ha-map.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notation. Registry contracts use `Ha` for the source macro `\(\Ha\)`.
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

SOURCE CHECK (BYTE-VERBATIM) — `KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1146-1149`

```
Let $Q$ be a one-dimensional $\delta $-projection in an $\eps $-$C^*$ algebra $\calA $. For any $\delta
$-projections $P,R\in \calA $, we define a map $\Ha ^{Q}_{P,R}\colon \calS _{P,R}\to \Bo (\calS _{R,Q},\calS
_{P,Q})$. (Here $\Bo (\calS _{R,Q},\calS _{P,Q})$ is the space of bounded linear maps from $\calS _{R,Q}$ to
$\calS _{P,Q}$ with the norm induced by the Euclidean norm $\|X\|_\Euc =\sqrt {\braket {X}{X}}$. Due to
inequality \eqref {1dQ_ip_norm}, the latter is equal to $\|X\|$ up to a $1\pm O(\eps +\delta )$ factor.) For
each $Z\in \calS _{P,R}$ and $X\in \calS _{R,Q}$, the element $\Ha ^{Q}_{P,R}(Z)(X)\in \calS _{P,Q}$ is
defined by the condition
\begin {equation}\label {Ha_def}
(Y^\dag \cdot Z)\cdot X+Y^\dag \cdot (Z\cdot X)
=2\,\bbraket {Y}{\Ha ^{Q}_{P,R}(Z)(X)}\,\wt {Q}\quad \text {for all }\,Y\in \calS _{P,Q}.
\end {equation}
```

Definition 27 (Lefschetz number and local fixed-point index (`def-lefschetz-fixed-point-data`))**.** For a smooth self-map f of a compact orientable manifold M: at a fixed point x with $\det(1-f'(x)) \neq 0$ (where $f'(x)\colon T_xM \to T_xM$ is the differential), the *fixed point index* is $\operatorname{ind}(f,x) = \operatorname{sgn}\det(1-f'(x))$; the *Lefschetz number* is $\Lambda(f) = \sum_k (-1)^k \operatorname{Tr} f^{*k}$, where $f^{*k}\colon H^k(M;\mathbb{R}) \to H^k(M;\mathbb{R})$ is the induced map on real cohomology.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** Lefschetz number; fixed point index; fixed-point index; ind(f,x); Lambda(f).
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:961-967`; SHA256 prefix `e7eb512a2ec2438d`; record: byte-match lock per definitions/README.md (cited criterion); provisioned by DESIGN-FUDW-DECOMP-v4.1.md §4.1 (kind "cited / lock after byte-match").
Used in this report by `lem-stage1-bound-inversion-isolation`, `lem-stage1-bound-quotient-index-data`, `lem-stage1-bound-quotient-local-index`, `lem-stage1-extra-fixed-class`, `lem-stage1-quotient-inversion-index-data`, `lem-topology-lefschetz-hopf` and 1 further reproduced result (7 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-lefschetz-fixed-point-data.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Definitions only (per the DESIGN-FUDW-DECOMP-v4.1 §4.1 register row: "pinned TeX 957–967, definitions only"): this shard carries the two displayed definitions and NO theorem content — in particular it does not assert the Lefschetz–Hopf identity $\sum_{x\in \mathrm{Fix}(f)} \operatorname{ind}(f,x) = \Lambda(f)$ (that is the separate result row `lem-topology-lefschetz-hopf`, to be `cited` from Arkowitz–Brown), nor the index-sign theorem (`lem-topology-local-index-sign`, Granas–Dugundji). The source's index formula is stated in the nondegenerate ($\det \neq 0$) smooth case; the general topological index the theorem rows use is the standard one in their own sources. See `def-h-space-left-inversion`.

SOURCE CHECK (BYTE-VERBATIM) — `KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:961-967`

```
The \emph {fixed point index} $\ind (f,x)$ and the \emph {Lefschetz number} $\Lambda (f)$ are defined as
follows:
\begin {alignat}{2}
\ind (f,x)&=\sgn \det (1-f'(x)),\qquad
```

```
& f'(x)\colon \Ta _xM&\to \Ta _xM,\\[2pt]
\Lambda (f)&=\sum _{k}(-1)^k\Tr f^{*k},\qquad
& f^{*k}\colon H^k(M;\RR )&\to H^k(M;\RR ),
\end {alignat}
```

Definition 28 (one-dimensional δ-projection and equivalence (`def-one-dimensional-delta-projection`))**.** When $P=Q$, the abbreviations C_P, $\mathcal{S}_P$ are used. It is clear that $\mathcal{S}_P=0$ if and only if P is sufficiently close to 0 as described by the first alternative in [`P_alternatives`]. (Recall that in the opposite case, we call P "nonvanishing".) Similarly, $\mathrm{C}_P=1$ and $\mathcal{S}_P=\mathcal{A}$ if and only if P is close to I. If $\dim\mathcal{S}_P=1$, we say that P is a *one-dimensional δ-projection.*

One-dimensional δ-projections P and Q are called *equivalent* if $\dim\mathcal{S}_{P,Q}=1$. This is indeed an equivalence relation; in particular, it is transitive due to Lemma [`lem_PQR`].

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** one-dimensional delta-projection; equivalent delta-projections.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1066,1187`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\Co`, `\calA`, `\calS`; source cross-reference keys shown in brackets. The byte-verbatim source is reproduced below as the second check.
Used in this report by `conj-hcb`, `lem-extcb-one-dimensional-corner-dimension`, `lem-extcb-one-dimensional-product`, `lem-maincb-corner-equivalence`, `lem-maincb-cross-class-merging-datum`, `lem-maincb-cross-datum-bijectivity` and 2 further reproduced results (8 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-one-dimensional-delta-projection.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Locked after the cited-definition byte-match criterion was user-ratified on 2026-07-24.
Provenance. Byte-verbatim from the pinned source at the recorded loci and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

Source check (byte-verbatim) — `kitaev-2405.02434, approximate_algebras.tex:1066,1187`

```
When $P=Q$, the abbreviations $\Co _{P}$, $\calS _P$ are used. It is clear that $\calS _P=0$ if and only if
$P$ is sufficiently close to $0$ as described by the first alternative in \eqref {P_alternatives}. (Recall
that in the opposite case, we call $P$ ``nonvanishing''.) Similarly, $\Co _P=1$ and $\calS _P=\calA $ if and
only if $P$ is close to $I$. If $\dim \calS _P=1$, we say that $P$ is a \emph {one-dimensional $\delta
$-projection}.
```

```
One-dimensional $\delta $-projections $P$ and $Q$ are called \emph {equivalent} if $\dim \calS _{P,Q}=1$. This
is indeed an equivalence relation; in particular, it is transitive due to Lemma~\ref {lem_PQR}.
```

Definition 29 (operator space (`def-operator-space`))**.** A complex vector space $\mathcal{L}$ is called an *operator space* if each space $M_n\otimes\mathcal{L}$ (for $n=1,2,\dots$) is equipped with a norm $\|\cdot\|_n$ satisfying the following axioms:

$$\|AXB\|_n\le\|A\|\,\|X\|_k\,\|B\|\qquad(A\in M_{n,k},\ B\in M_{k,n},\ X\in M_k\otimes\mathcal{L}),$$

$$\left\|\begin{pmatrix}X&0\\0&Y\end{pmatrix}\right\|_{k+n}=\max\bigl\{\|X\|_k,\|Y\|_n\bigr\}\qquad(X\in M_k\otimes\mathcal{L},\ Y\in M_n\otimes\mathcal{L}).$$

The norm on $\mathcal{L}$ itself is defined by identifying $\mathcal{L}$ with $M_1\otimes\mathcal{L}$. An operator space is called *self-adjoint* if it is equipped with a conjugate-linear involution $\dagger$ that preserves all norms $\|\cdot\|_n$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** self-adjoint operator space.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1453-1464 (Definition def:opspace)`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; byte-verified this date).
Used in this report by `lem-maincb-compressed-corner-unit-comparison`, `lem-maincb-direct-sum-inclusion-merge`, `lem-maincb-extended-inclusion-monotone`, `lem-maincb-full-corner-identification`, `lem-maincb-improvement-iteration`, `lem-maincb-improvement-one-step` and 10 further reproduced results (16 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-operator-space.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Byte-verbatim from the local source Definition block at `approximate_algebras.tex:1453-1464` ONLY (per `AUDIT-MAIN-STRUCTURE-v3.md` §2: the rectangular norms/inclusions constructed at `approximate_algebras.tex:1467-1475` are DERIVED notation/consequences, referenced in later result-row provenance but deliberately NOT primitive fields of this cited definition). Provisioned as the P0 vocabulary gate of the MAIN-CB structural design (`DESIGN-MAIN-STRUCTURE-v5.md` §1.1). Related: `def-epsilon-cstar-algebra`, `def-extended-epsilon-cstar-algebra`, `def-compressed-corner`.

Definition 30 (projection basis of a finite-dimensional commutative C^*-algebra (`def-projection-basis`))**.** The second application of Lemma [`lem_merging`] involves an auxiliary statement about δ-projections $P_1,\dots,P_p,Q_1,\dots,Q_q$ satisfying certain conditions, which will be formulated in terms of inclusions of finite-dimensional commutative C^* algebras. Such an algebra $\mathcal{B}$ is described by a *projection basis* $\{\Pi_1,\dots,\Pi_n\}\subseteq\mathcal{B}$ such that $\Pi_j^\dagger=\Pi_j,\ \Pi_j\Pi_k=\delta_{jk}\Pi_k$, and $\sum_{j=1}^{n}\Pi_j=I$.

Kind. cited (byte-matched to a pinned source under `refs/`); `status: locked`. **Aliases.** projection basis.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:1361`; SHA256 prefix `e7eb512a2ec2438d`; record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off: byte-match-to-source criterion for cited; delegated ratification for consensus/original).
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\calB`, `\dag`; source cross-reference keys shown in brackets. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-extcb-corner-dimension-additivity`, `lem-extcb1-cross-corner-dimension`, `lem-maincb-corner-nontriviality`, `lem-maincb-maximal-reset-selection`, `lem-maincb-stage1-maximality`, `lem-routef-f2-positive-unital-compression` and 1 further reproduced result (7 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-projection-basis.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Locked on the recorded user ratification of 2026-07-24 (see frontmatter `consensus:`; lock commit b9270ef4). An earlier draft-transcription note stood here until 2026-07-27 (stale-body drift caught by `AUDIT-F2-TYPING.md` correction 3).
Provenance. Byte-verbatim from the pinned source at the recorded locus and SHA256 prefix; provisioned by `DESIGN-FUDW-DECOMP-v3.md` §4.1 and admitted only for the safe subset by `VERDICT-FUDW-DECOMP-V3.md` §D.

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:1361

```
The second application of Lemma~\ref {lem_merging} involves an auxiliary statement about $\delta $-projections
$P_1,\dots ,P_p,Q_1,\dots ,Q_q$ satisfying certain conditions, which will be formulated in terms of inclusions
of finite-dimensional commutative $C^*$ algebras. Such an algebra $\calB $ is described by a \emph {projection
basis} $\{\Pi _1,\dots ,\Pi _n\}\subseteq \calB $ such that $\Pi _j^\dag =\Pi _j$,\, $\Pi _j\Pi _k=\delta
_{jk}\Pi _k$, and $\sum _{j=1}^{n}\Pi _j=I$.
```

Definition 31 (theta idempotent-approximation map (`def-theta-idempotent-approximation`))**.** Power-series functional calculus (tex:505-510):

$$f(X)=a_0I+\sum_{n=1}^{\infty}a_n(X-x_0I)^n\quad \text{if }\, \|X-x_0I\|<r,$$
$$\|f(X)-f(x_0I)\|\le \sum_{n=1}^{\infty}|a_n|\,\|X-x_0I\|^n.$$

Absolute value and sign (tex:517-520):

$$|X|=(X^2)^{1/2},\qquad \mathrm{sgn}(X)=X(X^2)^{-1/2}\qquad \quad \bigl(\|X^2-x_0^2I\|<x_0^2\bigr).$$

The theta map (tex:527-528, the definitional display of Proposition `prop_P`):

$$\widetilde{P}=\theta(2P-I),\qquad \text{where}\quad \theta(X)=\frac{1}{2}\bigl(I+\mathrm{sgn}(X)\bigr).$$

Kind. cited (byte-matched to a pinned source under `refs/`); `status: draft`. **Aliases.** theta map; functional-calculus idempotent approximation.
Provenance. `source: kitaev-2405.02434`; locus `approximate_algebras.tex:505-528 (displays Taylor_simple, Taylor_simple_bound, abs_sgn, and the theta display inside Proposition prop_P --- ONLY the definitional displays are cited; the Proposition's claims are NOT citable and must be re-proved)`; SHA256 prefix `e7eb512a2ec2438d`.
Typeset from the source. The statement above is the pinned source's own text, set by *macro-translation table v1* of `scripts/gen-report-defs.py` — spelling only, never meaning: `\sgn`, `\ts`, `\wt`; the shard quotes the naked body of a source display; restored to a display environment. The byte-verbatim source is reproduced below as the second check.
Used in this report by `lem-compcb-amplification-naturality`, `lem-compcb-entrywise-compression-naturality` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-theta-idempotent-approximation.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Scope note. This is the map `theta` referenced by the compression-map display `Co_{P, Q}=\theta(\La_P\Ra_Q+\Ra_Q\La_P-1)` in `def-compressed-corner` (tex:1057). The neighbouring Proposition's conclusions (idempotence of `wtP`, commutation, distance bound) are claims, not definitions — they enter the registry only as re-proved results.
Ratification. Draft pending recorded sign-off; expected to lock under the user's recorded byte-match criterion (2026-07-24: "if they match the source then they are valid").

SOURCE CHECK (BYTE-VERBATIM) — KITAEV-2405.02434, APPROXIMATE_ALGEBRAS.TEX:505-528 (DISPLAYS TAYLOR_SIMPLE, TAYLOR_SIMPLE_BOUND, ABS_SGN, AND THE THETA DISPLAY INSIDE PROPOSITION PROP_P --- ONLY THE DEFINITIONAL DISPLAYS ARE CITED; THE PROPOSITION'S CLAIMS ARE NOT CITABLE AND MUST BE RE-PROVED)

```
f(X)=a_0I+\sum _{n=1}^{\infty }a_n(X-x_0I)^n\quad \:
\text {if }\, \|X-x_0I\|<r,\\
\|f(X)-f(x_0I)\|\le \sum _{n=1}^{\infty }|a_n|\ts \|X-x_0I\|^n.
```

```
|X|=(X^2)^{1/2},\qquad \sgn (X)=X(X^2)^{-1/2}\qquad \quad
\bigl (\|X^2-x_0^2I\|<x_0^2\bigr ).
```

```
\wt {P}=\theta (2P-I),\qquad
\text {where}\quad \theta (X)=\frac {1}{2}\bigl (I+\sgn (X)\bigr ).
```

Project-internal packaging: hypothesis data and derived notation

Definition 32 (approximate unitary space (`def-approximate-unitary-space`)). Let $(\mathcal{X}, J, \cdot, \dagger)$ be an exact-unit ε-C^*-algebra (`def-epsilon-cstar-algebra`). Define:

- $\mathcal{H} = \{X \in \mathcal{X} : X^\dagger = X\}$ (Hermitian) and $i\mathcal{H} = \{X \in \mathcal{X} : X^\dagger = -X\}$ (anti-Hermitian);
- $\overline{\mathcal{U}}_\delta = \{X \in \mathcal{X} : \|X^\dagger \cdot X - J\| \leq 2\delta$ and X has a right inverse$\}$, and $\mathcal{U}_\delta$ the analogous set with the strict inequality $\|X^\dagger \cdot X - J\| < 2\delta$;
- the *unitaries* $\mathcal{U} = \overline{\mathcal{U}}_0$;
- $\mathcal{U}_e =$ the connected component of $\mathcal{U}$ containing J, and the scalar action $U \equiv cU$ for $c \in U(1)$ on it.

For a point V for which the left-multiplier L_V is invertible, reserve the *coordinate notation* $\phi_V(X) = L_V^{-1} \cdot (X - V)$ and its components $\phi_V^{\parallel}$ (anti-Hermitian part) and $\phi_V^{\perp}$ (Hermitian part). Reserve the symbols u, h (polar inverse components) and μ, σ (group operations) ONLY as partial notation on domains supplied by result rows.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** approximate unitaries; unitary set of an epsilon-C^*-algebra.
Provenance. `source: internal`; locus `adopted from refs/kitaev-2405.02434 approximate_algebras.tex: 692-706 (sets), 845-859 (u,h and group maps), 945 (U_e and the scalar quotient)`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; content per DESIGN-S1-POLAR-v6.md sect 8, audited AUDIT-S1-POLAR-v4/v5/v6).
Used in this report by `lem-stage1-approximate-group-laws`, `lem-stage1-approximate-group-laws-transport`, `lem-stage1-bound-inversion-isolation`, `lem-stage1-bound-quotient-index-data`, `lem-stage1-bound-quotient-left-inversion`, `lem-stage1-bound-quotient-local-index` and 27 further reproduced results (33 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-approximate-unitary-space.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Vocabulary/typing only — this shard asserts NO chart, inverse, estimate, smoothness, compactness, orientation, or isolation theorem (all of those are the contracts of the Stage-1 polar result rows, `DESIGN-S1-POLAR-v6.md` §3). Adopted from the local source's approximate- unitary discussion with harmonised notation; the source loci are recorded above for orientation, but this is a `consensus` adoption, not a byte-cited definition. Related: `def-epsilon-cstar-algebra`, `def-stage1-polar-witness-data`, `def-h-space-left-inversion`, `def-lefschetz-fixed-point-data`.

Definition 33 (canonical corner identifications (`def-canonical-corner-identifications`)). After identifying the one-dimensional corner $\mathcal{S}_{Q,Q}$ with $\mathbb{C}$ by the normalized vector q_0, the canonical column identification is

$$[J_{P,Q,n}(Z)c]_i = \sum_j Z_{ij} c_j,$$

and the canonical row identification is

$$J_{Q,P,n}(Z) = J_{P,Q,n}(Z^\dagger)^\dagger.$$

Kind. original (introduced by this project); `status: locked`. **Aliases.** canonical J maps; canonical column and row identifications.
Provenance. `source: internal`; locus `PROOF-W74F-E-HCB.md §8.1; DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `conj-hcb`, `lem-hcb4-canonical-closeness`, `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-inverse` (4 registry results import it in total; see `argument/INDEX.md`).

Canonical shard. `definitions/def-canonical-corner-identifications.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft packaging of notation only; no norm or inverse estimate is included, and ratification is required before locking.
Provenance. Transcribed from `PROOF-W74F-E-HCB.md` §8.1 and the provisioning row of `DESIGN-FUDW-DECOMP-v3.md` §4.1; admitted by `VERDICT-FUDW-DECOMP-V3.md` §D.

Definition 34 (compressed associator (`def-compressed-associator`))**.** For compatible compressed rectangular products, the *compressed associator* is

$$(A \cdot B) \cdot C - A \cdot (B \cdot C).$$

Kind. original (introduced by this project); `status: locked`. **Aliases.** rectangular compressed associator.
Provenance. `source: internal`; locus `PROOF-W74F-E-HCB.md §3; DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `lem-hcb0-compressed-associator`, `lem-hcb1-column-action` (2 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-compressed-associator.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft project name for the displayed source expression; it asserts no estimate and requires ratification before locking.
Provenance. Transcribed from `PROOF-W74F-E-HCB.md` §3 and the provisioning row of `DESIGN-FUDW-DECOMP-v3.md` §4.1; admitted by `VERDICT-FUDW-DECOMP-V3.md` §D.

Definition 35 (closed EXT-CB datum (`def-extcb-datum`))**.** A *closed EXT-CB datum* consists of an extended ε-C^*-algebra $\mathcal{A}$; δ-projections P, Q with $\|P + Q - I\| \leq \delta$; an extended δ-isomorphism $v : M_r \to S_P$; $\dim S_Q = 1$; and $S_{P,Q} \neq 0$, with $e = \delta + \varepsilon$.

Kind. original (introduced by this project); `status: locked`. **Aliases.** EXT-CB datum.
Provenance. `source: internal`; locus `DESIGN-FUDW-DECOMP-v3.md §4.1 exact proposed datum package`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `conj-extcb`, `lem-extcb1-close-corner-dimension`, `lem-extcb1-cross-corner-dimension`, `lem-maincb-stage2-call-envelope`, `lem-maincb-stage2-extcb-datum`, `lem-maincb-stage2-raw-extension` (6 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-extcb-datum.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft theorem-free hypothesis package; it includes no extension conclusion and requires ratification before locking.
Provenance. Byte-faithful transcription of the exact datum package in `DESIGN-FUDW-DECOMP-v3.md` §4.1, approved as definition provisioning by `VERDICT-FUDW-DECOMP-V3.md` §§6,D.

Definition 36 (extended δ-inclusion and isomorphism (`def-extended-delta-inclusion`))**.** An *extended δ-inclusion* is a linear map v for which every amplification $1_{M_n} \otimes v$ is a δ-inclusion: it is a δ-homomorphism and satisfies the two-sided $(1 \pm \delta)$ norm bounds. It is an *extended δ-isomorphism* when v is bijective.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** extended delta-inclusion; extended delta-isomorphism.
Provenance. `source: internal`; locus `harmonization of approximate_algebras.tex:443-456,1477-1484 prescribed by DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `conj-extcb`, `lem-compcb-single-compression-transfer`, `lem-extcb-corner-dimension-additivity`, `lem-extcb-four-corner-merge`, `lem-maincb-binary-block-merge`, `lem-maincb-compressed-corner-unit-comparison` and 31 further reproduced results (37 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-extended-delta-inclusion.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.

Status. Draft harmonization only; it contains no theorem and requires ratification before locking.
Provenance. Transcribed from the exact provisioning decision in `DESIGN-FUDW-DECOMP-v3.md` §4.1, which harmonizes the two pinned-source loci; `VERDICT-FUDW-DECOMP-V3.md` §§6,D admits it only for the safe subset.

Definition 37 (amplified four-corner merging datum (`def-four-corner-merging-datum`))**.** An *amplified four-corner merging datum* consists of two complementary source projections, two approximately complementary target projections, and four fixed level-one corner maps whose amplifications share one defect bound for: involution compatibility, compressed-product compatibility, diagonal-unit approximation, and two-sided norm control. The four maps are the same maps at every amplification. **Quantitative complementarity (amendment 2026-07-25, per the cited locus):** the target projections are delta-projections and satisfy $\|P_1 + P_2 - I\| \le \rho$ with ρ the same common defect bound — byte-faithful to the pinned source at `approximate_algebras.tex:1326` ("let P_1, P_2 be δ-projections in an ε-C^* algebra $\mathcal{A}$ such that $\|P_1 + P_2 - I\| \le \delta$").

Kind. original (introduced by this project); `status: locked`. **Aliases.** four-corner merging datum.
Provenance. `source: internal`; locus `project packaging of approximate_algebras.tex:1325-1345 prescribed by DESIGN-FUDW-DECOMP-v3.md §4.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers); quantitative-complementarity amendment user-ratified 2026-07-25 (tobiasosborne, in-session sign-off — transcription-fidelity correction to the cited locus; af challenge ch-4dd98c4d).
Used in this report by `lem-extcb-four-corner-merge`, `lem-extcb-four-corner-norm`, `lem-maincb-binary-block-merge`, `lem-maincb-cross-class-merging-datum`, `lem-maincb-cross-datum-bijectivity`, `lem-maincb-stage3-call-envelope` and 1 further reproduced result (7 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-four-corner-merging-datum.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Status. Draft project packaging of the four source hypotheses; it asserts no merge conclusion and requires ratification before locking.
Provenance. Transcribed from `DESIGN-FUDW-DECOMP-v3.md` §4.1 and the four conditions at the pinned-source locus; admitted only for the safe EXT front end by `VERDICT-FUDW-DECOMP-V3.md` §D.

Definition 38 (closed H-CB datum (`def-hcb-datum`))**.** A *closed H-CB datum* consists of an extended ε-C^*-algebra $\mathcal{A}$; one level-one one-dimensional δ-projection Q; δ-projections P, R, S; the corresponding compressed corners, amplified column spaces, and their defined sesquilinear forms; and $e = \delta + \varepsilon$.

Kind. original (introduced by this project); `status: locked`. **Aliases.** H-CB datum.
Provenance. `source: internal`; locus `DESIGN-FUDW-DECOMP-v3.md §4.1 exact proposed datum package`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers).
Used in this report by `lem-hcb-column-hilbert-squared`, `lem-hcb0-compressed-associator`, `lem-hcb1-column-action`, `lem-hcb1-variational-identity`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and 9 further reproduced results (15 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-hcb-datum.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Scope. This is data and notation only. It includes no column-norm comparison, Ha estimate, inverse, or parent H-CB conclusion.
Status. Draft theorem-free package; ratification is required before locking.
Provenance. Byte-faithful transcription of the exact datum package in `DESIGN-FUDW-DECOMP-v3.md` §4.1, approved as definition provisioning by `VERDICT-FUDW-DECOMP-V3.md` §§6,D.

Definition 39 (MAIN-CB current reset state (`def-maincb-reset-state`))**.** A *MAIN-CB current reset state* consists of: a current index union U; its compressed ambient A_U (a compressed corner); a recorded ambient defect ε_U; a named finite-dimensional C^*-algebra B_U; a supplied level-one map $v_U : B_U \to A_U$; the fixed amplification family $(I_n \otimes v_U)_{n \ge 1}$; a recorded map-defect number d_U; and a supplied tag saying whether the map is an extended inclusion or extended isomorphism.

Kind. original (introduced by this project); `status: locked`. **Aliases.** MAIN reset state.
Provenance. `source: internal`; locus `DESIGN-FUDW-DECOMP-v4.1.md:419-423; DESIGN-MAIN-STRUCTURE-v5.md sect 1.2`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; verbatim from the audited MAIN v5 design).
Used in this report by `lem-maincb-binary-block-merge`, `lem-maincb-cross-class-merging-datum`, `lem-maincb-cross-datum-bijectivity`, `lem-maincb-initial-reset-inclusion`, `lem-maincb-maximal-reset-selection`, `lem-maincb-one-class-extension` and 12 further reproduced results (18 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-maincb-reset-state.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.

NOT TYPESET. This paragraph could not be typeset: math uses \rm, which report/main.tex does not define and the translation table does not cover. Its bytes follow, unaltered.

```
**Notes / provenance.** The tag is hypothesis data, not an existence or
success assertion. This shard asserts NO theorem content (R35): no
$d_U\le c_0^{\rm cb}\varepsilon_U$ invariant (that is the proof obligation of
the result row ‘lem-maincb-reset-invariant-preservation‘ / M19-R), no
smallness, admissibility, preservation, or construction. Companion data
packages: [[def-maincb-partition-state]] (geometry),
[[def-maincb-raw-call]] (one literal attempted call). Provisioned as P0 of
‘DESIGN-MAIN-STRUCTURE-v5.md‘ (audited ‘AUDIT-MAIN-STRUCTURE-v4.md‘ \S{}2,
VALID-WITH-CORRECTIONS applied; ‘AUDIT-MAIN-STRUCTURE-v5.md‘ REPAIR-CONFIRMED).
```

Rendering note. 1 block(s) of this shard are flagged above; see `report/generated/defs/MANIFEST.md`.

Definition 40 (MAIN-CB partition state (`def-maincb-partition-state`))**.** A *MAIN-CB partition state* consists of: a finite atomic index set J; an ambient A; a supplied commutative map $w : \mathbb{C}^J \to A$; the images $P_j = w(e_j)$; the conditional relation $j \sim k \iff \dim S^A_{P_j,P_k} = 1$; when this relation is an equivalence, its class family $\mathcal{C}$; for $U \subseteq J$, $P_U = \sum_{j\in U} P_j$ and $A_U = S^A_{P_U}$ (a compressed corner); one current nonempty subset U of J; and a reference to one separately supplied `def-maincb-reset-state` for that subset.

Kind. original (introduced by this project); `status: locked`. **Aliases.** MAIN partition state.
Provenance. `source: internal`; locus `DESIGN-MAIN-STRUCTURE-v5.md sect 1.4`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; verbatim from the audited MAIN v5 design); AMENDED user-ratified 2026-07-30 in-session: 'one current union U of classes' -> 'one current nonempty subset U of J' (the strict class-union field made the M13/M19-S2/M25 partial-class hypotheses unsatisfiable — M13 validated VACUOUSLY, bead aism-usmn; class-union-ness, where needed, is a per-row hypothesis and M12/M26/M27 already state it explicitly).
Used in this report by `lem-maincb-binary-block-merge`, `lem-maincb-corner-equivalence`, `lem-maincb-cross-class-merging-datum`, `lem-maincb-cross-datum-bijectivity`, `lem-maincb-cross-union-zero-corners`, `lem-maincb-direct-corner-envelope` and 12 further reproduced results (18 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-maincb-partition-state.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. This shard records NEITHER the global ambient defect NOR a defect/unit tag for w: every consuming result that needs those bounds must quantify A as an extended ε-C^*-algebra and w as an extended inclusion explicitly, AND must tie the state to that same displayed (A, w) (“a supplied MAIN partition state for this same A, w” — the identity constraint of `AUDIT-MAIN-STRUCTURE-v4.md` §3.1, applied in MAIN v5). It asserts neither that $\sim$ is an equivalence nor that a current map exists (“when the relation is an equivalence” is conditional data, not an assertion — `AUDIT-MAIN-STRUCTURE-v3.md` §2). Its single current union cannot supply simultaneous U, V data: cross-class rows (M12, M19-S3) require two separately supplied reset states. Companions: `def-maincb-reset-state`, `def-maincb-raw-call`; `def-extended-delta-inclusion`, `def-extended-epsilon-cstar-algebra`.

Definition 41 (MAIN-CB raw call (`def-maincb-raw-call`))**.**

NOT TYPESET. This paragraph could not be typeset: math uses \rm, which report/main.tex does not define and the translation table does not cover. Its bytes follow, unaltered.

```
A *MAIN-CB raw call* consists of: a
literal call-type tag (global scalar, compressed-corner scalar, Stage 1,
Stage 2, or Stage 3); the supplied input reset states/maps
([[def-maincb-reset-state]]); the named finite-dimensional $C^*$-algebra
source and explicit target corner; a pre-helper base scale $t$; any
post-helper datum scale; the literal output level-one map and its fixed
amplification family; the recorded target ambient defect
$\varepsilon _{\rm target}$; and a recorded raw-defect number $d_{\rm raw}$.
```

Kind. original (introduced by this project); `status: locked`. **Aliases.** MAIN raw-call datum.
Provenance. `source: internal`; locus `DESIGN-FUDW-DECOMP-v4.1.md:424-428; DESIGN-MAIN-STRUCTURE-v5.md sect 1.3`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; verbatim from the audited MAIN v5 design).
Used in this report by `lem-maincb-binary-block-merge`, `lem-maincb-cross-class-merging-datum`, `lem-maincb-initial-raw-inclusion`, `lem-maincb-initial-reset-inclusion`, `lem-maincb-one-class-extension`, `lem-maincb-reset-constant-ledger` and 11 further reproduced results (17 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-maincb-raw-call.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.

NOT TYPESET. This paragraph could not be typeset: math uses \rm, which report/main.tex does not define and the translation table does not cover. Its bytes follow, unaltered.

```
**Notes / provenance.** A recorded number does NOT assert that the literal
map is an extended inclusion or isomorphism --- result rows consuming a raw
call must state that hypothesis explicitly (‘AUDIT-MAIN-STRUCTURE-v3.md‘
fatal defect C; the corrected M19-R does so). No hidden domain, smallness,
success, reset, preservation, or iteration clause is present (R35). The two
scalar tags are distinct because the global scalar call uses base scale
$t_0=\varepsilon$ whereas the compressed-corner scalar call at the one-class
induction base uses $t_{\rm atom}=K_{\rm call}\varepsilon$
(‘AUDIT-MAIN-STRUCTURE-v3.md‘ fatal defect D; MAIN v5 \S{}2). Companions:
[[def-maincb-partition-state]], [[def-maincb-reset-state]].
```

Rendering note. 2 block(s) of this shard are flagged above; see `report/generated/defs/MANIFEST.md`.

Definition 42 (MAIN-CB witness ledger (`def-maincb-witness-ledger`))**.** A *MAIN-CB witness ledger* is a tuple W of twelve named real scalar fields

$$W = (c0_{cb}, L, K1, K2, K3, K_{call}, e_{env}, e1, e_{s2}, e_{cross}, r_{reset}, epsilon_{MAIN}).$$

The first five fields are receiving coefficients, the sixth is a derived coefficient, the next four are receiving margins, and the last two are derived scales.

Kind. original (introduced by this project); `status: locked`. **Aliases.** MAIN witness tuple; MAIN-CB constants tuple.
Provenance. `source: internal`; locus `DESIGN-MAINCB-REPAIR-v2.md sect 2 (hostile-audited AUDIT-MAINCB-REPAIR.md, verdict VALID AS DATA)`; no source hash (not a literature transcription); record: user-ratified 2026-08-01 (tobiasosborne, in-session sign-off; delegated ratification for consensus/original tiers; aism-jl4g repair package).
Used in this report by `lem-maincb-binary-block-merge`, `lem-maincb-corner-nontriviality`, `lem-maincb-cross-class-merging-datum`, `lem-maincb-cross-datum-bijectivity`, `lem-maincb-initial-reset-inclusion`, `lem-maincb-maximal-reset-selection` and 15 further reproduced results (21 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-maincb-witness-ledger.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.

Notes / provenance. Pure data: this shard contains no inequality between fields, no existence, uniqueness, estimate, map, regularity, admissibility, or dimension-freeness assertion. The analytic-witness relation and scalar arithmetic are exported only by `lem-maincb-witness-arithmetic` and `lem-maincb-reset-constant-ledger`. This is the MAIN-CB instance of the typed-witness pattern in `def-stage1-polar-witness-data` and `DESIGN-S1-POLAR-v6.md` sects 2–3 and 8, motivated by `docs/LEARNINGS.md` 2026-07-28 laws (i)–(ii). Related: `def-maincb-reset-state`, `def-maincb-raw-call`, `def-maincb-partition-state`.

Definition 43 (positive approximate retract (`def-positive-approximate-retract`)). A *positive approximate retract* between finite commutative C^*-algebras is a pair of positive unital maps

$$A : \ell^\infty(k) \longrightarrow \ell^\infty(n), \qquad M : \ell^\infty(n) \longrightarrow \ell^\infty(k)$$

for which MA is close to I_k in the $\ell^\infty \to \ell^\infty$ operator norm. Its retract defect is

$$\varepsilon_{\rm ret} := \|MA - I_k\|_{\infty\to\infty}.$$

Equivalently, the matrices of A and M have probability-vector rows.

Kind. original (introduced by this project); `status: locked`. **Aliases.** positive retract pair; PRH datum.
Provenance. `source: internal`; locus `project formulation transcribed from docs/plans/2026-07-23-W74F-artifacts/PROOF-W74F-A-PRH.md §§1,4`; no source hash (not a literature transcription); record: user-ratified 2026-07-24 (tobiasosborne, in-session sign-off).
Used in this report by `cor-classical-sharpness`, `lem-prh`, `lem-prh-sharpness`, `lem-prh-sharpness-family-arithmetic`, `lem-routef-prh-finish` (5 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-positive-approximate-retract.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Scope. This project term packages the datum hardened by `lem-prh`; it does not assert that such a pair exists for any particular almost-idempotent map.
Provenance. Project formulation extracted from the W74F-A PRH artifact. User-ratified and locked 2026-07-24 (tobiasosborne, in-session sign-off).

Definition 44 (Stage-1 polar witness data (`def-stage1-polar-witness-data`)).
NOT TYPESET. This paragraph could not be typeset: math uses \rm, which report/main.tex does not define and the translation table does not cover. Its bytes follow, unaltered.

```
A *Stage-1 polar witness datum* is a
tuple $W$ of fourteen named scalar fields:
$$W=(C_{\rm rect},\,C_{\rm ch},\,C_{\rm pol},\,C_{\rm grp},\,C_{\rm path},\,C_{\rm der},\,
e_{\rm rect},\,\kappa _{\rm ch},\,\kappa _{\rm pol},\,\kappa _{\rm der},\,
\delta _*,\,\varepsilon _*^r,\,e_{\rm S1},\,r_{\rm iso}).$$
The first six are called coefficients, the next four margins, the last four
derived scales.
```

Kind. original (introduced by this project); `status: locked`. **Aliases.** polar witness tuple; Stage-1 polar constants tuple.
Provenance. `source: internal`; locus `DESIGN-S1-POLAR-v6.md sect 8 (unchanged from v2 sect 8; audited AUDIT-S1-POLAR-v3/v4/v5/v6)`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D2, docs/plans/2026-07-27-W78-ratification-package.md; verbatim from the audited polar design).
Used in this report by `lem-stage1-approximate-group-laws-transport`, `lem-stage1-bound-inversion-isolation`, `lem-stage1-bound-quotient-index-data`, `lem-stage1-bound-quotient-left-inversion`, `lem-stage1-bound-quotient-local-index`, `lem-stage1-extra-fixed-class` and 9 further reproduced results (15 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-stage1-polar-witness-data.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Pure data: this shard contains NO positivity, inequality, existence, uniqueness, estimate, map, regularity, admissibility, or topological assertion (R35; `AUDIT-S1-POLAR-v4.md` §7: “VALID AS DATA”). The analytic-witness relation — that one such tuple simultaneously witnesses the seven Stage-1 polar producer rows and

satisfies the scalar arithmetic — is exported ONLY by the result rows `lem-stage1-polar-scalar-arithmetic` and `lem-stage1-polar-constant-ledger` and their seven parameterized transport helpers (`DESIGN-S1-POLAR-v6.md` §§2–3), never by this definition. Related: `def-approximate-unitary-space`, `def-epsilon-cstar-algebra`.
Rendering note. 1 block(s) of this shard are flagged above; see `report/generated/defs/MANIFEST.md`.

Definition 45 (unital completely positive map (`def-ucp-map`))**.** Let A and B be unital C^*-algebras (in this repo's Route-F usage: finite-dimensional, typically ℓ^n_∞, M_n, or a finite-dimensional unital C^*-algebra $\mathcal{B}$). A linear map $\varphi : A \to B$ is *positive* if $\varphi(a) \geq 0$ whenever $a \geq 0$; *completely positive* (CP) if for every $r \geq 1$ the amplification $\mathrm{id}_{M_r} \otimes \varphi : M_r(A) \to M_r(B)$ is positive; and *unital* if $\varphi(1_A) = 1_B$. A *UCP map* is a unital completely positive map. Standard facts used silently at the BSc/MSc level: a UCP map is contractive; a positive map out of a commutative C^*-algebra is automatically CP; compositions of UCP maps are UCP.

Kind. consensus (project-internal, agreed; no single external source); `status: locked`. **Aliases.** UCP map; UCP; completely positive unital map.
Provenance. `source: internal`; locus `standard operator-algebra textbook notion (e.g. Paulsen, Completely Bounded Maps and Operator Algebras, ch. 2-3); adopted for the Route-F rows per AUDIT-F0-ASSEMBLY.md sect 1.1`; no source hash (not a literature transcription); record: user-ratified 2026-07-27 (W79 decision D3, docs/plans/2026-07-27-W78-ratification-package.md; def shard chosen over an L2 textbook exemption because the F0 lift row's contract concludes "Phi is UCP").
Used in this report by `cor-kitaev-diagonal-cpization`, `lem-routef-delta-normalization-closeness`, `lem-routef-delta-prime-closeness`, `lem-routef-f0-defect-identity`, `lem-routef-f0-ucp-lift`, `lem-routef-f2-positive-unital-compression` and 8 further reproduced results (14 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-ucp-map.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.
Notes / provenance. Provisioned per the F0 assembly audit (`AUDIT-F0-ASSEMBLY.md` §1.1: the acronym UCP appeared in proposed contracts with no canonical anchor in the definitions layer). Consensus adoption of the universal textbook notion — no project-specific content. Consumers: the F0 lift rows (`lem-routef-f0-ucp-lift`), the K-ledger factorization rows (UCP Δ, Υ), F2. Related: `def-stochastic` (row-stochastic = unital positive on ℓ^n_∞), `def-fd-cstar-diagonal`.

Definition 46 (Route-F raw-factor setting data (`def-routef-raw-factor-setting`))**.** *Route-F raw-factor setting data* have two levels: a scalar header `W_RF`, and a setting datum `S` over that header.

The scalar header has four receiving real-scalar fields

$$(\eta_A, C_A, C_E, \varepsilon_E).$$

The following symbols are derived notation, not independent witnesses:

$$C_\theta := 12(\sqrt{2}-1), \qquad \rho_\theta := \frac{1}{8}, \qquad \rho_{\mathrm{AI}} := \eta_A, \qquad \bar{C}_E := \max\{1, C_E\},$$

$$C_V := \bar{C}_E C_A, \qquad C_T := C_\theta + 3C_V,$$

$$\rho_T := \min\left\{\rho_\theta, \rho_{\mathrm{AI}}, \frac{\varepsilon_E}{C_A}, \frac{1}{4(1+C_\theta)}, \frac{1}{4(1+C_V)}\right\}. \qquad (1.1)$$

Continue, in the displayed order, with

$$\begin{aligned}
\rho_{\mathrm{unit}} &:= \rho_T,\\
\rho_{\mathrm{id}} &:= \min\{\rho_{\mathrm{AI}}, \varepsilon_E/C_A\},\\
\rho_{\mathrm{id}}^{\mathrm{corr}} &:= \min\{\rho_\theta, \rho_{\mathrm{AI}}, \varepsilon_E/C_A\},\\
\rho_{\mathrm{prod}} &:= \rho_T,\\
C_{\Delta'} &:= C_T + 4C_\theta,\\
\rho_{\Delta'} &:= \min\{\rho_T, \rho_{\mathrm{prod}}\},\\
C_\Delta &:= 6C_T + 7C_{\Delta'},\\
\rho_\Delta &:= \min\left\{\rho_{\mathrm{unit}}, \rho_{\Delta'}, [2(C_T + C_{\Delta'})]^{-1}\right\},\\
C_2 &:= C_{\Delta'} + 4C_\Delta,\\
\rho_2 &:= \min\{\rho_{\mathrm{prod}}, \rho_{\Delta'}, \rho_\Delta\},\\
\rho_{\Delta\Phi} &:= \min\{\rho_\theta, \rho_\Delta, \rho_2\},\\
C_3 &:= 10 + 20C_\Delta + 12C_\theta + 2C_{\Delta'},\\
\rho_3 &:= \min\{\rho_\theta, \rho_{\Delta'}, \rho_\Delta, \rho_2\}.
\end{aligned} \tag{1.2}$$

For the componentwise block, put

$$\begin{aligned}
C_N &:= C_V + C_\Delta,\\
C_R &:= C_N + C_2 = C_V + C_\Delta + C_2,\\
C_L &:= C_2 + C_3 + 2C_R,\\
C_{\Upsilon'} &:= 1 + C_\theta + 2C_\Delta + 2C_L,
\end{aligned} \tag{1.3}$$

$$\rho_{\Upsilon'} := \min\left\{\rho_T, \rho_{\mathrm{id}}, \rho_\Delta, \rho_2, \rho_3, (2C_R)^{-1}\right\}. \tag{1.4}$$

Then put

$$\begin{aligned}
C_\Upsilon &:= 6C_T + 7C_{\Upsilon'},\\
\rho_\Upsilon &:= \min\left\{\rho_{\mathrm{unit}}, \rho_{\Upsilon'}, [2(C_T + C_{\Upsilon'})]^{-1}\right\},\\
\rho_{\Delta\Upsilon} &:= \min\{\rho_\theta, \rho_T, \rho_{\mathrm{id}}, \rho_\Delta, \rho_\Upsilon\},\\
\rho_{\mathrm{mult}} &:= \min\{\rho_T, \rho_{\mathrm{id}}, \rho_{\Delta\Phi}, \rho_\Upsilon\},\\
\rho_{\Upsilon\Delta} &:= \min\{\rho_T, \rho_{\mathrm{id}}, \rho_\Delta, \rho_\Upsilon\}.
\end{aligned} \tag{1.5}$$

Finally,

$$K := \max\left\{1, C_\theta + C_\Delta + 2C_\Upsilon, C_\Upsilon + 2(C_2 + C_\theta + C_\Delta), C_\Upsilon + 2C_\Delta\right\}, \tag{1.6}$$

$$\rho_{\mathrm{fac}} := \min\{\rho_2, \rho_{\Delta\Upsilon}, \rho_{\mathrm{mult}}, \rho_{\Upsilon\Delta}\}, \tag{1.7}$$

$$\eta_K := \min\{\rho_{\mathrm{fac}}, (24K)^{-1}, 1\}. \tag{1.8}$$

A setting datum `S` over `W_RF` records the following typed data:

1. a nonzero finite-dimensional complex Hilbert space `H`, a UCP map `Phi:B(H)->B(H)`, and a real scalar `eta`; 2. a linear map `tilde-Phi:B(H)->B(H)`, its range `A:=Im(tilde-Phi)`, a bilinear operation `X star Y:=tilde-Phi(XY)` on `A`, and the scalar notation

$$r := \frac{3}{2}\big((1-4\eta)^{-1/2} - 1\big), \qquad \varepsilon_{\mathrm{AI}}(\eta) := \max\{r, 20\eta + 2((1+r)^5 - 1), 3r - r^2\};$$

3. a finite-dimensional unital C^*-algebra `B` and linear maps `v:B->A` and `u:A->B`; and 4. the notation

$$\widetilde{\Delta} := \iota_{\mathcal{A}\subseteq B(H)} \circ v, \qquad \widetilde{\Upsilon} := u \circ \widetilde{\Phi}.$$

The displayed map `tilde-Phi` is the notation

$$\widetilde{\Phi} := \frac{1}{2}\Big(I + (2\Phi - I)\big(I - 4(\Phi - \Phi^2)\big)^{-1/2}\Big).$$

For every linear map `T` occurring in a datum and every integer `q>=1`, `T_q:=id_{M_q} tensor T`. Registry ASCII `A` and `B` denote the two algebras above; `I_B` denotes the identity map of `B`. An unsubscripted `I` is the unit forced by the adjacent map types.

Kind. original (introduced by this project); `status: locked`. **Aliases.** finite Route-F raw-factor setting; Route-F raw-factor scalar header; raw-factor setting datum.
Provenance. `source: internal`; locus `DESIGN-LEDGER-SETTING-RESCOPE-V2.md sect-1 (hostile re-audit AUDIT-LEDGER-SETTING-RESCOPE-V2.md, deletion test CLEARED: data-and-typing only)`; no source hash (not a literature transcription); record: user-ratified 2026-08-05 (tobiasosborne, in-session sign-off of the audited LAND-WITH-EXACT-CORRECTIONS rescope package; corrections folded in at landing); I_B notation corrected to the identity map of B, user-ratified 2026-08-05 (fresh-verifier-caught type error in the ratified design text: rows 3/12 use I_B as a map, not the unit element).
Used in this report by `lem-routef-degree-three-estimate`, `lem-routef-degree-two-estimate`, `lem-routef-delta-normalization-closeness`, `lem-routef-delta-phi-product`, `lem-routef-delta-prime-closeness`, `lem-routef-delta-upsilon-telescope` and 17 further reproduced results (23 registry results import it in total; see `argument/INDEX.md`).
Canonical shard. `definitions/def-routef-raw-factor-setting.md`; twins: `definitions/INDEX.md`, `argument/DAG.md`.

NOT TYPESET. This paragraph could not be typeset: unmapped non-ASCII '“' (U+201C, LEFT DOUBLE QUOTATION MARK) in prose. Its bytes follow, unaltered.

```
**Notes / provenance.** This shard asserts only the shape and notation of a
record.  In particular, it asserts none of the following: that a header or
datum exists; that any scalar is positive, finite, universal, or independent
of input data; that ‘||Phi^2-Phi||_cb<=eta‘; that ‘tilde-Phi‘ is
idempotent; that ‘A‘ is an extended approximate C*-algebra; that ‘v‘ is an
[[def-extended-delta-inclusion|extended isomorphism]]; or any norm,
smallness, admissibility, CP, or UCP estimate for the recorded maps.  Those
assertions belong only to result rows, beginning with
‘lem-routef-raw-factor-setting-formation‘.  The terms \UCP map", \extended
epsilon-C*-algebra", and \extended delta-isomorphism" are referenced from
their canonical shards and are not redefined here.  The labels ‘u‘ and ‘v‘ do not assert an inverse
relation; that relation is a conclusion of the formation result.
```

Rendering note. 1 block(s) of this shard are flagged above; see `report/generated/defs/MANIFEST.md`.

1 further canonical definition in `definitions/` is outside the current proof strategy's dependency closure and is not rendered here. It remains canonical — the shard is the single source of truth either way — and reachable at the shard path and through `definitions/INDEX.md`:
`def-pivot`

3 The signed-stochastic bridge

Lemma 47 (`lem-classical-equiv`). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants. There is a universal $C > 0$ such that:*

1. *(stochastic $\to$ signed) if Q is row-stochastic with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta$, then $P = \theta(2Q-1)$ is a signed affine retraction (`def-signed-idempotent`) satisfying $\|P - Q\|_{\infty\to\infty} \le C\eta$ and negative mass $\delta \le C\eta$;*

2. *(signed $\to$ stochastic) conversely, row-normalising the positive parts p_i^+ of a signed idempotent P yields a row-stochastic Q with $\|P - Q\|_{\infty\to\infty} \le 2\delta$ and $\|Q^2 - Q\|_{\infty\to\infty} \le 6\delta + 4\delta^2$.*

Transcription note. *The contract writes the map norm as $\|\cdot\|$ without a subscript; per `CONVENTIONS.md` (b) it is the $\infty\to\infty$ map norm $\|\cdot\|_{\infty\to\infty}$. The operator θ is carried verbatim from the contract, which does not expand it; the validated tree instantiates it as the functional calculus $\theta(X) = \frac{1}{2}(I + \operatorname{sgn} X)$ of `def-theta-idempotent-approximation`.*

Registry contract (verbatim).

```
The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent
up to universal constants: Q row-stochastic with ||Q^2-Q|| <= eta gives P=theta(2Q-1) signed affine
retraction with ||P-Q|| <= C eta and neg mass delta <= C eta, and conversely row-normalising p_i^+
gives Q with ||P-Q|| <= 2 delta, ||Q^2-Q|| <= 6 delta+4 delta^2.
```

Proof (prose account of the af-validated tree). *Forward direction.* Put $X = 2Q - I$. Row-stochasticity gives $X\mathbf{1} = \mathbf{1}$ and $\|X\|_{\infty\to\infty} \le 3$, while the algebraic identity

$$X^2 - I = 4\,(Q^2 - Q) \tag{2}$$

converts the almost-idempotence hypothesis into $\|X^2 - I\|_{\infty\to\infty} \le 4\eta$. Write $Y = X^2 - I$. Two properties of Y drive everything: $Y\mathbf{1} = 0$ (because $X\mathbf{1} = \mathbf{1}$ forces $X^2\mathbf{1} = \mathbf{1}$), and $\|Y\|_{\infty\to\infty} \le 4\eta \le \frac{1}{2}$ once $\eta \le \frac{1}{8}$.

The smallness of Y puts the inverse square root inside its convergence radius: the binomial series

$$B = (I+Y)^{-1/2} = \sum_{m\ge 0} \binom{-\frac{1}{2}}{m} Y^m$$

converges absolutely in the map norm. Being a norm limit of polynomials in $Y = X^2 - I$, the operator B commutes with X; since $Y\mathbf{1} = 0$ kills every positive power, $B\mathbf{1} = \mathbf{1}$; and the absolute coefficient sum gives the quantitative bound $\|B - I\|_{\infty\to\infty} \le (1-t)^{-1/2} - 1 \le 2t$ at $t = \|Y\|_{\infty\to\infty} \le \frac{1}{2}$, hence $\|B - I\|_{\infty\to\infty} \le 8\eta$. The scalar identity $\big((1+z)^{-1/2}\big)^2(1+z) = 1$ transfers to the absolutely convergent series in Y, so $B^2X^2 = I$; commutation then makes $S = XB$ an exact involution,

$$S^2 = XBXB = X^2B^2 = I, \qquad S\mathbf{1} = XB\mathbf{1} = X\mathbf{1} = \mathbf{1},$$

with $S - X = X(B - I)$ and therefore $\|S - X\|_{\infty\to\infty} \le 3 \cdot 8\eta = 24\eta$.

Setting $P = \theta(X) = \frac{1}{2}(I + S)$ now gives $P\mathbf{1} = \mathbf{1}$ and $P^2 = \frac{1}{4}(I + 2S + S^2) = \frac{1}{2}(I + S) = P$, so P is an exact signed idempotent in the sense of `def-signed-idempotent`. Since $Q = \frac{1}{2}(I + X)$, the difference is $P - Q = \frac{1}{2}(S - X)$ and hence $\|P - Q\|_{\infty\to\infty} \le 12\eta$. Finally the negative mass: row-stochastic Q is entrywise nonnegative (`def-stochastic`), so for every entry $\max\{-P_{ij}, 0\} \le |P_{ij} - Q_{ij}|$ — trivially where $P_{ij} \ge 0$, and because $-P_{ij} \le Q_{ij} - P_{ij}$ where $P_{ij} < 0$. Summing over j and maximising over i gives, with δ as in `def-negative-mass`,

$$\delta(P) \;\le\; \|P - Q\|_{\infty\to\infty} \;\le\; 12\eta. \tag{3}$$

So $C = 12$ serves. The export records an instructive scope correction here: the negative-mass step was first blocked because `def-negative-mass` — although permitted by the registry shard — had not been registered inside the workspace, and the tree carried an explicit gap certificate rather than the estimate; registering the definition discharged it.

Converse direction. Let P be an exact signed idempotent with negative mass δ. Split each row into disjointly supported nonnegative parts, $p_i = p_i^+ - p_i^-$, and set $a_i = \sum_k (P_{ik})_- \le \delta$. Total mass one (`def-signed-idempotent`) gives $\sum_k p_{ik}^+ = 1 + a_i$, so

$$q_i = \frac{p_i^+}{1 + a_i}$$

is a probability vector and the matrix Q with rows q_i is row-stochastic. The correction $e_i = q_i - p_i$ is

$$e_i = p_i^- - \frac{a_i}{1 + a_i}\, p_i^+,$$

a difference of two disjointly supported vectors of ℓ^1-mass a_i each; hence $\|e_i\|_1 = 2a_i \le 2\delta$ and $\|P - Q\|_{\infty\to\infty} \le 2\delta$.

For the idempotence defect, write $D = Q - P$. The row-geometry clause of `def-signed-idempotent` gives $\|P\|_{\infty\to\infty} \le 1 + 2\delta$, hence $\|P - I\|_{\infty\to\infty} \le 2 + 2\delta$. Using $P^2 = P$ exactly, each row satisfies the exact decomposition

$$q_i Q - q_i = e_i(P - I) + q_i D. \tag{4}$$

Since q_i is a probability vector, $\|q_i D\|_1 \le \|D\|_{\infty\to\infty} \le 2\delta$, while $\|e_i(P - I)\|_1 \le 2\delta\,(2 + 2\delta)$. Adding, $\|Q^2 - Q\|_{\infty\to\infty} \le 6\delta + 4\delta^2$, exactly as contracted.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-classical-equiv`: a 29-node tree (21 live nodes validated, 8 superseded nodes archived), root `validated`, taint clean.

Role in the chain. This is the licence to move between the two pictures of §1, and the only such licence in the registry. It has no dependencies. Everything stated in the signed picture reaches the stochastic formulation of `op-classical` only through it, and only up to the universal constants above; `op-classical` itself is now discharged at the af-validated rung via Route F (§76).

4 Positive-retract hardening

Lemma 48 (lem-prh). *Let $k, n \geq 1$ and let*

$$A : \ell^\infty(k) \longrightarrow \ell^\infty(n), \qquad M : \ell^\infty(n) \longrightarrow \ell^\infty(k)$$

be positive unital maps — equivalently, maps whose matrices have probability-vector rows. If

$$\|MA - I_k\|_{\infty\to\infty} \;\leq\; \varepsilon, \qquad 0 \leq \varepsilon < \tfrac{1}{2},$$

then there is a stochastic idempotent $E : \ell^\infty(n) \to \ell^\infty(n)$ (def-stochastic) with

$$\|AM - E\|_{\infty\to\infty} \;\leq\; 2\sqrt{2\varepsilon}. \tag{5}$$

Transcription note. *The pair (A, M) is the datum of def-positive-approximate-retract, with retract defect $\varepsilon_{\mathrm{ret}} = \|MA - I_k\|_{\infty\to\infty}$. Following the validated tree, A is presented by rows $a_i \in \Delta_k$ indexed by $i \in \{1, \ldots, n\}$ and M by rows μ_s indexed by $s \in \{1, \ldots, k\}$, so that the s-th row of MA is the probability vector $\sum_i \mu_s(i)\, a_i$.*

Registry contract (verbatim).

```
Positive-retract hardening (PRH): let k,n >= 1 and let A:l-infinity(k)->l-infinity(n) and
M:l-infinity(n)->l-infinity(k) be positive unital maps (equivalently, have probability-vector
rows); if ||MA-I_k||_{infinity->infinity} <= epsilon with 0 <= epsilon < 1/2, then there is a
stochastic idempotent E:l-infinity(n)->l-infinity(n) with ||AM-E||_{infinity->infinity} <=
2*sqrt(2*epsilon).
```

Proof (prose account of the af-validated tree). The tree splits on the defect. If $\varepsilon = 0$ the hypothesis reads $MA = I_k$ exactly, and then $E := AM$ is already row-stochastic with $E^2 = A(MA)M = AM = E$, so (5) holds with error 0. The substance is the case $0 < \varepsilon < \frac{1}{2}$.

The scale and the cores. Fix

$$\lambda = \sqrt{\varepsilon/2}, \qquad C_s = \{\, i : \; a_{is} > 1 - \lambda \,\}, \qquad \beta_s = \mu_s(C_s^c).$$

The single quantitative input is a coordinate defect identity. The s-th row of MA is the probability vector $\sum_i \mu_s(i) a_i$, and for probability vectors the ℓ^1 distance to the basis vector e_s is governed by one coordinate:

$$\|(MA)_{s,*} - e_s\|_1 = 2\big(1 - (MA)_{ss}\big) = 2 \sum_i \mu_s(i)\,(1 - a_{is}) =: 2D_s, \tag{6}$$

so the norm hypothesis is exactly $D_s \leq \varepsilon/2$ for every s. Two consequences follow. First, the cores are pairwise disjoint: if $i \in C_s \cap C_t$ with $s \neq t$ then $a_{is} + a_{it} > 2(1 - \lambda) > 1$ (using $\lambda < \frac{1}{2}$, which is where $\varepsilon < \frac{1}{2}$ enters), contradicting that a_i is a probability vector. Second, off the core one has $1 - a_{is} \geq \lambda$, so dropping the on-core terms of the nonnegative sum D_s gives $\lambda\beta_s \leq D_s \leq \varepsilon/2$, whence

$$\beta_s \;\leq\; \frac{\varepsilon}{2\lambda} \;=\; \lambda \;<\; \tfrac{1}{2}. \tag{7}$$

The tree records a correction at this point: an earlier node asserted the strict inequality $\beta_s < \lambda$, which the threshold $a_{is} \leq 1 - \lambda$ off the core does not give; the weak form (7) is what holds and all that is used.

The exact construction. Condition each decoder row on its core and replace each encoder row inside a core by a basis vector:

$$\nu_s(i) = \frac{\mu_s(i)\,\mathbf{1}_{C_s}(i)}{1-\beta_s}, \qquad \widehat{A} \text{ has row } e_s \text{ on } C_s \text{ and row } a_i \text{ off } \textstyle\bigcup_s C_s.$$

Both are well defined: $\beta_s < 1$ makes ν_s a probability vector, and disjointness of the cores makes $\widehat{A}$ unambiguous. The construction is engineered so that the retraction becomes *exact*: the s-th row of $N\widehat{A}$ is $\sum_{i\in C_s}\mu_s(i)e_s/(1-\beta_s) = e_s$, i.e.

$$N\widehat{A} = I_k, \qquad \text{hence} \qquad E := \widehat{A}N \text{ satisfies } E^2 = \widehat{A}(N\widehat{A})N = \widehat{A}N = E. \tag{8}$$

As a product of positive unital maps, E is row-stochastic, so it is a stochastic idempotent in the sense of `def-stochastic`.

The two error terms. Write $AM - E = A(M-N) + (A-\widehat{A})N$ and bound the summands separately.

For the conditioning term, ν_s is μ_s conditioned on C_s, so the two rows differ by mass β_s removed from C_s^c and mass β_s redistributed on C_s; hence $\|\mu_s - \nu_s\|_1 = 2\beta_s \le \varepsilon/\lambda$ by (7) and $\|M-N\|_{\infty\to\infty} \le \varepsilon/\lambda$. Because A has probability rows it is an $\infty\to\infty$ contraction — concretely, $\sum_j |(AB)_{ij}| \le \sum_s a_{is}\sum_j |B_{sj}| \le \|B\|_{\infty\to\infty}$ — so $\|A(M-N)\|_{\infty\to\infty} \le \varepsilon/\lambda$.

For the core-replacement term, the rows of $(A-\widehat{A})N$ vanish outside $\bigcup_s C_s$. For $i \in C_s$ the corresponding row is $\sum_t a_{it}\nu_t - \nu_s$, and since every ν_t is a probability vector its ℓ^1 norm is at most $(1-a_{is}) + \sum_{t\ne s} a_{it} = 2(1-a_{is}) < 2\lambda$. Hence $\|(A-\widehat{A})N\|_{\infty\to\infty} \le 2\lambda$.

Optimisation. Adding the two bounds,

$$\|AM - E\|_{\infty\to\infty} \;\le\; \frac{\varepsilon}{\lambda} + 2\lambda,$$

and the choice $\lambda = \sqrt{\varepsilon/2}$ balances the two terms at the common value $\sqrt{2\varepsilon}$, giving (5). The square-root rate is intrinsic to this balance: conditioning costs ε/λ and core replacement costs 2λ, and no choice of threshold does better than their geometric mean.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-prh`: 14 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. `lem-prh` has no registry dependencies. It is the finishing step of the live route: it installs the reduction of `op-classical` to the *existence* of a positive approximate retract for a given almost-idempotent Q — stochastic A, M with $\|AM - Q\|_{\infty\to\infty} = O(\eta)$ and $\|MA - I\|_{\infty\to\infty} = O(\eta)$ — after which (5) delivers the exact idempotent at rate $\sqrt{\eta}$. Producing such a pair is what the COMP and H-CB tiers below are ultimately for, and that production is *not* established: `lem-prh` is a conditional finisher, and `op-classical` is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`. Downstream registry consumers are `lem-routef-prh-finish` (`proved`, `af: validated`) and the strengthened `lem-routef-k-ledger` (`proved`, `af: validated`).

The sharp square-root order

The upper bound above has a matching T0 lower-bound family. The four rows below separate the finite arithmetic, the general Markov-chain fact, the PRH conclusion, and its direct stochastic almost-idempotence corollary.

Lemma 49 (`lem-prh-sharpness-family-arithmetic`). *For $0 < \lambda < \frac{1}{2}$, put*

$$A_\lambda = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1-\lambda & \lambda \\ \lambda & 1-\lambda \end{pmatrix}, \qquad M_\lambda = \begin{pmatrix} 1-\lambda & 0 & \lambda & 0 \\ 0 & 1-\lambda & 0 & \lambda \end{pmatrix}.$$

Both matrices have nonnegative probability rows and hence represent positive unital maps $A_\lambda : \ell^\infty(2) \to \ell^\infty(4)$ and $M_\lambda : \ell^\infty(4) \to \ell^\infty(2)$. For every $B \in \mathbb{R}^{r\times s}$, $r, s \geq 1$, its induced norm is $\|B\|_{\infty\to\infty} = \max_{1\leq i\leq r} \sum_{j=1}^{s} |B_{ij}|$. Moreover,

$$M_\lambda A_\lambda = \begin{pmatrix} 1-\lambda^2 & \lambda^2 \\ \lambda^2 & 1-\lambda^2 \end{pmatrix}, \qquad \varepsilon_\lambda := \|M_\lambda A_\lambda - I_2\|_{\infty\to\infty} = 2\lambda^2 \longrightarrow 0,$$

and $P_\lambda := A_\lambda M_\lambda$ is row-stochastic with rows

$$\begin{pmatrix} 1-\lambda & 0 & \lambda & 0 \\ 0 & 1-\lambda & 0 & \lambda \\ (1-\lambda)^2 & \lambda(1-\lambda) & \lambda(1-\lambda) & \lambda^2 \\ \lambda(1-\lambda) & (1-\lambda)^2 & \lambda^2 & \lambda(1-\lambda) \end{pmatrix}, \qquad \|\mathrm{row}_1(P_\lambda) - \mathrm{row}_3(P_\lambda)\|_1 = 2\lambda.$$

Registry contract (verbatim).

```
Witness arithmetic for PRH sharpness: for every real lambda with 0 < lambda < 1/2, let A_lambda in
R^(4x2) have rows (1,0), (0,1), (1-lambda,lambda), (lambda,1-lambda), and let M_lambda in R^(2x4)
have rows (1-lambda,0,lambda,0), (0,1-lambda,0,lambda); A_lambda and M_lambda have nonnegative
entries and probability-vector rows and hence represent positive unital maps
A_lambda:l-infinity(2)->l-infinity(4) and M_lambda:l-infinity(4)->l-infinity(2); for every pair of
integers r,s >= 1 and every B in R^(r x s), the induced l-infinity operator norm is
||B||_{infinity->infinity}=max_{1<=i<=r} sum_{j=1}^s |B_ij|; M_lambda A_lambda has rows
(1-lambda^2,lambda^2), (lambda^2,1-lambda^2), so epsilon_lambda:=||M_lambda
A_lambda-I_2||_{infinity->infinity}=2*lambda^2 and epsilon_lambda tends to 0 as lambda tends to 0;
and P_lambda:=A_lambda M_lambda is row-stochastic with rows (1-lambda,0,lambda,0),
(0,1-lambda,0,lambda), ((1-lambda)^2,lambda*(1-lambda),lambda*(1-lambda),lambda^2),
(lambda*(1-lambda),(1-lambda)^2,lambda^2,lambda*(1-lambda)), and
||row_1(P_lambda)-row_3(P_lambda)||_1=2*lambda.
```

Proof (prose account of the af-validated tree). Positivity and the row sums are immediate from $0 < \lambda < \frac{1}{2}$. For the norm formula, the triangle inequality gives the maximum absolute row sum as an upper bound; choosing signs against a row attaining that maximum gives the reverse inequality, including the zero-row case. Direct row-by-column multiplication gives the displayed $M_\lambda A_\lambda$, whose two rows differ from those of I_2 by absolute row sum $2\lambda^2$. The same multiplication gives all four displayed rows of P_λ; they are probability rows, and subtracting its first and third rows gives exact ℓ^1 norm 2λ. This is the content of the validated 24-node export, including its repaired non-strict norm chain and explicit per-row product derivations.

Lemma 50 (`lem-prh-sharpness-row-coincidence`). *Let $F = (f_{ab})$ be an $n \times n$ row-stochastic idempotent. If $f_{ii} > 0$ and $f_{ij} > 0$, then $\mathrm{row}_i(F) = \mathrm{row}_j(F)$.*

Registry contract (verbatim).

```
Row coincidence for stochastic idempotents: for every integer n >= 1, every n-by-n row-stochastic
idempotent matrix F=(f_ab) over R, and all i,j in {1,...,n}, if f_ii>0 and f_ij>0 then
row_i(F)=row_j(F).
```

Proof (prose account of the af-validated tree). Set $p = \mathrm{row}_i(F)$ and $S = \{a : p_a > 0\}$. Idempotence makes p stationary; nonnegativity shows that S is closed. On the positive-entry graph of the restriction to S, a source component with an outgoing edge would lose strictly positive stationary mass, so no edge joins distinct strongly connected components. Since $f_{ia} = p_a > 0$ for every $a \in S$, the graph is strongly connected. The row $q = \mathrm{row}_j(F)$ is supported on S and stationary there. Finally, with $c = \min_a q_a/p_a$, the nonnegative stationary row $q - cp$ has a zero coordinate; strong connectivity forces every nonzero nonnegative stationary row to be strictly positive, so $c = 1$ and $q = p$. This follows the validated 19-node export; its cross-sibling repairs make the closure, stationarity, and uniqueness steps dependency-explicit.

Lemma 51 (`lem-prh-sharpness`). *For every $0 < \lambda < \frac{1}{2}$ there are positive unital maps $A : \ell^\infty(2) \to \ell^\infty(4)$ and $M : \ell^\infty(4) \to \ell^\infty(2)$ with $\varepsilon_\lambda = \|MA - I_2\|_{\infty\to\infty} = 2\lambda^2 \to 0$ such that every stochastic idempotent F on $\ell^\infty(4)$ satisfies*

$$\|AM - F\|_{\infty\to\infty} \geq \lambda = \sqrt{\varepsilon_\lambda/2}.$$

Thus the $\sqrt{\varepsilon}$ order in PRH is intrinsically sharp.

Registry contract (verbatim).

```
PRH square-root sharpness: for every 0 < lambda < 1/2 there are positive unital maps
A:l-infinity(2)->l-infinity(4) and M:l-infinity(4)->l-infinity(2) with
epsilon_lambda=||MA-I_2||_{infinity->infinity}=2*lambda^2 tending to 0 such that every stochastic
idempotent F on l-infinity(4) satisfies ||AM-F||_{infinity->infinity} >=
lambda=sqrt(epsilon_lambda/2); hence the sqrt(epsilon) order in PRH is intrinsically sharp.
```

Proof (prose account of the af-validated tree). Use $A_\lambda, M_\lambda, P_\lambda$ from Lemma 49. If a stochastic idempotent F had $\|P_\lambda - F\|_{\infty\to\infty} < \lambda$, the max-row norm formula would give $f_{11} > 1 - 2\lambda > 0$ and $f_{13} > 0$. The row-coincidence lemma would then make rows 1 and 3 of F equal, while the triangle inequality would put the two corresponding rows of P_λ at distance strictly less than 2λ, contradicting their exact distance. Hence every F is at least λ away; since $\varepsilon_\lambda = 2\lambda^2$, this is $\sqrt{\varepsilon_\lambda/2}$. The factored main tree validated first-pass with 12 nodes and zero challenges after the two earlier monolithic attempts ballooned.

Corollary 52 (`cor-classical-sharpness`). *For $0 < \lambda < \frac{1}{2}$, choose the preceding A_λ, M_λ and put $\eta_\lambda = 2\lambda^2$, $Q_\lambda = A_\lambda M_\lambda$. Then Q_λ is row-stochastic,*

$$\|Q_\lambda^2 - Q_\lambda\|_{\infty\to\infty} \leq \eta_\lambda, \qquad \|Q_\lambda - F\|_{\infty\to\infty} \geq \lambda = \sqrt{\eta_\lambda/2}$$

for every stochastic idempotent F. Consequently, for every $C > 0$, $\eta_0 > 0$, and $\beta > \frac{1}{2}$, some $0 < \eta < \min\{\eta_0, \frac{1}{4}\}$ and row-stochastic Q satisfy $\|Q^2 - Q\|_{\infty\to\infty} \leq \eta$ while every stochastic idempotent E satisfies $\|Q - E\|_{\infty\to\infty} > C\eta^\beta$. Thus no uniform exponent $\beta > \frac{1}{2}$ can replace $\frac{1}{2}$ in `op-classical`.

Registry contract (verbatim).

```
Classical square-root sharpness: for every 0 < lambda < 1/2, choose positive unital maps
A_lambda:l-infinity(2)->l-infinity(4) and M_lambda:l-infinity(4)->l-infinity(2) supplied by
lem-prh-sharpness, and put eta_lambda=2*lambda^2 and Q_lambda=A_lambda M_lambda; then Q_lambda is
row-stochastic, ||Q_lambda^2-Q_lambda||_{infinity->infinity} <= eta_lambda, and every stochastic
idempotent F on l-infinity(4) satisfies ||Q_lambda-F||_{infinity->infinity} >=
lambda=sqrt(eta_lambda/2). Consequently, for every C>0, eta_0>0, and beta>1/2 there exist
0<eta<min{eta_0,1/4} and a row-stochastic Q on l-infinity(4) with ||Q^2-Q||_{infinity->infinity} <=
eta such that every stochastic idempotent E satisfies ||Q-E||_{infinity->infinity} > C*eta^beta;
equivalently, no uniform exponent beta>1/2 can replace 1/2 in op-classical.
```

Proof (prose account of the af-validated tree). Products of probability-row matrices are row-stochastic and have operator norm one. With $Q_\lambda = A_\lambda M_\lambda$ defined at the root,

$$Q_\lambda^2 - Q_\lambda = A_\lambda (M_\lambda A_\lambda - I_2) M_\lambda,$$

so submultiplicativity gives the stated weak defect bound; the preceding lemma gives the distance lower bound. Given C, η_0, β, choose

$$0 < \lambda < \min\left\{ \frac{1}{2\sqrt{2}}, \sqrt{\frac{\eta_0}{2}}, (C2^\beta)^{-1/(2\beta-1)} \right\}.$$

Then $\eta = 2\lambda^2$ has the required range and $C\eta^\beta = C2^\beta\lambda^{2\beta} < \lambda \le \|Q_\lambda - E\|_{\infty\to\infty}$ for every E. This is the literal dimension-four counterexample to any proposed uniform (C, η_0, β) estimate. Run 1 ballooned at 26 live nodes; under the user-ratified, hostile-audited remedy-(b) linear-node design, the af tree validated first-pass with all five nodes clean, zero challenges, and only the validated PRH sharpness external.

5 Amplification naturality of the functional calculus

Lemma 53 (`lem-compcb-amplification-naturality`). *Let $\mathcal{B}$ be a unital Banach algebra and let $\iota_n : \mathcal{B} \to M_n(\mathcal{B})$, $X \mapsto 1_{M_n} \otimes X$, be the unital amplification, an isometric unital homomorphism. Let f be given on $\|X - x_0 I\| < r$ by the power series*

$$f(X) = a_0 I + \sum_{m \geq 1} a_m \,(X - x_0 I)^m. \tag{9}$$

Then for every $X \in \mathcal{B}$ with $\|X - x_0 I\| < r$ the element $f(\iota_n(X))$ is defined and

$$f(\iota_n(X)) = \iota_n(f(X)). \tag{10}$$

In particular, whenever $\|(2P - I)^2 - I\| < 1$,

$$\theta\big(\iota_n(2P - I)\big) = \iota_n\big(\theta(2P - I)\big), \qquad \theta(X) = \tfrac{1}{2}\big(I + \operatorname{sgn} X\big).$$

Transcription note. *θ and* sgn *are the power-series functional calculus of* ***`def-theta-idempotent-approximation`***; *only its definitional displays are in scope here. The norm is the ambient Banach-algebra norm.*

Registry contract (verbatim).

```
Amplification naturality of the power-series functional calculus: let B be a unital Banach algebra,
iota_n: B -> M_n(B) the unital amplification X -> 1_{M_n} tensor X (an isometric unital
homomorphism), and f given on ||X - x_0 I|| < r by the power series f(X) = a_0 I + sum_{m>=1} a_m
(X - x_0 I)^m; then for every X in B with ||X - x_0 I|| < r, f(iota_n(X)) is defined and
f(iota_n(X)) = iota_n(f(X)); in particular, whenever ||(2P-I)^2 - I|| < 1, theta(iota_n(2P-I)) =
iota_n(theta(2P-I)) with theta(X) = (I + sgn(X))/2.
```

Proof (prose account of the af-validated tree). *The general statement.* Three structural facts about ι_n do all the work. Unitality gives $\iota_n(I_{\mathcal{B}}) = I_{M_n(\mathcal{B})}$, hence $\iota_n(X) - x_0 I = \iota_n(X - x_0 I)$; isometry then transports the convergence hypothesis verbatim, $\|\iota_n(X) - x_0 I\| = \|X - x_0 I\| < r$, so the series (9) evaluated at $\iota_n(X)$ is inside its domain; and multiplicativity gives $(\iota_n(X) - x_0 I)^m = \iota_n\big((X - x_0 I)^m\big)$ for every $m \geq 0$. Consequently the N-th partial sums correspond exactly,

$$T_N = a_0 I + \sum_{m=1}^{N} a_m(\iota_n(X) - x_0 I)^m = \iota_n(S_N), \qquad S_N = a_0 I + \sum_{m=1}^{N} a_m(X - x_0 I)^m.$$

Since $S_N \to f(X)$ in $\mathcal{B}$ and ι_n is isometric, hence continuous, $T_N \to \iota_n(f(X))$. The power-series definition identifies the limit of the same sequence T_N with $f(\iota_n(X))$, so uniqueness of limits yields both definedness and (10). No property of ι_n beyond isometry, unitality and multiplicativity is used.

The θ specialization. Here the functional calculus is not a single series but a composite built from the inverse square root, so the tree first pins that branch down. With $c_m = \binom{-1/2}{m}$ one has $c_0 = 1$ and $c_m/c_{m-1} = -(2m-1)/(2m)$, whose modulus tends to 1; the ratio test therefore gives radius exactly one for $h(w) = \sum_{m \geq 0} c_m w^m$. On $|w| < 1$ termwise differentiation is legitimate and the coefficient recurrence is precisely the differential equation $2(1+w)h'(w) + h(w) = 0$; hence $\frac{d}{dw}\big[(1+w)h(w)^2\big] = 0$, and $h(0) = 1$ gives $(1+w)h(w)^2 = 1$. So h *is* the analytic inverse-square-root branch normalised at 1.

Transporting this to a unital Banach algebra: for $\|Z-I\|<1$ the series $H=\sum_{m\geq 0} c_m(Z-I)^m$ converges absolutely in norm, and absolute convergence licenses the same Cauchy products as in the scalar identity, so $ZH^2=I$; the normalisation $H=I+$ (terms of positive degree) identifies H with the branch $Z^{-1/2}$ used by the registered calculus. Applying the general statement to this series with $Z=Y^2$, $Y=2P-I$, gives $\big(\iota_n(Y)^2\big)^{-1/2}=\iota_n\big((Y^2)^{-1/2}\big)$ whenever $\|Y^2-I\|<1$; multiplicativity then propagates the identity through $\operatorname{sgn}(Y)=Y(Y^2)^{-1/2}$ and finally through $\theta=\frac{1}{2}(I+\operatorname{sgn})$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplification-naturality`: 12 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. This lemma exists because of a stall, and the shape of the campaign record is worth stating: three separate blocking challenges in the first orchestration of `lem-compcb-amplified-compression` named this one identity, so it was factored out into its own shard and validated first, bottom-up. It has no dependencies. It is imported by `lem-compcb-amplified-compression` (§6) and by `lem-compcb-entrywise-compression-naturality` (§8), and it is the reason every compression identity in the COMP tier holds at *every* amplification level rather than only at level one.

6 The amplified compression identity

Lemma 54 (`lem-compcb-amplified-compression`)**.** *There is a universal $e_{\mathrm{cmp}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{cmp}}$, every pair of δ-projections P, Q in an extended ε-C^*-algebra and every $n \ge 1$ satisfy*

$$1_{M_n} \otimes \mathrm{Co}_{P,Q} \;=\; \mathrm{Co}_{I_n \otimes P, I_n \otimes Q} \tag{11}$$

and

$$M_n \otimes \mathcal{S}_{P,Q} \;=\; \mathcal{S}_{I_n \otimes P, I_n \otimes Q}. \tag{12}$$

Transcription note. $\mathrm{Co}_{P,Q}$ *and* $\mathcal{S}_{P,Q}$ *are the compression map and compressed corner of* ***`def-compressed-corner`****;* *δ-projections are as in* ***`def-delta-projection`****, and "extended" is as in* ***`def-extended-epsilon-cstar-algebra`****,* *so that every $M_n \otimes \mathcal{A}$ is itself an ε-C^*-algebra. On the left of* (11) $1_{M_n} \otimes (\cdot)$ *denotes the entrywise amplification of an* operator *on $\mathcal{A}$; on the right $I_n \otimes P$ is the amplified* element.

Registry contract (verbatim).

```
Amplified compression identity: there is a universal e_cmp > 0 such that, whenever e=delta+epsilon
<= e_cmp, every pair of delta-projections P,Q in an extended epsilon-C*-algebra and every n >= 1
satisfy 1_{M_n} tensor Co_{P,Q}=Co_{I_n tensor P,I_n tensor Q} and M_n tensor S_{P,Q}=S_{I_n tensor
P,I_n tensor Q}.
```

Proof (prose account of the af-validated tree). *A uniform threshold.* Write $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Because $\mathcal{A}$ is *extended*, each $M_n \otimes \mathcal{A}$ is an ε-C^*-algebra with compatible matrix multiplication, involution, unit and amplification norm; since $P_n^\dagger = P_n$ and $\|P_n^2 - P_n\| = \|I_n \otimes (P^2 - P)\| = \|P^2 - P\| \le \delta$, the amplified elements are δ-projections *with the same δ, uniformly in n.* Next, the compression is the value of θ at a fixed operator: writing $\mathrm{A}_{P,Q} = \frac{1}{2}(L_P R_Q + R_Q L_P)$ and $T_{P,Q} = 2\mathrm{A}_{P,Q} - I$, `def-compressed-corner` supplies a single universal C with $\|\mathrm{A}_{P,Q}^2 - \mathrm{A}_{P,Q}\| \le Ce$ in *every* ε-C^*-algebra, and the exact operator identity

$$T_{P,Q}^2 - I = 4\big(\mathrm{A}_{P,Q}^2 - \mathrm{A}_{P,Q}\big) \tag{13}$$

converts this into $\|T_{P,Q}^2 - I\| \le 4Ce$. Since the same C applies inside every $M_n \otimes \mathcal{A}$ to the pair (P_n, Q_n), choosing e_{cmp} with $4Ce_{\mathrm{cmp}} < 1$ puts $T_{P,Q}$ and every T_{P_n,Q_n} simultaneously inside the θ/sgn domain of `def-theta-idempotent-approximation`. Uniformity of the threshold in n is exactly what (13) buys.

The amplification is entrywise, not diagonal. Left and right multiplication by the block-diagonal P_n, Q_n act slotwise: for $X = [x_{ij}]$ one has $L_{P_n} R_{Q_n}(X) = [P(x_{ij}Q)]$ and $R_{Q_n} L_{P_n}(X) = [(Px_{ij})Q]$, whence $T_{P_n,Q_n} = 1_{M_n} \otimes T_{P,Q}$. Here the tree records a correction that is genuinely load-bearing. The relevant map is *not* the diagonal embedding of $B(\mathcal{A})$ into $M_n\big(B(\mathcal{A})\big)$ but the entrywise operator amplification

$$\Gamma_n : B(\mathcal{A}) \to B(M_n \otimes \mathcal{A}), \qquad \Gamma_n(U)\big([x_{ij}]\big) = [U(x_{ij})],$$

and the tree proves exactly the properties Γ_n has and no more: coordinate compression gives $\|x_{ij}\| \le \|X\|$ and $\|E_{ij} \otimes y\| = \|y\|$, so $\|\Gamma_n(U)\| \le n^2 \|U\|$ — bounded at each fixed n, with *no* claim of isometry and *no* uniform-in-n operator bound; and entrywise evaluation gives $\Gamma_n(I) = I$ and $\Gamma_n(UV) = \Gamma_n(U)\Gamma_n(V)$. Boundedness at fixed n is all that is needed, because the θ series is a norm limit for each fixed n separately.

Transporting θ. Put $T = T_{P,Q}$ and $R = T^2 - I$. On $\|R\| < 1$ the registered calculus writes $\theta(T) = \frac{1}{2}\bigl(I + T\sum_{k\geq 0} b_k R^k\bigr)$ with b_k the Taylor coefficients of $(1+z)^{-1/2}$ — the branch identified in `lem-compcb-amplification-naturality`. Since Γ_n is a unital algebra homomorphism it commutes with every partial sum and satisfies $\Gamma_n(R) = \Gamma_n(T)^2 - I$; boundedness lets one pass to the norm limit; and the threshold above guarantees that the limit at $\Gamma_n(T) = T_{P_n,Q_n}$ is the value $\theta(T_{P_n,Q_n})$. Hence $\Gamma_n(\theta(T_{P,Q})) = \theta(T_{P_n,Q_n})$, which by $\mathrm{Co}_{P,Q} = \theta(T_{P,Q})$ is precisely (11).

Ranges. For any linear $C : \mathcal{A} \to \mathcal{A}$ and finite n one has $\mathrm{Im}(1_{M_n} \otimes C) = M_n \otimes \mathrm{Im}\, C$: every output has entries $C(x_{ij})$, and conversely finitely many entrywise preimages assemble into a single matrix preimage. Since $\mathcal{S}_{P,Q} = \mathrm{Im}\,\mathrm{Co}_{P,Q}$ by `def-compressed-corner`, applying this to (11) gives (12).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplified-compression`: 13 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplification-naturality`. This is the structural keystone of the COMP tier: every later result that says "uniformly in n" does so because (11)–(12) let a level-one fact be read at level n with no loss. It is imported by `lem-compcb-amplified-compression-identities`, `lem-compcb-entrywise-compression-naturality`, `lem-compcb-rectangular-product`, `lem-compcb-amplifie`
`lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-produc`
`lem-compcb-single-compression-transfer` and `lem-hcb2-product-defect`.

7 Idempotence and adjoint symmetry of amplified compressions

Lemma 55 (`lem-compcb-amplified-compression-identities`). *There is a universal* $e_{\mathrm{cmp}} > 0$ *such that, whenever* $\mathcal{A}$ *is an extended* ε*-*C^**-algebra,* $e = \delta + \varepsilon \le e_{\mathrm{cmp}}$*,* P, Q *are* δ*-projections in* $\mathcal{A}$*,* $n \ge 1$*, and* $X \in M_n \otimes \mathcal{A}$*, one has*

$$\mathrm{Co}_{P_n,Q_n}^2 = \mathrm{Co}_{P_n,Q_n}, \qquad \mathrm{Co}_{P_n,Q_n}(X)^\dagger = \mathrm{Co}_{Q_n,P_n}\big(X^\dagger\big), \tag{14}$$

where $P_n = I_n \otimes P$ *and* $Q_n = I_n \otimes Q$*.*

Registry contract (verbatim).

```
Amplified compression identities: there is a universal e_cmp > 0 such that, whenever A is an
extended epsilon-C*-algebra, e=delta+epsilon <= e_cmp, P,Q are delta-projections in A, n >= 1, and
X is in M_n tensor A, one has Co_{P_n,Q_n}^2=Co_{P_n,Q_n} and
Co_{P_n,Q_n}(X)^dagger=Co_{Q_n,P_n}(X^dagger), where P_n=I_n tensor P and Q_n=I_n tensor Q.
```

Proof (prose account of the af-validated tree). The argument is short because the amplification lemma has already done the structural work; what remains is to check that two *exact* level-one identities survive the passage to matrices.

Identification. Take e_{cmp} to be the universal constant supplied by `lem-compcb-amplified-compression`. For $e \le e_{\mathrm{cmp}}$ that lemma gives $\mathrm{Co}_{P_n,Q_n} = 1_{M_n} \otimes \mathrm{Co}_{P,Q}$, and — applied a second time to the *ordered* pair (Q, P), which is a distinct instance — also $\mathrm{Co}_{Q_n,P_n} = 1_{M_n} \otimes \mathrm{Co}_{Q,P}$. Both amplified compressions are therefore canonical amplifications of level-one maps.

The level-one identities. Writing $T = \mathrm{Co}_{P,Q}$ and $U = \mathrm{Co}_{Q,P}$, `def-compressed-corner` states exactly $T^2 = T$ and $T(x)^\dagger = U(x^\dagger)$ for all $x \in \mathcal{A}$. These are equalities, not estimates; no smallness is consumed by them, and the threshold above is needed only for the identification step.

Entrywise transfer. Let T, U be any linear maps on a self-adjoint operator space satisfying those two identities, and let $T_n = 1_{M_n} \otimes T$, $U_n = 1_{M_n} \otimes U$. For $X = [x_{ij}]$, idempotence is checked slotwise,

$$(T_n^2 X)_{ij} = T\big(T(x_{ij})\big) = (T^2)(x_{ij}) = T(x_{ij}) = (T_n X)_{ij},$$

so $T_n^2 = T_n$. Adjoint symmetry combines the matrix involution with the slotwise action: the (i,j) entry of $T_n(X)^\dagger$ is $T(x_{ji})^\dagger = U\big(x_{ji}^\dagger\big)$, while the (i,j) entry of $U_n(X^\dagger)$ is $U\big((X^\dagger)_{ij}\big) = U\big(x_{ji}^\dagger\big)$. The transposition of indices is what makes the *swap* $\mathrm{Co}_{P_n,Q_n} \mapsto \mathrm{Co}_{Q_n,P_n}$ in (14) the right statement. Combining the identification with the transfer gives the lemma.

A recorded contract amendment. The export's final node is a contract-reconciliation step, and it documents why the statement is shaped as it is. The originally codified contract left the ambient algebra unbound: a challenge observed that the proof used, but the contract never asserted, that $\mathcal{A}$ is an extended ε-C^*-algebra with P, Q in $\mathcal{A}$. The contract was amended to the prover's corrected root text — the version reproduced above — as a mechanical, verdict-driven typing repair. The added clauses are the intended and already-used hypotheses of the source argument, so nothing was strengthened or weakened.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplified-compression-identities`: 7 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression`. These two identities are the algebraic skeleton on which every quantitative COMP estimate is hung: idempotence is what makes "X lies in the corner" equivalent to "X is fixed by its compression", and the dagger swap is what lets an estimate proved on one side of a rectangular corner be read on the other. It is imported by `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`, `lem-hcb2-amplified-adjointness` and `lem-hcb1-variational-`

8 Entrywise compression naturality at a matrix unit

Lemma 56 (`lem-compcb-entrywise-compression-naturality`). *There is a universal $e_{\mathrm{nat}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{nat}}$, Q is a δ-projection in an extended ε-C^*-algebra $\mathcal{A}$, $n \ge 1$, and E_{11} is the $(1,1)$ matrix unit of M_n, every $Z \in \mathcal{A}$ satisfies*

$$\mathrm{Co}_{I_n \otimes Q, I_n \otimes Q}\big(E_{11} \otimes Z\big) \;=\; E_{11} \otimes \mathrm{Co}_Q(Z). \tag{15}$$

Transcription note. Co_Q *is the* $P = Q$ *abbreviation* $\mathrm{Co}_{Q,Q}$ *of* ***`def-compressed-corner`***. *The point of the statement is that the amplified compression does not merely preserve the corner but acts* in the single occupied slot*, leaving the zero blocks zero.*

Registry contract (verbatim).

```
Entrywise compression naturality: there is a universal e_nat > 0 such that, whenever e =
delta+epsilon <= e_nat, Q is a delta-projection in an extended epsilon-C*-algebra A, n >= 1, and
E_{11} is the (1,1) matrix unit of M_n, every Z in A satisfies Co_{I_n tensor Q, I_n tensor
Q}(E_{11} tensor Z) = E_{11} tensor Co_Q(Z).
```

Proof (prose account of the af-validated tree). Set $e_{\mathrm{nat}} := e_{\mathrm{cmp}}$, the universal threshold supplied by `lem-compcb-amplified-compression`. That lemma is an equality of *linear maps* on $M_n(\mathcal{A})$, and the proof consists of specializing it and then evaluating.

Specializing the imported identity to the pair $(P, Q) = (Q, Q)$ gives

$$\mathrm{Co}_{I_n \otimes Q, I_n \otimes Q} \;=\; 1_{M_n} \otimes \mathrm{Co}_{Q,Q} \;=\; 1_{M_n} \otimes \mathrm{Co}_Q,$$

the last equality being nothing but the abbreviation $\mathrm{Co}_{Q,Q} = \mathrm{Co}_Q$ recorded in `def-compressed-corner`. This is where the whole quantitative content of the statement sits: the threshold, the θ-domain control and the amplification argument are all inherited from `lem-compcb-amplified-compression`, which in turn rests on `lem-compcb-amplification-naturality`.

What remains is a one-line evaluation. For any linear $T : \mathcal{A} \to \mathcal{A}$ the tensor amplification acts on elementary tensors by $(1_{M_n} \otimes T)(X \otimes Z) = X \otimes T(Z)$; taking $T = \mathrm{Co}_Q$ and $X = E_{11}$ gives (15).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-entrywise-compression-naturality`: 4 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplification-naturality`. Like §5, this shard exists because of a recorded stall: the column-Hilbert workspace (§17) reached its third consecutive stall on exactly the identity (15), whose statement was then extracted mechanically from the blocking challenge and validated on its own. It is an explicit single-slot specialization of the $1_{M_n}\otimes$-clause: that clause identifies two maps, while (15) records its value on the single-slot element needed downstream. It has one registry consumer, `lem-hcb-column-hilbert-squared`.

9 The uniform rectangular compressed-product estimate

Lemma 57 (`lem-compcb-rectangular-product`). *There are universal $C_{\mathrm{co}} < \infty$ and $e_{\mathrm{co}} > 0$ such that, for $e = \delta + \varepsilon \le e_{\mathrm{co}}$, every compatible amplified rectangular pair A, B satisfies*

$$\|A \cdot B - AB\| \;\le\; C_{\mathrm{co}}\, e\, \|A\|\, \|B\|. \tag{16}$$

Transcription note. *The tree fixes "compatible amplified rectangular pair" as an explicit abbreviation, and the quantifier in* (16) *ranges over exactly these data and no others: an extended ε-C^*-algebra $\mathcal{A}$; an integer $n \ge 1$; δ-projections P, Q, R in $\mathcal{A}$; the amplifications $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$, $R_n = I_n \otimes R$; elements $A \in \mathcal{S}_{P_n,Q_n}$ and $B \in \mathcal{S}_{Q_n,R_n}$; and the compressed product $A \cdot B = \mathrm{Co}_{P_n,R_n}(AB)$ formed as in* `def-compressed-corner` *inside $M_n \otimes \mathcal{A}$. In particular both factors live in one* square *amplification $M_n \otimes \mathcal{A}$ — a restriction that later forces a separate row/column statement (§14).*

Registry contract (verbatim).

```
Uniform rectangular compressed-product estimate: there are universal C_co < infinity and e_co > 0
such that, for e=delta+epsilon <= e_co, every compatible amplified rectangular pair satisfies ||A
dot B-AB|| <= C_co*e*||A||||B||.
```

Proof (prose account of the af-validated tree). The shape of the argument is: unpack a level-one universal estimate into named constants, verify that the amplified data satisfy its hypotheses, and apply it inside $M_n \otimes \mathcal{A}$. No new analysis is done at level n; the whole point is that none is needed.

Unpacking the level-one estimate. `def-compressed-corner` states, for any triple U, V, W of δ-projections in an ε-C^*-algebra, that $X \cdot Y = \mathrm{Co}_{U,W}(XY)$ on $\mathcal{S}_{U,V} \times \mathcal{S}_{V,W}$ and that $\|X \cdot Y - XY\| \le O(\delta{+}\varepsilon)\|X\|\|Y\|$. The tree makes the big-O explicit: there are constants $C_{\mathrm{corner}} < \infty$ and $e_{\mathrm{corner}} > 0$, independent of the algebra, of the projections and of the elements, such that the bound reads $C_{\mathrm{corner}} e\|X\|\|Y\|$ whenever $e \le e_{\mathrm{corner}}$. This independence is the reason the final constant is uniform in n: the level-one estimate is a statement about *every* ε-C^*-algebra, and $M_n \otimes \mathcal{A}$ is one of them.

The amplified data are admissible. Because $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε-C^*-algebra for every n; and P_n is a δ-projection whenever P is, since $P_n^\dagger = P_n$, $P_n^2 - P_n = I_n \otimes (P^2 - P)$ and the block-diagonal norm satisfies $\|I_n {\otimes} Z\| = \|Z\|$ — likewise for Q_n, R_n. Applying `lem-compcb-amplified-compression` to the three pairs (P, Q), (Q, R), (P, R) identifies the amplified corners with $M_n {\otimes} \mathcal{S}_{P,Q}$, $M_n {\otimes} \mathcal{S}_{Q,R}$, $M_n {\otimes} \mathcal{S}_{P,R}$ and the amplified compressions with the entrywise amplifications; `lem-compcb-amplified-compression` supplies the idempotence and dagger identities for those maps. Hence a compatible amplified rectangular pair really is an ordinary rectangular pair in $M_n {\otimes} \mathcal{A}$, and its amplified compressed product is literally the ordinary compressed product there, $A \cdot B = \mathrm{Co}_{P_n,R_n}(AB)$, which lands in $\mathcal{S}_{P_n,R_n}$.

Assembly. Take

$$C_{\mathrm{co}} = C_{\mathrm{corner}}, \qquad e_{\mathrm{co}} = \min\bigl(e_{\mathrm{corner}}, e_{\mathrm{cmp}}\bigr),$$

with e_{cmp} the threshold of the amplification dependencies; both are universal, finite and positive. For $e \le e_{\mathrm{co}}$ the level-one estimate applies inside $M_n \otimes \mathcal{A}$ with $U = P_n$, $V = Q_n$, $W = R_n$, $X = A$, $Y = B$, and gives (16) with constants independent of n and of all the data.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-rectangular-product`: 12 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-i`
This is the workhorse estimate of the COMP tier — the licence to replace a compressed product by an ambient product at cost $O(e)$ — and it is invoked by almost everything downstream: `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-hcb0-compressed-associator`, `lem-compcb-corner-algebra`, `lem-hcb-column-hilbert-squared`, `lem-compcb-single-compression-transfe` and `lem-hcb2-product-defect`. Its square-amplification compatibility clause is a real restriction, not a formality: §14 exists precisely because a row/column pair fails it.

10 Amplified almost-containment of corners

Lemma 58 (`lem-compcb-amplified-almost-containment`)**.** *There are universal $C_{\mathrm{ac}} < \infty$ and $e_{\mathrm{ac}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{ac}}$, the elements P_1, P, Q_1, Q are δ-projections with*

$$\|P_1 P - P_1\| \le \delta, \qquad \|Q_1 Q - Q_1\| \le \delta, \tag{17}$$

and $n \ge 1$, every $X \in M_n \otimes \mathcal{S}_{P_1,Q_1}$ satisfies

$$\|\mathrm{Co}_{P_n,Q_n}(X) - X\| \;\le\; C_{\mathrm{ac}}\, e\, \|X\|, \tag{18}$$

with $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Transcription note. *Hypothesis* (17) *is the approximate form of "$P_1 \le P$ and $Q_1 \le Q$"; the conclusion says that the inner corner $\mathcal{S}_{P_1,Q_1}$ sits almost inside the outer one, uniformly at every amplification level.*

Registry contract (verbatim).

```
Amplified almost-containment: there are universal C_ac < infinity and e_ac > 0 such that, whenever
e=delta+epsilon <= e_ac, P_1,P,Q_1,Q are delta-projections with ||P_1 P-P_1||,||Q_1 Q-Q_1|| <=
delta, every n >= 1, and X in M_n tensor S_{P_1,Q_1}, one has ||Co_{P_n,Q_n}(X)-X|| <=
C_ac*e*||X||.
```

Proof (prose account of the af-validated tree). *A uniform ledger.* All four elements amplify to δ-projections in $M_n \otimes \mathcal{A}$, and the containment hypotheses amplify too, with the involution converting one side into the other: $P_n P_{1,n} - P_{1,n} = I_n \otimes (PP_1 - P_1) = I_n \otimes (P_1 P - P_1)^\dagger$ has norm at most δ, and $Q_{1,n} Q_n - Q_{1,n} = I_n \otimes (Q_1 Q - Q_1)$ likewise. A crude but uniform norm bound is also available: for $\varepsilon, \delta \le \frac{1}{2}$ the ε-C^* axioms and `def-delta-projection` give $(1-\varepsilon)\|R_n\|^2 \le \|R_n^2\| \le \|R_n\| + \delta$, hence $\|R_n\| \le B := 3$ for each of the four. Finally `def-compressed-corner` supplies one universal k_{co} with

$$\|\mathrm{Co}_{R,S}(Z) - R(ZS)\| \le k_{\mathrm{co}} e \|Z\|, \qquad \|\mathrm{Co}_{R,S}(Z) - (RZ)S\| \le k_{\mathrm{co}} e \|Z\|, \tag{19}$$

in every $M_n \otimes \mathcal{A}$. The threshold e_0 is the minimum of $\frac{1}{2}$, the threshold behind (19), and those of the two amplification dependencies.

Control in the source corner. By `lem-compcb-amplified-compression`, $M_n \otimes \mathcal{S}_{P_1,Q_1} = \mathcal{S}_{P_{1,n},Q_{1,n}}$, so X lies in the image of $\mathrm{Co}_{P_{1,n},Q_{1,n}}$; idempotence (`lem-compcb-amplified-compression-identities`) then upgrades membership to the fixed-point identity $\mathrm{Co}_{P_{1,n},Q_{1,n}}(X) = X$. Feeding this into (19) with $R = P_{1,n}$, $S = Q_{1,n}$, $Z = X$ replaces the compression by X itself and yields the two bracketings

$$\|X - P_{1,n}(XQ_{1,n})\| \le k_{\mathrm{co}} e \|X\|, \qquad \|X - (P_{1,n}X)Q_{1,n}\| \le k_{\mathrm{co}} e \|X\|. \tag{20}$$

Transfer to the outer projections. The claim is that X is almost invariant under P_n on the left and Q_n on the right, with a universal $L := k_{\mathrm{co}}(2B+1) + 2B(B^2+2)$. For the left estimate put $D = P_{1,n}(XQ_{1,n})$ and $Y = XQ_{1,n}$, and telescope

$$P_n X - X = P_n(X - D) + \big(P_n D - D\big) + (D - X).$$

The outer terms are controlled by (20) and submultiplicativity. The middle term is where the containment hypothesis and the *approximate* associativity of an ε-C^*-algebra both enter: $P_n D - D = P_n(P_{1,n}Y) - P_{1,n}Y$ is bounded by $\big[\varepsilon B^2 + (1+\varepsilon)\delta\big]\|Y\|$, since re-associating costs $\varepsilon\|P_n\|\|P_{1,n}\|\|Y\|$

and $(P_nP_{1,n} - P_{1,n})Y$ costs $(1+\varepsilon)\delta\|Y\|$. With $\|Y\| \le 2B\|X\|$ and $\varepsilon, \delta \le e$ the three terms sum to at most $Le\|X\|$. The right estimate is the mirror image, run with $D = (P_{1,n}X)Q_{1,n}$, $Y = P_{1,n}X$ and the hypothesis $\|Q_{1,n}Q_n - Q_{1,n}\| \le \delta$.

Assembly. Insert $P_n(XQ_n)$ and P_nX:

$$\|\mathrm{Co}_{P_n,Q_n}(X) - X\| \le \|\mathrm{Co}_{P_n,Q_n}(X) - P_n(XQ_n)\| + \|P_n(XQ_n - X)\| + \|P_nX - X\|.$$

The first term is $\le k_{\mathrm{co}}e\|X\|$ by (19); the second is $\le (1+\varepsilon)\|P_n\|\, Le\|X\| \le 2BLe\|X\|$; the third is $\le Le\|X\|$. Hence (18) holds with $C_{\mathrm{ac}} = k_{\mathrm{co}} + (2B+1)L$ and $e_{\mathrm{ac}} = e_0$, both finite, positive and universal.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-amplified-almost-containment`: 13 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-i`
It is the corner-inclusion primitive of the COMP tier: it says that once a corner is approximately contained in another, the outer compression is a near-identity on it, again with one constant at every level. Its registry consumer is `lem-compcb-single-compression-transfer` (§18).

11 The compressed-unit norm estimate

Lemma 59 (`lem-compcb-compressed-unit-norm`). *There are universal $C_{\mathrm{co}} < \infty$ and $e_{\mathrm{co}} > 0$ such that, for $e = \delta + \varepsilon \le e_{\mathrm{co}}$,*

$$\|u_T\| \;\le\; 1 + C_{\mathrm{co}}\, e \qquad \text{for every } \delta\text{-projection } T, \tag{21}$$

and

$$\big|\, \|u_T\| - 1 \,\big| \;\le\; C_{\mathrm{co}}\, e \qquad \text{for every nonvanishing } T. \tag{22}$$

Transcription note. *$u_T = \mathrm{Co}_T(T)$ is the compressed unit of* ***`def-compressed-corner`****; "nonvanishing" is the second branch of the alternative in* ***`def-delta-projection`****. The asymmetry is deliberate and load-bearing: the two-sided bound is* false *in general, since a δ-projection may fall in the small branch $\|T\| = O(\delta)$, in which case u_T is small rather than close to a unit.*

Registry contract (verbatim).

```
Compressed-unit norm estimate: there are universal C_co < infinity and e_co > 0 such that, for
e=delta+epsilon <= e_co, every delta-projection T satisfies ||u_T|| <= 1+C_co*e, and every
nonvanishing T satisfies abs(||u_T||-1) <= C_co*e.
```

Proof (prose account of the af-validated tree). This shard's tree is instructive twice over, because its first attempt was audited and found unsound, and the audit — not the estimate — is what is validated in the early nodes.

The blocked first attempt. The originally registered inputs (`def-extended-epsilon-cstar-algebra` together with `def-compressed-corner`) yield only the δ-projection dichotomy, the identity $u_T = \mathrm{Co}_T(T)$, and the comparison $\|\mathrm{Co}_T - L_T R_T\| \le ke$. They do *not* yield the quantitative multiplication bound $\|XY\| \le (1+\varepsilon)\|X\|\|Y\|$ — the extended definition says only that each $M_n \otimes \mathcal{A}$ *is* an ε-C^*-algebra, without supplying that term's quantitative axioms — nor any unit-norm axiom that would give (22) from "$\mathcal{S}_T$ is an $O(e)$-C^*-algebra with unit u_T". A closure audit node records this verdict explicitly and refuses the root: without those axioms there is no sound bridge from the permitted premises to the contract.

The repair. The missing quantitative content was then registered in the workspace as byte-matched externals — the ε-C^* norm axioms (`epsilon-banach-cstar-norm-axioms`) and the nonvanishing alternative — and each $O(\cdot)$ in them unpacked into a named universal constant after shrinking the threshold. That gives universal $a, b, k < \infty$ and $e_0 > 0$ such that for $e \le e_0$: (i) every δ-projection T satisfies $\|T\| \le a\delta$ or is nonvanishing; (ii) $\|XY\| \le (1+\varepsilon)\|X\|\|Y\|$; (iii) $u_T = \mathrm{Co}_T(T)$ with $\|\mathrm{Co}_T - L_T R_T\| \le ke$; and (iv) nonvanishing T obeys $|\|T\| - 1| \le be$.

Small branch. Suppose $\|T\| \le a\delta$ and $e \le \min(e_0, 1)$. Since $(L_T R_T)(T) = T(T^2)$, two applications of (ii) give $\|(L_T R_T)(T)\| \le (1+\varepsilon)^2 \|T\|^3 \le 4a^3 e^3 \le 4a^3 e$, while (iii) gives $\|u_T - (L_T R_T)(T)\| \le ke\|T\| \le kae^2 \le kae$. Hence $\|u_T\| \le De$ with $D = 4a^3 + ka$, and in particular $\|u_T\| \le 1 + De$. Note that here the compressed unit is *small*, which is why no two-sided statement can hold.

Nonvanishing branch. Shrink e_0 so that $e_0 \le 1$ and $be_0 \le 1$; then (iv) gives $\|T\| \le 2$. The compressed unit is compared directly with T rather than through any unit axiom: bilinearity gives $T(T^2) - T = T(T^2 - T) + (T^2 - T)$, so by (iii), (ii) and $\|T^2 - T\| \le \delta$,

$$\|u_T - T\| \le \|u_T - (L_T R_T)(T)\| + \|T(T^2) - T\| \le ke\|T\| + (1+\varepsilon)\|T\|\delta + \delta \le (2k+5)e.$$

The reverse triangle inequality together with (iv) then gives $\big|\|u_T\| - 1\big| \le (2k+5+b)e$, hence also the upper bound.

Assembly. With e_{co} the shrunken threshold and $C_{\mathrm{co}} = \max\bigl(4a^3+ka,\ 2k+5+b\bigr)$, the dichotomy routes every δ-projection to one of the two branches: the first gives (21), the second gives (21) and (22). The final assembly node states expressly that it uses neither the superseded audit node nor any compressed-unit norm axiom.

Status. `proved`; `af:  validated` (T0). Workspace `proofs/lem-compcb-compressed-unit-norm`: 15 nodes (12 `validated`, 3 superseded nodes archived), root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-i`
It supplies the unit-size input to `lem-compcb-corner-algebra` (§16) and to `lem-hcb-column-hilbert-squared` (§17); in the latter it is applied to the one-dimensional projection Q of an H-CB datum, which is why the nonvanishing hypothesis of (22) has to be discharged there by a separate bridge rather than assumed.

12 Exact adjointness of the amplified Ha map

Lemma 60 (`lem-hcb2-amplified-adjointness`). *There is a universal $e_{\text{adj}} > 0$ such that every H-CB datum with $e \le e_{\text{adj}}$, every $n \ge 1$ and every $Z \in M_n \otimes \mathcal{S}_{P,R}$ satisfy*

$$\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)^{\dagger} \;=\; \big(\mathrm{Ha}^{Q}_{R,P}\big)_n(Z^{\dagger}). \tag{23}$$

Transcription note. *The datum is that of* ***`def-hcb-datum`*** *and* $\mathrm{Ha}^{Q}_{P,R}$ *is the map of* ***`def-ha-map`****;* $e = \delta + \varepsilon$*. The two daggers in* (23) *are different operations: on the left it is the Hilbert-space adjoint of the operator* $\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)$ *between amplified column spaces, on the right it is the involution of the ambient algebra applied to* Z*. The identity is* exact *— no* $O(e)$ *error — and the threshold is needed only so that the sesquilinear forms involved are genuine inner products.*

Registry contract (verbatim).

```
Exact amplified Ha adjointness: there is a universal e_adj > 0 such that every H-CB datum with e <=
e_adj, every n >= 1, and every Z in M_n tensor S_{P,R} satisfy
(Ha^Q_{P,R})_n(Z)^dagger=(Ha^Q_{R,P})_n(Z^dagger).
```

Proof (prose account of the af-validated tree). *Threshold.* Set $e_{\text{adj}} = \min(e_{\text{cmp}}, e_{\text{ip}})$, where e_{cmp} comes from `lem-compcb-amplified-compression-identities` and e_{ip} is a universal smallness threshold below which the Euclidean sesquilinear forms of `def-ha-map` and of the column-Hilbert displays are genuine inner products — such a threshold exists because those forms agree with the norms up to a $1 \pm O(e)$ factor. Fix an arbitrary datum, n and Z.

Corner symmetries. Applying `lem-compcb-amplified-compression-identities` at $n = 1$: if $a \in \mathcal{S}_{A,B} = \operatorname{Im} \mathrm{Co}_{A,B}$ then $a = \mathrm{Co}_{A,B}(a)$, so $a^{\dagger} = \mathrm{Co}_{B,A}(a^{\dagger})$ and $a^{\dagger} \in \mathcal{S}_{B,A}$; and since $Q^{\dagger} = Q$, the compressed unit $\widetilde{Q} = \mathrm{Co}_{Q}(Q)$ is Hermitian.

Exact product reversal. The pivotal algebraic fact is

$$(a \cdot b)^{\dagger} = b^{\dagger} \cdot a^{\dagger} \qquad (a \in \mathcal{S}_{A,B},\ b \in \mathcal{S}_{B,C}), \tag{24}$$

and it is *exact*. The proof is three substitutions: the compressed-product display gives $a \cdot b = \mathrm{Co}_{A,C}(ab)$; the imported identities at $n = 1$ give $\mathrm{Co}_{A,C}(ab)^{\dagger} = \mathrm{Co}_{C,A}\big((ab)^{\dagger}\big)$; the involution axiom gives $(ab)^{\dagger} = b^{\dagger}a^{\dagger}$; and a second use of the display, legitimate because the reversed corner memberships were just established, gives $\mathrm{Co}_{C,A}(b^{\dagger}a^{\dagger}) = b^{\dagger} \cdot a^{\dagger}$. No associativity estimate and no approximation enter. The tree records a correction here worth one sentence: an earlier audit node concluded that (24) was *not* derivable, because the formula $a \cdot b = \mathrm{Co}_{A,C}(ab)$ had not been registered as an allowed input in the workspace; after the compressed-product display was registered, that negative conclusion — and the associated demand for further definition provisioning — was explicitly withdrawn.

Level one. Fix $z \in \mathcal{S}_{P,R}$ and put $T = \mathrm{Ha}^{Q}_{P,R}(z)$. The defining relation of `def-ha-map` reads, for all $x \in \mathcal{S}_{R,Q}$ and $y \in \mathcal{S}_{P,Q}$,

$$(y^{\dagger} \cdot z) \cdot x + y^{\dagger} \cdot (z \cdot x) = 2\,\langle y, Tx \rangle\, \widetilde{Q}. \tag{25}$$

Apply † to (25). On the left, (24) reverses each compressed product, turning the two summands into $x^{\dagger} \cdot (z^{\dagger} \cdot y)$ and $(x^{\dagger} \cdot z^{\dagger}) \cdot y$. On the right, $\widetilde{Q}$ is Hermitian and the scalar is conjugated, and conjugate symmetry together with the definition of the Hilbert-space adjoint gives $\overline{\langle y, Tx\rangle} = \langle Tx, y\rangle = \langle x, T^\dagger y\rangle$. Thus for all x, y,

$$(x^\dagger \cdot z^\dagger) \cdot y + x^\dagger \cdot (z^\dagger \cdot y) = 2\,\langle x, T^\dagger y\rangle\,\widetilde{Q}.$$

This is precisely the defining relation of `def-ha-map` for the pair (R, P), the symbol $z^\dagger$, the input y and the test vector x; and that relation determines its solution uniquely. Hence $\mathrm{Ha}^Q_{R,P}(z^\dagger)y = T^\dagger y$ for every y, i.e. $\mathrm{Ha}^Q_{R,P}(z^\dagger) = \mathrm{Ha}^Q_{P,R}(z)^\dagger$.

Amplification. Write $Z = [Z_{ij}]$. By the amplification convention of the contract, $\left(\mathrm{Ha}^Q_{P,R}\right)_n(Z)$ is the block operator $[\mathrm{Ha}^Q_{P,R}(Z_{ij})]$ from the n-fold Hilbert direct sum of $\mathcal{S}_{R,Q}$ to that of $\mathcal{S}_{P,Q}$; its Hilbert-space adjoint therefore has (i, j) block $\mathrm{Ha}^Q_{P,R}(Z_{ji})^\dagger$. On the other side, $(Z^\dagger)_{ij} = Z^\dagger_{ji}$, which lies in $\mathcal{S}_{R,P}$ by the corner symmetry above, so the (i, j) block of $\left(\mathrm{Ha}^Q_{R,P}\right)_n(Z^\dagger)$ is $\mathrm{Ha}^Q_{R,P}(Z^\dagger_{ji})$. The level-one identity equates the two blockwise, and equality of all blocks is (23). Since n and Z were arbitrary, the lemma follows.

Status. `proved`; `af:  validated` (T0). Workspace `proofs/lem-hcb2-amplified-adjointness`: 21 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression-identities`. Being exact rather than approximate, it is what allows a C^*-identity argument to be run on the amplified Ha map: it is used in `lem-hcb3-diagonal-upper-norm` (§23) to identify $\left(\mathrm{Ha}^Q_{P,P}\right)_n(Z)^\dagger$ with $\left(\mathrm{Ha}^Q_{P,P}\right)_n(Z^\dagger)$ inside $\|T\|^2 = \|T^\dagger T\|$, and the same role in `lem-hcb3-diagonal-lower-modulus` (§25). It is also a declared dependency of `lem-hcb3-offdiagonal-inverse` (§27) and `lem-hcb4-canonical-close` (§29), both now `af:  validated`.

13 The uniform compressed-unit action

Lemma 61 (`lem-compcb-compressed-unit-action`). *There are universal $C_{\rm co} < \infty$ and $e_{\rm co} > 0$ such that, for $e = \delta + \varepsilon \le e_{\rm co}$, every compatible amplified rectangular corner satisfies*

$$\|u_T \cdot A - A\| \le C_{\rm co}\, e\, \|A\|, \qquad \|A \cdot u_R - A\| \le C_{\rm co}\, e\, \|A\|. \tag{26}$$

Transcription note. *As spelled out in the tree, the datum is an extended ε-C^*-algebra $\mathcal{A}$, an $n \ge 1$, δ-projections T, R with $T_n = I_n \otimes T$, $R_n = I_n \otimes R$, and an element $A \in \mathcal{S}_{T_n,R_n}$; the compressed units are $u_{T,n} = \mathrm{Co}_{T_n,T_n}(T_n)$ and $u_{R,n} = \mathrm{Co}_{R_n,R_n}(R_n)$. By* `lem-compcb-amplified-compression` *these equal $I_n \otimes \mathrm{Co}_T(T)$ and $I_n \otimes \mathrm{Co}_R(R)$, which is the notation u_T, u_R of the contract.*

Registry contract (verbatim).

```
Uniform compressed-unit action: there are universal C_co < infinity and e_co > 0 such that, for
e=delta+epsilon <= e_co, every compatible amplified rectangular corner satisfies ||u_T dot A-A|| <=
C_co*e*||A|| and ||A dot u_R-A|| <= C_co*e*||A||.
```

Proof (prose account of the af-validated tree). The estimate is assembled in three layers: replace the compressed units by the projections, act with the projections in the *ambient* product, then replace the ambient product by the compressed one.

Primitive bounds. Since $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε-C^*-algebra and T_n, R_n are δ-projections there (adjoints and products amplify entrywise and $\|I_n \otimes Z\| = \|Z\|$). Unpacking the $O(\cdot)$ terms of `def-delta-projection` and `def-compressed-corner` gives universal $a, k < \infty$ and $e_0 \le 1$ such that every δ-projection S obeys $\|S\| \le a\delta$ or $\|S\| \le 1 + ae$ — so $\|T_n\|, \|R_n\| \le M := \max(a, 1+a)$ in either branch — and such that each compression is within ke of both $L \cdot R$ orderings. Idempotence of the amplified compression (`lem-compcb-amplified-compression-identities`) also gives the fixed-point identity $A = \mathrm{Co}_{T_n,R_n}(A)$.

Compressed unit versus projection. For $S \in \{T_n, R_n\}$ and $u_S = \mathrm{Co}_{S,S}(S)$ one has $(L_S R_S)(S) = S(S^2)$, and bilinearity gives $S(S^2) - S = S(S^2 - S) + (S^2 - S)$. With $\|S^2 - S\| \le \delta$, the multiplication bound, and $\|\mathrm{Co}_{S,S} - L_S R_S\| \le ke$,

$$\|u_S - S\| \;\le\; ke\|S\| + (1+\varepsilon)\|S\|\delta + \delta \;\le\; K_u e, \qquad K_u := kM + 2M + 1, \tag{27}$$

and hence $\|u_S\| \le M + K_u$, uniformly in n.

Ambient action of the projections. Put $D = T_n(AR_n)$. The fixed-point identity and the compression comparison give $\|A - D\| \le ke\|A\|$, and one telescopes

$$T_n A - A = T_n(A - D) + \big(T_n D - D\big) + (D - A).$$

The outer terms cost $2Mke\|A\|$ and $ke\|A\|$. For the middle term write $Y = AR_n$, so that $T_n D - D = T_n(T_n Y) - T_n Y$; approximate associativity contributes $\varepsilon M^2\|Y\|$ and $\|T_n^2 - T_n\| \le \delta$ contributes $2\delta\|Y\|$, and $\|Y\| \le 2M\|A\|$. Altogether $\|T_n A - A\| \le K_T e\|A\|$ with $K_T = 2Mk + 2M(M^2+2) + k$; the right-hand estimate $\|AR_n - A\| \le K_R e\|A\|$ is the mirror argument with $D = (T_n A)R_n$ and the same constant. Substituting the units via (27) — $\|(u_{T,n} - T_n)A\| \le 2K_u e\|A\|$ and symmetrically — gives the ambient form

$$\|u_{T,n} A - A\| \le K_a e\|A\|, \quad \|A\, u_{R,n} - A\| \le K_a e\|A\|, \qquad K_a := \max(K_T, K_R) + 2K_u. \tag{28}$$

From ambient to compressed. The pairs $(u_{T,n}, A)$ and $(A, u_{R,n})$ are compatible amplified rectangular pairs in the exact sense required by `lem-compcb-rectangular-product`, which therefore gives universal C_p with $\|u_{T,n} \cdot A - u_{T,n}A\| \le C_p e \|u_{T,n}\| \|A\|$ and its mirror. Combining with (28) and $\|u_S\| \le M + K_u$ yields (26) with

$$C_{\mathrm{co}} = K_a + C_p\,(M + K_u), \qquad e_{\mathrm{co}} = \min\bigl(e_0, e_p, e_{\mathrm{cmp}}, e_{\mathrm{id}}, 1\bigr),$$

all universal and independent of n, $\mathcal{A}$, T, R, A.

A bookkeeping remark: the later nodes of the export re-derive (27)–(28) in "dependency-closed" form, so that the final assembly rests only on validated premises and not on any node still pending at the time. The mathematics is unchanged; the restructuring is what let the tree close bottom-up.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-compressed-unit-action`: 20 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-iden` and `lem-compcb-rectangular-product`. It supplies the unit axiom needed by `lem-compcb-corner-algebra` (§16) — a corner cannot be an approximate C^*-algebra without an approximate unit — and it is the input that drives the diagonal unit estimate `lem-hcb3-diagonal-unit` (§22).

14 The row-column compressed-product estimate

Lemma 62 (`lem-compcb-row-column-product`)**.** *There are universal $C_{\mathrm{rc}} < \infty$ and $e_{\mathrm{rc}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{rc}}$, P, Q are δ-projections in an extended ε-C^*-algebra $\mathcal{A}$, $n \ge 1$, and $X, Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$, one has*

$$\|\mathrm{Co}_Q\big(Y^\dagger X\big) - Y^\dagger X\| \;\le\; C_{\mathrm{rc}}\, e\, \|Y\|\, \|X\|, \tag{29}$$

where $Y^\dagger X$ is the ambient product of the $1 \times n$ row $Y^\dagger$ with the $n \times 1$ column X. Transcription note. *The product $Y^\dagger X$ is a single element of $\mathcal{A}$, not a matrix: the row and column contract. This is exactly why `lem-compcb-rectangular-product` does not apply directly — its compatibility clause requires both factors inside one* square *amplification.*

Registry contract (verbatim).

```
Row-column compressed-product estimate: there are universal C_rc < infinity and e_rc > 0 such that,
whenever e = delta+epsilon <= e_rc, P,Q are delta-projections in an extended epsilon-C*-algebra A,
n >= 1, and X,Y are in M_{n,1} tensor S_{P,Q}, one has ||Co_Q(Y^dagger X) - Y^dagger X|| <=
C_rc*e*||Y||*||X||, where Y^dagger X is the ambient product of the 1-by-n row Y^dagger with the
n-by-1 column X.
```

Proof (prose account of the af-validated tree). Set $C_{\mathrm{rc}} = C_{\mathrm{co}}$ and $e_{\mathrm{rc}} = \min(e_{\mathrm{co}}, e_{\mathrm{cmp}})$ with the constants of `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`, and fix arbitrary admissible data. The whole proof is a change of ambient size: embed the row/column pair as off-diagonal blocks of one square amplification, apply the validated square estimate there, and read the result off the single occupied entry.

The embedding. Use the decomposition $\mathbb{C}^{n+1} = \mathbb{C}^n \oplus \mathbb{C}$ and let

$$C_X = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \qquad R_Y = \begin{pmatrix} 0 & 0 \\ Y^\dagger & 0 \end{pmatrix}$$

in $M_{n+1} \otimes \mathcal{A}$. Writing $X = (X_i)$ and $Y = (Y_i)$ with entries in $\mathcal{S}_{P,Q}$, the dagger identity of `def-compressed-corner` gives $Y_i^\dagger \in \mathcal{S}_{Q,P}$, so X has entries in $\mathcal{S}_{P,Q}$ and $Y^\dagger$ has entries in $\mathcal{S}_{Q,P}$ with the indicated orientations. Since $e \le e_{\mathrm{cmp}}$, `lem-compcb-amplified-compression` at level $n+1$ identifies $M_{n+1} \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{I_{n+1} \otimes P, I_{n+1} \otimes Q}$ and the transposed statement, so zero-padding puts C_X and R_Y in exactly those compatible corners. The padding is isometric: in the operator-space matrix norms, insertion by a scalar coordinate isometry and extraction by its adjoint are both contractive, hence equalities; and involution is isometric because $\mathcal{A}$ is self-adjoint. Therefore

$$\|C_X\| = \|X\|, \qquad \|R_Y\| = \|Y^\dagger\| = \|Y\|, \tag{30}$$

and (R_Y, C_X) is a compatible *square* amplified rectangular pair — the right projection of the first factor and the left projection of the second are both $I_{n+1} \otimes P$.

One occupied entry. Ordinary block multiplication gives

$$R_Y C_X = \mathrm{diag}\big(0_{n \times n},\, Y^\dagger X\big),$$

every other block containing a zero factor. On the compressed side, `def-compressed-corner` applied to the triple $(I_{n+1} \otimes Q, I_{n+1} \otimes P, I_{n+1} \otimes Q)$ gives $R_Y \cdot C_X = \mathrm{Co}_{I_{n+1} \otimes Q}(R_Y C_X)$, using the abbreviation $\mathrm{Co}_{S,S} = \mathrm{Co}_S$; and `lem-compcb-amplified-compression` applied to the pair (Q, Q) at level $n+1$ says $\mathrm{Co}_{I_{n+1}\otimes Q} = 1_{M_{n+1}} \otimes \mathrm{Co}_Q$, an entrywise map. Applying it to the diagonal block matrix above,

$$R_Y \cdot C_X = \mathrm{diag}\big(0_{n\times n},\, \mathrm{Co}_Q(Y^\dagger X)\big). \tag{31}$$

Conclusion. Since $e \le e_{\mathrm{co}}$ and (R_Y, C_X) is compatible, `lem-compcb-rectangular-product` yields $\|R_Y \cdot C_X - R_Y C_X\| \le C_{\mathrm{co}} e \|R_Y\| \|C_X\|$. By (31) the left-hand difference is the zero-padded element with single entry $\mathrm{Co}_Q(Y^\dagger X) - Y^\dagger X$, whose norm equals the norm of that entry by the same corner isometry as in (30); and (30) converts the right-hand side. This is (29) with $C_{\mathrm{rc}} = C_{\mathrm{co}}$. Both thresholds are positive universal constants, so $e_{\mathrm{rc}} > 0$, and universal generalisation over the arbitrary data completes the proof.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-row-column-product`: 20 nodes (18 `validated`, 2 superseded nodes archived), root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. This shard was factored out of the column-Hilbert workspace at a recorded third stall: the blocking challenge observed that the validated rectangular-product contract quantifies only over square amplifications, so the pair $(Y^\dagger, X)$ — a $1\times n$ row against an $n\times 1$ column — simply does not satisfy its hypotheses, and the gap had to be closed by a statement of its own rather than by an instantiation. It has one registry consumer, `lem-hcb-column-hilbert-squared` (§17), where $Y = X$ and the estimate becomes the bridge between the column inner product and the ambient norm.

15 The uniform compressed associator estimate

Lemma 63 (`lem-hcb0-compressed-associator`)**.** *There are universal $C_{\rm as} < \infty$ and $e_{\rm as} > 0$ such that every H-CB datum with $e \le e_{\rm as}$ and all compatible amplified rectangular A, B, C satisfy*

$$\|(A \cdot B) \cdot C - A \cdot (B \cdot C)\| \;\le\; C_{\rm as}\, e\, \|A\|\, \|B\|\, \|C\|. \tag{32}$$

Transcription note. *The left-hand side is the compressed associator of* `def-compressed-associator`*; "compatible amplified rectangular" is the datum fixed in §9, here for a composable triple. Both the compressed product and the ambient product are only* approximately *associative — the ambient one by the ε-associator axiom of an ε-C^*-algebra — so the content is that the two failures add up to a single $O(e)$.*

Registry contract (verbatim).

```
Uniform compressed associator: there are universal C_as < infinity and e_as > 0 such that every
H-CB datum with e <= e_as and all compatible amplified rectangular A,B,C satisfy ||(A dot B) dot
C-A dot (B dot C)|| <= C_as*e*||A||||B||||C||.
```

Proof (prose account of the af-validated tree). Let $C_{\rm co}, e_{\rm co}$ be the universal witnesses of `lem-compcb-rectangular-product` and set

$$e_{\rm as} = \min(e_{\rm co}, 1), \qquad C_{\rm as} = 2C_{\rm co}(2 + C_{\rm co}) + 4C_{\rm co} + 1.$$

A product-norm bound. For a compatible pair, the rectangular estimate plus the ambient bound $\|AB\| \le (1+\varepsilon)\|A\|\|B\|$ from the ε-C^* norm axioms give

$$\|A \cdot B\| \le \big(1 + \varepsilon + C_{\rm co}e\big)\|A\|\|B\|, \tag{33}$$

and likewise for $B \cdot C$. This is what keeps the outer terms of the telescope below from degrading.

The telescope. The compressed associator has the exact five-term expansion

$$\begin{aligned}(A \cdot B) \cdot C - A \cdot (B \cdot C) = \big[(A \cdot B) \cdot C - (A \cdot B)C\big] + \big[(A \cdot B)C - (AB)C\big] \\ + \big[(AB)C - A(BC)\big] + \big[A(BC) - A(B \cdot C)\big] + \big[A(B \cdot C) - A \cdot (B \cdot C)\big],\end{aligned} \tag{34}$$

which routes both associations through the ambient product and isolates the ambient associator in the middle.

Write $N = \|A\|\|B\|\|C\|$. The first and fifth terms are rectangular-product errors for the compatible pairs $(A\cdot B, C)$ and $(A, B\cdot C)$, so by (33) each is at most $C_{\rm co}e(1+\varepsilon+C_{\rm co}e)N$. The second and fourth are the rectangular-product errors for (A, B) and (B, C) multiplied by one ambient factor — bilinearity gives $(A\cdot B)C-(AB)C = (A\cdot B-AB)C$ and $A(BC)-A(B\cdot C) = A(BC-B\cdot C)$ — so each is at most $(1+\varepsilon)C_{\rm co}eN$. The third is precisely the ε-associator axiom, $\|(AB)C - A(BC)\| \le \varepsilon N$.

Summation. Adding and using $0 \le \varepsilon \le e \le 1$, so that $1 + \varepsilon + C_{\rm co}e \le 2 + C_{\rm co}$ and $1 + \varepsilon \le 2$, gives (32) with the stated $C_{\rm as}$.

Two pieces of hygiene in the tree are worth recording, since they shape the constant. The summation is written so that the ambient-associator term is bounded directly by eN rather than as $(\varepsilon/e) \cdot eN$, because at the endpoint $e = 0$ one has $\varepsilon = 0$ and every error vanishes — the quotient ε/e is never formed. And the upstream witness is normalised to $C_{\rm co} := \max\{C_{\rm co}, 0\}$, which cannot weaken the imported estimate since $e\|A\|\|B\| \geq 0$, but does make the sign-sensitive arithmetic above valid.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb0-compressed-associator`: 20 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-rectangular-product`. It is the base of the H-CB tier: the defining relation of the Ha map (`def-ha-map`) pairs an element against precisely the difference of the two associations $(Y^\dagger \cdot Z) \cdot X$ and $Y^\dagger \cdot (Z \cdot X)$, so (32) is exactly the estimate that turns that relation into a quantitative bound. It feeds `lem-hcb1-column-action` (§20) and `lem-hcb2-product-defect` (§21).

16 The compressed corner as an extended approximate C^*-algebra

Lemma 64 (`lem-compcb-corner-algebra`)**.** *There are universal $C_{\mathrm{ca}} < \infty$ and $e_{\mathrm{ca}} > 0$ such that, whenever $e = \delta + \varepsilon \le e_{\mathrm{ca}}$ and P is a nonvanishing δ-projection, the corner $\mathcal{S}_P$ — equipped with the compressed product, the inherited involution and the compressed unit $u_P = \mathrm{Co}_P(P)$ — is an extended $C_{\mathrm{ca}}e$-C^*-algebra.* Transcription note. *"Extended" is as in* ***`def-extended-epsilon-cstar-algebra`****: not merely that $\mathcal{S}_P$ satisfies the approximate C^* axioms, but that* every *$M_n \otimes \mathcal{S}_P$ does, with the* same *parameter $C_{\mathrm{ca}}e$, independent of n. That uniformity is the whole content.*

Registry contract (verbatim).

```
Uniform compressed-corner algebra: there are universal C_ca < infinity and e_ca > 0 such that,
whenever e=delta+epsilon <= e_ca and P is a nonvanishing delta-projection, S_P with the compressed
product, inherited involution, and compressed unit u_P=Co_P(P) is an extended C_ca*e-C*-algebra.
```

Proof (prose account of the af-validated tree). The tree is organised as one node per axiom family, over a shared constant ledger, with a preliminary step that moves everything to level n at once.

Constant ledger and amplified setup. Let C_r, e_r come from `lem-compcb-rectangular-product`; C_u, e_u from `lem-compcb-compressed-unit-action`; C_N, e_N from `lem-compcb-compressed-unit-norm`; and e_i from `lem-compcb-amplified-compression-identities`. Put

$$K_p = C_r + 1, \quad K_a = 2C_r(K_p + 3) + 1, \quad C_{\mathrm{ca}} = \max\{K_p, K_a, C_u, C_N\}, \quad e_{\mathrm{ca}} = \min\{e_r, e_u, e_N, e_i, 1\}.$$

With $P_n = I_n \otimes P$, the tree then *derives inside itself* — from `def-compressed-corner` and the registered θ calculus, rather than importing it — the naturality $\mathrm{Co}_{P_n} = \mathrm{id}_{M_n} \otimes \mathrm{Co}_P$: writing $T_P = L_P R_P + R_P L_P - 1$, multiplication by the block-diagonal P_n acts entrywise so $T_{P_n} = \mathrm{id}_{M_n} \otimes T_P$; powers of $T_{P_n} - x_0 I$ amplify by induction; so every partial sum of the θ series is the amplification of the corresponding level-one partial sum, and at fixed finite n the norm limits agree. Consequently $\mathcal{S}_{P_n} = M_n \otimes \mathcal{S}_P$, the P_n-compressed product of $[X_{ij}], [Y_{jk}]$ has entries $\mathrm{Co}_P(\sum_j X_{ij} Y_{jk}) = \sum_j X_{ij} \cdot Y_{jk}$ — exactly the matrix amplification of the compressed product — and $\mathrm{Co}_{P_n}(P_n) = I_n \otimes \mathrm{Co}_P(P)$. Also P_n is a δ-projection, so all four imported estimates apply at every level.

Operator space and completeness. `lem-compcb-amplified-compression-identities` gives $\mathrm{Co}_{P_n}^2 = \mathrm{Co}_{P_n}$, so $\mathcal{S}_{P_n} = \mathrm{Im}\, \mathrm{Co}_{P_n}$ is a closed subspace of the complete space $M_n \otimes \mathcal{A}$ and hence complete in the inherited norm. With the inherited matrix norms $\mathcal{S}_P$ is therefore a complete operator space.

Multiplication and associator. By `lem-compcb-rectangular-product` and the ambient bound $\|AB\| \le (1+\varepsilon)\|A\|\|B\|$, one gets $\|A \cdot B\| \le (1 + K_p e)\|A\|\|B\|$. For the associator, insert successively $(A \cdot B)C$, $(AB)C$, $A(BC)$ and $A(B \cdot C)$ between the two associations; the two outer rectangular errors cost $C_r e(1+K_p e)N$ each, the two errors obtained by multiplying $A \cdot B - AB$ or $BC - B \cdot C$ by an ambient factor cost $(1+\varepsilon)C_r eN$ each, and the ambient associator costs εN, where $N = \|A\|\|B\|\|C\|$. With $\varepsilon \le e \le 1$ the total is at most $K_a eN$.

Involution. If $A \in \mathcal{S}_{P_n}$ then $A = \mathrm{Co}_{P_n}(A)$, so the imported dagger identity gives $A^\dagger = \mathrm{Co}_{P_n}(A^\dagger)$ and the inherited isometric involution preserves the corner. Anti-multiplicativity is exact: $(A \cdot B)^\dagger = \mathrm{Co}_{P_n}\big((AB)^\dagger\big) = \mathrm{Co}_{P_n}(B^\dagger A^\dagger) = B^\dagger \cdot A^\dagger$. The unit is fixed, $u_{P,n}^\dagger = \mathrm{Co}_{P_n}(P_n^\dagger) = u_{P,n}$, since $P_n^\dagger = P_n$. Finally the lower C^*-norm bound: the rectangular estimate applied to $A^\dagger$ and A, isometry of the involution, and the ambient lower estimate give

$$\|A^\dagger \cdot A\| \;\ge\; \|A^\dagger A\| - C_r e\|A\|^2 \;\ge\; \big(1 - \varepsilon - C_r e\big)\|A\|^2 \;\ge\; \big(1 - (C_r + 1)e\big)\|A\|^2.$$

Unit. `lem-compcb-compressed-unit-action` applied to the amplified square corner gives $\|u_{P,n} \cdot A - A\| \le C_u e\|A\|$ and its right-hand mirror at every n. For the unit norm, each P_n is a δ-projection with $\|P_n\| = \|P\|$, so the *nonvanishing* alternative passes from P to P_n; that is exactly the hypothesis under which `lem-compcb-compressed-unit-norm` gives the two-sided $\big|\|u_{P,n}\| - 1\big| \le C_N e$, uniformly in n. This is where the nonvanishing hypothesis of the contract is used, and why it cannot be dropped.

Collecting the four families under the ledger constants gives the extended $C_{\mathrm{ca}}e$-C^* structure.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-corner-algebra`: 14 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression-identities`, `lem-compcb-rectangular-pro` `lem-compcb-compressed-unit-action` and `lem-compcb-compressed-unit-norm`. It is the output of the COMP tier: it converts a bundle of separate estimates into a single *structure* that later tiers may use as a black box. Among its registry consumers are `lem-compcb-single-compression-transfer` (§18), `lem-hcb3-diagonal-upper-norm` (§23) — where the matrix-uniform product axiom is precisely the input that a level-one statement could not supply — `lem-hcb3-uniform-square-lower` (§24), which uses the matching lower axiom, and `lem-hcb3-diagonal-lower-modulus` (§25).

17 The corrected amplified column-Hilbert estimate

Lemma 65 (`lem-hcb-column-hilbert-squared`). *There are universal $C_{\mathrm{col}} < \infty$ and $e_{\mathrm{col}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{col}}$, every $n \ge 1$ and every $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$\Big| \langle X, X\rangle_n - \|X\|_{n,1}^2 \Big| \;\le\; C_{\mathrm{col}}\, e\, \|X\|_{n,1}^2. \tag{35}$$

Transcription note. *$\langle\cdot,\cdot\rangle_n$ is the amplified column inner product of* ***`def-column-hilbert-corner`****, defined by $Y^\dagger \cdot X = \langle Y, X\rangle_n\, u_Q$ with $u_Q = \mathrm{Co}_Q(Q)$, and $\|\cdot\|_{n,1}$ is the ambient norm on the column space. The word "corrected" in the contract refers to the fact that this is the* squared *comparison: the unsquared display at the corresponding place in the pinned source is known to be false and is deliberately excluded from* ***`def-column-hilbert-corner`****.*

Registry contract (verbatim).

```
Corrected amplified column-Hilbert estimate: there are universal C_col < infinity and e_col > 0
such that every H-CB datum with e <= e_col, every n >= 1, and every X in M_{n,1} tensor S_{P,Q}
satisfy abs(<X,X>_n-||X||_{n,1}^2) <= C_col*e*||X||_{n,1}^2.
```

Proof (prose account of the af-validated tree). *Scalarisation.* Fix an admissible datum, n and X, and abbreviate

$$r = \|X\|_{n,1}, \qquad h = \langle X, X\rangle_n, \qquad a = \|u_Q\|, \qquad b = \|X^\dagger X\|.$$

The column space is a Hilbert space for its defined inner product, so $h \ge 0$ is real, and the defining amplified display of `def-column-hilbert-corner` at $Y = X$ reads

$$X^\dagger \cdot X = h\, u_Q. \tag{36}$$

The proof is then the interaction of three scalar estimates for h, a, b, r.

(i) The compressed-versus-ambient comparison. Applying `lem-compcb-row-column-product` with $Y = X$ gives $\|\mathrm{Co}_Q(X^\dagger X) - X^\dagger X\| \le C_{\mathrm{rc}} e\, r^2$; and $\mathrm{Co}_Q(X^\dagger X) = X^\dagger \cdot X = h\, u_Q$ by (36). Absolute homogeneity and the reverse triangle inequality therefore give

$$|ha - b| \;=\; \big| \|h\, u_Q\| - \|X^\dagger X\| \big| \;\le\; \|h\, u_Q - X^\dagger X\| \;\le\; C_{\mathrm{rc}} e\, r^2. \tag{37}$$

(ii) The C^* comparison. The ε-C^* lower norm axiom at the amplified rectangular level gives $b \ge (1-\varepsilon) r^2$, and submultiplicativity together with isometry of the involution gives $b \le (1+\varepsilon)\|X^\dagger\|\, r = (1+\varepsilon) r^2$. Hence

$$|b - r^2| \;\le\; \varepsilon\, r^2 \;\le\; e\, r^2. \tag{38}$$

(iii) The compressed unit is close to a unit. Applying `lem-compcb-compressed-unit-norm` with $T = Q$ gives $|a - 1| \le Ce$, and with $Ce \le \frac{1}{2}$ also $a \ge \frac{1}{2}$. This step needs Q *nonvanishing*, which the datum does not literally state — `def-hcb-datum` says only that Q is a *one-dimensional* δ-projection. The tree first records this as a genuine gap, then closes it by a registered bridge: one-dimensionality means $\dim \mathcal{S}_Q = 1$, so $\mathcal{S}_Q \ne 0$; and $\mathcal{S}_Q = 0$ characterises the first (vanishing) alternative of `def-delta-projection`; hence Q lies in the second alternative, i.e. is nonvanishing.

Synthesis. Enlarge the constant and shrink the threshold so that one pair $C \ge 1$, $e_0 \le 1/(2C)$ serves all three estimates. Then (37) and (38) give $|ha - r^2| \le (C+1) e r^2$. The exact scalar identity

$$a\,(h - r^2) = (ha - r^2) + r^2(1 - a) \tag{39}$$

combined with $|a-1| \le Ce$ then yields $a|h-r^2| \le (2C+1)er^2$, and dividing by $a \ge \frac{1}{2}$ gives (35) with $C_{\mathrm{col}} = 4C+2$ and $e_{\mathrm{col}} = e_0$.

The withdrawn route. A substantial part of this tree is a *negative* audit, and it is the reason two of the four dependencies exist. The natural first attempt was to square-pad the column, $X_c = Xe_1^*$ and $X_r = X_c^\dagger$, and apply `lem-compcb-rectangular-product` to the pair (X_r, X_c). The audit found that under this shard's allowed inputs none of the three needed identifications was available: the corner memberships $M_n \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{P_n,Q_n}$ and $\big(\mathcal{S}_{P,Q}\big)^\dagger \subseteq \mathcal{S}_{Q,P}$, the compressed-product identity $X_r \cdot X_c = E_{11} \otimes (X^\dagger \cdot X)$, and the matrix-norm identifications $\|X_c\| = r$ and $\|E_{11} \otimes z\| = \|z\|$. Each affirmative claim was withdrawn rather than repaired in place; a further over-claim — that the failure of this one construction proved an *exhaustive* impossibility — was also withdrawn when challenged. The actual repair was structural: register `lem-compcb-row-column-product` and `lem-compcb-entrywise-compression-naturality` as registry dependencies, after which step (i) above applies directly, with no square embedding and no E_{11} identification needed at this level.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb-column-hilbert-squared`: 50 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-rectangular-product`, `lem-compcb-compressed-unit-norm`, `lem-compcb-entrywise-compression-naturality` and `lem-compcb-row-column-product`. It is the analytic hinge of the H-CB tier: it makes the column-Hilbert norm q_P and the ambient norm interchangeable up to $1 \pm O(e)$, which is what lets estimates proved in one be read in the other. It feeds `lem-hcb1-variational-identity` (§19), `lem-hcb1-column-action` (§20) and `lem-hcb4-canonical-closeness` (§29), now `af: validated`, which consumes it at $n = 1$ to normalise the compressed unit.

18 Single-compression transfer of an extended inclusion

Lemma 66 (`lem-compcb-single-compression-transfer`)**.** *There are universal $C_{\mathrm{co}} < \infty$ and $e_{\mathrm{co}} > 0$ such that restricting an extended α-inclusion to one ideal and following it by one compatible amplified compression produces an extended $C_{\mathrm{co}}(\alpha+\varepsilon)$-inclusion whenever $\alpha + \varepsilon \leq e_{\mathrm{co}}$.*
Transcription note. *The tree fixes the datum: $\mathcal{A}$ an extended ε-C^*-algebra, $\mathcal{B}$ an exact unital C^*-algebra, $\mathcal{J} \subseteq \mathcal{B}$ a unital direct-summand ideal with unit q, and $v : \mathcal{B} \to \mathcal{A}$ an extended α-inclusion (`def-extended-delta-inclusion`). The "compatible single compression" is*

$$P := v(q), \qquad T := \mathrm{Co}_P \circ v|_{\mathcal{J}},$$

and throughout $e = \alpha + \varepsilon$.

Registry contract (verbatim).

```
Single-compression transfer: there are universal C_co < infinity and e_co > 0 such that restricting
an extended alpha-inclusion to one ideal and following it by one compatible amplified compression
produces an extended C_co*(alpha+epsilon)-inclusion whenever alpha+epsilon <= e_co.
```

Proof (prose account of the af-validated tree). *The projection and the ledger.* Since $q = q^\dagger = q^2$ with $\|q\| = 1$ in the exact C^*-algebra $\mathcal{B}$, the inclusion clauses give $P^\dagger = v(q^\dagger) = P$, $\|P^2 - P\| = \|v(q)v(q) - v(q^2)\| \leq \alpha$ and $1 - \alpha \leq \|P\| \leq 1 + \alpha$; so P is an α-projection and, for small e, a *nonvanishing* one, its norm lying in the second alternative of `def-delta-projection`. That is what licenses `lem-compcb-corner-algebra` to make $\mathcal{S}_P$ an extended $C_{\mathrm{ca}}e$-C^*-algebra — the target structure. The remaining ledger constants are k_{cmp} from the compression comparison $\|\mathrm{Co}_{P_n}(Z) - (P_n Z)P_n\| \leq k_{\mathrm{cmp}} e\|Z\|$, then $D = 2k_{\mathrm{cmp}} + 5$ and $K_N = D + 1$, together with C_r, e_r from `lem-compcb-rectangular-product`.

The exact clauses. By tensor functoriality $T_n = \mathrm{id}_{M_n} \otimes T = (\mathrm{id}_{M_n} \otimes \mathrm{Co}_P) \circ v_n$, and `lem-compcb-amplified-comp` rewrites this as $T_n = \mathrm{Co}_{P_n} \circ v_n$ with range $\mathrm{Im}\,\mathrm{Co}_{P_n} = \mathcal{S}_{P_n} = M_n \otimes \mathcal{S}_P$; linearity is inherited from v_n and Co_{P_n}. Two clauses hold *exactly*, with no error term. The star clause of the extended inclusion gives $v_n(x^\dagger) = v_n(x)^\dagger$, and `lem-compcb-amplified-compression-identities` for the square pair (P_n, P_n) gives $\mathrm{Co}_{P_n}(Z^\dagger) = \mathrm{Co}_{P_n}(Z)^\dagger$, so $T_n(x^\dagger) = T_n(x)^\dagger$. And the unit of $M_n \otimes \mathcal{J}$ is $q_n = I_n \otimes q$ with $v_n(q_n) = P_n$, so

$$T_n(q_n) = \mathrm{Co}_{P_n}(P_n) = I_n \otimes \mathrm{Co}_P(P) = I_n \otimes u_P :$$

the unit defect is zero.

Compression closeness. Because q is the unit of the ideal, $q_n x = x q_n = x$ for $x \in M_n \otimes \mathcal{J}$, so with $Z = v_n(x)$ the multiplicative clause of the inclusion gives $\|P_n Z - Z\| \leq \alpha\|x\|$ and $\|Z P_n - Z\| \leq \alpha\|x\|$, while its norm clause gives $\|P_n\| \leq 1 + \alpha$ and $\|Z\| \leq (1+\alpha)\|x\|$. Combining with the compression comparison and $e \leq \frac{1}{2}$,

$$\|T_n(x) - v_n(x)\| \leq \|\mathrm{Co}_{P_n}(Z) - (P_n Z)P_n\| + \|(P_n Z)P_n - Z\| \leq D\, e\, \|x\|, \tag{40}$$

uniformly in n; the second term is bounded by $\big[(1+\varepsilon)(1+\alpha)+1\big]\alpha\|x\| \leq 5e\|x\|$.

Norm distortion. The inclusion gives $(1-\alpha)\|x\| \leq \|v_n(x)\| \leq (1+\alpha)\|x\|$; with (40) the triangle and reverse-triangle inequalities give

$$(1 - K_N e)\|x\| \ \leq\ \|T_n(x)\| \ \leq\ (1 + K_N e)\|x\|, \qquad K_N = D + 1. \tag{41}$$

Multiplicativity. Insert $v_n(xy)$, $v_n(x)v_n(y)$ and $T_n(x)T_n(y)$ between $T_n(xy)$ and $T_n(x) \cdot T_n(y)$. The first term is $\leq De\|x\|\|y\|$ by (40), using that $M_n \otimes \mathcal{J}$ is an exact C^*-algebra — so its norm is genuinely submultiplicative, $\|xy\| \leq \|x\|\|y\|$. The second is $\leq \alpha\|x\|\|y\|$ by the multiplicative clause. The third, after inserting $T_n(x)v_n(y)$, is $\leq 2D(K_N+3)e\|x\|\|y\|$ by (40), (41) and the ε-product bound. The fourth needs a naturality observation: for $A, B \in M_n \otimes \mathcal{S}_P$ the amplified compressed product has entries $\sum_k \mathrm{Co}_P(a_{ik}b_{kj}) = \mathrm{Co}_P\big(\sum_k a_{ik}b_{kj}\big)$, i.e. it equals $\mathrm{Co}_{P_n}(AB)$ — so it *is* the rectangular compressed product for the compatible triple (P_n, P_n, P_n), not merely an element of the same space, and `lem-compcb-rectangular-product` applies, giving $\leq C_r(1+K_N)^2 e\|x\|\|y\|$. Summing, $K_M := D + 1 + 2D(K_N+3) + C_r(1+K_N)^2$ works.

Assembly. With $C_{\mathrm{co}} = \max\{1, C_{\mathrm{ca}}, K_N, K_M\}$ and e_{co} the positive minimum of all thresholds used, `def-extended-delta-inclusion` makes each T_n a $C_{\mathrm{co}}e$-inclusion; since n was arbitrary and every constant is universal, T is an extended $C_{\mathrm{co}}(\alpha+\varepsilon)$-inclusion, with no dependence on the amplification level or the ideal's dimension.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-compcb-single-compression-transfer`: 33 nodes (32 `validated`, 1 superseded node archived), root `validated`, taint clean.

Role in the chain. This is the single-compression transfer primitive aimed at compression-based EXT and Stage-1 assembly, where inclusions have to be pushed through compressions repeatedly. The EXT parent `conj-extcb` is now `proved` with `af: validated`, and so is the broader assembly carrier `lem-thmainext-conditional`. This lemma's registry consumers are `lem-maincb-stage1-raw-refinement` and `lem-maincb-stage1-call-envelope`.

19 The amplified Ha variational identity

Lemma 67 (`lem-hcb1-variational-identity`). *There is a universal $e_{\mathrm{var}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{var}}$, every $n \ge 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$, $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ and $Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$2\,\langle Y, \big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)X - Z\cdot X\rangle_n\,\mathrm{Co}_Q(Q) \;=\; \big(Y^\dagger\cdot Z\big)\cdot X - Y^\dagger\cdot\big(Z\cdot X\big). \tag{42}$$

Transcription note. *The identity is* exact*; the threshold only guarantees that the objects involved are defined. Its content is that the "Ha defect" — the difference between letting* $\mathrm{Ha}^{Q}_{P,R}(Z)$ *act on a column and simply multiplying by Z — is, after pairing, literally the compressed associator of* ***`def-compressed-associator`*** *for the triple* $(Y^\dagger, Z, X)$. Contract amendment. *The codified contract originally wrote the normalising factor as u_Q; the ratified* ***`def-ha-map`*** *identity and the column-Hilbert displays write it as $\widetilde{Q} = \mathrm{Co}_Q(Q)$. The contract now spells out* $\mathrm{Co}_Q(Q)$. *These are definitionally the same element — the compressed unit is $u_Q = \mathrm{Co}_Q(Q)$ — so this was pure notation typing, neither a strengthening nor a weakening.*

Registry contract (verbatim).

```
Amplified Ha variational identity: there is a universal e_var > 0 such that every H-CB datum with e
<= e_var, every n >= 1, Z in M_n tensor S_{P,R}, X in M_{n,1} tensor S_{R,Q}, and Y in M_{n,1}
tensor S_{P,Q} satisfy 2*<Y,(Ha^Q_{P,R})_n(Z)X-Z dot X>*Co_Q(Q)=(Y^dagger dot Z) dot X-Y^dagger dot
(Z dot X).
```

Proof (prose account of the af-validated tree). *Threshold.* Take $e_{\mathrm{var}} = \min(e_{\mathrm{cmp}}, e_{\mathrm{col}})$, the thresholds of `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-squared`, and fix an arbitrary datum and arbitrary n, Z, X, Y.

Entrywise dictionary. Write $Z = (Z_{ij})_{1\le i,j\le n}$, $X = (X_j)_j$, $Y = (Y_i)_i$. Matrix amplification, the matrix compressed product and the amplified sesquilinear form unfold as

$$\big(\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)X\big)_i = \sum_j \mathrm{Ha}^{Q}_{P,R}(Z_{ij})(X_j), \quad (Z\cdot X)_i = \sum_j Z_{ij}\cdot X_j,$$

$$\big(Y^\dagger\cdot Z\big)_j = \sum_i Y_i^\dagger\cdot Z_{ij}, \qquad \langle Y, W\rangle_n = \sum_i \langle Y_i, W_i\rangle.$$

Summing the level-one relation. For each pair (i,j), the defining relation of `def-ha-map`, applied to $Z_{ij} \in \mathcal{S}_{P,R}$, $X_j \in \mathcal{S}_{R,Q}$ and $Y_i \in \mathcal{S}_{P,Q}$, gives

$$2\,\langle Y_i, \mathrm{Ha}^{Q}_{P,R}(Z_{ij})(X_j)\rangle\,\mathrm{Co}_Q(Q) = \big(Y_i^\dagger\cdot Z_{ij}\big)\cdot X_j + Y_i^\dagger\cdot\big(Z_{ij}\cdot X_j\big).$$

Summing over i, j and using the dictionary together with sesquilinearity,

$$2\,\langle Y, \big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)X\rangle_n\,\mathrm{Co}_Q(Q) = \sum_{i,j}\Big[\big(Y_i^\dagger\cdot Z_{ij}\big)\cdot X_j + Y_i^\dagger\cdot\big(Z_{ij}\cdot X_j\big)\Big]. \tag{43}$$

The column display. The registered amplified column-Hilbert display $Y^\dagger\cdot W = \langle Y, W\rangle_n \mathrm{Co}_Q(Q)$, applied to $W = Z\cdot X$ and expanded entrywise, gives

$$\langle Y, Z\cdot X\rangle_n\,\mathrm{Co}_Q(Q) = Y^\dagger\cdot(Z\cdot X) = \sum_{i,j} Y_i^\dagger\cdot\big(Z_{ij}\cdot X_j\big). \tag{44}$$

Cancellation. Associating only as displayed — this is the point where the two sides differ — ordinary matrix multiplication gives $\big(Y^\dagger \cdot Z\big) \cdot X = \sum_{i,j}\big(Y_i^\dagger \cdot Z_{ij}\big) \cdot X_j$, whereas $Y^\dagger \cdot (Z \cdot X) = \sum_{i,j} Y_i^\dagger \cdot \big(Z_{ij} \cdot X_j\big)$. Subtracting twice (44) from (43) and using sesquilinearity in the second variable, the doubled $Y_i^\dagger \cdot (Z_{ij} \cdot X_j)$ sum cancels once and reverses sign, leaving exactly (42). Since the datum and n, Z, X, Y were arbitrary and e_{var} is universal and positive, the lemma follows.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb1-variational-identity`: 8 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-s` It is the conversion device of the H-CB tier: on its own it proves no estimate, but it turns every bound on the compressed associator into a bound on the Ha defect. Its registry consumers are `lem-hcb1-column-action` (§20), where it is combined with `lem-hcb1-combo-bound`, and the parent `conj-hcb` (§31).

20 The uniform Ha column action

Lemma 68 (`lem-hcb1-column-action`). *There are universal $C_{\mathrm{act}} < \infty$ and $e_{\mathrm{act}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{act}}$, every $n \ge 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$ and every $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ satisfy*

$$q_P\big(\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)X - Z \cdot X\big) \;\le\; C_{\mathrm{act}}\, e\, \|Z\|\, q_R(X). \tag{45}$$

Transcription note. *$q_P(U) = \langle U, U\rangle_n^{1/2}$ is the column-Hilbert norm of* ***`def-column-hilbert-corner`*** *on $M_{n,1} \otimes \mathcal{S}_{P,Q}$, and likewise q_R; $\|Z\|$ is the ambient norm.*

Registry contract (verbatim).

```
Uniform Ha column action: there are universal C_act < infinity and e_act > 0 such that every H-CB
datum with e <= e_act, every n >= 1, Z in M_n tensor S_{P,R}, and X in M_{n,1} tensor S_{R,Q}
satisfy q_P((Ha^Q_{P,R})_n(Z)X-Z dot X) <= C_act*e*||Z||*q_R(X).
```

Proof (prose account of the af-validated tree). Throughout put $D := \big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)X - Z\cdot X$, the object to be bounded.

Nondegeneracy of the compressed unit. Everything is divided by $\|u_Q\|$ at the end, so a lower bound is needed. Two ingredients combine. First, $\|u_Q - Q\| \le K_1 e$. This is proved *inside the associative operator algebra* $B(\mathcal{A})$, not in the ambient ε-C^*-algebra, which is only approximately associative — a distinction the tree makes explicitly. With $L = L_Q$, $R = R_Q$ and $\mathrm{A} = LR + RL - I$ one has $\mathrm{Co}_Q = \theta(\mathrm{A})$; the δ-projection dichotomy bounds $\|Q\|$, hence $\|L\|, \|R\|$, and the ε-associator and δ-projection bounds give $\|LR - RL\| = O(e)$, $\|L^2 - L\| = O(e)$, $\|R^2 - R\| = O(e)$; from these, $T = \frac{1}{2}(LR + RL)$ satisfies $\|T^2 - T\| \le C_T e$ and so $\|\mathrm{A}^2 - I\| = 4\|T^2 - T\| \le 4C_T e$. Writing $r = Q^2 - Q$ and keeping the displayed parentheses, $(\mathrm{A} - I)Q = 2r + Qr + rQ$, so $\|(\mathrm{A} - I)Q\| \le C_Q e$. Finally, below the threshold making $\|\mathrm{A}^2 - I\| < \frac{1}{2}$, the series $B = (\mathrm{A}^2)^{-1/2}$ converges and factors as $B - I = (\mathrm{A}^2 - I)K(\mathrm{A}^2)$; since A commutes with B and $K(\mathrm{A}^2)$, one gets the clean factorisation $\theta(\mathrm{A}) - I = (\mathrm{A} - I)H(\mathrm{A})$ with $\|H(\mathrm{A})\| \le C_H$, whence $\|u_Q - Q\| = \|(\theta(\mathrm{A}) - I)Q\| \le C_H C_Q e$. Second, $\|Q\| \ge 1 - K_0 e$, which holds *provided Q is nonvanishing.* The tree runs this branch strictly conditionally — it refuses to infer nonvanishing from one-dimensionality — and discharges the antecedent only later, from the registered bridge $\dim \mathcal{S}_Q = 1 \Rightarrow \mathcal{S}_Q \ne 0 \Rightarrow Q$ nonvanishing. Combining the two by the reverse triangle inequality and shrinking the threshold gives

$$\|u_Q\| \;\ge\; c_u := \tfrac{1}{2}. \tag{46}$$

Column-norm comparison. By `lem-hcb-column-hilbert-squared`, $\big|q_T(U)^2 - \|U\|_{n,1}^2\big| \le C_{\mathrm{col}} e \|U\|_{n,1}^2$ for $T = P$ and $T = R$; with $C_{\mathrm{col}} e \le \frac{1}{2}$ all terms are nonnegative and taking square roots gives $(1 - C_{\mathrm{col}} e)^{1/2}\|U\|_{n,1} \le q_T(U) \le (1 + C_{\mathrm{col}} e)^{1/2}\|U\|_{n,1}$, in particular $\|U\|_{n,1} \le \sqrt{2}\, q_T(U)$.

Defect pairing. By `lem-hcb1-variational-identity`, $2\,\langle Y, D\rangle_n\, u_Q = \big(Y^\dagger \cdot Z\big)\cdot X - Y^\dagger \cdot \big(Z \cdot X\big)$ for every compatible column Y. The right-hand side is precisely the compressed associator of $A = Y^\dagger$, $B = Z$, $C = X$ (`def-compressed-associator`), so `lem-hcb0-compressed-associator` bounds it by $C_{\mathrm{as}} e \|Y^\dagger\|\|Z\|\|X\|_{n,1}$, and the involution is isometric, so $\|Y^\dagger\| = \|Y\|_{n,1}$. Taking norms on the left and using absolute homogeneity,

$$2\,\big|\langle Y, D\rangle_n\big|\,\|u_Q\| \;\le\; C_{\mathrm{as}}\, e\, \|Y\|_{n,1}\, \|Z\|\, \|X\|_{n,1}. \tag{47}$$

Assembly by self-pairing. Insert (46) and the two norm comparisons into (47): for every compatible Y,

$$\big|\langle Y, D\rangle_n\big| \;\le\; 2C_{\mathrm{as}}\, e\, q_P(Y)\, \|Z\|\, q_R(X).$$

Now the finishing move avoids any duality argument: since both $\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)X$ and $Z \cdot X$ lie in $M_{n,1} \otimes \mathcal{S}_{P,Q}$, so does D, and $Y = D$ is an admissible test vector. Substituting it gives $q_P(D)^2 = \langle D, D\rangle_n \le 2C_{\mathrm{as}} e\|Z\| q_R(X)\, q_P(D)$; if $q_P(D) = 0$ the conclusion is immediate, and otherwise one divides. Hence (45) with $C_{\mathrm{act}} = 2C_{\mathrm{as}}$.

Two recorded refinements. The tree first proved the terminal cancellation only *conditionally*, explicitly declining to divide by an unavailable c_u and carrying a node that says so; the unconditional root appears only after the nonvanishing bridge is registered, and the earlier negative conclusion is marked stale rather than deleted. Separately, the threshold is set as $e_{\mathrm{act}} = t/2$ for $t = \min(e_u, e_{\mathrm{col}}, e_{\mathrm{var}}, e_{\mathrm{as}}, 1/(2\max(C_{\mathrm{col}}, 1)))$ so that the strict inequalities required upstream hold even at the endpoint, and the degenerate case $C_{\mathrm{col}} = 0$ is split off so that the reciprocal $1/(2C_{\mathrm{col}})$ is never formed.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb1-column-action`: 38 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-hcb-column-hilbert-squared`, `lem-hcb0-compressed-associator` and `lem-hcb1-variational-identity`. This is the first genuinely quantitative statement about the Ha map: it says that $\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z)$ is, to within $O(e)$, left multiplication by Z. Everything later in the tier is built on it — `lem-hcb2-product-defect` (§21), `lem-hcb3-diagonal-unit` (§22) — and it is a declared dependency of `lem-hcb4-canonical-closeness` (§29), now `af: validated`.

21 The uniform amplified Ha product defect

Lemma 69 (`lem-hcb2-product-defect`). *There are universal $C_{\mathrm{prod}} < \infty$ and $e_{\mathrm{prod}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{prod}}$, every $n \ge 1$, every $Z \in M_n \otimes \mathcal{S}_{P,S}$ and every $W \in M_n \otimes \mathcal{S}_{S,R}$ satisfy*

$$\big\|\big(\mathrm{Ha}^Q_{P,R}\big)_n(Z \cdot W) - \big(\mathrm{Ha}^Q_{P,S}\big)_n(Z)\,\big(\mathrm{Ha}^Q_{S,R}\big)_n(W)\big\| \;\le\; C_{\mathrm{prod}}\, e\,\|Z\|\,\|W\|. \tag{48}$$

Transcription note. *On the left the juxtaposition of the two amplified Ha maps is composition of operators between amplified column spaces, whereas $Z \cdot W$ is the compressed product in the ambient corner; the norm is the operator norm of* `def-ha-map`*. So the statement is that* $\mathrm{Ha}^Q_{\cdot,\cdot}$ *is multiplicative up to $O(e)$.*

Registry contract (verbatim).

```
Uniform amplified Ha product defect: there are universal C_prod < infinity and e_prod > 0 such that
every H-CB datum with e <= e_prod, every n >= 1, Z in M_n tensor S_{P,S}, and W in M_n tensor
S_{S,R} satisfy ||(Ha^Q_{P,R})_n(Z dot W)-(Ha^Q_{P,S})_n(Z)(Ha^Q_{S,R})_n(W)|| <=
C_prod*e*||Z||||W||.
```

Proof (prose account of the af-validated tree). *Calibration.* Three uniform facts are set up first. The whole-column identity $X^\dagger \cdot X = q_T(X)^2 u_Q$ of `def-column-hilbert-corner`, combined with `lem-compcb-rectangular-product` and the ε-C^* norm axioms, gives $\big|\,\|X^\dagger \cdot X\| - \|X\|^2\big| \le (C_{\mathrm{co}}+1)e\|X\|^2$; since Q is one-dimensional it is nonvanishing by the registered bridge, so u_Q is the unit of an $O(e)$-C^* corner and its norm is within $O(e)$ of 1. Shrinking the threshold, this calibrates the column norm against the ambient norm,

$$\tfrac{1}{2}\|X\| \;\le\; q_T(X) \;\le\; 2\|X\|. \tag{49}$$

Next, the rectangular estimate and $\|AB\| \le (1+\varepsilon)\|A\|\|B\|$ give $\|A \cdot B\| \le K_0\|A\|\|B\|$ with $K_0 = 2 + C_{\mathrm{co}}$; and for a square-by-column pair, (49) converts this into the multiplier bound $q_P(A \cdot X) \le K_{\mathrm{mul}}\|A\| q_R(X)$ with $K_{\mathrm{mul}} = 4K_0$. Finally, `lem-hcb1-column-action` together with the multiplier bound gives a uniform operator bound for the Ha map itself,

$$\big\|\big(\mathrm{Ha}^Q_{A,B}\big)_n(U)\big\| \;\le\; K_H\|U\|, \qquad K_H := K_{\mathrm{mul}} + C_{\mathrm{act}}. \tag{50}$$

The telescope. Abbreviate $H_{A,B}(U) := \big(\mathrm{Ha}^Q_{A,B}\big)_n(U)$. For every compatible column X the following identity is exact, obtained by adding and subtracting compressed multiplications:

$$\begin{aligned}\big[H_{P,R}(Z \cdot W) - H_{P,S}(Z)H_{S,R}(W)\big]X = {}&\big[H_{P,R}(Z \cdot W)X - (Z \cdot W)\cdot X\big]\\ &+ \big[(Z \cdot W)\cdot X - Z \cdot (W \cdot X)\big]\\ &+ Z \cdot \big[W \cdot X - H_{S,R}(W)X\big]\\ &+ \big[Z \cdot H_{S,R}(W)X - H_{P,S}(Z)H_{S,R}(W)X\big].\end{aligned} \tag{51}$$

Each bracket is one of the two estimates already available. The first is a column-action defect for the element $Z \cdot W$, so `lem-hcb1-column-action` and $\|Z \cdot W\| \le K_0\|Z\|\|W\|$ bound it by $C_{\mathrm{act}}K_0 e\|Z\|\|W\| q_R(X)$. The second is the compressed associator, bounded by `lem-hcb0-compressed-associator` in the ambient norm by $C_{\mathrm{as}} e\|Z\|\|W\|\|X\|$; converting both ends through (49) costs two factors of 2, giving $4C_{\mathrm{as}} e\|Z\|\|W\| q_R(X)$. The third is the column-action defect for W, pushed through the square-by-column multiplier bound for Z, giving $K_{\mathrm{mul}}C_{\mathrm{act}}e\|Z\|\|W\|q_R(X)$. The fourth is the column-action defect for Z evaluated on the column $H_{S,R}(W)X$, whose column norm is controlled by the operator bound (50), giving $C_{\mathrm{act}}K_He\|Z\|\|W\|q_R(X)$.

Conclusion. Applying the q_P triangle inequality to (51) and adding the four bounds gives the pointwise estimate with

$$C_* := C_{\mathrm{act}}K_0 + 4C_{\mathrm{as}} + C_{\mathrm{act}}K_{\mathrm{mul}} + C_{\mathrm{act}}K_H.$$

Taking the supremum over columns with $q_R(X) = 1$ converts it into the operator-norm statement (48), with $C_{\mathrm{prod}} = C_*$ and $e_{\mathrm{prod}} = \min\{e_{\mathrm{cmp}}, e_{\mathrm{co}}, e_{\mathrm{as}}, e_{\mathrm{act}}, e_q, 1\}$; `lem-compcb-amplified-compression` is what guarantees that all the corners and compressed products appearing at level n are the compatible amplified objects the imported estimates quantify over.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb2-product-defect`: 12 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-hcb0-compressed-associator`, `lem-hcb1-column-action`, `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. Together with the exact adjointness of §12 it makes $\mathrm{Ha}^{Q}_{\cdot,\cdot}$ an approximate $*$-homomorphism at every amplification level, which is exactly the input the norm estimates of the next tier consume: it feeds `lem-hcb3-diagonal-upper-norm` (§23) and `lem-hcb3-diagonal-lower-modulus` (§25), and it is a declared dependency of `lem-hcb3-offdiagonal-inverse` (§27), now `af: validated`.

22 The uniform diagonal Ha unit estimate

Lemma 70 (`lem-hcb3-diagonal-unit`). *There are universal $C_{\mathrm{unit}} < \infty$ and $e_{\mathrm{unit}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{unit}}$ and every $n \ge 1$ satisfy*

$$\big\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n\big(I_n \otimes u_P\big) - I\big\| \;\le\; C_{\mathrm{unit}}\, e. \tag{52}$$

Transcription note. $u_P = \mathrm{Co}_P(P)$ *is the compressed unit of the corner* $\mathcal{S}_P$*,* $I_n \otimes u_P$ *its diagonal amplification, and* I *the identity operator on the amplified column space* $M_{n,1} \otimes \mathcal{S}_{P,Q}$*; the norm is the operator norm.*

Registry contract (verbatim).

```
Uniform diagonal Ha unit estimate: there are universal C_unit < infinity and e_unit > 0 such that
every H-CB datum with e <= e_unit and every n >= 1 satisfy ||(Ha^Q_{P,P})_n(I_n tensor u_P)-I|| <=
C_unit*e.
```

Proof (prose account of the af-validated tree). Write $U = I_n \otimes u_P$. The estimate is pointwise on columns, then a supremum.

A uniform bound on the amplified unit. The obvious route — bound $\|u_P\|$ at level one and appeal to isometry of the diagonal amplification — is *not* taken, and the tree says why: no registered input supplies that isometry. Instead $\|U\|$ is bounded directly at level n by a C^*-style argument. Put $t = \|U\|$. The ε-C^* norm axioms at level n give $\|U^\dagger U\| \ge (1-\varepsilon)t^2 \ge (1-e)t^2$ and $\|U^\dagger\| = t$; the compressed-product display in the amplified P,P corner gives $\|U^\dagger U - U^\dagger \cdot U\| \le C_{\mathrm{pr}} e t^2$; and `lem-compcb-compressed-unit-action`, in its right-unit form with $A = U^\dagger$, gives $\|U^\dagger \cdot U - U^\dagger\| \le C_{\mathrm{co}} e t$. Chaining,

$$(1-e)t^2 \;\le\; \|U^\dagger U\| \;\le\; C_{\mathrm{pr}} e t^2 + (1 + C_{\mathrm{co}} e)t.$$

If $t = 0$ there is nothing to prove; otherwise divide by t to get $\big[1 - (1 + C_{\mathrm{pr}})e\big]t \le 1 + C_{\mathrm{co}} e$, and choosing $e_u \le 1/\big[2(1 + C_{\mathrm{pr}})\big]$ makes the bracket at least $\frac{1}{2}$, so

$$\|U\| \;\le\; K_u := 2\big(1 + C_{\mathrm{co}} e_u\big), \tag{53}$$

uniformly in n.

Two defects. For a column $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ split

$$\big(\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(U) - I\big)X = \big(\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(U)X - U \cdot X\big) + \big(U \cdot X - X\big). \tag{54}$$

The first bracket is a column-action defect: `lem-hcb1-column-action` with $R = P$ and $Z = U$ gives $q_P\big(\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(U)X - U \cdot X\big) \le C_{\mathrm{act}} e\|U\| q_P(X)$, which (53) turns into $C_{\mathrm{act}} K_u e q_P(X)$.

The second bracket says that the amplified compressed unit is an approximate identity on columns, and it is assembled coordinatewise. Multiplication by the block-diagonal $U = I_n \otimes u_P$ acts entrywise, so writing $X = (X_1, \ldots, X_n)^{\mathsf{T}}$ and applying `lem-compcb-compressed-unit-action` in each level-one P,Q corner gives $\|u_P \cdot X_j - X_j\| \le C_{\mathrm{co}} e\|X_j\|$ for every j. The level-one norm/inner-product comparison registered with `def-ha-map` and the column-Hilbert displays supplies a universal κ with $q_P(Y) \le \kappa\|Y\|$ and $\|Y\| \le \kappa q_P(Y)$, so each coordinate estimate becomes $q_P(u_P \cdot X_j - X_j) \le \kappa^2 C_{\mathrm{co}} e\, q_P(X_j)$. Finally the coordinate-sum display $q_P\big((Y_j)_j\big)^2 = \sum_j q_P(Y_j)^2$ lets the squared coordinate estimates be summed, giving $q_P(U \cdot X - X) \le \kappa^2 C_{\mathrm{co}} e\, q_P(X)$ uniformly in n; the existential constant C_{co} is then simply replaced by the larger universal $\kappa^2 C_{\mathrm{co}}$ and the notation retained.

Conclusion. Setting $e_0 = \frac{1}{2}\min(e_u, e_{\mathrm{act}}, e_{\mathrm{co}})$ and applying the q_P triangle inequality to (54),

$$q_P\Big(\big((\mathrm{Ha}^Q_{P,P})_n(U) - I\big)X\Big) \;\le\; \big(C_{\mathrm{act}}K_u + C_{\mathrm{co}}\big)e\, q_P(X),$$

and passing to the operator norm gives (52) with $C_{\mathrm{unit}} = C_{\mathrm{act}}K_u + C_{\mathrm{co}}$ and $e_{\mathrm{unit}} = e_0$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-unit`: 18 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-hcb1-column-action` and `lem-compcb-compressed-unit-action`. It is one of the estimates the parent contract `conj-hcb` collects — the clause asserting that the amplified Ha map is approximately unital — and its registry consumer is the parent contract itself (§31), now `proved` with `af: validated`; neither it nor this lemma bears on the status of `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

23 The uniform diagonal Ha upper norm

Lemma 71 (`lem-hcb3-diagonal-upper-norm`). *There are universal $C_{\mathrm{up}} < \infty$ and $e_{\mathrm{up}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{up}}$, every $n \ge 1$ and every $Z \in M_n \otimes \mathcal{S}_P$ satisfy*

$$\big\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(Z)\big\| \;\le\; \big(1 + C_{\mathrm{up}}\, e\big)\|Z\|. \tag{55}$$

Transcription note. *The left norm is the operator norm on maps of the amplified column space; $\|Z\|$ is the ambient norm of Z in $M_n \otimes \mathcal{S}_P$. The estimate is uniform in n, which is the whole difficulty: the level-one statement is a routine consequence of the corner structure, the matrix-uniform one is not.*

Registry contract (verbatim).

```
Uniform diagonal Ha upper norm: there are universal C_up < infinity and e_up > 0 such that every
H-CB datum with e <= e_up, every n >= 1, and every Z in M_n tensor S_P satisfy
||(Ha^Q_{P,P})_n(Z)|| <= (1+C_up*e)||Z||.
```

Proof (prose account of the af-validated tree). Write $h_n = \big(\mathrm{Ha}^{Q}_{P,P}\big)_n$ and $b_n = \|h_n\|$ for its operator norm; the goal is $b_n \le 1 + C_{\mathrm{up}}e$ for every n.

The matrix-uniform square estimate. The first ingredient is

$$\|Z^{\dagger} \cdot Z\| \;\le\; (1 + \kappa e)\|Z\|^2, \qquad Z \in M_n \otimes \mathcal{S}_P, \tag{56}$$

with $\kappa = C_{\mathrm{ca}}$ and threshold $e_{\mathrm{sq}} = e_{\mathrm{ca}}$. If $\mathcal{S}_P = 0$ then $M_n \otimes \mathcal{S}_P = 0$ for every n, so $Z = 0$ and (56) is trivial. Otherwise $\mathcal{S}_P \ne 0$ puts P in the nonvanishing branch of the δ-projection dichotomy, and `lem-compcb-corner-algebra` makes $\mathcal{S}_P$ an *extended* $C_{\mathrm{ca}}e$-C^*-algebra — so, by the meaning of “extended”, every $M_n \otimes \mathcal{S}_P$ is a $C_{\mathrm{ca}}e$-C^*-algebra with the *same* parameter, independent of n. Its product upper axiom together with isometry of the inherited involution then gives (56).

This is the shard’s instructive correction, and it explains the shape of the dependency list. An earlier audit of the tree found that the permitted inputs at that time — the compressed-product display, the level-one ε-C^* axioms, `def-hcb-datum` (data and notation only), and the two Ha lemmas — gave (56) at $n = 1$ and *nowhere else*: passing to arbitrary n was not an inference from any allowed premise but a genuinely missing amplified-corner theorem. Once `lem-compcb-corner-algebra` became an allowed dependency, the negative audit nodes became obsolete and are marked as such.

A quadratic recurrence for b_n. Finiteness of b_n is established without any finite-dimensionality assumption on $\mathcal{S}_P$. From the defining Ha relation and the level-one bounds — $\|A{\cdot}B\| \le K_{\mathrm{dot}}\|A\|\|B\|$, the two-sided comparison $c_q^{-1}\|X\| \le q(X) \le c_q\|X\|$, and $\|u_Q\| \ge c_u > 0$ (with Q nonvanishing because it is one-dimensional) — one gets $2|\langle Y, h(Z)X\rangle|\,\|u_Q\| \le 2K_{\mathrm{dot}}^2\|Y\|\|Z\|\|X\|$, hence $|\langle Y, h(Z)X\rangle| \le K_1 q(Y)\|Z\|q(X)$ with $K_1 = K_{\mathrm{dot}}^2 c_q^2/c_u$, and Hilbert-space duality in the q norm gives $\|h(Z)\| \le K_1\|Z\|$ at level one. For fixed finite n, $h_n(Z) = [h(Z_{ij})]$ on the Hilbert direct sum, so $\|h_n(Z)\| \le \big(\sum_{i,j}\|h(Z_{ij})\|^2\big)^{1/2} \le nK_1\|Z\|$, using contractivity of matrix-coordinate compressions. Crude in n, but finite — which is all that is needed to *form* b_n and take a supremum.

Now let $e \le \min(e_{\mathrm{sq}}, e_{\mathrm{prod}}, e_{\mathrm{adj}})$ and $Z \in E_n := M_n \otimes \mathcal{S}_P$. Put $T = h_n(Z)$. The operator C^*-identity gives $\|T\|^2 = \|T^{\dagger}T\|$; `lem-hcb2-amplified-adjointness` identifies $T^{\dagger} = h_n(Z^{\dagger})$ — exactly, which is what makes this move legitimate; and `lem-hcb2-product-defect` with all three corner indices equal to P gives $\|h_n(Z^\dagger)h_n(Z) - h_n(Z^\dagger \cdot Z)\| \le C_{\mathrm{prod}} e\|Z\|^2$. Chaining with boundedness of h_n and (56),

$$\|h_n(Z)\|^2 \;\le\; b_n\|Z^\dagger \cdot Z\| + C_{\mathrm{prod}} e\|Z\|^2 \;\le\; \big[(1+\kappa e)b_n + C_{\mathrm{prod}} e\big]\|Z\|^2.$$

If $E_n = \{0\}$ then $b_n = 0$ and the recurrence below holds trivially; otherwise the unit sphere of E_n is nonempty, b_n is a finite supremum over it, and $\sup\|h_n(Z)\|^2 = \big(\sup\|h_n(Z)\|\big)^2 = b_n^2$, so

$$b_n^2 \;\le\; (1+\kappa e)\, b_n + C_{\mathrm{prod}} e. \tag{57}$$

The root argument. It remains to solve (57). For nonnegative b, e, κ, C with $b^2 \le ab + Ce$ and $a = 1+\kappa e$, set $x = 1+(\kappa+C)e$ and $f(t) = t^2 - at - Ce$. Direct expansion gives $f(x) = C(\kappa+C)e^2 \ge 0$, and f is strictly increasing on $[x, \infty)$ because $2x - a = 1+(\kappa+2C)e > 0$. The hypothesis is $f(b) \le 0$; if $b > x$ then monotonicity would give $f(b) > f(x) \ge 0$, a contradiction. Hence $b \le x$.

Applying this with $b = b_n$ and $C = C_{\mathrm{prod}}$ gives $b_n \le 1 + (\kappa + C_{\mathrm{prod}})e$ for every n, which is (55) with $C_{\mathrm{up}} = \kappa + C_{\mathrm{prod}}$ and $e_{\mathrm{up}} = \min(e_{\mathrm{sq}}, e_{\mathrm{prod}}, e_{\mathrm{adj}})$; both are universal.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-upper-norm`: 30 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and `lem-compcb-corner-algebra`. It is the upper half of the norm control that the parent contract `conj-hcb` requires; the matching lower half is `lem-hcb3-diagonal-lower-modulus` (§25), which is validated but only *conditionally* — it assumes a level-one lower modulus of at least $\frac{1}{4}$, a hypothesis supplied only inside §37, and the diagonal inverse estimate `lem-hcb3-diagonal-inverse` (§26) inherits it. The off-diagonal inverse (§27), the canonical tier (§§28–30) and `conj-hcb` itself (§31) are now all `af: validated`; the conditional hypotheses named above are supplied only inside §37, and nowhere else here. Nothing here bears on the status of `op-classical`.

24 The uniform square lower estimate

Lemma 72 (`lem-hcb3-uniform-square-lower`)**.** *There are universal $K_{\mathrm{sq}} < \infty$ and $e_{\mathrm{sq}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{sq}}$, every $n \ge 1$ and every $Z \in M_n \otimes \mathcal{S}_P$ satisfy*

$$\|Z^\dagger \cdot Z\| \;\ge\; \bigl(1 - K_{\mathrm{sq}}\, e\bigr)\|Z\|^2. \tag{58}$$

Transcription note. *The datum is that of* `def-hcb-datum` *and $e = \delta + \varepsilon$; $\cdot$ is the compressed product on the corner $\mathcal{S}_P$ of* `def-compressed-corner`*, and † the inherited involution. No hypothesis is placed on P beyond its being one of the datum's δ-projections — in particular* no *nonvanishing assumption — which is what makes the statement usable as a black box downstream.*

Registry contract (verbatim).

```
Uniform square lower estimate: there are universal K_sq < infinity and e_sq > 0 such that every
H-CB datum with e <= e_sq, every n >= 1, and every Z in M_n tensor S_P satisfy ||Z^dagger dot Z||
>= (1-K_sq*e)||Z||^2.
```

Proof (prose account of the af-validated tree). The tree splits on the δ-projection dichotomy of `def-delta-projection`. The nonvanishing branch is a citation; the vanishing branch is the work, and it is settled by showing that the corner is then *literally zero.*

The vanishing branch. Suppose P is not nonvanishing, so the registered dichotomy puts it in the small alternative $\|P\| \le C_v\delta$ for a universal C_v and e below a universal threshold. Put $T = L_P R_P + R_P L_P$. The ε-Banach product bound gives

$$\|T\| \;\le\; 2(1+\varepsilon)^2\|P\|^2 \;\le\; 2(1+e)^2 C_v^2 e^2,$$

so after a further universal shrink of the threshold one may assume $\|T\| < \frac{1}{3}$.

At that point the θ calculus of `def-theta-idempotent-approximation` evaluates in closed form. Write $U = I - T$ and $X = T - I = -U$. Since $\|T\| < \frac{1}{3}$ the Neumann series $\sum_{k\ge 0} T^k$ converges absolutely to U^{-1}; and $X^2 - I = -2T + T^2$ has norm at most $2\|T\| + \|T\|^2 < \frac{7}{9} < 1$, so the registered binomial series defining $(X^2)^{-1/2}$ converges absolutely. The two absolutely convergent series are then identified through their scalar counterparts: on the disc $|z| < \frac{1}{3}$ one has $\mathrm{Re}(1-z) > 0$, so the branch of the square root normalised to 1 at $z = 0$ satisfies $\bigl((1-z)^2\bigr)^{-1/2} = (1-z)^{-1} = \sum_{k\ge 0} z^k$; the two sides therefore have identical Taylor coefficients at 0, and evaluating both at the single Banach-algebra element T gives $(X^2)^{-1/2} = U^{-1}$. Consequently

$$\operatorname{sgn}(X) = X\,(X^2)^{-1/2} = (-U)U^{-1} = -I, \qquad \theta(X) = \tfrac{1}{2}\bigl(I + \operatorname{sgn} X\bigr) = 0. \tag{59}$$

Since the registered compression construction is $\mathrm{Co}_P = \theta\bigl(L_P R_P + R_P L_P - I\bigr) = \theta(T - I)$, (59) gives $\mathrm{Co}_P = 0$ and hence $\mathcal{S}_P = \mathrm{Im}\,\mathrm{Co}_P = \{0\}$.

The nonvanishing branch. If P *is* nonvanishing and $e \le e_{\mathrm{ca}}$, then `lem-compcb-corner-algebra` makes $\mathcal{S}_P$, with the compressed product, the inherited involution and the compressed unit, an *extended* $C_{\mathrm{ca}}e$-C^*-algebra. By the meaning of "extended" in `def-extended-epsilon-cstar-algebra`, every $M_n \otimes \mathcal{S}_P$ is then a $C_{\mathrm{ca}}e$-C^*-algebra with the *same* parameter, independently of n, and its lower C^*-axiom applied to Z reads exactly $\|Z^\dagger \cdot Z\| \ge (1 - C_{\mathrm{ca}}e)\|Z\|^2$. This — and not any level-one estimate — is where the uniformity in n comes from.

Exhaustion. Set

$$K_{\mathrm{sq}} = C_{\mathrm{ca}}, \qquad e_{\mathrm{sq}} = \min\Big\{e_{\mathrm{ca}},\, e_v,\, \tfrac{1}{2\max\{1,C_{\mathrm{ca}}\}}\Big\} > 0.$$

The two branches exhaust the possibilities for P. In the nonvanishing branch the displayed axiom is (58); in the vanishing branch $M_n \otimes \mathcal{S}_P = \{0\}$, so the only admissible Z is 0 and (58) reads $0 \geq 0$. The tree also records an explicit endpoint check: since $e_{\mathrm{sq}} \leq e_{\mathrm{ca}}$, the nonvanishing branch applies even at $e = e_{\mathrm{sq}} = e_{\mathrm{ca}}$, so no boundary point is omitted.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-uniform-square-lower`: 18 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-compcb-corner-algebra`. This shard exists because of a recorded balloon abort: the first orchestration of `lem-hcb3-diagonal-lower-modulus` tripped the node cap on exactly this claim, so it was factored out of that tree — its statement extracted mechanically from the blocking challenges — and validated on its own, bottom-up. Its value to the parent is that it carries the vanishing case internally, so the downstream argument may use (58) without re-splitting on the dichotomy. Its registry consumers include `lem-hcb3-diagonal-lower-modulus` (§25).

25 Diagonal Ha lower-modulus propagation

Lemma 73 (`lem-hcb3-diagonal-lower-modulus`). *There are universal $C_{\mathrm{diag}} < \infty$ and $e_{\mathrm{diag}} > 0$ such that, for every H-CB datum with $e \le e_{\mathrm{diag}}$,* if the level-one lower modulus of $\mathrm{Ha}^{Q}_{P,P}$ is at least $\frac{1}{4}$, *then*

$$\big\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(Z)\big\| \;\ge\; \big(1 - C_{\mathrm{diag}}\, e\big)\|Z\| \qquad \text{for every } n \ge 1 \text{ and } Z \in M_n \otimes \mathcal{S}_P. \tag{60}$$

Transcription note. *The hypothesis is load-bearing and is* not *discharged here: the conclusion is asserted only for those data whose level-one map already has lower modulus $\ge \frac{1}{4}$. (This conditional shape is the corrected one; the corresponding clause of the parent contract* **`conj-hcb`** *was made conditional after an unconditional inverse claim was refuted by an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample.) The "lower modulus" of a bounded linear map is the largest c with $\|\cdot\| \ge c\|\cdot\|$, i.e. the infimum of the norm ratios over nonzero arguments.*

Registry contract (verbatim).

```
Diagonal Ha lower-modulus propagation: there are universal C_diag < infinity and e_diag > 0 such
that, for every H-CB datum with e <= e_diag, if the level-one lower modulus of Ha^Q_{P,P} is at
least 1/4, then ||(Ha^Q_{P,P})_n(Z)|| >= (1-C_diag*e)||Z|| for every n >= 1 and Z in M_n tensor
S_P.
```

Proof (prose account of the af-validated tree). Write $T_n = \big(\mathrm{Ha}^{Q}_{P,P}\big)_n$ throughout.

The root gap. Let $C_{\mathrm{prod}}, e_{\mathrm{prod}}$ come from `lem-hcb2-product-defect` — normalised to $C_{\mathrm{prod}} \ge 0$, which cannot weaken that estimate since $e\|Z\|\|W\| \ge 0$ — let $K_{\mathrm{sq}}, e_{\mathrm{sq}}$ come from `lem-hcb3-uniform-square-low` and put

$$C_{\mathrm{diag}} := K_{\mathrm{sq}} + 4C_{\mathrm{prod}}, \qquad A := 1 - K_{\mathrm{sq}} e, \qquad B := C_{\mathrm{prod}} e.$$

Choose $e_{\mathrm{diag}} > 0$ below e_{prod}, e_{sq} and e_{adj} (from `lem-hcb2-amplified-adjointness`) and small enough that $A \ge \frac{3}{4}$, $A^2 - 4B \ge 0$, $4B < \frac{1}{8}$ and $C_{\mathrm{diag}} e \le \frac{1}{2}$; each requirement is a continuity condition at $e = 0$, where $A = 1$ and $A^2 - 4B = 1$, and is vacuous if the relevant constant vanishes, so the intersection is a positive universal interval. Let $r_{\mp} = \frac{1}{2}\big(A \mp \sqrt{A^2 - 4B}\big)$ be the roots of $x^2 - Ax + B$. Rationalising the smaller root — legitimate because the denominator is at least $A > 0$, including when $B = 0$ — gives

$$0 \le r_- = \frac{2B}{A + \sqrt{A^2 - 4B}} \le \frac{2B}{A} \le \tfrac{8}{3}B \le 4B < \tfrac{1}{8}, \qquad r_+ = A - r_- \;\ge\; 1 - C_{\mathrm{diag}} e \;\ge\; \tfrac{1}{2}. \tag{61}$$

The gap between $r_- < \frac{1}{8}$ and $r_+ \ge \frac{1}{2}$ is the whole mechanism: any quantity known to satisfy the quadratic inequality and to exceed $\frac{1}{8}$ is forced up to r_+.

The zero-corner branch. If $\mathcal{S}_P = \{0\}$ then $M_m \otimes \mathcal{S}_P = \{0\}$ for every m, so the only admissible Z is 0; linearity gives $T_m(0) = 0$ and $(1 - C_{\mathrm{diag}} e)\|Z\| = 0$, so (60) holds at every level. The tree keeps this branch strictly separate, and *no* lower modulus a_m is defined in it — the infimum below would be over an empty set of ratios.

The dichotomy on the nonzero branch. Assume $\mathcal{S}_P \ne \{0\}$ and pick $0 \ne x \in \mathcal{S}_P$. For each n the element $E_{11} \otimes x$ is a nonzero member of $M_n \otimes \mathcal{S}_P$ — the operator-space direct-sum axiom gives $\|\mathrm{diag}(x, 0, \dots, 0)\| = \|x\| > 0$ — so the ratio set $R_n = \big\{\|T_n(Z)\|/\|Z\| : 0 \ne Z \in M_n \otimes \mathcal{S}_P\big\}$ is nonempty and $a_n := \inf R_n$ is a finite nonnegative real.

Fix Z. The C^*-identity in the bounded-operator target gives $\|T_n(Z)\|^2 = \|T_n(Z)^\dagger T_n(Z)\|$; `lem-hcb2-amplified-adjointness` identifies $T_n(Z)^\dagger = T_n(Z^\dagger)$ *exactly*; and `lem-hcb2-product-defect`, applied with all three corner indices equal to P and with the factors $Z^\dagger, Z$, gives

$$\|T_n(Z)\|^2 \;\geq\; \big\|T_n\big(Z^\dagger \cdot Z\big)\big\| - C_{\mathrm{prod}} e \|Z\|^2, \tag{62}$$

the involution being isometric. Applying the defining property $\|T_n(X)\| \geq a_n \|X\|$ at $X = Z^\dagger \cdot Z$ and then `lem-hcb3-uniform-square-lower` — multiplication by $a_n \geq 0$ preserving the inequality — turns (62) into $\|T_n(Z)\|^2 \geq \big[A\, a_n - B\big]\|Z\|^2$. Dividing by $\|Z\|^2$ and taking the infimum over nonzero Z, using that $\inf\{r^2 : r \in R\} = (\inf R)^2$ for a nonempty $R \subseteq [0, \infty)$ with finite infimum, gives

$$a_n^2 \;\geq\; A\, a_n - B, \qquad \text{i.e.} \qquad (a_n - r_-)(a_n - r_+) \geq 0, \tag{63}$$

so $a_n \leq r_-$ or $a_n \geq r_+$: the level-n lower modulus cannot sit in the gap.

Propagation upward. Two operator-space facts move the moduli between levels. First a halving estimate: writing $Z \in M_{2n} \otimes \mathcal{S}_P$ as a 2×2 matrix of $n \times n$ blocks and factoring $Z = \mathrm{A} \cdot \mathrm{diag}(Z_{11}, Z_{12}, Z_{21}, Z_{22}) \cdot \mathrm{B}$ with the explicit scalar matrices $\mathrm{A} = \left[\begin{smallmatrix} I & I & 0 & 0 \\ 0 & 0 & I & I \end{smallmatrix}\right]$ and $\mathrm{B} = \left[\begin{smallmatrix} I & 0 \\ 0 & I \\ I & 0 \\ 0 & I \end{smallmatrix}\right]$ of norm $\sqrt{2}$, the Ruan axioms give $\|Z\| \leq 2 \max_{i,j} \|Z_{ij}\|$; and since T_{2n} is the entrywise amplification of T_n, compression to the (i,j) target block gives $\|T_{2n}(Z)\| \geq \max_{i,j} \|T_n(Z_{ij})\|$. Hence $\|T_{2n}(Z)\| \geq a_n \max_{i,j} \|Z_{ij}\| \geq \frac{1}{2} a_n \|Z\|$, i.e. $a_{2n} \geq a_n/2$. Second a monotonicity: the zero-corner inclusion $Z \mapsto \mathrm{diag}(Z, 0_k)$ is isometric by the direct-sum axiom and is carried by T_m to $\mathrm{diag}(T_n(Z), 0_k)$ of the same norm, so $a_n \geq a_m$ whenever $n \leq m$.

Now the bootstrap. The hypothesis of the lemma is exactly $a_1 \geq \frac{1}{4}$, while (61) gives $r_- < \frac{1}{8}$; so $a_1 > r_-$ and (63) at $n = 1$ forces $a_1 \geq r_+$. Inductively, if $a_{2^k} \geq r_+$ then the halving estimate gives

$$a_{2^{k+1}} \;\geq\; \tfrac{1}{2} a_{2^k} \;\geq\; \tfrac{1}{2} r_+ \;\geq\; \tfrac{1}{2}\big(1 - C_{\mathrm{diag}} e\big) \;\geq\; \tfrac{1}{4} \;>\; r_-,$$

using $C_{\mathrm{diag}} e \leq \frac{1}{2}$; so the dichotomy at level 2^{k+1} again excludes the lower alternative and $a_{2^{k+1}} \geq r_+$. Ordinary induction gives $a_{2^k} \geq r_+$ for every $k \geq 0$. Note that the halving alone would degrade the bound geometrically; it is the *gap* that repairs it at each step.

Conclusion. For arbitrary n choose a power of two $m \geq n$. Monotonicity gives $a_n \geq a_m \geq r_+$, and (61) gives $r_+ \geq 1 - C_{\mathrm{diag}} e$. Hence $\|T_n(Z)\| \geq a_n \|Z\| \geq (1 - C_{\mathrm{diag}} e)\|Z\|$ for every Z, which together with the zero-corner branch is (60); the constants are universal and $e_{\mathrm{diag}} > 0$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-lower-modulus`: 29 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Imports `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. Together with `lem-hcb3-diagonal-upper` (§23) it pins the amplified diagonal Ha map between $1 \pm O(e)$ — but only under the level-one hypothesis, which nothing here supplies. It is the input to `lem-hcb3-diagonal-inverse` (§26), which inherits that hypothesis, and to `lem-hcb3-offdiagonal-inverse` (§27), which consumes it as its diagonal anchor. The parent `conj-hcb` (§31) is now `proved` with `af: validated`, and `op-classical` is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

26 Diagonal Ha inverse propagation

Lemma 74 (`lem-hcb3-diagonal-inverse`)**.** *There are universal $C_{\text{inv}} < \infty$ and $e_{\text{inv}} > 0$ such that, for every H-CB datum with $e \le e_{\text{inv}}$,* if $\mathrm{Ha}^Q_{P,P}$ has level-one lower modulus at least $\frac{1}{4}$ and is bijective at level one*, then every amplification is bijective and*

$$\big\|\big((\mathrm{Ha}^Q_{P,P})_n\big)^{-1}\big\| \;\le\; 1 + C_{\text{inv}}\, e. \tag{64}$$

Transcription note. ***Both*** *level-one hypotheses are load-bearing and neither is discharged here: the conclusion is asserted only for those data whose level-one map is* already *bijective* and *already has lower modulus at least $\frac{1}{4}$. The two are used at genuinely different points of the proof — bijectivity supplies the inverse, the lower modulus supplies its norm — and neither implies the other in this setting. The conditional shape is the corrected one, inherited from the same verdict that made the lower-modulus clause of* `conj-hcb` *conditional after an unconditional inverse claim was refuted by an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample.*

Registry contract (verbatim).

```
Diagonal Ha inverse propagation: there are universal C_inv < infinity and e_inv > 0 such that, for
every H-CB datum with e <= e_inv, if Ha^Q_{P,P} has level-one lower modulus at least 1/4 and is
bijective at level one, then every amplification is bijective and ||((Ha^Q_{P,P})_n)^(-1)|| <=
1+C_inv*e.
```

Proof (prose account of the af-validated tree). The tree is short — five nodes — because the analytic work has already been done by `lem-hcb3-diagonal-lower-modulus` (§25); what remains is an exact algebraic step and one scalar inequality. Write $T := \mathrm{Ha}^Q_{P,P}$ and $T_n := \big(\mathrm{Ha}^Q_{P,P}\big)_n$.

Constants. Let $C_{\text{diag}}, e_{\text{diag}}$ be universal witnesses supplied by `lem-hcb3-diagonal-lower-modulus`. Enlarging C_{diag} only weakens that external conclusion, so one may take $C_{\text{diag}} > 0$ and set

$$e_{\text{inv}} := \min\Big\{e_{\text{diag}},\ \tfrac{1}{2C_{\text{diag}}}\Big\}, \qquad C_{\text{inv}} := 2C_{\text{diag}}.$$

Both are universal, $e_{\text{inv}} > 0$, and the choice is made precisely so that

$$0 \le C_{\text{diag}} e \le \tfrac{1}{2} \qquad \text{whenever } 0 \le e \le e_{\text{inv}}. \tag{65}$$

Bijectivity amplifies exactly. Fix an admissible datum and an arbitrary $n \ge 1$. This step uses the *bijectivity* hypothesis and nothing else. By the meaning of matrix amplification, $T_n\big([Z_{ij}]\big) = [T(Z_{ij})]$ acts slotwise. Surjectivity therefore passes upward: for any target $[Y_{ij}]$ the matrix $[T^{-1}(Y_{ij})]$ is a preimage. So does injectivity: $T_n([Z_{ij}]) = 0$ forces $T(Z_{ij}) = 0$ for every i, j, hence $Z_{ij} = 0$ for every i, j and $[Z_{ij}] = 0$. Thus T_n is bijective and, moreover,

$$(T_n)^{-1} \text{ is the entrywise amplification of } T^{-1}. \tag{66}$$

Note what is *not* claimed: nothing here bounds $\|(T_n)^{-1}\|$ — entrywise inversion alone gives no uniform norm control, which is why the second hypothesis is needed.

The imported lower bound. Since $e \le e_{\text{inv}} \le e_{\text{diag}}$ and the level-one lower-modulus hypothesis is at least $\frac{1}{4}$, `lem-hcb3-diagonal-lower-modulus` applies verbatim and gives

$$\|T_n(Z)\| \;\ge\; \big(1 - C_{\text{diag}} e\big)\|Z\| \qquad \text{for every } Z \in M_n \otimes \mathcal{S}_P. \tag{67}$$

This is the only place the $\frac{1}{4}$ hypothesis enters, and it enters through the imported lemma rather than being re-proved.

The reciprocal estimate. Combine the two. For an arbitrary target element Y, put $Z = (T_n)^{-1}Y$, which exists by (66); then (67) reads $\|Y\| = \|T_n(Z)\| \geq (1 - C_{\mathrm{diag}}e)\|Z\|$. By (65) the quantity $x := C_{\mathrm{diag}}e$ lies in $[0, \frac{1}{2}]$, and on that interval the elementary identity

$$\frac{1}{1-x} = 1 + \frac{x}{1-x} \ \leq \ 1 + 2x$$

holds because $1 - x \geq \frac{1}{2}$. Therefore

$$\big\|(T_n)^{-1}Y\big\| \ \leq \ \big(1 - C_{\mathrm{diag}}e\big)^{-1}\|Y\| \ \leq \ \big(1 + 2C_{\mathrm{diag}}e\big)\|Y\| \ = \ \big(1 + C_{\mathrm{inv}}e\big)\|Y\|.$$

Since the datum and n were arbitrary, every amplification is bijective with the inverse bound (64), and the constants are universal. The restriction to $x \leq \frac{1}{2}$ is not cosmetic: it is what converts the reciprocal into a linear bound in e, and it is exactly what the choice of e_{inv} secures.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-diagonal-inverse`: 5 nodes, all `validated`, root `validated`, taint clean; first pass, two rounds, no challenges surviving.
Role in the chain. Imports `lem-hcb3-diagonal-lower-modulus`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. It completes the diagonal block of the H-CB tier: with §23 and §25 it says that, *under the two level-one hypotheses*, the amplified diagonal Ha map is a bijection whose norm and inverse norm are both within $O(e)$ of 1. The matching off-diagonal statement `lem-hcb3-offdiagonal-inverse` (§27), the hcb4 canonical tier (§§28–30) and the parent `conj-hcb` (§31) are now all `proved` with `af: validated`. Nothing here bears on the status of `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

27 Off-diagonal Ha inverse propagation

Lemma 75 (`lem-hcb3-offdiagonal-inverse`). *There are universal $C_{\mathrm{rect}} < \infty$ and $e_{\mathrm{rect}} > 0$ such that, for every H-CB datum with $e \le e_{\mathrm{rect}}$,* if $\mathrm{Ha}^Q_{P,R}$ is bijective at level one and $\mathrm{Ha}^Q_{R,R}$ has level-one lower modulus at least $\frac{1}{4}$, *then every amplification of* $\mathrm{Ha}^Q_{P,R}$ *is bijective with*

$$\big\|\big((\mathrm{Ha}^Q_{P,R})_n\big)^{-1}\big\| \;\le\; 1 + C_{\mathrm{rect}}\, e. \tag{68}$$

Transcription note. *The two level-one hypotheses are placed on* different *maps: bijectivity is assumed of the rectangular map* $\mathrm{Ha}^Q_{P,R}$ *itself, the lower-modulus hypothesis of the* diagonal *map* $\mathrm{Ha}^Q_{R,R}$ *at the target index. Neither is discharged here, and neither implies the other. This is the same conditional shape as §26, inherited from the verdict that made the corresponding clause of* `conj-hcb` *conditional.*

Registry contract (verbatim).

```
Off-diagonal Ha inverse propagation: there are universal C_rect < infinity and e_rect > 0 such
that, for every H-CB datum with e <= e_rect, if Ha^Q_{P,R} is bijective at level one and Ha^Q_{R,R}
has level-one lower modulus at least 1/4, then every amplification of Ha^Q_{P,R} is bijective with
inverse norm at most 1+C_rect*e.
```

Proof (prose account of the af-validated tree). Write $T_n := \big(\mathrm{Ha}^Q_{P,R}\big)_n$ and $D_n := \big(\mathrm{Ha}^Q_{R,R}\big)_n$. The tree has four movements: a constant ledger, a rectangular square estimate, the passage to a lower modulus for T_n, and a reciprocal bound. Only the third movement uses the diagonal hypothesis, and it uses it through `lem-hcb3-diagonal-lower-modulus` rather than re-proving it.

Constants. Let $C_{\mathrm{prod}}, e_{\mathrm{prod}}$ come from `lem-hcb2-product-defect`, let $C_{\mathrm{diag}}, e_{\mathrm{diag}}$ come from `lem-hcb3-diagonal-lower-modulus`, and let e_{adj} be the threshold of `lem-hcb2-amplified-adjointness`. The universal $O(e)$ constant of the compressed-product display (a workspace external, not a registry lemma) is unpacked as a finite C_{cp} valid below a positive e_{cp}. Put

$$B := 1 + C_{\mathrm{cp}}, \qquad A := \max\{1,\; C_{\mathrm{diag}} + B + C_{\mathrm{prod}}\}, \qquad C_{\mathrm{rect}} := 2A,$$

$$e_{\mathrm{rect}} := \min\Big\{e_{\mathrm{adj}}, e_{\mathrm{prod}}, e_{\mathrm{diag}}, e_{\mathrm{cp}}, \tfrac{1}{2A}\Big\} > 0.$$

The last entry is what makes $Ae \le \frac{1}{2}$ throughout, and every subsequent coefficient nonnegative.

The rectangular square estimate. Fix an admissible datum, an arbitrary $n \ge 1$ and $Z \in M_n \otimes \mathcal{S}_{P,R}$. Left and right multiplication by $I_n \otimes P$ and $I_n \otimes R$ act entrywise, so the operator to which the θ calculus is applied amplifies as $1_{M_n} \otimes (\cdot)$ and the registered power series gives $\mathrm{Co}_{I_n \otimes P, I_n \otimes R} = 1_{M_n} \otimes \mathrm{Co}_{P,R}$; hence Z lies in the rectangular corner of the amplified algebra, $Z^\dagger$ in the reversed one, and $Z^\dagger \cdot Z$ is the compressed product formed *inside* $M_n \otimes \mathcal{A}$. Comparing it with the ambient product and using the amplified lower C^* axiom,

$$\|Z^\dagger \cdot Z\| \;\ge\; \big(1 - \varepsilon - C_{\mathrm{cp}} e\big)\|Z\|^2 \;\ge\; \big(1 - Be\big)\|Z\|^2, \tag{69}$$

uniformly in n. Note that this is a *rectangular* statement and is derived here from the compressed-product display and the norm axioms; the diagonal statement `lem-hcb3-uniform-square-lower` is not the vehicle.

Lower modulus for the rectangular map. Instantiate `lem-hcb2-product-defect` at corner indices (R, P, R) with first factor $Z^\dagger$ and second factor Z; by `lem-hcb2-amplified-adjointness` the first image is $(\mathrm{Ha}^Q_{R,P})_n(Z^\dagger) = T_n(Z)^\dagger$, and the C^* identity for bounded operators between the amplified column spaces gives $\|T_n(Z)^\dagger T_n(Z)\| = \|T_n(Z)\|^2$. Hence

$$\|T_n(Z)\|^2 \;\geq\; \|D_n(Z^\dagger \cdot Z)\| - C_{\mathrm{prod}} e\|Z\|^2. \tag{70}$$

The diagonal hypothesis now enters exactly once: applying `lem-hcb3-diagonal-lower-modulus` at the index R to $X = Z^\dagger \cdot Z$ gives $\|D_n(X)\| \geq (1 - C_{\mathrm{diag}}e)\|X\|$, and $1 - C_{\mathrm{diag}}e \geq 1 - Ae \geq \frac{1}{2} > 0$ licenses multiplying (69) by it. Substituting into (70) and discarding the nonnegative quadratic remainder,

$$\|T_n(Z)\|^2 \;\geq\; \big[(1{-}C_{\mathrm{diag}}e)(1{-}Be){-}C_{\mathrm{prod}}e\big]\|Z\|^2 \;\geq\; \big[1{-}(C_{\mathrm{diag}}{+}B{+}C_{\mathrm{prod}})e\big]\|Z\|^2 \;\geq\; (1{-}Ae)\|Z\|^2.$$

Since $0 \leq Ae \leq \frac{1}{2}$ and $\sqrt{1-x} \geq 1 - x$ on $[0, 1]$,

$$\|T_n(Z)\| \;\geq\; (1 - Ae)\|Z\|. \tag{71}$$

Bijectivity and the reciprocal bound. Bijectivity amplifies algebraically and with no estimate: $\mathrm{Ha}^Q_{P,R}$ is linear with a two-sided linear inverse S by the first hypothesis, $T_n = 1_{M_n} \otimes \mathrm{Ha}^Q_{P,R}$ acts slotwise, and $1_{M_n} \otimes S$ is a two-sided inverse for it. For a target Y put $Z = T_n^{-1}Y$; then (71) reads $\|Y\| \geq (1 - Ae)\|Z\|$, and $1/(1 - x) \leq 1 + 2x$ for $x = Ae \in [0, \frac{1}{2}]$ gives $\|T_n^{-1}Y\| \leq (1 + 2Ae)\|Y\| = (1 + C_{\mathrm{rect}}e)\|Y\|$. The datum and n were arbitrary and the constants are universal, which is (68).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb3-offdiagonal-inverse`: 16 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Registry imports: `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-hcb3-diagonal-lower-modulus`, `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and `lem-hcb3-diagonal-inverse`; the exported tree visibly consumes the first three, the remaining three being available but unused in the validated derivation. Together with §§22–26 this completes the third layer of the H-CB tier: the diagonal maps are controlled on both sides, and the rectangular maps inherit an inverse bound from the diagonal anchor at their target index. It is the last clause consumed by the parent `conj-hcb` (§31). Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

28 The canonical corner Gram estimate

Lemma 76 (`lem-hcb4-canonical-gram`)**.** *There are universal $C_J < \infty$ and $e_J > 0$ such that every H-CB datum with $e \le e_J$, every $n \ge 1$ and every Z in either special P, Q corner satisfy*

$$(1 - C_J e)\,\|Z\| \;\le\; \big\|J_n(Z)\big\| \;\le\; (1 + C_J e)\,\|Z\|. \tag{72}$$

Transcription note. *The* special *corners are the two involving the one-dimensional projection of the datum: $Z \in M_n \otimes \mathcal{S}_{P,Q}$ with $J_n = J_{P,Q,n}$ the canonical column identification, or $Z \in M_n \otimes \mathcal{S}_{Q,P}$ with $J_n = J_{Q,P,n}$ the canonical row identification, both fixed by* ***`def-canonical-corner-identifications`****. The two sides of* (72) *carry* different *norms: $\|Z\|$ is the ambient operator-space norm of the amplified corner, while $\|J_n(Z)\|$ is the operator norm of the resulting map between amplified column spaces — the Euclidean norm of* ***`def-ha-map`****. The content of the lemma is therefore that the canonical identification, which is pure notation and carries no estimate of its own, is a $1 \pm O(e)$ isometry between those two norms, uniformly in n.*

Registry contract (verbatim).

```
Canonical corner Gram estimate: there are universal C_J < infinity and e_J > 0 such that every H-CB
datum with e <= e_J, every n >= 1, and every Z in either special P,Q corner satisfy (1-C_J*e)||Z||
<= ||J_n(Z)|| <= (1+C_J*e)||Z||.
```

Proof (prose account of the af-validated tree). The tree proves the column case and transfers the row case to it. The column case is one exact identity fed by two uniform inputs, followed by a scalar computation; there is no analysis beyond the two imports.

Exact Gram scalarisation. Fix an admissible datum, $n \ge 1$ and $Z \in M_n \otimes \mathcal{S}_{P,Q}$, and put $G := J_{P,Q,n}(Z)^\dagger J_{P,Q,n}(Z) \in M_n$. Reading the defining formula of `def-canonical-corner-identifications` against the column-Hilbert inner-product displays of `def-column-hilbert-corner` and the compressed-product display, the compressed square of Z is computed entrywise:

$$\big[Z^\dagger \cdot Z\big]_{ij} \;=\; \sum_k \langle Z_{ki}, Z_{kj}\rangle\, u_Q \;=\; G_{ij}\, u_Q, \qquad \text{that is} \qquad Z^\dagger \cdot Z \;=\; G \otimes u_Q. \tag{73}$$

This is an identity, not an estimate. Two norm facts turn it into one. The Ruan scalar-tensor invariance among the operator-space matrix-norm axioms gives $\|G \otimes u_Q\| = \|G\|\,\|u_Q\|$, and the Hilbert-space Gram identity gives $\|G\| = \|J_{P,Q,n}(Z)\|^2$. Hence

$$\big\|Z^\dagger \cdot Z\big\| \;=\; \big\|J_{P,Q,n}(Z)\big\|^2\,\|u_Q\|. \tag{74}$$

Everything after this point is bookkeeping on two scalars: the compressed square on the left and the compressed unit on the right.

The two uniform inputs. There are universal $A, B < \infty$, valid below a universal threshold, with

$$\big|\,\|Z^\dagger \cdot Z\| - \|Z\|^2\,\big| \le A\,e\,\|Z\|^2, \qquad \big|\,\|u_Q\| - 1\,\big| \le B\,e. \tag{75}$$

For the first, the ε-C^* norm axioms of `def-extended-epsilon-cstar-algebra` place the *ambient* square $\|Z^\dagger Z\|$ between $(1-\varepsilon)\|Z\|^2$ and $(1+\varepsilon)\|Z\|^2$ at level n, while `lem-compcb-rectangular-product` (§9) applied to the compatible amplified pair $(Z^\dagger, Z)$ gives $\|Z^\dagger \cdot Z - Z^\dagger Z\| \le C_r e \|Z\|^2$; since $\varepsilon \le e$, the reverse triangle inequality yields the first bound with $A = C_r + 1$, uniformly in n. For the second, `lem-compcb-compressed-unit-norm` (§11) gives $|\|u_Q\| - 1| \le C_u e$ once Q is known to be *nonvanishing*; the datum states only that Q is one-dimensional, and the gap is closed as elsewhere in this tier by the registered bridge $\dim \mathcal{S}_Q = 1 \Rightarrow \mathcal{S}_Q \neq 0 \Rightarrow Q$ nonvanishing. So $B = C_u$.

The scalar passage. Assume $Z \neq 0$ (for $Z = 0$ both sides of (72) vanish) and put $x := \|J_{P,Q,n}(Z)\|/\|Z\|$. Dividing (74) by $\|Z\|^2$ gives $x^2\|u_Q\| = \|Z^\dagger \cdot Z\|/\|Z\|^2$, so (75) confines

$$x^2 \;\in\; \left[\tfrac{1-Ae}{1+Be},\; \tfrac{1+Ae}{1-Be}\right].$$

Take e_{col} no larger than the two external thresholds, $1/(2\max\{B,1\})$ and $1/\max\{A+B,1\}$. Then the denominators are bounded away from 0 and the two ends simplify to $x^2 \ge 1-(A+B)e$ and $x^2 \le 1+2(A+B)e$. Finally $\sqrt{1-t} \ge 1-t$ for $t \in [0,1]$ and $\sqrt{1+t} \le 1+t/2$ for $t \ge 0$ convert these into

$$1-(A+B)e \;\le\; x \;\le\; 1+(A+B)e, \tag{76}$$

which is (72) for the column corner with $C_{\mathrm{col}} = A+B$. The asymmetric factor 2 in the upper bound for x^2 is exactly what the sharper square-root inequality absorbs, so the same constant serves both sides.

Row transfer. For $Z \in M_n \otimes \mathcal{S}_{Q,P}$ put $W := Z^\dagger \in M_n \otimes \mathcal{S}_{P,Q}$. The self-adjoint operator-space axiom gives $\|W\| = \|Z\|$ at every level; `def-canonical-corner-identifications` *defines* the row map by $J_{Q,P,n}(Z) = J_{P,Q,n}(W)^\dagger$; and Hilbert-space adjunction preserves the operator norm. Applying the column case to W therefore returns both row inequalities with the same constants, so $C_J = C_{\mathrm{col}}$ and $e_J = e_{\mathrm{col}}$ serve the root.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb4-canonical-gram`: 9 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Registry imports: `lem-compcb-rectangular-product`, `lem-compcb-compressed-unit-norm` `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first two, the remaining two being available but unused in the validated derivation. This is the base of the canonical layer of the H-CB tier. Canonical closeness (§29) uses it once, to convert $\|J_{P,Q,n}(Z)\|$ into $\|Z\|$; the canonical inverse estimate (§30) uses its lower half twice, once for the bound $\|J_n^{-1}\| \le \frac{4}{3}$ and once for the norm equivalence that makes the Neumann target complete. Through those two it reaches the canonical-identity clause of the parent `conj-hcb` (§31), which lists it among its sixteen dependencies. Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

29 Canonical Ha closeness

Lemma 77 (`lem-hcb4-canonical-closeness`)**.** *There are universal $C_{\mathrm{sp}} < \infty$ and $e_{\mathrm{sp}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{sp}}$ satisfies*

$$\max\{\|\big(\mathrm{Ha}^Q_{P,Q}\big)_n - J_{P,Q,n}\|,\ \|\big(\mathrm{Ha}^Q_{Q,P}\big)_n - J_{Q,P,n}\|\} \ \le\ C_{\mathrm{sp}}\, e \qquad \textit{for every } n \ge 1. \tag{77}$$

Transcription note. *Both differences are taken between maps defined on the same amplified corner — $M_n \otimes \mathcal{S}_{P,Q}$ in the first case, $M_n \otimes \mathcal{S}_{Q,P}$ in the second — and the outer norm is the induced one: the supremum over Z of the operator norm of the difference, divided by $\|Z\|$. The estimate is* unconditional*: unlike the inverse and lower-modulus results of §§25 –27 it assumes nothing about the level-one behaviour of any map.*

Registry contract (verbatim).

```
Canonical Ha closeness: there are universal C_sp < infinity and e_sp > 0 such that every H-CB datum
with e <= e_sp satisfies max{||(Ha^Q_{P,Q})_n-J_{P,Q,n}||,||(Ha^Q_{Q,P})_n-J_{Q,P,n}||} <= C_sp*e
for every n >= 1.
```

Proof (prose account of the af-validated tree). The tree has three movements: a scalar normalisation, the column estimate, and the same adjoint transfer used in §28. The organising observation is that the *entire* discrepancy between the special Ha map and the canonical map is carried by one number — the amount by which the compressed unit u_Q fails to be a unit vector for the column-Hilbert norm q_Q.

The normalised compressed unit. There are universal $A < \infty$ and $e_0 > 0$ such that, for $e \le e_0$, the scalar

$$\alpha := q_Q(u_Q) \tag{78}$$

is strictly positive, the normalised element $q_0 = \alpha^{-1} u_Q$ of `def-canonical-corner-identifications` satisfies $q_Q(q_0) = 1$ by absolute homogeneity, and $|\alpha - 1| \le Ae$. Two estimates combine. First, `lem-compcb-corner-algebra` (§16) applied with $P = Q$ — legitimate because Q is nonvanishing, by the registered one-dimensional-to-nonvanishing bridge — makes $\mathcal{S}_Q$ an extended γ-C^*-algebra with $\gamma = C_{\mathrm{ca}} e$ and unit u_Q, so the ε-Banach C^* norm axioms give $|\|u_Q\| - 1| \le \gamma$; after imposing $e \le \min\{e_{\mathrm{ca}}, 1/(2\max\{C_{\mathrm{ca}}, 1\})\}$ one has $\frac{1}{2} \le \|u_Q\| \le \frac{3}{2}$ and in particular $u_Q \ne 0$. Second, `lem-hcb-column-hilbert-squared` (§17) at $n = 1$, with its variable corner specialised to Q and $X = u_Q \in \mathcal{S}_Q$, compares the inner product with the norm: $\langle u_Q, u_Q\rangle = \alpha^2$, so $|\alpha^2 - \|u_Q\|^2| \le C_{\mathrm{col}} e \|u_Q\|^2$, and for $C_{\mathrm{col}} e \le \frac{1}{2}$ the inequality $|\sqrt{1+t} - 1| \le |t|$ on $|t| \le \frac{1}{2}$ gives $\alpha > 0$ and $|\alpha/\|u_Q\| - 1| \le C_{\mathrm{col}} e$. Adding the two through $\|u_Q\| \le \frac{3}{2}$,

$$|\alpha - 1| \le |\alpha - \|u_Q\|| + |\|u_Q\| - 1| \le \big(\tfrac{3}{2} C_{\mathrm{col}} + C_{\mathrm{ca}}\big) e,$$

so $A = 2C_{\mathrm{col}} + C_{\mathrm{ca}}$ works, with e_0 the minimum of the thresholds just used.

The column estimate. Fix $n \ge 1$ and $Z \in M_n \otimes \mathcal{S}_{P,Q}$ and put $H := \big(\mathrm{Ha}^Q_{P,Q}\big)_n(Z)$. The comparison is made *on the adjoint side*, where the column-action lemma is directly applicable. By `lem-hcb2-amplified-adjointness` (§12), $H^\dagger = \big(\mathrm{Ha}^Q_{Q,P}\big)_n(Z^\dagger)$, and by `def-canonical-corner-identification` $J_{P,Q,n}(Z)^\dagger = J_{Q,P,n}(Z^\dagger)$; so it suffices to compare those two. Applying `lem-hcb1-column-action` (§20) with corner variables (Q, P), matrix symbol $Z^\dagger$ and an arbitrary column X, and using $\|Z^\dagger\| = \|Z\|$,

$$q_Q\big(H^\dagger X - Z^\dagger \cdot X\big) \ \le\ C_{\mathrm{act}}\, e\, \|Z\|\, q_P(X). \tag{79}$$

The remaining term is exactly scalar. Put $d := J_{P,Q,n}(Z)^\dagger X \in \mathbb{C}^n$. Entrywise, the column-Hilbert displays and the defining formula for J give $[Z^\dagger \cdot X]_i = \sum_k Z^\dagger_{ki} \cdot X_k = d_i\, u_Q$; under the canonical identification by q_0 that vector *is* αd, whereas $J_{P,Q,n}(Z)^\dagger X$ is d. Hence

$$q_Q\big(Z^\dagger \cdot X - J_{P,Q,n}(Z)^\dagger X\big) \;=\; |\alpha - 1|\,\|d\|_2 \;\le\; A\, e\,(1 + C_J e)\,\|Z\|\, q_P(X), \tag{80}$$

the last step bounding $\|d\|_2$ by $\|J_{P,Q,n}(Z)\|\, q_P(X)$ and then invoking `lem-hcb4-canonical-gram` (§28) — its only use in this proof. Shrinking e below e_0, the two external thresholds, e_J and $1/\max\{C_J, 1\}$ makes the factor $(1 + C_J e)$ at most 2, so adding (79) and (80) by the triangle inequality and taking the supremum over nonzero X gives $\|H^\dagger - J_{P,Q,n}(Z)^\dagger\| \le C_* e\|Z\|$ with $C_* = C_{\mathrm{act}} + 2A$; Hilbert-space adjunction preserves the operator norm, so $\|H - J_{P,Q,n}(Z)\|$ obeys the same bound. No constant and no threshold depends on n.

Row transfer and completion. For $W \in M_n \otimes \mathcal{S}_{Q,P}$, adjointness gives $\big(\mathrm{Ha}^Q_{Q,P}\big)_n(W) = \big(\big(\mathrm{Ha}^Q_{P,Q}\big)_n(W^\dagger)\big)^\dagger$ and the definition gives $J_{Q,P,n}(W) = J_{P,Q,n}(W^\dagger)^\dagger$; with $\|W^\dagger\| = \|W\|$ and adjoint norm invariance, the column estimate transfers verbatim. Taking suprema over nonzero Z and W turns both pointwise bounds into map-norm bounds, so (77) holds with $C_{\mathrm{sp}} = C_*$ and e_{sp} the positive minimum of the thresholds collected above.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb4-canonical-closeness`: 11 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Registry imports: `lem-hcb-column-hilbert-squared`, `lem-hcb1-column-action`, `lem-hcb2-amplified-adjointness`, `lem-hcb4-canonical-gram`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first five. It is the "canonical-identity closeness" clause of the parent `conj-hcb` (§31) — one of the eight suppliers that parent's tree cites directly — and it supplies the hypothesis $\|A_n - J_n\| \le C_{\mathrm{sp}} e$ of the Neumann step in §30. Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

30 The canonical Ha inverse estimate

Lemma 78 (`lem-hcb4-canonical-inverse`)**.** *There are universal $C_{\mathrm{sp,inv}} < \infty$ and $e_{\mathrm{sp,inv}} > 0$ such that every H-CB datum with $e \le e_{\mathrm{sp,inv}}$ has the special maps $\mathrm{Ha}^Q_{P,Q}$ and $\mathrm{Ha}^Q_{Q,P}$ completely bijective, with their amplified inverses differing from the corresponding canonical inverses by at most*

$$\big\|\big((\mathrm{Ha}^Q_{P,Q})_n\big)^{-1} - J^{-1}_{P,Q,n}\big\| \;\le\; C_{\mathrm{sp,inv}}\, e, \tag{81}$$

and likewise for the row pair $\big((\mathrm{Ha}^Q_{Q,P})_n, J_{Q,P,n}\big)$. Transcription note. *"Completely bijective" means bijective at* every *amplification level $n \ge 1$, not merely at level one; and the estimate is uniform in n. Unlike the diagonal and off-diagonal inverse statements of §§26–27, this lemma is* unconditional*: no level-one hypothesis is assumed, because the comparison map J_n is bijective by construction and the smallness of e does the rest.*

Registry contract (verbatim).

```
Canonical Ha inverse estimate: there are universal C_sp,inv < infinity and e_sp,inv > 0 such that
every H-CB datum with e <= e_sp,inv has the special maps Ha^Q_{P,Q} and Ha^Q_{Q,P} completely
bijective, with their amplified inverses differing from the corresponding canonical inverses by at
most C_sp,inv*e.
```

Proof (prose account of the af-validated tree). The mathematics is a perturbation of an invertible map by a small one; the length of the tree is due almost entirely to justifying that the target space is Banach, which is where the Neumann series lives.

The ledger and the canonical inverse. Let C_J, e_J come from `lem-hcb4-canonical-gram` (§28) and $C_{\mathrm{sp}}, e_{\mathrm{sp}}$ from `lem-hcb4-canonical-closeness` (§29), and put

$$e_{\mathrm{sp,inv}} := \min\Big\{e_J,\ e_{\mathrm{sp}},\ \tfrac{1}{4(C_J + C_{\mathrm{sp}} + 1)}\Big\}, \qquad C_{\mathrm{sp,inv}} := 4\big(C_{\mathrm{sp}} + 1\big). \tag{82}$$

The last entry of the threshold forces both $C_J e \le \frac14$ and $C_{\mathrm{sp}} e \le \frac14$ simultaneously. For either special ordered corner and every n, the canonical map J_n is the amplified coefficient-space identification of `def-canonical-corner-identifications` — the column map, or the row map defined as its adjoint — hence an algebraic linear bijection onto the indicated operator corner, with the corresponding canonical inverse. Its inverse is bounded: the lower half of the Gram estimate reads $\|J_n Z\| \ge (1 - C_J e)\|Z\|$, and applying it to $Z = J_n^{-1} Y$ with $C_J e \le \frac14$ gives

$$\big\|J_n^{-1}\big\| \;\le\; \tfrac43. \tag{83}$$

The quantitative Neumann step. The abstract step used twice below is: if $J : X \to Y$ is a bounded bijection onto a *Banach* space Y with $\|J^{-1}\| \le \frac43$, and $A : X \to Y$ satisfies $\|A - J\| \le C_{\mathrm{sp}} e$ with $C_{\mathrm{sp}} e \le \frac14$, then A is bijective and $\|A^{-1} - J^{-1}\| \le \frac83 C_{\mathrm{sp}} e$. Put $T := (A - J)J^{-1} : Y \to Y$, so that $A = (I_Y + T)J$ and $\|T\| \le \|A - J\|\|J^{-1}\| \le \frac43 C_{\mathrm{sp}} e \le \frac13$. Since Y is Banach so is $B(Y)$, and the partial sums $S_N = \sum_{k=0}^N (-T)^k$ are Cauchy there by $\sum_{k\ge0} \|T\|^k \le (1 - \|T\|)^{-1}$; passing to the limit in the finite geometric identities $(I_Y + T)S_N = S_N(I_Y + T) = I_Y - (-T)^{N+1}$ and using $\|T\|^{N+1} \to 0$ identifies the limit as the two-sided inverse, with $\|(I_Y + T)^{-1}\| \le \frac32$. Then $A^{-1} = J^{-1}(I_Y + T)^{-1}$ and $(I_Y + T)^{-1} - I_Y = -(I_Y + T)^{-1} T$ give

$$\big\|A^{-1} - J^{-1}\big\| \;\le\; \tfrac43 \cdot \tfrac32 \cdot \tfrac43\, C_{\mathrm{sp}} e \;=\; \tfrac83\, C_{\mathrm{sp}}\, e. \tag{84}$$

Why the targets are Banach. This is not a formality, and the tree does not treat it as one. In an H-CB datum the ambient $\mathcal{A}$ is complete by `def-extended-epsilon-cstar-algebra`, and `def-compressed-corner` states that $\mathrm{Co}_{P,Q}$ is an exact bounded idempotent with $\mathcal{S}_{P,Q} = \mathrm{Im}\,\mathrm{Co}_{P,Q} = \mathrm{Ker}(1 - \mathrm{Co}_{P,Q})$; a kernel of a continuous map is closed, and a closed subspace of a Banach space is Banach, so $\mathcal{S}_{P,Q}$ is complete in the *inherited* norm. The norm actually carried by the target is the Euclidean one, and the two are equivalent by the Gram estimate at $n = 1$: there $J_{P,Q,1}(Z)c = Zc$, so $\|J_{P,Q,1}(Z)\| = \|Z\|_{\mathrm{Euc}}$ and $(1-C_J e)\|Z\| \leq \|Z\|_{\mathrm{Euc}} \leq (1+C_J e)\|Z\|$ with $C_J e \leq \frac{1}{4}$. Hence $(\mathcal{S}_{P,Q}, \|\cdot\|_{\mathrm{Euc}})$ is complete, its finite Hilbertian sum $\mathbb{C}^n \otimes \mathcal{S}_{P,Q}$ is complete, and since $B(E,F)$ is Banach whenever F is, both actual targets $B(\mathbb{C}^n, \mathbb{C}^n \otimes \mathcal{S}_{P,Q})$ and $B(\mathbb{C}^n \otimes \mathcal{S}_{P,Q}, \mathbb{C}^n)$ — the column and row targets of `def-ha-map` and `def-canonical-corner-identifications` — are Banach.

Assembly. Fix $n \geq 1$ and either special corner, and let A_n be $\left(\mathrm{Ha}^{Q}_{P,Q}\right)_n$ or $\left(\mathrm{Ha}^{Q}_{Q,P}\right)_n$. Canonical closeness gives $\|A_n - J_n\| \leq C_{\mathrm{sp}}e$, (83) gives the bijection J_n with $\|J_n^{-1}\| \leq \frac{4}{3}$, and (82) gives $C_{\mathrm{sp}}e \leq \frac{1}{4}$; so the Neumann step applies and $\|A_n^{-1} - J_n^{-1}\| \leq \frac{8}{3}C_{\mathrm{sp}}e \leq 4(C_{\mathrm{sp}}+1)e = C_{\mathrm{sp,inv}}e$. The constants do not depend on n or on the corner, so both special maps are completely bijective with the uniform estimate (81).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-hcb4-canonical-inverse`: 19 nodes, all `validated`, root `validated`, taint clean. Six of the nineteen exported nodes form the completeness branch.

Role in the chain. Registry imports: `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-closeness`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the exported tree visibly consumes the first two. Its `defs` line records `def-ha-map`, `def-hcb-datum`, `def-canonical-corner-identificati` `def-compressed-corner` (the range/kernel identity) and `def-extended-epsilon-cstar-algebra` (completeness of $\mathcal{A}$); the last two are consumed essentially in the validated closed-corner completeness branch. Downstream, this lemma is one of the sixteen declared dependencies of `conj-hcb` (§31); the parent's own exported tree cites eight suppliers directly and this is not among them, so the parent consumes it through the registry rather than in its validated derivation. Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

31 The H-CB parent contract

Lemma 79 (`conj-hcb`)**.** *There are universal $C_H < \infty$ and $e_H > 0$ such that, whenever $e = \delta + \varepsilon \le e_H$, Q is a level-one one-dimensional δ-projection in an extended ε-C^*-algebra $\mathcal{A}$, and P, R, S are δ-projections, the maps $1_{M_n} \otimes \mathrm{Ha}^Q_{P,R}$ — read, under the* COL-HILB *identification, as operators from $\mathbb{C}^n \otimes \mathcal{S}_{R,Q}$ to $\mathbb{C}^n \otimes \mathcal{S}_{P,Q}$ — satisfy for every n the adjoint equality, a product defect at most $C_H e\|Z\|\|W\|$, and the uniform unit, upper-norm, homomorphism and canonical-identity closeness estimates required by `lem_extension`; moreover*

1. *if the level-one lower modulus of $\mathrm{Ha}^Q_{P,P}$ is at least $\frac{1}{4}$, then every amplification has lower modulus at least $1 - C_H e$;*
2. *if $\mathrm{Ha}^Q_{P,P}$ is* also *bijective at level one, then every amplification is bijective with inverse norm at most $1 + C_H e$;*
3. *the analogous off-diagonal inverse bound for $\mathrm{Ha}^Q_{P,R}$ is asserted* only *when $\mathrm{Ha}^Q_{P,R}$ is bijective at level one and $\mathrm{Ha}^Q_{R,R}$ satisfies the diagonal lower-modulus hypothesis;*

all constants independent of n, $\dim \mathcal{A}$, block count and block dimensions. Transcription note. *`lem_extension` is a label in the pinned source, not a registry identifier: it names the consumer whose required estimate list this contract is calibrated to. The three clauses above are conditional as written; nothing in this document establishes their level-one hypotheses for any particular datum.*

Registry contract (verbatim).

```
H-CB: there are universal C_H < infinity and e_H > 0 such that, whenever e=delta+epsilon <= e_H, Q
is a level-one one-dimensional delta-projection in an extended epsilon-C*-algebra A, and P,R,S are
delta-projections, the maps 1_{M_n} tensor Ha^Q_{P,R}, under the COL-HILB identification with
operators on C^n tensor S_{R,Q} and C^n tensor S_{P,Q}, satisfy for every n the adjoint equality,
product defect at most C_H*e*||Z||||W||, and the uniform unit, upper-norm, homomorphism, and
canonical-identity closeness estimates required by lem_extension; moreover, if the level-one lower
modulus of Ha^Q_{P,P} is at least 1/4, then every amplification has lower modulus at least 1-C_H*e,
and if Ha^Q_{P,P} is also bijective at level one then every amplification is bijective with inverse
norm at most 1+C_H*e; the analogous off-diagonal inverse bound for Ha^Q_{P,R} is asserted only when
Ha^Q_{P,R} is bijective at level one and Ha^Q_{R,R} satisfies that diagonal lower-modulus
hypothesis; all constants independent of n, dim A, block count, and block dimensions.
```

Proof (prose account of the af-validated tree). The tree is a *composition*: eleven nodes, one for the constant ledger, one per asserted clause, one for uniformity. No new analysis is done at this level, and that is the point — every estimate has already been proved matrix-uniformly in §§12–27 and in the canonical tier (§§28–30), so the parent's only work is to exhibit one constant pair that serves them all simultaneously.

The ledger. Let the named coefficients and thresholds be those supplied by `lem-hcb2-amplified-adjointness` (e_{adj}), `lem-hcb2-product-defect` ($C_{\mathrm{prod}}, e_{\mathrm{prod}}$), `lem-hcb3-diagonal-unit` ($C_{\mathrm{unit}}, e_{\mathrm{unit}}$), `lem-hcb3-diagonal-upp` ($C_{\mathrm{up}}, e_{\mathrm{up}}$), `lem-hcb3-diagonal-lower-modulus` ($C_{\mathrm{diag}}, e_{\mathrm{diag}}$), `lem-hcb3-diagonal-inverse` ($C_{\mathrm{inv}}, e_{\mathrm{inv}}$), `lem-hcb3-offdiagonal-inverse` ($C_{\mathrm{rect}}, e_{\mathrm{rect}}$) and `lem-hcb4-canonical-closeness` ($C_{\mathrm{sp}}, e_{\mathrm{sp}}$). Put

$$C_H := \max\{1, C_{\mathrm{prod}}, C_{\mathrm{unit}}, C_{\mathrm{up}}, C_{\mathrm{diag}}, C_{\mathrm{inv}}, C_{\mathrm{rect}}, C_{\mathrm{sp}}\}, \qquad e_H := \min\{1, e_{\mathrm{adj}}, e_{\mathrm{prod}}, e_{\mathrm{unit}}, e_{\mathrm{up}}, e_{\mathrm{diag}}, e_{\mathrm{inv}}, e_{\mathrm{rect}}, e_{\mathrm{sp}}\}. \tag{85}$$

A finite maximum of finite universal constants is finite and universal; a finite minimum of positive universal thresholds is positive and universal. Consequently $e \le e_H$ meets *every* imported threshold at once, and each imported error coefficient is dominated by C_H — which is the only step where the clauses interact.

The clauses. Each is now the corresponding import, read at $e \le e_H$ and with its coefficient enlarged to C_H. For $Z \in M_n \otimes \mathcal{S}_{P,R}$ the adjoint clause is the exact identity $\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z)^\dagger = \big(\mathrm{Ha}^{Q}_{R,P}\big)_n(Z^\dagger)$; for $Z \in M_n \otimes \mathcal{S}_{P,S}$ and $W \in M_n \otimes \mathcal{S}_{S,R}$ the product-defect clause is

$$\big\|\big(\mathrm{Ha}^{Q}_{P,R}\big)_n(Z \cdot W) - \big(\mathrm{Ha}^{Q}_{P,S}\big)_n(Z)\,\big(\mathrm{Ha}^{Q}_{S,R}\big)_n(W)\big\| \;\le\; C_H\, e\,\|Z\|\|W\|, \tag{86}$$

which is simultaneously the asserted uniform amplified homomorphism estimate; the unit clause is $\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(I_n \otimes u_P) - I\| \le C_H e$; the upper-norm clause is $\|\big(\mathrm{Ha}^{Q}_{P,P}\big)_n(Z)\| \le (1 + C_H e)\|Z\|$; and the canonical-identity clause is

$$\max\big\{\|\big(\mathrm{Ha}^{Q}_{P,Q}\big)_n - J_{P,Q,n}\|,\; \|\big(\mathrm{Ha}^{Q}_{Q,P}\big)_n - J_{Q,P,n}\|\big\} \;\le\; C_H\, e,$$

with J the maps fixed by `def-canonical-corner-identifications`. The three conditional clauses are, in order, `lem-hcb3-diagonal-lower-modulus`, `lem-hcb3-diagonal-inverse` (§26) and `lem-hcb3-offdiagon` (§27), each carrying its level-one hypotheses unchanged into the parent statement.

Uniformity. The witnesses (85) are finite maxima and minima of universal quantities only. No entry depends on n, on $\dim \mathcal{A}$, on the block count or on the block dimensions, and no clause involves an entrywise sum over the amplification index; the hostile verifier looked for an n-growth family and found none.

Status. `proved`; `af: validated` (T0). Workspace `proofs/conj-hcb`: 11 nodes, all `validated`, root `validated`, taint clean. The identifier keeps its historical `conj-` slug; registry ids are stable.
Contract amendment of record. The conjecture as originally transcribed asserted the inverse estimate *unconditionally*. That clause is false: an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample refutes it, and the hostile verifier confirmed the refutation as genuine. The contract above carries the verifier's replacement clause verbatim. The verifier also recorded that this does not weaken what the EXT tier consumes, because at the point of use the relevant map h_{11} is first shown to be a level-one $O(e)$-isomorphism, so both conditional hypotheses are met at a universal smallness threshold. Three further corrections were wording-level: the compressed-unit input splits into a general upper bound and a nonvanishing-only two-sided bound; the phrase "established separately" means a universal $1 - O(e)$ lower bound threshold-shrunk to $\frac{1}{4}$, not bare bijectivity; and bijectivity of the canonical corners comes from the Neumann condition, not from two-sided norm bounds alone.
Role in the chain. The registry shard lists sixteen dependencies — the whole COMP-fed H-CB tier including `lem-hcb0-compressed-associator`, `lem-hcb-column-hilbert-squared`, `lem-hcb1-variational-` `lem-hcb1-column-action`, `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and the canonical tier `lem-hcb4-canonical-gram` (§28), `lem-hcb4-canonical-closeness` (§29), `lem-hcb4-canonical` (§30); the exported tree cites eight of them directly, the others entering through those eight. All sixteen are `proved` with `af: validated`, and all sixteen are now reproduced in this document. Downstream, `conj-hcb` is the principal input to `conj-extcb` (§37), itself `proved` with `af: validated` and reproduced below; the conditional clauses above are consumed there, after their level-one hypotheses have been established. The registry shard additionally records the paper-proof's explicit relative witnesses $C_H = 4000c$ and $e_H = 1/(10000c)$, with c the maximum of the sanctioned COMP-CB and corrected COL-HILB constants; the validated tree does not use them, deriving (85) instead. Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

32 One-dimensional corner calculus

The EXT tier works with the corners of a *closed EXT-CB datum* (`def-extcb-datum`) and asks for one map on a matrix algebra of one larger size. The two results of this section are its arithmetic base: through a one-dimensional δ-projection the compressed product is multiplicative on norms up to $1 \pm O(e)$, and consequently a corner between two one-dimensional projections is at most a line. Both are level-one statements; the amplified content of the tier is carried elsewhere.

Lemma 80 (`lem-extcb-one-dimensional-product`). *There are universal $C_{PQR} < \infty$ and $e_{PQR} > 0$ such that, for $e = \delta + \varepsilon \le e_{PQR}$, if Q is one-dimensional then*

$$\big|\, \|X \cdot Y\| - \|X\|\|Y\| \,\big| \;\le\; C_{PQR}\, e\, \|X\|\|Y\| \qquad \textit{for } X \in \mathcal{S}_{P,Q},\ Y \in \mathcal{S}_{Q,R}. \tag{87}$$

Transcription note. *One-dimensionality is required of the* middle *index Q — the index shared by the two corners — and of nothing else; P and R are arbitrary δ-projections. "One-dimensional" means* $\dim \mathcal{S}_Q = 1$ *(`def-one-dimensional-delta-projection`), and the statement is level-one.*

Registry contract (verbatim).

```
Level-one one-dimensional corner product: there are universal C_PQR < infinity and e_PQR > 0 such
that, for e=delta+epsilon <= e_PQR, if Q is one-dimensional then abs(||X dot Y||-||X||||Y||) <=
C_PQR*e*||X||||Y|| for X in S_{P,Q} and Y in S_{Q,R}.
```

Proof (prose account of the af-validated tree). The tree proves the *squared* comparison and takes a square root at the end; the squared form is what the algebra produces, because the only handle on X is $X^\dagger \cdot X$.

A uniform ledger. One-dimensionality gives $\mathcal{S}_Q \neq 0$, and vanishing of the corner characterises the small alternative of `def-delta-projection`; hence Q is nonvanishing and `lem-compcb-corner-algebra` makes $\mathcal{S}_Q$ an extended $C_{\mathrm{ca}}e$-C^*-algebra with unit $u_Q = \mathrm{Co}_Q(Q)$ and $|\|u_Q\| - 1| \le C_{\mathrm{ca}}e$. The dichotomy bounds $\|P\|, \|Q\|, \|R\|$ by a universal B; the compressed-product display supplies a universal C_{dot} with $\|A \cdot B - AB\| \le C_{\mathrm{dot}}e\|A\|\|B\|$ on compatible pairs. One consequence is used repeatedly: u_Q is a left unit on $\mathcal{S}_{Q,R}$ up to $O(e)$,

$$\|u_Q Y - Y\| \;\le\; C_L\, e\, \|Y\|, \tag{88}$$

obtained by combining $\|u_Q - Q\| = O(e)$ (from $\mathrm{Co}_Q(A) = Q(AQ) + O(e)\|A\|$ and $\|Q^2 - Q\| \le \delta$) with $\|QY - Y\| = O(e)\|Y\|$ (from $Y = \mathrm{Co}_{Q,R}(Y) = Q(YR) + O(e)\|Y\|$ and one associator move). Write C_0 for the maximum of these constants and shrink the threshold below $1/(2C_0)$, so that $\|u_Q\| \ge \frac{1}{2}$.

Scalarisation. Fix $X \in \mathcal{S}_{P,Q}$. The compression adjoint identity puts $X^\dagger$ in $\mathcal{S}_{Q,P}$, so $a := X^\dagger \cdot X$ lies in $\mathcal{S}_Q$; since $\dim \mathcal{S}_Q = 1$ and $u_Q \neq 0$, the unit spans, and $a = h\, u_Q$ for a unique scalar h, real because $a^\dagger = a$ and $u_Q^\dagger = u_Q$. Comparing a with the ambient square and using the ε-C^* axioms, $|\|a\| - \|X\|^2| \le (C_0 + 1)e\|X\|^2$; and $\|a\| = |h|\, \|u_Q\|$ together with $|\|u_Q\| - 1| \le C_0 e$ and $\|u_Q\| \ge \frac{1}{2}$ converts this into

$$\big|\, |h| - \|X\|^2 \,\big| \;\le\; C_1\, e\, \|X\|^2, \qquad |h| \;\le\; C_h \|X\|^2. \tag{89}$$

The squared comparison. Put $r = \|X\|$, $s = \|Y\|$, $W = X \cdot Y$ and $Z = XY$ for the ambient product. Three estimates combine. First, $\|W - Z\| \le C_0 e\, rs$ and $|\|Z^\dagger Z\| - \|Z\|^2| \le \varepsilon \|Z\|^2$ give

$\big|\|W\|^2 - \|Z^\dagger Z\|\big| \le C_3 e\, r^2 s^2$. Second, exact anti-multiplicativity of the involution, two applications of the ε-associator axiom, the replacement of $X^\dagger X$ by $a = h\, u_Q$ and then (88) give

$$\big\|(XY)^\dagger(XY) - h\, Y^\dagger Y\big\| \;\le\; C_4\, e\, r^2 s^2. \tag{90}$$

Third, $\big|\|Y^\dagger Y\| - s^2\big| \le \varepsilon s^2$ and (89) give $\big|\|h\, Y^\dagger Y\| - r^2 s^2\big| \le C_5 e\, r^2 s^2$. The reverse triangle inequality then yields $\big|\|W\|^2 - r^2 s^2\big| \le C_2 e\, r^2 s^2$ with $C_2 = C_3 + C_4 + C_5$.

Square root. If $rs = 0$ then $X = 0$ or $Y = 0$, and bilinearity of the compressed product makes $W = 0$, so (87) holds exactly. Otherwise $\|W\| + rs \ge rs > 0$ and the factorisation $|\alpha - \beta| = |\alpha^2 - \beta^2|/(\alpha + \beta)$ for nonnegative α, β turns the squared comparison into (87) with $C_{PQR} = C_2$ and e_{PQR} the minimum of the finitely many universal thresholds.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-one-dimensional-product`: 25 nodes, all `validated`, root `validated`, taint clean.

Lemma 81 (`lem-extcb-one-dimensional-corner-dimension`). *For sufficiently small universal $\delta + \varepsilon$, if P and Q are one-dimensional δ-projections then* $\dim \mathcal{S}_{P,Q} \le 1$.

Registry contract (verbatim).

```
Level-one one-dimensional corner dimension: for sufficiently small universal delta+epsilon, if P
and Q are one-dimensional delta-projections then dim S_{P,Q} <= 1.
```

Proof (prose account of the af-validated tree). Set $e_0 = \min\{e_{PQR},\, 1/(2(C_{PQR}+1))\}$ and let $e \le e_0$. For nonzero $X, Y \in \mathcal{S}_{P,Q}$ the compression adjoint identity puts $X^\dagger$ in $\mathcal{S}_{Q,P}$ with $\|X^\dagger\| = \|X\|$, and $X^\dagger \cdot Y$ lies in $\mathcal{S}_Q$. Apply Lemma 80 to the triple (Q, P, Q) — whose middle index is P, one-dimensional by hypothesis — to get

$$\|X^\dagger \cdot Y\| \;\ge\; (1 - C_{PQR} e)\, \|X\|\|Y\| \;\ge\; \tfrac{1}{2}\|X\|\|Y\| \;>\; 0. \tag{91}$$

This is the whole analytic content; the rest is linear algebra. If $\mathcal{S}_{P,Q} = 0$ there is nothing to prove. Otherwise fix a nonzero X and let $T_X : \mathcal{S}_{P,Q} \to \mathcal{S}_Q$ be $T_X(Y) = X^\dagger \cdot Y$, linear because the compressed product is bilinear. By (91), $T_X(Y) \ne 0$ whenever $Y \ne 0$, so $\operatorname{Ker} T_X = 0$; an injective linear map into $\mathcal{S}_Q$, which is one-dimensional by hypothesis, forces $\dim \mathcal{S}_{P,Q} \le \dim \mathcal{S}_Q = 1$.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-one-dimensional-corner-dimension`: 8 nodes, all `validated`, root `validated`, taint clean.
Role in the chain. Both lemmas import `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the second imports the first. Together they are the approximate-C^* substitute for the exact fact that a corner through a rank-one projection is a line: the product statement is the result the pinned source invokes for transitivity of equivalence of one-dimensional δ-projections (`def-one-dimensional-delta-projection`), and the dimension statement is consumed by `lem-extcb1-cross-co` (§34), which is where that transitivity is actually used.

33 Corner-dimension additivity

Lemma 82 (`lem-extcb-corner-dimension-additivity`). *For two finite-dimensional commutative C^*-algebras with projection bases and non-unital sufficiently accurate inclusions v, w, the compressed corner $\mathcal{S}_{v(I),w(I)}$ is linearly bijective to*

$$\bigoplus_{j,k} \mathcal{S}_{v(\Pi_j),w(\Sigma_k)}. \tag{92}$$

Transcription note. *"Projection basis" is* `def-projection-basis` *($\Pi_j^\dagger = \Pi_j$, $\Pi_j \Pi_k = \delta_{jk}\Pi_k$, $\sum_j \Pi_j = I$, and likewise $\{\Sigma_k\}$); "inclusion" is* `def-extended-delta-inclusion`*, and* non-unital *means the unit of the source need not be carried to the unit of the ambient algebra — $v(I)$ is only an approximate projection there. "Sufficiently accurate" is a universal ceiling on the inclusion defect plus the ambient ε; the tree fixes it as the minimum of finitely many universal thresholds, independent of the two basis cardinalities. The conclusion is a* linear *bijection, bounded with bounded inverse; no isometry and no bound uniform in the number of blocks is asserted, and none is needed downstream, where only the dimension count is consumed.*

Registry contract (verbatim).

```
Level-one corner-dimension additivity: for two finite-dimensional commutative C*-algebras with
projection bases and non-unital sufficiently accurate inclusions v,w, the compressed corner
S_{v(I),w(I)} is linearly bijective to the direct sum over j,k of S_{v(Pi_j),w(Sigma_k)}.
```

Proof (prose account of the af-validated tree). The tree proves one binary splitting lemma, transports it across the second index by the involution, and then runs two finite inductions. Write $P_j = v(\Pi_j)$, $Q_k = w(\Sigma_k)$, $P_{[1,j]} = \sum_{r \le j} P_r$ and $Q_{[1,k]} = \sum_{s \le k} Q_s$.

Images of a projection basis. Each Π_j and each partial sum $\sum_{r<j} \Pi_r$ is an exact Hermitian projection of norm one in the source, and distinct basis projections annihilate each other. Since v is linear, star-preserving, δ-multiplicative and two-sidedly norm-bounded, every image is Hermitian with norm at most $1 + \delta$ and $\|P_j^2 - P_j\| \le \delta$; the partial-sum identities $P_{[1,j]} = P_{[1,j-1]} + P_j$ and $P_{[1,p]} = v(I)$ are *exact*, by linearity alone, while orthogonality survives only approximately,

$$\|P_{[1,j-1]} P_j\| \le \delta, \qquad \|P_j P_{[1,j-1]}\| \le \delta, \tag{93}$$

the second from the first by star-preservation. The same holds for w and the Q_k. This mixture — exact additivity, approximate orthogonality — is what the splitting lemma is built to consume.

Binary splitting. Let $R = R_0 + R_1$ and T be uniformly bounded approximate projections with R_0, R_1 approximately orthogonal, and let e dominate the ambient associativity and product errors, all projection defects and both orthogonality defects. Define

$$A(X_0, X_1) = \mathrm{Co}_{R,T}(X_0 + X_1), \qquad B(X) = \big(\mathrm{Co}_{R_0,T} X,\ \mathrm{Co}_{R_1,T} X\big),$$

bounded and linear because every compression is a bounded idempotent whose range is the displayed corner. The calculus of `def-compressed-corner` gives, uniformly, $X = U(XV) + O(e)\|X\|$ for $X \in \mathcal{S}_{U,V}$. Hence for $X_i \in \mathcal{S}_{R_i,T}$, expanding $R = R_0 + R_1$ reproduces X_i from the matching summand and leaves $O(e)\|X_i\|$ from the other — one reassociation plus $R_{1-i} R_i = O(e)$ — so $\mathrm{Co}_{R,T} X_i = X_i + O(e)\|X_i\|$; applying $\mathrm{Co}_{R_a,T}$ and using $R_a R_i = \delta_{ai} R_i + O(e)$ gives $\mathrm{Co}_{R_a,T}\mathrm{Co}_{R,T} X_i =$

$\delta_{ai}X_i + O(e)\|X_i\|$, i.e. $\|BA - I\| \leq Ce$ for the max norm on the sum. In the other order, for $X \in \mathcal{S}_{R,T}$ the same replacement gives $X = R(XT) + O(e)\|X\|$ and $\mathrm{Co}_{R_i,T}X = R_i(XT) + O(e)\|X\|$; adding the two and using $R_0 + R_1 = R$ *exactly* gives $\mathrm{Co}_{R_0,T}X + \mathrm{Co}_{R_1,T}X = X + O(e)\|X\|$, and applying the bounded $\mathrm{Co}_{R,T}$, which fixes X, gives $\|AB - I\| \leq Ce$. Shrink the universal threshold so that $Ce < 1$: the Neumann series invert BA and AB, so $(BA)^{-1}B$ is a left inverse of A and $B(AB)^{-1}$ a right inverse; a map with both is a bounded linear bijection.

The second index. By the compression adjoint identity and isometry of the involution, $X \mapsto X^\dagger$ is a conjugate-linear isometric bijection $\mathcal{S}_{U,V} \to \mathcal{S}_{V,U}$. Conjugating the first-index assembly for a split $T = T_0 + T_1$ by this map on domain and codomain gives a *complex*-linear bounded bijection — two conjugate-linear maps around a complex-linear one — and the adjoint identity identifies it with the canonical second-index assembly $\mathcal{S}_{R,T_0} \oplus \mathcal{S}_{R,T_1} \to \mathcal{S}_{R,T}$. No second calculation is needed.

Two inductions. Fix j. The base $k = 1$ is the identity, since $Q_{[1,1]} = Q_1$ exactly; for $k \geq 2$ the second-index splitting applies to $Q_{[1,k]} = Q_{[1,k-1]} + Q_k$, whose hypotheses are supplied uniformly by (93). Finite induction over the q basis projections, composing each step with the preceding bijection and using $Q_{[1,q]} = w(I)$, produces a bounded linear bijection $\mathcal{S}_{P_j,w(I)} \to \bigoplus_k \mathcal{S}_{P_j,Q_k}$. The first-index splitting run along $P_{[1,j]} = P_{[1,j-1]} + P_j$ produces, in the same way, $\mathcal{S}_{v(I),w(I)} \to \bigoplus_j \mathcal{S}_{P_j,w(I)}$. Composing the second with the direct sum of the first family and reassociating the finite double sum — an isometric relabelling — gives (92). Each direct sum of finitely many bounded bijections is bounded with bounded inverse, the bound being a maximum over a finite index set.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-corner-dimension-additivity`: 39 nodes, all `validated`, root `validated`, taint clean. Several of those nodes are *re-derivations*: where a child had been written to consume a still-pending sibling, the tree carries a sharper child that discharges the same step from validated material only. That is a validation-ordering artifact of the bottom-up protocol, not additional mathematics; the mathematical content is the four movements above.

Role in the chain. Imports `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. It is the counting engine of the EXT tier: a corner over a matrix-unit basis decomposes into its blocks, so a dimension over a composite projection is the sum of the dimensions over its parts. Its single registry consumer is `lem-extcb1-cross-corner-dimension` (§34), which applies it to the diagonal subalgebra of M_r and to the scalars, and reads off $\dim \mathcal{S}_{v(I_r),Q} = r\,d$.

34 Dimension selection in an EXT-CB datum

A closed EXT-CB datum (`def-extcb-datum`) carries δ-projections P, Q with $\|P+Q-I\| \le \delta$, an extended δ-isomorphism $v : M_r \to \mathcal{S}_P$, $\dim \mathcal{S}_Q = 1$ and $\mathcal{S}_{P,Q} \neq 0$. Nothing in the datum says how large the cross corner is. The two lemmas of this section settle it: the corner has dimension exactly r, and the diagonal Q-corner exactly 1 — the shape needed to build one map on M_{r+1}. The first lemma is the technical enabler; the second is the count.

Lemma 83 (`lem-extcb1-close-corner-dimension`)**.** *There is a universal $e_{\text{close}} > 0$ such that, in an EXT-CB datum with $e \le e_{\text{close}}$, the compression ranges $\mathcal{S}_{v(I_r),Q}$ and $\mathcal{S}_{P,Q}$ have the same dimension.* Transcription note. *$v(I_r)$ is the image of the* source *unit; it is an approximate projection close to P, not equal to it, and the two corners it and P cut out are literally different subspaces. Only their dimensions are claimed equal.*

Registry contract (verbatim).

```
Close-compression range invariance: there is a universal e_close > 0 such that, in an EXT-CB datum
with e <= e_close, the compression ranges S_{v(I_r),Q} and S_{P,Q} have the same dimension.
```

Proof (prose account of the af-validated tree). Write $R = v(I_r)$, $E = \mathrm{Co}_{R,Q}$ and $F = \mathrm{Co}_{P,Q}$.

R is close to P, and is itself an approximate projection. Bijectivity of v onto $\mathcal{S}_P$ makes that corner nonzero, so P is nonvanishing; $\dim \mathcal{S}_Q = 1$ makes Q nonvanishing; the nonvanishing alternative of `def-delta-projection` then bounds $\|P\|, \|Q\|$ by a universal M_1. By `lem-compcb-corner-algebra` the corner $\mathcal{S}_P$ is an extended $C_{\text{ca}}e$-C^*-algebra with unit $u_P = \mathrm{Co}_P(P)$, and the unit clause for the extended δ-isomorphism v gives $\|R - u_P\| \le \delta$; combining with $\|u_P - P(PP)\| = O(e)\|P\|$ and $\|P(PP) - P\| \le \big((1+\varepsilon)\|P\| + 1\big)\delta$ yields

$$\|R - P\| \;\le\; C_1 e. \tag{94}$$

Star-preservation gives $R^\dagger = R$, δ-multiplicativity gives $\|R \cdot R - R\| \le \delta$ inside $\mathcal{S}_P$, and the compressed-versus-ambient comparison converts this into $\|R^2 - R\| \le C_2 e$ in $\mathcal{A}$: R is a $(C_R e)$-projection there.

Two close bounded idempotents. With the common adjusted defect $\delta' = C_R e$, the pairs (R, Q) and (P, Q) both satisfy the hypotheses of `lem-compcb-amplified-compression-identities` at $n = 1$, so $E^2 = E$ and $F^2 = F$. Each compression is $O(e)$-close to the corresponding left-times-right multiplication, and $\big(L_R R_Q - L_P R_Q\big)(X) = (R-P)(XQ)$ has norm at most $(1+\varepsilon)^2 \|R-P\| \|Q\| \|X\|$; with (94) this gives universal C_0, M_0 with

$$\|E - F\| \;\le\; C_0 e, \qquad \|E\| \;\le\; M_0. \tag{95}$$

Range transport. If E, F are bounded idempotents on a Banach space with $\|E\| \le M$ and $\|E - F\|(2M+1) < 1$, put $W = FE + (I-F)(I-E)$. Idempotence gives the two exact identities

$$W - I = (F - E)(2E - I), \qquad WE = FW = FE, \tag{96}$$

so $\|W - I\| \le \|F - E\|(2\|E\| + 1) < 1$ and W is invertible by its Neumann series. The relation $WE = FW$ sends $\mathrm{Im}\, E$ into $\mathrm{Im}\, F$ and, read as $EW^{-1} = W^{-1}F$, sends $\mathrm{Im}\, F$ back; hence W restricts to a linear bijection $\mathrm{Im}\, E \to \mathrm{Im}\, F$ and the two ranges have equal dimension. Taking e_{close}

below the thresholds of the two preceding steps and below $1/[2C_0(2M_0+1)]$ makes (95) satisfy that hypothesis, and $\operatorname{Im}\mathrm{Co}_{R,Q} = \mathcal{S}_{R,Q}$, $\operatorname{Im}\mathrm{Co}_{P,Q} = \mathcal{S}_{P,Q}$ finish it.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb1-close-corner-dimension`: 16 nodes, all `validated`, root `validated`, taint clean. This lemma exists because a hostile verdict on the EXT-CB transcription demanded that the level-one close-idempotent normalisation be made explicit, and that the selection threshold cover it.

Lemma 84 (`lem-extcb1-cross-corner-dimension`). *There is a universal* $e_{\mathrm{sel}} > 0$ *such that every EXT-CB datum with* $e \le e_{\mathrm{sel}}$ *satisfies*

$$\big(\dim \mathcal{S}_{P,Q},\ \dim \mathcal{S}_{Q,Q}\big) = (r, 1). \tag{97}$$

Registry contract (verbatim).

```
EXT-CB cross-corner dimension: there is a universal e_sel > 0 such that every EXT-CB datum with e
<= e_sel satisfies (dim S_{P,Q},dim S_{Q,Q})=(r,1).
```

Proof (prose account of the af-validated tree). The count is produced by transporting the diagonal matrix units through v, showing that each transported unit is a one-dimensional projection equivalent to all the others, and then summing.

Two inclusions. Let $D_r \subseteq M_r$ be the diagonal subalgebra with projection basis $\Pi_a = E_{aa}$, let v_D be the restriction of v to D_r, put $R_a = v(E_{aa})$, and let $w : \mathbb{C} \to \mathcal{A}$ be $w(\lambda) = \lambda Q$, whose source has the one-element projection basis $\{1\}$. Restriction preserves linearity, star-preservation, δ-multiplicativity and the two-sided norm bounds, and the compressed product on $\mathcal{S}_P$ differs from the ambient one by $O(e)$ on bounded elements, so v_D is a non-unital $O(e)$-inclusion into $\mathcal{A}$ with $v_D(I) = v(I_r)$, uniformly in r. For w: nonvanishing of Q gives $|\|Q\| - 1| \le C_{\mathrm{nv}}e$, the operator-space axioms give the exact cross-norm identity $\|X \otimes Q\| = \|X\|\|Q\|$ at every level, and $\|(XY)\otimes Q - (X \otimes Q)(Y \otimes Q)\| = \|(XY) \otimes (Q - Q^2)\| \le \delta\|X\|\|Y\|$; so every amplification of w is an $O(e)$-inclusion as well.

The transported matrix units. For $a, b \le r$ let $c_{ab}(X) = E_{aa}XE_{bb}$ on M_r and put $T_{ab} = v\, c_{ab}\, v^{-1}\mathrm{Co}_P$. Since Co_P is an idempotent with range $\mathcal{S}_P = \operatorname{Im} v$, one has $\mathrm{Co}_P v = v$ and $v^{-1}\mathrm{Co}_P v = \mathrm{id}$; as $c_{ab}^2 = c_{ab}$ with one-dimensional range, T_{ab} is an *exact* bounded idempotent of rank one, with range $\mathbb{C}\, v(E_{ab})$. The tree then compares it with the corresponding compression: uniformly in a, b and r,

$$\|\mathrm{Co}_{R_a,R_b} - T_{ab}\| \ \le\ C_m e. \tag{98}$$

Three fixed-length estimates give this — each R_a is a Hermitian C_1e-projection (from $\|E_{aa}\| = 1$, δ-multiplicativity and the compressed-versus-ambient comparison); the subordination bound $\|R_a(XR_b) - R_a\big((\mathrm{Co}_P X)R_b\big)\| \le C_s e\|X\|$, a finite telescope using $R_i = \mathrm{Co}_P(R_i)$ and $\mathrm{Co}_P(Y) = P(YP) + O(e)\|Y\|$; and the transported-source comparison $\|T_{ab}X - R_a\big((\mathrm{Co}_P X)R_b\big)\| \le C_3 e\|X\|$, from two applications of δ-multiplicativity in M_r and the corner-algebra bounds — together with $\|\mathrm{Co}_{R_a,R_b}X - R_a(XR_b)\| \le C_4 e\|X\|$. Since all source matrix factors have norm one, no constant depends on r.

Rank invariance. If E, F are bounded idempotents with $\|E - F\| < 1$ then $x \in \operatorname{Im} E$ with $Fx = 0$ gives $x = Ex = (E - F)x$, hence $x = 0$; interchanging E and F gives the reverse, so the two ranges have equal dimension when either is finite-dimensional. Shrinking e so that $C_m e < 1$ and applying this to (98) gives $\dim \mathcal{S}_{R_a,R_b} = \operatorname{rank} T_{ab} = 1$ for all a, b: taking $a = b$, each

R_a is a one-dimensional projection; taking $a \neq b$, the R_a are pairwise equivalent in the sense of `def-one-dimensional-delta-projection`.

Dichotomy, additivity, selection. Put $d_a = \dim \mathcal{S}_{R_a,Q}$. By `lem-extcb-one-dimensional-corner-dimension` (§32), applied to the one-dimensional projections R_a and Q with the enlarged common accuracy, $d_a \leq 1$, so $d_a \in \{0, 1\}$. The values coincide: if some $d_b = 1$ then R_b is equivalent to Q, every R_a is equivalent to R_b, and transitivity of equivalence — the registered property whose source proof is the one-dimensional product lemma of §32 — gives $d_a = 1$ for all a; otherwise all $d_a = 0$. Write d for the common value. Now `lem-extcb-corner-dimension-additivity` (§33), applied to D_r and $\mathbb{C}$ with the inclusions v_D, w, gives a linear bijection $\mathcal{S}_{v(I_r),Q} \to \bigoplus_{a=1}^{r} \mathcal{S}_{R_a,Q}$ and hence $\dim \mathcal{S}_{v(I_r),Q} = \sum_a d_a = r\,d$; Lemma 83 turns this into $\dim \mathcal{S}_{P,Q} = r\,d$. The datum assumes $\mathcal{S}_{P,Q} \neq 0$ and $r \geq 1$, so $d \neq 0$, i.e. $d = 1$ and $\dim \mathcal{S}_{P,Q} = r$. Finally $\mathcal{S}_{Q,Q} = \mathcal{S}_Q$ has dimension 1 by hypothesis, and e_{sel} is the minimum of the finitely many universal thresholds used, giving (97).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb1-cross-corner-dimension`: 30 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. The close-corner lemma imports `lem-compcb-amplified-compression-identities`, `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`; the cross-corner lemma imports both of the preceding sections and the close-corner lemma. The pair is the *selection* step of the EXT tier: it is what licenses treating $\mathcal{S}_{P,Q}$ as an r-dimensional column and $\mathcal{S}_Q$ as a line, i.e. the block shape that the four-corner merge (§36) assembles into a single map on M_{r+1}. Its consumer is `conj-extcb` (§37), now `proved` with `af: validated`; the close-corner lemma is there because a hostile verdict on that parent demanded it.

35 The four-corner assembled norm estimate

Lemma 85 (`lem-extcb-four-corner-norm`). *There are universal $K_{\mathrm{norm}} < \infty$ and $e_{\mathrm{norm}} > 0$ such that, for every four-corner merging datum with $e = \rho + \varepsilon \le e_{\mathrm{norm}}$, every $n \ge 1$ and every $X \in M_n \otimes \mathcal{B}$, the assembled map $\gamma_n = \alpha_n \Gamma_n \mu_n$, where*

$$\mu_n(X) = \big((I_n \otimes \Pi_j)\, X\, (I_n \otimes \Pi_k)\big)_{j,k}, \qquad \Gamma_n = \bigoplus_{j,k} \gamma_{jk,n}, \qquad \alpha_n\big((Y_{jk})\big) = \sum_{j,k} Y_{jk},$$

satisfies

$$(1 - K_{\mathrm{norm}} e)\|X\| \;\le\; \|\gamma_n(X)\| \;\le\; (1 + K_{\mathrm{norm}} e)\|X\|. \tag{99}$$

Transcription note. *ρ is the* datum's *common defect bound of* ***`def-four-corner-merging-datum`*** *— involution compatibility, compressed-product compatibility, diagonal-unit approximation, two-sided norm control, and the quantitative complementarity $\|P_1 + P_2 - I\| \le \rho$ of the target projections — and not the signed-picture scale of the same name in* ***`CONVENTIONS.md`****. The four maps γ_{jk} are* fixed *at level one and reused at every amplification: γ_n is not a separately chosen family.*

Registry contract (verbatim).

```
Four-corner assembled norm estimate: there are universal K_norm < infinity and e_norm > 0 such
that, for every four-corner merging datum with e = rho+epsilon <= e_norm, every n >= 1, and every X
in M_n tensor B, the assembled map gamma_n = alpha_n Gamma_n mu_n (where mu_n(X) = ((I_n tensor
Pi_j) X (I_n tensor Pi_k))_{j,k}, Gamma_n is the direct sum of the four fixed amplifications
gamma_{jk,n}, and alpha_n((Y_jk)) = sum_{jk} Y_jk) satisfies (1-K_norm*e)||X|| <= ||gamma_n(X)|| <=
(1+K_norm*e)||X||.
```

Proof (prose account of the af-validated tree). The estimate is proved in three stages, and the order matters: crude two-sided bounds first, an approximate multiplicativity of the *square* second, and only then a scalar bootstrap that turns the crude constants into $1 \pm O(e)$. A direct triangle-inequality attack on (99) does not work, because the four blocks can cancel.

Stage 1: block conditioning. Put $\pi_j = I_n \otimes \Pi_j$ and $X_{jk} = \pi_j X \pi_k$. The source projections are exactly complementary, so $X = \sum_{jk} X_{jk}$, $\|X_{jk}\| \le \|X\|$ and

$$\max_{jk}\|X_{jk}\| \;\le\; \|X\| \;\le\; \sum_{jk}\|X_{jk}\| \;\le\; 4\max_{jk}\|X_{jk}\|. \tag{100}$$

On the target side put $p_j = I_n \otimes P_j$ and $C_{jk} = 1_{M_n} \otimes \mathrm{Co}_{P_j,P_k}$. The compression construction commutes with amplification — the operator $L_{P_j} R_{P_k} + R_{P_k} L_{P_j} - 1$ amplifies entrywise, every partial sum of the defining power series satisfies $f_N(T_n) = 1_{M_n} \otimes f_N(T)$, and matrix entries are bounded coordinate maps — so $C_{jk} = \mathrm{Co}_{p_j,p_k}$ is genuinely the compression built inside $M_n \otimes \mathcal{A}$ and $\mathrm{Im}\, C_{jk} = M_n \otimes \mathcal{S}_{P_j,P_k}$. Quantitative complementarity gives $\|p_j^2 - p_j\| \le \rho$ and $\|p_1 + p_2 - I\| \le \rho$, whence $\|p_j\| \le 2$ and, for $j \ne l$, $\|p_j p_l\| \le 4e$. Consequently C_{jk} fixes its own corner and almost annihilates the other three: $\|C_{jk}(Y_{lm})\| \le L_c e \|Y_{lm}\|$ for $(j,k) \ne (l,m)$, with universal L_c. With $\beta(Y) = (C_{jk} Y)_{jk}$ and the max norm on the fourfold sum this reads $\|\beta\alpha - I\| \le 3L_c e$ and $\|\beta\| \le M_c$. Feeding (100) and the datum's two-sided norm clause through this extraction gives universal $c_{\mathrm{blk}} > 0$ and $B_{\mathrm{blk}} < \infty$ — concretely $c_{\mathrm{blk}} = 1/(16M_c)$ and $B_{\mathrm{blk}} = 5$ — with

$$c_{\mathrm{blk}}\|X\| \;\le\; \|\gamma_n(X)\| \;\le\; B_{\mathrm{blk}}\|X\|, \qquad \text{uniformly in } n. \tag{101}$$

Stage 2: the square defect. For U in the jk target corner and V in the lm one, two cases occur. If $k = l$ the compressed-product estimate applies at matrix level: $\|UV - U \cdot V\| \le c_c e\|U\|\|V\|$. If $k \ne l$ the product is *small*: replacing U, V by their compressed representatives $p_j(Up_k)$ and $p_l(Vp_m)$ and moving the middle factors together by exactly three ε-associator steps exposes the factor $p_k p_l$, so $\|UV\| \le K_{\mathrm{inc}} e\|U\|\|V\|$. Expanding $\gamma_n(X^\dagger X)$ and $\gamma_n(X)^\dagger \gamma_n(X)$ over the blocks produces sixteen terms, eight compatible and eight incompatible; the compatible ones are matched using the datum's involution and compressed-product clauses, the incompatible ones are simply small. Hence

$$\big\|\gamma_n(X^\dagger X) - \gamma_n(X)^\dagger\gamma_n(X)\big\| \;\le\; D_{\mathrm{sq}} e\|X\|^2, \qquad D_{\mathrm{sq}} = 8\big(4 + 2c_c + 2K_{\mathrm{inc}}\big). \tag{102}$$

Stage 3: the scalar bootstrap. Fix n and let a_n and b_n be the infimum and supremum of $\|\gamma_n(X)\|/\|X\|$ over $X \ne 0$; by (101), $0 < c_{\mathrm{blk}} \le a_n \le b_n \le 5$. The source is an exact C^*-algebra, so $\|X^\dagger X\| = \|X\|^2$. For the upper bound, the target lower C^* inequality and (102) give $(1-\varepsilon)\|\gamma_n(X)\|^2 \le \|\gamma_n(X^\dagger X)\| + D_{\mathrm{sq}} e\|X\|^2 \le (b_n + D_{\mathrm{sq}} e)\|X\|^2$, so a maximising sequence yields $(1-\varepsilon)b_n^2 \le b_n + D_{\mathrm{sq}} e$; if $b_n > 1$ then $b_n - 1 \le b_n(b_n - 1) \le \varepsilon b_n^2 + D_{\mathrm{sq}} e \le (25 + D_{\mathrm{sq}})e$. For the lower bound, the same identity read in the other direction gives $a_n \le (1+\varepsilon)a_n^2 + D_{\mathrm{sq}} e$, and if $a_n < 1$ then $c_{\mathrm{blk}}(1 - a_n) \le a_n(1 - a_n) \le (25 + D_{\mathrm{sq}})e$. Therefore

$$b_n \le 1 + (25 + D_{\mathrm{sq}})e, \qquad a_n \ge 1 - \frac{25 + D_{\mathrm{sq}}}{c_{\mathrm{blk}}}\, e,$$

which is (99) with $K_{\mathrm{norm}} = \max\{25 + D_{\mathrm{sq}}, (25 + D_{\mathrm{sq}})/c_{\mathrm{blk}}\}$ and e_{norm} the minimum of the universal ceilings of Stages 1 and 2 (and, harmlessly, $1/(2K_{\mathrm{norm}})$). The crude constant c_{blk} survives only in K_{norm}: it is what prevents the bootstrap from being vacuous, not what the final estimate looks like.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-four-corner-norm`: 18 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-corner-algebra` and `lem-hcb3-uniform-square-lower`. This shard exists because of a recorded *third-stall* classification: three successive orchestrations of `lem-extcb-four-corner-merge` stalled on exactly this claim, so it was factored out of that tree — its statement extracted mechanically from the blocking challenges, with the assembly maps written out from an already-validated node — and validated on its own. Its single registry consumer is `lem-extcb-four-corner-merge` (§36), which uses it as the entire norm-control half of the merge.

36 The complete four-corner merge

Lemma 86 (`lem-extcb-four-corner-merge`). *There are universal $C_{\text{merge}} < \infty$ and $a_{\text{merge}} > 0$ such that four fixed bijective level-one corner maps satisfying* ***`def-four-corner-merging-datum`*** *with common defect ρ and $\rho + \varepsilon \le a_{\text{merge}}$ combine into one extended $C_{\text{merge}}(\rho + \varepsilon)$-isomorphism.* Transcription note. *The smallness hypothesis constrains the* total *defect $\rho + \varepsilon$, not ρ alone; that is a verdict-driven amendment of the transcribed contract, and the record of why it cannot be dropped is reproduced below. "Extended isomorphism" is* ***`def-extended-delta-inclusion`****: every amplification is a δ-homomorphism with two-sided $(1 \pm \delta)$ norm bounds, and the level-one map is bijective.*

Registry contract (verbatim).

```
Complete four-corner merge: there are universal C_merge < infinity and a_merge > 0 such that four
fixed bijective level-one corner maps satisfying def-four-corner-merging-datum with common defect
rho and rho+epsilon <= a_merge combine into one extended C_merge*(rho+epsilon)-isomorphism.
```

Proof (prose account of the af-validated tree). Write $\pi_j = I_n \otimes \Pi_j$, $X_{jk} = \pi_j X \pi_k$, and let $\mu_n, \Gamma_n, \alpha_n$ and $\gamma_n = \alpha_n \Gamma_n \mu_n$ be the assembly maps of §35. Set $e = \rho + \varepsilon$.

One map, not a family. Exact source complementarity gives $\pi_j \pi_k = \delta_{jk} \pi_j$, $\pi_1 + \pi_2 = I$ and $X = \sum_{jk} X_{jk}$; since the four $\gamma_{jk,n}$ are the amplifications of the same four level-one maps,

$$\gamma_n = 1_{M_n} \otimes \gamma_1, \qquad \gamma_1(X) = \sum_{j,k} \gamma_{jk}(\Pi_j X \Pi_k). \tag{103}$$

This is the structural point of the lemma: the conclusion is about *one* linear map and its amplifications, so nothing may be chosen level by level. Two clauses follow at once. Adjoints of source corners transpose indices, $(\pi_j X \pi_k)^\dagger = \pi_k X^\dagger \pi_j$, and the datum's involution clause matches them, so $\gamma_n(X^\dagger) = \gamma_n(X)^\dagger$ *exactly.* The source unit has only diagonal blocks, so $\gamma_n(I) - I$ splits into the two diagonal-unit errors and $1_{M_n} \otimes (P_1 + P_2 - I)$, whose norm equals its level-one value by the direct-sum norm axiom; hence

$$\|\gamma_n(I) - I\| \ \le\ 3\rho, \qquad \text{uniformly in } n. \tag{104}$$

Corner separation and the multiplicative defect. As in §35, compression commutes with amplification, so all matrix-level corner estimates are available with level-one constants; and from $p_k p_l = p_k(s - I) + (p_k I - p_k) - (p_k^2 - p_k)$ with $s = p_1 + p_2$ one gets $\|p_k p_l\| \le 5e$ for $k \ne l$ once $e \le \frac{1}{4}$. Hence a product of corner elements is compressed-product-close when the middle indices match ($\|UV - U \cdot V\| \le c_{\text{prod}} e \|U\| \|V\|$) and is itself small when they do not ($\|UV\| \le c_{\text{cross}} e \|U\| \|V\|$, by three associator moves that expose $p_k p_l$). Expanding $\gamma_n(XY)$ and $\gamma_n(X)\gamma_n(Y)$ over the blocks gives sixteen terms — eight with matching middle index, compared through the datum's compressed-product clause, and eight without, which are small — so

$$\|\gamma_n(XY) - \gamma_n(X)\gamma_n(Y)\| \ \le\ D_{\text{mult}}\, e\, \|X\| \|Y\|, \qquad D_{\text{mult}} = 8\big[1 + 2c_{\text{prod}} + 2c_{\text{cross}}\big], \tag{105}$$

independently of n and of all dimensions.

Norm control. The datum and the assembly (103) are exactly the hypotheses of `lem-extcb-four-corner-norm` (§35), which supplies the two-sided bound $(1 - K_{\text{norm}} e)\|X\| \le \|\gamma_n(X)\| \le (1 + K_{\text{norm}} e)\|X\|$ at every level simultaneously.

Coverage, and bijectivity at level one. Injectivity of γ_1 is immediate from the lower bound once $K_{\mathrm{norm}}e \le \frac{1}{2}$. Surjectivity is the delicate half, and it is obtained without assuming that the target corner sum is direct: with $\beta(Y) = (\mathrm{Co}_{P_j,P_k}Y)_{jk}$, each compression is $O(e)$-close to $Y \mapsto P_j(YP_k)$, so $\alpha_1\beta$ is $O(e)$-close to $Y \mapsto s(Ys)$ with $s = P_1 + P_2$ — an identity of bilinearity alone, no associativity used — and $\|s - I\| \le \rho$ together with the product and approximate-unit axioms gives

$$\|\alpha_1\beta - I\| \;\le\; C_{\mathrm{cov}}e. \tag{106}$$

Choosing e with $C_{\mathrm{cov}}e < 1$ makes $\alpha_1\beta$ invertible by its Neumann series, so $Y = \alpha_1\big(\beta((\alpha_1\beta)^{-1}Y)\big)$ for every Y and α_1 is onto. Now μ_1 is a bijection by exact source complementarity, Γ_1 is a bijection because all four fixed level-one maps are bijective by hypothesis, and α_1 is onto; therefore $\gamma_1 = \alpha_1\Gamma_1\mu_1$ is onto, hence bijective, and every $1_{M_n} \otimes \gamma_1$ is algebraically bijective with inverse $1_{M_n} \otimes \gamma_1^{-1}$.

Conclusion. Let a_{merge} be the minimum of the thresholds above — e_{corner}, e_{norm}, e_{cov}, $1/(2K_{\mathrm{norm}})$ and the universal compression radius — and put $C_{\mathrm{merge}} = \max\{3, D_{\mathrm{mult}}, K_{\mathrm{norm}}\}$. Then (104) and (105) make every amplification a $C_{\mathrm{merge}}e$-homomorphism, the norm lemma supplies the two-sided $(1 \pm C_{\mathrm{merge}}e)$ bounds, and level-one bijectivity supplies the isomorphism clause; `def-extended-delta-inclusion` then returns exactly the asserted extended $C_{\mathrm{merge}}(\rho + \varepsilon)$-isomorphism.

Why the hypothesis is on the total defect. The transcribed contract bounded only ρ, leaving ε free. That version is *false*, and the tree records an exact counterexample rather than a repair. Take $\mathcal{A} = \mathbb{C}^3$ with coordinatewise involution, the multiplication $(a,b,c)(a',b',c') = (aa', bb', 0)$ and designated unit $I = (1,1,0)$: at every matrix level this is an extended 1-C^*-algebra, the lower C^* inequality being vacuous at $\varepsilon = 1$. The elements $P_1 = (1,0,0)$ and $P_2 = (0,1,0)$ are exact projections with $P_1 + P_2 = I$, so $\rho = 0$; the registered θ formula gives $\mathrm{Co}_{P_j} = E_j$ and $\mathrm{Co}_{P_j,P_k} = 0$ for $j \ne k$, so the four target corners are $\mathbb{C}P_1, 0, 0, \mathbb{C}P_2$. With $\mathcal{B} = \mathbb{C}^2$ and its exact structure, the two diagonal identities and the two zero maps form a datum with common defect $\rho = 0$ whose four maps are bijective complete isometries; yet the assembled map is $\gamma(a,b) = (a,b,0)$, which is not surjective, and no linear bijection $\mathbb{C}^2 \to \mathbb{C}^3$ exists at all. So ρ may be arbitrarily small while the conclusion fails, and only a ceiling on $\rho + \varepsilon$ repairs it. The prover amended the root accordingly and the fresh verifiers validated the amended statement; the registry contract carries that amended root verbatim, and downstream consumption supplies a small ambient ε, so the conditional form is the consumable one.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-extcb-four-corner-merge`: 22 nodes `validated` plus 4 `archived`, root `validated`, taint clean. The archived nodes are dependency-gated duplicates superseded by children that discharge the same step from validated material only.

Role in the chain. Imports `lem-compcb-corner-algebra`, `lem-hcb3-uniform-square-lower` and `lem-extcb-four-corner-norm` (§35). It is the assembly primitive of the EXT tier: given the block shape selected in §34 and four corner maps matched to it, the merge produces the single map that the tier's conclusion is about. Its consumer is `conj-extcb` (§37), now `proved` with `af: validated`: this merge is the last step of that proof. Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

37 The EXT-CB parent contract

Lemma 87 (`conj-extcb`). *There are universal $C_{\mathrm{ext}} < \infty$ and $e_{\mathrm{ext}} > 0$ such that if $e = \delta+\varepsilon \le e_{\mathrm{ext}}$, P, Q are δ-projections in an extended ε-C^*-algebra $\mathcal{A}$ with $\|P + Q - I\| \le \delta$, $v : M_r \to \mathcal{S}_P$ is an extended δ-isomorphism, $\dim \mathcal{S}_Q = 1$ at level one, and $\mathcal{S}_{P,Q} \neq 0$, then there is one map $v_+ : M_{r+1} \to \mathcal{A}$ whose every amplification is a $C_{\mathrm{ext}}e$-isomorphism; the same level-one unitary and the same four corner maps carry all amplification levels, with constants independent of r, n and $\dim \mathcal{A}$.* Transcription note. *The hypothesis list is exactly* ***`def-extcb-datum`****; the conclusion's "isomorphism" is* ***`def-extended-delta-inclusion`*** *read at accuracy $C_{\mathrm{ext}}e$. The clause $\dim \mathcal{S}_Q = 1$ is used* only *at level one: no amplification of Q is anywhere called one-dimensional. The final clause is not decoration — it is the content of "CB": one unitary and four fixed level-one maps serve every n, so nothing is chosen level by level.*

Registry contract (verbatim).

```
EXT-CB: there are universal C_ext < infinity and e_ext > 0 such that if e=delta+epsilon <= e_ext,
P,Q are delta-projections in an extended epsilon-C*-algebra A with ||P+Q-I|| <= delta, v:M_r->S_P
is an extended delta-isomorphism, dim S_Q=1 at level one, and S_{P,Q} is nonzero, then there is one
map v_+:M_{r+1}->A whose every amplification is a C_ext*e-isomorphism; the same level-one unitary
and the same four corner maps carry all amplification levels, with constants independent of r, n,
and dim A.
```

Proof (prose account of the af-validated tree). Put $P_1 = P$, $P_2 = Q$, $H_j = \mathcal{S}_{P_j,Q}$, $K_1 = \mathbb{C}^r$, $K_2 = \mathbb{C}$ and $h_{jk} = \mathrm{Ha}^{Q}_{P_j,P_k}$. The hypotheses are a closed EXT-CB datum, so for e below the universal e_{sel} of `lem-extcb1-cross-corner-dimension` (§34) the cross corners have dimensions $\dim H_1 = r$ and $\dim H_2 = 1$. Fix once and for all a unitary $U_2 : K_2 \to H_2$.

A dimension-free exact-target correction. The tree proves in-workspace, from no external theorem, the following. *There are universal $a_{\mathrm{corr}} > 0$ and $C_{\mathrm{corr}} < \infty$ such that if $\mathcal{B}$ is a finite-dimensional C^*-algebra, H a finite-dimensional Hilbert space, and $T : \mathcal{B} \to B(H)$ is linear, †-preserving and satisfies $\|T_n(XY) - T_n(X)T_n(Y)\| \le a\|X\|\|Y\|$ and $\|T_n(I) - I\| \le a$ at every n, with $0 \le a \le a_{\mathrm{corr}}$, then one unital †-homomorphism $\mu : \mathcal{B} \to B(H)$ satisfies $\|\mu_n - T_n\| \le C_{\mathrm{corr}}a$ at every n — the same level-one μ at every amplification.* It is a Newton iteration on the multiplicative defect.

At $\|X\| = 1$, †-preservation and the defect estimate give $\|T_n(X)\|^2 \le \|T_n(X^\dagger X)\| + a$, so $M_n = \|T_n\|$ obeys $M_n^2 \le M_n + a$ and $\|T\|_{\mathrm{cb}} \le 2$ for $a \le 1$. Fixing a state ϕ on $\mathcal{B}$, $S_0(x) = T(x) + \phi(x)(I - T(I))$ is †-preserving and *exactly* unital, with $\|S_0 - T\|_{\mathrm{cb}} \le a$, $\|S_0\|_{\mathrm{cb}} \le 3$ and multiplicative defect $b_0 \le c_u a$, $c_u = 7$. Normalised Haar measure on the compact unitary group of $\mathcal{B}$ gives $D = \int U^\dagger \otimes U\, dU$, in finite dimension a finite convex combination $\sum_s p_s U_s^\dagger \otimes U_s$ of projective norm one, with $XD = DX$ and $\mathrm{mult}(D) = I$ by invariance. For a defect G of a map S, the correction $w'(x) = \sum_s S(p_s U_s^\dagger)G(U_s, x)$ then obeys $\|w'\|_{\mathrm{cb}} \le \|S\|_{\mathrm{cb}}\|G\|$, and at level n the same formula is $1_{M_n} \otimes w'$ entrywise, because the diagonal identity moves arbitrary matrix entries. This is the step that makes every correction *one level-one map* with n-free, dimension-free bounds.

Now the Newton step. Let S be exactly unital, †-preserving, $\|S\|_{\mathrm{cb}} \le 4$, with defect G of uniform amplified bilinear norm b; put $w''(x) = w'(x^\dagger)^\dagger$, $w = (w' + w'')/2$ and $S^+ = S + w$. Since $G(y, I) = G(I, y) = 0$ one has $w(I) = 0$, so S^+ is still exactly unital and †-preserving, with $\|w\|_{\mathrm{cb}} \le 4b$. Exact associativity in $B(H)$ gives $S(x)G(y, z) - G(xy, z) + G(x, yz) - G(x, y)S(z) = 0$, and expanding $S(x)w(y) - w(xy) + w(x)S(y)$ with $XD = DX$, $\mathrm{mult}(D) = I$ and $S(I) = I$ cancels

G exactly to first order, every surviving term carrying two defect factors; a triangle estimate gives $\|G_{S^+}\| \le K_N b^2$ at every amplification, for one numerical universal K_N, while $\|S^+ - S\|_{\mathrm{cb}} \le 4b$. Choosing a_{corr} with $K_N c_u a_{\mathrm{corr}} \le \frac{1}{2}$ and $8c_u a_{\mathrm{corr}} < 1$ and iterating from S_0, the recursion $b_{k+1} \le K_N b_k^2$ gives $b_k \le b_0 2^{-k}$, hence $\|S_k - S_0\|_{\mathrm{cb}} \le \sum_{j<k} 4b_j \le 8b_0 \le 8c_u a < 1$ and $\|S_k\|_{\mathrm{cb}} < 4$, so every iteration is licensed — including the degenerate case $b_0 = 0$, where every increment vanishes. The sequence converges in cb norm to one unital †-preserving μ; continuity of multiplication and $b_k \to 0$ give $\mu(xy) = \mu(x)\mu(y)$, and cb convergence means every μ_n is the amplification of that same level-one μ. Finally $\|\mu - T\|_{\mathrm{cb}} \le a + 8c_u a$, i.e. $C_{\mathrm{corr}} = 1 + 8c_u = 57$.

One spatial four-corner system. Let $T = h_{11} \circ v$. For $X, Y \in M_n \otimes M_r$, `def-extended-delta-inclusion` gives $\|v_n(XY) - v_n(X) \cdot v_n(Y)\| \le \delta\|X\|\|Y\|$, $\|v_n(X)\| \le (1+\delta)\|X\|$, exact †-preservation and the compressed unit estimate in $\mathcal{S}_P$; feeding these through the upper-norm, product, unit and adjoint clauses of `conj-hcb` (§31) yields

$$\|T_n(XY) - T_n(X)T_n(Y)\| \le \big[(1 + C_H e)\delta + C_H e(1+\delta)^2\big]\|X\|\|Y\|, \qquad \|T_n(I) - I\| \le A_0 e,$$

so below one universal ceiling both defects are at most $A_0 e$ with $A_0 = 4(C_H + 1)$, uniformly in n, r and $\dim \mathcal{A}$. The correction now applies, in two cases that exhaust $e \ge 0$: if $e = 0$ then $\delta = \varepsilon = 0$, both defects vanish, T is itself an exact unital †-homomorphism and one takes $\mu_{11} := T$; if $e > 0$, shrink the threshold so that $a = A_0 e < a_{\mathrm{corr}}$ and correct. Either way there is one level-one unital †-homomorphism $\mu_{11} : M_r \to B(H_1)$ with

$$\|\mu_{11,n} - h_{11,n} v_n\| \;\le\; \kappa e \quad \text{for every } n, \qquad \kappa = C_{\mathrm{corr}} A_0. \tag{107}$$

It is unital and nonzero, so its kernel is a two-sided ideal in the simple algebra M_r and vanishes; domain and codomain both have dimension r^2 by the level-one count, so it is onto. A unital †-isomorphism between full matrix algebras preserves matrix units, hence is spatial: there is one unitary $U_1 : \mathbb{C}^r \to H_1$ with $\mu_{11}(A) = U_1 A U_1^\dagger$. With the already fixed U_2, set $\mu_{jk}(A) = U_j A U_k^\dagger$. These four fixed level-one maps are exact, completely isometric, adjoint-compatible and multiplicative at every amplification via $1_{M_n} \otimes U_j$, with no level-dependent choice.

The conditional H-CB clauses, in the required order. Shrink the threshold so that $\kappa e < 1$ and $C_H e < \frac{1}{4}$. Since μ_{11} is invertible and $\|T - \mu_{11}\| < 1$, Neumann inversion makes T bijective at level one; v is bijective, so h_{11} is, and for $Z = v(A)$ one has $\|h_{11}(Z)\|/\|Z\| \ge (1 - \kappa e)/(1+\delta) > \frac{1}{4}$. The canonical-identity closeness clause of `conj-hcb` makes each of h_{12}, h_{21}, h_{22} a strict-norm perturbation of its canonical level-one bijection, hence level-one bijective by Neumann inversion, and gives h_{22} level-one lower modulus above $\frac{1}{4}$. *Only now* are the conditional clauses invoked: the diagonal clause for h_{11} and h_{22}, then the off-diagonal clause for h_{12} anchored at h_{22} and for h_{21} anchored at h_{11}. Every $h_{jk,n}$ is then bijective with

$$(1 - C_H e)\|Z\| \le \|h_{jk,n}(Z)\| \le (1 + C_H e)\|Z\|, \qquad \|h_{jk,n}^{-1}\| \le 1 + C_H e, \tag{108}$$

and $h_{jk,n}^{-1} = 1_{M_n} \otimes h_{jk}^{-1}$ by uniqueness of inverses, so no inverse is chosen afresh at level n. The order is the delicate point, and the hostile verifier checked it: the parent's inverse clauses are conditional (§31), and their level-one hypotheses are established *before* use, never assumed.

Four fixed corner maps form a merging datum. Put $\gamma_{11} = v$ and $\gamma_{jk} = h_{jk}^{-1} \circ \mu_{jk}$ for $(j,k) \ne (1,1)$, and $d_{jk,n} = h_{jk,n}\gamma_{jk,n} - \mu_{jk,n}$. Off $(1,1)$ the composition is exact, so $d_{jk,n} = 0$; at $(1,1)$, (107) gives $\|d_{11,n}\| \le \kappa e$. *Involution* is exact: off $(1,1)$, exact H-CB adjointness and spatial adjoint compatibility give $h_{kj,n}\big(\gamma_{jk,n}(X)^\dagger\big) = h_{kj,n}\gamma_{kj,n}(X^\dagger)$ and injectivity of $h_{kj,n}$ cancels it, while $\gamma_{11,n} = v_n$ preserves † exactly. For the *product*, (108) and isometry of μ give $\|\gamma_{jk,n}(X)\| \le$

$(1+C_He)(1+\kappa e)\|X\| \le \frac{25}{16}\|X\| \le 2\|X\|$; with $D = \gamma_{jl,n}(XY) - \gamma_{jk,n}(X)\cdot\gamma_{kl,n}(Y)$, the product-defect clause of `conj-hcb` supplies E with $\|E\| \le 4C_He\|X\|\|Y\|$ and the exact spatial multiplication $\mu_{jl,n}(XY) = \mu_{jk,n}(X)\mu_{kl,n}(Y)$ turns the comparison into

$$h_{jl,n}(D) = d_{jl,n}(XY) - \mu_{jk,n}(X)d_{kl,n}(Y) - d_{jk,n}(X)\mu_{kl,n}(Y) - d_{jk,n}(X)d_{kl,n}(Y) - E,$$

whence $\|h_{jl,n}(D)\| \le [3\kappa + \kappa^2 e + 4C_H]e\|X\|\|Y\| \le 4(C_H+\kappa)e\|X\|\|Y\|$ and, applying $\|h_{jl,n}^{-1}\| \le \frac{5}{4}$, $\|D\| \le 5(C_H+\kappa)e\|X\|\|Y\|$. For the *unit and norm* clauses, spatiality gives $\mu_{jj,n}(I) = I$, so the H-CB diagonal-unit estimate, $\|u_{P_j} - P_j\| \le C_He$ and (108) give $\|\gamma_{jj,n}(I) - I_n\otimes P_j\| \le 3(C_H+\kappa)e$, while isometry of μ and the d-bound give $(1-(C_H+\kappa)e)\|X\| \le \|\gamma_{jk,n}(X)\| \le (1+2(C_H+\kappa)e)\|X\|$. Finally $\gamma_{11} = v$ is bijective by hypothesis and every other γ_{jk} composes two bijections. Since the source block projections in $M_{r+1} = B(K_1\oplus K_2)$ are exactly complementary and $\|P_1+P_2-I\| \le \delta \le e$, every clause of `def-four-corner-merging-datum` holds with the one common defect $\rho = D_0e$, $D_0 = 5(C_H+\kappa)$ enlarged so that $D_0 \ge 1$, for the same four fixed level-one maps at every amplification.

Merge, and the root. Let e_{ext} be the positive minimum of the thresholds above, of $a_{\mathrm{merge}}/(D_0+1)$, and of the finitely many Neumann and $\frac{1}{4}$ ceilings. Then $\rho+\varepsilon \le (D_0+1)e \le a_{\mathrm{merge}}$, so `lem-extcb-four-corner-merge` (§36) — whose norm half is `lem-extcb-four-corner-norm` (§35) — applies and returns the single sum map $v_+ : M_{r+1} \to \mathcal{A}$ as an extended $C_{\mathrm{merge}}(\rho+\varepsilon)$-isomorphism, hence an extended $C_{\mathrm{ext}}e$-isomorphism with $C_{\mathrm{ext}} = C_{\mathrm{merge}}(D_0+1)$. Its n-th map is $1_{M_n}\otimes v_+$, assembled from $1_{M_n}\otimes U_1$, $1_{M_n}\otimes U_2$ and the same four γ_{jk}; every constant used is universal, so none depends on r, on n, on $\dim\mathcal{A}$ or on the block data. That is the root.

Status. `proved`; `af: validated` (T0). Workspace `proofs/conj-extcb`: 40 nodes `validated` plus 6 `archived`, root `validated`, taint clean. The archived nodes are dependency-gated duplicates superseded by siblings that discharge the same step from validated material only: they are not part of the validated spine, and nothing above rests on them. The identifier keeps its historical `conj-` slug; registry ids are stable.

Correction of record (proof-level, not contract-level). The registry shard records one verifier-mandated correction: the level-one close-idempotent normalisation in the selection step must be explicit — the added-dimension corner is stated for $\widehat{P} = v(I_r)$, with $\|\widehat{P} - P\| = O(e)$ and range identification by close idempotents — and the selection threshold must cover it, at a universal threshold with no r- or level-dependence. That is what `lem-extcb1-close-corner-dimension` (§34) supplies; the contract itself was not amended.

Role in the chain. The registry shard lists ten dependencies: `conj-hcb` (§31), the EXT front end (§§32–36), `lem-compcb-corner-algebra` (§16) and `lem-hcb3-uniform-square-lower` (§24); all ten are `proved` with `af: validated` and all ten are reproduced in this document. The exported tree cites four directly — `conj-hcb`, `lem-extcb1-cross-corner-dimension`, `lem-extcb-four-corner-merge`, `lem-extcb-four-corner-norm` — the rest entering through the selection lemma. The registry body additionally records the paper-proof's witness in the expanded form $C_{\mathrm{ext}} = C_{\mathrm{merge}}[1+5C_H+20C_{\mathrm{app}}(C_H+1)]$; the validated tree does not use it, deriving $C_{\mathrm{merge}}(D_0+1)$ instead. Downstream, `conj-extcb` with `conj-hcb` discharges both gap nodes of the extended assembly, whose registry carrier `lem-thmainext-conditional` is now `proved` with `af: validated` (**not** reproduced here); the Route-F closure above it — the strengthened `lem-routef-k-ledger` replacement with its three helper rows and `lem-routef-f0-assembly` — is likewise `proved` with `af: validated`. Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

38 The Route F finish edge

Lemma 88 (lem-routef-prh-finish). *Let $A : \ell^\infty(k) \to \ell^\infty(n)$ and $M : \ell^\infty(n) \to \ell^\infty(k)$ be positive unital maps and let $Q : \ell^\infty(n) \to \ell^\infty(n)$ be row-stochastic. If $K \geq 1$,*

$$0 \;\leq\; \eta \;\leq\; \min\bigl\{(24K)^{-1},\, 1\bigr\}, \tag{109}$$

and

$$\|Q - AM\|_{\infty\to\infty} \;\leq\; K\eta, \qquad \|MA - I\|_{\infty\to\infty} \;\leq\; \frac{3K\eta}{1-3K\eta}, \tag{110}$$

*then there is a stochastic idempotent E (**def-stochastic**) with*

$$\|Q - E\|_{\infty\to\infty} \;\leq\; \bigl(K + 4\sqrt{2K}\bigr)\sqrt{\eta}. \tag{111}$$

Transcription note. *Both displays in* (110) *are* hypotheses*: the pair* (A, M) *— the datum of* ***def-positive-approximate-retract*** *— is given, and nothing in this statement, or anywhere in this document, manufactures it from Q. The denominator in* (110) *is positive precisely because of the window* (109)*. The constant in* (111) *is inherited from the hypothesis constant K and is universal only to the extent that K is; no result reproduced here fixes a value of K.*

Registry contract (verbatim).

```
Route F PRH finish: let A:l-infinity(k)->l-infinity(n) and M:l-infinity(n)->l-infinity(k) be
positive unital maps and let Q be row-stochastic; if K >= 1, 0 <= eta <= min{(24*K)^(-1),1},
||Q-AM||_{infinity->infinity} <= K*eta, and ||MA-I||_{infinity->infinity} <= 3*K*eta/(1-3*K*eta),
then there is a stochastic idempotent E with ||Q-E||_{infinity->infinity} <=
(K+4*sqrt(2*K))*sqrt(eta).
```

Proof (prose account of the af-validated tree). The tree has three steps — a scalar preparation, one import of `lem-prh` (§4) guarded by a dimension split, and a triangle-inequality finish — and a synthesis node that assembles them.

Scalar preparation. Put $x = 3K\eta$ and

$$\varepsilon := \frac{3K\eta}{1-3K\eta} = \frac{x}{1-x}. \tag{112}$$

The window (109) with $K \geq 1$ gives $0 \leq x \leq \frac{1}{8}$, hence $1 - x \geq \frac{7}{8} > 0$, so (112) is well defined and

$$0 \;\leq\; \varepsilon \;\leq\; \tfrac{8}{7}x \;\leq\; \tfrac{1}{7} \;<\; \tfrac{1}{2}, \qquad \varepsilon \;\leq\; \tfrac{24}{7}K\eta \;\leq\; 4K\eta.$$

Separately, $0 \leq \eta \leq 1$ gives $\eta^2 \leq \eta$, i.e. $\eta \leq \sqrt{\eta}$, and multiplying by $K \geq 1$,

$$K\eta \;\leq\; K\sqrt{\eta}. \tag{113}$$

The two bounds on ε are the whole point of the preparation: the first puts ε inside the PRH window $[0, \frac{1}{2})$, the second is what converts a $\sqrt{\varepsilon}$ rate into a $\sqrt{\eta}$ rate.

The PRH import, with an exhaustive dimension split. By (112) the second hypothesis of (110) reads exactly $\|MA - I_k\|_{\infty\to\infty} \leq \varepsilon$ with $0 \leq \varepsilon < \frac{1}{2}$, which is the hypothesis of `lem-prh`. That lemma, however, is stated for $k, n \geq 1$, and the contract above assumes no dimension bound; the tree therefore splits, and the split is exhaustive. If $k, n \geq 1$, then `def-positive-approximate-retract` identifies A and M as positive unital maps with probability-vector rows, every premise of `lem-prh` is met, and it returns a stochastic idempotent $E : \ell^\infty(n) \to \ell^\infty(n)$ with

$$\|AM - E\|_{\infty\to\infty} \;\le\; 2\sqrt{2\varepsilon}. \tag{114}$$

The mixed degenerate cases cannot occur: if $k = 0 < n$ then each of the n rows of A is the empty vector, of coordinate sum 0 rather than 1, contradicting the probability-row requirement, and the same argument applied to M excludes $n = 0 < k$. There remains $k = n = 0$, where $\ell^\infty(0)$ is the zero space, its unique endomorphism is the empty 0×0 matrix, which is entrywise nonnegative, fixes the (empty) all-ones vector and is idempotent — a stochastic idempotent by `def-stochastic` — and satisfies $AM = E$, so (114) holds with error 0. In every admissible dimension, then, such an E exists, and `lem-prh` is never invoked outside its stated domain.

The finish. Writing $Q - E = (Q - AM) + (AM - E)$, the triangle inequality for $\|\cdot\|_{\infty\to\infty}$ with the first hypothesis of (110) and (114) gives $\|Q - E\|_{\infty\to\infty} \le K\eta + 2\sqrt{2\varepsilon}$. Monotonicity and multiplicativity of the square root on the nonnegative reals turn $\varepsilon \le 4K\eta$ into $2\sqrt{2\varepsilon} \le 2\sqrt{8K\eta} = 4\sqrt{2K}\sqrt{\eta}$, and with (113),

$$\|Q - E\|_{\infty\to\infty} \;\le\; K\sqrt{\eta} + 4\sqrt{2K}\sqrt{\eta} \;=\; \big(K + 4\sqrt{2K}\big)\sqrt{\eta},$$

which is (111).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-prh-finish`: 22 nodes created and 22 `validated`, root `validated`, taint clean; three challenges raised and three resolved, four node amendments. The workspace registers two definition imports, `def-positive-approximate-retract` and `def-stochastic`, and two external imports carrying the validated contract of `lem-prh`.
Correction of record (proof-level, not contract-level). All three challenges were the same defect, raised three times at three heights of the tree: the import node asserted a direct application of `lem-prh` while neither the root contract nor either definition supplies that lemma's premise $k, n \ge 1$, so the invocation was, as written, outside the external's stated domain — and the $k = n = 0$ branch is admissible under the root's hypotheses. The repair was not to weaken anything but to make the dichotomy explicit and to discharge the degenerate branch by hand, which is the form reproduced above. The degenerate case is a formality; the protocol nonetheless refused the import until it was discharged, which is exactly the guard one wants on a hypothesis-carrying external. The registry contract was not amended.
What this closes, and what it does not. This is the validated conditional PRH finish edge within Route F and it is now af-validated, but it is an *implication*. Its antecedent — a positive unital pair (A, M) with $\|Q - AM\|_{\infty\to\infty} \le K\eta$ and $\|MA - I\|_{\infty\to\infty} \le 3K\eta/(1 - 3K\eta)$ for a dimension-free K — is not established unconditionally anywhere in this document. Two of the registry rows that produce it, `lem-routef-f2-positive-unital-compression` (the positive unital pair) and `lem-routef-f3-retract-defect` (the retract-defect bound, whose conclusion is literally the second hypothesis here), are now `proved` with `af: validated` and are reproduced in §43, as are the seam and ledger rows upstream of them (§41, §42). The F2 row is itself an implication, but its antecedent is now discharged: the assembly carrier `lem-thmainext-conditional` and the strengthened `lem-routef-k-ledger` — the factorization row that fixes the dimension-free K — are both `proved` with `af: validated` (the latter not reproduced here; §76). Consequently the unconditional $\sqrt{\eta}$ bound for a row-stochastic Q holds through this edge, and `op-classical` is now **discharged** at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

Role in the chain. Registry `defs`: `def-positive-approximate-retract`, `def-stochastic`. Registry `deps`: `lem-prh` (§4), `proved` with `af: validated` and reproduced above — the single mathematical import, and the only place the square-root rate is produced. The registry records no result depending on this one; `lem-routef-f2-positive-unital-compression` and `lem-routef-f3-retract-defect` (§43) name it in their shard bodies as the consumer they feed, which is the direction of the eventual chain, not a current dependency edge.

39 Quantitative inverse-function control

Lemma 89 (`lem-stage1-quantitative-inverse-function`). *Let X, Y be Banach spaces, let $V : X \to Y$ be a Banach-space isomorphism, let $B_r(x_0) \subset X$ be the open ball of radius $r > 0$ about x_0, and let $f : B_r(x_0) \to Y$ be C^1 with*

$$\|V^{-1}Df(x) - I_X\| \;\le\; c \;<\; 1 \qquad \text{for every } x \in B_r(x_0). \tag{115}$$

Then f is injective,

$$(1-c)\|x_1 - x_2\| \;\le\; \|V^{-1}\big(f(x_1) - f(x_2)\big)\| \;\le\; (1+c)\|x_1 - x_2\| \qquad (x_1, x_2 \in B_r(x_0)), \tag{116}$$

and the image contains a definite ball transported by V:

$$f(x_0) + V\big(B_{(1-c)r}(0)\big) \;\subseteq\; f\big(B_r(x_0)\big). \tag{117}$$

Transcription note. *The comparison operator V is* fixed *and the bound* (115) *is uniform on the whole ball; nothing here is stated at, or centred on, $Df(x_0)$. The covering radius $(1-c)r$ is measured in the* source *and then pushed forward, so* (117) *guarantees an image of the V-transported ball, which is in general not a norm ball of Y. All three conclusions are quantitative in the single scalar c; no smallness beyond $c < 1$ is assumed.*

Registry contract (verbatim).

```
Quantitative inverse-function control: if V:X->Y is a Banach-space isomorphism and f:B_r(x_0)->Y is
C^1 with ||V^(-1)Df(x)-I|| <= c < 1, then f is injective, (1-c)||x_1-x_2|| <=
||V^(-1)(f(x_1)-f(x_2))|| <= (1+c)||x_1-x_2||, and f(B_r(x_0)) contains f(x_0)+V(B_{(1-c)r}(0)).
```

Proof (prose account of the af-validated tree). The tree is self-contained: it imports no definition shard and no external theorem, so the two classical ingredients it needs — a Banach-valued fundamental theorem of calculus and the contraction-mapping principle — are proved inside the workspace rather than cited. The argument runs in two independent halves, the secant estimate and the covering, joined only by a common normalisation.

Normalisation. Put $F(x) = V^{-1}\big(f(x) - f(x_0)\big)$ on $B_r(x_0)$. Then F is C^1 with $DF(x) = V^{-1}Df(x)$ and $F(x_0) = 0$, so the hypothesis (115) reads exactly $\|DF(x) - I_X\| \le c$: the map has been turned into a perturbation of the identity of X, and the target space no longer appears except through V^{-1}.

The secant-error estimate. The core step is: if a C^1 map F on the open ball satisfies $\|DF - I_X\| \le c$ throughout, then

$$\|F(x_1) - F(x_2) - (x_1 - x_2)\| \;\le\; c\|x_1 - x_2\| \qquad (x_1, x_2 \in B_r(x_0)). \tag{118}$$

Fix x_1, x_2, put $d = x_1 - x_2$ and $\gamma(t) = x_2 + td$; convexity of the ball keeps $\gamma([0,1])$ inside the domain, which is where openness and convexity — and nothing else about the domain — are used. For $q(t) = F(\gamma(t)) - F(x_2) - td$ the chain rule gives $q'(t) = \big(DF(\gamma(t)) - I_X\big)d$, hence $\|q'(t)\| \le c\|d\|$ on $[0,1]$. The tree then proves, for a C^1 curve $q : [0,1] \to X$, that $q(1) - q(0) = \int_0^1 q'(t)\,dt$: continuity of q' makes its tagged Riemann sums converge in norm to the Bochner/Riemann integral while the matching sums of increments telescope to $q(1) - q(0)$; the norm inequality for that integral gives $\|q(1) - q(0)\| \le \int_0^1 \|q'(t)\|\,dt$. With $q(1) - q(0) = F(x_1) - F(x_2) - d$ this is (118).

From the secant error to (116) *and injectivity.* The remaining step is a two-line normed-space deduction, isolated in the tree as its own node: if $\|a - d\| \le c\|d\|$ then the reverse and ordinary triangle inequalities give $(1-c)\|d\| \le \|a\| \le (1+c)\|d\|$, and when $c < 1$ the vanishing of a forces $d = 0$. Applying this with $d = x_1 - x_2$ and $a = F(x_1) - F(x_2) = V^{-1}(f(x_1) - f(x_2))$ yields (116); the strict positivity of $1 - c$ then yields injectivity of f.

The centred covering. Let $g(h) = V^{-1}\big(f(x_0+h) - f(x_0)\big)$ on $B_r(0)$, so $g(0) = 0$, $\|Dg(h) - I_X\| \le c$, and the segment argument above applies verbatim to give $\|(g(h) - g(k)) - (h - k)\| \le c\|h - k\|$. Fix z with $\|z\| < (1-c)r$ and choose ρ with $0 < \rho < r$ and $\|z\| < (1-c)\rho$ — the strict inequality is what buys the room for the next line. On the closed ball $K = \{h \in X : \|h\| \le \rho\}$, which is complete because X is, set

$$T(h) \;=\; z + h - g(h).$$

The secant bound gives $\|T(h) - T(k)\| \le c\|h-k\|$, and taking $k = 0$ with $g(0) = 0$ gives $\|h - g(h)\| \le c\|h\|$, whence $\|T(h)\| \le \|z\| + c\rho < \rho$. So T is a c-contraction of the complete metric space K into itself. The tree proves the fixed-point principle in place: the iterates $h_{n+1} = T(h_n)$ satisfy $d(h_{n+1}, h_n) \le c^n d(h_1, h_0)$, the geometric bound $d(h_m, h_n) \le c^n d(h_1, h_0)/(1-c)$ makes them Cauchy, completeness supplies a limit h, and $d(T(h), h) \le c\, d(h, h_n) + d(h_{n+1}, h) \to 0$ identifies it as a fixed point. Then $h \in K \subset B_r(0)$ and $T(h) = h$ says precisely $g(h) = z$.

The translation back. For *this* g the defining identity is $Vg(h) = f(x_0 + h) - f(x_0)$. Hence $g(h) = z$ gives $f(x_0 + h) = f(x_0) + Vz$, and every $y \in f(x_0) + V(B_{(1-c)r}(0))$ is $f(x_0 + h)$ with $x_0 + h \in B_r(x_0)$. That is (117), and with it the root. The fixed point lives in X; Y enters only through V^{-1}.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-stage1-quantitative-inverse-function`: 14 nodes created and 14 `validated`, root `validated`, taint clean; three challenges raised and three resolved, two node amendments. No definition imports and no external imports are registered in the workspace.

Corrections of record (proof-level, not contract-level). The tree reached its verdict only after three hostile challenges, and the shape of the final nodes is their residue. (i) A *dependencies* challenge at the Newton–Leibniz node: as first written it was a leaf that silently borrowed the curve q, its formula and its derivative bound from a *pending* sibling node, which the protocol forbids; it was repaired by a child node carrying the abstract estimate and its instantiation together. (ii) A *type-error* challenge at the fixed-point node: the universal statement was phrased for an arbitrary complete metric space but written with norm differences $\|h_{n+1} - h_n\|$, which is ill-typed there; it was amended to the metric d. (iii) A *scope* challenge at the covering node, with a concrete counterexample: as first written, that node asserted the translation "$g(h) = z$ implies $y = f(x_0+h)$" for an *arbitrary* map g satisfying only $g(0) = 0$ and the secant bound. Take $X = Y = \mathbb{R}$, $V = I$, $x_0 = 0$, $r = 1$, $c = \frac{1}{2}$ and $f(x) = \frac{3}{2}x$; then $g(h) = h$ is admissible in that sense, and for $z = h = \frac{1}{4}$ one has $y = f(0) + Vz = \frac{1}{4}$ while $f(0+h) = \frac{3}{8}$. The node was amended to state the covering for an arbitrary admissible g but the translation *only* for the specific g of the normalisation, and a further node makes that specialisation explicit. The registry contract itself was not amended.

Role in the chain. The registry shard lists no `deps` and no `defs`: this is a leaf of the knowledge DAG, an item of standard Banach-space calculus re-established here at this document's rigour bar rather than cited, because no matching local source sits in `refs/`. Its one registered registry consumer is `lem-stage1-exact-unit-rectification` (§40), whose workspace imports this validated contract as a byte-matched external — see the caveat recorded there about how that import is actually used. Both results belong to the Stage-1 front of the FUDW decomposition, which is otherwise not reproduced in this document. Nothing here bears on `op-classical`, which is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

40 Dimension-free exact-unit rectification

Lemma 90 (`lem-stage1-exact-unit-rectification`). *There are universal constants $C_{\mathrm{unit}} < \infty$ and $\varepsilon_{\mathrm{unit}} > 0$ such that every finite-dimensional ε_X-C^*-algebra $\mathcal{A}$ (`def-epsilon-cstar-algebra`) with $\varepsilon_X \le \varepsilon_{\mathrm{unit}}$ admits, on the same involutive normed space, an exact unit J and a bilinear product $\boldsymbol{\cdot}$ with*

$$\|J - I_X\| \;\le\; C_{\mathrm{unit}}\,\varepsilon_X, \qquad \|x \boldsymbol{\cdot} y - xy\| \;\le\; C_{\mathrm{unit}}\,\varepsilon_X\|x\|\|y\| \quad (x, y \in \mathcal{A}). \tag{119}$$

Transcription note. *"On the same involutive normed space" is a real restriction and not a formality: the underlying vector space, its norm and its involution are untouched, and only the multiplication and the distinguished unit are replaced. "Exact unit" has exactly the meaning of Definition 22; no alternative definition is introduced here. I_X denotes the given approximate unit, and the threshold written $\varepsilon_{\mathrm{unit}}$ here is the contract's* `e_unit` *— a ceiling on ε_X alone, not on a combined defect. What the statement asserts about $\boldsymbol{\cdot}$ is exactly* (119) *together with the exactness of J:* no *clause claims that $(\mathcal{A}, \boldsymbol{\cdot}, J)$ is again an ε'-C^*-algebra, and the validated tree estimates neither the associativity defect nor the product-norm constant of $\boldsymbol{\cdot}$.*

Registry contract (verbatim).

```
Dimension-free exact-unit rectification: there are universal C_unit < infinity and e_unit > 0 such
that every finite-dimensional epsilon_X-C*-algebra with epsilon_X <= e_unit admits on the same
involutive normed space an exact unit J and product bold-dot with ||J-I_X|| <= C_unit*epsilon_X and
||x bold-dot y-xy|| <= C_unit*epsilon_X*||x||||y||.
```

Proof (prose account of the af-validated tree). The construction is explicit and the constants are exhibited: the tree closes with the universal witnesses

$$\varepsilon_{\mathrm{unit}} = \tfrac{1}{2}, \qquad C_{\mathrm{unit}} = 8. \tag{120}$$

It proceeds in four steps — rescale the unit, produce a Hermitian norming functional, correct the product by three terms of rank one in each argument, and estimate.

A norm-one self-adjoint unit. Write $I = I_X$, $a = \|I\|$ and $J = I/a$. The approximate-unit clause of `def-epsilon-cstar-algebra` gives $|a - 1| \le \varepsilon_X$, so $\varepsilon_X \le \frac{1}{2}$ forces $a \ge \frac{1}{2} > 0$ and J is defined. Then $\|J\| = 1$; $J^\dagger = J$ because $I^\dagger = I$ and a is a positive real; and

$$\|J - I\| = |1 - a| \le \varepsilon_X. \tag{121}$$

This is the only place the self-adjointness of J is produced, and the next step consumes it.

A Hermitian norming functional at J. On the one-dimensional subspace $\mathbb{C}J$ define $g(\alpha J) = \alpha$; since $\|\alpha J\| = |\alpha|$ this has norm one, and complex Hahn–Banach extends it to a complex-linear $\psi : \mathcal{A} \to \mathbb{C}$ with $\|\psi\| = 1$ and $\psi(J) = 1$. Put $\psi^\sharp(x) = \overline{\psi(x^\dagger)}$. Because the involution is conjugate-linear, isometric and bijective, $\psi^\sharp$ is again complex-linear of norm one, and $J^\dagger = J$ gives $\psi^\sharp(J) = \overline{\psi(J^\dagger)} = \overline{\psi(J)} = 1$. Hence $\varphi = \frac{1}{2}(\psi + \psi^\sharp)$ is complex-linear with $\|\varphi\| \le 1$, $\varphi(J) = 1$ — so in fact $\|\varphi\| = 1$, since $1 = |\varphi(J)| \le \|\varphi\|\|J\|$ — and is Hermitian:

$$\varphi(x^\dagger) = \overline{\varphi(x)}, \qquad \varphi(J) = 1, \qquad \|\varphi\| = 1. \tag{122}$$

The corrected product. Define, bilinearly,

$$x \boldsymbol{\cdot} y \;:=\; xy + \varphi(x)\big(y - Jy\big) + \varphi(y)\big(x - xJ\big) - \varphi(x)\varphi(y)\big(J - JJ\big). \tag{123}$$

Substituting $x = J$ and using $\varphi(J) = 1$, the first correction term cancels Jy against y and the last two terms cancel each other, giving $J \cdot y = y$; symmetrically $x \cdot J = x$. With $\|J\| = 1$ and $J^\dagger = J$ this is exactly the exact-unit condition of `def-epsilon-cstar-algebra`. The involution is unchanged and remains compatible: Hermiticity (122), the axiom $(xy)^\dagger = y^\dagger x^\dagger$, $J^\dagger = J$ and $(JJ)^\dagger = JJ$ give $(x \cdot y)^\dagger = y^\dagger \cdot x^\dagger$ term by term, so the construction really does live on the same involutive normed space.

The estimate. For any z, the product-norm axiom, (121) and the approximate-unit axiom give

$$\|Jz - z\| \;\le\; \|(J-I)z\| + \|Iz - z\| \;\le\; \big((1+\varepsilon_X)\varepsilon_X + \varepsilon_X\big)\|z\| = (2+\varepsilon_X)\varepsilon_X\|z\|,$$

and likewise $\|zJ - z\| \le (2+\varepsilon_X)\varepsilon_X\|z\|$; taking $z = J$ and using $\|J\| = 1$ also gives $\|J - JJ\| \le (2+\varepsilon_X)\varepsilon_X$. Since $\|\varphi\| = 1$, the three correction terms of (123) are each bounded by $(2+\varepsilon_X)\varepsilon_X\|x\|\|y\|$, so

$$\|x \cdot y - xy\| \;\le\; 3(2+\varepsilon_X)\varepsilon_X\|x\|\|y\| \;\le\; \tfrac{15}{2}\,\varepsilon_X\|x\|\|y\| \;\le\; 8\,\varepsilon_X\|x\|\|y\| \qquad \text{for } \varepsilon_X \le \tfrac{1}{2},$$

while (121) gives $\|J - I_X\| \le \varepsilon_X \le 8\varepsilon_X$. That is (119) with the witnesses (120), and it is the root.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-stage1-exact-unit-rectification`: 6 nodes created and 6 `validated`, root `validated`, taint clean; one challenge raised and resolved. The workspace registers one definition import, `def-epsilon-cstar-algebra`, byte-matched to its pinned source, and two external imports naming the validated contract of `lem-stage1-quantitative-inverse-fu` (§39).

Correction of record (proof-level, not contract-level). The Hahn–Banach step drew a *gap* challenge with a counterexample. As first written, the node asserted only that J has norm one and then symmetrised ψ; but $\psi^\sharp(J) = 1$ needs $J^\dagger = J$, and without it the symmetrisation can destroy the value at J. The verifier's example: $\mathcal{A} = \mathbb{C}$ with $z^\dagger = \overline{z}$, $J = i$ and $\psi(z) = -iz$ — a complex-linear functional of norm one with $\psi(J) = 1$ — gives $\psi^\sharp(z) = iz$, hence $\psi^\sharp(J) = -1$ and $\varphi = 0$, so $\varphi(J) = 0$; indeed no Hermitian complex-linear functional takes the value 1 at i. The repair was a child node deriving $J^\dagger = J$ from the rescaling step and using it explicitly, which is why the account above records self-adjointness where it does. The registry contract was not amended.

Two hypotheses that are carried but not used. First, *finite dimension*: no node of the exported tree invokes it — the rescaling, Hahn–Banach, symmetrisation and norm estimates are all dimension-free — so it is carried from the contract, not consumed by the proof. Second, the *dependency*: the registry lists `lem-stage1-quantitative-inverse-function` as a `dep` and the workspace registers it as a byte-matched external, but no node of the exported tree invokes that contract; the rectification here is elementary. The dependency edge is the one fixed by the governing verdict for the Stage-1 package, and it is recorded rather than exercised. Both observations are about the exported tree, not about the contract, which stands as transcribed.

Role in the chain. Registry `defs`: `def-epsilon-cstar-algebra`. Registry `deps`: `lem-stage1-quantitative-in` (§39), which is `proved` with `af: validated` and reproduced in this document. Downstream, the Stage-1 decomposition rows that would consume an exactly unital model are *not* `af`-validated and are **not** reproduced here; in particular nothing in this document assembles this rectification into a statement about almost-idempotent maps. `op-classical` is now **discharged** at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

41 The F0 seam: a stochastic matrix as a unital completely positive map

Route F attacks `op-classical` with the approximate-C^* machinery of the pinned source, and that machinery speaks about unital completely positive maps of a matrix algebra, not about stochastic matrices. The two lemmas of this section are the seam between the two languages: the first says that the classical datum lifts to an admissible input for the machinery, the second says that the lift costs nothing in the one quantity the machinery consumes.

Notation for this section. Fix $n \geq 1$. Write M_n for the complex $n \times n$ matrices, $\mathbb{C}^n = \ell_n^\infty(\mathbb{C})$ for the commutative C^*-algebra of diagonals of M_n, $J : \mathbb{C}^n \to M_n$ for the diagonal inclusion and $D : M_n \to \mathbb{C}^n$, $D([a_{ij}]) = (a_{11}, \ldots, a_{nn})$, for diagonal extraction. The real self-adjoint part of $\mathbb{C}^n$ is written $\ell^\infty(n)$; a row-stochastic Q (`def-stochastic`, Definition 3) acts there, and $Q_{\mathbb{C}}$ denotes its canonical complex-linear extension to $\mathbb{C}^n$. "UCP" is `def-ucp-map` (Definition 45) throughout, and $\|\cdot\|_{\mathrm{cb}}$ is the completely bounded norm, the supremum over matrix levels $r \geq 1$ of the norms of the amplifications $T_r = \mathrm{id}_{M_r} \otimes T$.

Lemma 91 (`lem-routef-f0-ucp-lift`). *With the notation above, let Q be row-stochastic. Then*

$$\Phi \;:=\; J\, Q_{\mathbb{C}}\, D \;:\; M_n \longrightarrow M_n \tag{124}$$

is a unital completely positive map. Transcription note. *The typing is the content: D lands in the* complex *diagonal algebra $\mathbb{C}^n$, and it is $Q_{\mathbb{C}}$, not Q, that acts there. No smallness hypothesis on Q is assumed, and nothing is claimed about Φ beyond unitality and complete positivity.*

Registry contract (verbatim).

```
Route F F0 UCP lift: let n >= 1, let D: M_n -> C^n be diagonal extraction onto the complex diagonal
algebra C^n = l_inf^n(C) and J: C^n -> M_n the diagonal inclusion, let Q: l_inf^n -> l_inf^n be
row-stochastic, and let Q_C: C^n -> C^n be the canonical complex-linear extension of Q; then Phi :=
J Q_C D: M_n -> M_n is a unital completely positive map.
```

Proof (prose account of the af-validated tree). The tree factors the claim through the three composed maps and then closes by composition.

The middle map. By `def-stochastic` the matrix Q has nonnegative entries q_{ij} and satisfies $Q\mathbf{1} = \mathbf{1}$. If $x = (x_j)$ is positive in $\mathbb{C}^n$ then every x_j is a nonnegative real, so $(Q_{\mathbb{C}}x)_i = \sum_j q_{ij}x_j \geq 0$; and $Q_{\mathbb{C}}(\mathbf{1}) = \mathbf{1}$. Thus $Q_{\mathbb{C}}$ is positive and unital, and since its domain $\mathbb{C}^n$ is a commutative C^*-algebra it is automatically completely positive (`def-ucp-map`); hence UCP.

The two diagonal maps. D is unital, and for $r \geq 1$ and a positive block matrix $X = [X_{ab}] \in M_r(M_n)$ the amplification D_r returns the n-tuple of principal compressions $\big[(X_{ab})_{ii}\big]_{a,b=1}^r$, $i = 1, \ldots, n$, each of which is positive in M_r; so D is CP, hence UCP. Dually J is unital, and for $r \geq 1$ a positive element $(A_1, \ldots, A_n)$ of $M_r(\mathbb{C}^n) = \bigoplus_{i=1}^n M_r$ is sent by J_r, after the canonical permutation of tensor factors, to the positive block-diagonal matrix $\mathrm{diag}(A_1, \ldots, A_n) \in M_{rn}$; so J is CP, hence UCP.

Composition. Compositions of UCP maps are UCP (`def-ucp-map`), and (124) is such a composition.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f0-ucp-lift`: root and 9/9 nodes `validated`, taint clean, one challenge raised and resolved. Two definition imports are registered, `def-stochastic` and `def-ucp-map`; there are no external imports.

Correction of record (contract-level). The single mid-run challenge was a typing defect: the contract as first seeded routed D into the *real* space $\ell^\infty(n)$, while the diagonal of a complex matrix algebra lies in $\mathbb{C}^n$. The repair was an amendment of the contract to the text quoted above — the diagonal algebra made complex and Q replaced by $Q_\mathbb{C}$ — after which the tree was verifier-accepted. The same defect family is repaired in Lemma 96 below, where it was the cause of a first, abandoned elevation run.

Lemma 92 (`lem-routef-f0-defect-identity`)**.** *With the notation above and $\Phi = J\,Q_\mathbb{C}\,D$ as in* (124)*,*

$$\|\Phi^2 - \Phi\|_{\mathrm{cb}} \;=\; \|Q^2 - Q\|_{\infty\to\infty}. \tag{125}$$

Transcription note. *This is an* equality*, not a two-sided estimate with constants: the seam neither loses nor gains defect, and no $\eta \mapsto \varepsilon$ conversion happens here. The left side is a completely bounded norm on M_n, the right side the ℓ^∞ map norm of* `def-almost-idempotent` *(Definition 4); the identity is what licenses reading one as the other.*

Registry contract (verbatim).

```
Route F F0 defect identity: let n >= 1, let D: M_n -> C^n be diagonal extraction onto the complex
diagonal algebra C^n = l_inf^n(C) and J: C^n -> M_n the diagonal inclusion, let Q: l_inf^n ->
l_inf^n be row-stochastic with canonical complex-linear extension Q_C: C^n -> C^n, and put Phi := J
Q_C D; then ||Phi^2 - Phi||_cb = ||Q^2 - Q||_{infinity->infinity}.
```

Proof (prose account of the af-validated tree). Put $L := Q_\mathbb{C}^2 - Q_\mathbb{C}$. The proof is an algebraic reduction, then two matched inequalities at every matrix level, then a single scalar computation.

Reduction. For $x \in \mathbb{C}^n$ the matrix $J(x)$ is diagonal with diagonal x, and D extracts that diagonal, so $DJ = I_{\mathbb{C}^n}$. Hence

$$\Phi^2 - \Phi \;=\; J\,Q_\mathbb{C}(DJ)Q_\mathbb{C}\,D - J\,Q_\mathbb{C}\,D \;=\; J\big(Q_\mathbb{C}^2 - Q_\mathbb{C}\big)D \;=\; JLD. \tag{126}$$

The matrix levels. Fix $r \ge 1$ and identify $M_r(\mathbb{C}^n) = \bigoplus_{j=1}^n M_r$ with the norm $\max_j\|X_j\|$. After the canonical permutation of tensor factors $J_r(X) = \mathrm{diag}(X_1,\dots,X_n)$, so $\|J_rX\| = \max_j\|X_j\|$ and J_r is isometric; D_r returns the tuple of diagonal $n\times n$ blocks, each a compression, so D_r is contractive; and $D_rJ_r = I$. Submultiplicativity therefore gives $\|J_rL_rD_r\| \le \|L_r\|$, while testing on inputs of the form J_rX gives $\|J_rL_rD_r(J_rX)\| = \|J_rL_rX\| = \|L_rX\|$ and hence $\|J_rL_rD_r\| \ge \|L_r\|$. So $\|(\Phi^2-\Phi)_r\| = \|L_r\|$ for every r, by (126).

The scalar computation. Let (l_{ij}) be the coefficient matrix of L, which is the real matrix of $Q^2 - Q$. For $X = (X_j) \in M_r(\mathbb{C}^n)$ one has $(L_rX)_i = \sum_j l_{ij}X_j$, so $\|L_rX\| \le \big(\max_i\sum_j|l_{ij}|\big)\max_j\|X_j\|$. Conversely pick i_0 attaining the maximal row sum and take $X_j = \mathrm{sgn}(l_{i_0j})I_r$ (with $\mathrm{sgn}\,0 = 0$): then $\max_j\|X_j\| \le 1$ and the i_0-component of L_rX is $\big(\sum_j|l_{i_0j}|\big)I_r$. Hence $\|L_r\| = \max_i\sum_j|l_{ij}|$ for *every* r, and that row maximum is $\|Q^2-Q\|_{\infty\to\infty}$. Taking the supremum over r, which is the definition of the cb norm, gives (125).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f0-defect-identity`: root and 12/12 nodes `validated`, taint clean, *zero* challenges — the fresh verifiers confirmed both directions independently, including the zero-row edge case and the matrix-level supremum

identity. Three definition imports are registered, `def-stochastic`, `def-almost-idempotent` and `def-ucp-map`; there are no external imports, and in particular this proof does not use Lemma 91.
Role in the chain. Neither row has registry `deps`: both are leaves. Together they are the entry point of Route F — a row-stochastic Q with defect η in the sense of `def-almost-idempotent` becomes a UCP Φ with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \leq \eta$, which is exactly the input the ledger of §42 consumes. The registry records their consumer as the strengthened `lem-routef-k-ledger` replacement, now `proved` with `af: validated` together with its three helper projections (**not** reproduced here) — so the seam feeds a fully validated chain, and `op-classical` is now **discharged** at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

42 The approximate-algebra ledger for an almost-idempotent UCP map

The seam of §41 delivers a unital completely positive map Φ on a matrix algebra whose square is close to itself. This section reproduces the three `af`-validated rows that turn that datum into an *algebra*: an exact idempotent $\widetilde{\Phi}$ obtained by functional calculus, its image equipped with a corrected product, and the bookkeeping that shows the resulting structure is an extended ε-C^*-algebra whose defect is linear in η.

Notation for this section. $\mathcal{H}$ is a nonzero Hilbert space and $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ is UCP (`def-ucp-map`, Definition 45). All norms written $\|\cdot\|_{\mathrm{cb}}$ are completely bounded norms; I is the identity map, and products of maps are compositions, so that $CB(B(\mathcal{H}))$ is a unital Banach algebra. At amplification level n we write $T = 1_{M_n} \otimes \Phi$ and $D = T^2 - T$. Finally

$$r \;=\; \tfrac{3}{2}\big((1-4\eta)^{-1/2} - 1\big), \qquad M \;=\; 1 + r. \tag{127}$$

Lemma 93 (`lem-kitaev-almost-idemp-audit`). *Let $\mathcal{H} \neq 0$ and let $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ be UCP with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta < \frac{1}{4}$. Put*

$$\widetilde{\Phi} \;=\; \tfrac{1}{2}\Big(I + (2\Phi - I)\big(I - 4(\Phi - \Phi^2)\big)^{-1/2}\Big), \qquad \mathcal{A} = \mathrm{Im}(\widetilde{\Phi}), \qquad X \star Y = \widetilde{\Phi}(XY). \tag{128}$$

Then $\widetilde{\Phi}^2 = \widetilde{\Phi}$; both amplified associativity identities Φ_{assoc1} and Φ_{assoc2}, namely

$$\begin{aligned} \big\|T\big(T(T(X)T(Y))T(Z)\big) - T\big(T(X)T(Y)T(Z)\big)\big\| \;&\le\; 10\eta\|X\|\|Y\|\|Z\|, \\ \big\|T\big(T(X)T(T(Y)T(Z))\big) - T\big(T(X)T(Y)T(Z)\big)\big\| \;&\le\; 10\eta\|X\|\|Y\|\|Z\|, \end{aligned} \tag{129}$$

hold at every amplification after the local source type and index corrections recorded below; and for sufficiently small universal η the inherited operator-space norms, involution and unit make $\mathcal{A}$ an extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$-algebra (`def-extended-epsilon-cstar-algebra`, Definition 23) with

$$\varepsilon_{\mathrm{AI}}(\eta) \;=\; \max\big\{\, r,\; 20\eta + 2(M^5 - 1),\; 3r - r^2 \,\big\} \;=\; O(\eta), \tag{130}$$

r, M as in (127)*, dimension-free at every amplification.* Transcription note. *This row is an* audit*: it registers the corrected statement of the pinned source's `th_almost_idemp`, together with the explicit interface* (130)*. It claims nothing about the source's `th_main_ext`, and no optimality of the displayed constants is asserted. The threshold "sufficiently small universal η" is not made explicit by the contract and is not made explicit here.*

Registry contract (verbatim).

```
Corrected th_almost_idemp audit: let H be nonzero and let Phi:B(H)->B(H) be UCP with
||Phi^2-Phi||_cb <= eta < 1/4; for tilde-Phi=(1/2)(I+(2Phi-I)(I-4(Phi-Phi^2))^(-1/2)),
A=Im(tilde-Phi), and X star Y=tilde-Phi(XY), one has tilde-Phi^2=tilde-Phi, both amplified
associativity identities Phi_assoc1 and Phi_assoc2 have error at most 10*eta*||X||||Y||||Z|| after
the local source type/index corrections, and for sufficiently small universal eta the inherited
operator-space norms, involution, and unit make A an extended epsilon_AI(eta)-C*-algebra with
epsilon_AI=max{r,20eta+2(M^5-1),3r-r^2}=O(eta), r=(3/2)((1-4eta)^(-1/2)-1), M=1+r, dimension-free
at every amplification.
```

Proof (prose account of the af-validated tree). The tree has four branches: a spectral package, the first associativity estimate, its adjoint twin, and the packaging into the axioms.

The spectral package. In $CB(B(\mathcal{H}))$ set $F = \Phi - \Phi^2$ and $A_0 = 2\Phi - I$. Both are polynomials in Φ, so they commute, and $A_0^2 = I - 4F$ while $\|4F\|_{\mathrm{cb}} \le 4\eta < 1$. The binomial series $\mathcal{R} = \sum_{k\ge 0}\binom{2k}{k}F^k$ therefore converges in norm to $(I-4F)^{-1/2}$ and commutes with A_0; hence $S = A_0\mathcal{R}$ satisfies $S^2 = I$ and $\widetilde{\Phi} = (I+S)/2$ is idempotent, which is the first clause. Since $\widetilde{\Phi} - \Phi = \frac{1}{2}A_0(\mathcal{R} - I)$, $\|A_0\|_{\mathrm{cb}} \le 2\|\Phi\|_{\mathrm{cb}} + 1 \le 3$ and the series coefficients are nonnegative, $\|\mathcal{R} - I\|_{\mathrm{cb}} \le (1-4\eta)^{-1/2} - 1$, so $\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \le r$. Unitality and $*$-preservation of Φ pass through the real-coefficient series, so $\widetilde{\Phi}(I) = I$ and $\widetilde{\Phi}(X^\dagger) = \widetilde{\Phi}(X)^\dagger$; a bounded idempotent has $\mathrm{Im} = \mathrm{Ker}(I - \widetilde{\Phi})$, so $\mathcal{A}$ is closed, complete, self-adjoint and contains the unit, and every formula commutes with amplification.

The associativity estimate. This is the delicate branch, and the tree proves it without importing a Stinespring theorem. It first constructs, from complete positivity alone, an iterated GNS dilation: Hilbert spaces K_j, isometries $V_j : K_{j-1} \to K_j$ and contractive unital $*$-homomorphisms $u_j : B(K_{j-1}) \to B(K_j)$ with $T(H) = V_1^\dagger u_1(H) V_1$ and the one-step compression identity $u_j(V_j^\dagger U V_j) = V_{j+1}^\dagger u_{j+1}(U) V_{j+1}$ for $U \in B(K_j)$. Writing $u_{b,a} = u_b \circ \cdots \circ u_{a+1}$, $P_1 = V_1V_1^\dagger$ and $E_X = (I_1 - P_1)u_1(T(X))$, the elementary defect ledger $\|T\| \le 1$, $\|D\| \le \eta$ gives $\|T^2(U) - T(U)\| \le \eta\|U\|$ and $\|T^2(U)T^2(V) - T(U)T(V)\| \le 2\eta\|U\|\|V\|$, whence $G_X = T^2(T(X^\dagger)T(X)) - T(T^2(X^\dagger)T^2(X))$ obeys $\|G_X\| \le 3\eta\|X\|^2$. The "rectangles" $R_k(X) = u_{k+2,1}(E_X)V_{k+2}V_{k+1}$ then satisfy the exact identity $R_k(X)^\dagger R_k(X) = u_{k,0}(G_X)$, so $\|R_k(X)\| \le \sqrt{3\eta}\|X\|$. With $A = T(X)$, $B = T(Y)$, $C = T(Z)$ and $A_1 = T^2(X)$, $B_1 = T^2(Y)$, $C_1 = T^2(Z)$, the operator $W = V_1^\dagger R_1(X^\dagger)^\dagger u_{3,0}(B)V_3R_0(Z)$ expands *exactly* to $W = T(Q)$ with

$$Q = T\big(T(AB)C\big) - T^2(AB)C_1 - T(A_1B_1C) + T(A_1B_1)C_1, \tag{131}$$

and $\|W\| \le 3\eta\, p$ where $p = \|X\|\|Y\|\|Z\|$. Setting $F_1 = T(T(AB)C) - T(ABC)$, cancellation in (131) gives $Q - F_1 = [T(ABC) - T(A_1B_1C)] + [T(A_1B_1) - T^2(AB)]C_1$, and the perturbation bounds $\|A_1B_1 - AB\| \le 2\eta\|X\|\|Y\|$ and $\|T(A_1B_1) - T^2(AB)\| \le 3\eta\|X\|\|Y\|$ give $\|Q - F_1\| \le 5\eta p$. Contractivity of T transports this to $\|T(Q) - T(F_1)\| \le 5\eta p$, and since F_1 is a difference of two T-values the defect estimate applied to each gives $\|T(F_1) - F_1\| \le 2\eta p$. Hence $\|W - F_1\| \le 7\eta p$ and, with $\|W\| \le 3\eta p$, $\|F_1\| \le 10\eta p$: the first line of (129). The second follows by applying the first to $(Z^\dagger, Y^\dagger, X^\dagger)$ and taking adjoints, T being $*$-preserving.

The axioms. At each level write $P = 1_{M_n} \otimes \widetilde{\Phi}$, so $\|P - T\| \le r$ and $\|P\| \le M$. Bilinearity, range-valuedness, the exact unit and $(X \star Y)^\dagger = Y^\dagger \star X^\dagger$ are immediate from idempotence, unitality and $*$-preservation of P; $\|X \star Y\| \le M\|X\|\|Y\|$ is the submultiplicativity error r. For $X \in \mathrm{Im}(P)$, $\|T(X)\| \ge (1-r)\|X\|$, the Kadison–Schwarz inequality gives $\|T(X^\dagger X)\| \ge \|T(X)\|^2$, and hence $\|P(X^\dagger X)\| \ge (1 - 3r + r^2)\|X\|^2$: the C^* lower-norm error $3r - r^2$. Finally, replacing the five occurrences of P in $P(P(P(X)P(Y))P(Z))$ and in $P(P(X)P(P(Y)P(Z)))$ one at a time by T and using $\|T\| \le 1$, $\|P\| \le M$, $\|P - T\| \le r = M - 1$ bounds each telescoping sum by $r(1 + M + M^2 + M^3 + M^4)p = (M^5 - 1)p$; combined with the two estimates (129) this gives $\|(X \star Y) \star Z - X \star (Y \star Z)\| \le (20\eta + 2(M^5 - 1))p$ for $X, Y, Z \in \mathrm{Im}(P)$. The maximum of the three errors is (130), every input estimate is amplification-uniform, and $M^5 - 1 \le 31r$ for $0 \le r \le 1$ with $r = O(\eta)$, so $\varepsilon_{\mathrm{AI}} \to 0$ and a universal threshold makes it less than one.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-kitaev-almost-idemp-audit`; the registered definition import is `def-extended-epsilon-cstar-algebra`, and the dilation is built inside the tree rather than imported, which is why no Stinespring external appears.

Corrections of record (to the local source, carried by the contract). The audit's corrections are type- and index-level and are named in the contract itself: the operator variables at the source loci belong to the operator algebras $B(K_j)$ and not to the Hilbert spaces K_j, and the final isometry of the auxiliary compression identity is V_{1+k}, not V_1 — with V_1 the identity is ill-typed for $k > 0$, since V_1 has domain K_0. The canonical Stinespring construction remains a source dependency of the original argument; here it is replaced by the in-tree recursive dilation.

Lemma 94 (`lem-routef-functional-calculus-closeness`). *For $0 \le \eta \le \frac{1}{8}$ the exact functional-calculus projector satisfies*

$$\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \;\le\; C_\theta \eta, \qquad C_\theta = 12(\sqrt{2} - 1). \tag{132}$$

Transcription note. *The contract is elliptic by design: "the exact functional-calculus projector" is $\widetilde{\Phi}$ of* (128)*, and the ambient hypotheses — Φ UCP on $B(\mathcal{H})$ with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$ — are those of the imported Lemma 93, which is how the validated tree reads them. The statement is a* linear *bound valid on the stated window only; it says nothing for $\eta > \frac{1}{8}$.*

Registry contract (verbatim).

```
Functional-calculus closeness: for 0 <= eta <= 1/8, the exact functional-calculus projector
satisfies ||tilde-Phi-Phi||_cb <= C_theta*eta, where C_theta=12*(sqrt(2)-1).
```

Proof (prose account of the af-validated tree). The tree proves the estimate $\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \le r$ of (127) in its own right and then linearises r. For the first half it records that a UCP map is completely contractive — the matrix Kadison–Schwarz inequality $\Phi_m(X)^\dagger \Phi_m(X) \le \Phi_m(X^\dagger X)$ obtained from positivity of $[X^\dagger; I][X, I]$, combined with $X^\dagger X \le \|X\|^2 I$ — so $\|2\Phi - I\|_{\mathrm{cb}} \le 3$; then the algebraic identity $\widetilde{\Phi} - \Phi = \frac{1}{2}(2\Phi - I)\big((I - 4\Delta)^{-1/2} - I\big)$ with $\Delta = \Phi - \Phi^2$; then, since $\|4\Delta\|_{\mathrm{cb}} \le 4\eta \le \frac{1}{2} < 1$, the absolutely convergent binomial series with nonnegative coefficients gives $\|(I - 4\Delta)^{-1/2} - I\|_{\mathrm{cb}} \le (1 - 4\eta)^{-1/2} - 1$. For the second half, put $t = 4\eta \in [0, \frac{1}{2}]$ and $s = \sqrt{1 - t} \ge 1/\sqrt{2}$; rationalisation gives

$$\frac{(1 - t)^{-1/2} - 1}{t} = \frac{1}{s(1 + s)} \;\le\; \frac{1}{\frac{1}{\sqrt{2}}(1 + \frac{1}{\sqrt{2}})} = 2(\sqrt{2} - 1),$$

the case $t = 0$ being an equality in the inequality $(1 - t)^{-1/2} - 1 \le 2(\sqrt{2} - 1)t$ itself. Multiplying by t and then by $\frac{3}{2}$ turns $r \le \frac{3}{2} \cdot 2(\sqrt{2} - 1) \cdot 4\eta$ into (132).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-functional-calculus-closeness`; registry `defs`: `def-almost-idempotent`; registry `deps`: `lem-kitaev-almost-idemp-audit` (Lemma 93), from which the projector formula and the ambient hypotheses are imported.

Lemma 95 (`lem-routef-ai-defect-linearization`). *Set $C_\theta = 12(\sqrt{2} - 1)$. There are universal constants $C_A < \infty$ and $\eta_A > 0$, with $C_A = 20 + \frac{211}{8} C_\theta$, such that for every nonzero Hilbert space $\mathcal{H}$, every UCP map $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ and every η with $0 \le \eta \le \eta_A$ and $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$, the data $\widetilde{\Phi}, \mathcal{A}, \star$ of* (128) *together with r of* (127) *and $\varepsilon_{\mathrm{AI}}(\eta)$ of* (130) *make $\mathcal{A}$, with its inherited operator-space norms, involution and unit, an extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$-algebra with*

$$\varepsilon_{\mathrm{AI}}(\eta) \;\le\; C_A \eta. \tag{133}$$

Transcription note. *What this row adds to Lemma 93 is the* linearity (133) *and an explicit universal constant; the threshold η_A is asserted to exist and to be universal, and is not exhibited, because it inherits the unexhibited threshold of that lemma.*

Registry contract (verbatim).

```
Approximate-algebra defect linearization: set C_theta=12*(sqrt(2)-1). There are universal C_A <
infinity and eta_A > 0, with C_A=20+(211/8)*C_theta, such that for every nonzero Hilbert space H,
every UCP map Phi:B(H)->B(H), and every eta satisfying 0 <= eta <= eta_A and ||Phi^2-Phi||_cb <=
eta, if tilde-Phi=(1/2)*(I+(2*Phi-I)*(I-4*(Phi-Phi^2))^(-1/2)), A=Im(tilde-Phi), X star
Y=tilde-Phi(XY), r=(3/2)*((1-4*eta)^(-1/2)-1), and
epsilon_AI(eta)=max{r,20*eta+2*((1+r)^5-1),3*r-r^2}, then the inherited operator-space norms,
involution, and unit together with star make A an extended epsilon_AI(eta)-C*-algebra and
epsilon_AI(eta) <= C_A*eta.
```

Proof (prose account of the af-validated tree). Let $\eta_* > 0$ be the universal threshold of Lemma 93, shrunk below $\frac{1}{4}$, and put $\eta_A = \min\{\eta_*, \frac{7}{64}\}$, which is universal and positive. For $0 \le \eta \le \eta_A$ that lemma supplies the extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$-algebra structure and the formula (130), so only the three entries of the maximum have to be bounded. On $0 \le \eta \le \frac{7}{64}$ one has, with $t = \sqrt{1-4\eta} \ge 1/\sqrt{2}$, that $r/\eta = 6/(t(1+t)) \le C_\theta$, hence $0 \le r \le C_\theta \eta$ — the constant being exactly the one fixed by Lemma 94; also $1 - 4\eta \ge \frac{9}{16}$ gives $r \le \frac{1}{2}$. For $0 \le r \le \frac{1}{2}$ the binomial identity $2((1+r)^5 - 1) = 2r(5 + 10r + 10r^2 + 5r^3 + r^4)$ and monotonicity of the parenthesis give $2((1+r)^5 - 1) \le \frac{211}{8} r$. Therefore

$$20\eta + 2\big((1+r)^5 - 1\big) \;\le\; 20\eta + \tfrac{211}{8} C_\theta \eta \;=\; C_A \eta,$$

while $C_A - C_\theta = 20 + \frac{203}{8} C_\theta > 0$ and $C_A - 3C_\theta = 20 + \frac{187}{8} C_\theta > 0$ give $r \le C_A \eta$ and $3r - r^2 \le 3r \le C_A \eta$. Every entry of the maximum is at most $C_A \eta$, which is (133).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-ai-defect-linearization`: a 13-node tree, taint clean.

Correction of record (contract-level, banking-time). The prover expanded the seeded one-line contract into the self-contained root statement reproduced above, inlining the value of C_θ and the formula (128) from its imported dependencies; the fresh verifiers validated *that* statement, so the registry contract was mechanically replaced by the validated root verbatim at banking. No equivalence judgement was exercised by the orchestrator; the linker's contract-match refusal is what forced the reconciliation.

Role in the chain. Registry `defs`: `def-almost-idempotent`, `def-extended-epsilon-cstar-algebra`. Registry `deps`: `lem-kitaev-almost-idemp-audit` and `lem-routef-functional-calculus-closeness`, both reproduced above and both `af`-validated. The consumer of this ledger is the strengthened factorization row `lem-routef-k-ledger`, now `proved` with `af: validated` together with its three helper projections (**not** reproduced here; §76). Through that consumer and the F0 assembly, `op-classical` is now **discharged** at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

43 The F2/F3 block: manufacturing the positive unital retract pair

The finish edge of §38 consumes a pair of positive unital maps A, M between real ℓ^∞ spaces, subject to two estimates. This section reproduces the two `af`-validated rows that manufacture that pair, and the estimates, out of an approximate factorization of the F0 lift Φ of §41 through a finite-dimensional unital C^*-algebra.

Notation for this section. The notation of §41 is in force: M_n, the complex diagonal algebra $\mathbb{C}^n$, its real self-adjoint part $\ell^\infty(n)$, the maps $J, D, Q_\mathbb{C}$ and $\Phi = JQ_\mathbb{C}D$. The intermediate algebra of the contract is written $\mathcal{B}$, and M denotes a map, not a matrix algebra. All completely bounded norms are written $\|\cdot\|_{\mathrm{cb}}$, and $K \geq 1$ is a dimension-independent constant supplied with the hypotheses, not fixed here.

Lemma 96 (`lem-routef-f2-positive-unital-compression`). *Let $K \geq 1$ be dimension-independent, $n \geq 1$, $Q : \ell^\infty(n) \to \ell^\infty(n)$ row-stochastic, $\Phi = JQ_\mathbb{C}D$, let $\mathcal{B}$ be a finite-dimensional unital C^*-algebra and let $\Delta : \mathcal{B} \to M_n$ and $\Upsilon : M_n \to \mathcal{B}$ be UCP. If*

$$0 \;\leq\; \eta \;\leq\; \min\bigl\{(24K)^{-1},\, 1\bigr\} \tag{134}$$

and

$$\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}} \leq K\eta, \quad \|\Upsilon\Delta - I_\mathcal{B}\|_{\mathrm{cb}} \leq K\eta, \quad \|\Upsilon(\Delta x\, \Delta y) - xy\| \leq K\eta\|x\|\|y\| \;\; (x, y \in \mathcal{B}), \tag{135}$$

then $\mathcal{B}$ is commutative, and there are $k \geq 1$ and a unital $*$-isomorphism $\iota_\mathbb{C} : \mathbb{C}^k \to \mathcal{B}$ such that $D\Delta\iota_\mathbb{C}$ maps $\mathbb{R}^k$ into $\mathbb{R}^n$, $\iota_\mathbb{C}^{-1}\Upsilon J$ maps $\mathbb{R}^n$ into $\mathbb{R}^k$, and the resulting restrictions and corestrictions

$$A := (D\Delta\iota_\mathbb{C})|_{\mathbb{R}^k} : \ell^\infty(k) \to \ell^\infty(n), \qquad M := (\iota_\mathbb{C}^{-1}\Upsilon J)|_{\mathbb{R}^n} : \ell^\infty(n) \to \ell^\infty(k) \tag{136}$$

are positive unital maps satisfying

$$\|Q - AM\|_{\infty\to\infty} \leq K\eta, \qquad \|QA - A\|_{\infty\to\infty} \leq 2K\eta, \qquad \|Ax\|_\infty \geq (1 - 3K\eta)\|x\|_\infty \;\; (x \in \ell^\infty(k)). \tag{137}$$

Transcription note. *The three displays of* (135) *are* hypotheses*: the factorization datum $(\mathcal{B}, \Delta, \Upsilon)$ is given, and nothing here, or anywhere in this document, manufactures it from Q. Commutativity of $\mathcal{B}$ is* derived*, not assumed. The real/complex typing is load-bearing: Δ, Υ and $\iota_\mathbb{C}$ live over $\mathbb{C}$, and A, M are their restrictions and corestrictions to the real self-adjoint parts.*

Registry contract (verbatim).

```
Route F F2 positive-unital compression: let K >= 1 be a dimension-independent constant, n >= 1, Q:
l_inf^n -> l_inf^n row-stochastic, D: M_n -> C^n diagonal extraction onto the complex diagonal
algebra C^n = l_inf^n(C), J: C^n -> M_n diagonal inclusion, Q_C: C^n -> C^n the canonical
complex-linear extension of Q, and Phi = J Q_C D, B a finite-dimensional unital C*-algebra, and
Delta: B -> M_n, Upsilon: M_n -> B UCP maps; if 0 <= eta <= min{(24K)^{-1},1}, ||Delta Upsilon -
Phi||_cb <= K*eta, ||Upsilon Delta - I_B||_cb <= K*eta, and ||Upsilon(Delta x Delta y) - xy|| <=
K*eta*||x||*||y|| for all x,y in B, then B is commutative and there are k >= 1 and a unital
*-isomorphism iota_C: C^k = l_inf^k(C) -> B such that D Delta iota_C maps R^k into R^n, iota_C^{-1}
Upsilon J maps R^n into R^k, and the resulting restrictions and corestrictions A := (D Delta
iota_C)|_{R^k}: l_inf^k -> l_inf^n and M := (iota_C^{-1} Upsilon J)|_{R^n}: l_inf^n -> l_inf^k are
positive unital maps satisfying ||Q - AM||_{inf->inf} <= K*eta, ||QA - A||_{inf->inf} <= 2K*eta,
and ||Ax||_inf >= (1-3K*eta)*||x||_inf for every x in l_inf^k.
```

Proof (prose account of the af-validated tree). The tree runs in three movements: force commutativity, coordinatise, estimate.

Preliminaries. One shared node derives complete contractivity of a UCP map from 2-positivity: for $S = \mathrm{id}_{M_r} \otimes T$ the block matrix $[a, 1]^\dagger [a, 1]$ is positive, so $S(a)^\dagger S(a) \le S(a^\dagger a) \le \|a\|^2$, whence $\|T\|_{\mathrm{cb}} \le 1$; in particular $\|\Delta\|_{\mathrm{cb}}, \|\Upsilon\|_{\mathrm{cb}} \le 1$. A second node records the typing facts of the seam: Φ is UCP hence contractive, its range lies in $J(\mathbb{C}^n)$ so any two of its values commute, $DJ = I$, J is isometric, and every positive map is $*$-preserving.

Approximate invariance and approximate commutativity. For $b \in \mathcal{B}$,

$$\Delta b - \Phi\Delta b = \Delta\big(b - \Upsilon\Delta b\big) + (\Delta\Upsilon - \Phi)(\Delta b), \qquad \text{so} \qquad \|\Delta b - \Phi(\Delta b)\| \le 2K\eta\|b\|.$$

Put $a = \Delta x$, $b = \Delta y$ and $a_0 = \Phi a$, $b_0 = \Phi b$. Then $\|a - a_0\| \le 2K\eta\|x\|$, $\|b - b_0\| \le 2K\eta\|y\|$, $\|a\|, \|a_0\| \le \|x\|$, $\|b\|, \|b_0\| \le \|y\|$ and $[a_0, b_0] = 0$, so $\|[a, b]\| \le 8K\eta\|x\|\|y\|$. Feeding this through the third hypothesis of (135) for (x, y) and for (y, x), and using contractivity of Υ,

$$\|xy - yx\| \ \le\ 10K\eta\|x\|\|y\| \qquad (x, y \in \mathcal{B}). \tag{138}$$

The norm-two witness, and commutativity. The tree does *not* cite the finite-dimensional C^*-classification — it is not in the local sources — and proves instead: a noncommutative finite-dimensional unital C^*-algebra carries contractions x, y with $\|xy - yx\| = 2$. Decompose 1 into mutually orthogonal minimal projections $p_1, \ldots, p_m$ (a minimal projection p has $p\mathcal{B}p = \mathbb{C}p$, else a proper spectral projection of a non-scalar self-adjoint element of $p\mathcal{B}p$ would sit below p). If every off-diagonal corner $p_i\mathcal{B}p_j$ vanished, then $b = \sum_{i,j} p_i b p_j = \sum_i \lambda_i(b) p_i$ and $\mathcal{B}$ would be commutative; so pick $0 \ne z \in p\mathcal{B}q$ with $p \ne q$ minimal. Then $z^\dagger z \in q\mathcal{B}q$ and $zz^\dagger \in p\mathcal{B}p$, so the C^*-identity gives $z^\dagger z = \|z\|^2 q$ and $zz^\dagger = \|z\|^2 p$; with $v = z/\|z\|$ one has $v^\dagger v = q$, $vv^\dagger = p$ and $v^2 = (v^\dagger)^2 = 0$. The Pauli pair $x = p - q$, $y = v + v^\dagger$ consists of self-adjoint contractions with $xy - yx = 2(v - v^\dagger)$ of norm 2. Under (134) this contradicts (138), since $2 \le 10K\eta \le \frac{10}{24} = \frac{5}{12} < 2$; hence $\mathcal{B}$ is commutative.

Coordinates. A commutative finite-dimensional unital C^*-algebra carries a projection basis $\{\Pi_1, \ldots, \Pi_k\}$ — this, byte-matched to its pinned source, is the *only* external the tree is permitted (`def-projection-basis`, Definition 30). Then $k \ge 1$, the Π_j are nonzero and span, and $\iota_{\mathbb{C}}(z) = \sum_j z_j \Pi_j$ is a unital $*$-isomorphism $\mathbb{C}^k \to \mathcal{B}$. The compositions $D\Delta\iota_{\mathbb{C}}$ and $\iota_{\mathbb{C}}^{-1}\Upsilon J$ are UCP, hence positive, unital and $*$-preserving, so they carry self-adjoint parts to self-adjoint parts and (136) defines positive unital maps.

The three estimates. For $z \in \mathbb{R}^n$, $DJ = I$ and $\Phi = JQ_{\mathbb{C}}D$ give $Qz = D\Phi Jz$ while $AMz = D\Delta\Upsilon Jz$; contractivity of D and J then turns the first hypothesis of (135) into $\|Q - AM\|_{\infty\to\infty} \le K\eta$. For $x \in \mathbb{R}^k$, $D\Phi = Q_{\mathbb{C}}D$ and the restriction of $Q_{\mathbb{C}}$ to $\mathbb{R}^n$ give $QAx - Ax = D(\Phi\Delta\iota_{\mathbb{C}}x - \Delta\iota_{\mathbb{C}}x)$, so approximate invariance yields $\|QA - A\|_{\infty\to\infty} \le 2K\eta$. Finally $\|\Delta\iota_{\mathbb{C}}x\| \ge \|\Upsilon\Delta\iota_{\mathbb{C}}x\| \ge (1 - K\eta)\|x\|_\infty$ by the second hypothesis, approximate invariance costs a further $2K\eta\|x\|_\infty$, and $\|\Phi\Delta\iota_{\mathbb{C}}x\| = \|JQ_{\mathbb{C}}Ax\| \le \|Ax\|_\infty$; combining gives the lower modulus in (137).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f2-positive-unital-compression`: root and 22/22 nodes `validated`, taint clean; one genuine mid-run challenge, resolved. Registered imports: the definitions `def-stochastic`, `def-ucp-map`, `def-projection-basis`, and the single byte-verbatim projection-basis external.

Corrections of record. Two, at different levels. (i) *Contract-level*: the earlier text typed D into the real space $\ell^\infty(n)$ while the diagonal of M_n is complex — the same defect repaired on both F0 seam rows, and the cause of a first elevation run that was abandoned rather than resumed. The contract quoted above is the corrected text; every hypothesis, constant, threshold and estimate is unchanged by the correction. (ii) *Proof-level*: the challenge was the endpoint $\eta = 0$, where a strict inequality chain used in the commutativity node failed. The repair is the endpoint bridge reproduced implicitly above: at $\eta = 0$, (138) reads $\|xy - yx\| = 0$ and commutativity is immediate, so the conclusion holds on the whole closed window (134).

Lemma 97 (`lem-routef-f3-retract-defect`). *Let $K \geq 1$ be dimension-independent, $n, k \geq 1$, let $A : \ell^\infty(k) \to \ell^\infty(n)$ and $M : \ell^\infty(n) \to \ell^\infty(k)$ be positive unital maps, let $Q : \ell^\infty(n) \to \ell^\infty(n)$ be row-stochastic, and let $\eta \geq 0$ satisfy $3K\eta < 1$. If the three estimates* (137) *hold, then*

$$\|MA - I_k\|_{\infty\to\infty} \;\leq\; \frac{3K\eta}{1 - 3K\eta}. \tag{139}$$

Transcription note. *This row is stated entirely in the real picture; its hypotheses are exactly the three conclusions of Lemma 96, but they are* hypotheses *here, and the row does not import that lemma. The denominator is positive precisely because $3K\eta < 1$.*

Registry contract (verbatim).

```
Route F F3 retract defect: let K >= 1 be a dimension-independent constant, n,k >= 1, A: l_inf^k ->
l_inf^n and M: l_inf^n -> l_inf^k positive unital maps, Q: l_inf^n -> l_inf^n row-stochastic, and
eta >= 0 with 3K*eta < 1; if ||Q - AM||_{inf->inf} <= K*eta, ||QA - A||_{inf->inf} <= 2K*eta, and
||Ax||_inf >= (1-3K*eta)*||x||_inf for every x in l_inf^k, then ||MA - I_k||_{inf->inf} <=
3K*eta/(1-3K*eta).
```

Proof (prose account of the af-validated tree). First, A is contractive: for $x \in \ell^\infty(k)$ and $c = \|x\|_\infty$ one has $-c\mathbf{1}_k \leq x \leq c\mathbf{1}_k$ coordinatewise, positivity of A preserves the order and unitality sends $\mathbf{1}_k$ to $\mathbf{1}_n$, so $-c\mathbf{1}_n \leq Ax \leq c\mathbf{1}_n$ and $\|Ax\|_\infty \leq \|x\|_\infty$. Next, linearity gives the exact identity

$$A\big((MA - I_k)x\big) = (AM - Q)(Ax) + (QA - A)x. \tag{140}$$

Since $\|AM - Q\|_{\infty\to\infty} = \|Q - AM\|_{\infty\to\infty}$, the two operator-norm hypotheses and contractivity of A bound the right-hand side of (140) by $K\eta\|x\|_\infty + 2K\eta\|x\|_\infty = 3K\eta\|x\|_\infty$. Put $\alpha = 1 - 3K\eta > 0$ and apply the lower-modulus hypothesis to $y = (MA - I_k)x$: then $\alpha\|y\|_\infty \leq \|Ay\|_\infty \leq 3K\eta\|x\|_\infty$, i.e. $\|(MA - I_k)x\|_\infty \leq (3K\eta/\alpha)\|x\|_\infty$ for every x. Taking the supremum over $\|x\|_\infty \leq 1$ gives (139).

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f3-retract-defect`: a first-pass elevation, root and 11/11 nodes `validated`, taint clean, *zero* challenges. The registered definition import is `def-stochastic`; there are no external imports, and no dimension or norm loss occurs anywhere in the tree.

Role in the chain, and what is still missing. Neither row has registry `deps`. Their conclusions are, literally, the hypothesis list of the finish edge `lem-routef-prh-finish` (§38): (137) supplies $\|Q - AM\|_{\infty\to\infty} \leq K\eta$ and the input of (139), and (139) is verbatim that lemma's second hypothesis, at the same threshold $\eta \leq (24K)^{-1}$. What is *not* supplied anywhere in this document is the antecedent of Lemma 96 itself: the factorization datum $(\mathcal{B}, \Delta, \Upsilon)$ with the three estimates (135) and a dimension-free K. The registry row that supplies it, the strengthened `lem-routef-k-ledger`, is now `proved` with `af: validated` (**not** reproduced here), as is the assembly carrier `lem-thmainext-conditional`. The Route F chain is therefore closed at the af-validated rung: the unconditional $\sqrt{\eta}$ bound follows, and `op-classical` is now **discharged** via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

44 Rectified algebras and graph charts on the approximate unitaries

This section and the three that follow (§45, §46, §47) reproduce eleven `af`-validated Stage-1 results about the unitary set of an approximate C^*-algebra. Throughout, the ambient object is an *exact-unit* ε_r-C^*-algebra $(\mathcal{X}, J, \boldsymbol{\cdot}, \dagger)$ in the sense of Definition 22, and every set and symbol — $\mathcal{H}$, $i\mathcal{H}$, $\overline{\mathcal{U}}_\delta$, $\mathcal{U}_\delta$, $\mathcal{U}$, L_V, ϕ_V with its components $\phi_V^{\parallel}, \phi_V^{\perp}$, and the reserved letters u, h, μ, σ — is the vocabulary of Definition 32, used here and never restated. None of these results bears on `op-classical`, which is **open**.

Lemma 98 (`lem-stage1-rectified-cstar-control`). *There are universal $C_{\text{rect}} \ge 1$ and $e_{\text{rect}} \in (0, 1/C_{\text{rect}}]$ such that every finite-dimensional ε_X-C^*-algebra with $0 \le \varepsilon_X \le e_{\text{rect}}$ admits, on the same involutive normed space, a bilinear product $\boldsymbol{\cdot}$ and an element $J = J^\dagger$ for which $(\mathcal{X}, J, \boldsymbol{\cdot}, \dagger)$ satisfies* every *exact-unit ε_r-C^*-algebra axiom of Definition 22, including $\|J\| = 1$, where*

$$\varepsilon_r = C_{\text{rect}}\varepsilon_X, \qquad \|J - I_X\| \le C_{\text{rect}}\varepsilon_X, \qquad \|x \boldsymbol{\cdot} y - xy\| \le C_{\text{rect}}\varepsilon_X\|x\|\|y\|. \tag{141}$$

Transcription note. *This is strictly stronger than Lemma 90 (§40), which produces an exact unit and a close product but estimates neither the associativity defect nor the product-norm constant of $\boldsymbol{\cdot}$. Here the whole axiom list is transferred, at the inflated parameter $\varepsilon_r = C_{\text{rect}}\varepsilon_X$; that inflation is the price, and it is what makes the rectified algebra a legitimate input to everything below.*

Registry contract (verbatim).

```
Controlled exact-unit C*-rectification: there are universal C_rect >= 1 and e_rect in (0, 1/C_rect]
such that every finite-dimensional epsilon_X-C*-algebra with 0 <= epsilon_X <= e_rect admits, on
the same involutive normed space, a bilinear product bold-dot and J = J^dagger for which (calX, J,
bold-dot, dagger) satisfies EVERY exact-unit epsilon_r-C*-algebra axiom of
def-epsilon-cstar-algebra, including ||J|| = 1, where epsilon_r = C_rect*epsilon_X, and ||J - I_X||
<= C_rect*epsilon_X, ||x bold-dot y - xy|| <= C_rect*epsilon_X*||x||*||y||.
```

Proof (prose account of the af-validated tree). The witnesses are exhibited: the tree closes with

$$C_{\text{rect}} = 100, \qquad e_{\text{rect}} = \tfrac{1}{100}. \tag{142}$$

A Hermitian norming functional. Write $I = I_X$ and $a = \|I\|$, so $|a - 1| \le \varepsilon_X$ and $a > 0$. A finite-dimensional Hahn–Banach extension produces a complex-linear f with $\|f\| = 1$ and $f(I) = a$; since $\dagger$ is a conjugate-linear isometric involution and $I^\dagger = I$, the reflection $f^\sharp(x) = \overline{f(x^\dagger)}$ has the same norm and the same value at I, so $\varphi = (f + f^\sharp)/(2a)$ satisfies

$$\varphi(I) = 1, \qquad \|\varphi\| \le 1/a, \qquad \varphi(x^\dagger) = \overline{\varphi(x)}. \tag{143}$$

The corrected product. With $d_L(y) = y - Iy$, $d_R(x) = x - xI$ and $h = II - I$, put

$$m_0(x, y) = xy + \varphi(x)d_L(y) + \varphi(y)d_R(x) + \varphi(x)\varphi(y)h, \qquad J = I/a, \qquad x \boldsymbol{\cdot} y = a\, m_0(x, y). \tag{144}$$

Because $\varphi(I) = 1$ and $d_L(I) = d_R(I) = -h$, substitution gives $m_0(I, y) = y$ and $m_0(x, I) = x$, so J is an exact two-sided unit for $\boldsymbol{\cdot}$ with $J^\dagger = J$ and $\|J\| = 1$. Term by term, $d_L(y)^\dagger = d_R(y^\dagger)$, $d_R(x)^\dagger = d_L(x^\dagger)$, $h^\dagger = h$ and the last identity of (143) give $(x \boldsymbol{\cdot} y)^\dagger = y^\dagger \boldsymbol{\cdot} x^\dagger$ *exactly*, using only that a is a positive real.

The estimates and the axiom transfer. The approximate-unit axioms give $\|d_L(y)\| \le \varepsilon_X\|y\|$, $\|d_R(x)\| \le \varepsilon_X\|x\|$ and $\|h\| \le a\varepsilon_X$; with $\|\varphi\| \le 1/a$ this bounds $\|m_0(x,y) - xy\|$ by $3\varepsilon_X\|x\|\|y\|/a$, and after rescaling

$$\Delta(x,y) := x \cdot y - xy, \qquad \|\Delta(x,y)\| \le 5\,\varepsilon_X\|x\|\|y\| \qquad (\varepsilon_X \le \tfrac{1}{100}). \tag{145}$$

Every non-associative axiom now transfers by one triangle inequality: $\|x \cdot y\| \le (1+6\varepsilon_X)\|x\|\|y\|$ and $\|x^\dagger \cdot x\| \ge (1-6\varepsilon_X)\|x\|^2$, both comfortably inside the tolerance $100\varepsilon_X$. For the associator, writing m for the original product and $m' = m + \Delta$, the four-term expansion of $m'(m'(x,y),z) - m'(x,m'(y,z))$ is bounded by $[\varepsilon_X + 2(1+\varepsilon_X)\rho + 2(1+6\varepsilon_X)\rho]\|x\|\|y\|\|z\|$ with $\rho = 5\varepsilon_X$, i.e. by $(21\varepsilon_X + 70\varepsilon_X^2)\|x\|\|y\|\|z\| \le 100\varepsilon_X\|x\|\|y\|\|z\|$. Finally $\|J - I\| = |1-a| \le \varepsilon_X$, so (142) closes the root.

Lemma 99 (`lem-stage1-unitary-graph-control`)**.** *There are universal $C_{\mathrm{ch}} \ge 1$ and $\kappa_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{ch}}$, every $V \in \overline{\mathcal{U}}_\delta$ and every $A^\parallel \in B^{i\mathcal{H}}_{2\delta}(0)$, there is a unique $g_V(A^\parallel) \in B^{\mathcal{H}}_{2\delta}(0)$ with $f_V\big(A^\parallel + g_V(A^\parallel)\big) = 0$, and the corresponding $V \cdot \big(J + A^\parallel + g_V(A^\parallel)\big)$ lies in $\mathcal{U}$, where*

$$f_V(A) = \tfrac{1}{2}\Big(\big((J + A^\dagger) \cdot V^\dagger\big) \cdot \big(V \cdot (J + A)\big) - J\Big); \tag{146}$$

moreover

$$\|g_V(A^\parallel) + \tfrac{1}{2}(V^\dagger \cdot V - J)\| \le C_{\mathrm{ch}}(\varepsilon_r\delta + \delta^2), \quad \|Dg_V(A^\parallel)\| \le C_{\mathrm{ch}}(\varepsilon_r + \delta), \tag{147}$$

$$\|D_{A^\perp} f_V\big(A^\parallel + g_V(A^\parallel)\big) - I_{\mathcal{H}}\| \le C_{\mathrm{ch}}(\varepsilon_r + \delta) < 1, \tag{148}$$

and the resulting C^1 graph charts cover $\mathcal{U}$.

Registry contract (verbatim).

```
Uniform unitary graph control: there are universal C_ch >= 1, kappa_ch in (0, 1/2] such that for
every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0 with C_ch*(epsilon_r +
delta) <= kappa_ch, every V in calUbar_delta, and every A^par in B^{icalH}_{2delta}(0), there is a
unique g_V(A^par) in B^{calH}_{2delta}(0) with f_V(A^par + g_V(A^par)) = 0, and the corresponding V
bold-dot (J + A^par + g_V(A^par)) lies in calU, where f_V(A) = (1/2)*(((J + A^dagger) bold-dot
V^dagger) bold-dot (V bold-dot (J + A)) - J); moreover ||g_V(A^par) + (1/2)*(V^dagger bold-dot V -
J)|| <= C_ch*(epsilon_r*delta + delta^2), ||Dg_V(A^par)|| <= C_ch*(epsilon_r + delta), and
||D_{A^perp} f_V(A^par + g_V(A^par)) - I_{calH}|| <= C_ch*(epsilon_r + delta) < 1; the resulting
C^1 graph charts cover calU.
```

Proof (prose account of the af-validated tree). The closing witnesses are $C_{\mathrm{ch}} = 1024$ and $\kappa_{\mathrm{ch}} = \frac{1}{4}$; write $e = \varepsilon_r$ and $s = e + \delta$, so the guard gives $s \le 1/4096$.

Uniform multiplier control. For $V \in \overline{\mathcal{U}}_\delta$ with $q = V^\dagger \cdot V - J$ one has $\|q\| \le 2\delta$, and the C^*-lower bound gives $\|V\| \le 1 + 2s$. If Y is a right inverse of V then approximate associativity and exact unitality give $\|V^\dagger - Y\| \le 4s\|Y\|$, hence $\|Y\| \le 2$; the two Neumann estimates $\|L_V L_Y - I\| < 1$ and $\|L_{V^\dagger} L_V - I\| \le 4s < 1$ exhibit a right and a left inverse of L_V, which therefore coincide, with $\|L_V^{-1}\| \le 2$ and $\|L_V^{-1} - L_{V^\dagger}\| \le 8s$. For $\|A\| \le 4\delta$ and $X = V \cdot (J + A)$ the product-norm axiom gives $\|L_V^{-1}(L_X - L_V)\| \le 16s < 1$, so L_X is invertible and X has the right inverse $L_X^{-1}J$.

The graph equation. Put $F(P,H) = f_V(P + H)$ for $P \in i\mathcal{H}$, $H \in \mathcal{H}$ of norm $< 2\delta$; exact product reversal identifies $2F = X^\dagger \cdot X - J$, a degree-two real polynomial with values in $\mathcal{H}$, hence C^1. Expanding $2F$ with $V^\dagger \cdot V = J + q$ and reassociating *only* the two linear terms yields $2F = q + 2H + R$ with $\|R\| \le 128(e\delta + \delta^2)$, i.e.

$$\|F(P,H) - \big(\tfrac{q}{2} + H\big)\| \le 64\,(e\delta + \delta^2), \tag{149}$$

and the same expansion differentiated gives $\|D_H F - I_{\mathcal{H}}\| \le 64s$ and $\|D_P F\| \le 64s$.

Solving it. With $c = 64s \le \frac{1}{8}$, apply Lemma 89 (§39) to $H \mapsto F(P,H)$ on $B^{\mathcal{H}}_{2\delta}(0)$ with comparison isomorphism $I_{\mathcal{H}}$: it is injective and its image contains $F(P,0) + B^{\mathcal{H}}_{2\delta(1-c)}(0)$, while (149) and $\|q\| \le 2\delta$ give $\|F(P,0)\| \le \delta(1+c) < 2\delta(1-c)$. So 0 lies in the image and determines the unique $g_V(P)$. Substituting $H = g_V(P)$ in (149) gives the first bound of (147); the lower Lipschitz inequality of the same lemma together with $\|D_P F\| \le c$ makes g_V locally Lipschitz, after which $Dg_V(P) = -(D_H F)^{-1} D_P F$ gives $\|Dg_V(P)\| \le c/(1-c) \le 2c = 128s$ and the continuity of DF and of operator inversion makes g_V a C^1 map.

Unitarity and the cover. For $A = P + g_V(P)$, exact product reversal gives $X_V(P)^\dagger \cdot X_V(P) - J = 2F(P, g_V(P)) = 0$, and the multiplier step supplies the right inverse, so $X_V(P) \in \mathcal{U}$. Injectivity of L_V makes $P \mapsto X_V(P)$ injective with inverse $\phi_V^{\parallel}$, and conversely any unitary in the coordinate box solves $f_V = 0$ and is caught by uniqueness. Every $U \in \mathcal{U}$ lies in $\overline{\mathcal{U}}_\delta$ and has $f_U(0) = 0$, so $g_U(0) = 0$ and $U = X_U(0)$: the self-centred charts cover $\mathcal{U}$. Finally $64 \le C_{\mathrm{ch}}$, $128 \le C_{\mathrm{ch}}$ and $64s \le \frac{1}{4} < 1$ deliver (147)–(148).

Lemma 100 (`lem-stage1-maurer-cartan-trivialization`). *There are universal $C_{\mathrm{ch}} \ge 1$ and $\kappa_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{ch}}$, the graph maps supplied by Lemma 99 satisfy: every tangent space $T_U\mathcal{U}$ is the image of $L_U(I + Dg_U(0)) : i\mathcal{H} \to \mathcal{X}$, and*

$$\omega_U(Z) = \big(L_U^{-1} Z\big)^{\parallel} \;:\; T_U\mathcal{U} \longrightarrow i\mathcal{H} \tag{150}$$

is a global C^1 bundle trivialization with distortion at most $1 + C_{\mathrm{ch}}\varepsilon_r$, satisfying $\omega_{cU}(cZ) = \omega_U(Z)$ and $\omega_U(iU) = iJ$ for every $c \in U(1)$.

Registry contract (verbatim).

```
Uniform global tangent/Maurer-Cartan control: there are universal C_ch >= 1, kappa_ch in (0, 1/2]
such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and every delta > 0 with
C_ch*(epsilon_r + delta) <= kappa_ch, the graph maps supplied by lem-stage1-unitary-graph-control
satisfy: every tangent space T_U calU is the image of L_U(I + Dg_U(0)): icalH -> calX, and
omega_U(Z) = (L_U^{-1} Z)^par : T_U calU -> icalH is a global C^1 bundle trivialization with
distortion at most 1 + C_ch*epsilon_r, satisfying omega_{cU}(cZ) = omega_U(Z) and omega_U(iU) = iJ
for every c in U(1).
```

Proof (prose account of the af-validated tree). No graph-existence work is repeated: the tree fixes the witnesses of Lemma 99 and enlarges them.

Multiplier bounds at a unitary. For $U \in \mathcal{U}$ one has $U^\dagger \cdot U = J$ and $\|J\| = 1$, so the lower C^*-inequality gives $\|U\| \le (1-\varepsilon_r)^{-1/2}$ and the associator axiom gives $\|L_{U^\dagger} L_U - I\| \le \varepsilon_r \|U\|^2 \le \varepsilon_r/(1-\varepsilon_r) < 1$ once $\varepsilon_r \le \frac{1}{4}$. Neumann inversion in finite dimension makes L_U invertible with $\|L_U\|, \|L_U^{-1}\| \le 1 + K\varepsilon_r$ for a universal K; C_{ch} is then chosen at least $\max(C_0, K, 4)$, with C_0, κ_0 the graph witnesses, and $\kappa_{\mathrm{ch}} = \min(\kappa_0, \frac{1}{4})$.

The centred graph. Exact unitality and $U^\dagger \cdot U = J$ give $f_U(0) = 0$, so $g_U(0) = 0$. Comparing the graph germs at two admissible radii — they agree near 0 by pointwise uniqueness — and letting the radius tend to 0 sharpens the derivative bound of (147) to $\|Dg_U(0)\| \le C_0\varepsilon_r$. Differentiating the centred chart $\Psi_U(A) = U \cdot (J + A + g_U(A))$ at 0 gives $D\Psi_U(0) = L_U(I + Dg_U(0))$, whose image is exactly $T_U\mathcal{U}$ because these charts cover $\mathcal{U}$.

The trivialization. Let $P^{\|}(X) = (X - X^{\dagger})/2$, of norm at most 1 by isometry of the involution. Since $Dg_U(0)$ maps $i\mathcal{H}$ into $\mathcal{H}$, one has $P^{\|}(A + Dg_U(0)A) = A$, so ω_U of (150) inverts $A \mapsto L_U(I + Dg_U(0))A$ fibrewise; its two norms are at most $\|L_U^{-1}\|$ and $\|L_U\|(1 + \|Dg_U(0)\|)$, both absorbed into $1 + C_{\mathrm{ch}}\varepsilon_r$.

Regularity. $U \mapsto L_U$ is linear and inversion is C^1 on the invertibles, so ω is C^1. For the *inverse* the tree does not argue from chart regularity (see the correction below) but from the defining equation: with $B_U = D_{A^{\perp}} f_U(0)|_{\mathcal{H}}$ and $C_U = D_{A^{\|}} f_U(0)|_{i\mathcal{H}}$, both polynomial in U, (148) gives $\|B_U - I_{\mathcal{H}}\| < 1$, and differentiating $f_U(A + g_U(A)) = 0$ at $A = 0$ yields

$$Dg_U(0) = -B_U^{-1} C_U, \tag{151}$$

a C^1 function of U. Hence $(U, A) \mapsto (U, L_U(A + Dg_U(0)A))$ is a C^1 inverse.

Equivariance. $U \mapsto cU$ preserves $\mathcal{U}$ with derivative $Z \mapsto cZ$, and $L_{cU} = cL_U$, so $\omega_{cU}(cZ) = \omega_U(Z)$. The curve $t \mapsto e^{it}U$ lies in $\mathcal{U}$ with velocity iU, and complex bilinearity with $U \cdot J = U$ gives $L_U(iJ) = iU$; as $J^{\dagger} = J$ the element iJ is anti-Hermitian, so $\omega_U(iU) = iJ$.

Status. All three are `proved` with `af: validated` (T0). Workspaces `proofs/lem-stage1-rectified-cstar-co` (17 nodes created, 17 `validated`, taint clean, root `validated`), `proofs/lem-stage1-unitary-graph-control` (15 live nodes all `validated` plus 3 `archived`, taint clean on all 18), and `proofs/lem-stage1-maurer-cartan-triv` (15 nodes, 15 `validated`, taint clean).

Correction of record (proof-level, not contract-level). The trivialization tree drew a challenge at the regularity node: as first written it inferred C^1 regularity of the fibrewise *inverse* from the merely C^1 graph frame, which is not valid in general. The repair is (151) — an explicit implicit-function formula with invertible normal derivative — and the challenge's counterexample is inapplicable precisely because it has no such formula. No contract was amended.

Role in the chain. Registry `defs`: `def-epsilon-cstar-algebra`, `def-approximate-unitary-space`. Registry `deps`: `lem-stage1-exact-unit-rectification` (§40) and `lem-stage1-quantitative-inverse-functi` (§39) for the rectification; the rectification and the same inverse-function lemma for the graph control; the graph control alone for the trivialization. Downstream they carry §45–§47.

45 The closed polar retraction and its coherence

Lemma 101 (`lem-stage1-polar-retraction`)**.** *There are universal $C_{\mathrm{pol}} \geq 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{pol}}$, the map*

$$\Pi_\delta(U, H) = U \cdot H \tag{152}$$

is a C^1 diffeomorphism from $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ onto an open set S_δ; its inverse (u_δ, h_δ) obeys

$$X = u_\delta(X) \cdot h_\delta(X), \qquad u_\delta(U) = U, \qquad h_\delta(U) = J \quad (U \in \mathcal{U}); \tag{153}$$

and, writing $D = \varepsilon_r\delta + \delta^2$,

$$\mathcal{U}_{\delta - C_{\mathrm{pol}}D} \ \subseteq\ S_\delta \ \subseteq\ \mathcal{U}_{\delta + C_{\mathrm{pol}}D}. \tag{154}$$

Transcription note. *Only the* shrunken *inner set of* (154) *is claimed; no larger inner radius is asserted. Both $\mathcal{U}_\cdot$ are the strict-inequality sets of Definition 32, so membership always carries the right-inverse clause as well as the defect bound.*

Registry contract (verbatim).

```
Closed C^1 polar retraction: there are universal C_pol >= 1, kappa_pol in (0, 1/2] such that for
every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 with C_pol*(epsilon_r +
delta) <= kappa_pol, Pi_delta(U, H) = U bold-dot H is a C^1 diffeomorphism from calU x
B^{calH}_delta(J) onto an open S_delta, its inverse (u_delta, h_delta) obeys X = u_delta(X)
bold-dot h_delta(X), u_delta(U) = U, h_delta(U) = J, and calU_{delta - C_pol*(epsilon_r*delta +
delta^2)} subseteq S_delta subseteq calU_{delta + C_pol*(epsilon_r*delta + delta^2)}.
```

Proof (prose account of the af-validated tree). The constants $C_{\mathrm{pol}}, \kappa_{\mathrm{pol}}$ are fixed at the end as a finite maximum and a finite positive minimum of the thresholds used below, so they remain universal.

The coordinate map. Fix a base $V \in \overline{\mathcal{U}}_r$ with $r \leq c\delta$ ($c = 4$ is the only scale used). As in §44, $\|L_{V^\dagger}L_V - I\| \leq \varepsilon_r\|V\|^2 + 2(1+\varepsilon_r)r$, which the guard makes at most $\frac{1}{4}$; hence L_V is injective, therefore bijective in finite dimension, with a universal bound on $\|L_V^{-1}\|$. Writing $A(a) = a + g_V(a)$ for the graph map of Lemma 99 at scale $c\delta$, exact unitality and bilinearity give the *exact* identity

$$p_V(a,q) := L_V^{-1}\big(\Pi_\delta(V \cdot (J + A(a)),\, J + q) - V\big) = A(a) + q + L_V^{-1}\big((V \cdot A(a)) \cdot q\big), \tag{155}$$

with no reassociation. Differentiating (155) and using $\|Dg_V\| \leq C_{\mathrm{ch}}(\varepsilon_r + c\delta)$, $\|A(a)\| \leq K\delta$ and the product-norm axiom bounds

$$\sup\|J_0^{-1}\big(Dp_V - J_0\big)\| \ \leq\ C_0(\varepsilon_r + \delta) \ =:\ \theta \ \leq\ \tfrac{1}{4} \tag{156}$$

on the convex box $\|a\| < 2c\delta$, $\|q\| < \frac{3}{2}\delta$, where $J_0(\alpha, \beta) = \alpha + \beta$ is the real-linear isomorphism $i\mathcal{H} \oplus \mathcal{H} \to \mathcal{X}$ (of inverse norm at most 1, because the involution is isometric). At the base point, $p_V(0,0) = g_V(0)$, so (147) gives $\|p_V(0,0) + \frac{1}{2}(V^\dagger \cdot V - J)\| \leq C_0 D$.

Existence of the polar factorisation. For $V \in \mathcal{U}_{\delta - C_{\mathrm{pol}}D}$ the base-point estimate reads $\|p_V(0,0)\| < \delta - (C_{\mathrm{pol}} - K_b)D$; choosing $C_{\mathrm{pol}} > K_b + K_d$ and shrinking the guard so that $\theta < \frac{1}{4}$ makes this at most $(1-\theta)\delta$, so some $\rho < \delta$ satisfies $\|p_V(0,0)\|/(1-\theta) < \rho$. On the closed ρ-ball in the max norm, $F(z) = z - p_V(z)$ is a θ-contraction into itself; its unique fixed point $z = (a,q)$ solves $p_V(a,q) = 0$, i.e.

$$V = \big[V \cdot (J + a + g_V(a))\big] \cdot (J + q), \qquad \|q\| < \delta, \tag{157}$$

with the bracket in $\mathcal{U}$ by Lemma 99. Integrating (156) along segments makes p_V injective on the whole box, so this zero is the only one there.

Global injectivity and the diffeomorphism. Suppose $X = U_i \cdot (J + q_i)$ with $U_i \in \mathcal{U}$ and $\|q_i\| < \delta$ for $i = 1, 2$. Expanding $X^\dagger \cdot X - J$ around $q = 0$ without losing the q factor shows $X \in \overline{\mathcal{U}}_{4\delta}$ and L_X invertible, so one may take the graph chart *at X itself* at scale 4δ: with $A_i = L_X^{-1}(U_i - X)$, the vanishing of $f_X(A_i)$ and uniqueness force $A_i^\perp = g_X(A_i^\parallel)$, and the near-identity estimate for p_X on that box gives $(a_1, q_1) = (a_2, q_2)$, hence $U_1 = U_2$ and $q_1 = q_2$. Thus Π_δ is injective; by (156) its derivative is invertible at every point, so the finite-dimensional C^1 inverse function theorem makes it a local C^1 diffeomorphism, S_δ open, and the local inverses glue to a C^1 inverse (u_δ, h_δ) with $X = u_\delta(X) \cdot h_\delta(X)$. Exact unitality gives $U = U \cdot J$, whence $u_\delta(U) = U$ and $h_\delta(U) = J$.

The radius sandwich. For $X = U \cdot (J + q) \in S_\delta$, the star rule expands $X^\dagger \cdot X - J$ into two cross terms — each equal to q up to an associator of norm $O(\varepsilon_r \delta)$ — plus one quadratic term of norm $O(\delta^2)$, giving $\|X^\dagger \cdot X - J\| < 2\delta + C_{\mathrm{out}} D$; and L_X is a small perturbation of the invertible L_U, so X has a right inverse. Taking $C_{\mathrm{pol}} \geq C_{\mathrm{out}}/2$ gives the outer inclusion of (154). The inner inclusion is (157).

Lemma 102 (`lem-stage1-polar-coherence-naturality`)**.** *For every exact-unit algebra and every two polar data $(\delta_j, S_j, u_j, h_j)$, $j = 1, 2$, for which $\Pi_{\delta_j} : \mathcal{U} \times B^{\mathcal{H}}_{\delta_j}(J) \to S_j$, $(U, H) \mapsto U \cdot H$, is bijective with inverse (u_j, h_j), one has*

$$(u_1, h_1) = (u_2, h_2) \qquad \text{on } S_1 \cap S_2; \tag{158}$$

moreover, for $c \in U(1)$ and $X, cX \in S_j$,

$$u_j(cX) = c\, u_j(X), \qquad h_j(cX) = h_j(X). \tag{159}$$

Transcription note. *This row is* conditional*: the polar data are hypotheses and no free polar witness is produced. It is never used as an existence assertion — existence is Lemma 101 — but only to identify inverse components that have already been produced.*

Registry contract (verbatim).

```
Polar coherence and scalar naturality: for every exact-unit algebra and every two polar data
(delta_j, S_j, u_j, h_j), j = 1, 2, for which Pi_{delta_j}: calU x B^{calH}_{delta_j}(J) -> S_j,
(U, H) |-> U bold-dot H, is bijective with inverse (u_j, h_j), one has (u_1, h_1) = (u_2, h_2) on
S_1 intersect S_2; moreover, for c in U(1) and X, cX in S_j, u_j(cX) = c*u_j(X) and h_j(cX) =
h_j(X).
```

Proof (prose account of the af-validated tree). *Coherence.* Fix $X \in S_1 \cap S_2$ and put $p_j = (u_j(X), h_j(X))$, so $p_j \in \mathcal{U} \times B^{\mathcal{H}}_{\delta_j}(J)$ and $\Pi_{\delta_j}(p_j) = X$. Let k index the larger radius; then $B^{\mathcal{H}}_{\delta_l}(J) \subseteq B^{\mathcal{H}}_{\delta_k}(J)$ for $l \neq k$, so both p_1, p_2 lie in the domain of Π_{δ_k}. Every Π_δ has the *same* formula $(U, H) \mapsto U \cdot H$, so both have Π_{δ_k}-image X, and the assumed bijectivity — hence injectivity — of Π_{δ_k} gives $p_1 = p_2$. That is (158).

Scalar naturality. $\mathcal{U}$ is closed under the circle action: if $U^\dagger \cdot U = J$ and $U \cdot R = J$ then conjugate-linearity of $\dagger$, bilinearity and $\bar{c}c = 1$ give $(cU)^\dagger \cdot (cU) = J$ and $(cU) \cdot (c^{-1}R) = J$, using no associativity and no positivity. Hence for $p = (u_j(X), h_j(X))$ the pair $p_c = (c\, u_j(X), h_j(X))$ again lies in $\mathcal{U} \times B^{\mathcal{H}}_{\delta_j}(J)$, and bilinearity in the first variable gives $\Pi_{\delta_j}(p_c) = c\, X$. Since $q = (u_j(cX), h_j(cX))$ has the same image, injectivity gives $q = p_c$, which is (159).

Status. Both are `proved` with `af: validated` (T0). Workspace `proofs/lem-stage1-polar-retraction`: 29 nodes created, 29 `validated`, root `validated`, taint clean — above the soft node cap, so the argument linker carries a standing REFACTOR advisory for this row even though the run completed cleanly. Workspace `proofs/lem-stage1-polar-coherence-naturality`: 10 nodes, 10 `validated`, taint clean.

Corrections of record (proof-level, not contract-level). Two shaping challenges are visible in the retraction tree. (i) A *scope* challenge at the multiplier node: as first written it inferred a small Neumann error from $r \leq \frac{1}{4}$ alone, with r unrelated to δ; the repair ties the base radius to the scale, $r \leq c\delta$ with c one of the finitely many absolute constants actually used, and every later estimate cites that hypothesis. (ii) An *endpoint* challenge in the existence chain: the corrected chain is $B < \delta - K_d D \leq (1-\theta)\delta$, with the strictness carried by the first inequality alone, so the equality case the verifier exhibited is harmless. Several nodes were also rewritten to derive their estimates internally rather than from a then-pending sibling. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space` (both) and `def-epsilon-cstar-algebra` (the retraction). Registry `deps`: `lem-stage1-unitary-graph-control` (§44) for the retraction, and the retraction itself for the coherence row. Downstream they carry the approximate group laws (§46) and the path and inversion rows (§47). `op-classical` is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

46 Quantitative approximate group laws on the unitaries

The unitary set $\mathcal{U}$ of Definition 32 is not a group: $\boldsymbol{\cdot}$ is only approximately associative and $U^\dagger$ is only approximately an inverse. The three results below aim to turn it into an approximate group by *projecting* the naive product and adjoint back to $\mathcal{U}$ through the polar inverse u_δ of Lemma 101. The first two are `af`-validated; the third — the compound parent — is **retired in place** (below). Throughout, all three carry the same pair of guards

$$C_{\mathrm{pol}}(\varepsilon_r+\delta)\le\kappa_{\mathrm{pol}},\qquad C_{\mathrm{grp}}\varepsilon_r<\delta-C_{\mathrm{pol}}(\varepsilon_r\delta+\delta^2),\tag{160}$$

with universal $C_{\mathrm{grp}},C_{\mathrm{pol}}\ge 1$ and $\kappa_{\mathrm{pol}}\in(0,\frac{1}{2}]$.

Lemma 103 (`lem-stage1-group-domain-membership`)**.** *There exist universal $C_{\mathrm{grp}},C_{\mathrm{pol}}\ge 1$ and $\kappa_{\mathrm{pol}}\in(0,\frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta>0$ satisfying* (160)*, the inverse $u_\delta:S_\delta\to\mathcal{U}$ of the polar map is defined at $U\boldsymbol{\cdot}V$ and at $U^\dagger$ for every $U,V\in\mathcal{U}$; moreover $U\boldsymbol{\cdot}V$ and $U^\dagger$ each have a right inverse.*

Registry contract (verbatim).

```
Group-input polar-domain membership: there exist universal C_grp, C_pol >= 1, kappa_pol in (0, 1/2]
such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 satisfying
C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), the inverse u_delta:S_delta -> calU of the polar map is defined at U bold-dot V and
U^dagger for every U, V in calU; moreover, U bold-dot V and U^dagger each have a right inverse.
```

Proof (prose account of the af-validated tree). Take $C_{\mathrm{pol}},\kappa_{\mathrm{pol}}$ from Lemma 101 and $C_{\mathrm{grp}}=4$; abbreviate $e=\varepsilon_r$ and $t=\delta-C_{\mathrm{pol}}(e\delta+\delta^2)$. By the inner inclusion (154) it suffices to place $U\boldsymbol{\cdot}V$ and $U^\dagger$ in $\mathcal{U}_t$, right-inverse clause included.

In-scope smallness. Everything below cites one arithmetic node, derived from (160) alone: $C_{\mathrm{pol}}\ge 1$ and $\kappa_{\mathrm{pol}}\le\frac{1}{2}$ give $s:=e+\delta\le\frac{1}{2}$ and $\delta\le\frac{1}{2}$, while the second guard with $C_{\mathrm{grp}}=4$ gives $4e<t$, hence $t>0$ and $e<t\le\delta(1-s)$. Since $(s-e)(1-s)$ is increasing in s on $[e,\frac{1}{2}]$, this forces

$$e<\tfrac{1}{6}.\tag{161}$$

No threshold beyond (161) is available, and none is used.

Multiplier control. For $W\in\mathcal{U}$, $W^\dagger\boldsymbol{\cdot}W=J$ and the C^*-lower bound give $(1-e)\|W\|^2\le 1$, while the associator axiom gives $\|L_{W^\dagger}L_W-I\|\le e\|W\|^2\le e/(1-e)<\frac{1}{5}$ by (161). Hence L_W is injective, therefore bijective in finite dimension, with $\|L_{W^\dagger}\|\le\frac{7}{5}$ and $\|L_W^{-1}\|\le\frac{7}{4}$.

The product. Put $X=U\boldsymbol{\cdot}V$, so $X^\dagger=V^\dagger\boldsymbol{\cdot}U^\dagger$ by the star rule. Inserting $((V^\dagger\boldsymbol{\cdot}U^\dagger)\boldsymbol{\cdot}U)\boldsymbol{\cdot}V$ and $(V^\dagger\boldsymbol{\cdot}(U^\dagger\boldsymbol{\cdot}U))\boldsymbol{\cdot}V$ between $X^\dagger\boldsymbol{\cdot}X$ and J costs two associators and lands on $(V^\dagger\boldsymbol{\cdot}J)\boldsymbol{\cdot}V=V^\dagger\boldsymbol{\cdot}V=J$, so

$$\|X^\dagger\boldsymbol{\cdot}X-J\|\;\le\;\frac{2e(1+e)}{(1-e)^2}\;\le\;4e\;<\;t\;<\;2t.\tag{162}$$

Approximate associativity gives $\|L_X-L_UL_V\|\le e\|U\|\|V\|<\frac{1}{5}$, while $\|(L_UL_V)^{-1}\|\le\frac{49}{16}$; the Neumann perturbation is $<\frac{49}{80}<1$, so L_X is invertible and $L_X^{-1}J$ is a right inverse of X.

The adjoint. $U^\dagger\boldsymbol{\cdot}U=J$ says U is a right inverse of $U^\dagger$. If R is the right inverse of U supplied by Definition 32 then $R=L_U^{-1}J$ and $\|R\|\le\frac{7}{4}$; one associator gives $\|U^\dagger-R\|\le e\|U\|^2\|R\|$ and hence $\|U\boldsymbol{\cdot}U^\dagger-J\|\le(1+e)e\|U\|^3\|R\|\le\frac{441}{125}e<4e<t<2t$. So both inputs lie in $\mathcal{U}_t\subseteq S_\delta$.

Endpoint discipline. Every estimate above is non-strict and remains meaningful at $e = 0$; the single strict passage is $C_{\mathrm{grp}} e < t$, followed by $t < 2t$ because $t > 0$. That is exactly what membership in the strict set $\mathcal{U}_t$ needs.

Lemma 104 (`lem-stage1-group-closeness`). *There exist universal $C_{\mathrm{grp}}, C_{\mathrm{pol}} \geq 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ satisfying* (160)*, the inverse $u_\delta : S_\delta \to \mathcal{U}$ of the polar map satisfies*

$$\|u_\delta(U \cdot V) - U \cdot V\| \leq C_{\mathrm{grp}} \varepsilon_r, \qquad \|u_\delta(U^\dagger) - U^\dagger\| \leq C_{\mathrm{grp}} \varepsilon_r \qquad (U, V \in \mathcal{U}). \tag{163}$$

Registry contract (verbatim).

```
Group-input polar closeness: there exist universal C_grp, C_pol >= 1, kappa_pol in (0, 1/2] such
that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 satisfying
C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), the inverse u_delta:S_delta -> calU of the polar map satisfies ||u_delta(U bold-dot V) -
U bold-dot V|| <= C_grp*epsilon_r and ||u_delta(U^dagger) - U^dagger|| <= C_grp*epsilon_r for every
U, V in calU.
```

Proof (prose account of the af-validated tree). Take $\kappa_{\mathrm{pol}} \leq \frac{1}{16}$ and $C_{\mathrm{grp}} = 5$, and write $\rho = \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2)$, so $\varepsilon_r + \delta \leq \frac{1}{16}$ and $\rho > 5\varepsilon_r \geq 0$. This row deliberately does *not* import Lemma 103: the displayed expressions assert that both inputs lie in the domain of u_δ, and the tree re-derives that typing internally — $\|(U \cdot V)^\dagger \cdot (U \cdot V) - J\| \leq 3\varepsilon_r$ with a right inverse obtained from injectivity of $L_{U \cdot V}$ in finite dimension, and $\|U \cdot U^\dagger - J\| \leq 2\varepsilon_r$ with the exact right inverse U — so that $U \cdot V, U^\dagger \in \mathcal{U}_\rho \subseteq S_\delta$. The duplication is the price of removing every edge between the two children.

The estimate itself is one calculation, stated for any $X \in S_\delta$ with $\|X^\dagger \cdot X - J\| \leq 3\varepsilon_r$. Write $u = u_\delta(X)$, $h = h_\delta(X)$ and $a = h - J$, so $X = u \cdot h$, $u \in \mathcal{U}$, $h \in \mathcal{H}$ and $\|a\| < \delta$ by Lemma 101. Two associator comparisons of $(h \cdot u^\dagger) \cdot (u \cdot h)$ against $h \cdot h$ give

$$\|X^\dagger \cdot X - h \cdot h\| \ \leq\ 2\varepsilon_r(1 + \varepsilon_r)\|h\|^2\|u\|^2 \ \leq\ 3\varepsilon_r. \tag{164}$$

Exact unitality and bilinearity give $h \cdot h - J = 2a + a \cdot a$, so $2\|a\| \leq 6\varepsilon_r + (1 + \varepsilon_r)\|a\|^2$; since $\|a\| < \delta \leq \frac{1}{16}$ the quadratic term is at most $\frac{17}{256}\|a\|$ and absorbs, leaving $\|a\| \leq \frac{1536}{495}\varepsilon_r \leq 4\varepsilon_r$. Finally exact right unitality gives $\|u_\delta(X) - X\| = \|u \cdot (J - h)\| \leq (1 + \varepsilon_r)\|u\|\|a\| \leq 5\varepsilon_r$, which is (163).

Conjecture 105 (`lem-stage1-approximate-group-laws` — proof retracted 2026-07-28)**.** *There exist universal $C_{\mathrm{grp}}, C_{\mathrm{pol}} \geq 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ satisfying* (160)*, the inverse u_δ of the polar map defines C^1 maps*

$$\mu(U, V) = u_\delta(U \cdot V), \qquad \sigma(U) = u_\delta(U^\dagger) \tag{165}$$

on all of $\mathcal{U}$, with $\mu(J, U) = \mu(U, J) = U$, $\sigma(J) = J$, and

$$\|\mu(U, V) - U \cdot V\| \leq C_{\mathrm{grp}} \varepsilon_r, \qquad \|\sigma(U) - U^\dagger\| \leq C_{\mathrm{grp}} \varepsilon_r, \tag{166}$$

$$\|\mu(\mu(U, V), W) - \mu(U, \mu(V, W))\| \leq C_{\mathrm{grp}} \varepsilon_r, \tag{167}$$

$$\|\mu(\sigma(U), U) - J\| \leq C_{\mathrm{grp}} \varepsilon_r, \qquad \|\mu(U, \sigma(U)) - J\| \leq C_{\mathrm{grp}} \varepsilon_r. \tag{168}$$

Retraction note (2026-07-28). *The parent's* `af` *validation was RETRACTED by the Stage-1 binder sweep: its export node 1.1.2 invokes coherence-naturality on the two children's conclusions, whose opaque contracts export only an anaphorically bound u_δ (no typed companion, no displayed source map, no preimage identity), so the two-typed-data antecedent is unavailable (see* `docs/LEARNINGS.md`*, second entry of 2026-07-28). The two* children *above are certified SOUND by the same sweep and remain T0. The parent statement stands as a* stated *candidate; its repair (a direct explicit-binder derivation) is part of the W97 campaign.*

Registry contract (verbatim).

```
Quantitative approximate group laws: there exist universal C_grp, C_pol >= 1, kappa_pol in (0, 1/2]
such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0 satisfying
C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), the inverse u_delta of the polar map defines C^1 maps mu(U, V) = u_delta(U bold-dot V),
sigma(U) = u_delta(U^dagger) on all of calU, with mu(J, U) = mu(U, J) = U, sigma(J) = J, ||mu(U, V)
- U bold-dot V|| <= C_grp*epsilon_r, ||sigma(U) - U^dagger|| <= C_grp*epsilon_r, ||mu(mu(U, V), W)
- mu(U, mu(V, W))|| <= C_grp*epsilon_r, ||mu(sigma(U), U) - J|| <= C_grp*epsilon_r, and ||mu(U,
sigma(U)) - J|| <= C_grp*epsilon_r.
```

Historical record: the RETRACTED proof attempt (not a proof; defect marked). *Witness synchronisation — THE DEFECTIVE STEP.* If (G_d, P_d, k_d), (G_c, P_c, k_c) and (P_r, k_r) are witnesses for the two lemmas above and for Lemma 101, put $C_{\text{pol}} = \max(P_d, P_c, P_r)$, $\kappa_{\text{pol}} = \min(k_d, k_c, k_r, \frac{1}{16})$ and $C_{\text{grp}} = \max(G_d, 8G_c, 8)$. Both guards are monotone in these witnesses, so (160) implies each upstream guard, and $\varepsilon_r \le \frac{1}{16}$. The tree then claimed: every polar datum in play uses the *same* map Π_δ, so Lemma 102 identifies the inverse components. *This is the sweep-adjudicated defect*: the children's contracts export only an anaphoric first component — no typed companion, no displayed source map, no preimage identity — so coherence's two-typed-data antecedent is unavailable and no "one common u_δ" is obtained.

Regularity and basepoints. $U \cdot V$ and $U^\dagger$ lie in the open set S_δ by Lemma 103, and u_δ is C^1 there; the product is bilinear and $\dagger$ is real-linear, hence both are C^1 in finite dimension, so (165) defines C^1 maps on all of $\mathcal{U}$. Since $J \in \mathcal{U}$, $J \cdot U = U \cdot J = U$, $J^\dagger = J$ and u_δ fixes $\mathcal{U}$ pointwise ((153)), the three basepoint identities are exact. Estimate (166) is Lemma 104 together with $C_{\text{grp}} \ge 8G_c$.

The three defects. All of $\mathcal{U}$ obeys $\|T\| \le \sqrt{16/15} < \frac{4}{3}$, by the C^*-lower bound applied to $T^\dagger \cdot T = J$. For (167), insert $A \cdot W$, $(U \cdot V) \cdot W$, $U \cdot (V \cdot W)$ and $U \cdot B$ between the two iterated products, where $A = \mu(U, V)$ and $B = \mu(V, W)$: the two outer differences are each $\le G_c\varepsilon_r$, each replacement difference is $\le \frac{17}{12} G_c \varepsilon_r$ by the product-norm axiom, and the sole reassociation costs $\le \frac{64}{27}\varepsilon_r$; the total is $\le \frac{389}{54} G_c\varepsilon_r \le 8G_c\varepsilon_r$. For the first bound of (168), closeness at $(\sigma(U), U)$ plus $\|\sigma(U) \cdot U - U^\dagger \cdot U\| \le \frac{17}{12} G_c\varepsilon_r$ and $U^\dagger \cdot U = J$ give $\le 3G_c\varepsilon_r$. For the second, the same two steps land on $U \cdot U^\dagger$, and a local estimate — $R = L_U^{-1} J$ with $\|R\| \le \frac{3}{2}$, one associator for $\|R - U^\dagger\|$, then $U \cdot U^\dagger - J = U \cdot (U^\dagger - R)$ — gives $\|U \cdot U^\dagger - J\| \le \frac{34}{9}\varepsilon_r \le 4\varepsilon_r$, for a total $\le \frac{77}{12} G_c\varepsilon_r \le 8G_c\varepsilon_r$. All bounds are non-strict and hold at $\varepsilon_r = 0$.

Status. The two children are `proved` with `af: validated` (T0, re-certified by the 2026-07-28 sweep): workspaces `proofs/lem-stage1-group-domain-membership` (10 nodes, 10 `validated`, taint clean) and `proofs/lem-stage1-group-closeness` (12 nodes, 12 `validated`, taint clean). The parent `lem-stage1-approximate-group-laws` is `stated / af: seeded`: its validation (14 nodes) was RETRACTED 2026-07-28 on the sweep- adjudicated node-1.1.2 defect above (`docs/LEARNINGS.md`).
Retired in place; the live replacement path. The parent row is **not** being re-elevated. Its live content re-enters the record through the binder-closed rebuild of §49: the two explicit bridges

`lem-stage1-explicit-group-domain-membership` and `lem-stage1-explicit-group-closeness` (each carrying a contract-local typed (u_δ, h_δ) in place of the anaphoric one), together with the explicit smooth bridge `lem-stage1-explicit-smooth-unitary-operations`; and the *parameterized* approximate-group statement itself is now carried by `lem-stage1-approximate-group-laws-transport` (row 13e), `af`-validated 2026-07-29 on exactly those two explicit bridges plus Lemma 101, with the retired parent and `lem-stage1-polar-coherence-naturality` dropped from its `deps`. No downstream row imports the retired parent any longer.

Correction of record (proof-level, not contract-level). The parent's first elevation **aborted on the balloon tripwire** at 60 live nodes. Per that protocol the *proof* was factored — the contract above is byte-unchanged — into the two children reproduced here, chosen so that neither imports the other; the parent then re-elevated on the factored dependencies. Separately, the membership tree's first run aborted STUCK on prover indiscipline about smallness: unjustified thresholds such as $\varepsilon_r \leq \frac{1}{8}$ were being assumed. The second run carries the binding rule recorded above — derive (161) once from the guards, cite that node for every later smallness inference, and absorb the worse constants into the proof body rather than sharpening the threshold. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space`, `def-epsilon-cstar-algebra`. Registry `deps`: `lem-stage1-polar-retraction` (§45) for both children; that lemma together with `lem-stage1-polar-coherence-naturality` and the two children for the parent. The parent has **no** live dependents: the all-$\mathcal{U}$ adjoint domain once drawn from it in §47 is now supplied by `lem-stage1-explicit-group-closeness` (§49). Nothing here bears on `op-classical`, which is **open**.

47 Paths, the inversion derivative, and the smooth atlas

Lemma 106 (`lem-stage1-polar-path-admissibility`). *There exist universal $C_{\mathrm{path}}, C_{\mathrm{pol}} \geq 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{pol}}$, and $U_0, U_1 \in \mathcal{U}$: if $0 \leq q \leq 1$, $\|U_1 - U_0\| \leq q$, $C_{\mathrm{path}} q \leq \frac{1}{4}$ and*

$$C_{\mathrm{path}}(q + \varepsilon_r q + q^2) \ < \ \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2), \tag{169}$$

then, for $Z_t = (1-t)U_0 + tU_1$, every L_{Z_t} is invertible, every $Z_t \in \overline{\mathcal{U}}_{C_{\mathrm{path}}(q+\varepsilon_r q+q^2)}$, and

$$H(t, U_0, U_1) = u_\delta(Z_t) \tag{170}$$

is jointly continuous in all displayed variables, joins U_0 to U_1, and obeys $H(t, cU_0, cU_1) = c\,H(t, U_0, U_1)$ for $c \in U(1)$.

Registry contract (verbatim).

```
Joint projected-straight-path admissibility: there exist universal C_path, C_pol >= 1, kappa_pol in
(0, 1/2] such that, for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0
with C_pol*(epsilon_r + delta) <= kappa_pol, and U_0, U_1 in calU, if 0 <= q <= 1, ||U_1 - U_0|| <=
q, C_path*q <= 1/4, and C_path*(q + epsilon_r*q + q^2) < delta - C_pol*(epsilon_r*delta + delta^2),
then for Z_t = (1-t)*U_0 + t*U_1 every L_{Z_t} is invertible, every Z_t in calUbar_{C_path*(q +
epsilon_r*q + q^2)}, and H(t, U_0, U_1) = u_delta(Z_t) is jointly continuous in all displayed
variables, joins U_0 to U_1, and obeys H(t, cU_0, cU_1) = c*H(t, U_0, U_1) for c in U(1).
```

Proof (prose account of the af-validated tree). Take C_{pol} from Lemma 101, $\kappa_{\mathrm{pol}} = \min(\kappa^0_{\mathrm{pol}}, C^0_{\mathrm{pol}}/8)$ and $C_{\mathrm{path}} = 1$; the guard then gives $\varepsilon_r \leq \frac{1}{8}$, and for $U \in \mathcal{U}$ the usual chain $\|U\| \leq \sqrt{8/7}$, $\|L_{U^\dagger} L_U - I\| \leq \frac{1}{7}$ makes L_U invertible with $\|L_U^{-1}\| \leq \frac{3}{2}$.

An exact quadratic identity. With $D = U_1 - U_0$, bilinearity, the star rule and $U_i^\dagger \cdot U_i = J$ give, with no error term,

$$Z_t^\dagger \cdot Z_t - J \ = \ t(t-1)\big(D^\dagger \cdot D\big), \tag{171}$$

whence $\|Z_t^\dagger \cdot Z_t - J\| \ \leq \ t(1-t)(1+\varepsilon_r)\|D\|^2 \ \leq \ (1+\varepsilon_r)q^2/4$. Since $L_{Z_t} = L_{U_0} + tL_D$ and $\|L_D\| \leq (1+\varepsilon_r)\|D\|$, the bound $q \leq \frac{1}{4}$ gives $\|tL_{U_0}^{-1} L_D\| \leq \frac{27}{64} < 1$, so every L_{Z_t} is invertible by Neumann and $L_{Z_t}^{-1} J$ is a right inverse of Z_t. As $(1+\varepsilon_r)q^2/4 \leq 2(q + \varepsilon_r q + q^2)$, this is exactly $Z_t \in \overline{\mathcal{U}}_{q+\varepsilon_r q+q^2}$ in the sense of Definition 32.

Into the polar domain. With $\gamma = q + \varepsilon_r q + q^2$ and $\rho = \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2)$, hypothesis (169) reads $\gamma < \rho$, so $\|Z_t^\dagger \cdot Z_t - J\| \leq 2\gamma < 2\rho$ and $Z_t \in \mathcal{U}_\rho$; the inner inclusion of (154) puts $Z_t \in S_\delta$ for every t.

The projected path. $(t, U_0, U_1) \mapsto Z_t$ is continuous and u_δ is C^1, hence continuous, on the open set S_δ, so H of (170) is jointly continuous; $Z_0 = U_0$, $Z_1 = U_1$ and $u_\delta(U_i) = U_i$ give the endpoints. For $c \in U(1)$ the rotated pair again lies in $\mathcal{U}$ with the same separation, its segment is cZ_t, and both Z_t and cZ_t lie in S_δ; the scalar-naturality clause (159) of Lemma 102, applied to the polar datum $(\delta, S_\delta, u_\delta, h_\delta)$, gives $u_\delta(cZ_t) = c\,u_\delta(Z_t)$.

Lemma 107 (`lem-stage1-inversion-derivative-control`). *There exist universal $C_{\mathrm{der}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}} \geq 1$ and $\kappa_{\mathrm{der}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\varsigma \in \{+1, -1\}$ and every $0 < r \leq \delta$ satisfying*

$$C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}}, \qquad C_{\mathrm{pol}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{pol}}, \qquad C_{\mathrm{grp}}\varepsilon_r < \delta - C_{\mathrm{pol}}(\varepsilon_r \delta + \delta^2), \tag{172}$$

$$C_{\mathrm{der}}(\varepsilon_r + r) \leq \kappa_{\mathrm{der}}, \qquad (1+\varepsilon_r)\big(1 + C_{\mathrm{ch}}(\varepsilon_r + \delta)\big)r + C_{\mathrm{grp}}\varepsilon_r < 2\delta, \tag{173}$$

the globally defined $\sigma(U) = u_\delta(U^\dagger)$ maps $\chi_\varsigma\big(B_r^{i\mathcal{H}}(0)\big)$ into the same ςJ-graph chart, where $\chi_\varsigma(A) = \varsigma J \cdot \big(J + A + g_{\varsigma J}(A)\big)$, and $F_\varsigma(A) = \phi^{\|}_{\varsigma J}\big(\sigma(\chi_\varsigma(A))\big)$ satisfies

$$\|D(F_\varsigma - \mathrm{id})(A) + 2I_{i\mathcal{H}}\| \;\le\; C_{\mathrm{der}}(\varepsilon_r + r) \qquad \textit{for all } A \in B_r^{i\mathcal{H}}(0). \tag{174}$$

Transcription note. *The registry writes the sign as s; it is renamed ς here only to keep it apart from the scale variables. The second display of* (173) *is the* chart-retention guard*: it is what makes the coordinate identity $F_\varsigma(A) = B$ below legitimate, rather than a comparison of coordinates in two different charts.* Historical note (contract byte-unchanged throughout). *A first **af** validation of 2026-07-27 was RETRACTED on 2026-07-28: an independent adjudication found that its export node 1.3 identified the anaphorically bound group-law polar factor with the typed retraction inverse without a typed preimage witness (`docs/LEARNINGS.md`, entry 2026-07-28). The 2026-07-29 re-validation below replays the calculation on the typed explicit-closeness spine, where one polar pair is bound from the displayed retraction inverse and the closeness estimate speaks about that same pair; the defective substitution is absent by construction.*

Registry contract (verbatim).

```
Typed inversion derivative with chart retention: there exist universal C_der, C_ch, C_pol, C_grp >=
1 and kappa_der, kappa_ch, kappa_pol in (0, 1/2] such that for every finite-dimensional exact-unit
epsilon_r-C*-algebra, s in {+1, -1}, and 0 < r <= delta satisfying C_ch*(epsilon_r + delta) <=
kappa_ch, C_pol*(epsilon_r + delta) <= kappa_pol, C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta
+ delta^2), C_der*(epsilon_r + r) <= kappa_der, and (1 + epsilon_r)*(1 + C_ch*(epsilon_r +
delta))*r + C_grp*epsilon_r < 2*delta, the globally defined sigma(U) = u_delta(U^dagger) maps
chi_s(B_r^{icalH}(0)) into the same sJ-graph chart, where chi_s(A) = sJ bold-dot (J + A +
g_{sJ}(A)), and F_s(A) = phi_{sJ}^par(sigma(chi_s(A))) satisfies ||D(F_s - id)(A) + 2*I_{icalH}||
<= C_der*(epsilon_r + r) for all A in B_r^{icalH}(0).
```

Proof (prose account of the af-validated tree). *Provider alignment and typed synchronisation — before any calculation.* Fix the witnesses (C_g, k_g) of Lemma 99, (P_r, k_r) of Lemma 101 and (G_c, P_c, k_c) of Lemma 113, and set $C_{\mathrm{ch}} = \max\{1, C_g\}$, $\kappa_{\mathrm{ch}} = \min\{\frac{1}{2}, k_g\}$, $C_{\mathrm{pol}} = \max\{1, P_r, P_c\}$, $\kappa_{\mathrm{pol}} = \min\{\frac{1}{2}, k_r, k_c\}$ and $C_{\mathrm{grp}} = \max\{1, G_c\}$; the receiving guards (172) then imply each provider's own guard. Both polar providers *display* the identical map $\Pi_\delta(U, H) = U \cdot H$ on the identical domain $\mathcal{U} \times B_\delta^{\mathcal{H}}(J)$ with target its image S_δ, and both call its inverse unique; ordinary uniqueness of a set-theoretic inverse therefore identifies their displayed pairs (u_δ, h_δ). This is the step at which the retracted tree substituted one opaque factor for another; here there is a single typed pair, bound once. The pair $C_{\mathrm{der}}, \kappa_{\mathrm{der}}$ is chosen last, after the universal estimates below are fixed.

The scalar chart. Write $V = \varsigma J$, $G = g_V$ and $Z_T = T + G(T)$. Exact unitality, bilinearity, the involution laws and $\varsigma^2 = 1$ collapse (146) to $f_V(Z_T) = G(T) + \frac{1}{2} Z_T^\dagger \cdot Z_T$, so $G(0) = 0$ by the uniqueness clause of Lemma 99, and with $d = C_{\mathrm{ch}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{ch}} \le \frac{1}{2}$ its derivative bound gives $\|G(T)\| \le d\|T\|$. Differentiating the collapsed identity and absorbing the resulting DG term gives $\|DG(T)\| \le K_g r$ for $\|T\| \le 2r$, once $\varepsilon_r + r$ lies below a universal cutoff.

Typed adjoint factorisation. For $A \in B_r^{i\mathcal{H}}(0)$ the point $U = \chi_\varsigma(A)$ lies in $\mathcal{U}$ by Lemma 99. Applied to the *synchronised* pair, the C^1 inverse of Lemma 101 and the adjoint estimate of Lemma 113 speak about one and the same u_δ, so $W = \sigma(U) = u_\delta(U^\dagger)$ and $Q = h_\delta(U^\dagger) - J$ are C^1 in A with $W \in \mathcal{U}$, $Q \in \mathcal{H}$, $\|Q\| < \delta$,

$$U^\dagger = W \cdot (J + Q), \qquad \|W - U^\dagger\| \le C_{\mathrm{grp}}\varepsilon_r; \tag{175}$$

no unnamed or separately supplied polar factor occurs. Since $\varsigma J \in \mathcal{U}$ and $(\varsigma J)^\dagger = \varsigma J$, the retraction identity (153) also gives $\sigma(\varsigma J) = \varsigma J$.

The normal term. From (175), $U^\dagger - W = W \cdot Q$, so $\|W \cdot Q\| \le C_{\mathrm{grp}}\varepsilon_r$; approximate associativity together with $W^\dagger \cdot W = J$ and the C^*-lower bound $\|W\| \le (1-\varepsilon_r)^{-1/2}$ absorb the associator error into

$$\|Q\| \;\le\; K_q C_{\mathrm{grp}}\varepsilon_r. \tag{176}$$

Chart retention. The involution law and the product estimate give $\|U^\dagger - \varsigma J\| \le (1+\varepsilon_r)(1 + C_{\mathrm{ch}}(\varepsilon_r+\delta))\|A\|$, so the closeness estimate of (175) and the second guard of (173) give $\|W - \varsigma J\| < 2\delta$. As $L_{\varsigma J} = \varsigma I$ exactly, $C = \phi_{\varsigma J}(W)$ has norm below 2δ and contractive components; unitarity of W makes $f_{\varsigma J}(C) = 0$, so uniqueness in Lemma 99 forces $C^\perp = G(C^\parallel)$. Thus $W = \chi_\varsigma(B)$ with $B = C^\parallel = F_\varsigma(A)$ in the *same* chart, and these coordinate maps are C^1.

The coordinate identity and its derivative. Substituting the three factorisations and cancelling ς gives the exact identity $J - A + G(A) = (J + B + G(B)) \cdot (J + Q)$, whose anti-Hermitian part reads

$$-A \;=\; B + P^\parallel\big((B + G(B)) \cdot Q\big), \tag{177}$$

so $\|B\| \le 2r$ once the cutoff makes $(1+\varepsilon_r)(1+d)\|Q\| \le \frac{1}{2}$ in (176). Differentiating (177) in a unit direction ξ, with $P = DF_\varsigma(A)\xi$ and $R = DQ(A)\xi$, gives $-\xi + DG(A)\xi = P + DG(B)P + R + E$ with $E = (P + DG(B)P) \cdot Q + (B + G(B)) \cdot R$; typing by Hermitian and anti-Hermitian parts yields $P + \xi = -P^\parallel(E)$. The product bound with (176), $\|B\| \le 2r$ and $\|DG\| \le K_g r$ gives $\|E\| \le \alpha\|P\| + \beta\|R\|$ with $\alpha \le K_1\varepsilon_r$ and $\beta \le K_2 r$; a universal cutoff making $\alpha, \beta, K_g r \le \frac{1}{8}$ bounds $\|P\| \le 2$, then $\|R\| \le K_3(\varepsilon_r + r)$ and finally $\|P + \xi\| \le K_4(\varepsilon_r + r)$. Choosing $C_{\mathrm{der}} \ge \max\{1, K_4\}$ and $\kappa_{\mathrm{der}} \in (0, \frac{1}{2}]$ so that $C_{\mathrm{der}}(\varepsilon_r + r) \le \kappa_{\mathrm{der}}$ enforces every cutoff and absorption used above, homogeneity extends the last estimate to all ξ, and since $D(F_\varsigma - \mathrm{id}) + 2I_{i\mathcal{H}} = DF_\varsigma + I_{i\mathcal{H}}$, this is (174).

Lemma 108 (`lem-stage1-smooth-unitary-atlas`). *For every finite-dimensional exact-unit ε_r-C^*-algebra, if $\mathcal{U}$ is covered by the unique C^1 graph functions g_V of Lemma 99 and $D_{A^\perp} f_V$ is invertible at every graph point, then those same g_V are C^∞, and the same graph charts make $\mathcal{U}$ a smooth embedded manifold; no point or first derivative of a graph or chart is changed.* Transcription note. *The statement is qualitative and conditional: it introduces no smoothness threshold and no new coefficient, and it asserts nothing about* which δ *makes the hypothesis hold — that is* (148).

Registry contract (verbatim).

```
Smooth graph-atlas upgrade: for every finite-dimensional exact-unit epsilon_r-C*-algebra, if calU
is covered by the unique C^1 graph functions g_V of lem-stage1-unitary-graph-control and D_{A^perp}
f_V is invertible at every graph point, then those same g_V are C^infinity, and the same graph
charts make calU a smooth embedded manifold; no point or first derivative of a graph or chart is
changed.
```

Proof (prose account of the af-validated tree). *The defining map is polynomial.* By Definition 22, $\cdot$ is complex-bilinear and $\dagger$ conjugate-linear, hence real-bilinear and real-linear respectively; with the parenthesisation fixed as in (146), the map $\Phi_V(P, Q) = f_V(P + Q)$ is a real polynomial of degree at most two on $i\mathcal{H} \times \mathcal{H}$, so it is C^∞ in finite dimension. The exact involution axioms give $\Phi_V(P, Q) = \frac{1}{2}(B^\dagger \cdot B - J)$ with $B = V \cdot (J + P + Q)$, without reassociation, so Φ_V indeed takes values in $\mathcal{H}$.

Lee's implicit function theorem. Identify $i\mathcal{H}$ and $\mathcal{H}$ linearly with $\mathbb{R}^n$ and $\mathbb{R}^k$. At a graph point $(P, g_V(P))$ one has $\Phi_V = 0$, and the derivative of $Q \mapsto \Phi_V(P, Q)$ is exactly $D_{A^\perp} f_V(P + g_V(P))$, invertible by hypothesis. Applying, with $c = 0$, Theorem C.40 (Implicit Function Theorem) of J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer GTM 218, 2012 — registered in the workspace as the byte-matched external `GT-lee-2ed-thm-C.40` against the pinned local source at the recorded locus — produces neighbourhoods O_P, O_Q and a C^∞ map $F : O_P \to O_Q$ whose graph is exactly $\Phi_V^{-1}(0) \cap (O_P \times O_Q)$.

Identification with the given graph. g_V is C^1, hence continuous, takes values in the permitted $\mathcal{H}$-ball, solves $\Phi_V(P', g_V(P')) = 0$ and is the unique such branch; shrinking O_P so that $g_V(O_P) \subseteq O_Q$, the zero-set equivalence forces $g_V = F$ on O_P. Doing this at every P shows the *same* g_V is C^∞ on its original domain; because this is an equality with the pre-existing C^1 function, its values and its first derivative are literally unchanged.

The atlas. In each graph coordinate the parametrisation $P \mapsto (P, g_V(P))$ is now smooth and its inverse on the graph is the real-linear projection to $i\mathcal{H}$; the chart transports ϕ_V of Definition 32 are affine-linear. On an overlap the transition is a target projection composed with a source parametrisation, hence smooth, and symmetrically for its inverse. The cover is the cover already asserted by Lemma 99, and every set, function and coordinate map used is the old one, so $\mathcal{U}$ is a smooth embedded manifold with nothing replaced.

Status. All three rows of this section are `proved` with `af: validated` (T0). Workspaces: `proofs/lem-stage1-polar-path-admissibility` (12 nodes, 12 `validated`, taint clean, no challenges); `proofs/lem-stage1-smooth-unitary-atlas` (14 nodes, 14 `validated`, taint clean), which registers three imports — the validated contracts of `lem-stage1-rectified-cstar-control` and `lem-stage1-unitary-graph-control`, and the byte-matched external `GT-lee-2ed-thm-C.40`; and `proofs/lem-stage1-inversion-derivative-control`, af-**RE**-VALIDATED 2026-07-29 by a clean re-seed on the amended typed dependency spine (root validated, 10/10 live nodes, taint clean, 2 rounds, ZERO challenges), after its 2026-07-27 validation (also 10 nodes, no challenges) had been RETRACTED 2026-07-28 on the adjudicated synchronisation defect recorded above (`docs/LEARNINGS.md`). The re-validated tree is not a repair of the old one: the workspace was re-seeded from empty, so the defective node-1.3 substitution is absent by construction.

Correction of record (proof-level, not contract-level). The atlas tree drew one genuine challenge: the transition-map node leaned on a *pending* sibling for the C^∞ smoothness of g_V, which the protocol forbids. It was repaired in run by a self-contained smoothness bridge that re-derives the Lee argument and the identification $g_V = F$ inside the node itself. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space`, `def-epsilon-cstar-algebra`. Registry `deps`: `lem-stage1-polar-retraction` and `lem-stage1-polar-coherence-naturality` (§45) for the path row; `lem-stage1-unitary-graph-control` (§44), `lem-stage1-polar-retraction` and `lem-stage1-explicit-group-closeness` (§49) for the inversion row — the 2026-07-28 deps amendment that dropped the anaphoric `lem-stage1-approximate-group-laws` and `lem-stage1-polar-coherenc` in favour of the typed explicit provider, with the contract and `defs` byte-unchanged; `lem-stage1-rectified-cstar` and `lem-stage1-unitary-graph-control` for the atlas. Nothing in this document assembles the Stage-1 front into a statement about almost-idempotent maps, and `op-classical` is now **discharged** at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

48 The smooth polar inverse, smooth operations, and the polar arithmetic

The three rows of this section continue the Stage-1 polar block begun in §44. The first two are smoothness upgrades in the style of Lemma 108: they replace no map, no point and no first derivative, but promote the C^1 regularity of the polar inverse and of the approximate-group operations to C^∞ for the embedded smooth structure. Of these, only the first is `af`-validated; the operations row is **retired in place** and its live content is carried by the explicit smooth bridge of §49. The third row is pure scalar arithmetic: it packages, once and for all, the guard inequalities that every analytic row of the block demands, so that downstream consumers can instantiate the whole front from a single smallness assumption on ε_X.

Lemma 109 (`lem-stage1-smooth-polar-inverse`). *For every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$: if Lemma 108 gives the smooth embedded atlas and $\Pi_\delta : \mathcal{U} \times B^{\mathcal{H}}_\delta(J) \to S_\delta$ is the bijective C^1 local diffeomorphism of Lemma 101 onto an open set, then the same ambient-bilinear Π_δ is a smooth diffeomorphism and its same set-theoretic inverse (u_δ, h_δ) is smooth; no point or first derivative is changed.* Transcription note. *Like the atlas row, this is qualitative and conditional: both hypotheses are supplied by the two named lemmas, no new threshold is introduced, and the conclusion re-asserts nothing about* which δ *realises them.*

Registry contract (verbatim).

```
Smooth polar-inverse upgrade: for every finite-dimensional exact-unit epsilon_r-C*-algebra and
every delta > 0, if lem-stage1-smooth-unitary-atlas gives the smooth embedded atlas and Pi_delta:
calU x B^{calH}_delta(J) -> S_delta is the bijective C^1 local diffeomorphism of
lem-stage1-polar-retraction onto an open set, then the same ambient-bilinear Pi_delta is a smooth
diffeomorphism and its same set-theoretic inverse (u_delta, h_delta) is smooth; no point or first
derivative is changed.
```

Proof (prose account of the af-validated tree). *The two sides are smooth manifolds.* Lemma 108 equips $\mathcal{U}$ with the smooth embedded graph atlas whose charts and first derivatives are those of the C^1 atlas; $\mathcal{H}$ is a real linear subspace of the finite-dimensional ambient space (Definition 32), so $B^{\mathcal{H}}_\delta(J)$ is an open subset of a finite-dimensional real vector space, and $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ carries the product smooth structure; S_δ is open in the ambient space by Lemma 101 and carries its open-subset structure.

Π_δ *is smooth.* By Definition 22 the product $\cdot$ is bilinear, so in real coordinates each component of $(X, H) \mapsto X \cdot H$ is a quadratic polynomial, hence C^∞. In a product chart built from a graph chart of the atlas and the identity chart on the ball, the coordinate representative of Π_δ is this polynomial map composed with the smooth ambient parametrisation of the embedded chart, hence smooth; such charts cover the domain and the values lie in S_δ, so Π_δ is smooth.

Everywhere-invertible derivative. At any point p the C^1 local-diffeomorphism assertion of Lemma 101 provides a local C^1 inverse G; the chain rule applied to $G \circ \Pi_\delta = \mathrm{id}$ and $\Pi_\delta \circ G = \mathrm{id}$ shows $D\Pi_\delta(p)$ is a linear isomorphism.

Smooth local inverses, glued. Fix p, put $q = \Pi_\delta(p)$, and choose smooth charts α about p and β about q. The coordinate map $F = \beta \circ \Pi_\delta \circ \alpha^{-1}$ is smooth with invertible derivative at $\alpha(p)$; Theorem C.34 (Inverse Function Theorem) of J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer GTM 218, 2012 — registered in the workspace as the byte-matched external `GT-lee-2ed-thm-C.34` against the pinned local source at the recorded locus — makes F a smooth diffeomorphism between neighbourhoods, so Π_δ has a smooth local inverse near q. By the bijectivity assertion of Lemma 101 the global set-theoretic inverse exists; on each such target neighbourhood it coincides with the unique smooth local inverse there, and these neighbourhoods cover S_δ, so the global inverse is smooth because smoothness is local — exactly the gluing recorded in Corollary C.36(b) of the same source, the byte-matched external `GT-lee-2ed-cor-C.36`. A bijection has only one set-theoretic inverse, so this smooth inverse *is* the pair (u_δ, h_δ) already supplied.

Nothing is changed. The atlas upgrade is identity-on-data through first order, the map is literally the same $\Pi_\delta(U, H) = U \bullet H$ on the same domain and codomain, and the inverse is the same set map; the derivative of a C^1 function in unchanged first-order charts is unique, so proving these same functions C^∞ changes no point value and no first derivative.

Conjecture 110 (`lem-stage1-smooth-unitary-operations` — proof retracted 2026-07-28)**.** *Under Lemmas 105, 108 and 109, the scalar action $U(1) \times \mathcal{U} \to \mathcal{U}$, $(c, U) \mapsto cU$, and the same maps $\mu : \mathcal{U} \times \mathcal{U} \to \mathcal{U}$, $\mu(U, V) = u_\delta(U \bullet V)$, and $\sigma : \mathcal{U} \to \mathcal{U}$, $\sigma(U) = u_\delta(U^\dagger)$, are smooth as maps into the embedded manifold $\mathcal{U}$; they obey*

$$\mu(cU, dV) = cd\,\mu(U, V), \qquad \sigma(cU) = \bar{c}\,\sigma(U) \qquad (c, d \in U(1)), \tag{178}$$

and no point or first derivative is changed. Transcription note. *The registry* `deps` *also list Lemma 102, which the contract exercises through its scalar-naturality clause rather than naming it in the antecedent.* Retraction note (2026-07-28). *The* `af` *validation was RETRACTED by the Stage-1 binder sweep: export nodes 1.2.1–1.2.2, 1.3.1.2 and 1.3.2 attach the typed smooth polar inverse to the anaphorically bound group-laws maps without a typed witness (the scalar-action subargument alone is sound but does not certify the compound root; see* `docs/LEARNINGS.md`*, second entry of 2026-07-28). The statement stands as a* stated *candidate; an explicit-binder smooth-operations bridge is part of the W97 campaign.*

Registry contract (verbatim).

```
Smooth action/operations upgrade: under lem-stage1-approximate-group-laws,
lem-stage1-smooth-unitary-atlas, and lem-stage1-smooth-polar-inverse, the scalar action U(1) x calU
-> calU, (c, U) |-> cU, and the same maps mu: calU x calU -> calU, mu(U, V) = u_delta(U bold-dot
V), and sigma: calU -> calU, sigma(U) = u_delta(U^dagger), are smooth as maps into the embedded
manifold calU; they obey mu(cU, dV) = c*d*mu(U, V) and sigma(cU) = conj(c)*sigma(U), and no point
or first derivative is changed.
```

Historical record: the RETRACTED proof attempt (not a proof; defect marked). *The circle preserves $\mathcal{U}$ (this subargument is sound).* If $U^\dagger \bullet U = J$ and $U \bullet R = J$, then for $c \in U(1)$ conjugate-linearity of † and bilinearity give $(cU)^\dagger \bullet (cU) = \bar{c}c\,J = J$ and $(cU) \bullet (\bar{c}R) = J$, so $cU \in \mathcal{U}$ with the explicit right inverse $\bar{c}R$. This local derivation is repeated wherever membership is needed below, deliberately never citing a sibling node. *The defect lies downstream: the tree's smoothness and naturality steps attach the typed smooth polar inverse to the anaphorically bound group-laws maps without a typed witness (export nodes 1.2.1–1.2.2, 1.3.1.2, 1.3.2) — the sweep-adjudicated attachment defect; the compound root is therefore unproved.*

The scalar action is smooth. Since $J \in B_\delta^{\mathcal{H}}(J)$ and $\Pi_\delta(U, J) = U \bullet J = U$ by exact unitality, both U and cU lie in the open set S_δ; the ambient map $(c, U) \mapsto cU$, the restriction of complex scalar multiplication, is smooth into S_δ, and composing with the smooth $u_\delta : S_\delta \to \mathcal{U}$ of Lemma 109 gives a smooth $\mathcal{U}$-valued map. The scalar-naturality clause (159) of Lemma 102 gives $u_\delta(cU) = c\,u_\delta(U) = cU$, so this smooth composite *is* the action itself.

μ and σ are smooth. By Lemma 105 the inputs $U \cdot V$ and $U^\dagger$ lie in S_δ for all $U, V \in \mathcal{U}$ and the displayed formulas are the same globally defined C^1 maps. The ambient product is bilinear and $\dagger$ is conjugate-linear, hence both restrict to smooth S_δ-valued maps on the embedded domain; composing with the smooth u_δ makes μ and σ smooth into $\mathcal{U}$.

Covariance. By the local membership derivation, $cU, dV \in \mathcal{U}$, so Lemma 105 puts $X = U \cdot V$ and $Y = (cU) \cdot (dV)$ in S_δ; bilinearity gives $Y = (cd)X$ with $cd \in U(1)$, and (159) gives $u_\delta(Y) = cd\, u_\delta(X)$, the first identity of (178). Likewise $(cU)^\dagger = \bar{c}U^\dagger$ with $\bar{c} \in U(1)$, both $U^\dagger$ and $(cU)^\dagger$ lie in S_δ, and (159) with scalar $\bar{c}$ gives $\sigma(cU) = \bar{c}\,\sigma(U)$.

Nothing is changed. The atlas and the polar inverse retain their point values and first derivatives by Lemmas 108 and 109; the ambient scalar, product and adjoint maps are untouched; and μ, σ keep the identical formulas of Lemma 105. Uniqueness of derivatives through the unchanged component derivatives leaves every first derivative the pre-existing one.

Lemma 111 (`lem-stage1-polar-scalar-arithmetic`)**.** *For every $C_{\mathrm{rect}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}}, C_{\mathrm{path}}, C_{\mathrm{der}} \geq 1$, every $e_{\mathrm{rect}} \in (0, 1/C_{\mathrm{rect}}]$ and every $\kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}} \in (0, \frac{1}{2}]$, set*

$$\begin{aligned} \delta_* &= \min\Big\{\tfrac{1}{4},\ \tfrac{\kappa_{\mathrm{ch}}}{4C_{\mathrm{ch}}},\ \tfrac{\kappa_{\mathrm{pol}}}{4C_{\mathrm{pol}}}\Big\}, &\qquad \varepsilon_*^r &= \min\Big\{\tfrac{1}{4},\ \tfrac{\kappa_{\mathrm{ch}}}{4C_{\mathrm{ch}}},\ \tfrac{\kappa_{\mathrm{pol}}}{4C_{\mathrm{pol}}},\ \tfrac{\kappa_{\mathrm{der}}}{8C_{\mathrm{der}}},\ \tfrac{1}{C_{\mathrm{grp}}},\ \tfrac{\delta_*}{12C_{\mathrm{path}}C_{\mathrm{grp}}}\Big\}, \\ e_{S1} &= \min\Big\{e_{\mathrm{rect}},\ \tfrac{\varepsilon_*^r}{C_{\mathrm{rect}}}\Big\}, & r_{\mathrm{iso}} &= \min\Big\{\tfrac{\delta_*}{4},\ \tfrac{\kappa_{\mathrm{der}}}{8C_{\mathrm{der}}}\Big\}, \end{aligned} \tag{179}$$

and, given $0 \leq \varepsilon_X \leq e_{S1}$, put $\varepsilon_r = C_{\mathrm{rect}}\varepsilon_X$, $q = C_{\mathrm{grp}}\varepsilon_r$, $r_- = \delta_* - C_{\mathrm{pol}}(\varepsilon_r\delta_* + \delta_*^2)$ and $\eta = C_{\mathrm{path}}(q + \varepsilon_r q + q^2)$. Then every such ε_X satisfies

$$C_{\mathrm{ch}}(\varepsilon_r + \delta_*) \leq \kappa_{\mathrm{ch}}, \qquad C_{\mathrm{pol}}(\varepsilon_r + \delta_*) \leq \kappa_{\mathrm{pol}}, \qquad q < r_-, \qquad C_{\mathrm{path}}q \leq \tfrac{1}{4}, \tag{180}$$

$$\eta < r_-, \qquad C_{\mathrm{der}}(\varepsilon_r + r_{\mathrm{iso}}) \leq \kappa_{\mathrm{der}}, \qquad (1+\varepsilon_r)\big(1 + C_{\mathrm{ch}}(\varepsilon_r + \delta_*)\big)r_{\mathrm{iso}} + q < 2\delta_*; \tag{181}$$

moreover $r_- \geq 3\delta_*/4$, $\eta \leq \delta_*/4$ and $C_{\mathrm{der}}(r_{\mathrm{iso}} + \varepsilon_r) \leq \kappa_{\mathrm{der}}/4 < 1$. Transcription note. *The row is universal over every already-selected coefficient and margin tuple and carries* no *analytic content: its ten inputs and four derived scales are exactly the fields of the witness tuple of Definition 44, and the seven displayed guards are the hypotheses of the analytic rows of §§44–47 at the scales $\delta = \delta_*$ and $r = r_{\mathrm{iso}}$.*

Registry contract (verbatim).

```
Universal Stage-1 polar arithmetic: for every C_rect, C_ch, C_pol, C_grp, C_path, C_der >= 1,
e_rect in (0, 1/C_rect], and kappa_ch, kappa_pol, kappa_der in (0, 1/2], setting delta_* = min{1/4,
kappa_ch/(4*C_ch), kappa_pol/(4*C_pol)}, epsilon_*^r = min{1/4, kappa_ch/(4*C_ch),
kappa_pol/(4*C_pol), kappa_der/(8*C_der), 1/C_grp, delta_*/(12*C_path*C_grp)}, e_S1 = min{e_rect,
epsilon_*^r/C_rect}, r_iso = min{delta_*/4, kappa_der/(8*C_der)}, epsilon_r = C_rect*epsilon_X, q =
C_grp*epsilon_r, r_- = delta_* - C_pol*(epsilon_r*delta_* + delta_*^2), and eta = C_path*(q +
epsilon_r*q + q^2), every 0 <= epsilon_X <= e_S1 satisfies C_ch*(epsilon_r + delta_*) <= kappa_ch,
C_pol*(epsilon_r + delta_*) <= kappa_pol, q < r_-, C_path*q <= 1/4, eta < r_-, C_der*(epsilon_r +
r_iso) <= kappa_der, and (1 + epsilon_r)*(1 + C_ch*(epsilon_r + delta_*))*r_iso + q < 2*delta_*;
moreover r_- >= 3*delta_*/4, eta <= delta_*/4, and C_der*(r_iso + epsilon_r) <= kappa_der/4 < 1.
```

Proof (prose account of the af-validated tree). *Channel and polar guards.* From $e_{S1} \leq \varepsilon_*^r/C_{\mathrm{rect}}$ and $C_{\mathrm{rect}} \geq 1$, $0 \leq \varepsilon_r \leq \varepsilon_*^r$. Since both ε_*^r and δ_* are at most $\kappa_{\mathrm{ch}}/(4C_{\mathrm{ch}})$ and $\kappa_{\mathrm{pol}}/(4C_{\mathrm{pol}})$, the first two guards hold with value at most $\kappa/2$; and since $r_- = \delta_*\big[1 - C_{\mathrm{pol}}(\varepsilon_r + \delta_*)\big]$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta_*) \leq \kappa_{\mathrm{pol}}/2 \leq \frac{1}{4}$, one gets $r_- \geq 3\delta_*/4$.

Path guards. $\varepsilon_r \le \delta_*/(12C_{\mathrm{path}}C_{\mathrm{grp}})$ gives $0 \le q \le \delta_*/(12C_{\mathrm{path}}) \le \delta_*/12$, hence $C_{\mathrm{path}}q \le \delta_*/12 \le \frac{1}{48} < \frac{1}{4}$ (using $\delta_* \le \frac{1}{4}$) and $q < 3\delta_*/4 \le r_-$. From $\varepsilon_r \le \frac{1}{4}$ and $q \le \frac{1}{48}$, $1+\varepsilon_r+q \le \frac{61}{48} < 3$, so $\eta = C_{\mathrm{path}}q\,(1+\varepsilon_r+q) \le \delta_*/4 < 3\delta_*/4 \le r_-$.

Derivative guards. $\varepsilon_r \le \kappa_{\mathrm{der}}/(8C_{\mathrm{der}})$ and $r_{\mathrm{iso}} \le \kappa_{\mathrm{der}}/(8C_{\mathrm{der}})$ give $C_{\mathrm{der}}(\varepsilon_r + r_{\mathrm{iso}}) \le \kappa_{\mathrm{der}}/4$; and $0 < \kappa_{\mathrm{der}} \le \frac{1}{2}$ makes $\kappa_{\mathrm{der}}/4 \le \kappa_{\mathrm{der}}$ and $\kappa_{\mathrm{der}}/4 \le \frac{1}{8} < 1$.

The product guard. The minima give $\varepsilon_r \le \frac{1}{4}$, $C_{\mathrm{ch}}(\varepsilon_r+\delta_*) \le \frac{1}{4}$, $r_{\mathrm{iso}} \le \delta_*/4$ and $q \le \delta_*/12$, so

$$(1+\varepsilon_r)\big(1+C_{\mathrm{ch}}(\varepsilon_r+\delta_*)\big)r_{\mathrm{iso}} + q \;\le\; \tfrac{5}{4}\cdot\tfrac{5}{4}\cdot\tfrac{\delta_*}{4} + \tfrac{\delta_*}{12} \;=\; \tfrac{91}{192}\,\delta_* \;<\; 2\delta_*, \tag{182}$$

strictly, because $\delta_* > 0$.

Status. `lem-stage1-smooth-polar-inverse` and `lem-stage1-polar-scalar-arithmetic` are `proved` with `af: validated` (T0, re-certified by the 2026-07-28 sweep): workspaces `proofs/lem-stage1-smooth-` (21 nodes, 21 `validated`, taint clean, no challenges; registers the byte-matched externals `GT-lee-2ed-thm-C.34` and `GT-lee-2ed-cor-C.36` together with the validated contracts of its two registry deps) and `proofs/lem-stage1-polar-scalar-arithmetic` (15 nodes, 15 `validated`, taint clean, no challenges; a leaf with no registry deps). `lem-stage1-smooth-unitary-operations` is `stated` / `af: seeded`: its validation (15 nodes) was RETRACTED 2026-07-28 on the sweep-adjudicated attachment defect above (`docs/LEARNINGS.md`).

Retired in place; the live replacement path. The operations row is **not** being re-elevated. Its live content re-enters through the binder-closed `lem-stage1-explicit-smooth-unitary-operations` (§49), `af`-validated 2026-07-29, whose contract quantifies explicitly over the graph family, the smooth structure, the displayed Π_δ with both typed inverse identities and the C^1 maps $\alpha_{C^1}, \mu_{C^1}, \sigma_{C^1}$, so that every formerly anaphoric map has a contract-local typed referent. It uses neither the retired parent nor `lem-stage1-polar-coherence-naturality`.

Corrections of record (proof-level, not contract-level). The operations tree drew two genuine cross-sibling challenges: as first written, the covariance nodes leaned on the *pending* sibling that made the scalar action $\mathcal{U}$-valued, which the protocol forbids. Both were repaired in run by the self-contained local membership derivation reproduced above, so no covariance node cites the action node. Its first run also aborted on a *spurious* prover-overreach trip — the orchestrator's own controller bookkeeping, not the prover, had dirtied `.frontier/`; the guard was exempted for that path and the run repeated. No contract was amended.

Role in the chain. Registry `defs`: `def-approximate-unitary-space` and `def-epsilon-cstar-algebra` for the two upgrades; `def-stage1-polar-witness-data` for the arithmetic row. Registry `deps`: `lem-stage1-polar-retraction` (§45) and `lem-stage1-smooth-unitary-atlas` (§47) for the smooth polar inverse; those two conclusions together with `lem-stage1-polar-coherence-naturality` and `lem-stage1-approximate-group-laws` (§46) for the operations; the arithmetic row has none. The retired operations row has no live dependents. Nothing in this document assembles the Stage-1 front into a statement about almost-idempotent maps, and `op-classical` is now **discharged** at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

49 The explicit binder bridges of the Stage-1 polar block

The three rows of this section are the binder-closed rebuild that replaces the two rows retired in §§46 and 48.

What binder closure means here. The retracted proofs failed for one reason: a contract that merely says "the inverse u_δ of the polar map" exports an *anaphoric* first component — a symbol with no typed companion h_δ, no displayed source map and no preimage identity — so two such components, arriving from two different imports, cannot legitimately be identified with each other. A contract is *binder-closed* when every map it mentions has a contract-local typed referent. In the two group rows below this is achieved by *displaying* $\Pi_\delta(U, H) = U \cdot H$ on $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ with target its own image S_δ and binding the typed pair (u_δ, h_δ) as *the* inverse of that displayed map — before either group input is treated; ordinary uniqueness of a set-theoretic inverse then synchronises any two consumers automatically. In the smooth row it is achieved by quantifying explicitly over the graph family, the smooth structure, the displayed Π_δ with both inverse identities, and the C^1 maps themselves.

Lemma 112 (`lem-stage1-explicit-group-domain-membership`). *There exist universal $C_{\rm grp}, C_{\rm pol} \geq 1$ and $\kappa_{\rm pol} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ satisfying* (160)*, writing (u_δ, h_δ) for the unique inverse of*

$$\Pi_\delta : \mathcal{U} \times B^{\mathcal{H}}_\delta(J) \longrightarrow S_\delta := \Pi_\delta\big(\mathcal{U} \times B^{\mathcal{H}}_\delta(J)\big), \qquad \Pi_\delta(U, H) = U \cdot H, \tag{183}$$

the first inverse component $u_\delta : S_\delta \to \mathcal{U}$ is defined at $U \cdot V$ and at $U^\dagger$ for every $U, V \in \mathcal{U}$; moreover $U \cdot V$ and $U^\dagger$ each have a right inverse. Transcription note. *The statement of Lemma 103 is recovered word for word once its anaphoric "$u_\delta : S_\delta \to \mathcal{U}$ of the polar map" is replaced by the displayed* (183)*; the mathematical content is the same and the sound anaphoric row is evidence for the calculation, not an import of it.*

Registry contract (verbatim).

```
Explicit group-input polar-domain membership: there exist universal C_grp, C_pol >= 1, kappa_pol in
(0, 1/2] such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0
satisfying C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta -
C_pol*(epsilon_r*delta + delta^2), writing (u_delta, h_delta) for the unique inverse of Pi_delta:
calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H) = U
bold-dot H, the first inverse component u_delta:S_delta -> calU is defined at U bold-dot V and
U^dagger for every U, V in calU; moreover, U bold-dot V and U^dagger each have a right inverse.
```

Proof (prose account of the af-validated tree). Take $C_{\rm pol}, \kappa_{\rm pol}$ from Lemma 101 and $C_{\rm grp} = 4$. *Before any group input is chosen*, apply that lemma under the first guard to obtain the displayed map (183) and bind its unique typed inverse $(u_\delta, h_\delta) : S_\delta \to \mathcal{U} \times B^{\mathcal{H}}_\delta(J)$; write $e = \varepsilon_r$ and $t = \delta - C_{\rm pol}(e\delta + \delta^2)$, so that the inner inclusion of (154) reads $\mathcal{U}_t \subseteq S_\delta$. Only now fix $U, V \in \mathcal{U}$.

Guard arithmetic (cited by every later smallness step). With $s = e + \delta$, the first guard and $C_{\rm pol} \geq 1$, $\kappa_{\rm pol} \leq \frac{1}{2}$ give $s \leq \frac{1}{2}$ and $\delta \leq \frac{1}{2}$; the second gives $4e < t$. Since $t = \delta(1 - C_{\rm pol}s) \leq \delta(1 - s) = (s - e)(1 - s)$, whose maximum over $e \leq s \leq \frac{1}{2}$ is $(\frac{1}{2} - e)/2$, the chain $e < t \leq (\frac{1}{2} - e)/2$ forces $e < \frac{1}{6}$.

The common left multiplier. For $W \in \mathcal{U}$, $W^\dagger \cdot W = J$ with the C^*-lower bound gives $(1-e)\|W\|^2 \le 1$, hence $\|W\| \le \frac{6}{5}$, while the associator axiom gives $\|L_{W^\dagger} L_W - I\| \le e\|W\|^2 \le e/(1-e) < \frac{1}{5}$. So L_W is injective, therefore bijective in finite dimension, with $\|L_{W^\dagger}\| \le \frac{7}{5}$ and $\|L_W^{-1}\| \le \frac{7}{4}$.

The product. For $X = U \cdot V$ the star rule gives $X^\dagger = V^\dagger \cdot U^\dagger$; inserting $((V^\dagger \cdot U^\dagger) \cdot U) \cdot V$ and $(V^\dagger \cdot (U^\dagger \cdot U)) \cdot V$ costs two associators and lands on J, so

$$\|X^\dagger \cdot X - J\| \ \le\ 2e(1+e)\|U\|^2\|V\|^2 \ \le\ 4e \ <\ t \ <\ 2t. \tag{184}$$

Approximate associativity gives $\|L_X - L_U L_V\| \le e\|U\|\|V\| < \frac{1}{5}$ while $\|(L_U L_V)^{-1}\| \le \frac{49}{16}$, so the Neumann perturbation is $< \frac{49}{80} < 1$: L_X is invertible and $R_X = L_X^{-1} J$ is a right inverse of X.

The adjoint. $U^\dagger \cdot U = J$ exhibits U as a right inverse of $U^\dagger$. The right inverse R of U supplied by Definition 32 equals $L_U^{-1} J$, so $\|R\| \le \frac{7}{4}$; one associator gives $\|U^\dagger - R\| \le e\|U\|^2\|R\|$ and hence $\|U \cdot U^\dagger - J\| \le (1+e)e\|U\|^3\|R\| \le \frac{441}{125}e < 4e < t < 2t$. Both inputs therefore lie in $\mathcal{U}_t \subseteq S_\delta$, which is the domain of the map u_δ bound at the outset; since the algebra, δ, U and V were arbitrary, this proves every clause.

Lemma 113 (`lem-stage1-explicit-group-closeness`). *There exist universal $C_{\mathrm{grp}}, C_{\mathrm{pol}} \ge 1$ and $\kappa_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ satisfying* (160)*, the first component of the unique inverse (u_δ, h_δ) of the displayed map* (183) *satisfies*

$$\|u_\delta(U \cdot V) - U \cdot V\| \le C_{\mathrm{grp}}\varepsilon_r, \qquad \|u_\delta(U^\dagger) - U^\dagger\| \le C_{\mathrm{grp}}\varepsilon_r \qquad (U, V \in \mathcal{U}). \tag{185}$$

Registry contract (verbatim).

```
Explicit group-input polar closeness: there exist universal C_grp, C_pol >= 1, kappa_pol in (0,
1/2] such that for every finite-dimensional exact-unit epsilon_r-C*-algebra and delta > 0
satisfying C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta -
C_pol*(epsilon_r*delta + delta^2), writing (u_delta, h_delta) for the unique inverse of Pi_delta:
calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H) = U
bold-dot H, the first inverse component u_delta:S_delta -> calU satisfies ||u_delta(U bold-dot V) -
U bold-dot V|| <= C_grp*epsilon_r and ||u_delta(U^dagger) - U^dagger|| <= C_grp*epsilon_r for every
U, V in calU.
```

Proof (prose account of the af-validated tree). Take C_{pol} from Lemma 101, $\kappa_{\mathrm{pol}} = \min(\kappa^0_{\mathrm{pol}}, \frac{1}{16})$ and $C_{\mathrm{grp}} = 5$, and put $\rho = \delta - C_{\mathrm{pol}}(\varepsilon_r\delta + \delta^2)$, so that $\varepsilon_r + \delta \le \frac{1}{16}$ and $\rho > 5\varepsilon_r \ge 0$. Under these guards that lemma supplies the displayed (183) together with its unique typed pair: for every $X \in S_\delta$ one has $X = u_\delta(X) \cdot h_\delta(X)$ with $u_\delta(X) \in \mathcal{U}$ and $\|h_\delta(X) - J\| < \delta$, and $\mathcal{U}_\rho \subseteq S_\delta$.

Typing the two inputs. For $W \in \mathcal{U}$ the C^*-lower bound gives $\|W\|^2 \le 1/(1-\varepsilon_r) \le \frac{16}{15}$; with R the right inverse of W, exact unitality and one associator give $\|R - W^\dagger\| \le \varepsilon_r\|W\|^2\|R\|$, hence $\|R\| \le \frac{15}{14}\|W\|$ and $\|W \cdot W^\dagger - J\| = \|W \cdot (W^\dagger - R)\| \le 2\varepsilon_r$. Two associators as in (184) give $\|X^\dagger \cdot X - J\| \le \frac{544}{225}\varepsilon_r \le 3\varepsilon_r$ for $X = U \cdot V$, and if $X \cdot Y = 0$ then $\|Y\| \le \big((1+\varepsilon_r)3\varepsilon_r + \varepsilon_r\|X\|^2\big)\|Y\| < \frac{1}{2}\|Y\|$, so L_X is injective, hence surjective in finite dimension, and X has a right inverse. Both defects are strictly below 2ρ — *including at $\varepsilon_r = 0$*, where they vanish while $\rho > 0$ — so $U \cdot V, U^\dagger \in \mathcal{U}_\rho \subseteq S_\delta$ and the pair fixed at the outset may be evaluated at both.

The estimate. Let $X \in S_\delta$ with $\|X^\dagger \cdot X - J\| \le 3\varepsilon_r$, and write $u = u_\delta(X)$, $h = h_\delta(X)$, $a = h - J$, so $X = u \cdot h$ and $\|a\| < \delta$. Two associator comparisons of $(h \cdot u^\dagger) \cdot (u \cdot h)$ against $h \cdot h$ give

$$\|X^\dagger \cdot X - h \cdot h\| \ \le\ 2\varepsilon_r(1+\varepsilon_r)\|h\|^2\|u\|^2 \ \le\ 3\varepsilon_r. \tag{186}$$

Exact unitality and bilinearity give $h \cdot h - J = 2a + a \cdot a$, so $2\|a\| \le 6\varepsilon_r + (1+\varepsilon_r)\|a\|^2$; as $\|a\| < \delta \le \frac{1}{16}$ the quadratic term is at most $\frac{17}{256}\|a\|$ and absorbs, leaving $\|a\| \le \frac{1536}{495}\varepsilon_r \le 4\varepsilon_r$. Exact right unitality then gives $\|u_\delta(X) - X\| = \|u \cdot (J-h)\| \le (1+\varepsilon_r)\|u\|\|a\| \le \frac{17}{\sqrt{15}}\varepsilon_r \le 5\varepsilon_r$. Applying this to the two inputs typed above is (185).

Lemma 114 (`lem-stage1-explicit-smooth-unitary-operations`). *Fix a finite-dimensional exact-unit ε_r-C^*-algebra and $\delta > 0$, and suppose given: a family $g = (g_V)_{V \in \mathcal{U}}$ of C^1 maps $g_V : B^{i\mathcal{H}}_{2\delta}(0) \to B^{\mathcal{H}}_{2\delta}(0)$ such that $g_V(A^{\|})$ is the* unique *$A^{\perp} \in B^{\mathcal{H}}_{2\delta}(0)$ with $f_V(A^{\|} + A^{\perp}) = 0$, where $f_V(A) = \frac{1}{2}\big(((J + A^\dagger) \cdot V^\dagger) \cdot (V \cdot (J + A)) - J\big)$, and such that $\chi_V(A^{\|}) = V \cdot (J + A^{\|} + g_V(A^{\|}))$ form a C^1 graph atlas $\mathcal{A}_\delta = \{\chi_V\}_{V \in \mathcal{U}}$ covering $\mathcal{U}$; a smooth embedded-manifold structure $\mathcal{M}_\delta$ on the underlying set $\mathcal{U}$ for which $\mathcal{A}_\delta$ is a C^∞ atlas and every displayed g_V, χ_V is C^∞ with exactly its displayed point values and C^1 differentials; the displayed Π_δ of* (183)*, assumed a C^1 diffeomorphism onto the open S_δ with inverse (u_δ, h_δ) characterised by both inverse identities, smooth relative to $\mathcal{M}_\delta$ with the same point values and differentials; and the memberships $U \cdot V, U^\dagger \in S_\delta$ for all $U, V \in \mathcal{U}$. Then, writing $\alpha_{C^1}(c, U) = cU$, $\mu_{C^1}(U, V) = u_\delta(U \cdot V)$ and $\sigma_{C^1}(U) = u_\delta(U^\dagger)$ for the resulting C^1 maps, there exist maps α, μ, σ, smooth relative to $\mathcal{M}_\delta$, with $\alpha = \alpha_{C^1}$, $\mu = \mu_{C^1}$, $\sigma = \sigma_{C^1}$ pointwise, $D\alpha = D\alpha_{C^1}$, $D\mu = D\mu_{C^1}$, $D\sigma = D\sigma_{C^1}$, and*

$$\mu(cU, dV) = cd\,\mu(U, V), \qquad \sigma(cU) = \bar{c}\,\sigma(U) \qquad (c, d \in U(1)). \tag{187}$$

Transcription note. *This is the quantifier-closed* v3.2 *contract: every hypothesis above is displayed inside it, so the row asserts no existence and introduces no threshold. Which δ realises the hypotheses is the business of Lemmas 99, 101, 108, 109 and 112.*

Registry contract (verbatim).

```
Explicit smooth action/operations bridge: for every finite-dimensional exact-unit
epsilon_r-C*-algebra and every delta > 0, every family g = (g_V)_{V in calU} of C^1 maps
g_V:B_{2delta}^{icalH}(0) -> B_{2delta}^{calH}(0) such that, for every V in calU and every A^par in
B_{2delta}^{icalH}(0), g_V(A^par) is the unique A^perp in B_{2delta}^{calH}(0) satisfying f_V(A^par
+ A^perp) = 0, where f_V(A) = (1/2)*(((J + A^dagger) bold-dot V^dagger) bold-dot (V bold-dot (J +
A)) - J), and such that the maps chi_V:B_{2delta}^{icalH}(0) -> calU, chi_V(A^par) = V bold-dot (J
+ A^par + g_V(A^par)), form a C^1 graph atlas calA_delta = {chi_V}_{V in calU} covering calU, and
every smooth embedded-manifold structure calM_delta on the underlying set calU for which this
displayed calA_delta is a C^infinity atlas and every displayed g_V and chi_V is C^infinity with
exactly its displayed point values and C^1 differentials Dg_V and Dchi_V, suppose Pi_delta:calU x
B_delta^{calH}(J) -> S_delta := Pi_delta(calU x B_delta^{calH}(J)), Pi_delta(U,H) = U bold-dot H,
is a C^1 diffeomorphism onto the open set S_delta with inverse (u_delta,h_delta):S_delta -> calU x
B_delta^{calH}(J) characterized by Pi_delta(u_delta(X),h_delta(X)) = X for every X in S_delta and
(u_delta(Pi_delta(U,H)),h_delta(Pi_delta(U,H))) = (U,H) for every (U,H) in calU x
B_delta^{calH}(J), suppose U bold-dot V and U^dagger lie in S_delta for every U,V in calU, and
suppose the displayed Pi_delta and displayed inverse (u_delta,h_delta) are smooth relative to
calM_delta with the same displayed point values and with smooth differentials D[(U,H) |-> U
bold-dot H] and D[X |-> (u_delta(X),h_delta(X))] equal to their displayed C^1 differentials;
writing alpha_C1:U(1) x calU -> calU, alpha_C1(c,U) = cU, mu_C1:calU x calU -> calU, mu_C1(U,V) =
u_delta(U bold-dot V), and sigma_C1:calU -> calU, sigma_C1(U) = u_delta(U^dagger), for the
resulting C^1 maps, there exist maps alpha:U(1) x calU -> calU, mu:calU x calU -> calU, and
sigma:calU -> calU that are smooth relative to calM_delta and satisfy alpha(c,U) = alpha_C1(c,U) =
cU, mu(U,V) = mu_C1(U,V) = u_delta(U bold-dot V), sigma(U) = sigma_C1(U) = u_delta(U^dagger),
Dalpha = Dalpha_C1, Dmu = Dmu_C1, Dsigma = Dsigma_C1, mu(cU,dV) = c*d*mu(U,V), and sigma(cU) =
conj(c)*sigma(U) for every U,V in calU and c,d in U(1).
```

Proof (prose account of the af-validated tree). Fix arbitrary data satisfying every hypothesis; proving the conclusion for them permits universal generalisation. Because all of g, $\mathcal{A}_\delta$, $\mathcal{M}_\delta$, Π_δ, (u_δ, h_δ) and the three C^1 maps are *displayed* in the hypothesis, no smallness guard and no existence claim is inferred inside the proof, and every later occurrence of a symbol refers to the one datum fixed here.

Ambient smoothness. By Definition 22 the product $\cdot$ is continuous complex-bilinear and $\dagger$ is continuous conjugate-linear (indeed isometric); ambient scalar multiplication is continuous complex-bilinear by the Banach-space structure. All three are therefore smooth as real maps, and since $U(1)$ is a smooth embedded submanifold of $\mathbb{C}$, the restriction of scalar multiplication to $U(1) \times \mathcal{X}$ is smooth. Restricting further to products involving $\mathcal{M}_\delta$ gives smooth ambient-valued maps, whose values lie in $\mathcal{U}$ respectively in the open set S_δ by the stipulated codomain and the membership hypothesis.

The three maps are smooth. α_{C^1} is exactly the restriction of ambient scalar multiplication to $U(1) \times \mathcal{M}_\delta$ with image in the embedded submanifold, so the initial property of an embedded submanifold makes the corestriction smooth. The restricted product $\mathcal{M}_\delta \times \mathcal{M}_\delta \to \mathcal{X}$ is smooth with image in the open S_δ, hence smooth as an S_δ-valued map; composing with the stipulated smooth u_δ gives μ_{C^1}. The same argument with the involution in place of the product gives σ_{C^1}.

Nothing is changed. Define $\alpha = \alpha_{C^1}$, $\mu = \mu_{C^1}$ and $\sigma = \sigma_{C^1}$ as literal equalities of maps between the same domains and codomains. They are smooth by the previous step, their values are the stipulated formulas by definition, and equal C^1 maps between the same manifolds have equal differentials.

Covariance. Complex bilinearity of $\cdot$ and conjugate-linearity of $\dagger$ give $(cU)\cdot(dV) = (cd)(U \cdot V)$, $((cd)W) \cdot H = (cd)(W \cdot H)$ and $(cU)^\dagger = \bar{c}U^\dagger$, with $cd, \bar{c} \in U(1)$. Put $W = u_\delta(U \cdot V)$ and $H = h_\delta(U \cdot V)$, so $U \cdot V = \Pi_\delta(W, H)$ by the inverse identity; then $(cU) \cdot (dV) = \Pi_\delta((cd)W, H)$. The pairs $((cd)W, H)$ and the inverse pair of $(cU)\cdot(dV)$ both lie in the displayed domain and have the same Π_δ-image, so *ordinary injectivity of the one displayed* Π_δ makes them equal, which is the first identity of (187). The same argument at $U^\dagger$ gives the second. No associativity, positivity or approximate group law enters.

Status. All three rows are `proved` with `af: validated` (T0). `proofs/lem-stage1-explicit-group-domain-me` run 2, root validated, 12/12 live nodes plus one archived, taint clean, 5 rounds, two in-run challenges repaired; run 1 had ABORTED on the **balloon** tripwire at 20 live nodes against a hard cap of 14, and its ledger was discarded at the clean re-seed. `proofs/lem-stage1-explicit-group-closeness`: first-pass run, root validated, 16/16 live nodes (hard cap 16, not exceeded), taint clean, 10 rounds, two in-run challenges repaired — one restoring the final numerical chain at $\varepsilon_r = 0$, one restructuring the dependencies so that the product defect, the right inverse and domain membership are derived without leaning on a pending sibling. `proofs/lem-stage1-explicit-smooth-unitary-operations`: first-pass run, root validated, 12/12 live nodes (exactly the design target; hard cap 18), taint clean, 6 rounds, ZERO challenges.

Role in the chain. Registry `defs` for all three: `def-approximate-unitary-space`, `def-epsilon-cstar-algebra` Registry `deps`: `lem-stage1-polar-retraction` (§45) for the two group rows; for the smooth row, `lem-stage1-explicit-group-domain-membership`, `lem-stage1-unitary-graph-control` (§44), `lem-stage1-polar-retraction`, `lem-stage1-smooth-unitary-atlas` (§47) and `lem-stage1-smooth-polar-inv` (§48). Downstream, the closeness row is the typed provider of `lem-stage1-inversion-derivative-control` (§47), and the two group rows are the typed providers of the parameterized transport `lem-stage1-approximate-gr` None of this assembles the Stage-1 front into a statement about almost-idempotent maps; `op-classical` is now **discharged** at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

50 Parameterized transports of the Stage-1 polar layer

This section and the next (§51) reproduce six `af`-validated *transport* rows, which carry six fixed-constant Stage-1 polar lemmas of §44–§47 to the *parameterized* form of Definition 44: throughout, W is a Stage-1 polar witness datum with coefficient fields $C_{\mathrm{rect}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}}, C_{\mathrm{path}}, C_{\mathrm{der}}$ and margin fields $e_{\mathrm{rect}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}}$ (the four derived scales of W play no role in these six rows). Each row asserts universal *thresholds* — coefficient floors C^0_* and margin caps κ^0_* — such that *every* tuple W whose coefficients dominate the floors and whose margins are dominated by the caps witnesses the parent conclusion at its own fields. All ambient vocabulary ($\mathcal{H}$, $i\mathcal{H}$, $\mathcal{U}$, $\overline{\mathcal{U}}.$, $\mathcal{U}.$, L_V, ϕ_V) is that of Definition 32, never restated.

The common transport skeleton. Every tree in the family shares one skeleton, stated once. (i) Instantiate the universal witnesses of the fixed-constant parent lemma as the thresholds. (ii) Derive $\varepsilon_r \geq 0$ from the product-norm axiom of Definition 22 at $x = y = J$: exact unitality and $\|J\| = 1$ give $1 = \|J \cdot J\| \leq (1 + \varepsilon_r)$. Hence every guard combination in play ($\varepsilon_r + \delta$, $\varepsilon_r\delta + \delta^2$, $q + \varepsilon_r q + q^2$, $\varepsilon_r + r$) is nonnegative. (iii) Transport each scalar guard by monotonicity: a coefficient only *shrinks* from the field of W down to the threshold and a margin only *grows*, so, the multiplied quantities being nonnegative, the assumed field guard implies the parent's witness guard (e.g. $\hat{C}(\varepsilon_r + \delta) \leq C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}} \leq \hat{\kappa}$). (iv) Apply the parent lemma. (v) Weaken every conclusion coefficient from the witness back up to the field, again by nonnegativity, using also that the defect sets are monotone in the radius: for $r \leq s$, $\overline{\mathcal{U}}_r \subseteq \overline{\mathcal{U}}_s$ and $\mathcal{U}_r \subseteq \mathcal{U}_s$ (the right-inverse clause is unchanged and the defect bound only relaxes). Existence, uniqueness, regularity, coverage and equivariance clauses carry over verbatim.

Lemma 115 (`lem-stage1-rectified-cstar-transport`). *There exist $C^0_{\mathrm{rect}} \geq 1$ and $e^0_{\mathrm{rect}} \in (0, 1/C^0_{\mathrm{rect}}]$ such that for every witness datum W with $C_{\mathrm{rect}} \geq C^0_{\mathrm{rect}}$ and $0 < e_{\mathrm{rect}} \leq \min\{e^0_{\mathrm{rect}}, 1/C_{\mathrm{rect}}\}$, every finite-dimensional ε_X-C^*-algebra with $0 \leq \varepsilon_X \leq e_{\mathrm{rect}}$ admits, on the same involutive normed space, a bilinear product $\cdot$ and an element $J = J^\dagger$ for which $(\mathcal{X}, J, \cdot, \dagger)$ satisfies every exact-unit ε_r-C^*-algebra axiom of Definition 22, including $\|J\| = 1$, where $\varepsilon_r = C_{\mathrm{rect}}\varepsilon_X$, and for every $x, y \in \mathcal{X}$, $\|J - I_X\| \leq C_{\mathrm{rect}}\varepsilon_X$ and $\|x \cdot y - xy\| \leq C_{\mathrm{rect}}\varepsilon_X\|x\|\|y\|$.*

Registry contract (verbatim).

```
Parameterized rectification transport: there exist C_rect^0 >= 1 and e_rect^0 in (0, 1/C_rect^0]
such that, for every def-stage1-polar-witness-data tuple W with C_rect >= C_rect^0 and 0 < e_rect
<= min{e_rect^0, 1/C_rect}, for every finite-dimensional epsilon_X-C*-algebra (calX, I_X, .,
dagger) with 0 <= epsilon_X <= e_rect, there are on the same involutive normed space a bilinear
product bold-dot and an element J = J^dagger for which (calX, J, bold-dot, dagger) satisfies every
exact-unit epsilon_r-C*-algebra axiom of def-epsilon-cstar-algebra, including ||J|| = 1, where
epsilon_r = C_rect*epsilon_X, and for every x, y in calX, ||J - I_X|| <= C_rect*epsilon_X and ||x
bold-dot y - xy|| <= C_rect*epsilon_X*||x||*||y||.
```

Proof (prose account of the af-validated tree). The skeleton, seeded by Lemma 98 with witnesses (c, e) and thresholds $C^0_{\mathrm{rect}} = c$, $e^0_{\mathrm{rect}} = e$, plus one extra node: *parameter monotonicity of the axiom list itself.* If $0 \leq \alpha \leq \beta$ and $(\mathcal{X}, J, \cdot, \dagger)$ satisfies every exact-unit α-C^*-algebra axiom, then it satisfies every β-axiom: $\|x \cdot y\| \leq (1 + \alpha)\|x\|\|y\| \leq (1 + \beta)\|x\|\|y\|$, the associator bound $\alpha\|x\|\|y\|\|z\|$ is at most $\beta\|x\|\|y\|\|z\|$, and $\|x^\dagger \cdot x\| \geq (1 - \alpha)\|x\|^2 \geq (1 - \beta)\|x\|^2$; every remaining axiom is parameter-free. Since $\varepsilon_X \leq e_{\mathrm{rect}} \leq e$, the parent applies and produces the axioms at

$\alpha = c\varepsilon_X$; monotonicity passes to $\beta = C_{\mathrm{rect}}\varepsilon_X \geq \alpha$, and $c \leq C_{\mathrm{rect}}$ weakens the two closeness estimates.

Lemma 116 (`lem-stage1-unitary-graph-transport`). *There exist $C^0_{\mathrm{ch}} \geq 1$ and $\kappa^0_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{ch}} \geq C^0_{\mathrm{ch}}$ and $0 < \kappa_{\mathrm{ch}} \leq \kappa^0_{\mathrm{ch}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}}$, every $V \in \overline{\mathcal{U}}_\delta$ and every $A^\parallel \in B^{i\mathcal{H}}_{2\delta}(0)$, there is a unique $g_V(A^\parallel) \in B^{\mathcal{H}}_{2\delta}(0)$ with $f_V(A^\parallel + g_V(A^\parallel)) = 0$ — f_V exactly as in (146) — the element $V \cdot (J + A^\parallel + g_V(A^\parallel))$ lies in $\mathcal{U}$, the two graph estimates (147) and the strict normal-derivative bound (148) hold with the field C_{ch} of W in place of the universal constant, and the resulting C^1 graph charts cover $\mathcal{U}$.*

Registry contract (verbatim).

```
Parameterized unitary-graph transport: there exist C_ch^0 >= 1 and kappa_ch^0 in (0, 1/2] such
that, for every def-stage1-polar-witness-data tuple W with C_ch >= C_ch^0 and 0 < kappa_ch <=
kappa_ch^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra (calX, J, bold-dot,
dagger), every delta > 0 with C_ch*(epsilon_r + delta) <= kappa_ch, every V in calUbar_delta, and
every A^par in B^{icalH}_{2delta}(0), there is a unique g_V(A^par) in B^{calH}_{2delta}(0) such
that f_V(A^par + g_V(A^par)) = 0, where f_V(A) = (1/2)*(((J + A^dagger) bold-dot V^dagger) bold-dot
(V bold-dot (J + A)) - J), the element V bold-dot (J + A^par + g_V(A^par)) lies in calU,
||g_V(A^par) + (1/2)*(V^dagger bold-dot V - J)|| <= C_ch*(epsilon_r*delta + delta^2),
||Dg_V(A^par)|| <= C_ch*(epsilon_r + delta), and ||D_{A^perp} f_V(A^par + g_V(A^par)) - I_{calH}||
<= C_ch*(epsilon_r + delta) < 1, and these C^1 graph charts cover calU.
```

Proof (prose account of the af-validated tree). The skeleton verbatim, seeded by Lemma 99 with witnesses $(\hat{C}, \hat{\kappa})$. With $s = \varepsilon_r + \delta$ and $q = \varepsilon_r\delta + \delta^2 = \delta s$, step (ii) gives $s, q > 0$; step (iii) is $\hat{C}s \leq C_{\mathrm{ch}}s \leq \kappa_{\mathrm{ch}} \leq \hat{\kappa}$; the parent supplies the unique g_V, membership in $\mathcal{U}$, cover, and the three bounds at $\hat{C}$; step (v) weakens them to $C_{\mathrm{ch}}q$, $C_{\mathrm{ch}}s$, $C_{\mathrm{ch}}s$, and the strict clause is $C_{\mathrm{ch}}s \leq \kappa_{\mathrm{ch}} \leq \kappa^0_{\mathrm{ch}} \leq \frac{1}{2} < 1$.

Lemma 117 (`lem-stage1-maurer-cartan-transport`). *There exist $C^0_{\mathrm{ch}} \geq 1$ and $\kappa^0_{\mathrm{ch}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{ch}} \geq C^0_{\mathrm{ch}}$ and $0 < \kappa_{\mathrm{ch}} \leq \kappa^0_{\mathrm{ch}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\mathrm{ch}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{ch}}$, and every family $g = (g_U)_{U \in \mathcal{U}}$ of C^1 maps $g_U : B^{i\mathcal{H}}_{2\delta}(0) \to B^{\mathcal{H}}_{2\delta}(0)$ such that, for every U and $A^\parallel$, $g_U(A^\parallel)$ is the* unique *element of $B^{\mathcal{H}}_{2\delta}(0)$ with $f_U(A^\parallel + g_U(A^\parallel)) = 0$: every tangent space $T_U\mathcal{U}$ is the image of $L_U(I + Dg_U(0)) : i\mathcal{H} \to \mathcal{X}$, and $\omega_U(Z) = (L_U^{-1}Z)^\parallel$ is a global C^1 bundle trivialization with distortion at most $1 + C_{\mathrm{ch}}\varepsilon_r$, satisfying $\omega_{cU}(cZ) = \omega_U(Z)$ and $\omega_U(iU) = iJ$ for every $c \in U(1)$.* Transcription note. *Unlike Lemma 100, whose graph maps are the distinguished family supplied by Lemma 99, this row quantifies over* any *family with the displayed uniqueness property.*

Registry contract (verbatim).

```
Parameterized Maurer-Cartan transport: there exist C_ch^0 >= 1 and kappa_ch^0 in (0, 1/2] such
that, for every def-stage1-polar-witness-data tuple W with C_ch >= C_ch^0 and 0 < kappa_ch <=
kappa_ch^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0 with
C_ch*(epsilon_r + delta) <= kappa_ch, and every family g = (g_U)_{U in calU} of C^1 maps g_U:
B^{icalH}_{2delta}(0) -> B^{calH}_{2delta}(0) such that, for every U in calU and A^par in
B^{icalH}_{2delta}(0), g_U(A^par) is the unique element of B^{calH}_{2delta}(0) satisfying
f_U(A^par + g_U(A^par)) = 0, where f_U(A) = (1/2)*(((J + A^dagger) bold-dot U^dagger) bold-dot (U
bold-dot (J + A)) - J), every tangent space T_U calU is the image of L_U(I + Dg_U(0)): icalH ->
calX, and omega_U(Z) = (L_U^{-1} Z)^par : T_U calU -> icalH is a global C^1 bundle trivialization
```

```
with distortion at most 1 + C_ch*epsilon_r, satisfying omega_{cU}(cZ) = omega_U(Z) and omega_U(iU)
= iJ for every U in calU, Z in T_U calU, and c in U(1).
```

Proof (prose account of the af-validated tree). Thresholds $C^0_{\mathrm{ch}} = \bar{C}$, $\kappa^0_{\mathrm{ch}} = \bar{\kappa}$ from Lemma 100; step (iii) of the skeleton is $\bar{C}(\varepsilon_r + \delta) \le C_{\mathrm{ch}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{ch}} \le \bar{\kappa}$, which is exactly that parent's smallness guard. The tangent image is then derived for the arbitrary family g *itself*, using no property of — and no equality with — the distinguished graph family.

Exact zeros are unitary. The guard gives $\varepsilon_r < \frac{1}{2}$ (as $C_{\mathrm{ch}} \ge 1$ and $\delta > 0$). If $X^\dagger \cdot X = J$ then the C^*-lower bound gives $\|X\|^2 \le (1-\varepsilon_r)^{-1}$, and approximate associativity with exact unitality gives $\|L_{X^\dagger} L_X Y - Y\| \le \frac{\varepsilon_r}{1-\varepsilon_r}\|Y\|$, a coefficient strictly below 1. By Neumann, $L_{X^\dagger} L_X$ is invertible, so L_X is injective and — in finite dimension — surjective, whence X has a right inverse: every exact zero of the displayed f_U-equation lies in $\mathcal{U}$.

The tangent image. Exact unitality and $U^\dagger \cdot U = J$ give $f_U(0) = 0$, so uniqueness forces $g_U(0) = 0$. For $B \in i\mathcal{H}$ and small real t put $X_B(t) = U \cdot (J + tB + g_U(tB))$; because the involution reverses the product, the defining zero equation reads $X_B(t)^\dagger \cdot X_B(t) = J$, so the previous step places $X_B(t)$ in $\mathcal{U}$, with $X_B(0) = U$. Differentiating at $t = 0$ gives $L_U(B + Dg_U(0)B)$, hence $\operatorname{im} L_U(I + Dg_U(0)) \subseteq T_U\mathcal{U}$.

Equality of the two spaces. The parent supplies the fibrewise linear isomorphism $\omega_U(Z) = (L_U^{-1}Z)^{\parallel}$, so $\dim T_U\mathcal{U} = \dim i\mathcal{H}$. Writing $\Phi_g(B) = L_U(B + Dg_U(0)B)$, the displayed formula gives $\omega_U(\Phi_g(B)) = B$ because $Dg_U(0)B$ is Hermitian; thus Φ_g is injective and its image has dimension $\dim i\mathcal{H}$. An inclusion of finite-dimensional spaces of equal dimension is an equality.

Transfer of the remaining clauses. The identified tangent bundle is the same intrinsic bundle on which the parent already asserts ω, the distortion bound $1 + \bar{C}\varepsilon_r$ and the identities $\omega_{cU}(cZ) = \omega_U(Z)$, $\omega_U(iU) = iJ$; none of those conclusions mentions a graph family, so they hold verbatim. Step (v) weakens the distortion to $1 + C_{\mathrm{ch}}\varepsilon_r$ since $\bar{C} \le C_{\mathrm{ch}}$ and $\varepsilon_r \ge 0$.

History (condensed): retraction and the first in-ledger repair. The 2026-07-27 validation was RETRACTED 2026-07-28 by the Stage-1 binder sweep: its node 1.3.3 inferred $Dg_U(0) = D\bar{g}_U(0)$ from a pointwise equality with the distinguished family that the sole registered external does not supply — its own sibling node 1.3.2 records the antecedent as absent. On 2026-07-29 the row was re-validated by the repository's first *in-ledger* repair rather than a re-seed: a fresh repair verifier independently confirmed the defect and revoked node 1.3.3 together with its closure chain $\{1, 1.3, 1.3.7, 1.3.6, 1.3.5, 1.3.4\}$ dependent-before-prerequisite; only then was 1.3.3 archived; distinct fresh verifiers re-accepted the typed-family bypass $1.3.4 \to 1.3.5 \to 1.3.6 \to 1.3.7 \to 1.3 \to 1$ bottom-up with zero challenges. That bypass is the account above. In the regenerated export node 1.3.3 carries `Status: archived` and lies **outside** the live closure; nodes 1.3.2/1.3.2.1 survive as a correctly *conditional*, unused side branch. No contract was amended.

Lemma 118 (`lem-stage1-polar-retraction-transport`)**.** *There exist $C^0_{\mathrm{pol}} \ge 1$ and $\kappa^0_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{pol}} \ge C^0_{\mathrm{pol}}$ and $0 < \kappa_{\mathrm{pol}} \le \kappa^0_{\mathrm{pol}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta) \le \kappa_{\mathrm{pol}}$, the map $\Pi_\delta(U, H) = U \cdot H$ is a C^1 diffeomorphism from $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ onto the open set $S_\delta := \Pi_\delta(\mathcal{U} \times B^{\mathcal{H}}_\delta(J))$, its inverse (u_δ, h_δ) obeys the normalisation* (153)*, and the sandwich* (154) *holds with the field C_{pol} of W.*

Registry contract (verbatim).

```
Parameterized polar-retraction transport: there exist C_pol^0 >= 1 and kappa_pol^0 in (0, 1/2] such
that, for every def-stage1-polar-witness-data tuple W with C_pol >= C_pol^0 and 0 < kappa_pol <=
```

```
kappa_pol^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra and every delta > 0 with
C_pol*(epsilon_r + delta) <= kappa_pol, the map Pi_delta: calU x B^{calH}_delta(J) -> calX,
Pi_delta(U, H) = U bold-dot H, is a C^1 diffeomorphism onto the open set S_delta := Pi_delta(calU x
B^{calH}_delta(J)), its inverse (u_delta, h_delta): S_delta -> calU x B^{calH}_delta(J) satisfies X
= u_delta(X) bold-dot h_delta(X), u_delta(U) = U, and h_delta(U) = J for every X in S_delta and U
in calU, and calU_{delta - C_pol*(epsilon_r*delta + delta^2)} subseteq S_delta subseteq calU_{delta
+ C_pol*(epsilon_r*delta + delta^2)}.
```

Proof (prose account of the af-validated tree). The skeleton verbatim (the shortest tree of the family), seeded by Lemma 101 with witnesses (C_*, κ_*). The diffeomorphism, openness and normalisation clauses are coefficient-free and transfer unchanged; only the sandwich is weakened: with $A = \varepsilon_r\delta + \delta^2 \geq 0$ and $C_* \leq C_{\text{pol}}$, one has $\delta - C_{\text{pol}}A \leq \delta - C_*A$ and $\delta + C_*A \leq \delta + C_{\text{pol}}A$, so radius monotonicity of $\mathcal{U}$. gives $\mathcal{U}_{\delta - C_{\text{pol}}A} \subseteq \mathcal{U}_{\delta - C_*A} \subseteq S_\delta \subseteq \mathcal{U}_{\delta + C_*A} \subseteq \mathcal{U}_{\delta + C_{\text{pol}}A}$.

Status. All four rows are `proved` with `af: validated` (T0), tier *routine*, taint clean: workspaces `proofs/lem-stage1-rectified-cstar-transport` (7 nodes, 7 `validated`, first-pass, zero challenges), `proofs/lem-stage1-unitary-graph-transport` (9 nodes, 9 `validated`, first-pass, zero challenges), `proofs/lem-stage1-polar-retraction-transport` (5 nodes, 5 `validated`, first-pass, zero challenges) and `proofs/lem-stage1-maurer-cartan-transport`, RE-VALIDATED 2026-07-29 with 12/12 live nodes `validated` plus 1 archived, zero challenges in the repair rounds.

Correction of record (proof-level, not contract-level). The Maurer–Cartan row is the repository's first *in-ledger* repair: the defective node was revoked with its whole closure chain *before* being archived, and every node of the surviving typed-family bypass was re-accepted by a fresh verifier bottom-up. The conditional uniqueness bridge (nodes 1.3.2/1.3.2.1) is retained but unused. No contract was amended, here or in the earlier retraction.

Role in the chain. Registry `defs`: `def-stage1-polar-witness-data` and `def-epsilon-cstar-algebra` (all four) and `def-approximate-unitary-space` (all but the rectification row). Registry `deps`: each transport imports exactly its fixed-constant parent — Lemma 98, Lemma 99 and Lemma 100 (§44), and Lemma 101 (§45) respectively. The path and inversion-derivative transports follow in §51; `op-classical` is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

51 Parameterized transports of the Stage-1 polar layer, II: paths and the inversion derivative

The two rows below complete the `af`-validated transport family begun in §50; the witness datum W, its coefficient and margin fields, the threshold discipline and the *common transport skeleton* (steps (i)–(v)) are exactly those stated there and are not repeated. What distinguishes these two rows is that their roots carry *explicit definite descriptions* where their fixed-constant ancestors bound u_δ, $g_{\varsigma J}$ and σ as bare anaphors; each is therefore built on a spine of *typed* providers that display the very same Π_δ, so that ordinary uniqueness of an inverse — and nothing weaker — performs the identification.

Lemma 119 (`lem-stage1-polar-path-transport`). *There exist $C^0_{\text{path}}, C^0_{\text{pol}} \geq 1$ and $\kappa^0_{\text{pol}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\text{path}} \geq C^0_{\text{path}}$, $C_{\text{pol}} \geq C^0_{\text{pol}}$ and $0 < \kappa_{\text{pol}} \leq \kappa^0_{\text{pol}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra, every $\delta > 0$ with $C_{\text{pol}}(\varepsilon_r + \delta) \leq \kappa_{\text{pol}}$, every $U_0, U_1 \in \mathcal{U}$ and every $q \in [0, 1]$ with $\|U_1 - U_0\| \leq q$, $C_{\text{path}} q \leq \frac{1}{4}$ and $C_{\text{path}}(q + \varepsilon_r q + q^2) < \delta - C_{\text{pol}}(\varepsilon_r \delta + \delta^2)$: every L_{Z_t} is invertible and every $Z_t = (1-t)U_0 + tU_1$ lies in $\overline{\mathcal{U}}_{C_{\text{path}}(q+\varepsilon_r q+q^2)}$ for $t \in [0, 1]$; and, writing u_δ for the unique first component of the inverse of Π_δ (the explicit binder of Lemma 118), $H(t, U_0, U_1) = u_\delta(Z_t)$ is jointly continuous in (t, U_0, U_1), joins U_0 to U_1, and obeys $H(t, cU_0, cU_1) = c\,H(t, U_0, U_1)$ for every $c \in U(1)$.*

Registry contract (verbatim).

```
Parameterized polar-path transport: there exist C_path^0, C_pol^0 >= 1 and kappa_pol^0 in (0, 1/2]
such that, for every def-stage1-polar-witness-data tuple W with C_path >= C_path^0, C_pol >=
C_pol^0, and 0 < kappa_pol <= kappa_pol^0, for every finite-dimensional exact-unit
epsilon_r-C*-algebra, every delta > 0 with C_pol*(epsilon_r + delta) <= kappa_pol, every U_0, U_1
in calU, and every q in [0, 1] satisfying ||U_1 - U_0|| <= q, C_path*q <= 1/4, and C_path*(q +
epsilon_r*q + q^2) < delta - C_pol*(epsilon_r*delta + delta^2), every L_{Z_t} is invertible and
every Z_t = (1-t)*U_0 + t*U_1 lies in calUbar_{C_path*(q + epsilon_r*q + q^2)} for t in [0, 1],
and, writing u_delta for the unique first component of the inverse of Pi_delta: calU x
B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H) = U bold-dot H,
the map H(t, U_0, U_1) = u_delta(Z_t) is jointly continuous in (t, U_0, U_1), joins U_0 to U_1, and
satisfies H(t, cU_0, cU_1) = c*H(t, U_0, U_1) for every c in U(1).
```

Proof (prose account of the af-validated tree). *Witnesses first.* Take (A, B, k) from Lemma 106 and (P, ρ) from Lemma 118, and set $C^0_{\text{path}} = A$, $C^0_{\text{pol}} = \max\{B, P\}$, $\kappa^0_{\text{pol}} = \min\{k, \rho\}$ — all fixed *before* the receiving tuple W is quantified. With $\gamma = q + \varepsilon_r q + q^2 \geq 0$ and $y = \varepsilon_r \delta + \delta^2 \geq 0$, step (iii) of the skeleton transports each guard: $B(\varepsilon_r + \delta) \leq \kappa_{\text{pol}} \leq k$, $P(\varepsilon_r + \delta) \leq \kappa_{\text{pol}} \leq \rho$, $Aq \leq C_{\text{path}} q \leq \frac{1}{4}$, and — the only strict one — the mixed chain $A\gamma \leq C_{\text{path}}\gamma < \delta - C_{\text{pol}} y \leq \delta - By$, whose strictness is carried by the hypothesised middle inequality alone.

Binder-free conclusions from the parent. Lemma 106 is used for exactly two things: every L_{Z_t} is invertible, and $Z_t \in \overline{\mathcal{U}}_{A\gamma}$ — which radius monotonicity enlarges to $\overline{\mathcal{U}}_{C_{\text{path}}\gamma}$ with the same right inverse.

The typed inverse. Lemma 118, applied to the same tuple W under the non-strict guard $C_{\text{pol}}(\varepsilon_r + \delta) \leq \kappa_{\text{pol}}$, gives for the *very* map displayed in the root a C^1 diffeomorphism onto S_δ with unique inverse (u_δ, h_δ), the normalisations (153), and $\mathcal{U}_{\delta - C_{\text{pol}} y} \subseteq S_\delta$. Since $Z_t \in \overline{\mathcal{U}}_{C_{\text{path}}\gamma}$ gives $\|Z_t^\dagger \cdot Z_t - J\| \leq 2C_{\text{path}}\gamma < 2(\delta - C_{\text{pol}} y)$, Definition 32 puts $Z_t \in \mathcal{U}_{\delta - C_{\text{pol}} y} \subseteq S_\delta$; so $u_\delta(Z_t)$ is the well-typed first component of the inverse of that same displayed Π_δ.

The path. $(t, U_0, U_1) \mapsto Z_t$ is jointly continuous and u_δ is C^1, hence continuous, on S_δ, so H is jointly continuous; $Z_0 = U_0$, $Z_1 = U_1$ and $u_\delta(U) = U$ give the endpoints. For $c \in U(1)$ put $(u, h) = (u_\delta(Z_t), h_\delta(Z_t))$; conjugate-linearity of $\dagger$, bilinearity of $\cdot$ and $|c| = 1$ show directly from Definition 32 that $cu \in \mathcal{U}$ (its residual is unchanged and a right inverse rescales by $\bar{c}$), while $\Pi_\delta(cu, h) = c\,\Pi_\delta(u, h) = cZ_t$ and $(1-t)cU_0 + tcU_1 = cZ_t$. A bijection has exactly one inverse, so $u_\delta(cZ_t) = c\,u_\delta(Z_t)$, i.e. $H(t, cU_0, cU_1) = c\,H(t, U_0, U_1)$.

Lemma 120 (`lem-stage1-inversion-derivative-transport`)**.** *There exist* $C^0_{\rm der}, C^0_{\rm ch}, C^0_{\rm pol}, C^0_{\rm grp} \ge 1$ *and* $\kappa^0_{\rm der}, \kappa^0_{\rm ch}, \kappa^0_{\rm pol} \in (0, \frac{1}{2}]$ *such that for every witness datum* W *with* $C_{\rm der} \ge C^0_{\rm der}$*,* $C_{\rm ch} \ge C^0_{\rm ch}$*,* $C_{\rm pol} \ge C^0_{\rm pol}$*,* $C_{\rm grp} \ge C^0_{\rm grp}$*,* $0 < \kappa_{\rm der} \le \kappa^0_{\rm der}$*,* $0 < \kappa_{\rm ch} \le \kappa^0_{\rm ch}$ *and* $0 < \kappa_{\rm pol} \le \kappa^0_{\rm pol}$*, for every finite-dimensional exact-unit* ε_r*-*C^**-algebra, every* $\delta > 0$*, every* $\varsigma \in \{+1, -1\}$ *and every* $0 < r \le \delta$ *satisfying the five guards* (172)*–*(173) *at the fields of* W*: writing* u_δ *for the unique first component of the inverse of* Π_δ *(as in Lemma 118) and* $g_{\varsigma J}$ *for the unique* C^1 *map* $B^{i\mathcal{H}}_{2\delta}(0) \to B^{\mathcal{H}}_{2\delta}(0)$ *with* $f_{\varsigma J}(A + g_{\varsigma J}(A)) = 0$ *throughout, define* $\chi_\varsigma(A) = \varsigma J \cdot (J + A + g_{\varsigma J}(A))$ *and the global* C^1 *map* $\sigma(U) = u_\delta(U^\dagger)$*; then* σ *maps* $\chi_\varsigma(B^{i\mathcal{H}}_r(0))$ *into the same* ςJ*-graph chart, and* $F_\varsigma(A) = \phi^{\parallel}_{\varsigma J}(\sigma(\chi_\varsigma(A)))$ *satisfies* $\|D(F_\varsigma - {\rm id})(A) + 2I_{i\mathcal{H}}\| \le C_{\rm der}(\varepsilon_r + r)$ *for every* $A \in B^{i\mathcal{H}}_r(0)$*.* Transcription note. *The registry writes the sign as s; it is renamed* ς *here exactly as in §47. Where the fixed-constant control row binds* u_δ*,* $g_{\varsigma J}$ *and* σ *as bare anaphors, this row carries* explicit definite descriptions *for all three — the point of the transport.*

Registry contract (verbatim).

```
Parameterized inversion-derivative transport: there exist C_der^0, C_ch^0, C_pol^0, C_grp^0 >= 1
and kappa_der^0, kappa_ch^0, kappa_pol^0 in (0, 1/2] such that, for every
def-stage1-polar-witness-data tuple W with C_der >= C_der^0, C_ch >= C_ch^0, C_pol >= C_pol^0,
C_grp >= C_grp^0, 0 < kappa_der <= kappa_der^0, 0 < kappa_ch <= kappa_ch^0, and 0 < kappa_pol <=
kappa_pol^0, for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0, every s
in {+1, -1}, and every 0 < r <= delta satisfying C_ch*(epsilon_r + delta) <= kappa_ch,
C_pol*(epsilon_r + delta) <= kappa_pol, C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta +
delta^2), C_der*(epsilon_r + r) <= kappa_der, and (1 + epsilon_r)*(1 + C_ch*(epsilon_r + delta))*r
+ C_grp*epsilon_r < 2*delta, writing u_delta for the unique first component of the inverse of
Pi_delta: calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H)
= U bold-dot H, and g_{sJ}: B_{2delta}^{icalH}(0) -> B_{2delta}^{calH}(0) for the unique C^1 map
such that, for every A in B_{2delta}^{icalH}(0), f_{sJ}(A + g_{sJ}(A)) = 0, where f_{sJ}(B) =
(1/2)*(((J + B^dagger) bold-dot (sJ)^dagger) bold-dot (sJ bold-dot (J + B)) - J), define chi_s(A) =
sJ bold-dot (J + A + g_{sJ}(A)) and the global C^1 map sigma(U) = u_delta(U^dagger); then sigma
maps chi_s(B_r^{icalH}(0)) into the same sJ-graph chart and, with F_s(A) =
phi_{sJ}^par(sigma(chi_s(A))), one has ||D(F_s - id)(A) + 2*I_{icalH}|| <= C_der*(epsilon_r + r)
for every A in B_r^{icalH}(0).
```

Proof (prose account of the af-validated tree). *Four fixed providers, then the tuple.* Fix (G_d, P_d, k_d) from `lem-stage1-explicit-group-domain-membership`, (G_c, P_c, k_c) from `lem-stage1-explicit-gro` (P_r, k_r) from Lemma 101 and (C_g, k_g) from Lemma 99, and set $C^0_{\rm ch} = \max\{1, C_g\}$, $C^0_{\rm pol} = \max\{1, P_d, P_c, P_r\}$, $C^0_{\rm grp} = \max\{1, G_d, G_c\}$, $\kappa^0_{\rm ch} = \min\{\frac{1}{2}, k_g\}$, $\kappa^0_{\rm pol} = \min\{\frac{1}{2}, k_d, k_c, k_r\}$ — all *before* the receiving tuple enters, so that no step ever absorbs the unbounded receiving $C_{\rm grp}$. With $x = \varepsilon_r\delta + \delta^2 \ge 0$ the receiving guards imply every provider guard by monotonicity, including $G\varepsilon_r \le C_{\rm grp}\varepsilon_r < \delta - C_{\rm pol}x \le \delta - Px$ for $(G, P) \in \{(G_d, P_d), (G_c, P_c)\}$ and the chart-retention inequality at C_g, G_c. All polar providers and the root display the identical Π_δ on $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ with target its image, so inverse uniqueness identifies their inverse pair with the root-bound (u_δ, h_δ); uniqueness in Lemma 99 at $V = \varsigma J$ identifies its graph map with the root-bound $g_{\varsigma J}$.

The graph map at ςJ. Writing $G = g_{\varsigma J}$ and $Z_T = T + G(T)$, exact unitality and $(\varsigma J)^\dagger = \varsigma J$ collapse the graph equation to $0 = G(T) + \frac{1}{2} Z_T^\dagger \cdot Z_T$; $G = 0$ solves it at $T = 0$, so $G(0) = 0$ by uniqueness, and the parent's derivative bound $d = C_g(\varepsilon_r + \delta) \le k_g \le \frac{1}{2}$ gives $\|G(T)\| \le d\|T\|$. Differentiating the exact quadratic equation gives $\|DG(T)\| \le (1 + \varepsilon_r)\|Z_T\|(1 + \|DG(T)\|)$ with $\|Z_T\| \le 3r$ for $\|T\| \le 2r$, so one universal cutoff yields $\|DG(T)\| \le K_g r$ with $K_g = 7$.

The typed factorization. The root-bound u_δ is the C^1 first component supplied by Lemma 101; it is defined at every $U^\dagger$ by the explicit domain-membership bridge, and $\dagger$ is real-linear, so $\sigma(U) = u_\delta(U^\dagger)$ is globally C^1 on $\mathcal{U}$, with $\sigma(\varsigma J) = \varsigma J$. For $A \in B_r^{i\mathcal{H}}(0)$, $U = \chi_\varsigma(A) \in \mathcal{U}$; put $W = \sigma(U)$ and $Q = h_\delta(U^\dagger) - J \in \mathcal{H}$ with $\|Q\| < \delta$. Then $U^\dagger = W \cdot (J + Q)$, everything depending C^1 on A, and the explicit closeness bridge for this same u_δ gives $\|W - U^\dagger\| \le G_c\varepsilon_r$. Exact unitality turns this into $\|W \cdot Q\| \le G_c\varepsilon_r$; since $W^\dagger \cdot W = J$, the C^*-lower bound and one associator absorb the remainder into $\|Q\| \le K_q G_c \varepsilon_r$, an estimate in which only the *fixed* G_c occurs.

Chart retention and the coordinate identity. From $U^\dagger = \varsigma(J - A + G(A))$ one gets $\|U^\dagger - \varsigma J\| \le (1 + C_g(\varepsilon_r + \delta))r$, and adding the fixed closeness bound the chart-retention guard gives $\|W - \varsigma J\| < 2\delta$. As $L_{\varsigma J} = \varsigma I$, $C = \phi_{\varsigma J}(W)$ has norm below 2δ; put $B = C^\parallel$. Unitarity of W makes $f_{\varsigma J}(C) = 0$, so graph uniqueness forces $C^\perp = G(B)$: $W = \chi_\varsigma(B)$ in the *same* chart, with $B = \phi^\parallel_{\varsigma J}(W) = F_\varsigma(A)$ and F_ς of class C^1. Substituting the three factorizations gives the exact identity $J - A + G(A) = (J + B + G(B)) \cdot (J + Q)$, whose anti-Hermitian part is $-A = B + P^\parallel((B + G(B)) \cdot Q)$; with $\|Q\| \le K_q G_c \varepsilon_r$ and one universal cutoff this yields $\|B\| \le 2r$, so the sharp $\|DG\| \le K_g r$ applies at both A and B.

The derivative. Differentiating that identity along $\xi \in i\mathcal{H}$, with $P = DF_\varsigma(A)\xi$ and $R = DQ(A)\xi$, gives $-\xi + DG(A)\xi = P + DG(B)P + R + E$ with $E = (P + DG(B)P) \cdot Q + (B + G(B)) \cdot R$; typing by parts gives $P + \xi = -P^\parallel(E)$ and $R = DG(A)\xi - DG(B)P - P^\perp(E)$. The product bound with the three fixed estimates yields $\|E\| \le \alpha\|P\| + \beta\|R\|$, $\alpha \le K_1\varepsilon_r$, $\beta \le K_2 r$; universal cutoffs give $\|P\| \le 2$, then $\|R\| \le K_3(\varepsilon_r + r)$ and $\|(DF_\varsigma(A) + I)\xi\| \le D_0(\varepsilon_r + r)\|\xi\|$. Finally, collecting the finitely many raw cutoffs c_i and all fixed coefficients into $D = \max\{1, D_0, d_1, \ldots, d_m\}$ and setting $C^0_{\text{der}} = D$, $\kappa^0_{\text{der}} = \min\{\frac{1}{2}, c_1/2, \ldots, c_n/2, 1/(2D)\}$ makes the receiving guards $C_{\text{der}} \ge C^0_{\text{der}}$ and $C_{\text{der}}(\varepsilon_r + r) \le \kappa_{\text{der}} \le \kappa^0_{\text{der}}$ enforce every cutoff *strictly*, even at the permitted endpoints, and give $D_0(\varepsilon_r + r) \le C_{\text{der}}(\varepsilon_r + r)$. Since $D(F_\varsigma - \text{id}) + 2I_{i\mathcal{H}} = DF_\varsigma + I_{i\mathcal{H}}$, this is the root estimate.

Status. Both rows are `proved` with `af: validated` (T0), tier *routine*, taint clean, af-RE-VALIDATED 2026-07-29 on clean re-seeds (the retracted trees were *not* used as repair bases). `proofs/lem-stage1-polar-path-transport`: root validated, 11/11 live nodes (design target 10, hard cap 14), 10 rounds, two in-run challenges repaired — both strict-versus-non-strict guard transfers at permitted equality cases — and re-verified fresh. `proofs/lem-stage1-inversion-derivative-transport`: root validated, 25/25 live nodes (design target 22, hard cap 25, not exceeded), 7 rounds, three in-run challenges repaired — the open-domain boundary cases at $r = \delta$ for G and DG, and a strict cutoff transfer — and re-verified fresh.

Correction of record (deps-level, not contract-level). Both rows carried `af` validations that were RETRACTED 2026-07-28 by the Stage-1 binder sweep and its adjudications: each attached a parent's *anaphorically* bound u_δ to the root's explicitly displayed Π_δ inverse on the strength of shared notation, with no typed preimage witness. The repair was to widen or replace the registry `deps` with typed providers that display the same Π_δ — for the path row, the polar-retraction transport alongside the parent used only for its binder-free conclusions; for the inversion row, the two explicit group bridges together with the polar retraction and unitary graph control, the retracted control parent being dropped entirely. In both cases the `contract` stayed **byte-unchanged**; nothing was amended.

Role in the chain. Registry `defs` (both rows): `def-stage1-polar-witness-data`, `def-approximate-unitary-s` def-epsilon-cstar-algebra. Registry deps: the path row imports Lemma 106 (§47) and Lemma 118; the inversion row imports lem-stage1-explicit-group-domain-membership, lem-stage1-explicit-group-clo Lemma 101 (§45) and Lemma 99 (§44). With the group-law transport of §52 these seven rows are exactly the analytic inputs of the Stage-1 constant ledger; nothing in them assembles the Stage-1 front into a statement about almost-idempotent maps, and op-classical is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by cor-classical-sharpness.

52 The Stage-1 keystone: the group-law transport and the single-tuple constant ledger

This section closes the parameterized Stage-1 polar layer. The first row is the last of the seven transports of §50–§51; the second is the *keystone* that consumes all seven at one and the same witness tuple; the third is an algebra-independent finite-poset row that consumes none of them. The witness datum W and its fields are those of Definition 44; all ambient vocabulary is that of Definition 32 and Definition 22, never restated.

Lemma 121 (`lem-stage1-approximate-group-laws-transport`). *There exist $C^0_{\mathrm{grp}}, C^0_{\mathrm{pol}} \geq 1$ and $\kappa^0_{\mathrm{pol}} \in (0, \frac{1}{2}]$ such that for every witness datum W with $C_{\mathrm{grp}} \geq C^0_{\mathrm{grp}}$, $C_{\mathrm{pol}} \geq C^0_{\mathrm{pol}}$ and $0 < \kappa_{\mathrm{pol}} \leq \kappa^0_{\mathrm{pol}}$, for every finite-dimensional exact-unit ε_r-C^*-algebra and every $\delta > 0$ with $C_{\mathrm{pol}}(\varepsilon_r + \delta) \leq \kappa_{\mathrm{pol}}$ and $C_{\mathrm{grp}}\varepsilon_r < \delta - C_{\mathrm{pol}}(\varepsilon_r\delta + \delta^2)$: writing (u_δ, h_δ) for the unique inverse of Π_δ (the explicit binder of Lemma 118), the formulas $\mu(U, V) = u_\delta(U \cdot V)$ and $\sigma(U) = u_\delta(U^\dagger)$ define C^1 maps on all of $\mathcal{U} \times \mathcal{U}$ and $\mathcal{U}$, and for all $U, V, Z \in \mathcal{U}$ one has $\mu(J, U) = \mu(U, J) = U$, $\sigma(J) = J$, and the five estimates $\|\mu(U, V) - U \cdot V\|$, $\|\sigma(U) - U^\dagger\|$, $\|\mu(\mu(U, V), Z) - \mu(U, \mu(V, Z))\|$, $\|\mu(\sigma(U), U) - J\|$, $\|\mu(U, \sigma(U)) - J\| \leq C_{\mathrm{grp}}\varepsilon_r$.*

Registry contract (verbatim).

```
Parameterized approximate-group transport: there exist C_grp^0, C_pol^0 >= 1 and kappa_pol^0 in (0,
1/2] such that, for every def-stage1-polar-witness-data tuple W with C_grp >= C_grp^0, C_pol >=
C_pol^0, and 0 < kappa_pol <= kappa_pol^0, for every finite-dimensional exact-unit
epsilon_r-C*-algebra and every delta > 0 satisfying C_pol*(epsilon_r + delta) <= kappa_pol and
C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta + delta^2), writing (u_delta, h_delta) for the
unique inverse of Pi_delta: calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x
B^{calH}_delta(J)), Pi_delta(U, H) = U bold-dot H, the formulas mu(U, V) = u_delta(U bold-dot V)
and sigma(U) = u_delta(U^dagger) define C^1 maps on all of calU x calU and calU, respectively, and
for every U, V, Z in calU, mu(J, U) = mu(U, J) = U, sigma(J) = J, ||mu(U, V) - U bold-dot V|| <=
C_grp*epsilon_r, ||sigma(U) - U^dagger|| <= C_grp*epsilon_r, ||mu(mu(U, V), Z) - mu(U, mu(V, Z))||
<= C_grp*epsilon_r, ||mu(sigma(U), U) - J|| <= C_grp*epsilon_r, and ||mu(U, sigma(U)) - J|| <=
C_grp*epsilon_r.
```

Proof (prose account of the af-validated tree). *Witnesses first.* Fix (G_d, P_d, k_d) from `lem-stage1-explicit-group-domain-membership`, (G_c, P_c, k_c) from `lem-stage1-explicit-group-closeness` and (P_r, k_r) from Lemma 101, and set $C^0_{\mathrm{grp}} = \max\{G_d, 8G_c, 8\}$, $C^0_{\mathrm{pol}} = \max\{P_d, P_c, P_r\}$ and $\kappa^0_{\mathrm{pol}} = \min\{k_d, k_c, k_r, \frac{1}{16}\}$ — *before* the receiving tuple W is quantified. Writing $e = \varepsilon_r$ and $y = e\delta + \delta^2 \geq 0$, the receiving guards give $P_i(e + \delta) \leq \kappa_{\mathrm{pol}} \leq k_i$ for $i \in \{d, c, r\}$ and $G_i e < \delta - P_i y$ for $i \in \{d, c\}$; also $0 \leq e < \frac{1}{16}$.

Synchronization. All three providers display *literally* the same $\Pi_\delta(U, H) = U \cdot H$ on the same domain $\mathcal{U} \times B^{\mathcal{H}}_\delta(J)$ with the same image S_δ, so ordinary uniqueness of the inverse of one bijection identifies their three inverse pairs with each other and with the (u_δ, h_δ) named in the root. Coherence naturality is neither needed nor imported. The domain bridge then puts $U \cdot V$ and $U^\dagger$ in S_δ for all $U, V \in \mathcal{U}$, so μ and σ are globally defined with values in $\mathcal{U}$; u_δ is C^1 on S_δ and both input maps are C^1 as real maps in finite dimension, so μ, σ are C^1.

Uniform estimates. For $T \in \mathcal{U}$, $T^\dagger \cdot T = J$ and a right inverse R exist; the C^*-lower bound with $\|J\| = 1$ gives $\|T\| \leq a := (1 - e)^{-1/2}$, so for $e < \frac{1}{16}$ multiplying a perturbation by a unitary costs at most $(1 + e)a < 2$ and an associator of three unitaries at most $ea^3 \leq 2e$. Exact unitality with the associator bound gives $\|R - T^\dagger\| \le e\|T\|^2\|R\|$, hence $\|R\| \le \|T\|(1-e)/(1-2e)$ and $\|T \bullet T^\dagger - J\| \le e(1+e)/((1-e)(1-2e)) \le 2e$. These apply to $\mu(A,B)$ and $\sigma(A)$ too, since u_δ takes values in $\mathcal{U}$.

The laws. Exact unitality gives $J \bullet U = U \bullet J = U$ and $J^\dagger = J$, while the synchronized retraction identity gives $u_\delta(U) = U$ for $U \in \mathcal{U}$; hence $\mu(J,U) = \mu(U,J) = U$ and $\sigma(J) = J$. The closeness bridge gives both $\|\mu(U,V) - U \bullet V\| \le G_c e$ and $\|\sigma(U) - U^\dagger\| \le G_c e$. For associativity the two outer retraction errors contribute $G_c e$ each, replacing $\mu(U,V)$ by $U \bullet V$ and $\mu(V,Z)$ by $V \bullet Z$ inside one multiplication costs at most $2G_c e$ on each side, and the algebra associator at most $2e$: total $(6G_c+2)e \le C_{\mathrm{grp}}e$, since $C_{\mathrm{grp}} \ge 8G_c$ and $C_{\mathrm{grp}} \ge 8$. For the inverse laws, $\|\mu(\sigma(U),U) - \sigma(U) \bullet U\| \le G_c e$, $\|\sigma(U) \bullet U - U^\dagger \bullet U\| \le 2G_c e$ and $U^\dagger \bullet U = J$ give $\|\mu(\sigma(U),U) - J\| \le 3G_c e$; on the other side the extra term $\|U \bullet U^\dagger - J\| \le 2e$ gives $\|\mu(U,\sigma(U)) - J\| \le (3G_c+2)e$. Both are at most $C_{\mathrm{grp}}\varepsilon_r$.

Lemma 122 (`lem-stage1-polar-constant-ledger`)**.** *There exists* one *universal Stage-1 polar witness tuple* $W = (C_{\mathrm{rect}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}}, C_{\mathrm{path}}, C_{\mathrm{der}}, e_{\mathrm{rect}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}}, \delta_*, \varepsilon_*^r, e_{S1}, r_{\mathrm{iso}})$ *in the sense of Definition 44, with all six coefficients* ≥ 1, $0 < e_{\mathrm{rect}} \le 1/C_{\mathrm{rect}}$ *and* $0 < \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}} \le \frac{1}{2}$, *such that the eight clauses* (A_1)–(A_7) *and* (R) *hold* simultaneously *at that single tuple:* (A_1)–(A_7) *are, verbatim, the conclusions of the seven transport rows (Lemmas 115, 116, 117, 118, 121, 119 and 120) instantiated at* W*, and* (R) *is the scalar range clause of Lemma 111 defining* $\delta_*, \varepsilon_*^r, e_{S1}, r_{\mathrm{iso}}$ *and asserting that every* $0 \le \varepsilon_X \le e_{S1}$ *meets all seven guards with the quantitative slack* $r_- \ge 3\delta_*/4$ *and* $\eta \le \delta_*/4$. Transcription note. *The full object-level text of the eight clauses is the verbatim contract below; the statement above names them rather than duplicating displays already typeset in §50–§48. What the consumer receives is the single existential "there exists* W *with* $\mathrm{Analytic}(W) \wedge \mathrm{Arithmetic}(W)$*",* not *a conjunction of unrelated existentials.*

Registry contract (verbatim).

```
Compatible Stage-1 polar witnesses and range: there exists one universal
def-stage1-polar-witness-data tuple W = (C_rect, C_ch, C_pol, C_grp, C_path, C_der, e_rect,
kappa_ch, kappa_pol, kappa_der, delta_*, epsilon_*^r, e_S1, r_iso), with C_rect, C_ch, C_pol,
C_grp, C_path, C_der >= 1, 0 < e_rect <= 1/C_rect, and 0 < kappa_ch, kappa_pol, kappa_der <= 1/2,
such that all of the following hold simultaneously: (A_1) for every finite-dimensional
epsilon_X-C*-algebra (calX, I_X, ., dagger) with 0 <= epsilon_X <= e_rect, there are on the same
involutive normed space a bilinear product bold-dot and an element J = J^dagger for which (calX, J,
bold-dot, dagger) satisfies every exact-unit epsilon_r-C*-algebra axiom of
def-epsilon-cstar-algebra, including ||J|| = 1, where epsilon_r = C_rect*epsilon_X, and for every
x, y in calX, ||J - I_X|| <= C_rect*epsilon_X and ||x bold-dot y - xy|| <=
C_rect*epsilon_X*||x||*||y||; (A_2) for every finite-dimensional exact-unit epsilon_r-C*-algebra
(calX, J, bold-dot, dagger), every delta > 0 with C_ch*(epsilon_r + delta) <= kappa_ch, every V in
calUbar_delta, and every A^par in B^{icalH}_{2delta}(0), there is a unique g_V(A^par) in
B^{calH}_{2delta}(0) such that f_V(A^par + g_V(A^par)) = 0, where f_V(A) = (1/2)*(((J + A^dagger)
bold-dot V^dagger) bold-dot (V bold-dot (J + A)) - J), the element V bold-dot (J + A^par +
g_V(A^par)) lies in calU, ||g_V(A^par) + (1/2)*(V^dagger bold-dot V - J)|| <= C_ch*(epsilon_r*delta
+ delta^2), ||Dg_V(A^par)|| <= C_ch*(epsilon_r + delta), and ||D_{A^perp} f_V(A^par + g_V(A^par)) -
I_{calH}|| <= C_ch*(epsilon_r + delta) < 1, and these C^1 graph charts cover calU; (A_3) for every
finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0 with C_ch*(epsilon_r + delta)
<= kappa_ch, and every family g = (g_U)_{U in calU} of C^1 maps g_U: B^{icalH}_{2delta}(0) ->
B^{calH}_{2delta}(0) such that, for every U in calU and A^par in B^{icalH}_{2delta}(0), g_U(A^par)
is the unique element of B^{calH}_{2delta}(0) satisfying f_U(A^par + g_U(A^par)) = 0, where f_U(A)
= (1/2)*(((J + A^dagger) bold-dot U^dagger) bold-dot (U bold-dot (J + A)) - J), every tangent space
T_U calU is the image of L_U(I + Dg_U(0)): icalH -> calX, and omega_U(Z) = (L_U^{-1} Z)^par : T_U
calU -> icalH is a global C^1 bundle trivialization with distortion at most 1 + C_ch*epsilon_r,
satisfying omega_{cU}(cZ) = omega_U(Z) and omega_U(iU) = iJ for every U in calU, Z in T_U calU, and
c in U(1); (A_4) for every finite-dimensional exact-unit epsilon_r-C*-algebra and every delta > 0
with C_pol*(epsilon_r + delta) <= kappa_pol, the map Pi_delta: calU x B^{calH}_delta(J) -> calX,
```

Pi_delta(U, H) = U bold-dot H, is a C^1 diffeomorphism onto the open set S_delta := Pi_delta(calU x B^{calH}_delta(J)), its inverse (u_delta, h_delta): S_delta -> calU x B^{calH}_delta(J) satisfies X = u_delta(X) bold-dot h_delta(X), u_delta(U) = U, and h_delta(U) = J for every X in S_delta and U in calU, and calU_{delta - C_pol*(epsilon_r*delta + delta^2)} subseteq S_delta subseteq calU_{delta + C_pol*(epsilon_r*delta + delta^2)}; (A_5) for every finite-dimensional exact-unit epsilon_r-C*-algebra and every delta > 0 satisfying C_pol*(epsilon_r + delta) <= kappa_pol and C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta + delta^2), writing (u_delta, h_delta) for the unique inverse of Pi_delta: calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H) = U bold-dot H, the formulas mu(U, V) = u_delta(U bold-dot V) and sigma(U) = u_delta(U^dagger) define C^1 maps on all of calU x calU and calU, respectively, and for every U, V, Z in calU, mu(J, U) = mu(U, J) = U, sigma(J) = J, ||mu(U, V) - U bold-dot V|| <= C_grp*epsilon_r, ||sigma(U) - U^dagger|| <= C_grp*epsilon_r, ||mu(mu(U, V), Z) - mu(U, mu(V, Z))|| <= C_grp*epsilon_r, ||mu(sigma(U), U) - J|| <= C_grp*epsilon_r, and ||mu(U, sigma(U)) - J|| <= C_grp*epsilon_r; (A_6) for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0 with C_pol*(epsilon_r + delta) <= kappa_pol, every U_0, U_1 in calU, and every q in [0, 1] satisfying ||U_1 - U_0|| <= q, C_path*q <= 1/4, and C_path*(q + epsilon_r*q + q^2) < delta - C_pol*(epsilon_r*delta + delta^2), every L_{Z_t} is invertible and every Z_t = (1-t)*U_0 + t*U_1 lies in calUbar_{C_path*(q + epsilon_r*q + q^2)} for t in [0, 1], and, writing u_delta for the unique first component of the inverse of Pi_delta: calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H) = U bold-dot H, the map H(t, U_0, U_1) = u_delta(Z_t) is jointly continuous in (t, U_0, U_1), joins U_0 to U_1, and satisfies H(t, cU_0, cU_1) = c*H(t, U_0, U_1) for every c in U(1); (A_7) for every finite-dimensional exact-unit epsilon_r-C*-algebra, every delta > 0, every s in {+1, -1}, and every 0 < r <= delta satisfying C_ch*(epsilon_r + delta) <= kappa_ch, C_pol*(epsilon_r + delta) <= kappa_pol, C_grp*epsilon_r < delta - C_pol*(epsilon_r*delta + delta^2), C_der*(epsilon_r + r) <= kappa_der, and (1 + epsilon_r)*(1 + C_ch*(epsilon_r + delta))*r + C_grp*epsilon_r < 2*delta, writing u_delta for the unique first component of the inverse of Pi_delta: calU x B^{calH}_delta(J) -> S_delta := Pi_delta(calU x B^{calH}_delta(J)), Pi_delta(U, H) = U bold-dot H, and g_{sJ}: B_{2delta}^{icalH}(0) -> B_{2delta}^{calH}(0) for the unique C^1 map such that, for every A in B_{2delta}^{icalH}(0), f_{sJ}(A + g_{sJ}(A)) = 0, where f_{sJ}(B) = (1/2)*(((J + B^dagger) bold-dot (sJ)^dagger) bold-dot (sJ bold-dot (J + B)) - J), define chi_s(A) = sJ bold-dot (J + A + g_{sJ}(A)) and the global C^1 map sigma(U) = u_delta(U^dagger); then sigma maps chi_s(B_r^{icalH}(0)) into the same sJ-graph chart and, with F_s(A) = phi_{sJ}^par(sigma(chi_s(A))), one has ||D(F_s - id)(A) + 2*I_{icalH}|| <= C_der*(epsilon_r + r) for every A in B_r^{icalH}(0); and (R) delta_* = min{1/4, kappa_ch/(4*C_ch), kappa_pol/(4*C_pol)}, epsilon_*^r = min{1/4, kappa_ch/(4*C_ch), kappa_pol/(4*C_pol), kappa_der/(8*C_der), 1/C_grp, delta_*/(12*C_path*C_grp)}, e_S1 = min{e_rect, epsilon_*^r/C_rect}, r_iso = min{delta_*/4, kappa_der/(8*C_der)}, and for every 0 <= epsilon_X <= e_S1, on setting epsilon_r = C_rect*epsilon_X, q = C_grp*epsilon_r, r_- = delta_* - C_pol*(epsilon_r*delta_* + delta_*^2), and eta = C_path*(q + epsilon_r*q + q^2), one has C_ch*(epsilon_r + delta_*) <= kappa_ch, C_pol*(epsilon_r + delta_*) <= kappa_pol, q < r_-, C_path*q <= 1/4, eta < r_-, C_der*(epsilon_r + r_iso) <= kappa_der, (1 + epsilon_r)*(1 + C_ch*(epsilon_r + delta_*))*r_iso + q < 2*delta_*, r_- >= 3*delta_*/4, eta <= delta_*/4, and C_der*(r_iso + epsilon_r) <= kappa_der/4 < 1.

Proof (prose account of the af-validated tree). *One tuple, chosen by finite maxima and minima.* Each of the seven transport rows is an existential over *thresholds*: base coefficients that must be dominated and base margins that must dominate. Take those fourteen existential packages; define each of $C_{\mathrm{rect}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}}, C_{\mathrm{path}}, C_{\mathrm{der}}$ as the maximum of 1 and every base coefficient bearing that name; define e_{rect} as the minimum of the rectification margin and $1/C_{\mathrm{rect}}$; and define each of $\kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}}$ as the minimum of $\frac{1}{2}$ and every positive base margin bearing that name. These are finite maxima and minima, so the ranges ≥ 1, $(0, 1/C_{\mathrm{rect}}]$ and $(0, \frac{1}{2}]$ hold and *every* helper's coefficient floor and margin cap is met at once. Define $\delta_*, \varepsilon_*^r, e_{S1}, r_{\mathrm{iso}}$ by the four displayed formulas of (R); by Definition 44 these fourteen scalars are one tuple W.

Seven applications and one arithmetic application. Because W meets every threshold, each parameterized transport applies to W *verbatim* and returns its conclusion at the fields of W; those conclusions are, respectively, clauses (A_1)–(A_7). The tuple also has coefficients ≥ 1, $0 < e_{\mathrm{rect}} \leq 1/C_{\mathrm{rect}}$ and the three margins in $(0, \frac{1}{2}]$, and its four derived scales were *defined* by the displayed formulas, so Lemma 111 applies verbatim and returns clause (R) for every $0 \leq \varepsilon_X \leq e_{S1}$. Conjoining the tuple range, (A_1)–(A_7) and (R) — all predicated of the *same* W — is the root. This is the entire content of the row: no new analysis, only the fact that seven separately-quantified thresholds admit a common witness.

Lemma 123 (`lem-finite-polyhedron-maximal-simplex-placement`)**.** *Every point of a finite polyhedron lies in a maximal simplex of its defining finite simplicial complex; therefore every point of every finite fixed set does.* Transcription and amendment note. *The second clause originally read "therefore every finite fixed set does". A fresh verifier challenged the root as* ambiguous*: on the collective reading ("the entire finite set lies in ONE maximal simplex") the clause is* ***refuted*** *— two isolated vertices are a counterexample — and only the pointwise reading is entailed. Because a disambiguation is a contract change rather than a local proof repair, the run was stopped and returned to the user; the pointwise wording quoted above was* ***ratified by the user on 2026-07-29*** *and is the reading intended by the design prose and by the downstream consumer* `lem-stage1-extra-fixed-class`*. The workspace was cleanly re-seeded against the amended contract, and it is that amended contract — not the ambiguous original — that carries the validation.*

Registry contract (verbatim).

```
Every point of a finite polyhedron lies in a maximal simplex of its defining finite simplicial
complex; therefore every point of every finite fixed set does.
```

Proof (prose account of the af-validated tree). Let K be the defining finite simplicial complex and let $x \in |K|$. By the definition of the geometric realization, x lies in some simplex ρ of K. The simplices of K containing ρ form a nonempty finite partially ordered set under inclusion, hence have a maximal element τ; were τ properly contained in a simplex of K, that simplex would also contain ρ, contradicting maximality in this set. So τ is a maximal simplex of K and $x \in \tau$. Now let F be any finite fixed set inside that polyhedron and apply the pointwise conclusion separately to each $x \in F$: for every $x \in F$ there is a maximal simplex τ_x with $x \in \tau_x$, the simplex being permitted to depend on x. That dependence is precisely what the amended second clause asserts. The argument is algebra-independent and uses only finite-poset and simplicial facts, which are common knowledge and carry no definition shard.

Status. All three rows are `proved` with `af: validated` (T0), tier *routine*, taint clean, first-pass, validated 2026-07-29. `proofs/lem-stage1-approximate-group-laws-transport`: root validated, 16/16 live nodes (exactly the design target; hard cap 22), 6 rounds, three in-run challenges repaired — all the $\varepsilon_r = 0$ strict-inequality endpoint, the bounds being made non-strict — and re-verified fresh. This is the **first** validation of the 13e row ever: two earlier attempts stuck on a synchronization gap between the group-law u_δ and the polar u_δ, which the typed three-provider spine removes by construction. `proofs/lem-stage1-polar-constant-ledger`: root validated, 11/11 live nodes (exactly the design budget; hard cap 12), 3 rounds, *zero* challenges. `proofs/lem-finite-polyhedron-maximal-simplex-placement`: run 2, root validated, 3/3 live nodes, 2 rounds, zero challenges (run 1 aborted STUCK on the contract ambiguity recorded above, and its ledger was discarded at workspace restore).

Correction of record. For the ledger row a mechanical *consumer re-check* was performed before seeding: clauses (A_5), (A_6) and (A_7) are byte-identical to the contract bodies of the 13e, 13f and 13g rows (modulo the terminal period), so the keystone consumes exactly what those rows export. For the finite-polyhedron row the correction is **contract-level** and is the amendment quoted in the lemma; both other rows are byte-unchanged.
Role in the chain. Registry `defs`: the two Stage-1 rows import `def-stage1-polar-witness-data`, `def-approximate-unitary-space` and `def-epsilon-cstar-algebra`; the finite-polyhedron row imports none. Registry `deps`: the group-law transport imports `lem-stage1-explicit-group-domain-membership`, `lem-stage1-explicit-group-closeness` and Lemma 101 (§45); the constant ledger imports all seven transports together with Lemma 111 (§48); the finite-polyhedron row imports nothing. The ledger is the Stage-1 keystone, not an endpoint: it supplies a compatible constant tuple to the phase-4 consumers, none of which is on the paper track yet. Nothing here assembles the Stage-1 front into a statement about almost-idempotent maps, and `op-classical` is now discharged at the af-validated rung via Route F (§76); sharpness is now carried at T0 by `cor-classical-sharpness`.

53 The MAIN-CB raw-call chain: initial inclusion, one-step improvement, iteration, and the corner envelope

This section records the six af-validated rows forming the analytic front of the MAIN-CB raw-call chain. The vocabulary is that of Definitions 29, 36, and 23; the proof accounts report the validated witnesses, corrections, and estimates without adding any assertion to the registry contracts.

Lemma 124 (lem-maincb-initial-raw-inclusion). *There are universal constants $D_0 < \infty$ and $e_0 > 0$ such that, in every finite-dimensional extended ϵ-C^*-algebra with $\epsilon \le t \le e_0$, the scalar map*

$$v : \mathbb{C} \longrightarrow A, \qquad v(\lambda) = \lambda I_A,$$

is an extended $D_0 t$-inclusion. If $\dim_{\mathbb{C}} A = 1$, then v is bijective.

Registry contract (verbatim).

```
There are universal D_0 < infinity and e_0 > 0 such that, in every finite-dimensional extended
epsilon-C*-algebra with epsilon <= t <= e_0, the scalar map lambda |-> lambda*I_A is an extended
D_0*t-inclusion; if dim A = 1, it is bijective.
```

Proof (prose account of the af-validated tree). The tree takes $D_0 = 2$ and $e_0 = 1/4$. Its first branch derives the exact operator-space cross-norm identity $\|X \otimes a\|_n = \|X\|\|a\|$: rectangular multiplication and block diagonals give the upper bound, while compression by unit vectors ξ, η with $|\eta^* X \xi| = \|X\|$ gives the reverse inequality. Thus $v_n(X) = X \otimes I_A$ has norm $\|X\|\|I_A\|$ at every level. The extended ϵ-C^* axioms make $I_n \otimes I_A$ the designated amplified unit, give $I_A^\dagger = I_A$ and $|\|I_A\| - 1| \le \epsilon$, and yield

$$\|v_n(X)v_n(Y) - v_n(XY)\|_n \le \epsilon\|v_n(XY)\|_n \le 2\epsilon\|X\|\|Y\|$$

when $\epsilon \le 1/2$. Hence $\epsilon \le t \le 1/4$ gives the multiplicative error $2t$ and the two-sided $(1 \pm 2t)$ norm bounds, exactly the requirements for an extended $2t$-inclusion. At level one these bounds give $\|I_A\| = \|v(1)\| \ge 1 - 2t \ge 1/2$. Therefore $I_A \ne 0$, and in complex dimension one the scalar map is both injective and surjective. Three challenges in the recorded run exposed the missing extended-algebra vocabulary and sibling-only dependencies; the conditional nodes were archived, and the live replacement branch uses Definition 23 explicitly.

Lemma 125 (lem-maincb-improvement-one-step). *There are universal $K_{\mathrm{step}} \ge 1$ and $e_{\mathrm{step}} > 0$ such that, if B is a finite-dimensional C^*-algebra, A is an extended ϵ-C^*-algebra, and $v : B \to A$ is an extended d-inclusion with $d + \epsilon \le e_{\mathrm{step}}$, then there is one dagger-preserving level-one map v^+, with $v_n^+ = I_n \otimes v^+$, for which*

$$\sup_{n\ge1}\|v_n^+ - v_n\| \le K_{\mathrm{step}} d, \qquad v^+ \textit{ is an extended } d^+\textit{-inclusion}, \qquad d^+ \le K_{\mathrm{step}}(d^2 + \epsilon).$$

Registry contract (verbatim).

```
There are universal K_step >= 1 and e_step > 0 such that, if B is a finite-dimensional C*-algebra,
A an extended epsilon-C*-algebra, and v:B->A an extended d-inclusion with d+epsilon <= e_step, then
one dagger-preserving level-one map v^+, with v_n^+ = I_n tensor v^+, satisfies sup_n ||v_n^+ -
v_n|| <= K_step*d and is an extended d^+-inclusion for d^+ <= K_step*(d^2+epsilon).
```

Proof (prose account of the af-validated tree). Choose the norm-one diagonal of Definition 24 in the form $D = \sum_j A_j \otimes B_j$, where $A_j = p_j U_j^\dagger$, $B_j = U_j$, $p_j \geq 0$, and $\sum_j p_j = 1$. The tree constructs it blockwise from Weyl unitaries and kills cross-block terms by independent sign averaging. For $g(X,Y) = v(XY) - v(X)v(Y)$ set

$$w'(X) = \sum_j v(A_j) g(B_j, X), \quad w''(X) = w'(X^\dagger)^\dagger, \quad w = \tfrac{1}{2}(w' + w''), \quad v^+ = v + w.$$

Entrywise multiplication proves that the correction at every matrix level is exactly $I_n \otimes w$; the diagonal weight sum and the inclusion bounds give $\sup_n \|I_n \otimes w\| \leq 2d$. The cocycle identity for g, centrality of D, and its multiplication identity cancel the linear defect. Seven explicit associativity, multiplicativity, product, and unit remainders total at most $100(d^2 + \epsilon)$; symmetrizing preserves that estimate, and the quadratic term $w(X)w(Y)$ raises it to $105(d^2 + \epsilon)$. With $q = d^2 + \epsilon$, the element $x = v^+(I_B)$ has $\|x - I_A\| \leq 3d$ and $\|x^2 - x\| \leq 105q$. Absorbing the approximate-unit terms for $d + \epsilon \leq 1/16$ gives the amplified unit defect at most $424q$. After further shrinking to $d + \epsilon \leq e_{\mathrm{step}} := \min\{1/16, (8 \cdot 424)^{-1}\}$, the upper C^* estimate and the lower bound inherited from v give two-sided norm bounds, so v^+ is an extended d^+-inclusion with $d^+ \leq 1700q$; these constants furnish the universal witnesses. The ledger records six repaired challenges across two runs, including provisioning the base ϵ-C^* axioms and the norm-one standard diagonal; the contract did not change.

Lemma 126 (`lem-maincb-improvement-iteration`). *There are universal $e_{\mathrm{it}} > 0$, $K_{\mathrm{disp}} < \infty$, and $K_{\mathrm{floor}} < \infty$ such that, if B is a finite-dimensional C^*-algebra, A is an extended ϵ-C^*-algebra, and $v : B \to A$ is an extended d-inclusion with $d + \epsilon \leq e_{\mathrm{it}}$, then one dagger-preserving $\widetilde{v}$, with $\widetilde{v}_n = I_n \otimes \widetilde{v}$, satisfies*

$$\sup_{n \geq 1} \|\widetilde{v}_n - v_n\| \leq K_{\mathrm{disp}} d \quad \textit{and is an extended } K_{\mathrm{floor}}\epsilon\textit{-inclusion.}$$

For $\epsilon > 0$ it is reached after finitely many correction steps; for $\epsilon = 0$ it is their operator-norm limit.

Registry contract (verbatim).

```
There are universal e_it > 0, K_disp < infinity, and K_floor < infinity such that, if B is a
finite-dimensional C*-algebra, A is an extended epsilon-C*-algebra, and v:B->A is an extended
d-inclusion with d+epsilon <= e_it, then one dagger-preserving v_tilde, with v_tilde_n = I_n tensor
v_tilde, satisfies sup_n ||v_tilde_n - v_n|| <= K_disp*d and is an extended
K_floor*epsilon-inclusion; for epsilon > 0 it is reached after finitely many correction steps, and
for epsilon = 0 it is their operator-norm limit.
```

Proof (prose account of the af-validated tree). Let $K = K_{\mathrm{step}}$ and $e = e_{\mathrm{step}}$ be the witnesses of Lemma 125, and choose

$$e_{\mathrm{it}} = \min\{e, (4K)^{-1}\}, \qquad K_{\mathrm{disp}} = 4K, \qquad K_{\mathrm{floor}} = 2K.$$

Starting from $(v_0, d_0) = (v, d)$, one step produces (v_{j+1}, d_{j+1}) with $d_{j+1} \leq K(d_j^2 + \epsilon)$ and amplified displacement at most Kd_j. While $d_j > 2K\epsilon$, the smallness choice gives $Kd_j^2 \leq d_j/4$ and $K\epsilon < d_j/2$, hence $d_{j+1} \leq 3d_j/4$. If $\epsilon > 0$, this geometric descent reaches $d_N \leq 2K\epsilon$ after finitely many steps; when the initial defect is already below the floor, the prescribed single correction remains below it. Monotonicity of the defining error inequalities makes v_N an extended $K_{\mathrm{floor}}\epsilon$-inclusion, and telescoping gives $\sup_n \|v_{N,n} - v_n\| \leq K \sum_j d_j \leq 4Kd$. If $\epsilon = 0$, instead $d_{j+1} \leq d_j/4$, so the maps are uniformly Cauchy at every amplification. Completeness at level one and finite dimensionality of B give a level-one operator-norm limit $\widetilde{v}$ whose amplifications are the corresponding limits. Dagger preservation, exact multiplicativity, the unit, and exact norm bounds pass to the limit, yielding an extended 0-inclusion with the same displacement estimate. This is the strengthened contract validated after the earlier weaker defect-only wording was superseded; one threshold-boundary challenge in the clean re-seed was repaired without changing the strengthened statement.

Lemma 127 (`lem-maincb-error-improvement`). *There are universal $\epsilon^{\mathrm{cb}}_{max} > 0$, $\delta^{\mathrm{cb}}_{max} > 0$, and $c^{\mathrm{cb}}_0 < \infty$ such that every extended δ-inclusion $v : B \to A$ from a finite-dimensional C^*-algebra into an extended ϵ-C^*-algebra, with $0 \leq \epsilon \leq \epsilon^{\mathrm{cb}}_{max}$ and $0 \leq \delta \leq \delta^{\mathrm{cb}}_{max}$, can be replaced by an extended $c^{\mathrm{cb}}_0\epsilon$-inclusion $\widetilde{v} : B \to A$ that is bijective whenever v is bijective.*

Registry contract (verbatim).

```
Complete error improvement: there are universal epsilon_max^cb>0, delta_max^cb>0 and
c_0^cb<infinity such that every extended delta-inclusion v:B->A from a finite-dimensional
C*-algebra B into an extended epsilon-C*-algebra A with 0<=epsilon<=epsilon_max^cb and
0<=delta<=delta_max^cb can be replaced by an extended c_0^cb*epsilon-inclusion v_tilde:B->A that is
bijective whenever v is bijective.
```

Proof (prose account of the af-validated tree). Take the witnesses of Lemma 126 and set

$$\epsilon^{\mathrm{cb}}_{max} = e_{\mathrm{it}}/2, qquad\delta^{\mathrm{cb}}_{\mathrm{max}} = \min\{e_{\mathrm{it}}/2, [2(1 + |K_{\mathrm{disp}}|)]^{-1}\}, \qquad c^{\mathrm{cb}}_0 = K_{\mathrm{floor}}.$$

Then $\delta + \epsilon \leq e_{\mathrm{it}}$ and $1 - \delta - |K_{\mathrm{disp}}|\delta \geq 1/2$. Applying the iteration row with $d = \delta$ supplies a dagger-preserving $\widetilde{v}$ that is an extended $c^{\mathrm{cb}}_0\epsilon$-inclusion and satisfies at level one $\|\widetilde{v} - v\| \leq K_{\mathrm{disp}}\delta$. If v is bijective, its lower inclusion bound and this displacement imply, for every $x \in B$,

$$\|\widetilde{v}(x)\| \geq \|v(x)\| - \|(\widetilde{v} - v)(x)\| \geq (1 - \delta - |K_{\mathrm{disp}}|\delta)\|x\| \geq \tfrac{1}{2}\|x\|.$$

Thus $\widetilde{v}$ is injective. Bijectivity of v identifies the finite dimensions of A and B, so this injective linear map is also surjective. The registry records four runs: earlier verifier cohorts found that the then-weaker iteration contract supplied only a defect bound, and an internalized attempt ballooned; the user-ratified strengthening reproduced in Lemma 126 made the final 17-node consumer tree close. The present root contract itself was unchanged throughout that repair history.

Lemma 128 (`lem-extcb-exact-target-correction`). *There are universal $a_{\mathrm{corr}} > 0$ and $C_{\mathrm{corr}} < \infty$ with the following property. Let B be a finite-dimensional C^*-algebra, H a finite-dimensional Hilbert space, and $T : B \to B(H)$ linear and dagger-preserving. If $0 \leq a \leq a_{\mathrm{corr}}$ and, for every n,*

$$\|T_n(XY) - T_n(X)T_n(Y)\| \leq a\|X\|\|Y\|, \qquad \|T_n(I) - I\| \leq a,$$

then one unital dagger-homomorphism $\mu : B \to B(H)$ satisfies $\|\mu_n - T_n\| \leq C_{\mathrm{corr}}a$ at every amplification.

Registry contract (verbatim).

```
There are universal a_corr>0 and C_corr<infinity with the following property: if B is a
finite-dimensional C*-algebra, H a finite-dimensional Hilbert space, and T:B->B(H) is linear,
dagger-preserving, has ||T_n(XY)-T_n(X)T_n(Y)||<=a||X||||Y|| and ||T_n(I)-I||<=a at every n, where
0<=a<=a_corr, then one unital dagger-homomorphism mu:B->B(H) satisfies ||mu_n-T_n||<=C_corr*a at
every n.
```

Proof (prose account of the af-validated tree). For $a \le 1$, the C^* identity and the multiplicative defect give $M_n^2 \le M_n + a$ for $M_n = \|T_n\|$, hence $M_n < 2$ uniformly. Choose a state φ on B, put $e = I - T(I)$, and set $S_0(x) = T(x) + \varphi(x)e$. Then S_0 is exactly unital and dagger-preserving, $\|S_0 - T\|_{\mathrm{cb}} \le a$, $\|S_0\|_{\mathrm{cb}} < 3$, and its amplified bilinear defect b_0 is at most $7a$. Haar averaging on the compact unitary group gives a finite convex norm-one diagonal $D = \sum_s p_s U_s^\dagger \otimes U_s$. For any exactly unital dagger-preserving S with $\|S\|_{\mathrm{cb}} \le 4$ and defect b, the correction

$$w'(x) = \sum_s S(p_s U_s^\dagger) G(U_s, x), \qquad w(x) = \tfrac{1}{2}\big(w'(x) + w'(x^\dagger)^\dagger\big)$$

has $\|w\|_{\mathrm{cb}} \le 4b$ and $w(I) = 0$. The exact cocycle identity, centrality of D, and multiplication of D to I show that $S^+ = S + w$ has defect at most $18b^2$. Taking $a_{\mathrm{corr}} = \min\{(2 \cdot 18 \cdot 7)^{-1}, (16 \cdot 7)^{-1}\}$ makes $b_k \le b_0\, 2^{-k}$ under iteration and keeps every iterate below cb norm 4. The corrections have total norm at most $8b_0$, including the degenerate case $b_0 = 0$, so they converge in cb norm to one level-one unital dagger-homomorphism μ. Finally $\|\mu - T\|_{\mathrm{cb}} \le a + 8(7a) = 57a$, so $C_{\mathrm{corr}} = 57$ works uniformly at all levels. The six-node workspace validated on its first pass with no challenges or contract amendment.

Lemma 129 (`lem-maincb-direct-corner-envelope`). *There are universal $L \ge 1$ and $e_{\mathrm{env}} > 0$ such that, if A is a finite-dimensional extended ϵ-C^*-algebra, $0 \le \epsilon \le e_{\mathrm{env}}$, and $w : \mathbb{C}^m \to A$ is an extended $c_0^{\mathrm{cb}}\epsilon$-inclusion, then, writing $e_U = \sum_{j \in U} e_j$, $P_U = w(e_U)$, and $A_U = \mathcal{S}^A_{P_U}$, every nonempty U has P_U a $c_0^{\mathrm{cb}}\epsilon$-projection and A_U an extended $L\epsilon$-C^*-algebra. Moreover, whenever $U \subseteq R$ and $V = R \setminus U$, one has*

$$\max\{\|P_U + P_V - P_R\|, \|P_U P_R - P_U\|, \|P_R P_U - P_U\|, \|P_V P_R - P_V\|, \|P_R P_V - P_V\|, \|P_U P_V\|, \|P_V P_U\|\} \le L\epsilon.$$

Registry contract (verbatim).

```
There are universal L >= 1 and e_env > 0 such that, if A is a finite-dimensional extended
epsilon-C*-algebra, 0 <= epsilon <= e_env, and w:C^m->A is an extended c_0^cb*epsilon-inclusion,
then every nonempty U has P_U a c_0^cb*epsilon-projection, every A_U = S^A_{P_U} is an extended
L*epsilon-C*-algebra, and for U subseteq R all subordination and complementarity errors among P_U,
P_{R minus U}, P_R are at most L*epsilon.
```

Proof (prose account of the af-validated tree). Choose the witness c_0^{cb} of Lemma 127 at least 1, and let $C_{\mathrm{ca}}, e_{\mathrm{ca}}$ be the witnesses of Lemma 64. Put

$$K = c_0^{\mathrm{cb}} + 1, \qquad L = \max\{1, c_0^{\mathrm{cb}}, C_{\mathrm{ca}} K\}, \qquad e_{\mathrm{env}} = \min\{e_{\mathrm{ca}}/K, (2c_0^{\mathrm{cb}})^{-1}\}.$$

For $\delta = c_0^{\mathrm{cb}}\epsilon$ these choices give $\delta \le 1/2$, $\delta + \epsilon \le e_{\mathrm{ca}}$, and both δ and $C_{\mathrm{ca}}(\delta + \epsilon)$ at most $L\epsilon$. For nonempty U, the coordinate idempotent e_U is self-adjoint and has norm 1. Applying the level-one inclusion bounds to $P_U = w(e_U)$ gives $P_U^\dagger = P_U$, $\|P_U^2 - P_U\| \le \delta$, and $1 - \delta \le \|P_U\| \le 1 + \delta$; thus it is a nonvanishing δ-projection. Lemma 64 then makes its compressed corner an extended $C_{\mathrm{ca}}(\delta + \epsilon)$-$C^*$-algebra, hence an extended $L\epsilon$-C^*-algebra. Finally the exact coordinate identities $e_U e_R = e_U$, $e_V e_R = e_V$, $e_U e_V = 0$, and $e_U + e_V = e_R$ give the displayed six errors by δ-multiplicativity and the exact additive identity by linearity. A verifier challenged a hidden partition-state assumption; the repaired live bridge binds P_U and A_U directly from this same displayed pair (A, w), so no current-union or reset-state field is consumed.

Status. Each of the six rows is `proved` with `af: validated`; the workspace and repair particulars are recorded in the companion provenance sidecar.

54 MAIN-CB corner identification: merges, nested comparison, dimension transport, and corner equivalence

This section records the validated corner-identification block used by the MAIN-CB assembly. The notation and ambient hypotheses are those of Definitions 20, 21, 36, and 23; no definition is repeated here.

Lemma 130 (`lem-maincb-direct-sum-inclusion-merge`). *There are universal constants $C_{\mathrm{dir}} < \infty$ and $e_{\mathrm{dir}} > 0$ such that, if B_1, B_2 are finite-dimensional C^*-algebras, P_1, P_2 are target t-projections with $\|P_1 + P_2 - I\| \le t$, and $v_i : B_i \to \mathcal{S}_{P_i}$ are extended t-inclusions whose target ambient defect is at most $t \le e_{\mathrm{dir}}$, then*

$$V : B_1 \oplus B_2 \longrightarrow \mathcal{A}, \qquad V(x_1, x_2) = v_1(x_1) + v_2(x_2),$$

is an extended $C_{\mathrm{dir}}t$-inclusion. It is asserted to be bijective only when both v_i are bijective and both cross-corners $\mathcal{S}_{P_1,P_2}$ and $\mathcal{S}_{P_2,P_1}$ vanish.

Registry contract (verbatim).

```
There are universal C_dir < infinity and e_dir > 0 such that, if B_1,B_2 are finite-dimensional
C*-algebras, P_1,P_2 are target t-projections, ||P_1+P_2-I|| <= t, and v_i:B_i->S_{P_i} are
extended t-inclusions with target ambient defect at most t <= e_dir, then (x_1,x_2) |->
v_1(x_1)+v_2(x_2) is an extended C_dir*t-inclusion; bijectivity is asserted only if both v_i are
bijective and both target cross-corners vanish.
```

Proof (prose account of the af-validated tree). At amplification level n, put $p_i = I_n \otimes P_i$ and let C_{ij} denote the corresponding compression. The tree first bounds $\|p_i\| \le 2$ and, writing $r = p_1 + p_2 - I$, uses the exact bilinear identity $p_i p_j = p_i r + (p_i I - p_i) - (p_i^2 - p_i)$ to obtain $\|p_i p_j\| = O(t)$. The compression estimates and amplified idempotence then give uniformly bounded C_{ii}, small composites $C_{ii}C_{jj}$ for $i \ne j$, and $C_{ii}(a_1 + a_2) = a_i + O(t)\max_j \|a_j\|$ for $a_j \in \mathcal{S}_{p_j,p_j}$. For $V_n = I_n \otimes V$, these extraction estimates first yield level-independent coarse upper and lower norm bounds. Exact involution preservation, the two diagonal homomorphism estimates, and the exposed factors p_1p_2, p_2p_1 control the unit, involution, and product defects. A repeated-squaring C^*-bootstrap then sharpens the coarse bounds to $(1 - Ct)\|x\| \le \|V_n x\| \le (1 + Ct)\|x\|$. For the optional conclusion, the four-corner analysis and synthesis maps satisfy $\|\alpha\beta - I\| = O(t)$, so α is onto by a Neumann series. If the cross-corners vanish and both v_i are onto their diagonal corners, this makes V surjective; cross-extraction gives injectivity. The validation repaired five sibling-dependency challenges, including making the coarse-to-sharp bootstrap self-contained; the final tree has 14 clean live nodes and makes no bijectivity claim without all four extra hypotheses.

Lemma 131 (`lem-maincb-full-corner-identification`). *There is a universal $e_{\mathrm{full}} > 0$ such that, if R is a t-projection in an extended t-C^*-algebra and $\|R - I\| \le t \le e_{\mathrm{full}}$, then $\mathrm{Co}_R = I$ and $\mathcal{S}_R = \mathcal{A}$ at every amplification.*

Registry contract (verbatim).

```
There is a universal e_full > 0 such that, if R is a t-projection in an extended t-C*-algebra and
||R-I|| <= t <= e_full, then Co_R = I and S_R = A, at every amplification.
```

Proof (prose account of the af-validated tree). The compression-to-raw-product comparison gives $\|\mathrm{Co}_R - L_R R_R\| \le 2C_{\mathrm{co}} t$. For $\|X\| = 1$, the approximate unit and product estimates, expanded with $R = I + (R - I)$ and with the displayed association $R(XR)$, give $\|L_R R_R - I\| \le 21t$. Thus $\|\mathrm{Co}_R - I\| \le (21 + 2C_{\mathrm{co}})t < 1$ after one universal threshold reduction. Lemma 55 makes Co_R idempotent. The tree then applies the elementary rigidity argument: if $E^2 = E$ and $\|E - I\| = q < 1$, then for $Y = (I - E)X$ one has $(I - E)Y = Y$, whence $\|Y\| \le q\|Y\|$ and therefore $Y = 0$. Consequently $E = I$, so here $\mathrm{Co}_R = I_{\mathcal{A}}$ and its range $\mathcal{S}_R$ is all of $\mathcal{A}$. Finally Lemma 54 identifies $\mathrm{Co}_{R_n,R_n} = I_n \otimes \mathrm{Co}_R$ and $\mathcal{S}_{R_n} = M_n \otimes \mathcal{S}_R$, transporting the conclusion to every amplification. One verifier challenge was repaired by making the rigidity node self-contained; the validated closure has 8 live nodes and 2 archived nodes.

Lemma 132 (`lem-maincb-nested-corner-comparison`)**.** *There are universal $C_{\mathrm{nest}} < \infty$ and $e_{\mathrm{nest}} > 0$ such that, whenever R, P, Q are t-projections in a finite-dimensional extended t-C^*-algebra, R is nonvanishing, P, Q are subordinate to R with all four left and right errors at most $t \le e_{\mathrm{nest}}$, and $\mathcal{A}_R = \mathcal{S}^{\mathcal{A}}_R$, $P^R = \mathrm{Co}^{\mathcal{A}}_R(P)$, $Q^R = \mathrm{Co}^{\mathcal{A}}_R(Q)$, then P^R, Q^R are $C_{\mathrm{nest}}t$-projections in $\mathcal{A}_R$ and, at every amplification,*

$$\|F^R_{P,Q}(\mathrm{Co}^{\mathcal{A}}_R X) - X\| \le C_{\mathrm{nest}} t \|X\| \qquad (X \in \mathcal{S}^{\mathcal{A}}_{P,Q}),$$
$$\|\mathrm{Co}^{\mathcal{A}}_{P,Q} Y - Y\| \le C_{\mathrm{nest}} t \|Y\| \qquad (Y \in \mathcal{S}^{\mathcal{A}_R}_{P^R,Q^R}).$$

Registry contract (verbatim).

```
There are universal C_nest < infinity and e_nest > 0 such that, whenever R,P,Q are t-projections in
a finite-dimensional extended t-C*-algebra, R is nonvanishing, P,Q are subordinate to R with all
four left/right subordination errors at most t <= e_nest, A_R = S^A_R, P^R = Co^A_R(P), and Q^R =
Co^A_R(Q), then P^R,Q^R are C_nest*t-projections in A_R and, at every amplification,
||F^R_{P,Q}(Co^A_R X) - X|| <= C_nest*t*||X|| for X in S^A_{P,Q}, while ||Co^A_{P,Q} Y - Y|| <=
C_nest*t*||Y|| for Y in S^{A_R}_{P^R,Q^R}.
```

Proof (prose account of the af-validated tree). Smallness first bounds every amplified t-projection by 2. Subordination and the raw compression comparisons give $P^R = P + O(t)$ and $Q^R = Q + O(t)$, uniformly at every matrix level. Lemma 64 equips $\mathcal{A}_R$ with defect $O(t)$; amplified compression identities preserve the involution, while the preceding perturbations and idempotence of Co_R show that P^R, Q^R have $O(t)$ compressed idempotence defects. For the forward telescope, take $X \in \mathcal{S}^{\mathcal{A}}_{P,Q}$ and put $Z = \mathrm{Co}^{\mathcal{A}}_R X$. Lemma 58 gives $Z = X + O(t)\|X\|$ and places Z in the amplified R-corner. The inner compression $F^R_{P,Q}(Z)$ is replaced by its raw P^R–Q^R product; two uses of Lemma 57 replace the internal products by ambient ones; then P^R, Q^R, Z are replaced by P, Q, X. Since the ambient P–Q compression fixes X, the remaining raw product differs from X by $O(t)\|X\|$. The reverse telescope starts from the exact range identity $F^R_{P,Q}(Y) = Y$ and runs the same fixed replacements backward, ending with $\mathrm{Co}^{\mathcal{A}}_{P,Q}(Y) = Y + O(t)\|Y\|$. All constants are amplification-independent. Five verifier-requested bridges made both telescopes explicit; run 1 ballooned at 18 live nodes against cap 16, was classified as transparent repair growth, and the validated resumed tree has 18 clean live nodes under the recorded scoped cap 22.

Lemma 133 (`lem-maincb-nested-corner-dimension-transport`)**.** *There is a universal $e_{\mathrm{ncd}} > 0$ such that, under the hypotheses of Lemma 132 with all four subordination errors at most $t \le e_{\mathrm{ncd}}$, one has*

$$\dim \mathcal{S}^{\mathcal{A}}_{P,Q} = \dim \mathcal{S}^{\mathcal{A}_R}_{P^R,Q^R}.$$

Registry contract (verbatim).

```
There is a universal e_ncd > 0 such that, whenever R,P,Q are t-projections in a finite-dimensional
extended t-C*-algebra, R is nonvanishing, all four left/right subordination errors of P,Q to R are
at most t <= e_ncd, A_R = S^A_R, P^R = Co^A_R(P), and Q^R = Co^A_R(Q), one has dim S^A_{P,Q} = dim
S^{A_R}_{P^R,Q^R}.
```

Proof (prose account of the af-validated tree). Choose $C_* = \max\{C_{\mathrm{nest}}, 1\}$ and $e_{\mathrm{ncd}} = \min\{e_{\mathrm{nest}}, (2C_*)^{-1}\}$. The two maps supplied by Lemma 132 are

$$T(X) = F^R_{P,Q}(\mathrm{Co}^A_R X), \qquad S(Y) = \mathrm{Co}^A_{P,Q} Y,$$

from the ambient corner to the nested corner and back. Their validated estimates read $\|T(X) - X\| \le C_* t\|X\|$ and $\|S(Y) - Y\| \le C_* t\|Y\|$, with $C_* t \le \frac{1}{2}$. Thus $T(X) = 0$ forces $\|X\| \le \frac{1}{2}\|X\|$ and $X = 0$; the same argument makes S injective. Both spaces are finite-dimensional subspaces of the given finite-dimensional algebra. The two injections therefore give the two opposite dimension inequalities, and hence equality. This was a first-pass validation with 4 clean live nodes and zero challenges; the proof adds no estimate beyond the two comparison inequalities.

Lemma 134 (`lem-maincb-outer-compression-transfer`). *There are universal $C_{\mathrm{out}} < \infty$ and $e_{\mathrm{out}} > 0$ such that, whenever R, P are t-projections in a finite-dimensional extended t-C^*-algebra, R is nonvanishing, both subordination errors of P to R are at most t, $v : B \to \mathcal{S}^A_P$ is an extended t-isomorphism, $\mathcal{A}_R = \mathcal{S}^A_R$, $P^R = \mathrm{Co}^A_R(P)$, and $t \le e_{\mathrm{out}}$, the map*

$$T = \mathrm{Co}^{\mathcal{A}_R}_{P^R} \circ \mathrm{Co}^A_R \circ v : B \longrightarrow \mathcal{S}^{\mathcal{A}_R}_{P^R}$$

is an extended $C_{\mathrm{out}} t$-isomorphism and $T_n = I_n \otimes T$ for every n.

Registry contract (verbatim).

```
There are universal C_out < infinity and e_out > 0 such that, whenever R,P are t-projections in a
finite-dimensional extended t-C*-algebra, R is nonvanishing, both subordination errors of P to R
are at most t, v:B->S^A_P is an extended t-isomorphism, A_R = S^A_R, P^R = Co^A_R(P), and t <=
e_out, the explicitly defined map T = Co^{A_R}_{P^R} o Co^A_R o v : B->S^{A_R}_{P^R} is an extended
C_out*t-isomorphism and T_n = I_n tensor T for every n.
```

Proof (prose account of the af-validated tree). Set $a = P^R$ and specialize Lemmas 132 and 133 to $Q = P$. Their forward map $W_n = \mathrm{Co}^{\mathcal{A}_R}_{a_n} \circ \mathrm{Co}^A_{R_n}$ is $O(t)$-close to the identity on $M_n \otimes \mathcal{S}^A_P$, and the source and target diagonal corners have equal dimension. The tree checks that P and a are nonvanishing: otherwise their idempotent compression operators would have norm below one and hence vanish, contradicting the nonzero unital source and the dimension equality. Corner algebra estimates then supply the approximate C^*-structures on all three corners. Amplification naturality gives the exact identity $T_n = W_n \circ v_n = I_n \otimes T$, and the two compressions preserve the involution. The comparison estimate and the near-isometry of v_n yield sharp two-sided norm bounds for T_n. For multiplication, the tree telescopes through the product in the P-corner, the ambient product, the R-corner product, and finally the a-corner product, using the rectangular-product estimate at each bridge. A separate unit telescope compares the two compressed units. At level one, $\|W_1 - I\| < 1$

makes W_1 injective; dimension equality makes it onto, and bijectivity of v makes T bijective. Run 1 was stopped on a tangled composite architecture; a clean re-seed made both rectangular-product bridges and exact compression naturality explicit, and the resulting 12-node tree validated with zero challenges.

Lemma 135 (`lem-maincb-corner-equivalence`)**.** *There is a universal $e_{\mathrm{sim}} > 0$ such that, for every finite family of one-dimensional t-projections $P_1, \ldots, P_m$ in an extended t-C^*-algebra with $t \le e_{\mathrm{sim}}$, the relation*

$$j \sim k \quad \Longleftrightarrow \quad \dim \mathcal{S}_{P_j,P_k} = 1$$

is an equivalence relation.

Registry contract (verbatim).

```
There is a universal e_sim > 0 such that, for every finite family of one-dimensional t-projections
P_1,...,P_m in an extended t-C*-algebra with t <= e_sim, the relation j ~ k iff dim S_{P_j,P_k} = 1
is an equivalence relation.
```

Proof (prose account of the af-validated tree). Let C_{PQR}, e_{PQR} be the constants from Lemma 80, and let e_{dim} be the threshold from Lemma 81. The validated tree fixes $e_{\mathrm{sim}} = \min\{e_{PQR}/2, e_{\mathrm{dim}}/2, (4C_{PQR})^{-1}\}$, so for $e = 2t$ one has $C_{PQR}e \le \frac{1}{2}$. Reflexivity is the diagonal one-dimensionality of each P_j. Symmetry follows because the involution restricts to a conjugate-linear bijection $\mathcal{S}_{P_j,P_k} \to \mathcal{S}_{P_k,P_j}$. For transitivity, suppose $j \sim k$ and $k \sim \ell$, and choose nonzero $X \in \mathcal{S}_{P_j,P_k}$ and $Y \in \mathcal{S}_{P_k,P_\ell}$. Their compressed product lies in $\mathcal{S}_{P_j,P_\ell}$ and satisfies $\|X \cdot Y\| \ge (1 - C_{PQR}e)\|X\|\|Y\| \ge \frac{1}{2}\|X\|\|Y\| > 0$. The corner-dimension bound gives $\dim \mathcal{S}_{P_j,P_\ell} \le 1$, so this nonzero corner has dimension exactly one. Hence $j \sim \ell$. The root's original strict threshold inference failed at $t = e_{\mathrm{sim}}$; one challenge repaired it by installing the non-strict threshold above. The final validation has 8 clean live nodes.

55 MAIN-CB stage extensions and raw moves

This section records five local MAIN-CB moves: class separation gives zero cross-corners, Stage 1 refines an inclusion, Stages 2 and 3 execute raw calls, and the last row builds the closed EXT-CB datum. The packages are Definitions 40, 39, 41, and 42.

Lemma 136 (`lem-maincb-cross-union-zero-corners`). *There is a universal $e_{\mathrm{zero}} > 0$ such that, when A is a finite-dimensional extended ϵ_A-C^*-algebra with $\epsilon_A \le t$, a MAIN partition state supplies a non-unital extended t-inclusion $w : \mathbb{C}^m \to A$ with one-dimensional images P_j, and U, V are disjoint nonempty unions sharing no class, then for $R = U \cup V$ and $t \le e_{\mathrm{zero}}$,*

$$\dim S^A_{P_U,P_V} = \dim S^A_{P_V,P_U} = 0, \qquad \dim S^{A_R}_{P^R_U,P^R_V} = \dim S^{A_R}_{P^R_V,P^R_U} = 0.$$

Registry contract (verbatim).

```
There is a universal e_zero > 0 such that, if A is a finite-dimensional extended
epsilon_A-C*-algebra with epsilon_A <= t, a supplied MAIN partition state has w:C^m->A a non-unital
extended t-inclusion with one-dimensional images P_j, U,V are disjoint nonempty unions sharing no
equivalence class, R = U union V, and t <= e_zero, then dim S^A_{P_U,P_V} = dim S^A_{P_V,P_U} = 0
and dim S^{A_R}_{P_U^R,P_V^R} = dim S^{A_R}_{P_V^R,P_U^R} = 0.
```

Proof (prose account of the af-validated tree). Choose e_{zero} below the universal thresholds for Lemma 135, corner-dimension additivity (Lemma 82), nested dimension transport (Lemma 133), and the rank-one argument below. The coordinate identities $e_j^2 = e_j$ and the defect bounds for w make every P_j a t-projection. Corner equivalence therefore makes $j \sim k \iff \dim S^A_{P_j,P_k} = 1$ an equivalence relation. For $j \in U$, $k \in V$, separation of classes excludes dimension one in both orientations; the tree explicitly does not mistake this exclusion for vanishing. To rule out larger dimensions, take $P = P_j$, $Q = P_k$ and, if $S^A_{P,Q} \ne 0$, choose $X \in S^A_{P,Q}$ of norm one. With $\widetilde{p} = \mathrm{Co}_P(P)$ and $\widetilde{q} = \mathrm{Co}_Q(Q)$, the compression, approximate-unit and associator estimates show that $\widetilde{p}$ and $\widetilde{q}$ act as units up to $O(t)$. Since the diagonal corners are one-dimensional, $X \cdot X^\dagger = \beta\widetilde{p}$ and $X^\dagger \cdot Z = \gamma(Z)\widetilde{q}$. The lower C^* estimate keeps $|\beta|$ uniformly away from zero, while associativity gives $\|\beta Z - \gamma(Z)X\| \le Ct\|Z\|$. Thus the rank-one map $Z \mapsto \beta^{-1}\gamma(Z)X$ is within distance < 1 of the identity after shrinking e_{zero}; the Neumann lemma makes it invertible, so the corner has dimension at most one. Adjointness gives the same conclusion in the reverse direction. Hence all atomwise cross-corners vanish, and additivity gives both ambient union-corner equalities. Finally $P_R = w(e_R) = P_U + P_V$ is a t-projection and $1 - t \le \|P_R\| \le 1 + t$, which is the nonvanishing norm alternative of Definition 21. The coordinate products give all four subordination errors at most t. Defining locally $P^R_U = \mathrm{Co}^A_{P_R}(P_U)$ and $P^R_V = \mathrm{Co}^A_{P_R}(P_V)$, nested dimension transport in the two orientations carries the ambient zero dimensions into A_R. The ledger records eight repaired challenges, chiefly the nonvanishing/type bookkeeping and the missing atomwise dimension bound; the latter is exactly the in-tree rank-one argument above.

Lemma 137 (`lem-maincb-stage1-raw-refinement`). *There are universal $D_1 < \infty$ and $e_1 > 0$ such that an explicit Stage-1 raw-call datum with complementary target t-projections, old inclusion $\mathbb{C}^{m-1} \to S_{P_{\mathrm{old}}}$ for $m > 1$, fresh inclusion $\mathbb{C}^2 \to S_{P_{\mathrm{fresh}}}$, fixed amplifications and all defects at most $t \le e_1$, has sum map an extended D_1t-inclusion $\mathbb{C}^{m+1} \to A$; for $m = 1$ the old side is absent and the conclusion is the supplied fresh inclusion.*

Registry contract (verbatim).

```
There are universal D_1 < infinity and e_1 > 0 such that, if an explicit Stage-1 raw-call datum
supplies complementary target t-projections, an old extended t-inclusion C^{m-1}->S_{P_old} when m
> 1, a fresh extended t-inclusion C^2->S_{P_fresh}, fixed amplification families, and every
projection, complementarity, map, and target-ambient defect is at most t <= e_1, then their sum map
is an extended D_1*t-inclusion C^{m+1}->A; when m = 1, the old side is absent and the conclusion is
the supplied fresh inclusion.
```

Proof (prose account of the af-validated tree). Let $C_{\mathrm{dir}}, e_{\mathrm{dir}}$ be the universal witnesses of Lemma 130, and set $D_1 = \max\{1, C_{\mathrm{dir}}\}$ and $e_1 = e_{\mathrm{dir}}$. The first tree branch notes that, for a fixed amplification family and codomain, every defining homomorphism and two-sided norm inequality weakens when its nonnegative defect parameter is enlarged. If $m > 1$, apply the direct-sum merge to $B_1 = \mathbb{C}^{m-1}$, $B_2 = \mathbb{C}^2$, $P_1 = P_{\mathrm{old}}$ and $P_2 = P_{\mathrm{fresh}}$. The raw-call hypotheses supply its complementarity and defect conditions, so the sum map is an extended $C_{\mathrm{dir}}t$-inclusion on $\mathbb{C}^{m-1} \oplus \mathbb{C}^2 = \mathbb{C}^{m+1}$ into A; monotonicity replaces this by $D_1 t$. If $m = 1$, no merge or codomain transfer is asserted: the contract's separate branch returns the already supplied fresh inclusion, whose source is $\mathbb{C}^2 = \mathbb{C}^{m+1}$. This last wording is the repair to the ledger's sole challenge, which correctly rejected an unsupported passage from the compressed-corner codomain to the ambient algebra.

Lemma 138 (`lem-maincb-stage2-raw-extension`)**.** *Fix witnesses C^0_{s2}, e^0_{s2} of Lemma 140, $C_{\mathrm{ext}}, e_{\mathrm{ext}}$ of Lemma 87, and $C_{\mathrm{iso}}, e_{\mathrm{iso}}$ of Lemma 141. There are universal $D_2 < \infty$, $e_2 > 0$ such that, if a witness ledger W satisfies*

$$W.e_{s2} \le \min\{e^0_{s2}, e_2\}, \quad C^0_{s2} e_2 \le e_{\mathrm{ext}}, \quad (C_{\mathrm{ext}} + 1) C^0_{s2} e_2 \le e_{\mathrm{iso}},$$

every closed Stage-2 EXT-CB raw-call datum in A_R of total defect at most $C^0_{s2} t$, $0 \le t \le W.e_{s2}$, admits an extended $D_2 t$-isomorphism $v_+ : M_{r+1} \to A_R$ with $\|v_+(I_{M_{r+1}}) - u_{A_R}\| \le D_2 t$.

Registry contract (verbatim).

```
After first fixing the universal C_s2^0,e_s2^0 witnesses of lem-maincb-stage2-extcb-datum,
C_ext,e_ext witnesses of conj-extcb, and C_iso_unit,e_iso_unit witnesses of
lem-maincb-isomorphism-unit-control, there are universal D_2 < infinity and e_2 > 0 such that for
every def-maincb-witness-ledger datum W with W.e_s2 <= min{e_s2^0,e_2}, C_s2^0*e_2 <= e_ext, and
(C_ext+1)*C_s2^0*e_2 <= e_iso_unit, every explicit Stage-2 raw-call closed EXT-CB datum in A_R with
total post-helper defect at most C_s2^0*t and 0 <= t <= W.e_s2 admits an extended D_2*t-isomorphism
v_+:M_{r+1}->A_R satisfying ||v_+(I_{M_{r+1}})-u_{A_R}|| <= D_2*t.
```

Proof (prose account of the af-validated tree). The tree chooses

$$e_2 = \tfrac{1}{2} \min\left\{1, e^0_{s2}, \frac{e_{\mathrm{ext}}}{C^0_{s2}}, \frac{e_{\mathrm{iso}}}{C^0_{s2}(|C_{\mathrm{ext}} + 1| + 1)}\right\}, \qquad D_2 = 1 + C^0_{s2}\big(|C_{\mathrm{ext}}| + |C_{\mathrm{iso}}|(|C_{\mathrm{ext}}| + 1)\big).$$

For a datum with $e = \delta + \epsilon_{A_R} \le C^0_{s2} t$, the closed EXT-CB clauses and $e \le e_{\mathrm{ext}}$ allow Lemma 87 to supply one map $v_+ : M_{r+1} \to A_R$ that is an extended $C_{\mathrm{ext}} e$-isomorphism. Enlarging the defect bound makes this an extended $D_2 t$-isomorphism. The same map must satisfy the unit estimate.

The repaired tree separates $e = 0$, where $\epsilon_{A_R} = 0$ and unit control gives zero error, from $e > 0$. In the latter case the level-one two-sided bounds on the nonzero matrix identity force $C_{\text{ext}} \geq 0$. Consequently $0 \leq C_{\text{ext}}e + \epsilon_{A_R} \leq (C_{\text{ext}}+1)e \leq e_{\text{iso}}$, and Lemma 141 gives the stated D_2t bound. The split repairs the ledger challenge about multiplying by an otherwise unsigned $C_{\text{ext}}+1$. The registry contract is the user-ratified two-defect/witness-ledger amendment that superseded the 2026-07-30 form; the validated tree is a run-plus-resume.

Lemma 139 (`lem-maincb-stage3-raw-merge`). *Fix $C^0_{\text{cross}}, e^0_{\text{cross}}$ furnished by Lemmas 144 and 145, $C_{\text{merge}}, a_{\text{merge}}$ from Lemma 86, and unit-control witnesses $C_{\text{iso}}, e_{\text{iso}}$. There are universal $D_3 < \infty$, $e_3 > 0$ such that, under the witness-ledger bounds*

$$W.e_{\text{cross}} \leq \min\{e^0_{\text{cross}}, e_3\}, \quad (C^0_{\text{cross}} + 1)e_3 \leq a_{\text{merge}}, \quad (C_{\text{merge}}(C^0_{\text{cross}} + 1) + 1)e_3 \leq e_{\text{iso}},$$

every explicit Stage-3 amplified four-corner datum in A_R furnished by Lemma 145, with source $B_U \oplus B_V$, with B_U, B_V finite-dimensional C^*-algebras, with A_R a finite-dimensional extended ϵ_{A_R}-C^*-algebra, and with bijective level-one corner maps, $0 \leq \rho \leq C^0_{\text{cross}}t$ and $0 \leq \epsilon_{A_R} \leq t \leq W.e_{\text{cross}}$, yields an extended D_3t-isomorphism $v : B_U \oplus B_V \to A_R$ satisfying $\|v(I_{B_U \oplus B_V}) - u_{A_R}\| \leq D_3t$.

Registry contract (verbatim).

```
After first fixing the universal C_cross^0,e_cross^0 witnesses of
lem-maincb-cross-class-merging-datum furnished by lem-maincb-cross-datum-bijectivity,
C_merge,a_merge witnesses of lem-extcb-four-corner-merge, and C_iso_unit,e_iso_unit witnesses of
lem-maincb-isomorphism-unit-control, there are universal D_3 < infinity and e_3 > 0 such that for
every def-maincb-witness-ledger datum W with W.e_cross <= min{e_cross^0,e_3}, (C_cross^0+1)*e_3 <=
a_merge, and (C_merge*(C_cross^0+1)+1)*e_3 <= e_iso_unit, every explicit Stage-3 amplified
four-corner datum in A_R furnished by lem-maincb-cross-class-merging-datum with source B_U oplus
B_V, with B_U and B_V finite-dimensional C*-algebras, with A_R a finite-dimensional extended
epsilon_{A_R}-C*-algebra, whose four fixed level-one corner maps are bijective, and with 0 <= rho
<= C_cross^0*t and 0 <= epsilon_{A_R} <= t <= W.e_cross yields an extended D_3*t-isomorphism v:B_U
oplus B_V->A_R satisfying ||v(I_{B_U oplus B_V})-u_{A_R}|| <= D_3*t.
```

Proof (prose account of the af-validated tree). Take the merge and unit-control coefficients nonnegative and set $K = C_{\text{merge}}(C^0_{\text{cross}} + 1)$, $D_3 = \max\{K, C_{\text{iso}}(K + 1)\}$ and $e_3 = \min\{1, a_{\text{merge}}/(C^0_{\text{cross}} + 1), e_{\text{iso}}/(K + 1)\}$. The displayed ledger bounds follow immediately. For an arbitrary supplied datum,

$$\rho + \epsilon_{A_R} \leq (C^0_{\text{cross}} + 1)t \leq a_{\text{merge}},$$

so Lemma 86 supplies $v : B_U \oplus B_V \to A_R$ with defect $\delta = C_{\text{merge}}(\rho + \epsilon_{A_R})$. Thus $0 \leq \delta \leq Kt \leq D_3t$, and monotonicity of the fixed-amplification inequalities gives the required extended D_3t-isomorphism. Moreover $0 \leq \delta + \epsilon_{A_R} \leq (K + 1)t \leq e_{\text{iso}}$; unit control therefore gives $\|v(I) - u_{A_R}\| \leq C_{\text{iso}}(K + 1)t \leq D_3t$ for the same v. The six-node typed re-seed had no challenges. It validates the user-ratified amended contract above; two earlier untyped runs remain parked as a diagnostic obstruction record.

Lemma 140 (`lem-maincb-stage2-extcb-datum`). *There are universal $C_{s2} \geq 1$ and $e_{s2} > 0$, absorbing the corner-algebra constants, such that the following form the explicit closed Stage-2 EXT-CB raw-call datum in A_R. Let A be a finite-dimensional extended ϵ_A-C^*-algebra with $\epsilon_A \leq t$, let a MAIN partition state arise from a non-unital extended t-inclusion $w : \mathbb{C}^m \to A$ with one-dimensional P_j, and assume $\varnothing \neq U$, $j \notin U$, $\dim S^A_{P_k,P_j} = 1$ for $k \in U$, and $R = U \cup \{j\}$. If* *a reset isomorphism $v_U : M_{|U|} \to A_U$ satisfies $\epsilon_U, d_U \leq t \leq e_{s2}$ and $d_U \leq c_0^{\mathrm{cb}}\epsilon_U$, then A_R is an extended ϵ_{A_R}-C^*-algebra, P_U^R, P_j^R are quantitative projections there, and the outer-compressed v_U gives all clauses of the closed EXT-CB datum: approximate complementarity to I_{A_R}, $\dim S^{A_R}_{P_j^R} = 1$, $S^{A_R}_{P_U^R,P_j^R} \neq 0$, and total error $e = \delta + \epsilon_{A_R} \leq C_{s2}t$.*

Registry contract (verbatim).

```
There are universal C_s2 >= 1 and e_s2 > 0, with the e_ca threshold and universal C_ca coefficient
of lem-compcb-corner-algebra absorbed into e_s2 and C_s2, such that, if A is a finite-dimensional
extended epsilon_A-C*-algebra with epsilon_A <= t, a supplied MAIN partition state comes from a
non-unital extended t-inclusion w:C^m->A with one-dimensional images P_j, has nonempty U, j notin
U, dim S^A_{P_k,P_j} = 1 for every k in U, and R = U union {j}, and a supplied current reset state
v_U:M_{|U|}->A_U is an extended isomorphism satisfying epsilon_U, d_U <= t <= e_s2 and d_U <=
c_0^cb*epsilon_U, then lem-compcb-corner-algebra makes A_R an extended epsilon_{A_R}-C*-algebra,
lem-maincb-nested-corner-comparison makes P_U^R, P_j^R quantitative projections in A_R, and
together with the lem-maincb-outer-compression-transfer outer-compressed isomorphism they satisfy
every def-extcb-datum clause - approximate complementarity to I_{A_R}, one-dimensional S_{P_j^R},
nonzero S_{P_U^R,P_j^R}, and total error e = delta + epsilon_{A_R} - with e <= C_s2*t, forming the
explicit Stage-2 raw-call closed EXT-CB datum in A_R.
```

Proof (prose account of the af-validated tree). The universal ledger takes the maximum of the coefficients for nested-corner comparison, outer compression and the corner algebra, and the minimum of their thresholds together with those for dimension transport, corner equivalence and one fixed-source additivity application. Thus $t + \epsilon_A \leq 2t$ lies in the corner-algebra range and all remaining providers are simultaneously available. Writing $e_U = \sum_{k\in U} e_k$ and $e_R = e_U + e_j$, the inclusion gives $P_R = P_U + P_j$; P_U, P_j, P_R are t-projections, P_R satisfies the nonvanishing norm alternative, and all left/right subordination errors to P_R are at most t. Hence $A_R = S^A_{P_R}$ has the compressed extended-algebra structure from Lemma 64, with $\epsilon_{A_R} = C_{\mathrm{ca}}(t + \epsilon_A) \leq 2C_{\mathrm{ca}}t$, while Lemma 132 makes P_U^R, P_j^R quantitative projections. Lemma 134 sends the reset map to

$$T = \mathrm{Co}^{A_R}_{P_U^R} \circ \mathrm{Co}^{A}_{P_R} \circ v_U : M_{|U|} \longrightarrow S^{A_R}_{P_U^R},$$

an extended $C_{\mathrm{out}}t$-isomorphism with its fixed amplifications. Nested dimension transport preserves the one-dimensional P_j corner. For the cross corner choose $k_0 \in U$. If $U = \{k_0\}$ the hypothesis is immediate; otherwise split $U = \{k_0\} \sqcup V$ and apply corner-dimension additivity only to the fixed source pair $\mathbb{C}^2, \mathbb{C}$. Its summand $S^A_{P_{k_0},P_j}$ is one-dimensional, so $S^A_{P_U,P_j} \neq 0$, and nested transport carries nonvanishing into A_R. Linearity of compression gives the exact identity $P_U^R + P_j^R = I_{A_R}$. With $\delta = C_\delta t$ these facts are precisely the clauses of Definition 35, and $e = \delta + \epsilon_{A_R} \leq (C_\delta + 2C_{\mathrm{ca}})t \leq C_{s2}t$. This is the twelve-node, non-vacuous run 4 tree. It supersedes an earlier vacuous validation under the pre-amendment partition-state definition; the contract itself is unchanged in the validated non-vacuous run.

56 MAIN-CB interface bridges: unit control, witness arithmetic, unit comparison, bijectivity, the merging datum, and inclusion monotonicity

This section records six interface rows used by the MAIN-CB assembly: two scalar and unit bridges, the compressed-corner unit telescope, the fixed-map bijectivity bridge, the cross-class four-corner packet itself, and the monotonicity needed when defects are enlarged. The relevant data are those of Definitions 42, 40, and 37.

Lemma 141 (`lem-maincb-isomorphism-unit-control`). *There are universal constants $C_{\mathrm{iso,unit}} < \infty$ and $e_{\mathrm{iso,unit}} > 0$ such that, if B is a finite-dimensional C^*-algebra, A is a finite-dimensional extended ε-C^*-algebra, and $v : B \to A$ is an extended δ-isomorphism in the sense of Definition 36, with $0 \le \delta + \varepsilon \le e_{\mathrm{iso,unit}}$, then*

$$\|v(I_B) - I_A\| \le C_{\mathrm{iso,unit}}(\delta + \varepsilon).$$

The witnesses are independent of dimension, amplification, block data, and the source and target.

Registry contract (verbatim).

```
There are universal C_iso_unit < infinity and e_iso_unit > 0 such that if B is a finite-dimensional
C*-algebra, A is a finite-dimensional extended epsilon-C*-algebra, and v:B->A is an extended
delta-isomorphism with 0 <= delta+epsilon <= e_iso_unit, then ||v(I_B)-I_A|| <=
C_iso_unit*(delta+epsilon); the witnesses are independent of dimension, amplification, block data,
and the particular source and target.
```

Proof (prose account of the af-validated tree). The tree chooses the numerical witnesses $C_{\mathrm{iso,unit}} = 1$ and $e_{\mathrm{iso,unit}} = 1$, before any algebra or map is fixed. For admissible $B, A, v, \delta, \varepsilon$, the definition of an extended δ-isomorphism first makes v an extended δ-inclusion. Specialising its amplification requirement to $n = 1$ makes the level-one map a δ-homomorphism. The unit clause in that notion then gives directly $\|v(I_B) - I_A\| \le \delta$. At the same matrix level, the extended ε-C^* structure and its underlying approximate-algebra axioms give $|\|I_A\| - 1| \le \varepsilon$; in particular $\varepsilon \ge 0$. Together with the assumed $\delta \ge 0$, this yields $\delta \le \delta+\varepsilon$. Hence $\|v(I_B)-I_A\| \le \delta+\varepsilon = C_{\mathrm{iso,unit}}(\delta+\varepsilon)$. Because both witnesses were fixed numerical constants, none depends on a matrix level, dimension, block decomposition, or on the particular B, A, v.

Lemma 142 (`lem-maincb-witness-arithmetic`). *Fix positive finite universal provider witnesses $D_0, D_1, D_2, D_3, e_0, e_1, e_2, e_3, \varepsilon^{\mathrm{cb}}_{\mathrm{max}}, \delta^{\mathrm{cb}}_{max}, e_{\mathrm{it}}, K_{\mathrm{disp}}, K_{\mathrm{floor}}, \varepsilon_{\mathrm{unit}}, \delta_{\mathrm{unit}}, a_{\mathrm{unit}}, L, c_0^{\mathrm{cb}}, K_1, K_2, K_3, e_{\mathrm{env}}, e_{s2}, e_{\mathrm{cross}}, e_{\mathrm{sim}}, e_{\mathrm{full}}$ with $K_2, K_3 \ge \max\{1, L, c_0^{\mathrm{cb}} L\}$, and put $D_* := \max\{1, D_0, D_1, D_2, D_3\}$. Then there is a MAIN-CB witness ledger W (Definition 42) whose remaining fields are the correspondingly named witnesses and whose three derived fields are*

$$W.r_{\mathrm{reset}} = \min\{e_0, e_1, e_2, e_3, \varepsilon^{\mathrm{cb}}_{\mathrm{max}}, \delta^{\mathrm{cb}}_{\mathrm{max}}/D_*, e_{\mathrm{it}}/(D_*+1), \varepsilon_{\mathrm{unit}}, \delta_{\mathrm{unit}}/\max\{1, K_{\mathrm{floor}}\}, a_{\mathrm{unit}}/((1+K_{\mathrm{disp}})D_*), [2(1+$$

$$W.K_{\mathrm{call}} = \max\{1, L+1, c_0^{\mathrm{cb}}, K_1, K_2, K_3\},$$

$$W.\varepsilon_{\mathrm{MAIN}} = \min\{e_{\mathrm{env}}, e_1/K_1, e_{s2}/K_2, e_{\mathrm{cross}}/K_3, W.r_{\mathrm{reset}}/W.K_{\mathrm{call}}, e_{\mathrm{sim}}/W.K_{\mathrm{call}}, e_{\mathrm{full}}/W.K_{\mathrm{call}}, [2\max\{1, c_0^{\mathrm{cb}} W.K_{\mathrm{call}}\}]^{-}$$

Every field is positive, finite, universal, and independent of dimension, amplification, block data, class count, and stage index.

Registry contract (verbatim).

```
After first fixing positive finite universal provider witnesses
D_0,D_1,D_2,D_3,e_0,e_1,e_2,e_3,epsilon_max^cb,delta_max^cb,e_it,K_disp,K_floor,epsilon_unit,delta_unit,a_unit,L,c0
with K_2,K_3 >= max{1,L,c0_cb*L}, set D_* = max{1,D_0,D_1,D_2,D_3}; then there is a
def-maincb-witness-ledger datum W whose fields satisfy W.r_reset =
min{e_0,e_1,e_2,e_3,epsilon_max^cb,delta_max^cb/D_*,e_it/(D_*+1),epsilon_unit,delta_unit/max{1,K_floor},a_unit/((1+
W.K_call = max{1,L+1,c0_cb,K_1,K_2,K_3}, W.epsilon_MAIN =
min{e_env,e_1/K_1,e_s2/K_2,e_cross/K_3,W.r_reset/W.K_call,e_sim/W.K_call,e_full/W.K_call,[2*max{1,c0_cb*W.K_call}]^
and the remaining fields equal the correspondingly named receiving witnesses; in particular every
field is positive, finite, universal, and independent of dimension, amplification, block data,
class count, and stage index.
```

Proof (prose account of the af-validated tree). The proof is finite max–min arithmetic, performed only after all provider witnesses have been fixed. Positivity and finiteness of $D_0, \dots, D_3$ make $D_* \geq 1$ positive, finite, universal, and independent of the structural data. Every one of the eleven displayed reset candidates is therefore positive and finite: all denominators are positive, including $D_* + 1$, $\max\{1, K_{\text{floor}}\}$, and $2(1 + K_{\text{disp}})D_*$. Their finite minimum is thus the asserted r_{reset} with the same universal independence. Likewise the finite maximum $K_{\text{call}} = \max\{1, L + 1, c_0^{\text{cb}}, K_1, K_2, K_3\}$ is positive, finite, universal, and at least one. These properties make each of the eight $\varepsilon_{\text{MAIN}}$ candidates positive and finite, including the three ratios with K_{call} and the final reciprocal. Their finite minimum has the asserted properties. Finally the tree defines W to be the twelve-field tuple $(c_0^{\text{cb}}, L, K_1, K_2, K_3, K_{\text{call}}, e_{\text{env}}, e_1, e_{s2}, e_{\text{cross}}, r_{\text{reset}}, \varepsilon_{\text{MAIN}})$. The field typing in Definition 42 finishes the claim.

Lemma 143 (`lem-maincb-compressed-corner-unit-comparison`)**.** *There are universal $C_{\text{corner,unit}} < \infty$ and $e_{\text{corner,unit}} > 0$ such that, for every $0 \leq t \leq e_{\text{corner,unit}}$, the following hold at every $n \geq 1$. If P is a t-projection in a finite-dimensional extended t-C^*-algebra A, then*

$$\|I_n \otimes \mathrm{Co}_P(P) - I_n \otimes P\| \leq C_{\text{corner,unit}} t.$$

Under the hypotheses of Lemma 134, if $\|v(I_B) - u_{\mathcal{S}_P}\| \leq t$ and $T = \mathrm{Co}^{A_R}_{P^R} \circ \mathrm{Co}^A_R \circ v$, then

$$\|T_n(I_n \otimes I_B) - I_n \otimes P^R\| \leq C_{\text{corner,unit}} t.$$

Registry contract (verbatim).

```
There are universal C_corner_unit < infinity and e_corner_unit > 0 such that both of the following
hold: if P is a t-projection in a finite-dimensional extended t-C*-algebra A with 0 <= t <=
e_corner_unit, then the compressed-corner unit u_{S_P}=Co_P(P) satisfies ||I_n tensor u_{S_P}-I_n
tensor P|| <= C_corner_unit*t for every n >= 1; and, under the hypotheses of
lem-maincb-outer-compression-transfer with 0 <= t <= e_corner_unit, if ||v(I_B)-u_{S_P}|| <= t then
its explicit outer-compressed map T=Co^{A_R}_{P^R} o Co^A_R o v satisfies ||T_n(I_n tensor I_B)-I_n
tensor P^R|| <= C_corner_unit*t for every n >= 1.
```

Proof (prose account of the af-validated tree). The tree first fixes uniform bounds for a small defective projection S: the C^* lower bound and $\|S^2 - S\| \leq \delta$ give $\|S\| \leq 2$ after shrinking the universal threshold, while Definition 20 supplies $\|\mathrm{Co}_{S,S} - L_S R_S\| \leq K_{\text{cmp}}(\delta + \varepsilon)$. At level

n, $S_n = I_n \otimes S$ has the same norm and defect, and amplified compression identifies $I_n \otimes \mathrm{Co}_S(S)$ with $\mathrm{Co}_{S_n}(S_n)$. The exact split $S_n(S_n^2) - S_n = S_n(S_n^2 - S_n) + (S_n^2 - S_n)$ then gives a universal $K_u(\delta+\varepsilon)$ estimate. Applied directly to P_n, the live first-clause branch records the explicit coefficient $4K_{\mathrm{cmp}} + 7/2$. For the outer clause, write $a = P^R = \mathrm{Co}^A_R(P)$ and $A_R = \mathcal{S}^A_R$. The corner-algebra and amplified identities make A_R an extended $K_R t$-C^*-algebra and a a $K_a t$-projection; applying the preceding comparison inside A_R yields $\|I_n \otimes u_{\mathcal{S}^{A_R}_{P^R}} - I_n \otimes P^R\| \le K_{\mathrm{tar}} t$. The final live branch uses the assumed $\|v(I_B) - u_{\mathcal{S}_P}\| \le t$ and bounded outer compressions to telescope from $T_n(I_n \otimes I_B)$ to that target unit. This replacement repaired the challenged unsupported unit-preservation inference; finite maxima of coefficients and minima of rescaled thresholds give the two witnesses.

Lemma 144 (`lem-maincb-cross-datum-bijectivity`)**.** *Choose c_0 for Lemma 127, its direct-corner-envelope witnesses, and the witnesses of Lemma 143. There are universal $C^0_{\mathrm{cross}} \ge 1$ and $0 < e^0_{\mathrm{cross}} \le e_{\mathrm{corner,unit}}$, valid also for Lemma 145, such that the following holds. Let $W.c_0^{\mathrm{cb}} = c_0$, $W.e_{\mathrm{cross}} \le e^0_{\mathrm{cross}}$, and let A, w, U, V, R, v_U, v_V satisfy all hypotheses of that merging lemma, including $R = U \cup V$, the bounds $\varepsilon_U, \varepsilon_V, d_U, d_V \le t \le W.e_{\mathrm{cross}}$, $d_X \le W.c_0^{\mathrm{cb}} \varepsilon_X$, and $\|v_X(I_{B_X}) - u_{A_X}\| \le t$ for $X = U, V$. Put $B_R = B_U \oplus B_V$, $q_U = (I_{B_U}, 0)$, $q_V = (0, I_{B_V})$, and $P^R_X = \mathrm{Co}^A_{P_R}(P_X)$. Define*

$$\gamma_{UU}(b_U, 0) = \mathrm{Co}^{A_R}_{P^R_U}(\mathrm{Co}^A_{P_R}(v_U(b_U))), \qquad \gamma_{VV}(0, b_V) = \mathrm{Co}^{A_R}_{P^R_V}(\mathrm{Co}^A_{P_R}(v_V(b_V))),$$

and let γ_{UV}, γ_{VU} be the unique maps between the corresponding zero cross-corners. These are the four fixed level-one maps of the explicit Stage-3 datum, and all four maps are bijective.

Registry contract (verbatim).

```
After first choosing a universal c0 witness for which lem-maincb-error-improvement remains valid,
fixing the corresponding lem-maincb-direct-corner-envelope witnesses, and fixing the
C_corner_unit,e_corner_unit witnesses of lem-maincb-compressed-corner-unit-comparison, there are
universal C_cross^0 >= 1 and 0 < e_cross^0 <= e_corner_unit which are valid witnesses of
lem-maincb-cross-class-merging-datum such that for every def-maincb-witness-ledger datum W with
W.c0_cb = c0 and W.e_cross <= e_cross^0, if A is a finite-dimensional extended epsilon_A-C*-algebra
with epsilon_A <= t, a supplied MAIN partition state comes from a non-unital extended t-inclusion
w:C^m->A with one-dimensional images P_j, has disjoint nonempty unions U,V sharing no class and R=U
union V, and supplied current reset isomorphisms v_U:B_U->A_U and v_V:B_V->A_V satisfy
epsilon_U,epsilon_V,d_U,d_V <= t <= W.e_cross, d_U <= W.c0_cb*epsilon_U, d_V <= W.c0_cb*epsilon_V,
||v_U(I_{B_U})-u_{A_U}|| <= t, and ||v_V(I_{B_V})-u_{A_V}|| <= t, then, writing B_R:=B_U oplus B_V,
q_U:=(I_{B_U},0), q_V:=(0,I_{B_V}), P_X^R:=Co^A_{P_R}(P_X) for X in {U,V}, gamma_UU:q_U B_R
q_U->S^{A_R}_{P_U^R} defined by gamma_UU((b_U,0)):=Co^{A_R}_{P_U^R}(Co^A_{P_R}(v_U(b_U))),
gamma_VV:q_V B_R q_V->S^{A_R}_{P_V^R} defined by
gamma_VV((0,b_V)):=Co^{A_R}_{P_V^R}(Co^A_{P_R}(v_V(b_V))), gamma_UV:q_U B_R
q_V={0}->S^{A_R}_{P_U^R,P_V^R}={0} the unique map, and gamma_VU:q_V B_R
q_U={0}->S^{A_R}_{P_V^R,P_U^R}={0} the unique map, the explicit Stage-3 amplified four-corner datum
in A_R furnished by lem-maincb-cross-class-merging-datum has these four fixed level-one maps and
gamma_UU,gamma_UV,gamma_VU,gamma_VV are bijective.
```

Proof (prose account of the af-validated tree). The tree enlarges c_0 to at least one and fixes the providers, then sets $K = \max\{c_0, L, 1\}$ and shrinks the merging threshold by $e^0_{\mathrm{cross}} = \min\{e_*, e_{\mathrm{zero}}, e_{\mathrm{env}}, e_{\mathrm{out}}/K, (2K)^{-1}\}$. Thus Lemma 145 supplies its explicit datum. For the U diagonal, put $s = Kt$. The direct-corner envelope makes P_R, P_U s-projections with both subordination errors at most s; the two-sided norm bound for $w(1_R)$ makes P_R nonvanishing, and v_U weakens to an extended s-isomorphism. Outer compression therefore gives a bijection $T_U = \mathrm{Co}^{A_R}_{P^R_U} \circ \mathrm{Co}^A_{P_R} \circ v_U$.

The coordinate bijection $\iota_U(b_U) = (b_U, 0)$ satisfies $\gamma_{UU} \circ \iota_U = T_U$, proving bijectivity and identifying the fixed map. The V branch repeats the same construction with $\iota_V(b_V) = (0, b_V)$. The zero-cross-corner lemma makes both source and target off-diagonal corners zero, so each unique zero-to-zero map is its own bijective inverse. One verifier challenge exposed an omitted premise in the U fixed-map identification; the live tree repairs it by declaring the validated datum-construction node explicitly. The four identifications and bijectivity conclusions then assemble at the root.

Lemma 145 (`lem-maincb-cross-class-merging-datum`). *Choose c_0 for Lemma 127, fix the corresponding direct-corner-envelope witnesses, and fix the unit-comparison witnesses above. There are universal $C^0_{\mathrm{cross}} \geq 1$ and $0 < e^0_{\mathrm{cross}} \leq e_{\mathrm{corner,unit}}$ such that, for every ledger W with $W.c_0^{\mathrm{cb}} = c_0$ and $W.e_{\mathrm{cross}} \leq e^0_{\mathrm{cross}}$, the following holds. Let A be a finite-dimensional extended ε_A-C^*-algebra with $\varepsilon_A \leq t$, let the supplied partition state arise from a non-unital extended t-inclusion $w : \mathbb{C}^m \to A$ with one-dimensional images P_j, and let U, V be disjoint nonempty unions sharing no class, with $R = U \cup V$. If the reset isomorphisms $v_X : B_X \to A_X$ satisfy $\varepsilon_X, d_X \leq t \leq W.e_{\mathrm{cross}}$, $d_X \leq W.c_0^{\mathrm{cb}} \varepsilon_X$, and $\|v_X(I_{B_X}) - u_{A_X}\| \leq t$ for $X = U, V$, then the compressed-unit comparison, nested-corner, outer-compression, and zero-cross-corner constructions form in A_R the explicit Stage-3 amplified four-corner datum with common defect $\rho \leq C^0_{\mathrm{cross}} t$.*

Registry contract (verbatim).

```
After first choosing a universal c0 witness for which lem-maincb-error-improvement remains valid,
fixing the corresponding lem-maincb-direct-corner-envelope witnesses, and fixing the
C_corner_unit,e_corner_unit witnesses of lem-maincb-compressed-corner-unit-comparison, there are
universal C_cross^0 >= 1 and 0 < e_cross^0 <= e_corner_unit such that for every
def-maincb-witness-ledger datum W with W.c0_cb = c0 and W.e_cross <= e_cross^0, if A is a
finite-dimensional extended epsilon_A-C*-algebra with epsilon_A <= t, a supplied MAIN partition
state comes from a non-unital extended t-inclusion w:C^m->A with one-dimensional images P_j, has
disjoint nonempty unions U,V sharing no class and R=U union V, and supplied current reset
isomorphisms v_U:B_U->A_U and v_V:B_V->A_V satisfy epsilon_U,epsilon_V,d_U,d_V <= t <= W.e_cross,
d_U <= W.c0_cb*epsilon_U, d_V <= W.c0_cb*epsilon_V, ||v_U(I_{B_U})-u_{A_U}|| <= t, and
||v_V(I_{B_V})-u_{A_V}|| <= t, then lem-maincb-compressed-corner-unit-comparison and the
nested-corner, outer-compression, and zero-cross-corner constructions form the explicit Stage-3
amplified four-corner datum in A_R with common defect rho <= C_cross^0*t.
```

Proof (prose account of the af-validated tree). The provider ledger fixes all constants first and chooses $C^0_{\mathrm{cross}} = \max\{1, C_{\mathrm{nest}}, C_{\mathrm{out}}, C_{\mathrm{corner,unit}}, 2C_{\mathrm{ca}}\}$, then takes a positive finite minimum of every required threshold and its fixed rescaling. In $\mathbb{C}^m$, the projections $e_U, e_V, e_R = e_U + e_V$ map under w to $P_U, P_V, P_R = P_U + P_V$. The inclusion bounds make these defective projections and give both subordination errors to P_R. The challenged nonvanishing step is repaired in the live tree by $1 - t \leq \|P_R\| \leq 1 + t$, hence $|\|P_R\| - 1| \leq t$. The corner-algebra row makes $A_R = \mathcal{S}^A_{P_R}$ an extended $2C_{\mathrm{ca}}t$-C^*-algebra, while nested comparison gives P^R_U, P^R_V as $C_{\mathrm{nest}}t$-projections. Outer compression of each v_X supplies the two fixed diagonal maps and all their amplified algebra and norm estimates; Lemma 143 supplies their unit estimates. The zero-cross-corner row makes the two off-diagonal maps uniquely zero at every amplification. Finally $P^R_U + P^R_V = \mathrm{Co}_{P_R}(P_R) = u_R$ gives target complementarity. All clauses are bounded by the common defect $C^0_{\mathrm{cross}}t$. This is the amended M12 contract, superseding its earlier pre-unit-bridge form.

Lemma 146 (`lem-maincb-extended-inclusion-monotone`). *Let B be a finite-dimensional C^*-algebra, let A be a finite-dimensional extended ε-C^*-algebra, let $v : B \to A$ be linear, and suppose*

$0 \le \delta \le \delta'$. *Every extended δ-inclusion is an extended δ'-inclusion; every extended δ-isomorphism is an extended δ'-isomorphism and, in particular, an extended δ'-inclusion.*

Registry contract (verbatim).

```
If B is a finite-dimensional C*-algebra, A is a finite-dimensional extended epsilon-C*-algebra,
v:B->A is linear, and 0 <= delta <= delta', then if v is an extended delta-inclusion it is an
extended delta'-inclusion, and if v is an extended delta-isomorphism it is an extended
delta'-isomorphism and in particular an extended delta'-inclusion.
```

Proof (prose account of the af-validated tree). Fix an amplification $v_n = I_n \otimes v$ of an extended δ-inclusion. Definition 36 supplies the two-sided bounds $(1-\delta)\|X\| \le \|v_n(X)\| \le (1+\delta)\|X\|$ and makes v_n a δ-homomorphism. The byte-matched base definition provisioned after a verifier challenge spells out bounded linearity, exact involution preservation, $\|v_n(I) - I\| \le \delta$, and $\|v_n(XY) - v_n(X)v_n(Y)\| \le \delta\|X\|\|Y\|$. Since $0 \le \delta \le \delta'$ and all norm factors are nonnegative, the latter two inequalities remain true with δ'; linearity and involution preservation do not change. Likewise $(1-\delta')\|X\| \le (1-\delta)\|X\|$ and $(1+\delta)\|X\| \le (1+\delta')\|X\|$. Thus every amplification is a δ'-homomorphism with the required norm bounds, so v is an extended δ'-inclusion. If v was an extended δ-isomorphism, its bijectivity is independent of the defect parameter; combining that unchanged bijectivity with the first conclusion proves both remaining claims.

57 MAIN-CB reset states and ledgers: invariant preservation, the constant and structural-domain ledgers, and reset output typing

The four rows below isolate the bookkeeping around a MAIN-CB reset. They first keep one literal improvement iterate and its typing, then choose a single universal ledger for all four raw-call forms, and finally place every call scale inside the structural domains. The reset states, raw calls, and witness tuple are those of Definitions 39, 41, and 42; no existence or estimate is read into those data definitions.

Lemma 147 (`lem-maincb-reset-invariant-preservation`)**.** *Fix the universal witnesses $e_{\text{it}}, K_{\text{disp}}, K_{\text{floor}}$ of Lemma 126 and $\epsilon_{\max}^{\text{cb}}, \delta_{\max}^{\text{cb}}, c_0^0$ of Lemma 127. There are universal $0 < C_{\text{unit}} < \infty$ and $\epsilon_{\text{unit}}, \delta_{\text{unit}}, a_{\text{unit}} > 0$, furnished by the third clause of* ***prop_delta_hominc*** *and its implicit quantifier convention, with the following property. If $D \geq 1$, W is a MAIN-CB witness ledger,*

$$c_0^{\text{cb}} \geq \max\{c_0^0, K_{\text{floor}}, C_{\text{unit}}(K_{\text{floor}}{+}1)\}, \quad r_{\text{reset}} \leq \min\Bigg\{\epsilon_{\max}^{\text{cb}}, \frac{\delta_{\max}^{\text{cb}}}{D}, \frac{e_{\text{it}}}{D+1}, \epsilon_{\text{unit}}, \frac{\delta_{\text{unit}}}{\max\{1, K_{\text{floor}}\}}, \frac{a_{\text{unit}}}{(1+K_{\text{disp}})D}, \frac{\quad}{2(1+$$

and an explicit raw call at $0 \leq t \leq r_{\text{reset}}$ has target an extended ϵ_R-C^*-corner A_R, literal extended Dt-inclusion $u_R : B_R \to A_R$, $\|u_R(I_{B_R}) - u_{A_R}\| \leq Dt$, and $\epsilon_R \leq t$, then it admits a map $v_R : B_R \to A_R$ with

$$d_R \leq c_0^{\text{cb}} \epsilon_R, \qquad \|v_R(I_{B_R}) - u_{A_R}\| \leq c_0^{\text{cb}} \epsilon_R.$$

The map v_R is bijective whenever u_R is, and the source, target corner, and amplification form are unchanged.

Registry contract (verbatim).

```
After first fixing the universal e_it,K_disp,K_floor witnesses of lem-maincb-improvement-iteration
and epsilon_max^cb,delta_max^cb,c0^0 witnesses of lem-maincb-error-improvement, there are universal
0 < C_unit < infinity and epsilon_unit,delta_unit,a_unit > 0 furnished by the third clause of
prop_delta_hominc and its implicit quantifier convention such that, for every D >= 1 and every
def-maincb-witness-ledger datum W with W.c0_cb >= max{c0^0,K_floor,C_unit*(K_floor+1)} and
W.r_reset <=
min{epsilon_max^cb,delta_max^cb/D,e_it/(D+1),epsilon_unit,delta_unit/max{1,K_floor},a_unit/((1+K_disp)*D),[2*(1+K_d
every explicit raw call into an extended epsilon_R-C*-corner A_R at scale 0 <= t <= W.r_reset whose
literal map u_R:B_R->A_R is an extended D*t-inclusion and satisfies ||u_R(I_{B_R})-u_{A_R}|| <= D*t
and epsilon_R <= t admits an error-improved map v_R:B_R->A_R satisfying d_R <= W.c0_cb*epsilon_R
and ||v_R(I_{B_R})-u_{A_R}|| <= W.c0_cb*epsilon_R, preserving bijectivity when u_R is bijective and
leaving the source, target corner, and amplification form unchanged.
```

Proof (prose account of the af-validated tree). Choose $C_{\text{unit}}, \epsilon_{\text{unit}}, \delta_{\text{unit}}, a_{\text{unit}}$ from the registered near-unit clause, with a_{unit} strictly below its positive threshold. For a raw call put $d = Dt$. The bounds $\epsilon_R \leq t$ and $t \leq e_{\text{it}}/(D+1)$ give $d + \epsilon_R \leq e_{\text{it}}$, so Lemma 126 produces one fixed map v_R with $(v_R)_n = I_n \otimes v_R$, displacement at most $K_{\text{disp}} Dt$, and extended-inclusion defect $K_{\text{floor}} \epsilon_R$. At level one the triangle inequality gives

$$\|v_R(I_{B_R}) - u_{A_R}\| \leq (1 + K_{\text{disp}}) Dt \leq a_{\text{unit}}.$$

The other radius bounds give $\epsilon_R \leq \epsilon_{\text{unit}}$ and $K_{\text{floor}} \epsilon_R \leq \delta_{\text{unit}}$. Applying the near-unit clause to this same map yields the unit error $C_{\text{unit}}(K_{\text{floor}} + 1)\epsilon_R \leq c_0^{\text{cb}} \epsilon_R$, while its recorded defect is at most $K_{\mathrm{floor}}\epsilon_R \le c_0^{\mathrm{cb}}\epsilon_R$. If u_R is bijective, then $\|v_R(x)\| \ge [1-(1+K_{\mathrm{disp}})Dt]\|x\| \ge \|x\|/2$; thus v_R is injective, and equality of the finite dimensions of B_R and A_R makes it surjective. The proof deliberately retains the iteration output rather than substituting the separate existential map of Lemma 127; hence its displayed signature and amplification identity preserve the source, target, and amplification form. The quoted contract is the user-ratified two-defect repair, superseding its 2026-07-30 form.

Lemma 148 (`lem-maincb-reset-constant-ledger`)**.** *Fix, in the order stated in the registry contract below, the witnesses exported by the iteration, error-improvement, reset-invariance, corner-envelope, corner-identification, corner-equivalence, and four raw-call producer rows, and put $D_* := \max\{1, D_0, D_1, D_2, D_3\}$. Then Lemma 142 supplies one universal MAIN-CB witness ledger W such that*

$$c_0^{\mathrm{cb}} = c_0, \quad L \ge L^0, \quad K_1 \ge K_1^0, \quad K_2 \ge \max\{K_2^0, 1, L, c_0^{\mathrm{cb}}L\}, \quad K_3 \ge \max\{K_3^0, 1, L, c_0^{\mathrm{cb}}L\},$$
$$e_{\mathrm{env}} \le e_{\mathrm{env}}^0, \quad e_1 \le e_1^{\mathrm{provider}}, \quad e_{s2} \le \min\{e_{s2}^0, e_2\}, \quad e_{\mathrm{cross}} \le \min\{e_{\mathrm{cross}}^0, e_3\}.$$

Under the respective producer hypotheses at base scale $0 \le t \le W.r_{\mathrm{reset}}$ and target ambient defect at most t, let u_0, u_1, u_2, u_3 be the literal maps furnished respectively by `lem-maincb-initial-raw-inclusion`, `lem-maincb-stage1-call-envelope` with `lem-maincb-stage1-raw-refinement`, `lem-maincb-stage2-call-env` with `lem-maincb-stage2-raw-extension`, and `lem-maincb-stage3-call-envelope` with `lem-maincb-stage3-r` Then $u_0 : \mathbb{C} \to A$ and $u_1 : \mathbb{C}^{m+1} \to A$ are extended D_*t-inclusions, while $u_2 : M_{r+1} \to A_R$ and $u_3 : B_U \oplus B_V \to A_R$ are extended D_*t-isomorphisms. Their respective unit errors are at most D_*t, so all four are eligible for Lemma 147. Every selected witness is universal and independent of dimension, amplification, block data, class count, and stage index.

Registry contract (verbatim).

```
After first fixing e_it,K_disp,K_floor from lem-maincb-improvement-iteration,
epsilon_max^cb,delta_max^cb,c0^0 from lem-maincb-error-improvement,
C_unit,epsilon_unit,delta_unit,a_unit from lem-maincb-reset-invariant-preservation, a valid
enlarged c0 >= max{c0^0,K_floor,C_unit*(K_floor+1)}, L^0,e_env^0 from
lem-maincb-direct-corner-envelope for this c0, e_full from lem-maincb-full-corner-identification,
e_sim from lem-maincb-corner-equivalence, D_0,e_0 from lem-maincb-initial-raw-inclusion,
D_1,e_1,K_1^0 from lem-maincb-stage1-call-envelope, C_s2^0,e_s2^0 from
lem-maincb-stage2-extcb-datum, D_2,e_2 from lem-maincb-stage2-raw-extension, K_2^0 from
lem-maincb-stage2-call-envelope, C_cross^0,e_cross^0 from lem-maincb-cross-class-merging-datum,
D_3,e_3 from lem-maincb-stage3-raw-merge, and K_3^0 from lem-maincb-stage3-call-envelope, set D_* =
max{1,D_0,D_1,D_2,D_3}; then there exists one def-maincb-witness-ledger datum W supplied by
lem-maincb-witness-arithmetic with W.c0_cb=c0, W.L>=L^0, W.K1>=K_1^0,
W.K2>=max{K_2^0,1,W.L,W.c0_cb*W.L}, W.K3>=max{K_3^0,1,W.L,W.c0_cb*W.L}, W.e_env<=e_env^0,
W.e1<=e_1, W.e_s2<=min{e_s2^0,e_2}, and W.e_cross<=min{e_cross^0,e_3}, such that under the
respective producer hypotheses at base scale 0 <= t <= W.r_reset and target ambient defect at most
t, the literal maps u_0:C->A furnished by lem-maincb-initial-raw-inclusion and u_1:C^{m+1}->A
furnished by lem-maincb-stage1-call-envelope with lem-maincb-stage1-raw-refinement are extended
D_*t-inclusions, the literal maps u_2:M_{r+1}->A_R furnished by lem-maincb-stage2-call-envelope
with lem-maincb-stage2-raw-extension and u_3:B_U oplus B_V->A_R furnished by
lem-maincb-stage3-call-envelope with lem-maincb-stage3-raw-merge are extended D_*t-isomorphisms,
and ||u_0(I_C)-I_A||,||u_1(I_{C^{m+1}})-I_A||,||u_2(I_{M_{r+1}})-u_{A_R}||,||u_3(I_{B_U oplus
B_V})-u_{A_R}|| <= D_*t, so each satisfies the M02/M03 and near-unit thresholds and is eligible for
lem-maincb-reset-invariant-preservation; all selected witnesses are universal and independent of
dimension, amplification, block data, class count, and stage index.
```

Proof (prose account of the af-validated tree). The tree chooses $L = L^0$, $K_1 = K_1^0$ and $K_i = \max\{K_i^0, 1, L^0, c_0L^0\}$ for $i = 2, 3$, together with $e_{\text{env}} = e^0_{\text{env}}$, $e_{s2} = \min\{e^0_{s2}, e_2\}$ and $e_{\text{cross}} = \min\{e^0_{\text{cross}}, e_3\}$. Applying Lemma 142 records these exact choices in one W and makes r_{reset} no larger than every provider threshold and every radius required by Lemma 147 with $D = D_*$. The initial scalar map $u_0(\lambda) = \lambda I_A$ has zero unit error. Lemmas 151 and 137 give u_1 with defect and unit error D_1t. Monotonicity (Lemma 146) weakens both map bounds to D_*t. For Stage 2 the scale is $t = K_2\epsilon$: Lemma 152 provides the closed EXT-CB datum with $\epsilon_R = L\epsilon \le t$, and Lemma 138 supplies a chosen isomorphism v_+; the composite record defines $u_2 := v_+$ rather than identifying opaque outputs. Stage 3 similarly takes $t = K_3\epsilon$, obtains the amplified four-corner datum, applies Lemma 139 to the datum supplied by Lemma 153, and defines its output to be u_3. A verifier challenged this last invocation; the repaired live tree unpacks "the respective producer hypotheses" to retain the same fixed cross witnesses and the four required bijective level-one corner maps, without inferring bijectivity from the call envelope. Weakening D_it to D_*t preserves the two isomorphism tags and unit bounds. All four calls now meet the common reset radius and near-unit hypotheses, so Lemma 147 applies. This is the user-ratified amended contract; its later suspension and re-flip were mechanical dependency-status propagation, not a change to the preserved certificate.

Lemma 149 (`lem-maincb-structural-domain-ledger`). *Fix particular universal $e_{\text{sim}}, e_{\text{full}} > 0$ furnished by Lemmas 135 and 131, and fix a witness ledger W furnished by Lemma 148 using those same witnesses. If $0 \le \epsilon \le \epsilon_{\text{MAIN}}$, then*

$$\epsilon \le e_{\text{env}}, \qquad \epsilon \le \frac{e_1}{K_1}, \qquad \epsilon \le \frac{e_{s2}}{K_2}, \qquad \epsilon \le \frac{e_{\text{cross}}}{K_3}.$$

The scales ϵ, $K_{\text{call}}\epsilon$, $K_1\epsilon$, $K_2\epsilon$, and $K_3\epsilon$ are all at most $K_{\text{call}}\epsilon$, and

$$K_{\text{call}}\epsilon \le r_{\text{reset}}, e_{\text{sim}}, e_{\text{full}}, \qquad L\epsilon \le K_{\text{call}}\epsilon, \qquad c_0^{\text{cb}}K_{\text{call}}\epsilon \le \tfrac{1}{2}.$$

Registry contract (verbatim).

```
After first fixing a particular universal e_sim>0 witness furnished by
lem-maincb-corner-equivalence and a particular universal e_full>0 witness furnished by
lem-maincb-full-corner-identification, fix one def-maincb-witness-ledger datum W whose existence is
furnished by lem-maincb-reset-constant-ledger instantiated with those same e_sim,e_full witnesses;
then 0 <= epsilon <= W.epsilon_MAIN implies epsilon <= W.e_env, epsilon <= W.e1/W.K1, epsilon <=
W.e_s2/W.K2, and epsilon <= W.e_cross/W.K3, while the global scalar scale epsilon, atomic scalar
scale W.K_call*epsilon, and Stage-1, Stage-2, and Stage-3 scales
W.K1*epsilon,W.K2*epsilon,W.K3*epsilon are all at most W.K_call*epsilon <= W.r_reset,e_sim,e_full;
moreover W.L*epsilon <= W.K_call*epsilon and W.c0_cb*W.K_call*epsilon <= 1/2.
```

Proof (prose account of the af-validated tree). For this same fixed $W, e_{\text{sim}}, e_{\text{full}}$, the phrase "supplied by Lemma 142" in the constant-ledger conclusion yields positivity of every field and the exact formulas

$$K_{\text{call}} = \max\{1, L + 1, c_0^{\text{cb}}, K_1, K_2, K_3\},$$

and ϵ_{MAIN} as the minimum of $e_{\text{env}}, e_1/K_1, e_{s2}/K_2, e_{\text{cross}}/K_3, r_{\text{reset}}/K_{\text{call}}, e_{\text{sim}}/K_{\text{call}}, e_{\text{full}}/K_{\text{call}}$ and $[2\max\{1, c_0^{\text{cb}}K_{\text{call}}\}]^{-1}$. The first four entries give the four displayed domain bounds. The maximum formula gives $K_{\text{call}} \ge 1, K_1, K_2, K_3$, so multiplication by $\epsilon \ge 0$ compares the global, atomic, and stage scales. The next three minimum entries, multiplied by positive K_{call}, place their common upper bound inside $r_{\text{reset}}, e_{\text{sim}}, e_{\text{full}}$. Also $K_{\text{call}} \ge L + 1 \ge L$. Finally put $x = c_0^{\text{cb}}K_{\text{call}}$ and

$M = \max\{1, x\}$; then $\epsilon \le 1/(2M)$ and $x \le M$ give $x\epsilon \le 1/2$. The live tree records that the root already fixes W with the complete provider provenance; it does not derive omitted witnesses from an abbreviated dependency list. Challenges about that scope and about using a pending aggregate node were repaired by the explicit scope correction and by making the two scalar branches depend directly on the validated formula node. The amended contract also carries the recorded-field repair; its suspension and re-flip likewise followed the Stage-3 dependency status, with the workspace preserved.

Lemma 150 (`lem-maincb-reset-output-typing`). *Under the same fixed witnesses and radius hypotheses as Lemma 147, every indicated raw call admits one error-improved map $v_R : B_R \to A_R$ satisfying*

$$d_R \le c_0^{\mathrm{cb}} \epsilon_R, \qquad \|v_R(I_{B_R}) - u_{A_R}\| \le c_0^{\mathrm{cb}} \epsilon_R,$$

which is an extended $c_0^{\mathrm{cb}}\epsilon_R$-inclusion, is an extended $c_0^{\mathrm{cb}}\epsilon_R$-isomorphism when u_R is bijective, and leaves the source, target corner, and amplification form unchanged.

Registry contract (verbatim).

```
After first fixing the universal e_it,K_disp,K_floor witnesses of lem-maincb-improvement-iteration
and epsilon_max^cb,delta_max^cb,c0^0 witnesses of lem-maincb-error-improvement, there are universal
0 < C_unit < infinity and epsilon_unit,delta_unit,a_unit > 0 furnished by the third clause of
prop_delta_hominc and its implicit quantifier convention such that, for every D >= 1 and every
def-maincb-witness-ledger datum W with W.c0_cb >= max{c0^0,K_floor,C_unit*(K_floor+1)} and
W.r_reset <=
min{epsilon_max^cb,delta_max^cb/D,e_it/(D+1),epsilon_unit,delta_unit/max{1,K_floor},a_unit/((1+K_disp)*D),[2*(1+K_d
every explicit raw call into an extended epsilon_R-C*-corner A_R at scale 0 <= t <= W.r_reset whose
literal map u_R:B_R->A_R is an extended D*t-inclusion and satisfies ||u_R(I_{B_R})-u_{A_R}|| <= D*t
and epsilon_R <= t admits an error-improved map v_R:B_R->A_R that satisfies d_R <=
W.c0_cb*epsilon_R and ||v_R(I_{B_R})-u_{A_R}|| <= W.c0_cb*epsilon_R, is an extended
W.c0_cb*epsilon_R-inclusion, is an extended W.c0_cb*epsilon_R-isomorphism when u_R is bijective,
and leaves the source, target corner, and amplification form unchanged.
```

Proof (prose account of the af-validated tree). Unpack the same near-unit clause and fix its four universal constants independently of all algebras, maps, D, W, t. For one raw call set $q = (1 + K_{\mathrm{disp}})Dt$. The ledger radius gives every threshold needed by the iteration and near-unit steps, in particular $q \le \min\{a_{\mathrm{unit}}, 1/2\}$. Lemma 126 produces a single dagger-preserving v_R with amplifications $I_n \otimes v_R$, displacement at most $K_{\mathrm{disp}}Dt$, and extended-inclusion defect $K_{\mathrm{floor}}\epsilon_R$; this same map is retained also at $\epsilon_R = 0$, where the provider specifies its operator-norm limit. At the unit, the displacement plus the raw-call unit error is at most q, so the near-unit clause gives the sharper bound $C_{\mathrm{unit}}(K_{\mathrm{floor}} + 1)\epsilon_R \le c_0^{\mathrm{cb}}\epsilon_R$. Record $d_R = K_{\mathrm{floor}}\epsilon_R$. Since $K_{\mathrm{floor}}\epsilon_R \le c_0^{\mathrm{cb}}\epsilon_R$, monotonicity (Lemma 146) types this very map as the asserted extended inclusion. If u_R is bijective, the lower estimate $\|v_R(x)\| \ge (1 - q)\|x\| \ge \|x\|/2$ and finite-dimensionality make v_R bijective; monotonicity preserves the resulting extended-isomorphism tag. No separate existential output of Lemma 127 is substituted, so the literal source, target corner, and amplification family remain unchanged. This is the user-ratified standalone strengthened reset-output row supplied by the consumer repair.

58 The three MAIN-CB call envelopes: stage 1, stage 2, and stage 3

The three rows below are the typed interfaces between a MAIN-CB witness ledger (Definition 42) and the three raw-call constructions of Definition 41. Each chooses one universal receiving scale, then checks that every producer error is dominated at that scale; the rows do not perform the later reset or improvement steps.

Lemma 151 (`lem-maincb-stage1-call-envelope`). *First choose a universal c_0 for Lemma 127, then fix the corresponding direct-corner witnesses, the witnesses $C_{\mathrm{cu}}, e_{\mathrm{cu}}$ of Lemma 143, and the witnesses D_1^0, e_1^0 of Lemma 137. There are universal $K_1^0 \geq 1$, $D_1 \geq D_1^0$ and $0 < e_1 \leq \min\{e_1^0, e_{\mathrm{cu}}\}$ absorbing all Stage-1 prerequisites. Indeed, for every ledger W with $W.c0_{\mathrm{cb}} = c_0$, $W.K1 \geq K_1^0$ and $W.e1 \leq e_1$, let A be a finite-dimensional extended ε-C^*-algebra, let $0 \leq \varepsilon \leq W.e1/W.K1$, and let $w : \mathbb{C}^m \to A$ be an extended $W.c0_{\mathrm{cb}}\varepsilon$-inclusion satisfying $\|w(I_{\mathbb{C}^m}) - I_A\| \leq W.c0_{\mathrm{cb}}\varepsilon$. If $P_j = w(e_j)$ has $\dim \mathcal{S}_{P_j,P_j} > 1$, then at $t_1 = W.K1\varepsilon$ the three Stage-1 producers, corner-unit comparison, and literal old-side compression furnish the explicit raw call whose map $u_1 : \mathbb{C}^{m+1} \to A$ is an extended D_1t_1-inclusion and obeys $\|u_1(I_{\mathbb{C}^{m+1}}) - I_A\| \leq D_1t_1$. When $m = 1$, u_1 is the supplied fresh map $\mathbb{C}^2 \to A_{\mathrm{fresh}} = \mathcal{S}_{P_{\mathrm{fresh}},P_{\mathrm{fresh}}}$ followed by its canonical amplified linear embedding into A; rectangular-product control, corner-unit comparison, $P_{\mathrm{fresh}} = w(I_{\mathbb{C}})$, and the incoming unit estimate give the asserted ambient inclusion and unit bounds.*

Registry contract (verbatim).

```
After first choosing a universal c0 witness for which lem-maincb-error-improvement remains valid,
fixing the corresponding lem-maincb-direct-corner-envelope witnesses, fixing the
C_corner_unit,e_corner_unit witnesses of lem-maincb-compressed-corner-unit-comparison, and fixing
the D_1^0,e_1^0 witnesses of lem-maincb-stage1-raw-refinement, there are universal receiving
witnesses K_1^0 >= 1, D_1 >= D_1^0, and e_1 > 0 with e_1 <= min{e_1^0,e_corner_unit} and every
Stage-1 producer prerequisite absorbed, such that for every def-maincb-witness-ledger datum W with
W.c0_cb = c0, W.K1 >= K_1^0, and W.e1 <= e_1, if A is a finite-dimensional extended
epsilon-C*-algebra, 0 <= epsilon <= W.e1/W.K1, w:C^m->A is a supplied extended
W.c0_cb*epsilon-inclusion satisfying ||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon, and some P_j=w(e_j) has
dim S_{P_j}>1, then the three Stage-1 producers, lem-maincb-compressed-corner-unit-comparison, and
literal old-side compression furnish the explicit Stage-1 raw call at t_1=W.K1*epsilon whose
literal map u_1:C^{m+1}->A is an extended D_1*t_1-inclusion and satisfies ||u_1(I_{C^{m+1}})-I_A||
<= D_1*t_1, when m=1, u_1 is the supplied fresh C^2->A_fresh=S_{P_fresh} inclusion followed by the
canonical amplified linear embedding A_fresh->A; lem-compcb-rectangular-product,
lem-maincb-compressed-corner-unit-comparison, P_fresh=w(I_C), and the displayed incoming unit
estimate furnish the asserted A-valued inclusion and unit bounds.
```

Proof (prose account of the af-validated tree). Fix the provider constants as in the root and write L, e_{env} for the direct-corner pair, $C_{\mathrm{sc}}, e_{\mathrm{sc}}$ for the single-compression pair, and $C_{\mathrm{rect}}, e_{\mathrm{rect}}$ for the rectangular-product pair. The tree chooses K_1^0 at least

$$\max\{1, c_0 + 1, L, C_{\mathrm{sc}}(c_0 + 1), C_{\mathrm{proj}}L, C_{\mathrm{np}}L, C_{\mathrm{pair}}L\},$$

chooses e_1 below all producer thresholds (in particular below $e_1^0, e_{\mathrm{cu}}, e_{\mathrm{env}}, e_{\mathrm{sc}}, e_{\mathrm{rect}}/2$), and lets D_1 dominate the finite coefficients in the two unit telescopes and the $m = 1$ transfer. Thus $t_1 = W.K1\varepsilon$ dominates $\varepsilon, c_0\varepsilon, L\varepsilon$.

Set $P_{\mathrm{fresh}} = P_j$ and, for $m > 1$, $P_{\mathrm{old}} = \sum_{k \neq j} w(e_k)$. Lemma 129 makes their corners extended t_1-C^*-algebras and supplies the required subordination and complementarity estimates. The three fresh-side producers Lemmas 173, 174, and 175 give one coherent pair and a map $v_{\mathrm{fresh}} : \mathbb{C}^2 \to A_{\mathrm{fresh}}$; the proof never identifies separately existential pairs. For $m > 1$, restricting w off j and compressing by P_{old} gives v_{old}, and Lemma 66 makes it an extended t_1-inclusion. Lemma 137 then gives the literal sum map with defect $D_1^0 t_1$, while the two corner-unit errors plus the incoming unit error are at most $(2C_{\mathrm{cu}} + 1)t_1$.

For $m = 1$ there is no old side. The canonical embedding preserves involution and all matrix norms; Lemma 57 bounds the extra ambient product error by $4C_{\mathrm{rect}}t_1$, and corner-unit comparison bounds the unit error by $(C_{\mathrm{cu}} + 1)t_1$. Taking D_1 above these coefficients yields the recorded raw call in both cases. The validated re-seed has 11/11 clean nodes and three rounds; three earlier runs stalled on the anaphoric c_0^{cb}, after which the witness-ledger contract was amended and re-seeded.

Lemma 152 (`lem-maincb-stage2-call-envelope`). *First choose a universal c_0 for Lemma 127, fix the corresponding L^0, e^0_{env} of Lemma 129, and fix C^0_{s2}, e^0_{s2} of Lemma 140. There is a universal $K^0_2 \geq 1$ absorbing all Stage-2 prerequisites. For every ledger W with $W.c0_{\mathrm{cb}} = c_0$, $W.L \geq L^0$, $W.K2 \geq \max\{K^0_2, 1, W.L, W.c0_{\mathrm{cb}}W.L\}$ and $W.e_{s2} \leq e^0_{s2}$, let A be a finite-dimensional extended ε-C^*-algebra and let $w : \mathbb{C}^m \to A$ be an extended $W.c0_{\mathrm{cb}}\varepsilon$-inclusion with one-dimensional atomic images and $\|w(I_{\mathbb{C}^m}) - I_A\| \leq W.c0_{\mathrm{cb}}\varepsilon$. Suppose a partition state has $\varnothing \neq U$ contained in one class, $j \notin U$ in that class, and $R = U \cup \{j\}$. If $0 \leq \varepsilon \leq W.e_{s2}/W.K2$ and the supplied reset isomorphism $v_U : M_{|U|} \to A_U$ has $\varepsilon_U \leq W.L\varepsilon$, $d_U \leq W.c0_{\mathrm{cb}}\varepsilon_U$, and $\|v_U(I_{M_{|U|}}) - u_{A_U}\| \leq W.c0_{\mathrm{cb}}\varepsilon_U$, then with $\varepsilon_R = W.L\varepsilon$ and $t_2 = W.K2\varepsilon$ the direct-corner envelope certifies A_R, every datum error is at most t_2, and the Stage-2 EXT-CB provider furnishes the explicit raw-call datum with total defect at most $C^0_{s2}t_2$.*

Registry contract (verbatim).

```
After first choosing a universal c0 witness for which lem-maincb-error-improvement remains valid,
fixing the corresponding L^0,e_env^0 witnesses of lem-maincb-direct-corner-envelope, and fixing the
C_s2^0,e_s2^0 witnesses of lem-maincb-stage2-extcb-datum, there is a universal K_2^0 >= 1 with
every Stage-2 prerequisite absorbed such that for every def-maincb-witness-ledger datum W with
W.c0_cb = c0, W.L >= L^0, W.K2 >= max{K_2^0,1,W.L,W.c0_cb*W.L}, and W.e_s2 <= e_s2^0, if A is a
finite-dimensional extended epsilon-C*-algebra, w:C^m->A is an extended W.c0_cb*epsilon-inclusion
satisfying ||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon with one-dimensional atomic images, a supplied
MAIN partition state for A,w has nonempty U contained in one equivalence class, j notin U in that
same class, and R=U union {j}, 0 <= epsilon <= W.e_s2/W.K2, and a supplied current reset
isomorphism v_U:M_{|U|}->A_U has recorded ambient field epsilon_U <= W.L*epsilon and satisfies d_U
<= W.c0_cb*epsilon_U and ||v_U(I_{M_{|U|}})-u_{A_U}|| <= W.c0_cb*epsilon_U, then
lem-maincb-direct-corner-envelope certifies A_R with the Stage-2 raw-call target ambient record
epsilon_R := W.L*epsilon, and t_2=W.K2*epsilon dominates epsilon_U,d_U,epsilon_R, the reset unit
error, and every other datum error, so lem-maincb-stage2-extcb-datum furnishes the explicit Stage-2
EXT raw-call datum with total defect at most C_s2^0*t_2.
```

Proof (prose account of the af-validated tree). Starting with the coefficient supplied by Lemma 127, enlarge it if necessary to a nonnegative c_0; Lemma 146 preserves the resulting inclusion conclusion. Then set $K^0_2 = \max\{1, e^0_{s2}/e^0_{\mathrm{env}}\}$. For receiving data put $t_2 = W.K2\varepsilon$. The ledger inequalities give

$$0 \leq \varepsilon \leq e^0_{\mathrm{env}}, \qquad 0 \leq \varepsilon \leq t_2 \leq W.e_{s2} \leq e^0_{s2}.$$

Applying Lemma 129 to $U \subseteq R$ gives A_R at ambient tolerance $L^0\varepsilon$, projection defects at the separate scale $c_0\varepsilon$, and subordination and complementarity errors at scale $L^0\varepsilon$. Increasing only the ambient tolerance to $\varepsilon_R = W.L\varepsilon$ preserves every defining inequality of the extended C^*-algebra package: upper bounds weaken, the lower C^* bound weakens in the right direction, and exact clauses are unchanged at every amplification.

Since $W.K2 \ge \max\{W.L, c_0 W.L\}$, the quantities $\varepsilon_U, d_U, \varepsilon_R$, the reset unit error, both direct-envelope scales, and the incoming global unit error are all at most t_2. Moreover $0 \le c_0\varepsilon \le t_2$, so Lemma 146 promotes the same w to the non-unital extended t_2-inclusion required by Lemma 140; no exact unitality is inferred. The partition-state relation says that each $k \in U$ has $\dim \mathcal{S}_{P_k,P_j} = 1$, supplying the remaining same-class hypotheses. The EXT-CB datum provider now returns the explicit datum in the same A_R with total defect $e \le C^0_{s2} t_2$.

The current tree has 10/10 clean nodes and three rounds. Its record preserves two earlier failures: the pre-recorded-field contract was refuted by a validated M_2 countermodel, and a later certificate was demoted for an unregistered monotonicity premise. The re-validated tree repairs the latter by importing the monotonicity row explicitly and keeps the $c_0\varepsilon$ and $L^0\varepsilon$ scales separate.

Lemma 153 (`lem-maincb-stage3-call-envelope`). *First choose a universal c_0 for Lemma 127, fix the corresponding L^0, e^0_{env} of the direct-corner envelope, and fix $C^0_{\mathrm{cross}}, e^0_{\mathrm{cross}}$ of Lemma 145. There is a universal $K^0_3 \ge 1$ absorbing every Stage-3 prerequisite. For every ledger W with $W.c0_{\mathrm{cb}} = c_0$, $W.L \ge L^0$, $W.K3 \ge \max\{K^0_3, 1, W.L, W.c0_{\mathrm{cb}} W.L\}$ and $W.e_{\mathrm{cross}} \le e^0_{\mathrm{cross}}$, let A be a finite-dimensional extended ε-C^*-algebra and $w : \mathbb{C}^m \to A$ an extended $W.c0_{\mathrm{cb}}\varepsilon$-inclusion with one-dimensional atomic images and unit error at most $W.c0_{\mathrm{cb}}\varepsilon$. Let a partition state have disjoint nonempty unions U, V sharing no class, with $R = U \cup V$. If $0 \le \varepsilon \le W.e_{\mathrm{cross}}/W.K3$ and the two reset isomorphisms $v_U : B_U \to A_U$, $v_V : B_V \to A_V$ have $\varepsilon_U, \varepsilon_V \le W.L\varepsilon$, defects $d_U \le W.c0_{\mathrm{cb}}\varepsilon_U$, $d_V \le W.c0_{\mathrm{cb}}\varepsilon_V$, and unit bounds $\|v_U(I_{B_U}) - u_{A_U}\| \le W.c0_{\mathrm{cb}}\varepsilon_U$ and $\|v_V(I_{B_V}) - u_{A_V}\| \le W.c0_{\mathrm{cb}}\varepsilon_V$, then with $\varepsilon_R = W.L\varepsilon$ and $t_3 = W.K3\varepsilon$ the direct-corner envelope certifies A_R, every supplied and derived datum error is at most t_3, and the cross-class provider furnishes the explicit four-corner raw-call datum with $\rho \le C^0_{\mathrm{cross}} t_3$.*

Registry contract (verbatim).

```
After first choosing a universal c0 witness for which lem-maincb-error-improvement remains valid,
fixing the corresponding L^0,e_env^0 witnesses of lem-maincb-direct-corner-envelope, and fixing the
C_cross^0,e_cross^0 witnesses of lem-maincb-cross-class-merging-datum, there is a universal K_3^0
>= 1 with every Stage-3 prerequisite absorbed such that for every def-maincb-witness-ledger datum W
with W.c0_cb = c0, W.L >= L^0, W.K3 >= max{K_3^0,1,W.L,W.c0_cb*W.L}, and W.e_cross <= e_cross^0, if
A is a finite-dimensional extended epsilon-C*-algebra, w:C^m->A is an extended
W.c0_cb*epsilon-inclusion satisfying ||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon with one-dimensional
atomic images, a supplied MAIN partition state for A,w has disjoint nonempty unions U,V sharing no
class and R=U union V, 0 <= epsilon <= W.e_cross/W.K3, and supplied current reset isomorphisms
v_U:B_U->A_U and v_V:B_V->A_V have recorded ambient fields epsilon_U,epsilon_V <= W.L*epsilon and
satisfy d_U <= W.c0_cb*epsilon_U, d_V <= W.c0_cb*epsilon_V, ||v_U(I_{B_U})-u_{A_U}|| <=
W.c0_cb*epsilon_U, and ||v_V(I_{B_V})-u_{A_V}|| <= W.c0_cb*epsilon_V, then
lem-maincb-direct-corner-envelope certifies A_R with the Stage-3 raw-call target ambient record
epsilon_R := W.L*epsilon, and t_3=W.K3*epsilon dominates epsilon_U,epsilon_V,d_U,d_V,epsilon_R,
both displayed unit norms, and every other datum error, so lem-maincb-cross-class-merging-datum
furnishes the explicit Stage-3 four-corner raw-call datum with rho <= C_cross^0*t_3.
```

Proof (prose account of the af-validated tree). The tree first proves that the fixed error-improvement coefficient is nonnegative: apply Lemma 127 to the identity of $\mathbb{C}$ inside exact matrix algebras at a positive tolerance ε_*. The level-one two-sided norm bounds give $1 - c_0\varepsilon_* \le \|\widetilde{v}(1)\| \le 1 + c_0\varepsilon_*$, hence $c_0 \ge 0$. Retaining the already fixed cross-class witnesses, it sets

$K_3^0 = \max\{1, e^0_{\text{cross}}/e^0_{\text{env}}\}$. Thus, for $t_3 = W.K3\varepsilon$,

$$0 \le \varepsilon \le e^0_{\text{env}}, \qquad 0 \le \varepsilon \le t_3 \le W.e_{\text{cross}} \le e^0_{\text{cross}}.$$

The direct-corner envelope applied to U, V, R gives P_U, P_V, P_R at projection scale $c_0\varepsilon$, the target A_R at ambient scale $L^0\varepsilon$, and the subordination and complementarity estimates at the latter scale. Clause-by-clause monotonicity enlarges only the ambient record to $\varepsilon_R = W.L\varepsilon$ and leaves the distinct projection scale intact. The inequalities $W.K3 \ge \max\{W.L, c_0 W.L\}$ then bound $\varepsilon_U, \varepsilon_V, d_U, d_V$, ε_R, both reset-unit norms, and every direct-envelope and global-unit error by t_3. Lemma 146 promotes the same $w : \mathbb{C}^m \to A$ from tolerance $c_0\varepsilon$ to t_3.

All hypotheses of Lemma 145 are now met, so it returns in A_R the amplified four-corner datum with common defect $\rho \le C^0_{\text{cross}} t_3$. The tree records carefully that this is the *input datum* for the Stage-3 raw call, not itself the raw-call record's downstream output map. The present 19/19-node clean tree was re-validated in eight rounds after six dependency and scope challenges were repaired, including pending-sibling uses, the proof of $c_0 \ge 0$, and the scope of the fixed cross-class witnesses. It supersedes a certificate demoted for an unregistered monotonicity premise and a subsequent re-seed that stopped at the balloon cap; the contract itself was never refuted.

59 MAIN-CB selection rows: reset, refinement, and maximality

This section records the selection mechanism attached to the single witness ledger W of Definition 42. Extended inclusions are understood in the sense of Definition 36, and the coordinate elements e_j are the projection basis of Definition 30. The first three rows construct and refine admissible maps; the last two supply the integer squeeze that identifies every corner chosen at maximal rank.

Lemma 154 (`lem-maincb-initial-reset-inclusion`). *Fix the witness-ledger datum W supplied by Lemma 148. Every finite-dimensional extended ϵ-C^*-algebra A with $0 \le \epsilon \le W.\epsilon_{\mathrm{MAIN}}$ admits an extended $W.c_0^{\mathrm{cb}}\epsilon$-inclusion $v : \mathbb{C} \to A$ such that $\|v(I_{\mathbb{C}}) - I_A\| \le W.c_0^{\mathrm{cb}}\epsilon$.*

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; every
finite-dimensional extended epsilon-C*-algebra A with 0 <= epsilon <= W.epsilon_MAIN admits an
extended W.c0_cb*epsilon-inclusion v:C->A satisfying ||v(I_C)-I_A|| <= W.c0_cb*epsilon.
```

Proof (prose account of the af-validated tree). The tree first fixes all provider witnesses coherently, including one reset-output tuple, and uses that same tuple in Lemma 148. With $D_* := \max\{1, D_0, D_1, D_2, D_3\}$, the resulting ledger has all the reset thresholds required by the typed reset row, while Lemma 149 gives $\epsilon \le W.r_{\mathrm{reset}}$. Set $t = \epsilon$. Lemma 124 supplies the literal scalar map $u_0(\lambda) = \lambda I_A$; the initial-call clause of the ledger types this same map as an extended D_*t-inclusion and gives $\|u_0(I_{\mathbb{C}}) - I_A\| \le D_*t$. The raw-call record keeps source $\mathbb{C}$, target A, scale t, target defect ϵ, the fixed amplification family, and the explicit recorded field $d_{\mathrm{raw}} = D_*t$. Applying Lemma 150 to that exact map and the already-fixed reset witnesses yields $v : \mathbb{C} \to A$ with defect $W.c_0^{\mathrm{cb}}\epsilon$ and the asserted near-unit estimate. The recorded v2 contract amends and supersedes the earlier v5 form. During validation a type challenge observed that the raw-defect field had not been recorded; the added child choosing $d_{\mathrm{raw}} = D_*t$ is part of the live 6-node tree reproduced here.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-maincb-initial-reset-inclusion`.

Lemma 155 (`lem-maincb-maximal-reset-selection`). *Fix the witness-ledger datum W supplied by Lemma 148. If A is a finite-dimensional extended ϵ-C^*-algebra with $0 \le \epsilon \le W.\epsilon_{\mathrm{MAIN}}$, then the nonempty set of integers m for which there is an extended $W.c_0^{\mathrm{cb}}\epsilon$-inclusion $w : \mathbb{C}^m \to A$ satisfying $\|w(I_{\mathbb{C}^m}) - I_A\| \le W.c_0^{\mathrm{cb}}\epsilon$ has a maximum: its lower norm is positive, and every such m obeys $m \le \dim_{\mathbb{C}} A$.*

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; if A is a
finite-dimensional extended epsilon-C*-algebra with 0 <= epsilon <= W.epsilon_MAIN, then the
nonempty set of m admitting an extended W.c0_cb*epsilon-inclusion w:C^m->A with ||w(I_{C^m})-I_A||
<= W.c0_cb*epsilon has a maximum because the lower norm is positive and m <= dim_C A.
```

Proof (prose account of the af-validated tree). Choose the ledger's freely enlargable coefficient c_0 nonnegative and then fix the one resulting W. For $0 \le \epsilon \le W.\epsilon_{\mathrm{MAIN}}$, the structural ledger gives $\epsilon \le W.K_{\mathrm{call}}\epsilon$ and $W.c_0^{\mathrm{cb}}W.K_{\mathrm{call}}\epsilon \le \frac{1}{2}$. Thus, for $d = W.c_0^{\mathrm{cb}}\epsilon$, one has $0 \le d \le \frac{1}{2}$ and $1 - d > 0$.

Lemma 154 supplies the case $m = 1$, so the admissible-index set is nonempty. For any admissible m, the level-one lower norm from Definition 36 gives $\|w(x)\| \geq (1-d)\|x\| \geq \frac{1}{2}\|x\|$. Hence w is injective, and rank–nullity yields $m = \dim_{\mathbb{C}} \mathbb{C}^m \leq \dim_{\mathbb{C}} A$. The admissible indices therefore form a nonempty subset of the finite set $\{0, \ldots, \dim_{\mathbb{C}} A\}$ and have a maximum. This is the amended v2 contract, superseding the earlier v5 wording; its 9-node validated tree records no challenge or further proof repair.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-maincb-maximal-reset-selection`.

Lemma 156 (`lem-maincb-stage1-strict-refinement`)**.** *Fix the witness-ledger datum W supplied by Lemma 148. Let A be a finite-dimensional extended ϵ-C^*-algebra with $0 \leq \epsilon \leq W.\epsilon_{\mathrm{MAIN}}$, and let $w : \mathbb{C}^m \to A$ be an extended $W.c_0^{\mathrm{cb}}\epsilon$-inclusion with $\|w(I_{\mathbb{C}^m}) - I_A\| \leq W.c_0^{\mathrm{cb}}\epsilon$. If some $P_j = w(e_j)$ has $\dim \mathcal{S}_{P_j} > 1$, then there is an extended $W.c_0^{\mathrm{cb}}\epsilon$-inclusion $w_+ : \mathbb{C}^{m+1} \to A$ satisfying $\|w_+(I_{\mathbb{C}^{m+1}}) - I_A\| \leq W.c_0^{\mathrm{cb}}\epsilon$.*

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; if A is a
finite-dimensional extended epsilon-C*-algebra with 0 <= epsilon <= W.epsilon_MAIN and an extended
W.c0_cb*epsilon-inclusion w:C^m->A satisfies ||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon and has some
P_j=w(e_j) with dim S_{P_j}>1, then there is an extended W.c0_cb*epsilon-inclusion w_+:C^{m+1}->A
satisfying ||w_+(I_{C^{m+1}})-I_A|| <= W.c0_cb*epsilon.
```

Proof (prose account of the af-validated tree). The proof fixes one reset-output witness tuple and one Stage–1 envelope tuple before constructing W, so every later occurrence denotes the same witnesses. Put $t_1 = W.K_1\epsilon$. The structural ledger gives $\epsilon \leq W.e_1/W.K_1$ and $0 \leq t_1 \leq W.K_{\mathrm{call}}\epsilon \leq W.r_{\mathrm{reset}}$; since $W.K_1 \geq 1$, it also gives $\epsilon \leq t_1$. Under the corner hypothesis, Lemma 151, using Lemma 137, furnishes a particular explicit Stage–1 raw call with literal map $u_1 : \mathbb{C}^{m+1} \to A$. The Stage–1 clause of the reset ledger types that same u_1 as an extended $D_* t_1$-inclusion and bounds its unit error by $D_* t_1$. The typed reset row, applied with source $\mathbb{C}^{m+1}$, target A, target defect ϵ, and the exact literal map u_1, returns one map v_1 with unchanged source and target, defect $W.c_0^{\mathrm{cb}}\epsilon$, and the required near-unit bound; take $w_+ = v_1$. The v2 contract supersedes its earlier v5 form. Two dependency challenges in the 11-node tree were repaired by making the fixed-witness and literal-map provenance explicit child dependencies, removing both attempted uses of pending siblings.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-maincb-stage1-strict-refinement`.

Lemma 157 (`lem-maincb-stage1-maximality`)**.** *Fix the witness-ledger datum W supplied by Lemma 148. If A is a finite-dimensional extended ϵ-C^*-algebra with $0 \leq \epsilon \leq W.\epsilon_{\mathrm{MAIN}}$ and $w : \mathbb{C}^m \to A$ has maximum source dimension among all extended $W.c_0^{\mathrm{cb}}\epsilon$-inclusions satisfying $\|w(I_{\mathbb{C}^m}) - I_A\| \leq W.c_0^{\mathrm{cb}}\epsilon$, then every projection-basis image $P_j = w(e_j)$ satisfies $\dim \mathcal{S}_{P_j} = 1$.*

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; if A is a
finite-dimensional extended epsilon-C*-algebra with 0 <= epsilon <= W.epsilon_MAIN and w:C^m->A has
maximum source dimension among all extended W.c0_cb*epsilon-inclusions satisfying
||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon, then every projection-basis image P_j=w(e_j) satisfies dim
S_{P_j}=1.
```

Proof (prose account of the af-validated tree). For the fixed W, A, ϵ, let $D(A, W, \epsilon)$ be the source dimensions of all maps satisfying the displayed extended-inclusion and near-unit conditions. Lemma 155 makes this set nonempty with a maximum, and the hypothesis says $m = \max D(A, W, \epsilon)$; hence no admissible map from $\mathbb{C}^{m+1}$ exists. Fix an arbitrary projection-basis element e_j and write $P_j = w(e_j)$. Lemma 158, applied to this exact W, w, e_j, gives a nonzero element of $\mathcal{S}_{P_j}$, so $\dim \mathcal{S}_{P_j} \geq 1$. If instead $\dim \mathcal{S}_{P_j} > 1$, Lemma 156 produces an admissible $w_+ : \mathbb{C}^{m+1} \to A$, contradicting maximality. Since the dimension is a nonnegative integer, the lower bound and the exclusion of values greater than one force $\dim \mathcal{S}_{P_j} = 1$. No reset provider, second ledger, or second maximal map enters this argument. The contract is the amended v2 form, byte-unchanged during the later repair. Its first elevation was stopped because no allowed input supplied the lower bound; the user-ratified deps-only amendment added Lemma 158. The cleanly reseeded 5-node tree was then validated on its first post-repair pass with zero challenges, and it is that tree which this account reproduces.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-maincb-stage1-maximality`.

Lemma 158 (`lem-maincb-corner-nontriviality`). *Fix the witness-ledger datum W supplied by Lemma 148. If A is a finite-dimensional extended ϵ-C^*-algebra, $0 \leq \epsilon \leq W.\epsilon_{\mathrm{MAIN}}$, $w : \mathbb{C}^m \to A$ is an extended $W.c_0^{\mathrm{cb}}\epsilon$-inclusion, and e_j is a projection-basis element, then $P_j = w(e_j)$ is a $W.c_0^{\mathrm{cb}}\epsilon$-projection satisfying $\big|\|P_j\| - 1\big| \leq W.c_0^{\mathrm{cb}}\epsilon$ and hence is nonvanishing; moreover, $\mathcal{S}_{P_j}$ contains a nonzero element, so $\dim \mathcal{S}_{P_j} \geq 1$.*

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; if A is a
finite-dimensional extended epsilon-C*-algebra, 0 <= epsilon <= W.epsilon_MAIN, w:C^m->A is an
extended W.c0_cb*epsilon-inclusion, and e_j is any projection-basis element of C^m, then P_j=w(e_j)
is a W.c0_cb*epsilon-projection satisfying | ||P_j||-1 | <= W.c0_cb*epsilon and hence is
nonvanishing, while S_{P_j} contains a nonzero element and therefore dim S_{P_j} >= 1.
```

Proof (prose account of the af-validated tree). In the construction of the fixed W, retain the particular direct-corner witnesses L^0, e^0_{env} chosen beforehand. The ledger gives $W.L \geq L^0$ and $W.e_{\mathrm{env}} \leq e^0_{\mathrm{env}}$, while the structural ledger yields $0 \leq L^0\epsilon \leq W.L\epsilon \leq W.K_{\mathrm{call}}\epsilon \leq W.r_{\mathrm{reset}}$. Set $d = W.c_0^{\mathrm{cb}}\epsilon$. The projection-basis element e_j is a nonzero self-adjoint idempotent of norm one. At amplification one, the extended d-inclusion clauses give $P_j^\dagger = P_j$, $\|P_j^2 - P_j\| \leq d$, and $1 - d \leq \|P_j\| \leq 1 + d$; thus Definition 21 gives the asserted projection and nonvanishing conclusions. Apply Lemma 129 to the singleton $\{j\}$. It equips the exact space $\mathcal{S}_{P_j}$ with an extended $L^0\epsilon$-C^*-algebra structure whose unit $I_{\mathcal{S}_{P_j}}$ is an element of that very space. Lemma 142 places $L^0\epsilon$ below $W.r_{\mathrm{reset}} < \frac{1}{2}$; the approximate-unit norm axiom therefore gives $\|I_{\mathcal{S}_{P_j}}\| > \frac{1}{2}$. This nonzero element proves $\dim \mathcal{S}_{P_j} \geq 1$. The row was introduced as the additive repair for the maximality gap. Its one verifier challenge identified missing base homomorphism clauses; registering the byte-matched external and adding child 1.2.1 repaired the 7-node tree without changing the contract.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-maincb-corner-nontriviality`.

60 The MAIN-CB assembly: one-class extension, binary block merge, stage-3 finite recombination, and the structural assembly

This section reproduces the four af-validated assembly rows. Starting from the single witness ledger W of Definition 42, they extend a reset through one class, merge class unions, recombine the finite class family, and finally identify the resulting full corner with the original algebra.

Lemma 159 (lem-maincb-one-class-extension). *Fix the datum W supplied by Lemma 148. Let A be a finite-dimensional extended ε-C^*-algebra with $0 \le \varepsilon \le W.\varepsilon_{\mathrm{MAIN}}$, and suppose a supplied MAIN partition state comes from an extended $W.c_{0,\mathrm{cb}}\varepsilon$-inclusion $w : \mathbb{C}^m \to A$ satisfying $\|w(I_{\mathbb{C}^m}) - I_A\| \le W.c_{0,\mathrm{cb}}\varepsilon$. If all atomic images are one-dimensional and C is one equivalence class, then there is a current reset isomorphism*

$$v_C : M_{|C|} \longrightarrow A_C$$

whose recorded ambient field ε_C makes A_C an extended ε_C-C^*-algebra and satisfies

$$\varepsilon_C \le W.L\varepsilon \le W.K_{\mathrm{call}}\varepsilon, qquadd_C \le W.c_{0,\mathrm{cb}}\varepsilon_C, qquad\|v_C(I_{M_{|C|}}) - u_{A_C}\| \le W.c_{0,\mathrm{cb}}\varepsilon_C.$$

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; if A is a
finite-dimensional extended epsilon-C*-algebra, 0 <= epsilon <= W.epsilon_MAIN, a supplied MAIN
partition state comes from an extended W.c0_cb*epsilon-inclusion w:C^m->A satisfying
||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon, all atomic images are one-dimensional, and C is one
equivalence class, then there is a current reset isomorphism v_C:M_{|C|}->A_C whose recorded
ambient field epsilon_C is selected so that A_C is an extended epsilon_C-C*-algebra and epsilon_C
<= W.L*epsilon <= W.K_call*epsilon, and which satisfies d_C <= W.c0_cb*epsilon_C and
||v_C(I_{M_{|C|}})-u_{A_C}|| <= W.c0_cb*epsilon_C.
```

Proof (prose account of the af-validated tree). The tree first fixes one reset-output witness package and uses that same package in Lemma 148; no second reset witness or improved map is introduced. Lemma 149 then puts the atomic and Stage-2 scales below the reset radius and gives $W.L\varepsilon \le W.K_{\mathrm{call}}\varepsilon$. Lemma 129 permits the uniform recorded choice $\varepsilon_U = W.L\varepsilon$ for every nonempty $U \subseteq C$. For a singleton $U = \{j\}$, Lemma 124 supplies the scalar map $u_0 : \mathbb{C} = M_1 \to A_U$. The ledger types this literal bijection as the required raw inclusion with its near-unit bound, and Lemma 150, used alone, selects one map v_U carrying simultaneously its source, target, amplification family, isomorphism tag, defect bound and unit bound. For the successor step, the tree fixes the exact preceding state on U, adjoins $j \in C \setminus U$, and constructs a matching partition-state record whose current subset and reference are precisely that U and that same v_U. The Stage-2 envelope (Lemma 152) supplies the raw datum in $A_{U\cup\{j\}}$. The combined producer clause of the constant ledger—not an unsupported separate call to the raw-extension theorem—types its literal output u_2 as an extended isomorphism. A single application of Lemma 150 selects $v_{U\cup\{j\}}$ and preserves the full induction invariant. Finally enumerate $C = \{j_1, \ldots, j_q\}$ and iterate that exact successor output from the singleton base. The strengthened induction records at every stage the source M_r, the isomorphism tag, $\varepsilon_{U_r} = W.L\varepsilon$, and both estimates; at $U_q = C$ this is the asserted v_C. The current 20-node tree repairs five ledger challenges by making the matching state and isomorphism tag explicit, using the combined Stage-2 producer, and stating the strengthened induction and its dependencies literally. This revalidation replaces the demoted 2026-08-01 certificate; the contract itself was not refuted.

Lemma 160 (`lem-maincb-binary-block-merge`). *Fix the datum W supplied by Lemma 148. Let A be a finite-dimensional extended ε-C^*-algebra with $0 \le \varepsilon \le W.\varepsilon_{\mathrm{MAIN}}$, and suppose a supplied MAIN partition state comes from an extended $W.c_{0,\mathrm{cb}}\varepsilon$-inclusion $w : \mathbb{C}^m \to A$ with one-dimensional atomic images and $\|w(I_{\mathbb{C}^m}) - I_A\| \le W.c_{0,\mathrm{cb}}\varepsilon$. Let U, V be disjoint nonempty unions sharing no class, and let current reset isomorphisms $v_U : B_U \to A_U$ and $v_V : B_V \to A_V$ have recorded fields $\varepsilon_U, \varepsilon_V \le W.L\varepsilon$ and satisfy, for $T \in \{U, V\}$,*

$$d_T \le W.c_{0,\mathrm{cb}}\varepsilon_T, qquad\|v_T(I_{B_T}) - u_{A_T}\| \le W.c_{0,\mathrm{cb}}\varepsilon_T.$$

Then there is a current reset isomorphism $v_{U\cup V} : B_U \oplus B_V \to A_{U\cup V}$ whose recorded field makes the target an extended $\varepsilon_{U\cup V}$-C^*-algebra and satisfies

$$\varepsilon_{U\cup V} \le W.L\varepsilon, \quad d_{U\cup V} \le W.c_{0,\mathrm{cb}}\varepsilon_{U\cup V}, \quad \|v_{U\cup V}(I_{B_U\oplus B_V}) - u_{A_{U\cup V}}\| \le W.c_{0,\mathrm{cb}}\varepsilon_{U\cup V}.$$

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; if A is a
finite-dimensional extended epsilon-C*-algebra, 0 <= epsilon <= W.epsilon_MAIN, a supplied MAIN
partition state comes from an extended W.c0_cb*epsilon-inclusion w:C^m->A satisfying
||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon with one-dimensional atomic images and has disjoint nonempty
unions U,V sharing no class, and current reset isomorphisms v_U:B_U->A_U and v_V:B_V->A_V have
recorded ambient fields epsilon_U,epsilon_V <= W.L*epsilon and satisfy d_U <= W.c0_cb*epsilon_U,
d_V <= W.c0_cb*epsilon_V, ||v_U(I_{B_U})-u_{A_U}|| <= W.c0_cb*epsilon_U, and
||v_V(I_{B_V})-u_{A_V}|| <= W.c0_cb*epsilon_V, then there is a current reset isomorphism v_{U union
V}:B_U oplus B_V->A_{U union V} whose recorded ambient field epsilon_{U union V} is selected so
that A_{U union V} is an extended epsilon_{U union V}-C*-algebra and epsilon_{U union V} <=
W.L*epsilon, and which satisfies d_{U union V} <= W.c0_cb*epsilon_{U union V} and ||v_{U union
V}(I_{B_U oplus B_V})-u_{A_{U union V}}|| <= W.c0_cb*epsilon_{U union V}.
```

Proof (prose account of the af-validated tree). Fix the ledger witnesses once, put $t_3 = W.K_3\varepsilon$ and $\varepsilon_R = W.L\varepsilon$ for $R = U \cup V$. The domain ledger gives $0 \le t_3 \le W.r_{\mathrm{reset}}, W.e_{\mathrm{cross}}$ and $\varepsilon_R \le t_3$. Lemma 153 supplies in A_R the explicit amplified four-corner raw datum, bounds its error by $C^0_{\mathrm{cross}}t_3$, and dominates all ambient, reset-defect and unit errors by t_3. Since $W.c_{0,\mathrm{cb}}\varepsilon \le t_3$, Lemma 146 types the same w as an extended t_3-inclusion without adding unitality. The hypotheses of Lemma 144 now match literally: the unions are disjoint and share no class, the images are one-dimensional, and the datum and all its errors are controlled at t_3. Hence its four fixed level-one corner maps are bijective. Lemma 139 merges those maps into the literal $u_3 : B_U \oplus B_V \to A_R$; the constant-ledger producer types u_3 as an extended D_*t_3-isomorphism and supplies its near-unit bound. One use of Lemma 150 then selects a single v_R, preserves source, target and amplification family, and returns exactly the recorded-field, defect and unit bounds displayed above. The corrected registry contract was validated in an 11-node first-pass tree with no challenge entries; the registry also records that an earlier mechanical mis-landing was replaced before this validation.

Lemma 161 (`lem-maincb-stage3-finite-recombination`). *Fix the datum W supplied by Lemma 148. Let A be a finite-dimensional extended ε-C^*-algebra with $0 \le \varepsilon \le W.\varepsilon_{\mathrm{MAIN}}$, and suppose a supplied MAIN partition state comes from an extended $W.c_{0,\mathrm{cb}}\varepsilon$-inclusion $w : \mathbb{C}^m \to A$ with one-dimensional* *atomic images, $\|w(I_{\mathbb{C}^m})-I_A\| \le W.c_{0,\mathrm{cb}}\varepsilon$, and classes $C_1,\dots,C_q$. Suppose each initial current reset isomorphism $v_{C_a}: B_{C_a} \to A_{C_a}$ has recorded field $\varepsilon_{C_a} \le W.L\varepsilon$ and satisfies*

$$d_{C_a} \le W.c_{0,\mathrm{cb}}\varepsilon_{C_a}, qquad\|v_{C_a}(I_{B_{C_a}}) - u_{A_{C_a}}\| \le W.c_{0,\mathrm{cb}}\varepsilon_{C_a}.$$

Then there is a current reset isomorphism

$$v : \bigoplus_a B_{C_a} \longrightarrow A_{\bigcup_a C_a}$$

whose recorded field and two errors satisfy

$$\varepsilon_{\bigcup_a C_a} \le W.L\varepsilon, \quad d_{\bigcup_a C_a} \le W.c_{0,\mathrm{cb}}\varepsilon_{\bigcup_a C_a}, \quad \|v(I_{\bigoplus_a B_{C_a}}) - u_{A_{\bigcup_a C_a}}\| \le W.c_{0,\mathrm{cb}}\varepsilon_{\bigcup_a C_a}.$$

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; if A is a
finite-dimensional extended epsilon-C*-algebra, 0 <= epsilon <= W.epsilon_MAIN, a supplied MAIN
partition state comes from an extended W.c0_cb*epsilon-inclusion w:C^m->A satisfying
||w(I_{C^m})-I_A|| <= W.c0_cb*epsilon with one-dimensional atomic images and has classes
C_1,...,C_q, and each initial current reset isomorphism v_{C_a}:B_{C_a}->A_{C_a} has recorded
ambient field epsilon_{C_a} <= W.L*epsilon and satisfies d_{C_a} <= W.c0_cb*epsilon_{C_a} and
||v_{C_a}(I_{B_{C_a}})-u_{A_{C_a}}|| <= W.c0_cb*epsilon_{C_a}, then there is a current reset
isomorphism v:oplus_a B_{C_a}->A_{union_a C_a} whose recorded ambient field epsilon_{union_a C_a}
satisfies epsilon_{union_a C_a} <= W.L*epsilon, d_{union_a C_a} <= W.c0_cb*epsilon_{union_a C_a},
and ||v(I_{oplus_a B_{C_a}})-u_{A_{union_a C_a}}|| <= W.c0_cb*epsilon_{union_a C_a}.
```

Proof (prose account of the af-validated tree). No witness is reselected. The structural domain ledger fixes every scalar premise needed by each later use of the binary merge. Write $U_k = \bigcup_{a=1}^k C_a$, set $B^{(1)} = B_{C_1}$ and $B^{(k+1)} = B^{(k)} \oplus B_{C_{k+1}}$, and let $P(k)$ assert the existence of one current reset isomorphism $v_k : B^{(k)} \to A_{U_k}$ with its own recorded field and the two stated bounds. The partition state has a nonempty finite atomic index set, so its equivalence classes form a nonempty family and $q \ge 1$. The root-supplied v_{C_1} witnesses $P(1)$ without any change of map. If $P(k)$ is witnessed by the exact output selected at the preceding index, then U_k and C_{k+1} are disjoint nonempty unions sharing no class. Apply Lemma 160 to that exact v_k and the root-supplied $v_{C_{k+1}}$. Its selected output has source $B^{(k)} \oplus B_{C_{k+1}} = B^{(k+1)}$, target $A_{U_{k+1}}$, and precisely the three invariant bounds, so it alone witnesses $P(k+1)$. Ordinary finite induction yields $P(q)$, where $U_q = \bigcup_a C_a$ and $B^{(q)} = \bigoplus_a B_{C_a}$. The registry records that the one-dimensional-image hypothesis was restored before elevation; the resulting 7-node tree validated on its first pass and contains no challenge entry.

Lemma 162 (`lem-maincb-structural-assembly`)**.** *Fix the datum W supplied by Lemma 148. Every finite-dimensional extended ε-C^*-algebra A with $0 \le \varepsilon \le W.\varepsilon_{\mathrm{MAIN}}$ admits a finite-dimensional C^*-algebra and an extended isomorphism*

$$B = \bigoplus_C M_{|C|}, \qquad v : B \longrightarrow A, qquad\|v(I_B) - I_A\| \le W.c_{0,\mathrm{cb}}W.K_{\mathrm{call}}\varepsilon,$$

whose defect is $W.c_{0,\mathrm{cb}}W.K_{\mathrm{call}}\varepsilon$. Consequently $C_{\mathrm{struct}} = W.c_{0,\mathrm{cb}}W.K_{\mathrm{call}}$ and $e_{\mathrm{struct}} = W.\varepsilon_{\mathrm{MAIN}}$ are finite positive universal witnesses.

Registry contract (verbatim).

```
Fix the def-maincb-witness-ledger datum W supplied by lem-maincb-reset-constant-ledger; every
finite-dimensional extended epsilon-C*-algebra A with 0 <= epsilon <= W.epsilon_MAIN admits a
finite-dimensional C*-algebra B=oplus_C M_{|C|} and an extended
W.c0_cb*W.K_call*epsilon-isomorphism v:B->A satisfying ||v(I_B)-I_A|| <= W.c0_cb*W.K_call*epsilon;
hence C_struct=W.c0_cb*W.K_call and e_struct=W.epsilon_MAIN are finite positive universal
witnesses.
```

Proof (prose account of the af-validated tree). Lemma 155 first selects a maximal-source extended inclusion $w : \mathbb{C}^m \to A$ with near-unit image $R = w(I_{\mathbb{C}^m})$. Put $t = W.K_{\text{call}}\varepsilon$ and $P_j = w(e_j)$. Witness arithmetic gives $W.K_{\text{call}} \geq \max\{1, W.c_{0,\text{cb}}\}$, so the amplified axioms weaken from ε to t and the images P_j, R are t-projections. Stage-1 maximality (Lemma 157) makes every P_j one-dimensional, while Lemma 135 makes $j \sim k \iff \dim \mathcal{S}_{P_j,P_k} = 1$ an equivalence relation with finite nonempty class family $\mathcal{C}$. At the full set J, Lemma 131 gives $\mathrm{Co}_R = \mathrm{Id}_A$ and $A_J = \mathcal{S}_R = A$; Definition 20 gives the compressed unit $u_{A_J} = R$. The tree then constructs the MAIN partition state explicitly, with current subset J and the original map w as its recorded reset reference. For each $C \in \mathcal{C}$, Lemma 159 supplies $v_C : M_{|C|} \to A_C$. Lemma 161 merges these exact witnesses into $v : \bigoplus_{C\in\mathcal{C}} M_{|C|} \to A_J$, with recorded field at most $W.L\varepsilon$ and both errors bounded by $W.c_{0,\text{cb}}$ times that field. It remains to change the distinguished target unit from $R = u_{A_J}$ to I_A. The tree does not invoke monotonicity across these different units. Instead, at every amplification, $A_J = A$ and $\mathrm{Co}_R = \mathrm{Id}_A$ identify the multiplication, involution and matrix norms, so all non-unit clauses weaken directly to $D = W.c_{0,\text{cb}}W.K_{\text{call}}\varepsilon$. The unit clause satisfies

$$\|(I_n \otimes v)(I_n \otimes I_B) - I_n \otimes I_A\| \leq d_J^{\text{out}} + \|R - I_A\| \leq W.c_{0,\text{cb}}(W.L + 1)\varepsilon \leq D,$$

using $W.K_{\text{call}} \geq W.L + 1$. Bijectivity is unchanged, hence $v : B \to A$ is the asserted extended D-isomorphism. The ledger fields make C_{struct} and e_{struct} finite, positive and universal. The banked 20-node tree follows an earlier ballooned elevation caused by missing workspace vocabulary; the root contract was not challenged. Four challenges in the current ledger were repaired by validating the explicit partition-state and recombination dependencies in order and, most notably, replacing the ill-typed cross-unit monotonicity step by the direct amplification-wise estimate just displayed.

61 S1-ENDGAME cohomological rows: coproduct, exterior cohomology, and left inversion

This section reproduces the four af-validated cohomological rows of the S1-ENDGAME chain. The ambient H-space and left-inversion data are those of Definition 25; the argument passes from the coproduct tail, through the exterior-algebra structure, to the associated-graded action and its trace.

Lemma 163 (lem-stage1-hspace-coproduct-tail)**.** *Let M be a connected CW complex with $\dim_{\mathbb{R}} H^*(M;\mathbb{R}) < \infty$, and let (M, μ, e) be an H-space. Set*

$$A = H^*(M;\mathbb{R}), \qquad A^+ = \bigoplus_{k>0} A^k, qquad\Delta = (\times)^{-1} \circ \mu^*,$$

where $\times$ is the cohomological cross product. Then A is a finite-dimensional graded-commutative associative unital algebra with $A^0 = \mathbb{R}1$, and $\Delta : A \to A \otimes_{\mathbb{R}} A$ is a degree-preserving unital algebra homomorphism with $\Delta(1) = 1 \otimes 1$. Moreover, for every homogeneous $a \in A^+$ there are a finite set J_a and homogeneous $a'_j, a''_j \in A^+$, $j \in J_a$, such that

$$\Delta(a) = a \otimes 1 + 1 \otimes a + \sum_{j \in J_a} a'_j \otimes a''_j.$$

Registry contract (verbatim).

```
H-space coproduct-tail package over the real field: if M is a connected CW complex with dim_reals
H^*(M;reals) < infinity and (M,mu,e) is an H-space, set A=H^*(M;reals), A^+=direct_sum_{k>0} A^k,
and Delta=(cross product)^(-1) o mu^*; then A is a finite-dimensional graded-commutative
associative unital algebra with A^0=reals*1, Delta:A->A tensor_reals A is a degree-preserving
unital algebra homomorphism with Delta(1)=1 tensor 1, and for every homogeneous a in A^+ there
exist a finite set J_a and homogeneous a'_j,a''_j in A^+ for j in J_a such that Delta(a)=a tensor
1+1 tensor a+sum_{j in J_a} a'_j tensor a''_j.
```

Proof (prose account of the af-validated tree). The cup product first makes A a finite-dimensional graded-commutative associative unital real algebra, while connectedness gives $A^0 = H^0(M;\mathbb{R}) = \mathbb{R}1$. Finite total dimension makes every $H^q(M;\mathbb{R})$ a finitely generated free real module, so Lemma 181, applied with $X = Y = M$, makes the cross product $\kappa : A \otimes_{\mathbb{R}} A \to H^*(M \times M;\mathbb{R})$ a degree-preserving unital ring isomorphism. Thus $\Delta = \kappa^{-1}\mu^*$ is a degree-preserving unital algebra homomorphism and $\Delta(1) = 1 \otimes 1$.

The tree establishes the needed multiplicativity of pullback directly at cochain level. For a continuous f, composition defines a chain map $f_\#$ and hence a cochain map $f^\#$; the Alexander–Whitney formula commutes with the front and back face restrictions, so $f^\#(\alpha \smile \beta) = f^\#\alpha \smile f^\#\beta$, and the degree-zero unit is fixed. This is the repaired branch: a verifier challenged the earlier unsupported appeal to functoriality, and the two cochain-level nodes replaced it without changing the contract.

Finally let $\epsilon = e^* : A \to \mathbb{R}$. The two H-space unit homotopies, pulled back along $x \mapsto (x, e)$ and $x \mapsto (e, x)$, give $(\mathrm{id} \otimes \epsilon)\Delta = \mathrm{id}$ and $(\epsilon \otimes \mathrm{id})\Delta = \mathrm{id}$. For homogeneous $a \in A^n$, $n > 0$, degree preservation and $A^0 = \mathbb{R}1$ therefore force the edge components to be $a \otimes 1$ and $1 \otimes a$. The remaining finitely many bidegrees have both entries positive, and expanding their algebraic tensors into finite sums of pure tensors produces the asserted set J_a and the positive-degree factors.

Status. `proved`; `af: validated` (14-node tree, taint clean; one challenge repaired by the Alexander–Whitney cochain argument above).

Lemma 164 (`lem-stage1-exterior-cohomology`)**.** *Under the hypotheses and notation of Lemma 163, the entire coproduct-tail package holds, and A is isomorphic as a graded algebra to an exterior algebra on a finite family of homogeneous generators of odd positive degree.*

Registry contract (verbatim).

```
Exterior cohomology of a finite H-space over the real field: if M is a connected CW complex with
dim_reals H^*(M;reals) < infinity and (M,mu,e) is an H-space, set A=H^*(M;reals),
A^+=direct_sum_{k>0} A^k, and Delta=(cross product)^(-1) o mu^*; then A is a finite-dimensional
graded-commutative associative unital algebra with A^0=reals*1, Delta:A->A tensor_reals A is a
degree-preserving unital algebra homomorphism with Delta(1)=1 tensor 1, for every homogeneous a in
A^+ there exist a finite set J_a and homogeneous a'_j,a''_j in A^+ for j in J_a such that
Delta(a)=a tensor 1+1 tensor a+sum_{j in J_a} a'_j tensor a''_j, and A is isomorphic as a graded
algebra to an exterior algebra on a finite family of odd-positive-degree homogeneous generators.
```

Proof (prose account of the af-validated tree). Lemma 163 supplies the complete algebra and coproduct package, including connectedness, finite-dimensionality, and the positive-positive tail. Over the characteristic-zero field $\mathbb{R}$, these are exactly the weak Hopf-algebra hypotheses used by the validated Hatcher structure external; finite total dimension also ensures that every graded piece is finite-dimensional. That external yields an algebra isomorphism

$$\phi : \Lambda_{\mathbb{R}}(x_i \mid i \in I) \otimes_{\mathbb{R}} \mathbb{R}[y_j \mid j \in J] \longrightarrow A,$$

with the x_i mapped to homogeneous odd-degree classes and the y_j to homogeneous even-degree classes. Giving each formal generator the degree of its image makes ϕ degree-preserving on monomials, and hence a graded-algebra isomorphism.

No polynomial generator can occur: if some y_j existed, the powers $1, y_j, y_j^2, \ldots$ would be linearly independent, and tensoring with the exterior unit would contradict the finite dimension of A. Nor can the exterior family be infinite, since the distinct word-length-one monomials x_i would again give infinitely many linearly independent elements. Thus $J = \varnothing$ and I is finite, leaving precisely an exterior algebra on finitely many odd positive-degree homogeneous generators, together with the already established coproduct-tail clauses.

Status. `proved`; `af: validated` (8-node tree, taint clean, zero challenges).

Lemma 165 (`lem-stage1-left-inversion-associated-graded`)**.** *Let M and (M, μ, e) be as above, and let $\sigma : M \to M$ be a left inversion. Put*

$$A = H^*(M;\mathbb{R}), \quad A^+ = \bigoplus_{k>0} A^k, \quad F^{p,q} = (A^+)^p \cap A^{p+q}, \quad E^{p,q} = F^{p,q}/F^{p+1,q-1}.$$

Then σ^* preserves every $F^{p,q}$ and induces $(-1)^{p+q}\,\mathrm{id}$ on every $E^{p,q}$ for $p \geq 0$ and $p + q \geq 0$.

Registry contract (verbatim).

```
Associated-graded action of a left inversion over the real field: if M is a connected CW complex
with dim_reals H^*(M;reals) < infinity, (M,mu,e) is an H-space, and sigma:M->M is a left inversion,
set A=H^*(M;reals), A^+=direct_sum_{k>0} A^k, F^{p,q}=(A^+)^p intersect A^{p+q}, and
```

```
E^{p,q}=F^{p,q}/F^{p+1,q-1}; then sigma^* preserves every F^{p,q} and induces (-1)^(p+q)*id on
every E^{p,q} for p >= 0 and p+q >= 0.
```

Proof (prose account of the af-validated tree). Write $I = A^+$. Functoriality makes σ^* a degree-preserving unital algebra homomorphism, so it preserves I^p and hence every $F^{p,q}$. For homogeneous $a \in I$, let $\ell(x) = \mu(\sigma(x), x)$. The left-inversion homotopy makes ℓ homotopic to the constant map at e, whence $\ell^*(a) = 0$. Combining naturality, the diagonal cup-product formula, and the coproduct expansion from Lemma 164 gives

$$0 = \sigma^*(a) + a + \sum_{j \in J_a} \sigma^*(a_j')a_j''.$$

Every summand in the tail lies in I^2, so $\sigma^*(a) \equiv -a \pmod{I^2}$ and σ^* induces $-\operatorname{id}$ on I/I^2.

For $p \geq 1$, expand a product $a_1 \cdots a_p$ modulo I^{p+1} after writing $\sigma^*(a_i) = -a_i + r_i$ with $r_i \in I^2$. Every term except the all-leading term has at least $p + 1$ factors from I, so the induced action on I^p/I^{p+1} is $(-1)^p \operatorname{id}$; for $p = 0$, unitality gives the same conclusion on $A/I = \mathbb{R}1$. By Lemma 164, I^p/I^{p+1} is spanned by exterior monomials of length exactly p. Their generators all have odd degree, so a nonzero degree-n component can occur only when $n \equiv p \pmod 2$. Taking $n = p + q$ therefore changes $(-1)^p$ to $(-1)^{p+q}$ on $E^{p,q}$; if that component is zero, the asserted scalar action is automatic.

Status. `proved`; `af:` `validated` (9-node tree, first-pass single run, taint clean, zero challenges).

Lemma 166 (`lem-stage1-left-inversion-trace`). *Under the preceding hypotheses, for every $k \geq 0$ one has*

$$\operatorname{Tr}\Big(\sigma^{*k} : H^k(M;\mathbb{R}) \to H^k(M;\mathbb{R})\Big) = (-1)^k \dim_{\mathbb{R}} H^k(M;\mathbb{R}).$$

Registry contract (verbatim).

```
Left-inversion trace over the real field: if M is a connected CW complex with dim_reals
H^*(M;reals) < infinity, (M,mu,e) is an H-space, and sigma:M->M is a left inversion, then
Tr(sigma^{*k}:H^k(M;reals)->H^k(M;reals))=(-1)^k*dim_reals H^k(M;reals) for every k >= 0.
```

Proof (prose account of the af-validated tree). Fix $k \geq 0$, let $V = A^k$, and use the decreasing filtration

$$W_p = F^{p,k-p} = (A^+)^p \cap A^k \qquad (0 \leq p \leq k+1).$$

Here $W_0 = V$, while $W_{k+1} = 0$ because a product of $k + 1$ positive-degree classes has degree at least $k + 1$. The space V is finite-dimensional, and Lemma 165 says that the degree-k map σ^{*k} preserves every W_p. It also identifies

$$W_p/W_{p+1} = E^{p,k-p}$$

and gives the induced action there as $(-1)^{p+(k-p)} \operatorname{id} = (-1)^k \operatorname{id}$.

Choose a basis of V adapted to the filtration. The matrix of σ^{*k} is block triangular, with diagonal blocks the induced maps on W_p/W_{p+1}, so its trace is the sum of their traces. Each such trace is $(-1)^k \dim_{\mathbb{R}}(W_p/W_{p+1})$. Meanwhile the dimension identities $\dim W_p = \dim(W_p/W_{p+1}) + \dim W_{p+1}$ telescope from $W_0 = V$ to $W_{k+1} = 0$. Consequently the quotient dimensions sum to $\dim_{\mathbb{R}} V$, and the trace is $(-1)^k \dim_{\mathbb{R}} H^k(M;\mathbb{R})$. Since k was arbitrary, the formula holds in every degree.

Status. `proved`; `af:` `validated` (6-node tree, first-pass single run, taint clean, zero challenges).

62 S1-ENDGAME bound rows: quotient left inversion, quotient local index, inversion isolation, and quotient index data

These four bound rows form the parameterized B0 chain. Throughout, the witness tuple is that of Definition 44, and the ambient unitary and exact-unit vocabulary is that of Definitions 32 and 22. To keep the mathematical rendering legible, each statement names the long clauses $(A_2), (A_4)$–(A_7) and (R); their full, unabridged ASCII text is preserved immediately below each statement in the verbatim registry anchor.

Lemma 167 (`lem-stage1-bound-quotient-left-inversion`). *For every universal witness tuple W satisfying the ranges and scale identities displayed in the contract below, there are universal $e^r_{\mathrm{quot}} > 0$ and $\epsilon^r_B = \min\{\epsilon^r_*, e^r_{\mathrm{quot}}\} > 0$ such that every finite-dimensional exact-unit ϵ_r-C^*-algebra with $0 \le \epsilon_r \le \epsilon^r_B$ and $1 < \dim_{\mathbb{C}} \mathcal{X} < \infty$, equipped with the displayed data satisfying $(A_2), (A_4), (A_5), (A_6)$ and (R), has*

$$\breve{\mathcal{U}} = \mathcal{U}_e/\mathrm{U}(1), \quad \breve{e} = [J], \quad \breve{\mu}([U],[V]) = [\mu(U,V)], \quad \breve{\sigma}([U]) = [\sigma(U)].$$

Here $(\breve{\mathcal{U}}, \breve{\mu}, \breve{e})$ is a connected H-space, $\breve{\sigma}$ is a smooth left inversion, $\sigma(cU) = \bar{c}\,\sigma(U)$ and $\|\sigma(U) - U^\dagger\| \le C_{\mathrm{grp}}\epsilon_r$; moreover $\breve{\mathcal{U}}$ is a connected compact orientable smooth manifold without boundary of real dimension $\dim_{\mathbb{C}} \mathcal{X} - 1$.

Registry contract (verbatim).

```
Parameterized bound quotient H-space package: for every universal def-stage1-polar-witness-data
tuple
W=(C_rect,C_ch,C_pol,C_grp,C_path,C_der,e_rect,kappa_ch,kappa_pol,kappa_der,delta_*,epsilon_*^r,e_S1,r_iso)
with C_rect,C_ch,C_pol,C_grp,C_path,C_der>=1, 0<e_rect<=1/C_rect,
0<kappa_ch,kappa_pol,kappa_der<=1/2, delta_*=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol)},
epsilon_*^r=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol),kappa_der/(8*C_der),1/C_grp,delta_*/(12*C_path*C_grp)},
e_S1=min{e_rect,epsilon_*^r/C_rect}, and r_iso=min{delta_*/4,kappa_der/(8*C_der)}, there exist
universal e_quot^r>0 and epsilon_B^r>0 with epsilon_B^r=min{epsilon_*^r,e_quot^r} such that, for
every finite-dimensional exact-unit epsilon_r-C*-algebra with 0<=epsilon_r<=epsilon_B^r and 1<dim_C
calX<infinity, and for displayed data satisfying all of the following: (A_2) for every V in
calUbar_delta_* and A^par in B_{2delta_*}^{icalH}(0) there is a unique g_V(A^par) in
B_{2delta_*}^{calH}(0) with f_V(A^par+g_V(A^par))=0, where f_V(A)=(1/2)*(((J+A^dagger) bold-dot
V^dagger) bold-dot (V bold-dot (J+A))-J), chi_V(A^par)=V bold-dot (J+A^par+g_V(A^par)) lies in
calU, ||Dg_V(A^par)||<=C_ch*(epsilon_r+delta_*),
||D_{A^perp}f_V(A^par+g_V(A^par))-I_calH||<=C_ch*(epsilon_r+delta_*)<1, and these C^1 charts cover
calU; (A_4) set Pi_delta_*:calU x B_{delta_*}^{calH}(J)->calX by Pi_delta_*(U,K):=U bold-dot K and
set S_delta_*:=Pi_delta_*(calU x B_{delta_*}^{calH}(J)); Pi_delta_* is a C^1 diffeomorphism onto
S_delta_* with unique inverse maps u_delta:S_delta_*->calU and
h_delta:S_delta_*->B_{delta_*}^{calH}(J) satisfying X=u_delta(X) bold-dot h_delta(X) for every X in
S_delta_* and u_delta(U)=U and h_delta(U)=J for every U in calU; (A_5) the displayed maps mu:calU x
calU->calU and sigma:calU->calU are defined by mu(U,V):=u_delta(U bold-dot V) and
sigma(U):=u_delta(U^dagger), are global C^1 and, for every U,V,Z in calU, mu(J,U)=mu(U,J)=U,
sigma(J)=J, ||mu(U,V)-U bold-dot V||<=C_grp*epsilon_r, ||sigma(U)-U^dagger||<=C_grp*epsilon_r,
||mu(mu(U,V),Z)-mu(U,mu(V,Z))||<=C_grp*epsilon_r, ||mu(sigma(U),U)-J||<=C_grp*epsilon_r, and
||mu(U,sigma(U))-J||<=C_grp*epsilon_r; (A_6) for every U_0,U_1 in calU and q in [0,1] satisfying
||U_1-U_0||<=q, C_path*q<=1/4, and
C_path*(q+epsilon_r*q+q^2)<delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), the path
H(-,U_0,U_1):[0,1]->calU given by H(t,U_0,U_1):=u_delta((1-t)*U_0+t*U_1) is defined, is jointly
continuous in its displayed variables, and joins U_0 to U_1, and satisfies
H(t,cU_0,cU_1)=c*H(t,U_0,U_1) for every c in U(1) and t in [0,1]; and (R) set q_*:=C_grp*epsilon_r,
set r_-:=delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), and set
eta_*:=C_path*(q_*+epsilon_r*q_*+q_*^2); the guards C_ch*(epsilon_r+delta_*)<=kappa_ch,
```

```
C_pol*(epsilon_r+delta_*)<=kappa_pol, q_*<r_-, C_path*q_*<=1/4, and eta_*<r_- hold; then, for those
same u_delta,h_delta,mu,sigma,H, there exist a space breve-calU and maps breve-mu:breve-calU x
breve-calU->breve-calU and breve-sigma:breve-calU->breve-calU such that breve-calU=calU_e/U(1), set
breve-e:=[J], breve-mu([U],[V])=[mu(U,V)], breve-sigma([U])=[sigma(U)],
(breve-calU,breve-mu,breve-e) is a connected H-space, breve-sigma is a smooth left inversion,
sigma(cU)=conj(c)*sigma(U), and ||sigma(U)-U^dagger||<=C_grp*epsilon_r for every c in U(1) and U in
calU, and breve-calU is a connected compact orientable smooth manifold without boundary of real
dimension (dim_C calX)-1.
```

Proof (prose account of the af-validated tree). Fix W and the displayed algebra. Positivity of the entries in the defining minimum gives $\epsilon_*^r > 0$. Lemma 184 supplies the universal $e^r_{mathrmquot}$; taking $\epsilon_B^r = \min\{\epsilon_*^r, e^r_{mathrmquot}\}$ provides the connected compact orientable quotient manifold with the asserted dimension. The graph derivative in (A_2) differs from the identity by less than one, so the smooth atlas upgrade applies to the same g_V and χ_V, without changing values or first derivatives. For the polar map, the tree supplies the ambient-openness argument explicitly: the involution splits $\mathcal{X}_{\mathbb{R}} = \mathcal{H} \oplus i\mathcal{H}$, the graph charts give $\dim_{\mathbb{R}} \mathcal{U} = \dim_{\mathbb{R}} i\mathcal{H}$, and therefore the domain of Π_{δ_*} and $\mathcal{X}_{\mathbb{R}}$ have equal dimension. Its injective ambient differential is thus an isomorphism, so the inverse-function theorem makes S_{δ_*} ambient-open and the same inverse (u_δ, h_δ) smooth. The smooth-operations row then upgrades the same scalar action, μ, and σ and retains their equivariance and the (A_5) error estimate.

On the identity component, the scalar action is free and proper. The quotient theorem therefore gives a smooth surjective submersion; the tree deliberately chooses this quotient structure and transports compactness, dimension, boundarylessness, orientation, and connectedness topologically, rather than identifying it with the package atlas by a misapplied uniqueness clause. Equivariance makes $\breve{\mu}$ and $\breve{\sigma}$ representative-independent, while local quotient sections prove smoothness. Finally set $q_* = C_{\mathrm{grp}}\epsilon_r$ and use (A_6) on $U_0 = \mu(\sigma(U), U)$ and $U_1 = J$. The path $[H(t, \mu(\sigma(U), U), J)]$ is representative-independent, continuous, and basepoint preserving, hence is the required left-inversion homotopy. The live tree has thirteen nodes; three archived nodes record the repaired atlas-identification route. Fresh verifiers also forced the ambient-openness proof and an explicit dependency for the homotopy node; all three challenges were repaired without changing the contract.

Lemma 168 (`lem-stage1-bound-quotient-local-index`). *Under the stronger parameterized hypotheses $(A_2), (A_4)$–(A_7) and (R) quoted below, at the same universal bound ϵ_B^r, the quotient package of Lemma 167 has charts $\chi_s : B^{i\mathcal{H}}_{r_{\mathrm{iso}}}(0) \to \mathcal{U}$, $s \in \{\pm 1\}$, with $\chi_s(0) = sJ$ and inverse $\psi_s = \phi^{\mathrm{par}}_{sJ}$ on the image, and each chart image is retained by σ, and*

$$F_s = \psi_s \circ \sigma \circ \chi_s, qquad \|D(F_s - \mathrm{id})(A) + 2I\| < 1, qquad \|A - B\| \leq \|\chi_s(A) - \chi_s(B)\|.$$

The chart contains $\mathcal{U} \cap B_{r_{\mathrm{iso}}}(sJ)$. Moreover $\breve{e} = [J]$ is an isolated fixed point of $\breve{\sigma}$, the line $i\mathbb{R}J$ is $D\sigma_J$-invariant,

$$\|D\breve{\sigma}_{\breve{e}} + I\|_{\mathrm{quot}} < 1, qquad \det(I - D\breve{\sigma}_{\breve{e}}) > 0, qquad \mathrm{ind}(\breve{\sigma}, \breve{e}) = +1.$$

Registry contract (verbatim).

```
Parameterized same-map quotient local-index and ambient-chart package: for every universal
def-stage1-polar-witness-data tuple
W=(C_rect,C_ch,C_pol,C_grp,C_path,C_der,e_rect,kappa_ch,kappa_pol,kappa_der,delta_*,epsilon_*^r,e_S1,r_iso)
with C_rect,C_ch,C_pol,C_grp,C_path,C_der>=1, 0<e_rect<=1/C_rect,
```

```
0<kappa_ch,kappa_pol,kappa_der<=1/2, delta_*=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol)},
epsilon_*^r=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol),kappa_der/(8*C_der),1/C_grp,delta_*/(12*C_path*C_grp)},
e_S1=min{e_rect,epsilon_*^r/C_rect}, and r_iso=min{delta_*/4,kappa_der/(8*C_der)}, there exist
universal e_quot^r>0 and epsilon_B^r>0 with epsilon_B^r=min{epsilon_*^r,e_quot^r} such that, for
every finite-dimensional exact-unit epsilon_r-C*-algebra with 0<=epsilon_r<=epsilon_B^r and 1<dim_C
calX<infinity, and for displayed data satisfying all of the following: (A_2) for every V in
calUbar_delta_* and A^par in B_{2delta_*}^{icalH}(0) there is a unique g_V(A^par) in
B_{2delta_*}^{calH}(0) with f_V(A^par+g_V(A^par))=0, where f_V(A)=(1/2)*(((J+A^dagger) bold-dot
V^dagger) bold-dot (V bold-dot (J+A))-J), chi_V(A^par)=V bold-dot (J+A^par+g_V(A^par)) lies in
calU, ||Dg_V(A^par)||<=C_ch*(epsilon_r+delta_*),
||D_{A^perp}f_V(A^par+g_V(A^par))-I_calH||<=C_ch*(epsilon_r+delta_*)<1, and these C^1 charts cover
calU; (A_4) set Pi_delta_*:calU x B_{delta_*}^{calH}(J)->calX by Pi_delta_*(U,K):=U bold-dot K and
set S_delta_*:=Pi_delta_*(calU x B_{delta_*}^{calH}(J)); Pi_delta_* is a C^1 diffeomorphism onto
S_delta_* with unique inverse maps u_delta:S_delta_*->calU and
h_delta:S_delta_*->B_{delta_*}^{calH}(J) satisfying X=u_delta(X) bold-dot h_delta(X) for every X in
S_delta_* and u_delta(U)=U and h_delta(U)=J for every U in calU; (A_5) the displayed maps mu:calU x
calU->calU and sigma:calU->calU are defined by mu(U,V):=u_delta(U bold-dot V) and
sigma(U):=u_delta(U^dagger), are global C^1 and, for every U,V,Z in calU, mu(J,U)=mu(U,J)=U,
sigma(J)=J, ||mu(U,V)-U bold-dot V||<=C_grp*epsilon_r, ||sigma(U)-U^dagger||<=C_grp*epsilon_r,
||mu(mu(U,V),Z)-mu(U,mu(V,Z))||<=C_grp*epsilon_r, ||mu(sigma(U),U)-J||<=C_grp*epsilon_r, and
||mu(U,sigma(U))-J||<=C_grp*epsilon_r; (A_6) for every U_0,U_1 in calU and q in [0,1] satisfying
||U_1-U_0||<=q, C_path*q<=1/4, and
C_path*(q+epsilon_r*q+q^2)<delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), the path
H(-,U_0,U_1):[0,1]->calU given by H(t,U_0,U_1):=u_delta((1-t)*U_0+t*U_1) is defined, is jointly
continuous in its displayed variables, and joins U_0 to U_1, and satisfies
H(t,cU_0,cU_1)=c*H(t,U_0,U_1) for every c in U(1) and t in [0,1]; (A_7) for every s in {+1,-1}, set
chi_s:B_{r_iso}^{icalH}(0)->calU by chi_s(A):=sJ bold-dot (J+A+g_{sJ}(A)), let
phi_{sJ}^par:chi_s(B_{r_iso}^{icalH}(0))->B_{r_iso}^{icalH}(0) be its inverse, and set
F_s:B_{r_iso}^{icalH}(0)->icalH by F_s(A):=phi_{sJ}^par(sigma(chi_s(A))); then sigma maps
chi_s(B_{r_iso}^{icalH}(0)) into itself and ||D(F_s-id)(A)+2I||<=C_der*(epsilon_r+r_iso) for every
A in B_{r_iso}^{icalH}(0); and (R) set q_*:=C_grp*epsilon_r, set
r_-:=delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), and set eta_*:=C_path*(q_*+epsilon_r*q_*+q_*^2);
the guards C_ch*(epsilon_r+delta_*)<=kappa_ch, C_pol*(epsilon_r+delta_*)<=kappa_pol, q_*<r_-,
C_path*q_*<=1/4, eta_*<r_-, C_der*(epsilon_r+r_iso)<=kappa_der/4<1, and
(1+epsilon_r)*(1+C_ch*(epsilon_r+delta_*))*r_iso+q_*<2delta_* hold; then, for those same
u_delta,h_delta,mu,sigma,H,chi_s,F_s, there exist a space breve-calU, maps breve-mu:breve-calU x
breve-calU->breve-calU and breve-sigma:breve-calU->breve-calU, and maps
psi_s:chi_s(B_{r_iso}^{icalH}(0))->B_{r_iso}^{icalH}(0) for s in {+1,-1} such that
breve-calU=calU_e/U(1), set breve-e:=[J], breve-mu([U],[V])=[mu(U,V)], breve-sigma([U])=[sigma(U)],
(breve-calU,breve-mu,breve-e) is a connected H-space, breve-sigma is a smooth left inversion,
sigma(cU)=conj(c)*sigma(U), ||sigma(U)-U^dagger||<=C_grp*epsilon_r, breve-calU is a connected
compact orientable smooth manifold without boundary of real dimension (dim_C calX)-1,
chi_s:B_{r_iso}^{icalH}(0)->calU has chi_s(0)=sJ and inverse psi_s=phi_{sJ}^par on its image, sigma
retains that image, F_s=psi_s o sigma o chi_s and ||D(F_s-id)(A)+2I||<1, calU intersect
B_{r_iso}(sJ) is contained in chi_s(B_{r_iso}^{icalH}(0)), and ||A-B||<=||chi_s(A)-chi_s(B)|| for
A,B in B_{r_iso}^{icalH}(0); for these same maps, breve-e is an isolated fixed point of
breve-sigma, i*reals*J is D-sigma_J-invariant, ||D-breve-sigma_{breve-e}+I||<1 in the quotient
norm, det(I-D-breve-sigma_{breve-e})>0, and the local index of breve-e is +1.
```

Proof (prose account of the af-validated tree). The first branch imports Lemma 167 with the same $u_\delta, h_\delta, \mu, \sigma, H$ and repeats the smooth upgrades. Its repaired ambient-openness step uses the real splitting $\mathcal{X} = \mathcal{H} \oplus i\mathcal{H}$, equality of source and target dimensions, and the quantitative inverse-function lemma to show that every point of S_{δ_*} has an ambient-open image neighborhood. For $s = \pm 1$, exact unitality gives $f_{sJ}(0) = 0$; uniqueness in (A_2) gives $g_{sJ}(0) = 0$, hence $\chi_s(0) = sJ$. The inverse is the anti-Hermitian part $\psi_s = \phi_{sJ}^{\mathrm{par}}$. If $U \in \mathcal{U}$ lies within r_{iso} of sJ, then $\phi_{sJ}(U) = s(U - sJ)$

has Hermitian and anti-Hermitian parts inside the graph domain; uniqueness forces the Hermitian part to equal g_{sJ}, placing U in the chart. Contractivity of the anti-Hermitian projection gives the stated lower Lipschitz estimate.

Scalar equivariance differentiates to $D\sigma_J(iJ) = -iJ$. Thus, for $V = i\mathbb{R}J$ and $R = DF_+(0) + I$, one has $R|_V = 0$, so R induces $\overline{R} = D\breve{\sigma}_{\breve{e}} + I$ on $i\mathcal{H}/V$. The repaired quotient estimate is

$$\|\overline{R}[x]\|_{\mathrm{quot}} \leq \|R(x+v)\| \leq \|R\|\,\|x+v\| \quad (v \in V),$$

followed by the infimum over v; no false equality with an ambient norm is used. Hence $\|D\breve{\sigma}_{\breve{e}} + I\|_{\mathrm{quot}} < 1$ by the fully non-strict chain in (A_7) and (R). Therefore $I - D\breve{\sigma}_{\breve{e}}$ is invertible. In an arbitrary smooth chart the tree transports this invertibility by conjugacy, rather than transporting the norm bound, and applies the quantitative inverse-function lemma to $x \mapsto x - \breve{\sigma}(x)$; this isolates $\breve{e}$. Writing $E = D\breve{\sigma}_{\breve{e}} + I$, the path $2I - tE$ stays invertible for $0 \leq t \leq 1$, so its determinant retains the positive sign of $2I$. The local-index sign formula, Lemma 179, gives index $+1$. The nineteen-node tree records five repaired verifier challenges: ambient openness, the quotient-norm step, the coordinate-norm inference, a strict-inequality edge, and a missing dependency. None changed the registry contract; the hard cap alone was ratified upward from fifteen to twenty to retain these visible repairs.

Lemma 169 (`lem-stage1-bound-inversion-isolation`). *Under the same parameterized hypotheses and at the same ϵ_B^r, there is a radius $r_{\mathrm{bidx}} > 0$, depending only on W, with $r_{\mathrm{bidx}} = r_{\mathrm{iso}}$, for which all conclusions of Lemma 168 hold and*

$$\mathrm{Fix}(\sigma) \cap B_{r_{\mathrm{bidx}}}(J) = \{J\}, \qquad \mathrm{Fix}(\sigma) \cap B_{r_{\mathrm{bidx}}}(-J) = \{-J\},$$

where the intersections are taken inside $\mathcal{U}$.

Registry contract (verbatim).

```
Parameterized same-map actual-isolation package: for every universal def-stage1-polar-witness-data
tuple
W=(C_rect,C_ch,C_pol,C_grp,C_path,C_der,e_rect,kappa_ch,kappa_pol,kappa_der,delta_*,epsilon_*^r,e_S1,r_iso)
with C_rect,C_ch,C_pol,C_grp,C_path,C_der>=1, 0<e_rect<=1/C_rect,
0<kappa_ch,kappa_pol,kappa_der<=1/2, delta_*=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol)},
epsilon_*^r=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol),kappa_der/(8*C_der),1/C_grp,delta_*/(12*C_path*C_grp)},
e_S1=min{e_rect,epsilon_*^r/C_rect}, and r_iso=min{delta_*/4,kappa_der/(8*C_der)}, there exist
universal e_quot^r>0 and epsilon_B^r>0 with epsilon_B^r=min{epsilon_*^r,e_quot^r} such that, for
every finite-dimensional exact-unit epsilon_r-C*-algebra with 0<=epsilon_r<=epsilon_B^r and 1<dim_C
calX<infinity, and for displayed data satisfying all of the following: (A_2) for every V in
calUbar_delta_* and A^par in B_{2delta_*}^{icalH}(0) there is a unique g_V(A^par) in
B_{2delta_*}^{calH}(0) with f_V(A^par+g_V(A^par))=0, where f_V(A)=(1/2)*(((J+A^dagger) bold-dot
V^dagger) bold-dot (V bold-dot (J+A))-J), chi_V(A^par)=V bold-dot (J+A^par+g_V(A^par)) lies in
calU, ||Dg_V(A^par)||<=C_ch*(epsilon_r+delta_*),
||D_{A^perp}f_V(A^par+g_V(A^par))-I_calH||<=C_ch*(epsilon_r+delta_*)<1, and these C^1 charts cover
calU; (A_4) set Pi_delta_*:calU x B_{delta_*}^{calH}(J)->calX by Pi_delta_*(U,K):=U bold-dot K and
set S_delta_*:=Pi_delta_*(calU x B_{delta_*}^{calH}(J)); Pi_delta_* is a C^1 diffeomorphism onto
S_delta_* with unique inverse maps u_delta:S_delta_*->calU and
h_delta:S_delta_*->B_{delta_*}^{calH}(J) satisfying X=u_delta(X) bold-dot h_delta(X) for every X in
S_delta_* and u_delta(U)=U and h_delta(U)=J for every U in calU; (A_5) the displayed maps mu:calU x
calU->calU and sigma:calU->calU are defined by mu(U,V):=u_delta(U bold-dot V) and
sigma(U):=u_delta(U^dagger), are global C^1 and, for every U,V,Z in calU, mu(J,U)=mu(U,J)=U,
sigma(J)=J, ||mu(U,V)-U bold-dot V||<=C_grp*epsilon_r, ||sigma(U)-U^dagger||<=C_grp*epsilon_r,
||mu(mu(U,V),Z)-mu(U,mu(V,Z))||<=C_grp*epsilon_r, ||mu(sigma(U),U)-J||<=C_grp*epsilon_r, and
||mu(U,sigma(U))-J||<=C_grp*epsilon_r; (A_6) for every U_0,U_1 in calU and q in [0,1] satisfying
```

```
||U_1-U_0||<=q, C_path*q<=1/4, and
C_path*(q+epsilon_r*q+q^2)<delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), the path
H(-,U_0,U_1):[0,1]->calU given by H(t,U_0,U_1):=u_delta((1-t)*U_0+t*U_1) is defined, is jointly
continuous in its displayed variables, and joins U_0 to U_1, and satisfies
H(t,cU_0,cU_1)=c*H(t,U_0,U_1) for every c in U(1) and t in [0,1]; (A_7) for every s in {+1,-1}, set
chi_s:B_{r_iso}^{icalH}(0)->calU by chi_s(A):=sJ bold-dot (J+A+g_{sJ}(A)), let
phi_{sJ}^par:chi_s(B_{r_iso}^{icalH}(0))->B_{r_iso}^{icalH}(0) be its inverse, and set
F_s:B_{r_iso}^{icalH}(0)->icalH by F_s(A):=phi_{sJ}^par(sigma(chi_s(A))); then sigma maps
chi_s(B_{r_iso}^{icalH}(0)) into itself and ||D(F_s-id)(A)+2I||<=C_der*(epsilon_r+r_iso) for every
A in B_{r_iso}^{icalH}(0); and (R) set q_*:=C_grp*epsilon_r, set
r_-:=delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), and set eta_*:=C_path*(q_*+epsilon_r*q_*+q_*^2);
the guards C_ch*(epsilon_r+delta_*)<=kappa_ch, C_pol*(epsilon_r+delta_*)<=kappa_pol, q_*<r_-,
C_path*q_*<=1/4, eta_*<r_-, C_der*(epsilon_r+r_iso)<=kappa_der/4<1, and
(1+epsilon_r)*(1+C_ch*(epsilon_r+delta_*))*r_iso+q_*<2delta_* hold; then, for those same
u_delta,h_delta,mu,sigma,H, there exist r_bidx>0 depending only on W, a space breve-calU, and maps
breve-mu:breve-calU x breve-calU->breve-calU and breve-sigma:breve-calU->breve-calU such that
r_bidx=r_iso and breve-calU=calU_e/U(1), set breve-e:=[J], breve-mu([U],[V])=[mu(U,V)],
breve-sigma([U])=[sigma(U)], (breve-calU,breve-mu,breve-e) is a connected H-space, breve-sigma is a
smooth left inversion, sigma(cU)=conj(c)*sigma(U), ||sigma(U)-U^dagger||<=C_grp*epsilon_r,
breve-calU is a connected compact orientable smooth manifold without boundary of real dimension
(dim_C calX)-1, breve-e is an isolated fixed point of breve-sigma with local index +1, and J and -J
are the only sigma-fixed points in their respective ambient r_bidx-balls.
```

Proof (prose account of the af-validated tree). Apply Lemma 168 to obtain the same quotient, maps, charts, local index, and ambient-chart containment. It remains only to prove actual isolation at the two representatives. For each $s \in \{\pm 1\}$ set $G_s = F_s - \mathrm{id}$ on $B^{i\mathcal{H}}_{r_{\mathrm{iso}}}(0)$. Scalar equivariance gives $\sigma(sJ) = sJ$, and since $\chi_s(0) = sJ$ one has $G_s(0) = 0$. Apply Lemma 89 with

$$X = Y = i\mathcal{H}, \qquad V = -2I, \qquad x_0 = 0, \qquad r = r_{\mathrm{iso}}, \qquad f = G_s.$$

Since $V^{-1} = -\frac{1}{2}I$, clauses (A_7) and (R) yield, for every A in the ball,

$$\|V^{-1}DG_s(A) - I\| \le \tfrac{1}{2}C_{\mathrm{der}}(\epsilon_r + r_{\mathrm{iso}}) \le \kappa_{\mathrm{der}}/8 \le 1/16 < 1.$$

Thus G_s is injective on the whole chart ball. If $U \in \mathcal{U} \cap B_{r_{\mathrm{iso}}}(sJ)$ is fixed by σ, the ambient containment from the preceding lemma writes $U = \chi_s(A)$. Then $F_s(A) = \psi_s(\sigma(\chi_s(A))) = A$, so $G_s(A) = 0 = G_s(0)$; injectivity gives $A = 0$ and hence $U = sJ$. Finally $r_{\mathrm{iso}} = \min\{\delta_*/4, \kappa_{\mathrm{der}}/(8C_{\mathrm{der}})\}$ is positive and depends only on W, so take $r_{\mathrm{bidx}} = r_{\mathrm{iso}}$. The ten-node tree received one verifier challenge: its first version used strict inequalities where (A_7) and (R) provide non-strict bounds. The displayed non-strict chain is the repaired live proof; the contract was unchanged.

Lemma 170 (`lem-stage1-bound-quotient-index-data`). *Under the same parameterized hypotheses and at the same ϵ^r_B, all conclusions of Lemma 169 hold; in addition, $\breve{\mathcal{U}}$ is homeomorphic to a finite simplicial complex. For every fixed class $\breve{V}$ of $\breve{\sigma}$ there are $U_0 \in \mathcal{U}_e$ and $c, a \in \mathrm{U}(1)$ with*

$$[U_0] = \breve{V}, \qquad \sigma(U_0) = cU_0, qquad a^2 = c, qquad \sigma(aU_0) = aU_0, qquad \sigma(-aU_0) = -aU_0.$$

Registry contract (verbatim).

```
Parameterized complete quotient package: for every universal def-stage1-polar-witness-data tuple
W=(C_rect,C_ch,C_pol,C_grp,C_path,C_der,e_rect,kappa_ch,kappa_pol,kappa_der,delta_*,epsilon_*^r,e_S1,r_iso)
with C_rect,C_ch,C_pol,C_grp,C_path,C_der>=1, 0<e_rect<=1/C_rect,
```

```
0<kappa_ch,kappa_pol,kappa_der<=1/2, delta_*=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol)},
epsilon_*^r=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol),kappa_der/(8*C_der),1/C_grp,delta_*/(12*C_path*C_grp)},
e_S1=min{e_rect,epsilon_*^r/C_rect}, and r_iso=min{delta_*/4,kappa_der/(8*C_der)}, there exist
universal e_quot^r>0 and epsilon_B^r>0 with epsilon_B^r=min{epsilon_*^r,e_quot^r} such that, for
every finite-dimensional exact-unit epsilon_r-C*-algebra with 0<=epsilon_r<=epsilon_B^r and 1<dim_C
calX<infinity, and for displayed data satisfying all of the following: (A_2) for every V in
calUbar_delta_* and A^par in B_{2delta_*}^{icalH}(0) there is a unique g_V(A^par) in
B_{2delta_*}^{calH}(0) with f_V(A^par+g_V(A^par))=0, where f_V(A)=(1/2)*(((J+A^dagger) bold-dot
V^dagger) bold-dot (V bold-dot (J+A))-J), chi_V(A^par)=V bold-dot (J+A^par+g_V(A^par)) lies in
calU, ||Dg_V(A^par)||<=C_ch*(epsilon_r+delta_*),
||D_{A^perp}f_V(A^par+g_V(A^par))-I_calH||<=C_ch*(epsilon_r+delta_*)<1, and these C^1 charts cover
calU; (A_4) set Pi_delta_*:calU x B_{delta_*}^{calH}(J)->calX by Pi_delta_*(U,K):=U bold-dot K and
set S_delta_*:=Pi_delta_*(calU x B_{delta_*}^{calH}(J)); Pi_delta_* is a C^1 diffeomorphism onto
S_delta_* with unique inverse maps u_delta:S_delta_*->calU and
h_delta:S_delta_*->B_{delta_*}^{calH}(J) satisfying X=u_delta(X) bold-dot h_delta(X) for every X in
S_delta_* and u_delta(U)=U and h_delta(U)=J for every U in calU; (A_5) the displayed maps mu:calU x
calU->calU and sigma:calU->calU are defined by mu(U,V):=u_delta(U bold-dot V) and
sigma(U):=u_delta(U^dagger), are global C^1 and, for every U,V,Z in calU, mu(J,U)=mu(U,J)=U,
sigma(J)=J, ||mu(U,V)-U bold-dot V||<=C_grp*epsilon_r, ||sigma(U)-U^dagger||<=C_grp*epsilon_r,
||mu(mu(U,V),Z)-mu(U,mu(V,Z))||<=C_grp*epsilon_r, ||mu(sigma(U),U)-J||<=C_grp*epsilon_r, and
||mu(U,sigma(U))-J||<=C_grp*epsilon_r; (A_6) for every U_0,U_1 in calU and q in [0,1] satisfying
||U_1-U_0||<=q, C_path*q<=1/4, and
C_path*(q+epsilon_r*q+q^2)<delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), the path
H(-,U_0,U_1):[0,1]->calU given by H(t,U_0,U_1):=u_delta((1-t)*U_0+t*U_1) is defined, is jointly
continuous in its displayed variables, and joins U_0 to U_1, and satisfies
H(t,cU_0,cU_1)=c*H(t,U_0,U_1) for every c in U(1) and t in [0,1]; (A_7) for every s in {+1,-1}, set
chi_s:B_{r_iso}^{icalH}(0)->calU by chi_s(A):=sJ bold-dot (J+A+g_{sJ}(A)), let
phi_{sJ}^par:chi_s(B_{r_iso}^{icalH}(0))->B_{r_iso}^{icalH}(0) be its inverse, and set
F_s:B_{r_iso}^{icalH}(0)->icalH by F_s(A):=phi_{sJ}^par(sigma(chi_s(A))); then sigma maps
chi_s(B_{r_iso}^{icalH}(0)) into itself and ||D(F_s-id)(A)+2I||<=C_der*(epsilon_r+r_iso) for every
A in B_{r_iso}^{icalH}(0); and (R) set q_*:=C_grp*epsilon_r, set
r_-:=delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), and set eta_*:=C_path*(q_*+epsilon_r*q_*+q_*^2);
the guards C_ch*(epsilon_r+delta_*)<=kappa_ch, C_pol*(epsilon_r+delta_*)<=kappa_pol, q_*<r_-,
C_path*q_*<=1/4, eta_*<r_-, C_der*(epsilon_r+r_iso)<=kappa_der/4<1, and
(1+epsilon_r)*(1+C_ch*(epsilon_r+delta_*))*r_iso+q_*<2delta_* hold; then, for those same
u_delta,h_delta,mu,sigma,H, there exist r_bidx>0 depending only on W, a space breve-calU, and maps
breve-mu:breve-calU x breve-calU->breve-calU and breve-sigma:breve-calU->breve-calU such that
r_bidx=r_iso and breve-calU=calU_e/U(1), set breve-e:=[J], breve-mu([U],[V])=[mu(U,V)],
breve-sigma([U])=[sigma(U)], (breve-calU,breve-mu,breve-e) is a connected H-space, breve-sigma is a
smooth left inversion, sigma(cU)=conj(c)*sigma(U), ||sigma(U)-U^dagger||<=C_grp*epsilon_r,
breve-calU is a connected compact orientable smooth manifold without boundary of real dimension
(dim_C calX)-1 and is homeomorphic to a finite simplicial complex, breve-e is an isolated fixed
point of breve-sigma with local index +1, J and -J are the only sigma-fixed points in their
respective ambient r_bidx-balls, and for every breve-sigma-fixed class breve-V there exist U_0 in
calU_e and c,a in U(1) such that [U_0]=breve-V, sigma(U_0)=c*U_0, a^2=c, sigma(a*U_0)=a*U_0, and
sigma(-a*U_0)=-a*U_0.
```

Proof (prose account of the af-validated tree). Fix the tuple, algebra, and displayed data. Lemma 169 supplies the universal constants and, for the same $u_\delta, h_\delta, \mu, \sigma, H$, the quotient H-space, smooth left inversion, equivariance, manifold package, local index, and the two ambient isolation statements. Since the algebra is finite-dimensional and the quotient is a compact smooth manifold without boundary, Lemma 185 makes $\breve{\mathcal{U}}$ homeomorphic to a finite simplicial complex.

It remains to lift a fixed quotient class. Let $\breve{V}$ satisfy $\breve{\sigma}(\breve{V}) = \breve{V}$ and choose $U_0 \in \mathcal{U}_e$ representing it. The identity $\breve{\sigma}([U]) = [\sigma(U)]$ gives $[\sigma(U_0)] = [U_0]$; by the defining scalar-orbit equivalence there is $c \in \mathrm{U}(1)$ with $\sigma(U_0) = cU_0$. Choose a square root in the circle: write $c = e^{i\theta}$ and take $a = e^{i\theta/2}$, so $a^2 = c$. Scalar equivariance now yields

$$\sigma(aU_0) = \overline{a}\, cU_0 = aU_0, \qquad \sigma(-aU_0) = -\overline{a}\, cU_0 = -aU_0,$$

because $\overline{a}\, c = \overline{a}\, a^2 = a$. These are the two required actual fixed representatives of the same phase-adjusted orbit. The proof tree has seven live nodes and was validated on its first pass with no challenges; no amendment or supersession is recorded.

Status. All four rows are `proved` with `af: validated` (T0), validated 2026-07-30. Their workspaces are, in the order above, `proofs/lem-stage1-bound-quotient-left-inversion`, `proofs/lem-stage1-` `proofs/lem-stage1-bound-inversion-isolation`, and `proofs/lem-stage1-bound-quotient-index-data`.

63 S1-ENDGAME fixed-class rows: the extra fixed class, the projection bridges, the complementary pair, and the fresh two-point inclusion

This section follows one fixed class from the Stage-1 quotient to the first two-point operator-space inclusion. The first row is the topological keystone; the next three successively turn its phase-corrected fixed unitary into projections for the rectified and original products; the last row amplifies the complementary pair uniformly. The ambient structures are those of Definitions 44, 32, and 23.

Lemma 171 (`lem-stage1-extra-fixed-class`). *There is one universal Stage-1 polar witness tuple*

$$W = (C_{\mathrm{rect}}, C_{\mathrm{ch}}, C_{\mathrm{pol}}, C_{\mathrm{grp}}, C_{\mathrm{path}}, C_{\mathrm{der}}, e_{\mathrm{rect}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}}, \delta_*, \varepsilon_*^r, e_{S1}, r_{\mathrm{iso}})$$

with the six coefficients at least 1, $0 < e_{\mathrm{rect}} \le C_{\mathrm{rect}}^{-1}$, $0 < \kappa_{\mathrm{ch}}, \kappa_{\mathrm{pol}}, \kappa_{\mathrm{der}} \le \frac{1}{2}$, *and*

$$\begin{aligned}
\delta_* &= \min\{\tfrac{1}{4}, \kappa_{\mathrm{ch}}/(4C_{\mathrm{ch}}), \kappa_{\mathrm{pol}}/(4C_{\mathrm{pol}})\},\\
\varepsilon_*^r &= \min\{\tfrac{1}{4}, \kappa_{\mathrm{ch}}/(4C_{\mathrm{ch}}), \kappa_{\mathrm{pol}}/(4C_{\mathrm{pol}}), \kappa_{\mathrm{der}}/(8C_{\mathrm{der}}), C_{\mathrm{grp}}^{-1}, \delta_*/(12C_{\mathrm{path}}C_{\mathrm{grp}})\},\\
e_{S1} &= \min\{e_{\mathrm{rect}}, \varepsilon_*^r/C_{\mathrm{rect}}\}, \qquad r_{\mathrm{iso}} = \min\{\delta_*/4, \kappa_{\mathrm{der}}/(8C_{\mathrm{der}})\}.
\end{aligned}$$

There are also universal $e_{\mathrm{quot}}^r, \varepsilon_B^r, C_{\mathrm{fix}}, e_{\mathrm{fix}}^r, r_{\mathrm{bidx}} > 0$ *with* $C_{\mathrm{fix}} < \infty$, $\varepsilon_B^r = \min\{\varepsilon_*^r, e_{\mathrm{quot}}^r\}$, $0 < e_{\mathrm{fix}}^r \le \varepsilon_B^r$, $C_{\mathrm{fix}} \ge C_{\mathrm{grp}}$, *such that every finite-dimensional exact-unit* ε_r*-*C^**-algebra with* $0 \le \varepsilon_r \le e_{\mathrm{fix}}^r$ *and* $1 < \dim_{\mathbb{C}} \mathcal{X} < \infty$ *admits the following common data. There are charts* $g_V : B_{2\delta_*}^{i\mathcal{H}}(0) \to B_{2\delta_*}^{\mathcal{H}}(0)$ *for* $V \in \overline{\mathcal{U}}_{\delta_*}$, *maps* $u_\delta, h_\delta, \mu, \sigma, H$, *the quotient* $\breve{\mathcal{U}} = \mathcal{U}_e/U(1)$ *with* $\breve{\mu}, \breve{\sigma}$, *a* $\breve{\sigma}$*-fixed class* $\breve{U}$, *and* $U_0, U \in \mathcal{U}_e$, $c, a \in U(1)$, *satisfying:*

(A_2) $g_V(A^\parallel)$ *is the unique point of* $B_{2\delta_*}^{\mathcal{H}}(0)$ *with* $f_V(A^\parallel + g_V(A^\parallel)) = 0$, *where*

$$f_V(A) = \tfrac{1}{2}\big(((J + A^\dagger) \cdot V^\dagger) \cdot (V \cdot (J + A)) - J\big), \qquad \chi_V(A^\parallel) = V \cdot (J + A^\parallel + g_V(A^\parallel)) \in \mathcal{U};$$

the C^1 *charts cover* $\mathcal{U}$, *and* $\|Dg_V(A^\parallel)\| \le C_{\mathrm{ch}}(\varepsilon_r + \delta_*)$ *and* $\|D_{A^\perp} f_V(A^\parallel + g_V(A^\parallel)) - I_{\mathcal{H}}\| \le C_{\mathrm{ch}}(\varepsilon_r + \delta_*) < 1$.

(A_4) $\Pi_{\delta_*}(V, K) = V \cdot K$ *is a* C^1 *diffeomorphism from* $\mathcal{U} \times B_{\delta_*}^{\mathcal{H}}(J)$ *onto its image* S_{δ_*}, *with unique inverse* (u_δ, h_δ) *satisfying* $X = u_\delta(X) \cdot h_\delta(X)$ *and* $(u_\delta(V), h_\delta(V)) = (V, J)$ *for* $V \in \mathcal{U}$.

(A_5) The global C^1 *maps* $\mu(V_1, V_2) = u_\delta(V_1 \cdot V_2)$ *and* $\sigma(V) = u_\delta(V^\dagger)$ *have unit* J, $\sigma(J) = J$, *and, for* $V, V_1, V_2, V_3 \in \mathcal{U}$,

$$\begin{aligned}
\|\mu(V_1, V_2) - V_1 \cdot V_2\|,\ \|\sigma(V) - V^\dagger\| &\le C_{\mathrm{grp}}\varepsilon_r,\\
\|\mu(\mu(V_1, V_2), V_3) - \mu(V_1, \mu(V_2, V_3))\| &\le C_{\mathrm{grp}}\varepsilon_r,\\
\|\mu(\sigma(V), V) - J\|,\ \|\mu(V, \sigma(V)) - J\| &\le C_{\mathrm{grp}}\varepsilon_r.
\end{aligned}$$

(A_6) If $\|V_1 - V_0\| \le q \le 1$, $C_{\mathrm{path}} q \le \frac{1}{4}$, *and* $C_{\mathrm{path}}(q + \varepsilon_r q + q^2) < \delta_* - C_{\mathrm{pol}}(\varepsilon_r \delta_* + \delta_*^2)$, *then* $H(t, V_0, V_1) = u_\delta((1-t)V_0 + tV_1)$ *is a jointly continuous path from* V_0 *to* V_1 *and* $H(t, c_0 V_0, c_0 V_1) = c_0 H(t, V_0, V_1)$ *for* $c_0 \in U(1)$.

(A_7) For $s = \pm 1$, *the chart* $\chi_s(A) = sJ \cdot (J + A + g_{sJ}(A))$ *on* $B_{r_{\mathrm{iso}}}^{i\mathcal{H}}(0)$ *is preserved by* σ, *and, with* $F_s = (\phi_{sJ}^\parallel \circ \sigma \circ \chi_s)$, $\|D(F_s - \mathrm{id})(A) + 2I\| \le C_{\mathrm{der}}(\varepsilon_r + r_{\mathrm{iso}})$.

(R) For $q_* = C_{\mathrm{grp}}\varepsilon_r$, $r_- = \delta_* - C_{\mathrm{pol}}(\varepsilon_r\delta_* + \delta_*^2)$ and $\eta_* = C_{\mathrm{path}}(q_* + \varepsilon_r q_* + q_*^2)$, all the displayed graph, polar, path, and derivative guards hold: in particular $q_* < r_-$, $C_{\mathrm{path}}q_* \le \frac{1}{4}$, $C_{\mathrm{ch}}(\varepsilon_r + \delta_*) \le \kappa_{\mathrm{ch}}$, $C_{\mathrm{pol}}(\varepsilon_r + \delta_*) \le \kappa_{\mathrm{pol}}$, $\eta_* < r_-$, $C_{\mathrm{der}}(\varepsilon_r + r_{\mathrm{iso}}) \le \kappa_{\mathrm{der}}/4 < 1$, and $(1+\varepsilon_r)(1+C_{\mathrm{ch}}(\varepsilon_r+\delta_*))r_{\mathrm{iso}} + q_* < 2\delta_*$.

Moreover $r_{\mathrm{bidx}} = r_{\mathrm{iso}}$, $\breve{e} = [J]$, $\breve{\mu}([V_1],[V_2]) = [\mu(V_1,V_2)]$, $\breve{\sigma}([V]) = [\sigma(V)]$, and $(\breve{\mathcal{U}}, \breve{\mu}, \breve{e})$ is a connected H-space with smooth left inversion $\breve{\sigma}$. The quotient is a connected compact orientable smooth manifold without boundary of real dimension $\dim_{\mathbb{C}} \mathcal{X} - 1$, homeomorphic to a finite simplicial complex; $\breve{e}$ is an isolated fixed point of local index $+1$; and $J, -J$ are the only σ-fixed points in their respective ambient r_{bidx}-balls. Finally

$$\breve{U} \neq \breve{e}, \quad [U_0] = \breve{U}, \quad \sigma(U_0) = cU_0, \quad a^2 = c, \quad U = aU_0, \quad [U] = \breve{U}, \quad \sigma(U) = U,$$

$\sigma(c_0V) = \overline{c_0}\sigma(V)$ and $\|\sigma(V) - V^\dagger\| \le C_{\mathrm{grp}}\varepsilon_r$, while $\|U - U^\dagger\| \le C_{\mathrm{fix}}\varepsilon_r$ and $\|U - J\|, \|U + J\| \ge r_{\mathrm{bidx}}$.

Registry contract (verbatim).

```
Bound extra fixed class with one ledger selection: there exist one def-stage1-polar-witness-data
tuple
W=(C_rect,C_ch,C_pol,C_grp,C_path,C_der,e_rect,kappa_ch,kappa_pol,kappa_der,delta_*,epsilon_*^r,e_S1,r_iso)
such that C_rect,C_ch,C_pol,C_grp,C_path,C_der>=1, 0<e_rect<=1/C_rect,
0<kappa_ch,kappa_pol,kappa_der<=1/2, delta_*=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol)},
epsilon_*^r=min{1/4,kappa_ch/(4*C_ch),kappa_pol/(4*C_pol),kappa_der/(8*C_der),1/C_grp,delta_*/(12*C_path*C_grp)},
e_S1=min{e_rect,epsilon_*^r/C_rect}, r_iso=min{delta_*/4,kappa_der/(8*C_der)}, and there are
universal e_quot^r>0, epsilon_B^r>0, C_fix<infinity, 0<e_fix^r<=epsilon_B^r, and r_bidx>0 with
epsilon_B^r=min{epsilon_*^r,e_quot^r} and C_fix>=C_grp such that, for every finite-dimensional
exact-unit epsilon_r-C*-algebra with 0<=epsilon_r<=e_fix^r and 1<dim_C calX<infinity, there exist a
family g=(g_V:B_{2delta_*}^{icalH}(0)->B_{2delta_*}^{calH}(0))_{V in calUbar_delta_*}, maps
u_delta,h_delta,mu,sigma,H, a space breve-calU, maps breve-mu:breve-calU x breve-calU->breve-calU
and breve-sigma:breve-calU->breve-calU, a breve-sigma-fixed class breve-U, U_0 in calU_e, c,a in
U(1), and U in calU_e such that all of the following hold: (A_2) for every V in calUbar_delta_* and
A^par in B_{2delta_*}^{icalH}(0), g_V(A^par) is the unique element of B_{2delta_*}^{calH}(0) with
f_V(A^par+g_V(A^par))=0, where f_V(A)=(1/2)*(((J+A^dagger) bold-dot V^dagger) bold-dot (V bold-dot
(J+A))-J), chi_V(A^par)=V bold-dot (J+A^par+g_V(A^par)) lies in calU,
||Dg_V(A^par)||<=C_ch*(epsilon_r+delta_*),
||D_{A^perp}f_V(A^par+g_V(A^par))-I_calH||<=C_ch*(epsilon_r+delta_*)<1, and these C^1 charts cover
calU; (A_4) set Pi_delta_*:calU x B_{delta_*}^{calH}(J)->calX by Pi_delta_*(V,K):=V bold-dot K and
set S_delta_*:=Pi_delta_*(calU x B_{delta_*}^{calH}(J)); Pi_delta_* is a C^1 diffeomorphism onto
S_delta_* with unique inverse maps u_delta:S_delta_*->calU and
h_delta:S_delta_*->B_{delta_*}^{calH}(J) satisfying X=u_delta(X) bold-dot h_delta(X) for every X in
S_delta_* and u_delta(V)=V and h_delta(V)=J for every V in calU; (A_5) the displayed maps mu:calU x
calU->calU and sigma:calU->calU are defined by mu(V_1,V_2):=u_delta(V_1 bold-dot V_2) and
sigma(V):=u_delta(V^dagger), are global C^1 and, for every V,V_1,V_2,V_3 in calU,
mu(J,V)=mu(V,J)=V, sigma(J)=J, ||mu(V_1,V_2)-V_1 bold-dot V_2||<=C_grp*epsilon_r,
||sigma(V)-V^dagger||<=C_grp*epsilon_r,
||mu(mu(V_1,V_2),V_3)-mu(V_1,mu(V_2,V_3))||<=C_grp*epsilon_r,
||mu(sigma(V),V)-J||<=C_grp*epsilon_r, and ||mu(V,sigma(V))-J||<=C_grp*epsilon_r; (A_6) for every
V_0,V_1 in calU and q in [0,1] satisfying ||V_1-V_0||<=q, C_path*q<=1/4, and
C_path*(q+epsilon_r*q+q^2)<delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), the path
H(-,V_0,V_1):[0,1]->calU given by H(t,V_0,V_1):=u_delta((1-t)*V_0+t*V_1) is defined, is jointly
continuous in its displayed variables, and joins V_0 to V_1, and satisfies
H(t,c_0*V_0,c_0*V_1)=c_0*H(t,V_0,V_1) for every c_0 in U(1) and t in [0,1]; (A_7) for every s in
{+1,-1}, set chi_s:B_{r_iso}^{icalH}(0)->calU by chi_s(A):=sJ bold-dot (J+A+g_{sJ}(A)), let
phi_{sJ}^par:chi_s(B_{r_iso}^{icalH}(0))->B_{r_iso}^{icalH}(0) be its inverse, and set
F_s:B_{r_iso}^{icalH}(0)->icalH by F_s(A):=phi_{sJ}^par(sigma(chi_s(A))); then sigma maps
chi_s(B_{r_iso}^{icalH}(0)) into itself and ||D(F_s-id)(A)+2I||<=C_der*(epsilon_r+r_iso) for every
```

```
A in B_{r_iso}^{icalH}(0); (R) set q_*:=C_grp*epsilon_r, set
r_-:=delta_*-C_pol*(epsilon_r*delta_*+delta_*^2), and set eta_*:=C_path*(q_*+epsilon_r*q_*+q_*^2);
the guards C_ch*(epsilon_r+delta_*)<=kappa_ch, C_pol*(epsilon_r+delta_*)<=kappa_pol, q_*<r_-,
C_path*q_*<=1/4, eta_*<r_-, C_der*(epsilon_r+r_iso)<=kappa_der/4<1, and
(1+epsilon_r)*(1+C_ch*(epsilon_r+delta_*))*r_iso+q_*<2delta_* hold; moreover r_bidx=r_iso and
breve-calU=calU_e/U(1), set breve-e:=[J], breve-mu([V_1],[V_2])=[mu(V_1,V_2)],
breve-sigma([V])=[sigma(V)], (breve-calU,breve-mu,breve-e) is a connected H-space, breve-sigma is a
smooth left inversion, sigma(c_0*V)=conj(c_0)*sigma(V) and ||sigma(V)-V^dagger||<=C_grp*epsilon_r
for every c_0 in U(1) and V in calU, breve-calU is a connected compact orientable smooth manifold
without boundary of real dimension (dim_C calX)-1 and is homeomorphic to a finite simplicial
complex, breve-e is an isolated fixed point of breve-sigma with local index +1, J and -J are the
only sigma-fixed points in their respective ambient r_bidx-balls, breve-U!=breve-e, [U_0]=breve-U,
sigma(U_0)=c*U_0, a^2=c, U=a*U_0, [U]=breve-U, sigma(U)=U, ||U-U^dagger||<=C_fix*epsilon_r,
||U-J||>=r_bidx, and ||U+J||>=r_bidx.
```

Proof (prose account of the af-validated tree). The tree first selects the tuple W from Lemma 122 exactly once. For this same tuple, Lemma 170 supplies the conditional quotient package. The scalar branch then derives all its guards directly from the minimum formulas: $C_{\mathrm{ch}}(\varepsilon_r+\delta_*), C_{\mathrm{pol}}(\varepsilon_r+\delta_*)$ are at most half their respective margins, $r_- \geq 3\delta_*/4$, $q_* \leq \delta_*/12 < r_-$, and $\eta_* \leq 61\delta_*/576 < r_-$. It also obtains the derivative bound and $(1+\varepsilon_r)(1+C_{\mathrm{ch}}(\varepsilon_r+\delta_*))r_{\mathrm{iso}}+q_* < 2\delta_*$. Thus one may take $e^r_{\mathrm{fix}} = \varepsilon^r_B$ and $C_{\mathrm{fix}} = C_{\mathrm{grp}}$ and instantiate the analytic and topological data on common objects.

Suppose now that $\breve{e}$ were the only fixed class. The finite simplicial model and Lemma 123 put that point in a maximal simplex. Lemmas 178 and 179 then give $L(\breve{\sigma}) = 1$. On the other hand, Lemma 166 gives $\mathrm{Tr}(\breve{\sigma}^{*k}) = (-1)^k b_k$, so Definition 27 yields $L(\breve{\sigma}) = \sum_k b_k$. Connectedness gives $b_0 = 1$, while orientability and positive dimension give $b_d \geq 1$ by Lemma 180; hence $L(\breve{\sigma}) \geq 2$, a contradiction.

For an extra fixed class $\breve{U}$, the phase-lift clause returns $[U_0] = \breve{U}$, $\sigma(U_0) = cU_0$ and $a^2 = c$. Scalar covariance makes $U = aU_0$ fixed and gives $\|U - U^\dagger\| \leq C_{\mathrm{fix}}\varepsilon_r$. The two local isolation statements force $\|U - J\|, \|U + J\| \geq r_{\mathrm{bidx}}$, since either strict inequality would put U in the scalar class. Three recorded challenges were repaired: the bound package was restricted to its actual scope, the Lefschetz step was phrased on the same finite-polyhedron presentation rather than through an unstated conjugacy-invariance claim, and a sibling dependency was made explicit.

Lemma 172 (`lem-stage1-fixed-unitary-projection-bridge`). *There are universal $C_{\mathrm{bridge}} < \infty$ and $e^r_{\mathrm{bridge}} > 0$ such that every finite-dimensional exact-unit ε_r-C^*-algebra with $0 \leq \varepsilon_r \leq e^r_{\mathrm{bridge}}$ and $1 < \dim_{\mathbb{C}} \mathcal{X} < \infty$ contains a nontrivial $C_{\mathrm{bridge}}\varepsilon_r$-projection for the product $\cdot$ and unit J.*

Registry contract (verbatim).

```
Fixed-unitary projection bridge: there are universal C_bridge<infinity and e_bridge^r>0 such that
every finite-dimensional exact-unit epsilon_r-C*-algebra with 0<=epsilon_r<=e_bridge^r and 1<dim_C
calX<infinity contains a nontrivial C_bridge*epsilon_r-projection P for the product bold-dot and
unit J.
```

Proof (prose account of the af-validated tree). Apply Lemma 171 once, obtaining C_{fix}, e^r_{fix}, $r = r_{\mathrm{bidx}}$, and U with $\|U - U^\dagger\| \leq C_{\mathrm{fix}}\varepsilon_r$ and $\|U \pm J\| \geq r$. The tree fixes

$$K = 3 + 4C_{\mathrm{fix}}/r, \quad C_{\mathrm{bridge}} = \max\{C_{\mathrm{fix}}, K\}, \quad e^r_{\mathrm{bridge}} = \min\{e^r_{\mathrm{fix}}, \tfrac{1}{2}, C^{-1}_{\mathrm{fix}}, r/C_{\mathrm{fix}}\}.$$

Set $A = (U + U^\dagger)/2$, $P = (J + A)/2$, and $Q = J - P$. The unitary identity $U^\dagger \cdot U = J$ and the lower C^*-bound give $\|U\| < 2$, while $P \cdot P - P = (A \cdot A - J)/4$ and $Q \cdot Q - Q = P \cdot P - P$. Expanding around $U^\dagger \cdot U$ gives $\|A \cdot A - J\| \le 3\|U - U^\dagger\|$, hence both defects are at most $C_{\mathrm{bridge}}\varepsilon_r$.

The exact identities $U + J = 2P + (U - A)$ and $U - J = -2Q + (U - A)$, together with $\|U - A\| \le r/2$, imply $\|P\|, \|Q\| \ge r/4$. If R is either P or Q, the upper and lower C^* estimates applied to $R \cdot R$ bound $\|R\| < 3$ and then $|\|R\| - 1| \le (3 + 4C_{\mathrm{fix}}/r)\varepsilon_r$. Thus Definition 21 recognizes P as nontrivial. The endpoint $\varepsilon_r = 0$ uses only non-strict bounds. Two verifier challenges found an unbound Q in leaf statements; both were repaired by defining $Q = J - P$ locally before the affected estimates.

Lemma 173 (`lem-stage1-rectified-nontrivial-projection`). *There are universal $C_{\mathrm{proj}} < \infty$ and $e_{\mathrm{proj}} > 0$ such that every finite-dimensional extended ε_X-C^*-algebra $(\mathcal{X}, I_X, \cdot, \dagger)$ with $0 \le \varepsilon_X \le e_{\mathrm{proj}}$ and $1 < \dim_{\mathbb{C}} \mathcal{X} < \infty$ contains a nontrivial $C_{\mathrm{proj}}\varepsilon_X$-projection P_0 for the original product and unit I_X.*

Registry contract (verbatim).

```
There are universal C_proj<infinity and e_proj>0 such that every finite-dimensional extended
epsilon_X-C*-algebra (calX,I_X,.,dagger) with 0<=epsilon_X<=e_proj and 1<dim_C calX<infinity
contains a nontrivial C_proj*epsilon_X-projection P_0 for the original product and original unit
I_X.
```

Proof (prose account of the af-validated tree). Choose the rectification constants $C_{\mathrm{rect}}, e_{\mathrm{rect}}$ and the bridge constants $C_{\mathrm{bridge}}, e^r_{\mathrm{bridge}}$. With $D = \max\{1, C_{\mathrm{bridge}}\}$, the tree fixes a universal complement coefficient K, then $B = K(D+1)C_{\mathrm{rect}}$, $A = C_{\mathrm{rect}}(D+4)$, $C_{\mathrm{proj}} = A + 6$, and $e_{\mathrm{proj}} = \min\{e_{\mathrm{rect}}, e^r_{\mathrm{bridge}}/C_{\mathrm{rect}}, 1, B^{-1}\}$. Lemma 98 puts an exact-unit product $\cdot$ and unit J on the same involutive normed space, with $\varepsilon_r = C_{\mathrm{rect}}\varepsilon_X$, $\|J - I_X\| \le C_{\mathrm{rect}}\varepsilon_X$, and product discrepancy at most $C_{\mathrm{rect}}\varepsilon_X\|x\|\|y\|$. Lemma 172 then supplies a nontrivial rectified projection P.

Its nonvanishing estimates give $|\|P\| - 1|, |\|J - P\| - 1| \le B\varepsilon_X$ and hence $\|P\| \le 2$. Product comparison transfers the defect: $\|P^2 - P\| \le C_{\mathrm{rect}}\varepsilon_X\|P\|^2 + C_{\mathrm{bridge}}C_{\mathrm{rect}}\varepsilon_X \le A\varepsilon_X$. For $R = I_X - P$, the original approximate-unit errors contribute at most $6\varepsilon_X$, so $\|R^2 - R\| \le C_{\mathrm{proj}}\varepsilon_X$. Finally unit closeness transfers the nonvanishing estimate from $J - P$ to $I_X - P$. Thus $P_0 = P$ is nontrivial in the sense of Definition 21. The seven-node tree validated first-pass with no substantive challenge; its only transient blocked verdict cleared when dependencies did.

Lemma 174 (`lem-stage1-original-complementary-pair`). *There are universal $C_{\mathrm{np}} < \infty$ and $e_{\mathrm{np}} > 0$ such that every finite-dimensional extended ε_X-C^*-algebra $(\mathcal{X}, I_X, \cdot, \dagger)$ with $0 \le \varepsilon_X \le e_{\mathrm{np}}$ and $1 < \dim_{\mathbb{C}} \mathcal{X} < \infty$ contains nonvanishing $C_{\mathrm{np}}\varepsilon_X$-projections P', P'' for the original product satisfying*

$$P' + P'' = I_X, \qquad \|P'P''\|, \|P''P'\| \le C_{\mathrm{np}}\varepsilon_X.$$

Registry contract (verbatim).

```
There are universal C_np<infinity and e_np>0 such that every finite-dimensional extended
epsilon_X-C*-algebra (calX,I_X,.,dagger) with 0<=epsilon_X<=e_np and 1<dim_C calX<infinity contains
nonvanishing C_np*epsilon_X-projections P' and P'' for the original product such that P'+P''=I_X
and ||P'P''||,||P''P'||<=C_np*epsilon_X.
```

Proof (prose account of the af-validated tree). Take P_0 from Lemma 173 and put $P' = P_0$, $P'' = I_X - P_0$. Their sum is I_X. If $\delta = C_{\text{proj}}\varepsilon_X$, Definition 21 gives the complement defect as $\delta + O(\varepsilon_X)$. The tree exposes the universal coefficient of that term as C_{cmp} and enlarges to $\overline{C} = \max\{C_{\text{proj}}, C_{\text{proj}} + C_{\text{cmp}}\}$; both P' and P'' are then nonvanishing $\overline{C}\varepsilon_X$-projections. Export node 1.4 asserts the exact identities $P_0 I_X = I_X P_0 = P_0$ and hence the exact expansion $P_0 - P_0^2$; however, the registry amendment records that this exact-unit move was rejected and that the cross-products equal $P_0 - P_0^2$ only up to controlled unit error. These two canonical records are inconsistent, so this row is not a reconciled proof account. Taking $C_{\text{np}} = \overline{C}$ and $e_{\text{np}} = e_{\text{proj}}$ completes the quantified assembly, including $\varepsilon_X = 0$.

A major verifier challenge rejected the original leaf's claim that the original unit was automatically an exact two-sided unit. The repaired live tree factors that leaf into the general-unit complement-defect clause and a separate nonvanishing clause, and enlarges C_{np} accordingly; fresh verifiers then accepted all nine live nodes. This repair changed the proof ledger, not the registry contract.

Lemma 175 (`lem-stage1-fresh-two-point-inclusion`)**.** *There are universal $C_{\text{pair}} < \infty$ and $e_{\text{pair}} > 0$ such that every finite-dimensional extended ε_X-C^*-algebra $(\mathcal{X}, I_X, \cdot, \dagger)$ with $0 \le \varepsilon_X \le e_{\text{pair}}$ and $1 < \dim_{\mathbb{C}} \mathcal{X} < \infty$ contains nonvanishing $C_{\text{pair}}\varepsilon_X$-projections P', P'' with $P' + P'' = I_X$ for which*

$$v^{(2)} : \mathbb{C}^2 \longrightarrow \mathcal{X}, \qquad v^{(2)}(\lambda, \mu) = \lambda P' + \mu P'',$$

is an extended $C_{\text{pair}}\varepsilon_X$-inclusion, satisfies $v^{(2)}(1,1) = I_X$, and sends the standard projection basis Π', Π'' to P', P'', respectively.

Registry contract (verbatim).

```
There are universal C_pair<infinity and e_pair>0 such that every finite-dimensional extended
epsilon_X-C*-algebra (calX,I_X,.,dagger) with 0<=epsilon_X<=e_pair and 1<dim_C calX<infinity
contains nonvanishing C_pair*epsilon_X-projections P',P'' with P'+P''=I_X for which the linear map
v^(2):C^2->calX, v^(2)(lambda,mu)=lambda*P'+mu*P'', is an extended C_pair*epsilon_X-inclusion,
satisfies v^(2)(1,1)=I_X, and sends the standard projection basis Pi',Pi'' to P',P''.
```

Proof (prose account of the af-validated tree). Let q_1, q_2 be the pair from Lemma 174 and set

$$d = \max\{\|q_1^2 - q_1\|, \|q_2^2 - q_2\|, \|q_1 q_2\|, \|q_2 q_1\|\} \le C_{\text{np}}\varepsilon_X.$$

After one universal shrink, nonvanishing gives $\|q_i\| \ge \frac{1}{2}$. Define the single level-one map $v^{(2)}(\lambda, \mu) = \lambda q_1 + \mu q_2$ and only its canonical amplifications $v_n(A, B) = A \otimes q_1 + B \otimes q_2$. Definition 29 gives $\|T \otimes z\|_n = \|T\|\|z\|$: the upper bound is the rectangular matrix inequality and the reverse bound comes from compression by norming unit vectors.

Bilinearity now yields the exact four-term defect identity

$$v_n(x)v_n(y) - v_n(xy) = AC \otimes (q_1^2 - q_1) + AD \otimes q_1 q_2 + BC \otimes q_2 q_1 + BD \otimes (q_2^2 - q_2),$$

so every amplification has multiplicative defect at most $4d$. If $\max\{\|A\|, \|B\|\} = 1$, multiplication by $I_n \otimes q_1$ or $I_n \otimes q_2$ gives $\|v_n(A, B)\| \ge (1 - 4d)/(1 + \varepsilon_X)$; after shrinking, this is at least $1/4$. The tree then applies its recorded delta-homomorphism external with $\delta_n = 4d$ and lower modulus $1/4$. A repaired smallness leaf chooses $e_{\text{up}} \le \delta_{\max}/(4\max\{C_{\text{np}}, 1\})$ and $e_{\text{up}} \le 1/(64\max\{C_{\text{np}}, 1\})$, ensuring both $\delta_n \le \delta_{\max}$ and $2\delta_n < 1/4$.

The external's upper and lower $O(\delta_n + \varepsilon_X)$ estimates therefore admit one universal coefficient K, independent of n. Definition 36 makes $v^{(2)}$ an extended $K\varepsilon_X$-inclusion; exact linearity also gives the unit and standard-basis values from Definition 30. Taking $C_{\text{pair}} \geq \max\{C_{\text{np}}, K, 4C_{\text{np}}\}$ and $e_{\text{pair}} = e_{\text{up}}$ closes the row. Two challenges were repaired: the separate delta-smallness hypothesis was made explicit, and the level-uniform K was placed behind a hard dependency on that smallness node. The final tree has twelve live nodes. **Status.** All five rows are `proved` with `af: validated` (T0), taint clean.

64 The algebraic-topology toolbox

This section collects the seven `af`-validated topology rows used by the Stage-1 fixed-point and cohomological arguments. Each account reports the live proof tree, including the source and scope repairs recorded during adversarial verification.

Lemma 176 (`lem-topology-quotient-manifold`). *Let a Lie group G act smoothly, freely and properly on a smooth manifold M. Then M/G is a topological manifold of dimension $\dim M - \dim G$, and it has a unique smooth structure for which the quotient map $\pi : M \to M/G$ is a smooth submersion.*

Registry contract (verbatim).

```
Quotient manifold theorem: if a Lie group G acts smoothly, freely, and properly on a smooth
manifold M, then the orbit space M/G is a topological manifold of dimension dim M - dim G with a
unique smooth structure for which the quotient map pi:M->M/G is a smooth submersion.
```

Proof (prose account of the af-validated tree). The tree is a direct application of Lee's Quotient Manifold Theorem. Its registered external has precisely the smoothness, freeness and properness hypotheses above and returns the orbit-space dimension, the unique smooth structure and the submersion property in one step. The only elaboration is documentary: the extracted source writes the orbit notation, minus sign and quotient-map symbol with OCR control glyphs. The validated children identify those glyphs with M/G, $\dim M - \dim G$ and $\pi : M \to M/G$, while leaving the byte-matched external unchanged. Three verifier challenges observed that merely naming the theorem did not yet place it in the formal claim context. The repaired leaf therefore invokes the registered citation alias explicitly and records why the interface's empty optional context display does not cancel that binding. With the external formally in scope, its conclusion is exactly the registry contract. The resulting four-node tree is taint clean, and the root adds no new assertion. This is precisely the direct-source leaf recorded in the export. No further geometric argument or additional premise is used.

Lemma 177 (`lem-topology-finite-triangulation`). *Every compact smooth manifold without boundary is homeomorphic to the realization of a finite simplicial complex.*

Registry contract (verbatim).

```
Finite triangulation of compact smooth manifolds: every compact smooth manifold without boundary is
homeomorphic to a finite simplicial complex.
```

Proof (prose account of the af-validated tree). Let M satisfy the hypotheses. Smoothness implies the C^1 regularity used in the registered Munkres theorem, while the pinned terminology external identifies absence of boundary with "non-bounded." Munkres therefore supplies a simplicial complex K and a C^1 triangulation $f : |K| \to M$. The two registered definition excerpts make the important next step literal: such a triangulation is a homeomorphism onto M. Hence $|K|$ is compact, by applying the continuous inverse of f to compact M. For each vertex v, the live tree takes the union of the relative interiors of the simplices containing v; these open stars cover $|K|$, and a vertex w belongs to the star centered at v exactly when $w = v$. A finite subcover therefore captures every vertex among finitely many centers. There are then only finitely many simplices, since each is a finite subset of the finite vertex set. Thus K is finite. The validated workspace is the clean re-seeded Munkres route: the earlier Cairns route was retired because its missing Veblen–Whitehead definitions could not be grounded locally, and a later corrupted-locus re-seed was discarded rather than inherited.

Lemma 178 (`lem-topology-lefschetz-hopf`). *Let $f : X \to X$ be a map of a finite polyhedron with finitely many fixed points, each lying in a maximal simplex of X. Then*

$$L(f) = \sum_{x \in \mathrm{Fix}(f)} \mathrm{ind}(f, x).$$

Registry contract (verbatim).

```
Lefschetz-Hopf formula (maximal-simplex form): if f:X->X is a map of a finite polyhedron with a
finite set of fixed points, each of which lies in a maximal simplex of X, then the Lefschetz number
L(f) is the sum of the indices of all the fixed points of f.
```

Proof (prose account of the af-validated tree). The two-node tree applies Arkowitz–Brown, Theorem 1.2, verbatim. The registered external assumes a self-map of a finite polyhedron, a finite fixed set, and placement of every fixed point in a maximal simplex. Its conclusion is exactly the displayed sum of the local indices, with the notation of Definition 27. The proof tree changes no hypothesis and introduces no intermediate construction. In particular, the maximal-simplex condition is retained: the earlier design wording with only isolated fixed points was broader than the pinned theorem and was not validated. The sole child quotes the registered source statement literally. The root then takes that child by direct theorem application. There is no deformation, approximation or index comparison inside this row. The validated conclusion is therefore the exact pinned source statement. Thus any downstream use must establish maximal-simplex placement separately; this row does not infer it from finiteness or isolation.

Lemma 179 (`lem-topology-local-index-sign`). *Let x be an isolated fixed point of a smooth self-map f of a compact orientable manifold. If* $\det(I - Df_x) \neq 0$*, then*

$$\mathrm{ind}(f, x) = \operatorname{sgn} \det(I - Df_x).$$

Registry contract (verbatim).

```
Nondegenerate local fixed-point index: if x is an isolated fixed point of a smooth self-map f of a
compact orientable manifold with det(I-Df_x) != 0, then its local fixed-point index is sgn
det(I-Df_x).
```

Proof (prose account of the af-validated tree). The decisive live leaf applies the nondegenerate clause of Definition 27 directly: the smooth compact orientable scope and the determinant hypothesis are exactly its registered hypotheses, and 1 and I denote the same tangent-space identity. The remaining live branches audit the classical local calculation. Writing $A = Df_x$, invertibility of $I - A$ excludes the eigenvalue 1. After shrinking one chart ball so that its closed ball remains in the coordinate domain, the Leray–Schauder formula reduces $J(f, x)$ to $J(A, 0)$. For $h = I - A$, the

Brouwer-degree theorem gives $d(h, V) = \operatorname{sgn}\det(I - A)$. Independently, the linear Leray–Schauder formula gives $J(A, 0) = (-1)^\beta$, where β counts characteristic values in $(0, 1)$. Reciprocation identifies these with real eigenvalues of A larger than 1; conjugate nonreal pairs contribute positively to $\det(I - A)$, so $(-1)^\beta = \operatorname{sgn}\det(I - A)$. Verification repaired nine challenges concerning chart compactness, an out-of-scope use on a noncompact vector space, cutoff inequalities and unsupported locality bridges. The final scope was user-ratified from the original unqualified C^1 wording to the smooth compact orientable contract quoted above.

Lemma 180 (`lem-topology-orientable-top-cohomology`). *If M is a connected compact orientable d-manifold without boundary, then $H^d(M; \mathbb{R}) \neq 0$.*

Registry contract (verbatim).

```
Top cohomology of a closed orientable manifold: if M is a connected compact orientable d-manifold
without boundary, then H^d(M;R) != 0.
```

Proof (prose account of the af-validated tree). Compactness without boundary is the "closed" hypothesis in the registered Hatcher theorem. The live tree first obtains real orientability without importing an unregistered coefficient-change result. Hatcher's registered top-cohomology corollary supplies a nonzero class in $H^d(M; \mathbb{R})$; the field-coefficient universal coefficient theorem then forces $H_d(M; \mathbb{R}) \neq 0$. If M were not $\mathbb{R}$-orientable, Theorem 3.26(b) would inject this group into $\mathbb{R}$ with image $\{r : 2r = 0\} = \{0\}$, a contradiction. Theorem 3.26(a) now gives $H_d(M; \mathbb{R}) \cong \mathbb{R}$. Applying the same universal coefficient theorem yields

$$H^d(M; \mathbb{R}) \cong \operatorname{Hom}_{\mathbb{R}}(H_d(M; \mathbb{R}), \mathbb{R}),$$

and the right-hand side contains any chosen isomorphism $H_d(M; \mathbb{R}) \to \mathbb{R}$, hence is nonzero. A verifier challenge rejected the tree's first change-of-coefficients shortcut; the preceding Corollary–UCT–Theorem route is the validated replacement.

Lemma 181 (`lem-topology-kunneth-cross-product`). *Let R be a commutative coefficient ring and let X, Y be CW complexes such that each $H^k(Y; R)$ is a finitely generated free R-module. Then the cross product is a ring isomorphism*

$$H^*(X; R) \otimes_R H^*(Y; R) \xrightarrow{\cong} H^*(X \times Y; R).$$

In particular, this holds over $R = \mathbb{R}$ for finite-CW spaces with finite-dimensional cohomology.

Registry contract (verbatim).

```
Cohomological Kunneth isomorphism over R: for CW complexes X and Y with each H^k(Y;R) a finitely
generated free R-module, the cross product H*(X;R) (x)_R H*(Y;R) -> H*(X x Y;R) is an isomorphism
of rings; in particular this holds over the field R = reals for finite-CW spaces with
finite-dimensional cohomology.
```

Proof (prose account of the af-validated tree). Hatcher's registered Theorem 3.15 states the first assertion for CW complexes over a commutative coefficient ring when every $H^k(Y;R)$ is finitely generated and free. The live tree applies that theorem directly. The commutative-ring wording records the scope correction forced by verification, since the provisional "arbitrary ring" reading was broader than the source. For a noncommutative ring the displayed tensor product of two left modules need not even be available in the asserted form. For the special case, a finite-CW space is in particular a CW complex. Over the field $\mathbb{R}$, every finite-dimensional cohomology group has a finite basis, hence is a finitely generated free $\mathbb{R}$-module. These are exactly the hypotheses of the general clause, so its cross product gives the stated ring isomorphism for the finite-CW pair. The final tree node is this specialization by modus ponens. No finiteness conclusion beyond those explicit hypotheses, and no assertion for a noncommutative coefficient ring, is made.

Lemma 182 (`lem-topology-hopf-structure`)**.** *Every finite-dimensional connected graded-commutative bialgebra over a field of characteristic zero is an exterior algebra on odd-degree homogeneous generators, where "connected graded" uses the nonnegative grading convention.*

Registry contract (verbatim).

```
Hopf structure theorem in the form consumed by Stage 1: a finite-dimensional connected
graded-commutative bialgebra over a characteristic-zero field is an exterior algebra on odd-degree
homogeneous generators.
```

Proof (prose account of the af-validated tree). Let $A = \bigoplus_{n\geq 0} A_n$ be the bialgebra over the characteristic-zero field F. Connectedness gives $A_0 = F1$. For homogeneous $a \in A_n$, $n > 0$, degree preservation and the counit identities force the coproduct decomposition

$$\Delta a = a \otimes 1 + 1 \otimes a + \sum_i a_i' \otimes a_i'', \qquad |a_i'|, |a_i''| > 0.$$

Thus the hypotheses match Hatcher's registered Hopf-algebra convention, and total finite-dimensionality makes every graded piece finite-dimensional. The structure theorem gives an algebra isomorphism $A \cong \Lambda_F(V_{\text{odd}}) \otimes_F F[V_{\text{even}}]$. If V_{even} contained a polynomial generator x, the independent monomials $1, x, x^2, \ldots$ would make its polynomial algebra infinite-dimensional. The map $p \mapsto 1 \otimes p$ is injective, split by the exterior augmentation, so the tensor product and hence A would also be infinite-dimensional. Therefore the polynomial factor has no generators and $A \cong \Lambda_F(V_{\text{odd}})$. Verification repaired one scope gap by making explicit that "connected graded" carries the source's nonnegative convention; negatively graded counterexamples are not claimed.

65 The Stage-1 quotient package: uniform inversion isolation, the quotient manifold package, finite CW structure, quotient left inversion, and quotient index data

This section reproduces the five `af`-validated rows that pass from the smooth Stage-1 unitary space of Definition 32 to its scalar quotient. They isolate inversion before quotienting, establish the quotient's manifold and finite-CW structure, descend the H-space operations, and finally determine the local index of the scalar fixed class. The ambient approximate algebra is that of Definition 22.

Lemma 183 (`lem-stage1-uniform-inversion-isolation`)**.** *There are universal constants $e^r_{\mathrm{iso}} > 0$ and $r_{\mathrm{iso}} > 0$ such that, for every finite-dimensional exact-unit ε_r-C^*-algebra with $0 \le \varepsilon_r \le e^r_{\mathrm{iso}}$, the elements J and $-J$ are the only fixed points of the smooth map σ in their respective ambient r_{iso}-balls.*

Registry contract (verbatim).

```
There are universal e_iso^r > 0, r_iso > 0 such that, for every finite-dimensional exact-unit
epsilon_r-C*-algebra with 0 <= epsilon_r <= e_iso^r, J and -J are the only fixed points of the
smooth sigma in their respective ambient r_iso-balls.
```

Proof (prose account of the af-validated tree). Fix the universal tuple W of the Stage-1 constant ledger and set $e^r_{\mathrm{iso}} = C_{\mathrm{rect}} e_{S1}$ and $\rho = r_{\mathrm{iso}}$. For $\varepsilon_r \le e^r_{\mathrm{iso}}$, put $\varepsilon_X = \varepsilon_r / C_{\mathrm{rect}}$. Clause (R) then supplies every guard in (A_7) at $\delta = \delta_*$ and $r = \rho$, including $c = C_{\mathrm{der}}(\varepsilon_r + \rho) \le \kappa_{\mathrm{der}}/4 < 1$. The ledger charts, the smooth atlas and smooth polar inverse synchronize one smooth map $\sigma(U) = u_{\delta_*}(U^\dagger)$; scalar covariance shows that it fixes both J and $-J$. For $s \in \{+1, -1\}$ let F_s be the retained same-chart map from (A_7) and set $G_s = F_s - I$ on $B^{i\mathcal{H}}_\rho(0)$. The derivative estimate reads $\|DG_s(A) + 2I\| \le c$, so with $V = -2I$ one has $\|V^{-1} DG_s(A) - I\| \le c/2 < 1$. Lemma 89 therefore makes G_s injective. At the chart centre, $f_{sJ}(0) = 0$ and graph uniqueness gives $g_{sJ}(0) = 0$; hence $G_s(0) = 0$, making zero its unique zero in the chart ball. If $\sigma(U) = U$ and U lies in the ambient ρ-ball about sJ, decompose $\phi_{sJ}(U)$ into anti-Hermitian and Hermitian parts. Their norms are below ρ, and the unitary equation plus graph uniqueness writes $U = \chi_s(A)$. Chart retention and fixedness give $G_s(A) = 0$, so $A = 0$, then $g_{sJ}(0) = 0$ yields $U = sJ$. The single verifier challenge was proof-level: the negative-chart branch had an undeclared dependency, repaired by inserting the dependency-backed bridge now live as node 1.5.1; the contract was unchanged.

Lemma 184 (`lem-stage1-quotient-manifold-package`)**.** *There is a universal $e^r_{\mathrm{quot}} > 0$ such that, whenever $0 \le \varepsilon_r \le e^r_{\mathrm{quot}}$ and $1 < N = \dim_{\mathbb{C}} \mathcal{X} < \infty$, the scalar quotient $\breve{\mathcal{U}} = \mathcal{U}_e/U(1)$ is a connected compact orientable smooth manifold without boundary of real dimension $N - 1$.*

Registry contract (verbatim).

```
There is a universal e_quot^r > 0 such that, for every finite-dimensional exact-unit
epsilon_r-C*-algebra with 0 <= epsilon_r <= e_quot^r and 1 < N = dim_C calX < infinity, breve-calU
= calU_e/U(1) is a connected compact orientable smooth manifold without boundary of real dimension
N - 1.
```

Proof (prose account of the af-validated tree). From the universal ledger tuple choose $e^r_{\text{quot}} = \min\{\varepsilon^r_*, \frac{1}{4}, \kappa_{\text{ch}}/(4C_{\text{ch}})\}$ and $\delta = \kappa_{\text{ch}}/(4C_{\text{ch}})$. The graph clause, the smooth-atlas upgrade and the Maurer–Cartan trivialization then make $\mathcal{U}$ a smooth boundaryless manifold with globally trivial tangent bundle. The tree proves locally, without importing an extra smooth-operations row, that $a(c, U) = cU$ is a smooth action on the identity component $\mathcal{U}_e$: the unitary equations and the right inverse $\bar{c}R$ preserve the space, while the path $t \mapsto e^{it\theta}J$ and multiplication by c preserve its component. The action is free because $cU = U$ and $U \cdot R = J$ imply $cJ = J$, hence $c = 1$; it is proper by compactness of $U(1)$ and the closed-subset criterion. Lemma 176 now gives the smooth boundaryless quotient. The Maurer–Cartan map identifies $T_U\mathcal{U}$ with $i\mathcal{H}$, a real space of dimension N, so quotienting the one-dimensional scalar orbit gives dimension $N-1$. Connectedness is inherited from the continuous image of the component $\mathcal{U}_e$. For compactness, $(1-\varepsilon_r)\|U\|^2 \le 1$ bounds the unitary locus, while $U \cdot X = 0$ implies $\|X\| \le \varepsilon_r\|U\|^2\|X\|$; the coefficient is below one, so L_U is injective and hence surjective in finite dimension. Thus $\mathcal{U}$ is a closed zero locus in a bounded set, its component is compact, and so is its quotient. Finally equivariance of the Maurer–Cartan trivialization and $\omega_U(iU) = iJ$ descend it to $T\breve{\mathcal{U}} \cong \breve{\mathcal{U}} \times (i\mathcal{H}/\mathbb{R}iJ)$, which supplies an orientation. This first-pass tree recorded no challenges.

Lemma 185 (`lem-stage1-quotient-finite-cw`)**.** *For every finite-dimensional exact-unit ε_r-C^*-algebra, if $\breve{\mathcal{U}} = \mathcal{U}_e/U(1)$ is a compact smooth manifold without boundary, then $\breve{\mathcal{U}}$ is homeomorphic to a finite simplicial complex and hence has finite CW type.*

Registry contract (verbatim).

```
For every finite-dimensional exact-unit epsilon_r-C*-algebra, if breve-calU = calU_e/U(1) is a
compact smooth manifold without boundary, then breve-calU is homeomorphic to a finite simplicial
complex and hence has finite CW type.
```

Proof (prose account of the af-validated tree). Fix the algebra and write $M = \breve{\mathcal{U}}$. The root assumes directly that M is a compact smooth manifold without boundary. In the intended small-error range this antecedent is supplied by Lemma 184 when $N > 1$. For $N = 1$, exact unitality instead gives $\mathcal{X} = \mathbb{C}J$, $(zJ)^\dagger = \bar{z}J$ and $(zJ) \cdot (wJ) = zwJ$. Consequently $\mathcal{U} = \mathcal{U}_e = U(1)J$ and M is a single point, again a compact smooth manifold without boundary. Only the stated root antecedent is used from this point onward. Lemma 177 supplies a finite simplicial complex K and a homeomorphism $h : |K| \to M$. Thus M is homeomorphic to a finite simplicial complex, the first conclusion. The realization $|K|$ is a finite CW complex, with one open cell for the relative interior of each simplex and the usual face attachments. Transporting this CW structure along h makes M a finite CW complex; equivalently, h exhibits M as having finite CW type. This was a first-pass four-node validation with no challenges.

Lemma 186 (`lem-stage1-quotient-left-inversion`)**.** *There is a universal $e^r_H > 0$ such that, for every finite-dimensional exact-unit ε_r-C^*-algebra with $0 \le \varepsilon_r \le e^r_H$, the scalar-equivariant maps μ, σ and the jointly continuous projected straight paths descend to $\breve{\mathcal{U}}$; the descended multiplication makes it a connected H-space, and the descended smooth map $\breve{\sigma}$ is a left inversion.*

Registry contract (verbatim).

```
There is a universal e_H^r > 0 such that, for every finite-dimensional exact-unit
epsilon_r-C*-algebra with 0 <= epsilon_r <= e_H^r, the scalar-equivariant mu, sigma and the jointly
```

```
continuous projected straight paths descend to breve-calU; the descended multiplication makes it a
connected H-space, and the descended smooth map breve-sigma is a left inversion.
```

Proof (prose account of the af-validated tree). Fix the ledger tuple and set $e^r_H = \min\{C_{\mathrm{rect}} e_{S1}, e^r_{\mathrm{quot}}\}$, $q = C_{\mathrm{grp}}\varepsilon_r$ and $\eta = C_{\mathrm{path}}(q + \varepsilon_r q + q^2)$. Clause (R) supplies the guards for (A_5) and (A_6), including $q < r_-$, $C_{\mathrm{path}} q \le \frac{1}{4}$ and $\eta < r_-$. Thus (A_5) gives global $\mu(U, V) = u_{\delta_*}(U \cdot V)$ and $\sigma(U) = u_{\delta_*}(U^\dagger)$ with exact unit identities and $\|\mu(\sigma(U), U) - J\| \le q$; (A_6) gives the jointly continuous paths $H(t, U_0, U_1) = u_{\delta_*}((1-t)U_0 + tU_1)$. The atlas and polar upgrades identify these same maps as smooth, and the explicit operations row gives $\mu(cU, dV) = cd\mu(U, V)$ and $\sigma(cU) = \bar{c}\sigma(U)$. Their connected images remain in $\mathcal{U}_e$, so these covariance identities define $\breve{\mu}([U], [V]) = [\mu(U, V)]$, $\breve{\sigma}([U]) = [\sigma(U)]$ and the descended paths. The exact identities at J make both unit maps literally the identity, hence $(\breve{\mathcal{U}}, [J], \breve{\mu})$ is an H-space in the sense of Definition 25; it is connected as the quotient of $\mathcal{U}_e$. The verifier challenged the claimed smooth descent, so the repaired live tree constructs local slices explicitly: for a complement E_U to $\mathbb{R}(iU)$, $(\theta, v) \mapsto e^{i\theta}\psi(v)$ has invertible derivative at $(0, 0)$. Freeness and compactness of $U(1)$ let one shrink the slice, producing quotient charts, a smooth submersion and smooth local sections; there $\breve{\sigma} = p \circ \sigma \circ s_U$ is smooth. Finally $A(U) = \mu(\sigma(U), U)$ is phase invariant and lies within q of J. The homotopy $[H(t, A(U), J)]$ is therefore well-defined, basepoint-preserving, and joins $x \mapsto \breve{\mu}(\breve{\sigma}(x), x)$ to the constant $[J]$, proving left inversion.

Lemma 187 (`lem-stage1-quotient-inversion-index-data`)**.** *There is a universal $e^r_{\mathrm{idx}} > 0$ such that, whenever $0 \le \varepsilon_r \le e^r_{\mathrm{idx}}$ and $1 < N = \dim_{\mathbb{C}} \mathcal{X} < \infty$, $\breve{e} = [J]$ is an isolated fixed point of the smooth $\breve{\sigma}$, the line $i\mathbb{R}J$ is $D\sigma_J$-invariant, $\|D\breve{\sigma}_{\breve{e}} + I\| < 1$ in the quotient norm, and $\det(I - D\breve{\sigma}_{\breve{e}}) > 0$, so its local index is $+1$. More precisely, there is a quotient neighborhood $\mathcal{N}$ of $[J]$ such that a fixed $[U] \in \mathcal{N}$ has a representative U_0 close to J with $\sigma(U_0) = cU_0$; choosing $a \in U(1)$ with $a^2 = c$ gives $\sigma(aU_0) = \bar{a}\sigma(U_0) = aU_0$; the two actual fixed lifts $\pm aU_0$ lie in the respective isolation balls, hence equal $\pm J$, and therefore $[U] = [J]$.*

Registry contract (verbatim).

```
There is a universal e_idx^r > 0 such that, for every finite-dimensional exact-unit
epsilon_r-C*-algebra with 0 <= epsilon_r <= e_idx^r and 1 < N = dim_C calX < infinity, the scalar
class breve-e = [J] is an isolated fixed point of the smooth breve-sigma, the vertical line iR*J is
D-sigma_J-invariant, ||D-breve-sigma_{breve-e} + I|| < 1 in the quotient norm, and det(I -
D-breve-sigma_{breve-e}) > 0, so its local index is +1; more precisely, there is a quotient
neighborhood calN of [J] such that if [U] in calN is fixed, choose a representative U_0 close to J
and c in U(1) with sigma(U_0) = c*U_0, choose a in U(1) with a^2 = c, and use sigma(a*U_0) =
conj(a)*sigma(U_0) = a*U_0: the two actual fixed lifts +-a*U_0 lie in the J- and -J-isolation
balls, hence equal J and -J, so [U] = [J].
```

Proof (prose account of the af-validated tree). Choose $e^r_{\mathrm{idx}} = \min\{C_{\mathrm{rect}} e_{S1}, e^r_{\mathrm{iso}}, e^r_H, e^r_{\mathrm{quot}}\}$ from the four validated packages. The explicit minimum, inserted after a verifier challenge, proves that every isolation, H-space and quotient cutoff applies simultaneously. They synchronize one smooth scalar-equivariant σ and the smooth compact orientable manifold $M = \breve{\mathcal{U}}$ with fixed point $\breve{e} = [J]$. At J, the scalar orbit has tangent $i\mathbb{R}J$. The second repaired derivative subbranch builds local slice charts, in which the quotient map is $(t, k) \mapsto k$; hence $T_{\breve{e}}M \cong T_J\mathcal{U}_e/(i\mathbb{R}J)$ and differentiating $p \circ \sigma = \breve{\sigma} \circ p$ identifies the descended differential. Covariance gives $D\sigma_J(iJ) = -iJ$, so the vertical line is invariant. In the ledger graph coordinates, (A_7) gives $\|DF_+(0) + I\| \leq C_{\mathrm{der}}(\varepsilon_r + r_{\mathrm{iso}}) < 1$. Passing to the quotient norm cannot increase this bound, yielding $\|D\breve{\sigma}_{\breve{e}} + I\| < 1$. For $T = D\breve{\sigma}_{\breve{e}}$ put $B = (T + I)/2$; then $\|B\| < 1/2$, so every $I - tB$ is invertible and determinant continuity gives $\det(I - T) = 2^{\dim M} \det(I - B) > 0$. For isolation, shrink a quotient neighborhood so a representative U_0 is close to J. If its class is fixed, $\sigma(U_0) = cU_0$, and continuity makes c close to 1. Choose the square root a close to 1; covariance makes both aU_0 and $-aU_0$ actual fixed points lying in the J- and $-J$-isolation balls. Lemma 183 forces them to be J and $-J$, so the fixed class is $[J]$. Finally the positive determinant and Lemma 179 give local index $+1$ in the sense of Definition 27.

66 Route-F raw factors: setting formation, factor norms, factor identities, the product estimate, and factor units

This section forms the raw Route-F factorization datum and records its first four consequences. Definition 46 supplies the vocabulary; the first lemma supplies existence, and the remaining rows keep its one global scalar header fixed.

Lemma 188 (`lem-routef-raw-factor-setting-formation`). *There is one scalar header W_{RF}, independent of all input, dimension, amplification, and block data, with $C_\theta = 12(\sqrt{2}-1)$, $C_A = 20 + \frac{211}{8}C_\theta$, and fixed witnesses $\eta_A > 0$, $C_E < \infty$, $\varepsilon_E > 0$ from Lemmas 95 and 214. Put $\rho_\theta = \frac{1}{8}$, $\rho_{\mathrm{AI}} = \eta_A$, and define the remaining scalars by (1.1)–(1.8) of that definition. If $\mathcal{H} \neq 0$ is finite-dimensional, $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ is UCP, and*

$$0 \le \eta \le \rho_{\mathrm{id}}^{\mathrm{corr}}, \qquad \|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta,$$

then there are a finite-dimensional unital C^*-algebra B, an extended $C_E\varepsilon_{\mathrm{AI}}(\eta)$-isomorphism $v : B \to \mathcal{A}$, and a setting datum S over this same header, with the canonical fields and $u = v^{-1}$, such that $\widetilde{\Phi}^2 = \widetilde{\Phi}$, $\mathcal{A}$ is an extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$-algebra, and $0 \le \varepsilon_{\mathrm{AI}}(\eta) \le C_A\eta \le \varepsilon_E$.

Registry contract (verbatim).

```
Route F raw-factor setting formation: there exists one choice W_RF of the scalar header of
def-routef-raw-factor-setting, independent of H, Phi, eta, dimension, amplification level, and
block data, with C_theta=12*(sqrt(2)-1), C_A=20+(211/8)*C_theta, eta_A>0 and (C_A,eta_A) the fixed
witnesses of lem-routef-ai-defect-linearization, C_E<infinity and epsilon_E>0 the fixed witnesses
of lem-thmainext-conditional, rho_theta:=1/8, rho_AI:=eta_A, and all remaining named scalar
quantities defined by (1.1)-(1.8), such that for every nonzero finite-dimensional Hilbert space H,
every UCP map Phi:B(H)->B(H), and every eta with 0 <= eta <= rho_id^corr and ||Phi^2-Phi||_cb <=
eta, there exist a finite-dimensional unital C*-algebra B, an extended
C_E*epsilon_AI(eta)-isomorphism v:B->A, and a def-routef-raw-factor-setting datum S over this same
W_RF whose fields are the displayed H,Phi,eta,B,v,u=v^(-1) and the canonical
tilde-Phi,A,star,epsilon_AI(eta),tilde-Delta,tilde-Upsilon notation, with tilde-Phi^2=tilde-Phi, A
an extended epsilon_AI(eta)-C*-algebra, and 0 <= epsilon_AI(eta) <= C_A*eta <= epsilon_E.
```

Proof (prose account of the af-validated tree). Choose once (C_A, η_A) from Lemma 95 and (C_E, ε_E) from Lemma 214, before every input quantifier. The tree assembles W_{RF} in order (1.1)–(1.8); its denominators are positive because $C_A = 20 + \frac{211}{8}\,12(\sqrt{2}-1) > 0$, $\eta_A, \varepsilon_E > 0$, and $\overline{C}_E = \max\{1, C_E\} \ge 1$, so the header is universal and well typed.

For admissible $(\mathcal{H}, \Phi, \eta)$, the corrected radius gives $\eta \le \frac{1}{8} < \frac{1}{4}$, $\eta \le \eta_A$, and $C_A\eta \le \varepsilon_E$. Lemma 93 makes the canonical $\widetilde{\Phi}$ idempotent; Lemma 95 gives the extended $\varepsilon_{\mathrm{AI}}(\eta)$-$C^*$ structure and its nonnegative defect with $\varepsilon_{\mathrm{AI}}(\eta) \le C_A\eta$. Since $B(\mathcal{H})$ is finite-dimensional, so is $\mathcal{A} = \mathrm{Im}(\widetilde{\Phi})$.

Lemma 214 now supplies B and the extended $C_E\varepsilon_{\mathrm{AI}}(\eta)$-isomorphism $v : B \to \mathcal{A}$. Bijectivity gives $u = v^{-1}$, and the canonical compositions package these outputs into S over the fixed header without reselecting a global witness. Generalization closes the quantifier order. The registry records this as the user-ratified 2026-08-05 setting-rescope formation, not as a conclusion inferred from the data definition alone.

Lemma 189 (`lem-routef-raw-factor-norms`). *Fix the global W_{RF} and, for an admissible input, a supplied datum S. For $n \ge 1$, $X \in M_n(B)$, and $0 \le \eta \le \rho_T$, the scalars of* (1.1) *satisfy*

$$(1 - C_V\eta)\|X\| \le \|\widetilde{\Delta}_n X\| \le (1 + C_V\eta)\|X\|, \qquad \max\{\|\widetilde{\Delta}\|_{\mathrm{cb}}, \|\widetilde{\Upsilon}\|_{\mathrm{cb}}\} \le 1 + C_T\eta.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result; for every integer n >= 1 and every X in M_n(S.B), writing the fields of (W_RF,S) as the
unqualified symbols below: Raw factor-map norms: with C_V, C_T, rho_T from (1.1), for 0 <= eta <=
rho_T, every amplification satisfies (1-C_V*eta)*||X|| <= ||tilde-Delta_n X|| <= (1+C_V*eta)*||X||
and max{||tilde-Delta||_cb, ||tilde-Upsilon||_cb} <= 1+C_T*eta.
```

Proof (prose account of the af-validated tree). Put $d = C_E\varepsilon_{\mathrm{AI}}(\eta)$. Formation and linearization give $0 \le d \le \overline{C}_E C_A \eta = C_V\eta$, and the conditional structure theorem gives the extended d-isomorphism v carried by S. At every amplification, Definition 36 gives $(1-d)\|X\| \le \|v_n X\| \le (1+d)\|X\|$. Since $\widetilde{\Delta}$ is v followed by the isometric inclusion into $B(\mathcal{H})$, enlarging d to $C_V\eta$ proves the two-sided factor estimate and $\|\widetilde{\Delta}\|_{\mathrm{cb}} \le 1 + C_V\eta$.

For the inverse branch, amplified Kadison–Schwarz gives $\|\Phi\|_{\mathrm{cb}} \le 1$. Because $\rho_T \le \rho_\theta = \frac{1}{8}$, Lemma 94 yields $\|\widetilde{\Phi}\|_{\mathrm{cb}} \le 1 + C_\theta\eta$. The lower bound for every bijective v_n, together with $C_V\eta \le \frac{1}{4}$, gives $\|v_n^{-1}\| \le (1 - C_V\eta)^{-1}$. Thus

$$\|\widetilde{\Upsilon}\|_{\mathrm{cb}} \le \frac{1 + C_\theta\eta}{1 - C_V\eta}.$$

Writing $a = C_\theta\eta$, $b = C_V\eta$, the radius guards give $0 \le a, b \le \frac{1}{4}$ and $1 + a + 3b - (1+a)/(1-b) = b(2-a-3b)/(1-b) \ge 0$. Thus the quotient is at most $1 + (C_\theta + 3C_V)\eta = 1 + C_T\eta$, and $C_V \le C_T$ finishes the estimate. The validated repairs made the ambient hypotheses and dependency children explicit, and replaced a false strict dimension bound by the correct non-strict rank bound.

Lemma 190 (`lem-routef-raw-factor-identities`). *With the same fixed-header convention, for* $0 \le \eta \le \rho_{\mathrm{id}}^{\mathrm{corr}} = \min\{\rho_\theta, \rho_{\mathrm{AI}}, \varepsilon_E/C_A\}$ *one has*

$$\widetilde{\Delta}\widetilde{\Upsilon} = \widetilde{\Phi}, \qquad \widetilde{\Upsilon}\widetilde{\Delta} = I_B.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, writing the fields of (W_RF,S) as the unqualified symbols below: Raw factor-map identities:
for 0 <= eta <= rho_id^corr := min{rho_theta, rho_AI, epsilon_E/C_A}, tilde-Delta tilde-Upsilon =
tilde-Phi and tilde-Upsilon tilde-Delta = I_B.
```

Proof (prose account of the af-validated tree). Formation gives $\mathcal{A} = \mathrm{Im}(\widetilde{\Phi})$, $\widetilde{\Phi}^2 = \widetilde{\Phi}$, $u = v^{-1}$, $\widetilde{\Delta} = \iota_{\mathcal{A}\subseteq B(\mathcal{H})}v$, and $\widetilde{\Upsilon} = u\widetilde{\Phi}$. Thus for $x \in B(\mathcal{H})$,

$$(\widetilde{\Delta}\widetilde{\Upsilon})(x) = \iota\, v(u(\widetilde{\Phi}(x))) = \widetilde{\Phi}(x),$$

because $\widetilde{\Phi}(x) \in \mathcal{A}$ and $vu = I_{\mathcal{A}}$. This gives the first identity. For $b \in B$, the other composition is $u(\widetilde{\Phi}(\iota v(b)))$. Every $a \in \mathcal{A}$ has the form $a = \widetilde{\Phi}(x)$, so idempotence gives $\widetilde{\Phi}(a) = a$ after the canonical inclusion: the projector restricts to $I_{\mathcal{A}}$ on its range. At $v(b)$ the expression is $u(v(b)) = b$, so extensionality yields $\widetilde{\Upsilon}\widetilde{\Delta} = I_B$. The proof uses the supplied datum throughout; the 2026-08-05 rescope changed only the ambient binding prefix that makes this datum and its fixed global header explicit.

Lemma 191 (`lem-routef-raw-product-estimate`)**.** *With the same fixed-header convention, for $n \geq 1$, $X, Y \in M_n(B)$, and $0 \leq \eta \leq \rho_{\mathrm{prod}} := \rho_T$,*

$$\|\widetilde{\Phi}_n(\widetilde{\Delta}_n X\, \widetilde{\Delta}_n Y) - \widetilde{\Delta}_n(XY)\| \leq C_T \eta \|X\| \|Y\|.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result; for every integer n >= 1 and all X, Y in M_n(S.B), writing the fields of (W_RF,S) as the
unqualified symbols below: Raw tilde-Delta-product estimate: for 0 <= eta <= rho_prod := rho_T,
every amplification and all X, Y satisfy ||tilde-Phi_n(tilde-Delta_n X tilde-Delta_n Y) -
tilde-Delta_n(XY)|| <= C_T*eta*||X||*||Y||.
```

Proof (prose account of the af-validated tree). Again set $d = C_E \varepsilon_{\mathrm{AI}}(\eta)$. Formation makes $v : B \to \mathcal{A}$ an extended d-isomorphism and gives $d \leq \overline{C}_E C_A \eta = C_V \eta$. At level n, the homomorphism clause in the underlying extended inclusion therefore says

$$\|v_n(XY) - v_n(X) \star_n v_n(Y)\| \leq d \|X\| \|Y\|.$$

Under $\mathcal{A} \subseteq B(\mathcal{H})$, $v_n(XY) = \widetilde{\Delta}_n(XY)$. Entrywise matrix multiplication, linearity of $\widetilde{\Phi}$, and $a \star b = \widetilde{\Phi}(ab)$ identify the other term with $\widetilde{\Phi}_n(\widetilde{\Delta}_n X\, \widetilde{\Delta}_n Y)$. The inherited norm is the ambient norm, so reversing the difference changes nothing. Finally $C_T = C_\theta + 3C_V \geq C_V$ gives $d \leq C_T \eta$, uniformly in n.

A verifier challenged the original branch because the permitted inputs had not made the base multiplicative-defect clause available. The repaired tree provisioned the byte-verbatim external `GT-kitaev-def-delta-homomorphism`, then rebuilt and freshly validated the two children invoking that clause. The statement itself was unchanged.

Lemma 192 (`lem-routef-raw-factor-units`)**.** *With the same fixed-header convention, for $0 \leq \eta \leq \rho_{\mathrm{unit}} := \rho_T$,*

$$\max\{\|\widetilde{\Delta}(e_B) - I\|, \|\widetilde{\Upsilon}(I) - e_B\|\} \leq C_T \eta,$$

where e_B is the unit element of B and I is the ambient unit; this is the type-forced reading of the unsubscripted units in the registry contract.

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, writing the fields of (W_RF,S) as the unqualified symbols below: Raw factor-map units: for
0 <= eta <= rho_unit := rho_T, max{||tilde-Delta(I)-I||, ||tilde-Upsilon(I)-I||} <= C_T*eta.
```

Proof (prose account of the af-validated tree). Put $b = C_V\eta$. The radius guard gives $b \le C_V/[4(1+C_V)] \le \frac{1}{4}$, while $C_T = C_\theta + 3C_V$ gives $C_V \le C_T$ and $3C_V \le C_T$. For $d = C_E\varepsilon_{\mathrm{AI}}(\eta)$, formation gives an extended d-isomorphism $v : B \to \mathcal{A}$ and $d \le \overline{C}_E C_A\eta = C_V\eta$. The ground-truth unit clause for its underlying d-homomorphism gives $\|v(e_B) - I\| \le d$. Since the inclusion of $\mathcal{A}$ is isometric,

$$\|\widetilde{\Delta}(e_B) - I\| \le C_V\eta \le C_T\eta.$$

For the inverse factor, Lemma 189 at level one gives $\|v(x)\| \ge (1-b)\|x\|$, hence $\|u\| \le (1-b)^{-1}$. The ambient unit lies in $\mathcal{A} = \mathrm{Im}(\widetilde{\Phi})$ and idempotence gives $\widetilde{\Phi}(I) = I$. Therefore

$$\widetilde{\Upsilon}(I) - e_B = u(I) - u(v(e_B)) = u(I - v(e_B)),$$

so its norm is at most $b/(1-b) \le \frac{4}{3}b \le 3C_V\eta \le C_T\eta$.

The verifier history records two repairs. Because I_B denotes the identity *map*, the proof replaced ill-typed $v(I_B)$ by $v(e_B)$ while retaining the contract's type-forced notation. The unit-defect estimate was also absent from the permitted inputs; the byte-verbatim `GT-kitaev-def-delta-homomorphism` external supplied it. An intermediate blocking child and its $v = -I$ countermodel were superseded because $-I$ violates the newly available unit and multiplicativity clauses; the repaired branches were freshly validated.

67 Route-F normalization closeness: the Delta-prime and normalization estimates, the degree-two and degree-three estimates, and the Delta-Phi product

This section follows the raw factor package of Lemmas 188– 192. Throughout, the global witness package W_{RF} is chosen before the input, and the input-dependent datum S and all unqualified fields are those of Definition 46. The five rows first make the raw factor completely positive and unital, and then propagate its linear error through products of degrees two and three; every estimate is uniform over matrix amplifications.

Lemma 193 (`lem-routef-delta-prime-closeness`). *First fix one global package W_{RF} supplied by Lemma 188. For every admissible input $(\mathcal{H}, \Phi, \eta)$, fix a datum S over that same package supplied by the same lemma and write its fields without qualification. Set $C_{\Delta'} = C_T + 4C_\theta$ and $\rho_{\Delta'} = \min\{\rho_T, \rho_{\mathrm{prod}}\}$. If $0 \le \eta \le \rho_{\Delta'}$, the repaired norm-one diagonal produces a completely positive map Δ' satisfying*

$$\|\Delta' - \widetilde{\Delta}\|_{\mathrm{cb}} \le C_{\Delta'}\eta.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, writing the fields of (W_RF,S) as the unqualified symbols below: Delta-prime CP closeness:
with C_Delta' := C_T+4*C_theta and rho_Delta' := min{rho_T, rho_prod}, for 0 <= eta <= rho_Delta',
the repaired norm-one diagonal produces a CP map Delta' with ||Delta' - tilde-Delta||_cb <=
C_Delta'*eta.
```

Proof (prose account of the af-validated tree). The tree first fixes W_{RF} and then S, and reads from the scalar header that $\eta \le \rho_T, \rho_{\mathrm{prod}}, \rho_\theta = 1/8$ and $C_V\eta \le 1/4$. Lemma 207 supplies a finite phase-balanced diagonal $D = \sum_t q_t W_t^\dagger \otimes W_t$ with unitary W_t, nonnegative q_t summing to one, exact centrality, $\pi(D) = 1_B$, and its norm-one representation. The factor $\widetilde{\Delta} = \iota \circ v$ preserves involution: v does so as the isomorphism in Definition 36, and the involution on its range is the inherited ambient adjoint. The diagonal cpization row therefore makes $\Delta'(X) = \sum_t q_t \Phi(\widetilde{\Delta}(XW_t^\dagger)\widetilde{\Delta}(W_t))$ completely positive. At level n, put $U_t = I_n \otimes W_t$, $X_t = XU_t^\dagger$, $Y_t = U_t$ and $Z_t = \widetilde{\Delta}_n(X_t)\widetilde{\Delta}_n(Y_t)$. Then $X_tY_t = X$, while the raw norm bound and $C_V\eta \le 1/4$ give $\|Z_t\| \le 4\|X\|$. Functional-calculus closeness costs $4C_\theta\eta\|X\|$, and the raw product estimate costs $C_T\eta\|X\|$. Averaging with $q_t \ge 0$ and $\sum_t q_t = 1$ yields the displayed bound at every n; taking the supremum gives the cb estimate. Before this final run, verifier challenges added the diagonal-repair and inherited-involution dependencies without changing the contract. The live tree also repaired the ill-typed $\pi(D) = I_B$ to $\pi(D) = 1_B$, the algebra unit.

Lemma 194 (`lem-routef-delta-normalization-closeness`). *With W_{RF}, S as above, let Δ' be supplied by Lemma 193, and let $X \in S.B$. Set*

$$C_\Delta = 6C_T + 7C_{\Delta'}, \qquad \rho_\Delta = \min\{\rho_{\mathrm{unit}}, \rho_{\Delta'}, [2(C_T + C_{\Delta'})]^{-1}\}.$$

For $0 \le \eta \le \rho_\Delta$, the element $a = \Delta'(I)$ is invertible and $\Delta(X) = a^{-1/2}\Delta'(X)a^{-1/2}$ is UCP, with $\|\Delta - \widetilde{\Delta}\|_{\mathrm{cb}} \le C_\Delta\eta$.

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result and for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, and for
every X in S.B, writing the fields of (W_RF,S) as the unqualified symbols below: Delta UCP
normalization: with C_Delta := 6*C_T+7*C_Delta' and rho_Delta := min{rho_unit, rho_Delta',
[2*(C_T+C_Delta')]^(-1)}, for 0 <= eta <= rho_Delta, a = Delta'(I) is invertible and Delta(X) =
a^(-1/2)*Delta'(X)*a^(-1/2) is UCP with ||Delta - tilde-Delta||_cb <= C_Delta*eta.
```

Proof (prose account of the af-validated tree). Write $D = C_{\Delta'}$ and $d = (C_T + D)\eta$. The unit estimate and the preceding closeness row give positivity of $a = \Delta'(I)$ and $\|a - I\| \le d$; the radius definition gives $0 \le d \le 1/2$. Finite-dimensional spectral calculus places the spectrum of a in $[1-d, 1+d] \subset [1/2, 3/2]$, so a is invertible. For $c = a^{1/2}$ and $b = a^{-1/2}$, $bc = cb = I$, and the scalar identity $|\sqrt{\lambda} - 1| = |\lambda - 1|/(\sqrt{\lambda} + 1)$ gives $s := \|c - I\| \le d$ and $\|c\| \le 1 + s$. Define $\Delta(X) = b\Delta'(X)b$. Conjugation by $I_q \otimes b$ preserves positivity at every matrix level, so Δ is completely positive, and $\Delta(I) = bab = I$ makes it UCP and hence completely contractive. Writing $c_q = I_q \otimes c$, the exact relation $\Delta'_q(Y) = c_q \Delta_q(Y) c_q$ gives $\|\Delta'_q(Y) - \Delta_q(Y)\| \le s(2+s)\|Y\| \le 3d\|Y\|$. Thus $\|\Delta' - \Delta\|_{\mathrm{cb}} \le 3d$, and the triangle inequality yields $(3C_T + 4D)\eta \le (6C_T + 7D)\eta = C_\Delta \eta$. The ledger amendments only made the supplied-Δ' binder, prime notation, and node dependencies explicit; they changed neither the contract nor this normalization argument.

Lemma 195 (`lem-routef-degree-two-estimate`)**.** *With $W_{\mathrm{RF}}, S, \Delta', \Delta$ supplied as above, for every $n \ge 1$ and $X, Y \in M_n(S.B)$ set $C_2 = C_{\Delta'} + 4C_\Delta$ and $\rho_2 = \min\{\rho_{\mathrm{prod}}, \rho_{\Delta'}, \rho_\Delta\}$. For $0 \le \eta \le \rho_2$,*

$$\|\Phi_n(\Delta_n X\, \Delta_n Y) - \Delta_n(XY)\| \le C_2 \eta \|X\|\|Y\|.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, and every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness; for every integer n >=
1 and all X, Y in M_n(S.B), writing the fields of (W_RF,S) as the unqualified symbols below: Route
F degree-two estimate: with C_2 := C_Delta'+4*C_Delta and rho_2 := min{rho_prod, rho_Delta',
rho_Delta}, for 0 <= eta <= rho_2, every amplification satisfies ||Phi_n(Delta_n X Delta_n Y) -
Delta_n(XY)|| <= C_2*eta*||X||*||Y||.
```

Proof (prose account of the af-validated tree). The common radius makes the raw norm, raw product, functional-calculus, and normalization estimates available and gives $C_V \eta \le 1/4$. Both Φ_n and Δ_n are contractive. First expand $\Delta_n X \Delta_n Y - \widetilde{\Delta}_n X \widetilde{\Delta}_n Y$ by replacing one factor at a time. Normalization closeness, contractivity of Δ_n, and $\|\widetilde{\Delta}_n T\| \le (1 + C_V\eta)\|T\|$ bound its image under Φ_n by $3C_\Delta \eta \|X\|\|Y\|$. Next insert $\widetilde{\Phi}_n$ at the raw product. Functional-calculus closeness costs at most $C_\theta \eta (1 + C_V \eta)^2 \|X\|\|Y\| \le 2C_\theta \eta \|X\|\|Y\|$, and the raw product row costs $C_T \eta \|X\|\|Y\|$. Finally replace $\widetilde{\Delta}_n(XY)$ by $\Delta_n(XY)$ at cost $C_\Delta \eta \|XY\| \le C_\Delta \eta \|X\|\|Y\|$. The three comparisons total $(4C_\Delta + C_T + 2C_\theta)\eta\|X\|\|Y\|$; since $C_{\Delta'} = C_T + 4C_\theta$, this is at most $(C_{\Delta'} + 4C_\Delta)\eta\|X\|\|Y\|$. The ledger amendments made the ρ_θ consequence and the final node references explicit; the contract was unchanged.

Lemma 196 (`lem-routef-delta-phi-product`)**.** *With the same supplied data, for every $n \geq 1$ and $X, Y \in M_n(S.B)$ set $\rho_{\Delta\Phi} = \min\{\rho_\theta, \rho_\Delta, \rho_2\}$. If $0 \leq \eta \leq \rho_{\Delta\Phi}$, then*

$$\|\widetilde{\Phi}_n(\Delta_n X\, \Delta_n Y) - \widetilde{\Delta}_n(XY)\| \leq (C_2 + C_\theta + C_\Delta)\eta\|X\|\|Y\|.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, and every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness; for every integer n >=
1 and all X, Y in M_n(S.B), writing the fields of (W_RF,S) as the unqualified symbols below:
Normalized Delta product: for rho_DeltaPhi := min{rho_theta, rho_Delta, rho_2} and 0 <= eta <=
rho_DeltaPhi, every amplification satisfies ||tilde-Phi_n(Delta_n X Delta_n Y) -
tilde-Delta_n(XY)|| <= (C_2+C_theta+C_Delta)*eta*||X||*||Y||.
```

Proof (prose account of the af-validated tree). The radius gives functional-calculus closeness, normalization closeness, and the preceding degree-two estimate simultaneously. Because Δ is UCP, $\|\Delta_n X \Delta_n Y\| \leq \|X\|\|Y\|$, while $\|XY\| \leq \|X\|\|Y\|$ in $M_n(S.B)$. The tree uses the exact three-term decomposition

$$\begin{aligned}\widetilde{\Phi}_n(\Delta_n X \Delta_n Y) - \widetilde{\Delta}_n(XY) &= (\widetilde{\Phi}_n - \Phi_n)(\Delta_n X \Delta_n Y)\\ &\quad + [\Phi_n(\Delta_n X \Delta_n Y) - \Delta_n(XY)] + (\Delta_n - \widetilde{\Delta}_n)(XY).\end{aligned}$$

The first term is at most $C_\theta \eta\|X\|\|Y\|$, the middle term is at most $C_2\eta\|X\|\|Y\|$, and the last is at most $C_\Delta\eta\|X\|\|Y\|$. Their sum is precisely the asserted coefficient. All maps and estimates are taken at the same arbitrary amplification, so generalization in n, X, Y completes the root. The exported tree and ledger record no challenge or amendment.

Lemma 197 (`lem-routef-degree-three-estimate`)**.** *With the same supplied data, for every $n \geq 1$ and $X, Y, Z \in M_n(S.B)$ set*

$$C_3 = 10 + 20C_\Delta + 12C_\theta + 2C_{\Delta'}, \qquad \rho_3 = \min\{\rho_\theta, \rho_{\Delta'}, \rho_\Delta, \rho_2\}.$$

For $0 \leq \eta \leq \rho_3$, $\|\Phi_n(\Delta_n X \Delta_n Y \Delta_n Z) - \Delta_n(XYZ)\| \leq C_3\eta\|X\|\|Y\|\|Z\|$.

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, and every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness; for every integer n >=
1 and all X, Y, Z in M_n(S.B), writing the fields of (W_RF,S) as the unqualified symbols below:
Route F degree-three estimate: with C_3 := 10+20*C_Delta+12*C_theta+2*C_Delta' and rho_3 :=
min{rho_theta, rho_Delta', rho_Delta, rho_2}, for 0 <= eta <= rho_3, every amplification satisfies
||Phi_n(Delta_n X Delta_n Y Delta_n Z) - Delta_n(XYZ)|| <= C_3*eta*||X||*||Y||*||Z||.
```

Proof (prose account of the af-validated tree). The tree first proves the one-factor estimate $\|\Phi_n(\Delta_n T) - \Delta_n T\| \leq d\|T\|$ with $d = 2(C_\theta + C_\Delta)\eta$. Indeed, $\widetilde{\Phi}_n\widetilde{\Delta}_n = \widetilde{\Delta}_n$ exactly, $\|\widetilde{\Delta}\|_{\mathrm{cb}} \leq 2$, and inserting both tilded maps costs $2(C_\theta + C_\Delta)\eta$. Replacing the three factors in $\Phi_n(\Delta_n X \Delta_n Y \Delta_n Z)$ by their Φ_n-images costs $3d\|X\|\|Y\|\|Z\|$. The amplified first associativity estimate from Lemma 93 then changes the bracketing at cost $10\eta\|X\|\|Y\|\|Z\|$. A two-factor telescope inside the inner Φ_n, followed by replacement of the third factor, costs a further $3d\|X\|\|Y\|\|Z\|$. Two applications of Lemma 195, first to X, Y and then to XY, Z, cost $2C_2\eta\|X\|\|Y\|\|Z\|$ and arrive at $\Delta_n(XYZ)$. Finally $6d + 10\eta + 2C_2\eta$ equals $(10 + 20C_\Delta + 12C_\theta + 2C_{\Delta'})\eta$ because $C_2 = C_{\Delta'} + 4C_\Delta$. Ledger amendments only named the intermediate terms $P_1, \ldots, P_5$ and separated this coefficient arithmetic; the contract was unchanged.

68 Route-F Upsilon rows: component construction, the left inverse, and the two closeness estimates

This section reproduces the four `af`-validated rows that build the Υ' branch from the raw-factor setting, prove its approximate left-inverse property, compare it with $\widetilde{\Upsilon}$, and normalize it to a UCP map Υ. The common ambient datum is that of Definition 46; UCP is used in the sense of Definition 45.

Lemma 198 (`lem-routef-upsilon-prime-component-construction`). *Fix first one global package W_{RF} supplied by Lemma 188, and for every admissible (H, Φ, η) fix over it a raw-factor datum S, writing its fields without qualification. For every Δ' supplied by Lemma 193 and every Δ supplied from it by Lemma 194, with C_N, C_R from (1.3), put*

$$\rho_{\Upsilon'} = \min\{\rho_T, \rho_{\mathrm{id}}, \rho_\Delta, \rho_2, \rho_3, (2C_R)^{-1}\}.$$

If $0 \le \eta \le \rho_{\Upsilon'}$, then there are $m \ge 1$, nonzero finite-dimensional L_j, E_j $(1 \le j \le m)$, a finite-dimensional F, operators $W_j : H \to L_j \otimes E_j$, and an isometry $V : H \to H \otimes F$, such that

$$B = \bigoplus_{j=1}^{m} B(L_j), \quad \sum_j W_j^* W_j = I_H, \quad \Delta(X) = \sum_j W_j^*(X_j \otimes I_{E_j})W_j, \quad \Phi(T) = V^*(T \otimes I_F)V.$$

For each j there are a nonempty finite Σ_j, unitaries U_{js}, weights $p_{js} \ge 0$ summing to one, a positive contraction C_j on E_j, and a unit vector $\xi_j \in E_j$, with

$$\sum_s p_{js}(U_{js}^* \otimes I)Z(U_{js} \otimes I) = I_{L_j} \otimes \frac{\mathrm{Tr}_{L_j}(Z)}{\dim L_j}, \qquad R_j := \sum_s p_{js}(U_{js}^* \otimes I)W_j W_j^*(U_{js} \otimes I) = I_{L_j} \otimes C_j,$$

where the partial trace is unnormalized, and $0 \le C_j \le I_{E_j}$, $\|C_j\| \ge 1 - C_R\eta$, and $1 - \langle \xi_j, C_j^* C_j \xi_j \rangle \le 2C_R\eta$. Writing $\widehat{U}_{js}$ for U_{js} in block j and zero elsewhere and $\iota_{js}(z) = U_{js}z \otimes \xi_j$, the operators and maps

$$\Lambda_j := \sum_s p_{js}(\Delta(\widehat{U}_{js}^*) \otimes I_F)VW_j^*\iota_{js}, \quad \Upsilon_j'(T) := \Lambda_j^*(\Phi(T) \otimes I_F)\Lambda_j, \quad \Upsilon'(T) := (\Upsilon_j'(T))_{j=1}^m$$

satisfy: every Λ_j is a contraction, every Υ_j' and Υ' is CP, and $\|\Upsilon'\|_{\mathrm{cb}} \le 1$.

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, writing the fields of (W_RF,S) as the unqualified symbols below: for every Delta' supplied
for (W_RF,S) by lem-routef-delta-prime-closeness and every Delta supplied from that same Delta' by
lem-routef-delta-normalization-closeness, with C_N, C_R from (1.3) and rho_Upsilon' := min{rho_T,
rho_id, rho_Delta, rho_2, rho_3, (2*C_R)^(-1)}, for 0 <= eta <= rho_Upsilon' there exist an integer
m >= 1, nonzero finite-dimensional Hilbert spaces L_j and E_j for 1 <= j <= m, a finite-dimensional
Hilbert space F, operators W_j:H->L_j tensor E_j, an isometry V:H->H tensor F, nonempty finite
index sets Sigma_j, unitaries U_js on L_j and weights p_js for s in Sigma_j, positive contractions
C_j on E_j, unit vectors xi_j in E_j, contractions Lambda_j:L_j->H tensor F, CP maps
Upsilon'_j:B(H)->B(L_j), and a CP map Upsilon':B(H)->B such that B=direct-sum_{j=1}^m B(L_j), sum_j
W_j^*W_j=I_H, Delta(X)=sum_j W_j^*(X_j tensor I_Ej)W_j for every X=(X_1,...,X_m) in B, Phi(T)=V^*(T
tensor I_F)V for every T in B(H), p_js >= 0, sum_{s in Sigma_j}p_js=1, sum_{s in
```

```
Sigma_j}p_js*(U_js^* tensor I_Ej)Z(U_js tensor I_Ej)=I_Lj tensor (Tr_Lj(Z)/dim(L_j)) for every Z in
B(L_j tensor E_j), where Tr_Lj denotes the unnormalized partial trace over L_j, R_j:=sum_{s in
Sigma_j}p_js*(U_js^* tensor I_Ej)W_jW_j^*(U_js tensor I_Ej)=I_Lj tensor C_j, 0 <= C_j <= I_Ej,
||C_j|| >= 1-C_R*eta, 1-<xi_j,C_j^*C_j xi_j> <= 2*C_R*eta, if hat-U_js denotes U_js in the j-th
block of B and zero in every other block and iota_js(z):=U_js z tensor xi_j then Lambda_j:=sum_{s
in Sigma_j}p_js*(Delta(hat-U_js^*) tensor I_F)V W_j^*iota_js, Upsilon'_j(T):=Lambda_j^*(Phi(T)
tensor I_F)Lambda_j, Upsilon'(T):=(Upsilon'_j(T))_{j=1}^m, and ||Upsilon'||_cb <= 1.
```

Proof (prose account of the af-validated tree). The tree first applies finite-dimensional C^*-algebra structure and canonical Stinespring dilation. It writes B as the displayed direct sum, decomposes the representation as $L_j \otimes E_j$, and obtains the W_j formula; the same matrix-unit construction for $B(H)$ gives F, V and the formula for Φ. For $d_j = \dim L_j$, it chooses the explicit Weyl family $\Sigma_j = \{0, \dots, d_j - 1\}^2$, equal weights d_j^{-2}, and $U_j(a,b)e_k = e^{2\pi i a k/d_j} e_{k+b}$. Summing in a removes off-diagonal matrix units, and summing in b averages the diagonal ones, yielding the partial-trace twirl. Applied to $W_j W_j^*$, this gives $R_j = I_{L_j} \otimes C_j$ with $0 \le C_j \le I$.

For the lower bound the tree identifies $W_j^* R_j W_j = \sum_s p_{js} \Delta(\widehat{U}_{js}^*) \Delta(\widehat{U}_{js})$. The raw-factor norm and Δ-normalization bounds give $\|\Delta(e_j)\| \ge 1 - C_N \eta$; degree two and contractivity of Φ give $\|C_j\| \ge 1 - (C_N + C_2)\eta = 1 - C_R \eta$. The cutoff $(2C_R)^{-1}$ makes this at least $1/2$, proving $E_j \ne 0$ before choosing ξ_j. Norm attainment gives $\|C_j \xi_j\| = \|C_j\|$ and hence $1 - \|C_j\|^2 \le 2C_R \eta$. Finally the summands defining Λ_j are typed and contractive, so $\|\Lambda_j\| \le 1$. Tensoring, compression, and direct sums preserve CP; the amplified direct-sum norm is a maximum, so $\|\Upsilon'\|_{\mathrm{cb}} \le 1$ without an m- or amplification-factor. This is the construction factor introduced after the original monolithic row exceeded the registry's brittleness cap.

Lemma 199 (`lem-routef-upsilon-prime-left-inverse`). *In the setting of Lemma 198, put $C_L = C_2 + C_3 + 2C_R$ and use the same $\rho_{\Upsilon'}$. For every $q \ge 1$, $Y = (Y_1, \dots, Y_m) \in M_q(B)$, and j,*

$$\|(\Upsilon'_j)_q(\Delta_q(Y)) - Y_j\| \le C_L \eta \|Y\|.$$

Consequently $\|(\Upsilon'\Delta - I_B)_q(Y)\| \le C_L \eta \|Y\|$ and $\|\Upsilon'\Delta - I_B\|_{\mathrm{cb}} \le C_L \eta$.

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, writing the fields of (W_RF,S) as the unqualified symbols below: for every Delta' supplied
for (W_RF,S) by lem-routef-delta-prime-closeness, every Delta supplied from that same Delta' by
lem-routef-delta-normalization-closeness, and every componentwise package
(m,(L_j,E_j,W_j,Sigma_j,U_js,p_js,C_j,xi_j,Lambda_j,Upsilon'_j)_{j=1}^m,F,V,Upsilon') supplied for
(W_RF,S,Delta',Delta) by lem-routef-upsilon-prime-component-construction, with C_L:=C_2+C_3+2*C_R
from (1.3) and rho_Upsilon' := min{rho_T, rho_id, rho_Delta, rho_2, rho_3, (2*C_R)^(-1)}, for 0 <=
eta <= rho_Upsilon', every integer q >= 1, and every Y=(Y_1,...,Y_m) in M_q(B)=direct-sum_{j=1}^m
M_q(B(L_j)), ||(Upsilon'_j)_q(Delta_q(Y))-Y_j|| <= C_L*eta*||Y|| for every j, and consequently
||(Upsilon' Delta-I_B)_q(Y)|| <= C_L*eta*||Y|| and ||Upsilon' Delta-I_B||_cb <= C_L*eta.
```

Proof (prose account of the af-validated tree). Fix q, Y, j and expand both occurrences of Λ_j in $(\Upsilon'_j)_q(\Delta_q(Y))$. The resulting double probability average contains

$$A_{st} = \Phi_q\Big(\Delta_q(\widehat{U}_{js}) \Phi_q(\Delta_q(Y)) \Delta_q(\widehat{U}_{jt}^*)\Big).$$

Degree two with an identity input replaces the middle $\Phi_q(\Delta_q(Y))$ by $\Delta_q(Y)$ at cost $C_2\eta\|Y\|$. Lemma 197 then replaces the three-factor expression by $\Delta_q(\widehat{U}_{js}Y\widehat{U}_{jt}^*)$ at cost $C_3\eta\|Y\|$. Contractive sandwiches and probability weights preserve the sum of these errors. Call the resulting double average $P_{jq}(Y)$. Substituting the block formula for Δ_q and factoring the sums places the amplified twirl $I_q \otimes R_j$ on both sides of $Y_j \otimes I_{E_j}$. Since $R_j = I_{L_j} \otimes C_j$, compression by $z \mapsto z \otimes \xi_j$ gives exactly $P_{jq}(Y) = \alpha_j Y_j$, $\alpha_j = \langle \xi_j, C_j^* C_j \xi_j \rangle$. Thus $0 \le 1 - \alpha_j \le 2C_R\eta$ and $\|P_{jq}(Y) - Y_j\| \le 2C_R\eta\|Y\|$; the triangle inequality gives $C_2 + C_3 + 2C_R = C_L$. Taking the maximum component norm in $M_q(B)$ and then the supremum over $q, Y \neq 0$ proves the cb bound without multiplicity loss. The ledger records two repaired scope challenges: A_{st} was rebound locally rather than imported from a pending sibling, and $P_{jq}(Y)$ was defined again in its own branch before fresh validation.

Lemma 200 (`lem-routef-upsilon-prime-closeness`)**.** *With the same globally fixed W_{RF}, admissible datum S, and maps Δ', Δ, let $C_N, C_R, C_L, C_{\Upsilon'}$ be as in (1.3) and let $\rho_{\Upsilon'}$ be as above. If $0 \le \eta \le \rho_{\Upsilon'}$, every Choi multiplicity space in the component construction is nonzero, that construction produces a completely positive Υ', and*

$$\|\Upsilon' - \widetilde{\Upsilon}\|_{\mathrm{cb}} \le C_{\Upsilon'}\eta.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, and every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness, writing the fields of
(W_RF,S) as the unqualified symbols below: Upsilon-prime CP closeness: with C_N, C_R, C_L,
C_Upsilon' from (1.3) and rho_Upsilon' := min{rho_T, rho_id, rho_Delta, rho_2, rho_3,
(2*C_R)^(-1)}, for 0 <= eta <= rho_Upsilon', every Choi multiplicity space used below is nonzero
and the componentwise construction produces CP Upsilon' with ||Upsilon' - tilde-Upsilon||_cb <=
C_Upsilon'*eta.
```

Proof (prose account of the af-validated tree). Lemma 198 supplies the nonzero multiplicity spaces, CP Υ', and $\|\Upsilon'\|_{\mathrm{cb}} \le 1$. With $K_j(T) = \Lambda_j^*(T \otimes I_F)\Lambda_j$ and $K = (K_j)_j$, the tree has $\|K\|_{\mathrm{cb}} \le 1$ and $\Upsilon' = K\Phi$. Because $\rho_{\Upsilon'} \le \rho_T \le 1/8$, Lemma 94 applies, and

$$\Phi - \Phi\widetilde{\Phi} = (\Phi - \Phi^2) + \Phi(\Phi - \widetilde{\Phi})$$

gives $\|\Upsilon'(I - \widetilde{\Phi})\|_{\mathrm{cb}} \le (1 + C_\theta)\eta$. The Δ estimate and Lemma 199 give $\|\Upsilon'\widetilde{\Delta} - I_B\|_{\mathrm{cb}} \le (C_\Delta + C_L)\eta$, while the raw norm bound gives $\|\widetilde{\Upsilon}\|_{\mathrm{cb}} \le 2$. The identity $\widetilde{\Delta}\widetilde{\Upsilon} = \widetilde{\Phi}$ yields

$$\Upsilon' - \widetilde{\Upsilon} = \Upsilon'(I - \widetilde{\Phi}) + (\Upsilon'\widetilde{\Delta} - I_B)\widetilde{\Upsilon}.$$

The preceding estimates bound this by $[1 + C_\theta + 2C_\Delta + 2C_L]\eta = C_{\Upsilon'}\eta$, the ledger entry (1.3). Factoring added the preceding two dependencies while leaving this row's registry contract byte-unchanged.

Lemma 201 (`lem-routef-upsilon-normalization-closeness`)**.** *With the same fixed data and every Υ' supplied by Lemma 200, put*

$$C_\Upsilon = 6C_T + 7C_{\Upsilon'}, \qquad \rho_\Upsilon = \min\{\rho_{\mathrm{unit}}, \rho_{\Upsilon'}, [2(C_T + C_{\Upsilon'})]^{-1}\}.$$

For $0 \le \eta \le \rho_\Upsilon$, $b = \Upsilon'(I)$ *is invertible and, for every* $X \in B(H)$,

$$\Upsilon(X) = b^{-1/2}\Upsilon'(X)b^{-1/2}$$

defines a UCP map satisfying $\|\Upsilon - \widetilde{\Upsilon}\|_{\text{cb}} \le C_\Upsilon \eta$.

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness, and every Upsilon'
supplied from that same pair by lem-routef-upsilon-prime-closeness, and for every X in B(H),
writing the fields of (W_RF,S) as the unqualified symbols below: Upsilon UCP normalization: with
C_Upsilon := 6*C_T+7*C_Upsilon' and rho_Upsilon := min{rho_unit, rho_Upsilon',
[2*(C_T+C_Upsilon')]^(-1)}, for 0 <= eta <= rho_Upsilon, b = Upsilon'(I) is invertible and
Upsilon(X) = b^(-1/2)*Upsilon'(X)*b^(-1/2) is UCP with ||Upsilon - tilde-Upsilon||_cb <=
C_Upsilon*eta.
```

Proof (prose account of the af-validated tree). Set $d = (C_T + C_{\Upsilon'})\eta$. The radius gives $0 \le d \le 1/2$, and Lemma 200 with the raw unit estimate gives

$$\|b - I\| \le \|\Upsilon'(I) - \widetilde{\Upsilon}(I)\| + \|\widetilde{\Upsilon}(I) - I\| \le d.$$

Since Υ' is CP, b is positive. Its spectrum lies in $[1-d, 1+d] \subset [1/2, 3/2]$, so b is invertible and $c = b^{-1/2}$ satisfies $\|b\|, \|c\| \le 3/2$. On this interval, $|t^{-1/2} - 1| = |t-1|/[\sqrt{t}(1+\sqrt{t})]$, so functional calculus gives $\|c - I\| \le \|b - I\| \le d$. Conjugation of Υ' by c is CP at every amplification, while $cbc = I$ makes Υ unital; hence it is UCP with cb norm one. The rearrangement $\Upsilon'(Y) = b^{1/2}\Upsilon(Y)b^{1/2}$ gives $\|\Upsilon'\|_{\text{cb}} \le \|b\| \le 3/2$. Finally

$$\Upsilon - \Upsilon' = (c - I)\Upsilon' c + \Upsilon'(c - I)$$

gives $\|\Upsilon - \Upsilon'\|_{\text{cb}} \le (15/4)d \le 6d$. Adding the $C_{\Upsilon'}\eta$ prime-closeness bound yields $\|\Upsilon - \widetilde{\Upsilon}\|_{\text{cb}} \le (6C_T + 7C_{\Upsilon'})\eta$. The ledger's sole challenge was a sibling-scope gap in the cb-norm-one step; the live tree repairs it with a self-contained amplification argument before fresh validation. The quoted contract already carries the earlier rescope amendment inserting the explicit "for every $X \in B(H)$" binder; its mathematical suffix was otherwise unchanged.

Status. All four rows are `proved` with `af: validated` (T0), and the prose above reports only their live exported trees.

69 Route-F telescopes and the threshold: the three telescope estimates, K-finiteness, and the threshold minimum

Fix first the single global package W_{RF} supplied by Lemma 188. For each admissible (H, Φ, η), fix its datum S from that lemma and then quantify successively over every Δ', Δ, Υ', and Υ supplied by Lemmas 193, 194, 200, and 201. This is the common nested conditional setting below, with the fields of Definition 46 written without qualification; it does not assert that those maps exist outside their provider domains. The first three rows telescope the normalized factors, and the last two close the coefficient and radius ledger needed by the later factor packets.

Lemma 202 (`lem-routef-delta-upsilon-telescope`). *In the common setting, put*

$$\rho_{\Delta\Upsilon} = \min\{\rho_\theta, \rho_T, \rho_{\mathrm{id}}, \rho_\Delta, \rho_\Upsilon\}.$$

If $0 \le \eta \le \rho_{\Delta\Upsilon}$, then

$$\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}} \le (C_\theta + C_\Delta + 2C_\Upsilon)\eta.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness, every Upsilon' supplied
from that same pair by lem-routef-upsilon-prime-closeness, and every Upsilon supplied from that
same triple by lem-routef-upsilon-normalization-closeness, writing the fields of (W_RF,S) as the
unqualified symbols below: Delta-Upsilon telescope: for rho_DeltaUpsilon := min{rho_theta, rho_T,
rho_id, rho_Delta, rho_Upsilon} and 0 <= eta <= rho_DeltaUpsilon, ||Delta Upsilon - Phi||_cb <=
(C_theta+C_Delta+2*C_Upsilon)*eta.
```

Proof (prose account of the af-validated tree). The radius minimum places η simultaneously in the functional-calculus, raw-factor, and two normalization ranges. Thus Lemma 94 gives $\|\widetilde{\Phi} - \Phi\|_{\mathrm{cb}} \le C_\theta\eta$; Lemma 190 gives $\widetilde{\Delta}\widetilde{\Upsilon} = \widetilde{\Phi}$; and the two normalization rows give $\|\Delta - \widetilde{\Delta}\|_{\mathrm{cb}} \le C_\Delta\eta$ and $\|\Upsilon - \widetilde{\Upsilon}\|_{\mathrm{cb}} \le C_\Upsilon\eta$, with Υ UCP and hence $\|\Upsilon\|_{\mathrm{cb}} = 1$. The norm row gives $\|\widetilde{\Delta}\|_{\mathrm{cb}} \le 1 + C_T\eta$. Since $C_T = C_\theta + 3C_V$ and ρ_T is bounded by both reciprocal ledger entries, $C_T\eta \le \frac{1}{4} + \frac{3}{4} = 1$, so this norm is at most 2. At every amplification level q, the exact identity yields

$$\Delta_q\Upsilon_q - \Phi_q = (\Delta_q - \widetilde{\Delta}_q)\Upsilon_q + \widetilde{\Delta}_q(\Upsilon_q - \widetilde{\Upsilon}_q) + (\widetilde{\Phi}_q - \Phi_q).$$

Triangle inequality and submultiplicativity bound the three terms by $C_\Delta\eta$, $2C_\Upsilon\eta$, and $C_\theta\eta$; taking the supremum over q proves the claim. During construction node 1.3 was amended only to state these hypotheses locally; the contract was unchanged. The registry status is `proved`, `af: validated`.

Lemma 203 (`lem-routef-multiplicative-telescope`). *In the common setting, let $\rho_{\mathrm{mult}} = \min\{\rho_T, \rho_{\mathrm{id}}, \rho_{\Delta\Phi}, \rho_\Upsilon\}$. If $0 \le \eta \le \rho_{\mathrm{mult}}$, then for every $n \ge 1$ and $X, Y \in M_n(S.B)$,*

$$\|\Upsilon_n\big(\Delta_n(X)\Delta_n(Y)\big) - XY\| \le \big[C_\Upsilon + 2(C_2 + C_\theta + C_\Delta)\big]\eta\|X\|\|Y\|.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness, every Upsilon' supplied
from that same pair by lem-routef-upsilon-prime-closeness, and every Upsilon supplied from that
same triple by lem-routef-upsilon-normalization-closeness; for every integer n >= 1 and all X, Y in
M_n(S.B), writing the fields of (W_RF,S) as the unqualified symbols below: Multiplicative
telescope: for rho_mult := min{rho_T, rho_id, rho_DeltaPhi, rho_Upsilon} and 0 <= eta <= rho_mult,
every amplification satisfies ||Upsilon_n(Delta_n X Delta_n Y) - XY|| <=
[C_Upsilon+2*(C_2+C_theta+C_Delta)]*eta*||X||*||Y||.
```

Proof (prose account of the af-validated tree). The radius hypotheses activate the raw identities, the $\Delta\Phi$ product estimate, and both normalization estimates. The normalized map Δ is UCP; amplification-level Kadison–Schwarz therefore makes every Δ_n contractive. Hence, for $Z = \Delta_n(X)\Delta_n(Y)$, one has $\|Z\| \leq \|X\|\|Y\|$. The same scalar calculation as above gives $\|\widetilde{\Upsilon}\|_{\mathrm{cb}} \leq 2$. Set $E = \widetilde{\Phi}_n(Z) - \widetilde{\Delta}_n(XY)$. Amplifying $\widetilde{\Delta}\widetilde{\Upsilon} = \widetilde{\Phi}$ and $\widetilde{\Upsilon}\widetilde{\Delta} = I_B$ gives $\widetilde{\Upsilon}_n\widetilde{\Phi}_n = \widetilde{\Upsilon}_n$ and $\widetilde{\Upsilon}_n\widetilde{\Delta}_n(XY) = XY$, whence the exact decomposition

$$\Upsilon_n(Z) - XY = (\Upsilon_n - \widetilde{\Upsilon}_n)(Z) + \widetilde{\Upsilon}_n(E).$$

The first term is at most $C_\Upsilon\eta\|X\|\|Y\|$, while Lemma 196 bounds E by $(C_2{+}C_\theta{+}C_\Delta)\eta\|X\|\|Y\|$, and the factor norm contributes the displayed 2. A verifier challenged a pending-sibling dependency in the first tree; the repair factored the local estimates, identity, and conclusion into nodes 1.3.1–1.3.3, which were freshly validated. No contract amendment occurred; status is `proved`, `af: validated`.

Lemma 204 (`lem-routef-upsilon-delta-telescope`). *In the common setting, put $\rho_{\Upsilon\Delta} = \min\{\rho_T, \rho_{\mathrm{id}}, \rho_\Delta, \rho_\Upsilon\}$. If $0 \leq \eta \leq \rho_{\Upsilon\Delta}$, then*

$$\|\Upsilon\Delta - I_B\|_{\mathrm{cb}} \leq (C_\Upsilon + 2C_\Delta)\eta.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness, every Upsilon' supplied
from that same pair by lem-routef-upsilon-prime-closeness, and every Upsilon supplied from that
same triple by lem-routef-upsilon-normalization-closeness, writing the fields of (W_RF,S) as the
unqualified symbols below: Upsilon-Delta telescope: for rho_UpsilonDelta := min{rho_T, rho_id,
rho_Delta, rho_Upsilon} and 0 <= eta <= rho_UpsilonDelta, ||Upsilon Delta - I_B||_cb <=
(C_Upsilon+2*C_Delta)*eta.
```

Proof (prose account of the af-validated tree). Unfolding the radius gives access to the raw identity $\widetilde{\Upsilon}\widetilde{\Delta} = I_B$, the two closeness bounds, and the raw norm estimate. Because $\rho_T \leq \rho_{\mathrm{id}}^{\mathrm{corr}}$, the identity applies; because $\eta \leq \rho_\Delta, \rho_\Upsilon$, normalization gives $\|\Delta - \widetilde{\Delta}\|_{\mathrm{cb}} \leq C_\Delta\eta$ and $\|\Upsilon - \widetilde{\Upsilon}\|_{\mathrm{cb}} \leq C_\Upsilon\eta$. Moreover Δ is UCP, so every Δ_n is contractive. The raw norm bound and the two reciprocal entries in ρ_T again imply $\|\widetilde{\Upsilon}\|_{\mathrm{cb}} \le 1 + C_T\eta \le 2$. At amplification level n there is therefore the exact two-term telescope

$$(\Upsilon\Delta - I_B)_n = (\Upsilon_n - \widetilde{\Upsilon}_n)\Delta_n + \widetilde{\Upsilon}_n(\Delta_n - \widetilde{\Delta}_n).$$

Its terms have norms at most $C_\Upsilon\eta$ and $2C_\Delta\eta$, uniformly in n; the supremum is the claimed cb bound. A verifier found that the original supremum node referred to a pending sibling. The repaired tree added an explicit dependent child, node 1.2.3.1, and revalidated the closure without changing the contract. The registry status is `proved`, `af: validated`.

Lemma 205 (`lem-routef-k-finiteness`). *In the common setting, K in equation (1.6) of Definition 46 is finite and universal, while ρ_{fac} in equation (1.7) is positive and is a common domain for Lemma 195 and the three telescope estimates above.*

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness, every Upsilon' supplied
from that same pair by lem-routef-upsilon-prime-closeness, and every Upsilon supplied from that
same triple by lem-routef-upsilon-normalization-closeness, writing the fields of (W_RF,S) as the
unqualified symbols below: Route F common coefficient/domain: K in (1.6) is finite and universal,
and rho_fac in (1.7) is positive and is a common domain for the degree-two estimate and the three
Route-F factorization estimates.
```

Proof (prose account of the af-validated tree). The tree first fixes the quantifier discipline: the long prefix is a nested conditional binder, whereas on $0 \le \eta \le \rho_{\mathrm{fac}}$ the radius inequalities activate the four providers in sequence. It then expands the scalar ledger serially. The formation row makes $C_\theta, C_A, \overline{C}_E, C_V, C_T$ finite positive and input-independent, and makes every initial radius positive. The operations in equations (1.2)–(1.5) are only finite sums, positive integer multiples, maxima, minima, and reciprocals with positive denominators. Consequently $C_{\Delta'}, C_\Delta, C_2, C_3, C_N, C_R, C_L, C_{\Upsilon'}, C_\Upsilon$ remain finite positive and universal, and every derived radius through $\rho_{\Upsilon\Delta}$ remains positive. Equation (1.6) takes a finite maximum of expressions in $C_\theta, C_\Delta, C_\Upsilon, C_2$, so K is finite and universal. Equation (1.7) is the minimum of the four positive radii $\rho_2, \rho_{\Delta\Upsilon}, \rho_{\mathrm{mult}}, \rho_{\Upsilon\Delta}$, hence is positive. Its order properties put every smaller η in the degree-two domain and all three telescope domains. A verifier challenged an earlier child that asserted map existence throughout the formation range; the repair made the nested-conditional reading explicit and proved sequential availability only on the common domain. The contract stayed fixed, and the registry records `proved`, `af: validated`.

Lemma 206 (`lem-routef-threshold-minimum`). *In the common setting, let $\eta_K = \min\{\rho_{\mathrm{fac}}, (24K)^{-1}, 1\}$. Then $\eta_K > 0$, and every $0 \le \eta \le \eta_K$ satisfies*

$$\eta \le \rho_{\mathrm{fac}}, \qquad 0 \le \eta \le \min\{(24K)^{-1}, 1\}, \qquad 3K\eta \le \tfrac{1}{8} < 1,$$

and

$$\frac{3K\eta}{1 - 3K\eta} \le 4K\eta \le \tfrac{1}{6} < \tfrac{1}{2}.$$

Registry contract (verbatim).

```
After first fixing one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation, for every input (H,Phi,eta) to which that formation result
applies, fix one def-routef-raw-factor-setting datum S over that same W_RF supplied by the same
result, for every Delta' supplied for (W_RF,S) by lem-routef-delta-prime-closeness, every Delta
supplied from that same Delta' by lem-routef-delta-normalization-closeness, every Upsilon' supplied
from that same pair by lem-routef-upsilon-prime-closeness, and every Upsilon supplied from that
same triple by lem-routef-upsilon-normalization-closeness, writing the fields of (W_RF,S) as the
unqualified symbols below: Route F scalar threshold: let eta_K := min{rho_fac, (24*K)^(-1), 1};
then eta_K > 0, and every 0 <= eta <= eta_K satisfies eta <= rho_fac, 0 <= eta <=
min{(24*K)^(-1),1}, 3*K*eta <= 1/8 < 1, and 3*K*eta/(1-3*K*eta) <= 4*K*eta <= 1/6 < 1/2.
```

Proof (prose account of the af-validated tree). Lemma 205 supplies $\rho_{\text{fac}} > 0$ and finite universal K, while equation (1.6) defines K by a maximum containing 1, so $K \geq 1 > 0$. Thus all three entries in the definition of η_K are strictly positive and their finite minimum is positive. If $0 \leq \eta \leq \eta_K$, the minimum immediately gives $\eta \leq \rho_{\text{fac}}$, $\eta \leq (24K)^{-1}$, and $\eta \leq 1$. Multiplication by $3K$ and $4K$ yields respectively $3K\eta \leq 1/8 < 1$ and $4K\eta \leq 1/6 < 1/2$. For the remaining comparison put $x = 3K\eta$. Then $0 \leq x \leq 1/8$, so $1 - x > 0$ and $1 - 4x \geq 1/2$. Since $4K\eta = 4x/3$, common-denominator arithmetic gives

$$4K\eta - \frac{x}{1-x} = \frac{x(1-4x)}{3(1-x)} \geq 0,$$

which is the asserted rational bound. The build amended three nodes to make $K \geq 1$ and their root hypotheses explicit; there was no contract change in that run. At the registry level, however, the 2026-08-05 rescope had already replaced an unsupported F2/F3/PRH admissibility suffix by exactly these scalar inequalities. This revised contract is the one validated here; status is `proved`, `af: validated`.

70 The Kitaev diagonal repair and the entrywise CP-ization corollary

The two validated rows below repair the finite-dimensional diagonal used in the Route F factor package and then record its entrywise positivity consequence. The diagonal is the object of Definition 24; complete positivity and the UCP hypothesis are understood as in Definition 45. The first row supplies the exact phase-balanced witness consumed by the second.

Lemma 207 (`lem-kitaev-diagonal-repair`). *The direct-sum diagonal formulas printed at* `approximate_algebr` *and* `approximate_algebras.tex:2780--2783` *are false, already for $\mathcal{B} = \mathbb{C} \oplus \mathbb{C}$. Nevertheless, every finite-dimensional C^*-algebra*

$$\mathcal{B} = \bigoplus_{r=1}^{m} M_{d_r}$$

has a finite phase-balanced diagonal

$$D = \sum_t q_t W_t^\dagger \otimes W_t, \qquad q_t \geq 0, \qquad \sum_t q_t = 1,$$

in which every W_t is unitary, $ZD = DZ$ for every $Z \in \mathcal{B}$, $\pi(D) = I_{\mathcal{B}}$, and

$$\|D\|_\pi = \sum_t q_t \|W_t^\dagger\| \|W_t\| = 1.$$

These conclusions are independent of the number and dimensions of the blocks.

Registry contract (verbatim).

```
Kitaev diagonal repair: the direct-sum diagonal formula printed at approximate_algebras.tex:1254
and :2780-2783 is false (already for B=C direct-sum C), but every finite-dimensional C*-algebra
B=direct-sum_{r=1}^m M_{d_r} has a finite phase-balanced diagonal D=sum_t q_t W_t^dagger tensor W_t
with unitary W_t, q_t >= 0, sum_t q_t=1, ZD=DZ for every Z in B, pi(D)=I_B, and projective norm
||D||_pi=sum_t q_t||W_t^dagger||||W_t||=1, independently of block count and block dimensions.
```

Proof (prose account of the af-validated tree). The printed defect is exposed on $\mathbb{C} \oplus \mathbb{C}$ by taking the one-term diagonal $1 \otimes 1$ in each scalar summand. Either printed prescription then returns $I_{\mathcal{B}} \otimes I_{\mathcal{B}}$. For $Z = (1, 0)$, its left and right module actions are $Z \otimes I_{\mathcal{B}}$ and $I_{\mathcal{B}} \otimes Z$, which differ in the $(1, 2)$ tensor block; thus the printed element is not central.

For the repair, fix $d \geq 1$, put $\omega = e^{2\pi i/d}$, index matrix units modulo d, and define $U_{ab} = \sum_k \omega^{bk} E_{k+a,k}$. The action $U_{ab} e_k = \omega^{bk} e_{k+a}$ shows that each U_{ab} is unitary. Expanding the tensor and using the finite Fourier sum gives the exact Weyl average

$$\frac{1}{d^2} \sum_{a,b} U_{ab}^\dagger \otimes U_{ab} = D_d := \frac{1}{d} \sum_{i,j} E_{ij} \otimes E_{ji}.$$

For the block algebra, take $\gamma = ((a_r, b_r))_{r=1}^m$ and $\sigma \in \{\pm 1\}^m$, and set $W_{\gamma,\sigma} = \bigoplus_r \sigma_r U^{(r)}_{a_r b_r}$ and $q_{\gamma,\sigma} = 2^{-m} \prod_r d_r^{-2}$. The $W_{\gamma,\sigma}$ are unitary, the family is finite, and its positive coefficients sum to one. The phase moment $2^{-m} \sum_\sigma \sigma_r \sigma_s = \delta_{rs}$ deletes every cross-block tensor term; the surviving Weyl averages give $D = \sum_r (\iota_r \otimes \iota_r)(D_{d_r})$. For a matrix unit $E_{ab} \in M_d$, direct multiplication yields $E_{ab}D_d = d^{-1}\sum_j E_{aj} \otimes E_{jb} = D_d E_{ab}$, and hence blockwise centrality. Likewise, $\pi(D_d) = d^{-1}\sum_{i,j} E_{ij}E_{ji} = I_{M_d}$, so $\pi(D) = I_{\mathcal{B}}$. The displayed convex-unitary representation has cost one and therefore $\|D\|_\pi \le 1$; conversely $\|\pi(C)\| \le \|C\|_\pi$ for every projective tensor C, so normalization forces $\|D\|_\pi \ge 1$.

The retained ledger records two major challenges in the normalization branch: two leaves had improperly invoked centrality from a sibling. Both were amended to prove only $\pi(D) = I_{\mathcal{B}}$; the parent then combines that normalization with its separately validated centrality child. No contract amendment or supersession was made.

Status. `proved`; `af: validated` (T0). Workspace `proofs/lem-kitaev-diagonal-repair`: 20/20 live nodes validated, taint clean; the two repaired challenges are the dependency-scope corrections just described.

Corollary 208 (`cor-kitaev-diagonal-cpization`)**.** *Let $D = \sum_t q_t W_t^\dagger \otimes W_t$ be the finite phase-balanced diagonal supplied by Lemma 207. If $\widetilde{\Delta} : \mathcal{B} \to B(\mathcal{H})$ is an involution-preserving linear map and $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$ is UCP, then*

$$\Delta'(X) = \sum_t q_t \, \Phi\Big(\widetilde{\Delta}(XW_t^\dagger)\widetilde{\Delta}(W_t)\Big)$$

defines a completely positive map. Its complete positivity uses the exact centrality of D and does not require $\widetilde{\Delta}$ to be exactly multiplicative.

Registry contract (verbatim).

```
Entrywise CP-ization from the repaired diagonal: for the finite phase-balanced diagonal D=sum_t q_t
W_t^dagger tensor W_t supplied by lem-kitaev-diagonal-repair, every involution-preserving linear
map tilde-Delta:B->B(H) and every UCP map Phi define a completely positive map Delta'(X)=sum_t q_t
Phi(tilde-Delta(X W_t^dagger) tilde-Delta(W_t)); complete positivity uses exact centrality of D and
does not require exact multiplicativity of tilde-Delta.
```

Proof (prose account of the af-validated tree). Linearity of Δ' follows term by term from linearity of right multiplication, $\widetilde{\Delta}$, operator multiplication by the fixed $\widetilde{\Delta}(W_t)$, and Φ. Fix $n \ge 1$ and $Y \in M_n(\mathcal{B})$, and write

$$Z_t = \widetilde{\Delta}_n\big((I_n \otimes W_t)Y\big).$$

Exact centrality of D, applied entrywise and then left-multiplied by $Y_{ca}^\dagger$, gives

$$\sum_t q_t Y_{ca}^\dagger Y_{cb} W_t^\dagger \otimes W_t = \sum_t q_t Y_{ca}^\dagger W_t^\dagger \otimes W_t Y_{cb}.$$

Apply to this tensor identity the linear map induced by the bilinear rule $A \otimes B \mapsto \widetilde{\Delta}(A)\widetilde{\Delta}(B)$. Involution preservation, and not multiplicativity, changes its right-hand side into

$$\sum_t q_t \, \widetilde{\Delta}(W_t Y_{ca})^\dagger \widetilde{\Delta}(W_t Y_{cb}).$$

Summing over c and applying Φ therefore yields the exact matrix identity

$$\Delta_n'(Y^\dagger Y) = \sum_t q_t \Phi_n(Z_t^\dagger Z_t).$$

Each $Z_t^\dagger Z_t$ is positive, complete positivity of Φ makes each amplified image positive, and the sum stays positive because the phase-balanced family is finite and $q_t \geq 0$. Every positive element of $M_n(\mathcal{B})$ is $Y^\dagger Y$ for its positive square root; hence Δ'_n is positive for every n, proving complete positivity.

The ledger records two repaired major challenges. First, an exhaustive "uses only" claim had omitted finiteness and nonnegativity of the q_t; a validated child now states those coefficient hypotheses explicitly. Second, the frozen displayed formula left the type of Φ implicit. The registry's user-ratified typing scope and the validated final child read it as $\Phi : B(\mathcal{H}) \to B(\mathcal{H})$, which makes every product and amplification above well typed. Neither repair adds multiplicativity of $\widetilde{\Delta}$, and the contract itself was not amended.

Status. `proved`; `af: validated` (T0). Workspace `proofs/cor-kitaev-diagonal-cpization`: 22/22 live nodes validated, taint clean; the two repaired challenges are the coefficient and typing corrections recorded above.

71 The strengthened K-ledger package: scalar header positivity, the factor-map and factor-estimate packets, and the strengthened K-ledger

This section reproduces the four `af`-validated rows that package the Route-F factorization. The scalar header and setting datum are those of Definition 46; the first row fixes their universal quantifier order, the next two preserve one serial packet, and the last performs the positive-retract finish without reselecting any witness.

Lemma 209 (`lem-routef-scalar-header-positivity`). *There is one global witness package W_{RF} supplied by Lemma 188 such that, for its scalars K in (1.6), ρ_{fac} in (1.7), and $\eta_K = \min\{\rho_{\mathrm{fac}}, (24K)^{-1}, 1\}$ in (1.8), $K < \infty$, $K \geq 1$, $\rho_{\mathrm{fac}} > 0$, and $\eta_K > 0$. These scalars are universal and independent of $\mathcal{H}, \Phi, \eta, n$, the amplification level, the simple-block count, and the block dimensions, and*

$$\eta_K \leq \rho_{\mathrm{fac}} \leq \rho_2 \leq \rho_T \leq \rho_{\mathrm{id}}^{\mathrm{corr}}, \quad \rho_2 \leq \rho_{\Delta'}, \rho_{\Delta}, \quad \rho_{\mathrm{fac}} \leq \rho_{\Delta\Upsilon} \leq \rho_{\Upsilon} \leq \rho_{\Upsilon'}, \quad \rho_{\mathrm{fac}} \leq \rho_{\mathrm{mult}}, \rho_{\Upsilon\Delta}.$$

Registry contract (verbatim).

```
Route F scalar-header positivity: there exists one global witness package W_RF supplied by
lem-routef-raw-factor-setting-formation such that, writing K for its scalar (1.6), rho_fac for its
scalar (1.7), and eta_K := min{rho_fac, (24*K)^(-1), 1} for its scalar (1.8), K is finite with K >=
1, rho_fac > 0, and eta_K > 0, these scalars are universal and independent of H, Phi, eta, n,
amplification level, simple-block count, and block dimensions, and eta_K <= rho_fac <= rho_2 <=
rho_T <= rho_id^corr, rho_2 <= rho_Delta', rho_2 <= rho_Delta, rho_fac <= rho_DeltaUpsilon <=
rho_Upsilon <= rho_Upsilon', rho_fac <= rho_mult, and rho_fac <= rho_UpsilonDelta.
```

Proof (prose account of the af-validated tree). The tree first selects the scalar header supplied, before every input quantifier, by Lemma 188. Its primitive fields give $C_\theta = 12(\sqrt{2}-1) > 0$, the positive finite C_A, and $\bar{C}_E = \max\{1, C_E\}$; hence C_V, C_T are positive and finite. Every entry in the minimum defining ρ_T is then positive, including the two reciprocal entries. Proceeding through (1.2), each new coefficient is a finite sum of earlier coefficients and each new radius is a finite minimum of positive terms. The same induction through (1.3)–(1.5) gives positive finite $C_N, C_R, C_L, C_{\Upsilon'}, C_\Upsilon$ and positive $\rho_{\Upsilon'}, \rho_\Upsilon, \rho_{\Delta\Upsilon}, \rho_{\mathrm{mult}}$ and $\rho_{\Upsilon\Delta}$. Thus (1.6) makes K the maximum of 1 and three finite expressions, so $K < \infty$ and $K \geq 1$; (1.7) makes ρ_{fac} the minimum of the four positive radii $\rho_2, \rho_{\Delta\Upsilon}, \rho_{\mathrm{mult}}$ and $\rho_{\Upsilon\Delta}$; and (1.8) gives $\eta_K > 0$. The displayed inequalities are exactly the coordinate inequalities of these successive minima. Finally, all three scalars depend only on the pre-input header, proving their asserted universality. Two verifier challenges were repaired inside this proof tree: one removed a pending-sibling dependency by rebuilding the positivity step with validated support, and one corrected the count in (1.6) from five purported entries to its actual four. Two superseded helper nodes were archived; neither repair altered the registry contract.

Lemma 210 (`lem-routef-factor-map-packet`). *Fix the preceding W_{RF} and its $K, \rho_{\mathrm{fac}}, \eta_K$. For every $n \geq 1$, row-stochastic $Q : \ell^\infty(n) \to \ell^\infty(n)$, and $0 \leq \eta \leq \eta_K$ with $\|Q^2 - Q\|_{\infty\to\infty} \leq \eta$, let $D : M_n \to \mathbb{C}^n$ be diagonal extraction, $J : \mathbb{C}^n \to M_n$ diagonal inclusion, $Q_{\mathbb{C}}$ the canonical complex-linear extension of Q, and $\Phi = JQ_{\mathbb{C}}D$. Then Φ is UCP and $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \leq \eta$, and there are a finite-dimensional unital C^*-algebra B, one setting datum S over this same W_{RF} for $(\mathcal{H}, \Phi, \eta) = (\mathbb{C}^n, \Phi, \eta)$, CP maps $\Delta' : B \to M_n$, $\Upsilon' : M_n \to B$, and UCP maps $\Delta : B \to M_n$,*

$\Upsilon : M_n \to B$. *Here Δ' is supplied by Lemma 193, Δ from that same Δ' by Lemma 194, Υ' from that same (Δ', Δ) by Lemma 200, and Υ from that same triple by Lemma 201.*

Registry contract (verbatim).

```
Relative Route F factor-map packet: after first fixing one global witness package W_RF supplied by
lem-routef-scalar-header-positivity from lem-routef-raw-factor-setting-formation, writing K,
rho_fac, and eta_K for its scalars (1.6)-(1.8), for every n >= 1, every row-stochastic Q: l_inf^n
-> l_inf^n, and every 0 <= eta <= eta_K with ||Q^2-Q||_{infinity->infinity} <= eta, let D: M_n ->
C^n be diagonal extraction onto the complex diagonal algebra C^n = l_inf^n(C), let J: C^n -> M_n be
the diagonal inclusion, let Q_C: C^n -> C^n be the canonical complex-linear extension of Q, and put
Phi := J Q_C D; then Phi is UCP with ||Phi^2-Phi||_cb <= eta, and there exist a finite-dimensional
unital C*-algebra B, one def-routef-raw-factor-setting datum S over this same W_RF supplied by
lem-routef-raw-factor-setting-formation for the same (H:=C^n,Phi,eta) whose B-field is B, CP maps
Delta':B->M_n and Upsilon':M_n->B, and UCP maps Delta:B->M_n and Upsilon:M_n->B such that Delta' is
supplied for (W_RF,S) by lem-routef-delta-prime-closeness, Delta is supplied from that same Delta'
by lem-routef-delta-normalization-closeness, Upsilon' is supplied from that same (Delta',Delta) by
lem-routef-upsilon-prime-closeness, and Upsilon is supplied from that same (Delta',Delta,Upsilon')
by lem-routef-upsilon-normalization-closeness.
```

Proof (prose account of the af-validated tree). Fix the header from Lemma 209. Its radius chain gives $\eta_K \le \rho^{\mathrm{corr}}_{\mathrm{id}}, \rho_{\Delta'}, \rho_\Delta, \rho_{\Upsilon'}, \rho_\Upsilon$. For arbitrary n, Q, eta, the diagonal-lift row Lemma 91 makes $\Phi = JQ_{\mathbb{C}}D$ UCP, while Lemma 92 identifies its completely bounded defect with $\|Q^2 - Q\|_{\infty\to\infty}$. Since $\mathbb{C}^n$ is nonzero, the formation row applies at this exact triple and is invoked once, producing one algebra B and one datum S over the already fixed header. The proof then preserves that datum rigidly. First Lemma 193 supplies the CP map Δ'; then Lemma 194 consumes that particular map and supplies the UCP map Δ. Next Lemma 200 consumes the same (S, Δ', Δ) and supplies the CP map Υ', after which Lemma 201 consumes that exact triple and supplies the UCP map Υ. Thus the witnesses have the serial provenance claimed at the root; no component packet is opened and no witness is reselected. The exported tree is first-pass and clean, with no recorded challenge or amendment.

Lemma 211 (`lem-routef-factor-estimate-packet`). *Fix the preceding W_{RF} and its $K, \rho_{\mathrm{fac}}, \eta_K$. For every $n \ge 1$, row-stochastic $Q : \ell^\infty(n) \to \ell^\infty(n)$, and $0 \le \eta \le \eta_K$ with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta$, form $D, J, Q_{\mathbb{C}}$ and $\Phi = JQ_{\mathbb{C}}D$ as in Lemma 210. For every same-datum packet $(B, S, \Delta', \Delta, \Upsilon', \Upsilon)$ supplied there,*

$$\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}} \le K\eta, \qquad \|\Upsilon\Delta - I_B\|_{\mathrm{cb}} \le K\eta,$$

and, for every $r \ge 1$ and $X, Y \in M_r(B)$,

$$\|\Upsilon_r(\Delta_r X\, \Delta_r Y) - XY\| \le K\eta\|X\|\|Y\|.$$

Moreover $0 \le \eta \le \min\{(24K)^{-1}, 1\}$ and

$$3K\eta \le \tfrac{1}{8} < 1, \qquad \frac{3K\eta}{1 - 3K\eta} \le 4K\eta \le \tfrac{1}{6} < \tfrac{1}{2}.$$

Registry contract (verbatim).

```
Relative Route F factor-estimate packet: after first fixing one global witness package W_RF
supplied by lem-routef-scalar-header-positivity from lem-routef-raw-factor-setting-formation,
```

```
writing K, rho_fac, and eta_K for its scalars (1.6)-(1.8), for every n >= 1, every row-stochastic
Q: l_inf^n -> l_inf^n, and every 0 <= eta <= eta_K with ||Q^2-Q||_{infinity->infinity} <= eta, let
D: M_n -> C^n be diagonal extraction onto the complex diagonal algebra C^n = l_inf^n(C), let J: C^n
-> M_n be the diagonal inclusion, let Q_C: C^n -> C^n be the canonical complex-linear extension of
Q, and put Phi := J Q_C D; for every same-datum packet (B,S,Delta',Delta,Upsilon',Upsilon) supplied
for this (W_RF,n,Q,eta,D,J,Q_C,Phi) by lem-routef-factor-map-packet, ||Delta Upsilon-Phi||_cb <=
K*eta, ||Upsilon Delta-I_B||_cb <= K*eta, and for every integer r >= 1 and all X,Y in M_r(B),
||Upsilon_r(Delta_r X Delta_r Y)-XY|| <= K*eta*||X||*||Y||; moreover 0 <= eta <=
min{(24*K)^(-1),1}, 3*K*eta <= 1/8 < 1, and 3*K*eta/(1-3*K*eta) <= 4*K*eta <= 1/6 < 1/2.
```

Proof (prose account of the af-validated tree). Fix the header and an arbitrary packet supplied by Lemma 210. The serial provenance of that packet is precisely the input required by all three telescope rows, while Lemma 209 gives $0 \le \eta \le \eta_K \le \rho_{\mathrm{fac}}$ and places ρ_{fac} below their three domain radii. Lemma 202 first yields $\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}} \le (C_\theta + C_\Delta + 2C_\Upsilon)\eta$; its coefficient is an entry in the maximum (1.6), so it is at most K. Lemma 204 similarly gives coefficient $C_\Upsilon + 2C_\Delta \le K$. At every amplification, Lemma 203 gives coefficient $C_\Upsilon + 2(C_2 + C_\theta + C_\Delta) \le K$, and multiplication by the nonnegative $\eta\|X\|\|Y\|$ preserves the bound. Lemma 205 records the common domain and coefficient ledger used in those three applications. Finally, Lemma 206 applies the formula $\eta_K = \min\{\rho_{\mathrm{fac}}, (24K)^{-1}, 1\}$ to give the displayed smallness and rational estimates, including positivity of $1-3K\eta$. All assertions concern the same arbitrary packet; the exported tree is first-pass and clean, with no recorded challenge.

Lemma 212 (`lem-routef-k-ledger`)**.** *There is one global W_{RF} supplied by Lemma 188 such that, writing K for (1.6), ρ_{fac} for (1.7), and $\eta_K = \min\{\rho_{\mathrm{fac}}, (24K)^{-1}, 1\}$ for (1.8), $K \ge 1$ and $\eta_K > 0$ are universal and independent of n, amplification level, simple-block count and block dimensions. For every $n \ge 1$, row-stochastic $Q : \ell^\infty(n) \to \ell^\infty(n)$, and $0 \le \eta \le \eta_K$ with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta$, let $D : M_n \to \mathbb{C}^n$ be diagonal extraction, $J : \mathbb{C}^n \to M_n$ diagonal inclusion, $Q_{\mathbb{C}}$ the canonical complex-linear extension, and $\Phi = JQ_{\mathbb{C}}D$. Then there are a finite-dimensional unital C^*-algebra B and UCP maps $\Delta : B \to M_n$, $\Upsilon : M_n \to B$ satisfying*

$$\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}},\ \|\Upsilon\Delta - I_B\|_{\mathrm{cb}} \le K\eta, \qquad \|\Upsilon_r(\Delta_r X\,\Delta_r Y) - XY\| \le K\eta\|X\|\|Y\|$$

for every $r \ge 1$ and $X, Y \in M_r(B)$. The same Q admits a stochastic idempotent E such that

$$\|Q - E\|_{\infty\to\infty} \le \big(K + 4\sqrt{2K}\big)\sqrt{\eta}.$$

Registry contract (verbatim).

```
Relative Route F factorization-and-finish ledger: there exists one global witness package W_RF
supplied by lem-routef-raw-factor-setting-formation such that, writing K for its scalar (1.6),
rho_fac for its scalar (1.7), and eta_K := min{rho_fac, (24*K)^(-1), 1} for its scalar (1.8), K >=
1 and eta_K > 0 are universal and independent of n, amplification level, simple-block count, and
block dimensions, and for every n >= 1, every row-stochastic Q: l_inf^n -> l_inf^n, and every 0 <=
eta <= eta_K with ||Q^2-Q||_{infinity->infinity} <= eta, let D: M_n -> C^n be diagonal extraction
onto the complex diagonal algebra C^n = l_inf^n(C), let J: C^n -> M_n be the diagonal inclusion,
let Q_C: C^n -> C^n be the canonical complex-linear extension of Q, and put Phi := J Q_C D; then
there exist a finite-dimensional unital C*-algebra B and UCP maps Delta: B -> M_n and Upsilon: M_n
-> B such that ||Delta Upsilon-Phi||_cb <= K*eta, ||Upsilon Delta-I_B||_cb <= K*eta, and for every
integer r >= 1 and all X,Y in M_r(B), ||Upsilon_r(Delta_r X Delta_r Y)-XY|| <= K*eta*||X||*||Y||,
and the same Q admits a stochastic idempotent E satisfying ||Q-E||_{infinity->infinity} <=
(K+4*sqrt(2*K))*sqrt(eta).
```

Proof (prose account of the af-validated tree). Choose the single global header by Lemma 209; this supplies the pre-input, universal $K \geq 1$ and $\eta_K > 0$. For arbitrary admissible n, Q, η, Lemma 210 supplies one algebra B, its setting datum, and the serial maps $\Delta', \Delta, \Upsilon', \Upsilon$. On that very packet, Lemma 211 supplies the two completely bounded factor estimates, the amplified multiplicative estimate, and all threshold inequalities. The $r = 1$ specialization of the latter is the level-one hypothesis required by Lemma 96. That lemma proves that B is commutative, selects $k \geq 1$ and a unital $*$-isomorphism $\mathbb{C}^k \to B$, and produces positive unital maps $A : \ell^\infty(k) \to \ell^\infty(n)$ and $M : \ell^\infty(n) \to \ell^\infty(k)$ with $\|Q - AM\|_{\infty\to\infty} \leq K\eta$, $\|QA - A\|_{\infty\to\infty} \leq 2K\eta$, and $\|Ax\|_\infty \geq (1 - 3K\eta)\|x\|_\infty$. Since $3K\eta \leq 1/8 < 1$, Lemma 97 turns these into $\|MA - I_k\|_{\infty\to\infty} \leq 3K\eta/(1 - 3K\eta)$. The bounds $\eta \leq \min\{(24K)^{-1}, 1\}$ now meet the hypotheses of Lemma 88, which returns the stated stochastic idempotent for the same Q with constant $K + 4\sqrt{2K}$. This seven-node tree validated on its first pass. It proves the strengthened statement as a new obligation: no status was inherited from the narrower `proved-mod-audit` W74F paper ledger that this registry result supersedes.

72 The F0 assembly and the conditional main-extension theorem

This section records two terminal interfaces of the validated assembly. The first specializes the strengthened Route-F ledger to the stochastic conclusion, with no repetition of the factorization below it. The second hides the MAIN-CB witness ledger behind two universal constants. The relevant vocabulary is that of Definition 3, Definition 4, Definition 23, and Definition 36; none is restated here.

Lemma 213 (`lem-routef-f0-assembly`)**.** *There are universal constants $\eta_0, C > 0$, independent of n, such that the following holds. Let $n \geq 1$, let $Q : \ell_\infty^n \to \ell_\infty^n$ be row-stochastic, and suppose that $0 \leq \eta \leq \eta_0$ and*

$$\|Q^2 - Q\|_{\infty\to\infty} \leq \eta.$$

Let $D : M_n \to \mathbb{C}^n$ be diagonal extraction onto the complex diagonal algebra $\mathbb{C}^n = \ell_\infty^n(\mathbb{C})$, let $J : \mathbb{C}^n \to M_n$ be the diagonal inclusion, let $Q_\mathbb{C} : \mathbb{C}^n \to \mathbb{C}^n$ be the canonical complex-linear extension of Q, and put $\Phi = JQ_\mathbb{C}D$. Then the same Q admits a stochastic idempotent E satisfying

$$\|Q - E\|_{\infty\to\infty} \leq C\sqrt{\eta}.$$

For the universal K and η_K supplied by Lemma 212, one may take

$$\eta_0 = \eta_K, \qquad C = K + 4\sqrt{2K}.$$

Registry contract (verbatim).

```
Route F F0 assembly: there are universal eta_0,C > 0, independent of n, such that for every n >= 1,
every row-stochastic Q: l_inf^n -> l_inf^n, and every 0 <= eta <= eta_0 with
||Q^2-Q||_{infinity->infinity} <= eta, let D: M_n -> C^n be diagonal extraction onto the complex
diagonal algebra C^n = l_inf^n(C), let J: C^n -> M_n be the diagonal inclusion, let Q_C: C^n -> C^n
be the canonical complex-linear extension of Q, and put Phi := J Q_C D; then the same Q admits a
stochastic idempotent E satisfying ||Q-E||_{infinity->infinity} <= C*sqrt(eta); for the universal K
and eta_K supplied by lem-routef-k-ledger, one may take eta_0 := eta_K and C := K+4*sqrt(2*K).
```

Proof (prose account of the af-validated tree). The tree first invokes Lemma 212 and fixes its universal scalars K and η_K. That imported row gives $K \geq 1$ and $\eta_K > 0$, with both scalars independent of n, amplification level, simple-block count, and block dimensions. Define $\eta_0 := \eta_K$ and $C := K + 4\sqrt{2K}$. Positivity of η_0 is immediate. Since $K \geq 1$, the principal square root $\sqrt{2K}$ is nonnegative, and therefore $C \geq K \geq 1 > 0$. Both new constants are fixed expressions in the imported universal scalars, so they retain the required dimension independence.

Now fix arbitrary n, Q, η satisfying the hypotheses of the statement, and form $D, J, Q_\mathbb{C}$ and Φ exactly as displayed there. The identity $\eta_0 = \eta_K$ changes the bound $0 \leq \eta \leq \eta_0$ into the smallness hypothesis required by the imported ledger; every other hypothesis, including the canonical complex-linear typing of $Q_\mathbb{C}$, is literally the parent row's input. Lemma 212 therefore returns, for the same Q, a stochastic idempotent E with

$$\|Q - E\|_{\infty\to\infty} \leq \left(K + 4\sqrt{2K}\right)\sqrt{\eta} = C\sqrt{\eta}.$$

This substitution is the whole assembly: the factor maps, their estimates, and the stochastic finish have already been discharged inside the single imported ledger and are not counted again at this interface.

Status and record note. `proved`; `af: validated` (T0). Workspace `proofs/lem-routef-f0-assembly`: all seven nodes are validated and clean. Run 1 reached its round limit with six of seven nodes validated and zero challenges; a verify-phase resume validated the root. The contract carries the canonical-complexification typing correction. An earlier strengthened-parent package was rejected, but this F0 contract was byte-identical and retained for validation against the repaired parent. This is an upper-bound assembly only and imports no sharpness claim.

Lemma 214 (`lem-thmainext-conditional`). *There are universal constants $C_E < \infty$ and $\varepsilon_E > 0$ such that every finite-dimensional extended ε-C^*-algebra $\mathcal{A}$, for $0 \le \varepsilon \le \varepsilon_E$, is carried by one extended $C_E\varepsilon$-isomorphism*

$$v : \mathcal{B} \longrightarrow \mathcal{A}$$

from a finite-dimensional C^*-algebra $\mathcal{B}$. The constants are independent of dimension, amplification level, and block data.

Registry contract (verbatim).

```
Extended th_main_ext assembly: there are universal C_E < infinity and epsilon_E > 0 such that every
finite-dimensional extended epsilon-C*-algebra A, for 0 <= epsilon <= epsilon_E, is carried by one
extended C_E*epsilon-isomorphism v:B->A from a finite-dimensional C*-algebra, with constants
independent of dimension, amplification level, and block data.
```

Proof (prose account of the af-validated tree). Invoke Lemma 162 and fix its universal MAIN-CB witness-ledger datum W. The tree chooses the two witnesses

$$C_E := W.c0_{\mathrm{cb}} W.K_{\mathrm{call}}, \qquad \varepsilon_E := W.\varepsilon_{\mathrm{MAIN}}.$$

The final clause of the imported assembly states that these quantities are finite, positive, and universal, and that they are independent of dimension, amplification level, block count, and block dimensions. Thus they have exactly the scalar properties claimed at the root.

Let $\mathcal{A}$ be an arbitrary finite-dimensional extended ε-C^*-algebra with $0 \le \varepsilon \le \varepsilon_E$. Since $\varepsilon_E = W.\varepsilon_{\mathrm{MAIN}}$, the structural assembly supplies

$$\mathcal{B} = \bigoplus_{\mathcal{C}} M_{|\mathcal{C}|}$$

and a single extended $W.c0_{\mathrm{cb}} W.K_{\mathrm{call}}\varepsilon$-isomorphism $v : \mathcal{B} \to \mathcal{A}$. It also supplies a unit estimate, but that extra conclusion is not used by this interface. Substitution of the definition of C_E turns the displayed map into the required extended $C_E\varepsilon$-isomorphism.

Finally, Definition 36 says that this is one bijective linear map whose every amplification is a $C_E\varepsilon$-inclusion. Hence the same v works at all amplification levels with the same universal constant. Together with the independence already carried by W, and the arbitrariness of $\mathcal{A}$ and ε, this proves the claimed interface.

Status and record note. `proved`; `af: validated` (T0). Workspace `proofs/lem-thmainext-conditional`: the four-node tree was validated on its first pass, taint clean, with zero challenges. Before elevation, a user-ratified re-scope removed from the contract a documentary clause describing the corrected squared COL-HILB, H-CB, EXT-CB, and Stage-1 reset provenance; the mathematical existential statement was unchanged. The registry deliberately retains all seven dependency edges as the construction record, while the validated tree itself is the thin interface projection of Lemma 162 reproduced above.

73 The discharge of `op-classical`: the root theorem at the af-validated rung

This section records the discharged root of the classical upper-bound campaign. Its validated tree is deliberately short: the quantitative work has already been completed by the Route-F assembly, and the root imports that conclusion and closes the remaining vocabulary interface. The statement is the upper stability bound only; it contains no claim of square-root sharpness.

Theorem 215 (`op-classical`)**.** *There are universal constants $\eta_0, C > 0$, independent of the dimension n, such that every row-stochastic matrix Q satisfying*

$$\|Q^2 - Q\|_{\infty\to\infty} \le \eta \le \eta_0 \tag{188}$$

admits a stochastic idempotent E for which

$$\|Q - E\|_{\infty\to\infty} \le C\sqrt{\eta}. \tag{189}$$

This is the commutative case of `op-npps`*; the general* `op-npps` *problem is out of scope and is not a T0 result in this repo.*

Registry contract (verbatim).

```
Classical projection stability: there are universal eta_0,C>0 (n-free) such that every
row-stochastic Q with ||Q^2-Q||_{inf->inf} <= eta <= eta_0 admits a stochastic idempotent E with
||Q-E||_{inf->inf} <= C sqrt(eta) (the commutative case of op-npps).
```

Proof (prose account of the af-validated tree). The tree has one substantive import and one vocabulary branch, with five live nodes in all. It constructs no replacement constants at the root. Instead, node 1.1 invokes Lemma 213, which supplies universal constants $\eta_0, C > 0$ independent of n for every row-stochastic map $Q : \ell_n^\infty \to \ell_n^\infty$ satisfying $\|Q^2 - Q\|_{\infty\to\infty} \le \eta \le \eta_0$. That external conclusion produces a stochastic idempotent E with $\|Q - E\|_{\infty\to\infty} \le C\sqrt{\eta}$. Thus the witness pair (η_0, C) and the final estimate are taken over unchanged; there is no threshold shrinkage, constant loss, or auxiliary approximation in this root tree.

Node 1.2 checks that this imported conclusion is literally the conclusion requested here. Its first child uses Definition 3 and Definition 4 to identify a row-stochastic matrix with the corresponding operator on ℓ_n^∞, and to identify the two notations $\|\cdot\|_{\mathrm{inf}\to\mathrm{inf}}$ and $\|\cdot\|_{\infty\to\infty}$ for the same induced norm. Under that identification, "independent of n" is exactly "n-free", while the imported output is exactly a stochastic idempotent. The second child uses Definition 14 only to read the parenthetical description "the commutative case of `op-npps`": it adds neither a hypothesis nor a further conclusion. Combining the imported Route-F assertion with these two vocabulary matches gives the registry contract verbatim.

Record note. The registry records a user-ratified root rewire on 2026-08-08: the validated Route-F assembly became one branch of an OR-routes block, while the legacy signed-geometry route (`thm-classical-factorization` and `prop-approx-simplex`, each `proved-mod-audit` with `af: seeded`) was retained as a non-T0 independent alternative, and the former "OPEN" marker was removed on discharge. The earlier user-ratified D1 split leaves sharpness outside this contract. These are contract-and-routing changes, not repairs made inside the exported five-node proof; the ledger records all five nodes as validated and clean and contains no challenge event.

Status. `proved`; `af: validated` (T0). Workspace `proofs/op-classical`; 5/5 live nodes, all clean. Registry `defs`: `def-stochastic`, `def-almost-idempotent` and `def-near-positive-projection`. The validating branch is Lemma 213.

74 Argument-DAG atlas: the Route-F proof chain for `op-classical`

This section is **generated** by `scripts/gen-report-dag.py` directly from the registry shards `argument/lemmas/*.md` (parsed with `scripts/argument.py`'s own parser) and is re-checked by `scripts/check-all.sh`: if a shard's `status`, `af` state or `deps` change and this section is not regenerated, the gate fails. It is therefore a live view of the argument, not a snapshot. Its browsable twin is `argument/DAG.md` (Mermaid, also generated from the same shards, whole registry); this atlas is the print/PDF view, restricted to the landing chain.

What is drawn. Not the whole registry (374 results) — only the sub-DAG that carries the argument landing the north star: the transitive closure of `op-classical` along `deps:` and `routes:` edges, *together with* the Route-F families named by the live strategy `docs/plans/CURRENT.md`. That is **221 results** and 645 edges. The Route-F chain is deliberately included even though it does not yet reach the root through registry edges: that missing composition is the point, and it is drawn as 2 dashed **design-pending (GAP)** links.

Honest reading. Of the 221 results drawn, **176 are rigorous** in this repo's sense (`af: validated`, or `cited` and byte-matched to `refs/`). Everything else — including every node coloured blue (`status: proved`) — is a prose proof that has *not* cleared an independent `af` validation, and the north star itself is `open`. A path of coloured boxes from left to right is *not* a proof; it is a map of what would have to be discharged.

Crosslinking (the anchor scheme). Every drawn result carries the hypertarget `dag:<registry-id>` on its node, and every imported definition carries `dag:<def-id>` on its index row. From anywhere in this lab-book, write `\dagref{lem-prh}` or `\dagdef{def-height}` to jump to the atlas, and `\dagphase{comp}{...}` to jump to a phase page. In the other direction, a node links back (§) to its typeset statement whenever the registry shard names one via its `provenance: report <label>` token — the same join key `scripts/check-provenance.py` uses.

How to read the atlas

The colour of a node is its *rigour rung* (`CLAUDE.md` L0), computed by `scripts/argument.py`'s `proof_class()` — the same function that colours the browsable Mermaid twin `argument/DAG.md`. Colour is never the only channel: each node also carries a glyph and the spelled-out status, so the atlas survives greyscale printing and every form of colour-vision deficiency.

	class	rung	what it means
	■ af-validated (176)	RIGOROUS (L0 rung b)	`af: validated` — root + every node validated by fresh codex verifiers, taint clean. The only rung this repo currently reaches.
	◆ cited (0)	RIGOROUS (L0 rung a)	`status: cited` — byte-verbatim string-matched to a local published source under `refs/`. (No result in this subgraph currently holds this rung.)
	• proved (21)	NOT rigorous	`status: proved` with `af: none/seeded` — a proof written in-repo that has NOT cleared an independent `af` validation. Solid, but not L0.
	◇ seeded (2)	NOT rigorous	`af: seeded` — an `af` workspace exists and the contract is pinned, but no validated tree yet.

	class	rung	what it means
	▲ mod-audit / conj. (18)	NOT rigorous (flagged)	`proved-mod-audit` · `conjecture` · `heuristic` · `numerical` — the workhorse rungs of the inherited campaign. Content, no independent check.
	○ stated (2)	NOT rigorous	`status: stated` — transcribed from an upstream artifact, unchecked here.
	★ OPEN (2)	OPEN	`status: open` or `kind: open-problem` — a live open question. `op-classical` is the north star and is one of these.

Edges. ⟶ a registry `deps:` edge (the dependency *points at* the result that uses it); ⇢ a member of an `routes:` OR-route (*any one* route discharges the shard — here `op-hlc`); ⇢ a **design-pending (GAP) composition**: *not* a registry edge, reserved by the live strategy (`docs/plans/CURRENT.m` and not yet wired. Small dashed grey chips are *boundary copies* of a node that really lives on an earlier band or in another phase; their id is abbreviated to the shortest unambiguous prefix and each is a hyperlink to the real node.

The design-pending (GAP) links

The atlas exists to show *where the chain is still open*. These links are drawn but are **not** registry `deps:` edges — nothing below may be read as an established implication:

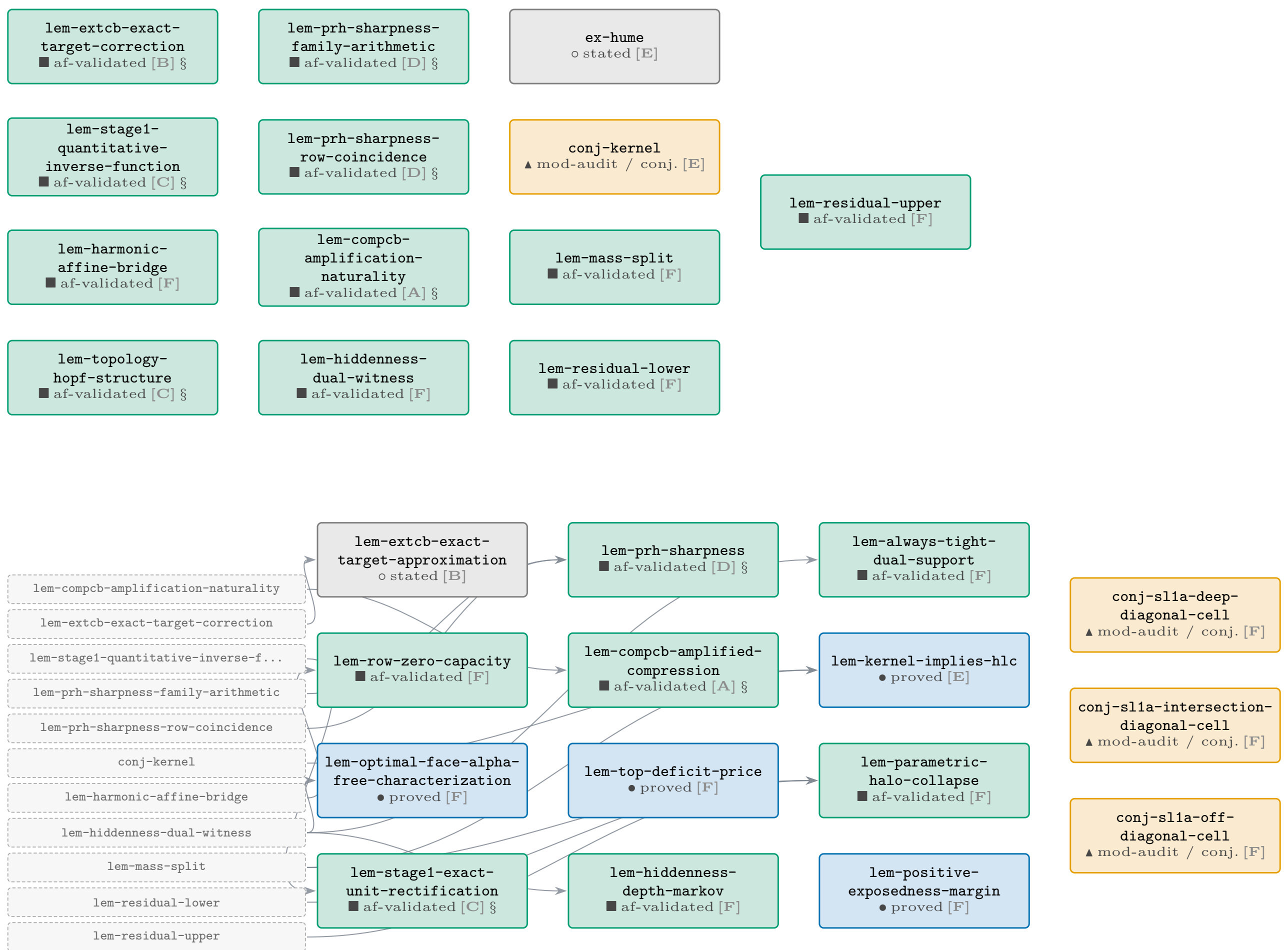
- `lem-routef-k-ledger` ⇢ `op-classical` — phase 5 (bd aism-y81y): F0 codification + root composition; docs/plans/CURRENT.md (v33 §The live campaign frontier: PHASE 4 preliminaries, elevation order and phase-5 root composition)
- `lem-routef-prh-finish` ⇢ `lem-routef-k-ledger` — the registered F2/F3 rows close the stochastic-retract bridge and "compose literally with lem-routef-prh-finish"; docs/plans/CURRENT.md (v33 §UNCHANGED)

Statistics of the drawn subgraph

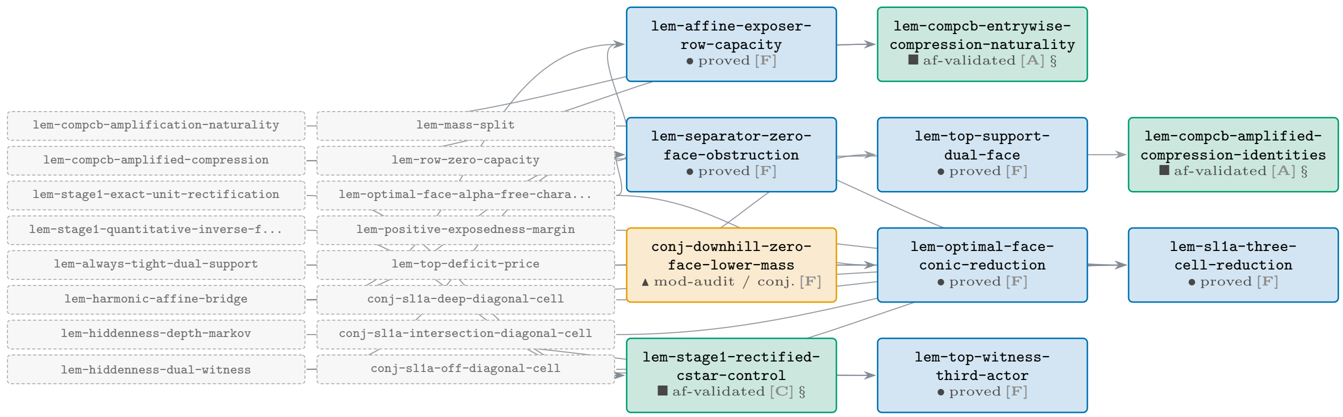
quantity	value	note
results drawn	221	of 374 in the registry
registry edges drawn	643	`deps:` 637, OR-route members 6
design-pending (GAP) links	2	drawn dashed; *not* registry edges
rigorous (L0)	**176**	`af: validated` or `status: cited` — 79.6% of the chain
non-rigorous	45	everything else, flagged loudly by colour and glyph
phase COMP/H-CB	27	COMP-CB / H-CB corner tower
phase EXT-CB	10	EXT-CB corner extension
phase Stage-1/topology	94	Stage-1 reset, topology leaves & the Kitaev repair
phase Route F	36	Route F bridges, PRH & the K-ledger
phase root	12	root, classical reduction & the exposed-hull spine
phase trunk	42	signed-geometry trunk (ancestors of the exposed-hull route)
`kind: corollary`	1	
`kind: lemma`	211	
`kind: obstruction`	1	
`kind: open-problem`	3	
`kind: proposition`	2	
`kind: theorem`	3	

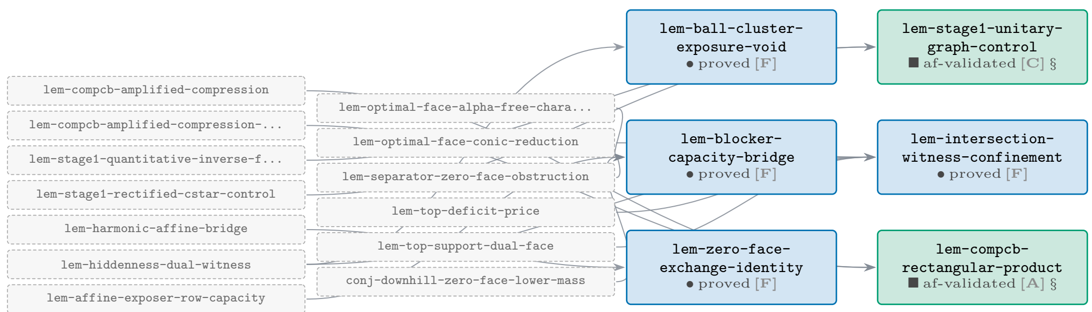
The Route-F proof chain for `op-classical` — complete sub-DAG

The complete Route-F landing chain: every registry result in the transitive `deps:/routes:` closure of `op-classical` together with the Route-F families. Dependencies flow **left to right**: a node sits one layer to the right of its deepest prerequisite, so the north star `op-classical` is on the far right of the last band. Bands read like text — finish a band, drop to the next.

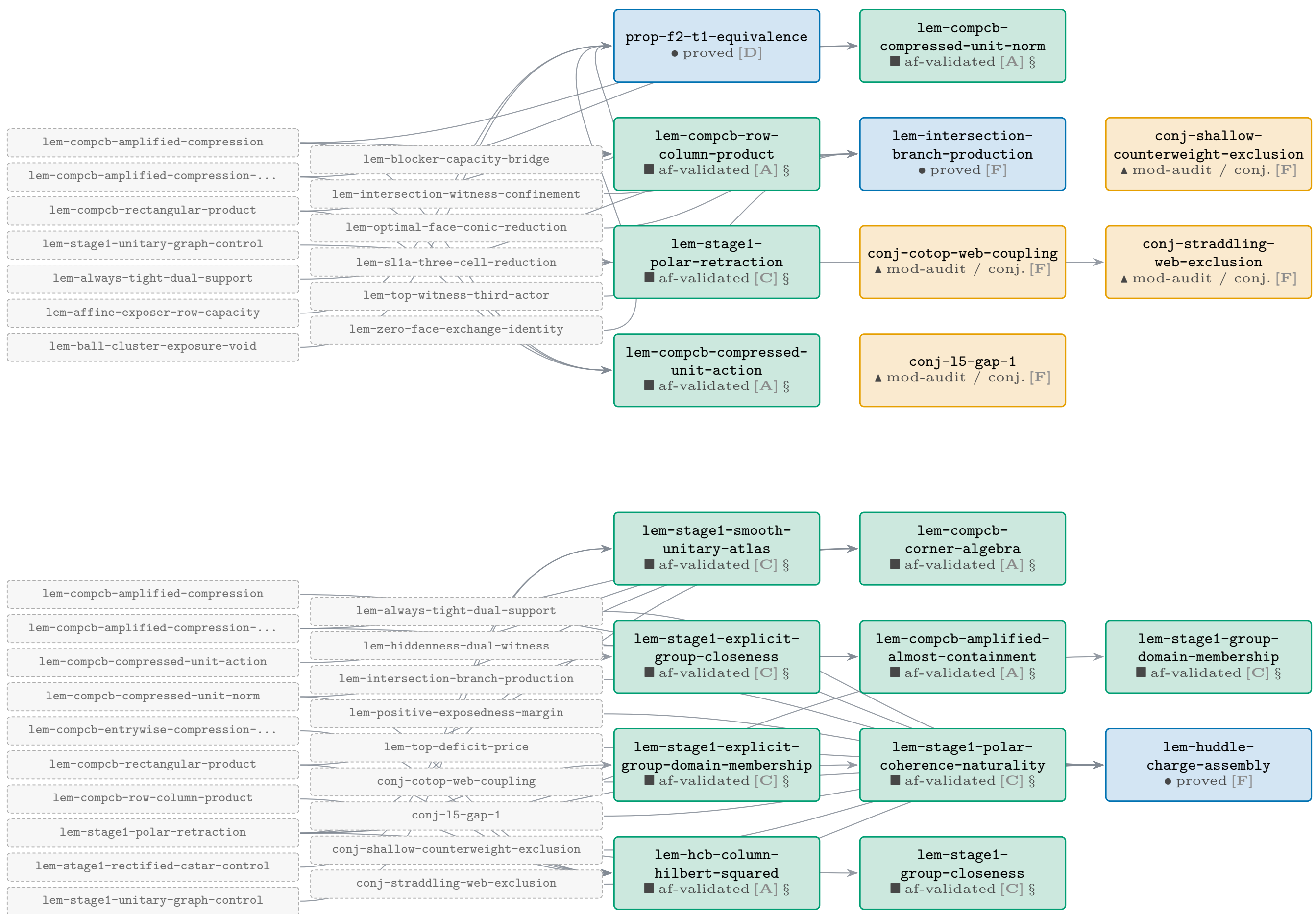


The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*

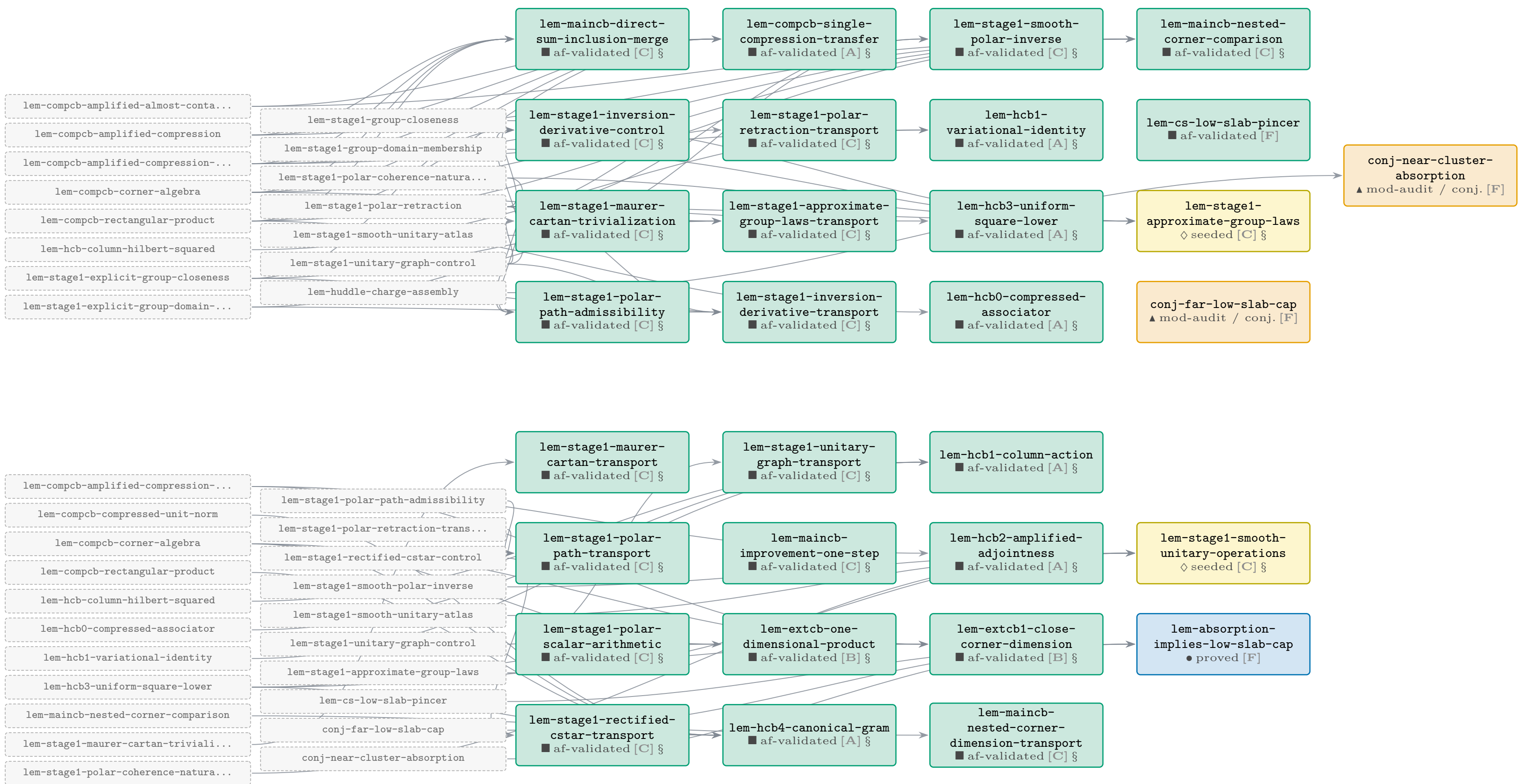




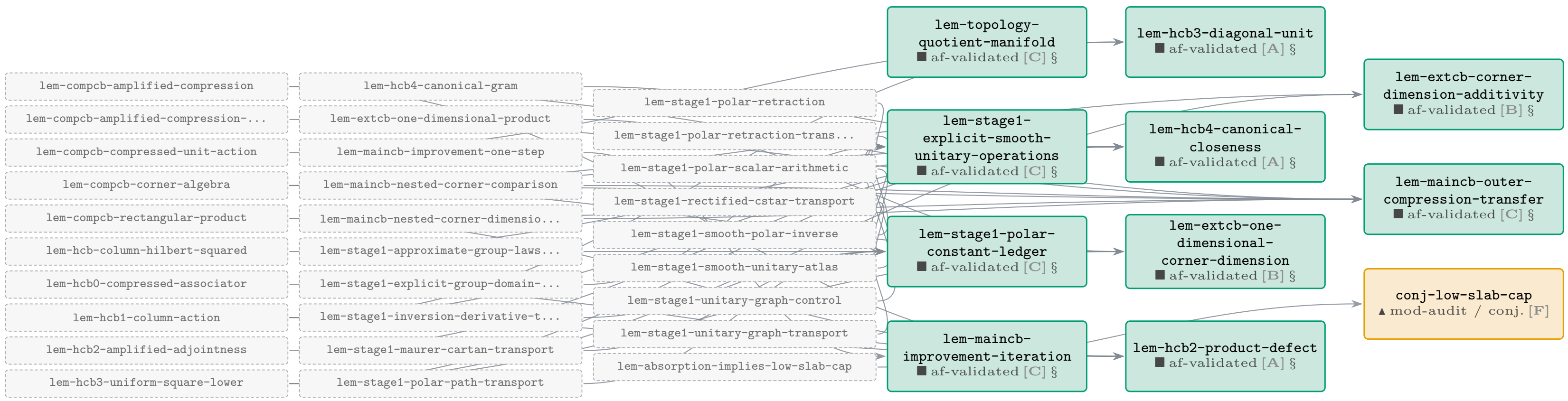
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*

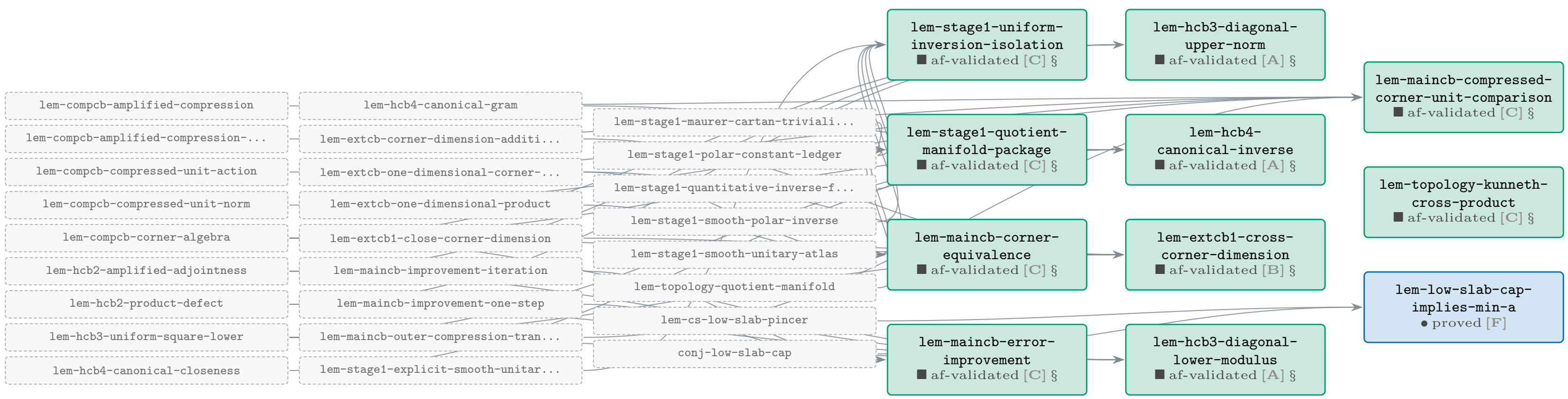


The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*

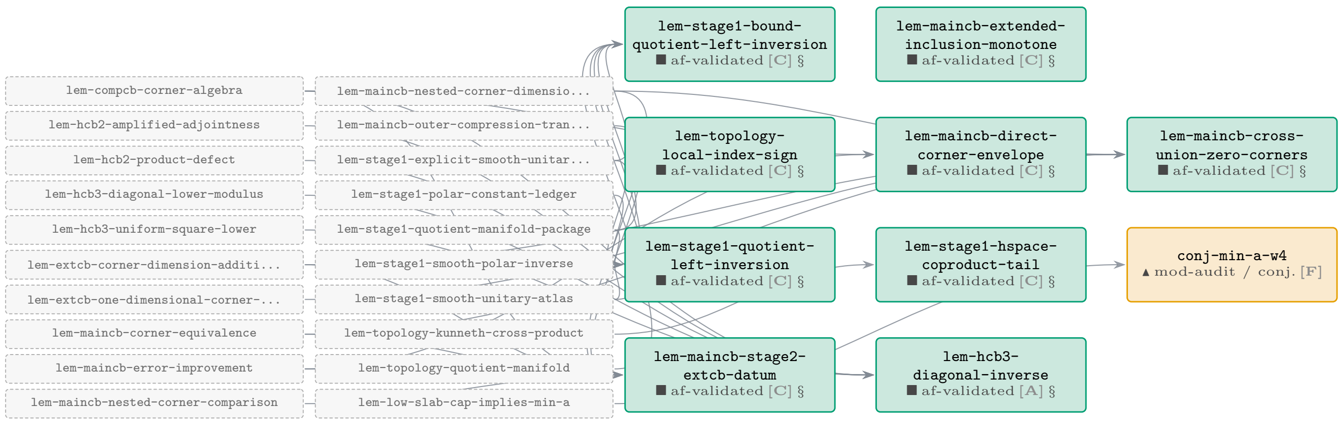


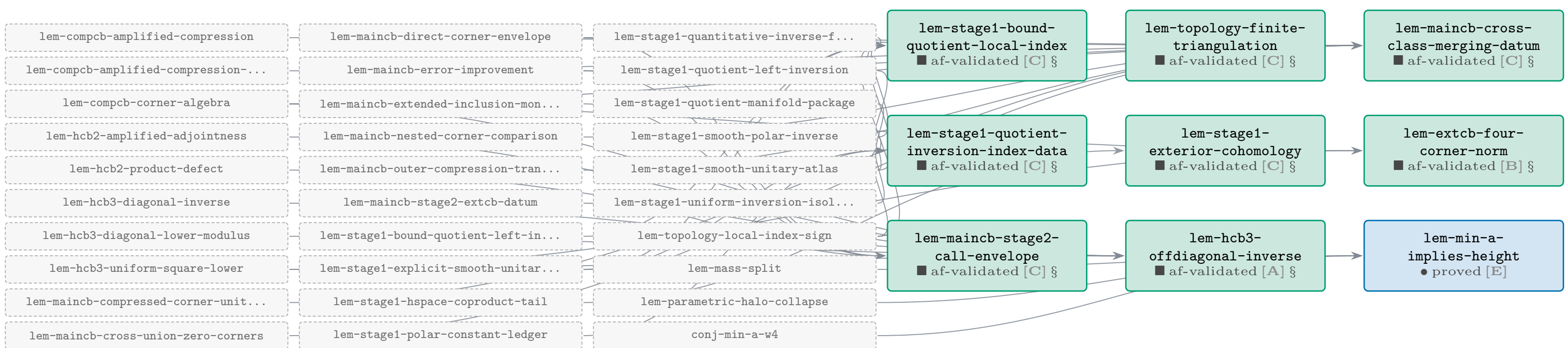
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



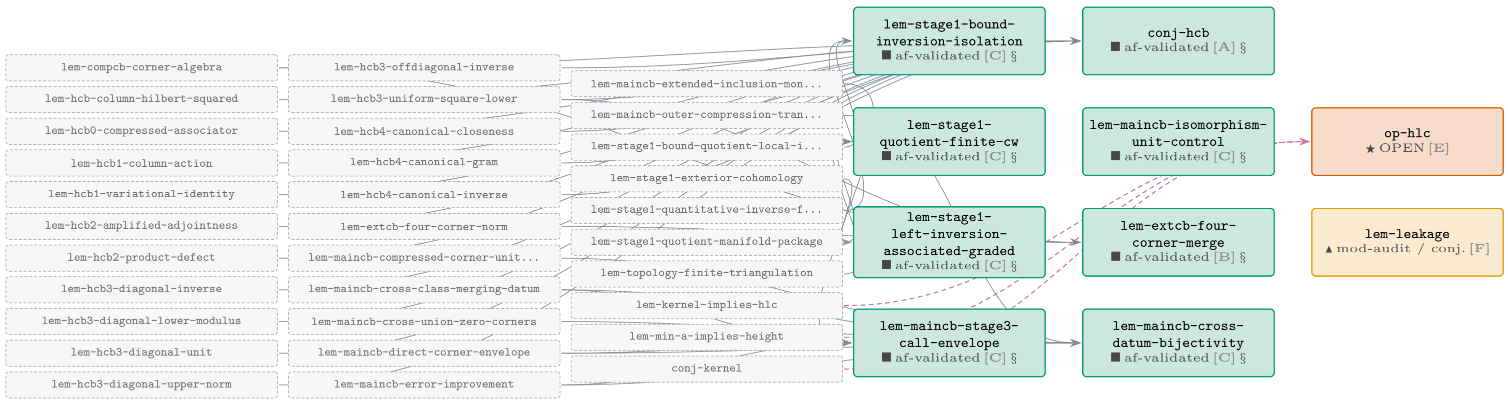


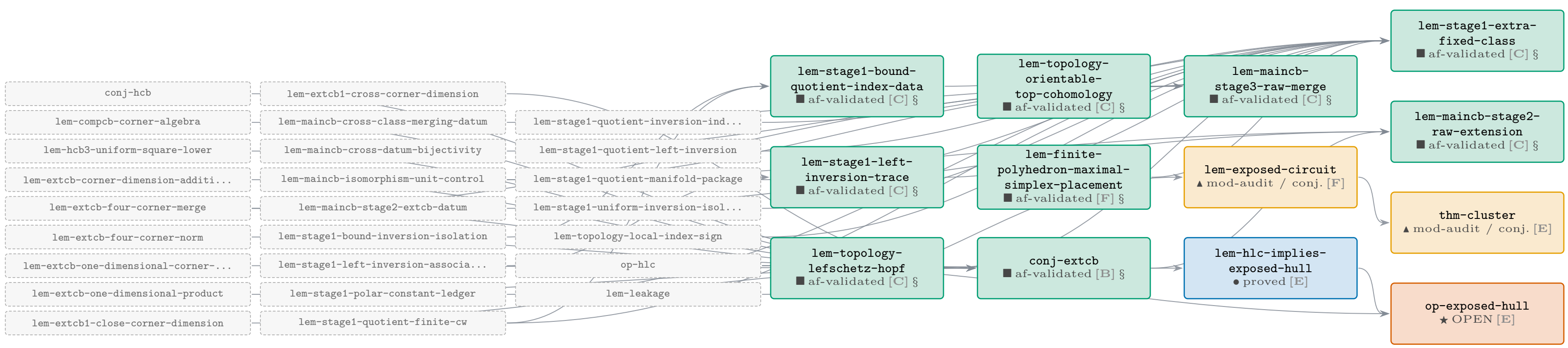
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



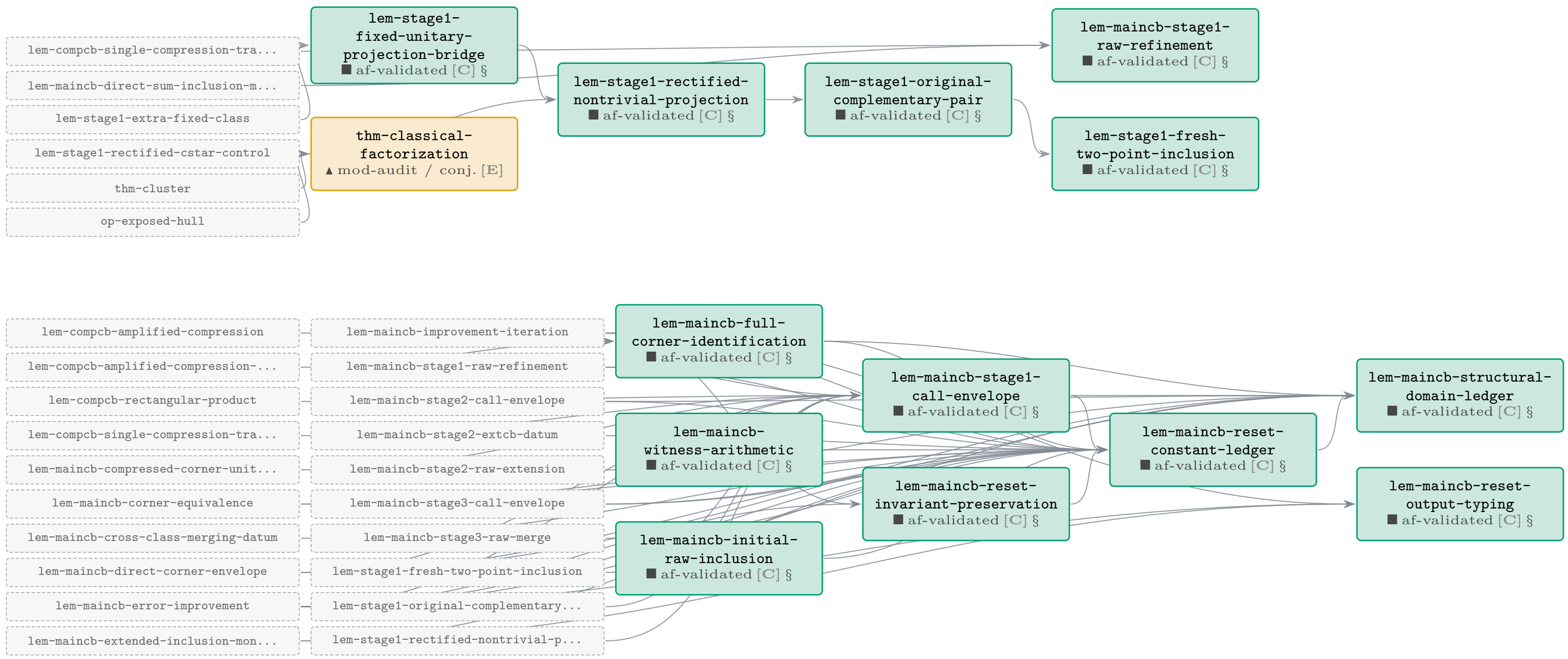


The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*

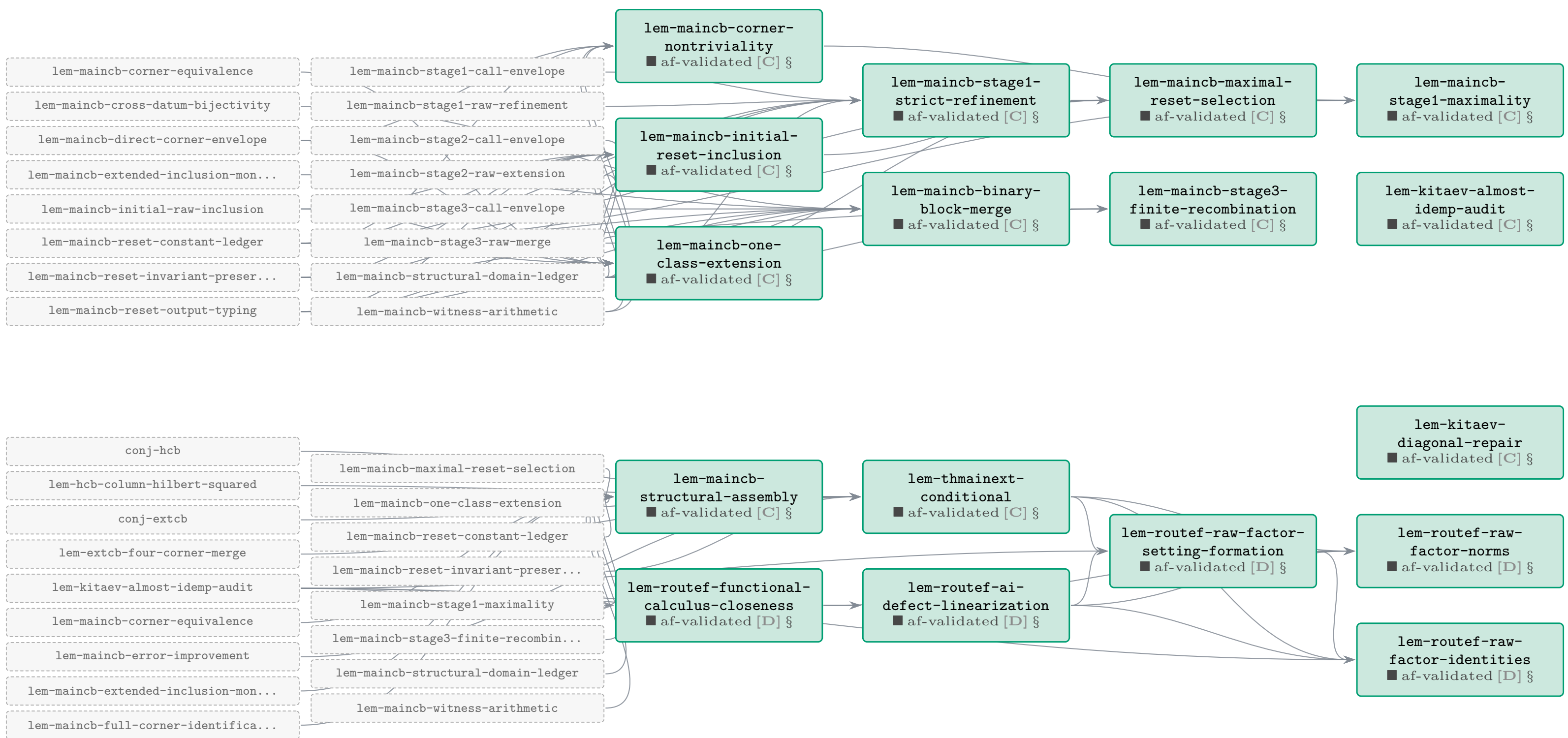




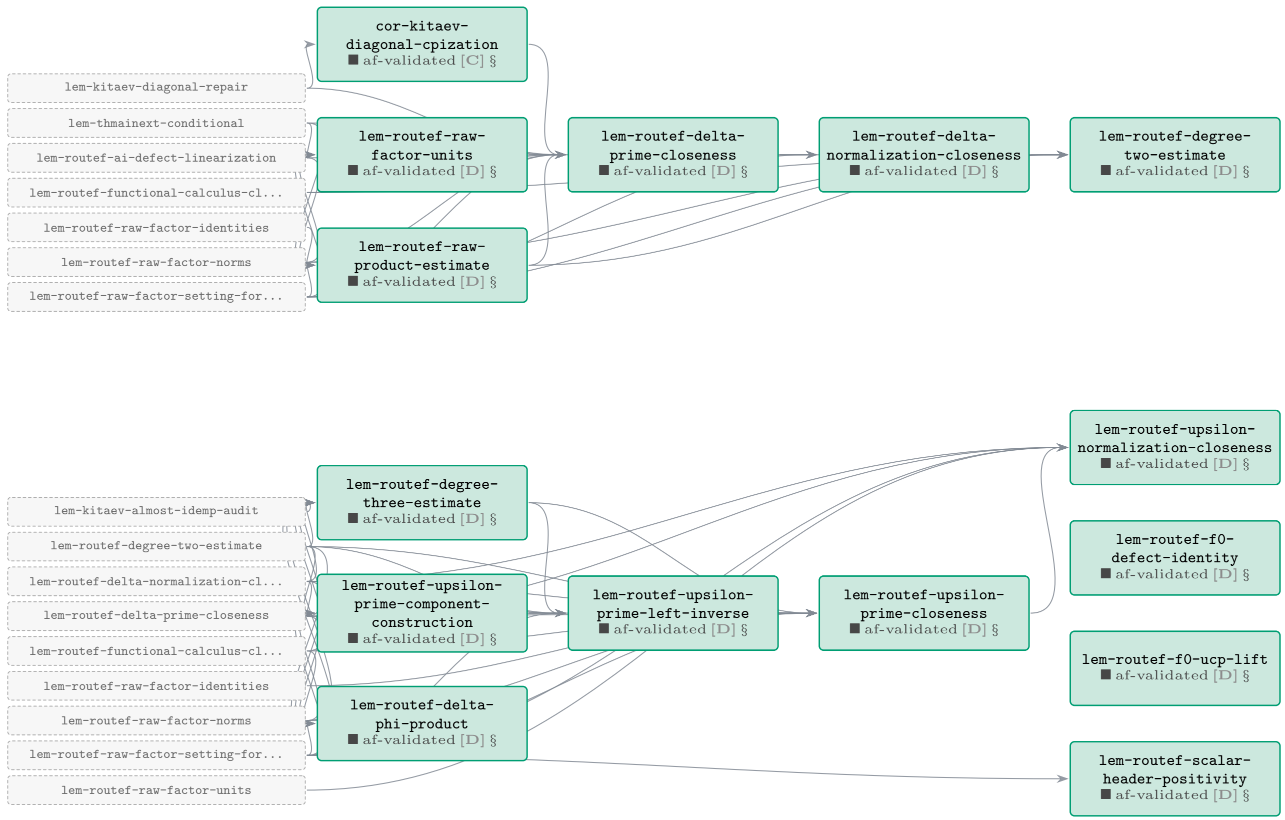
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



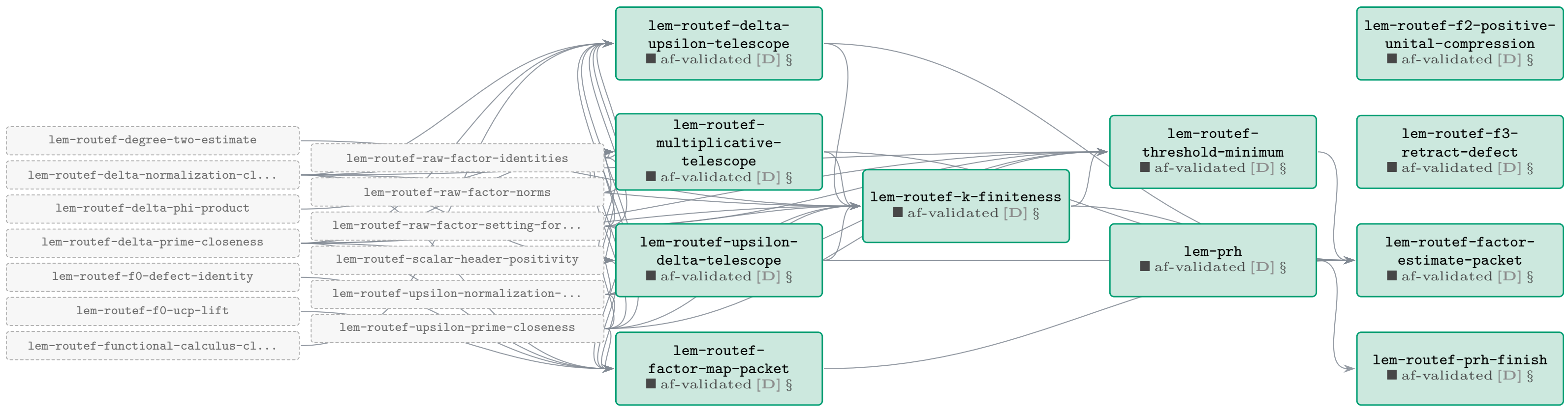
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*

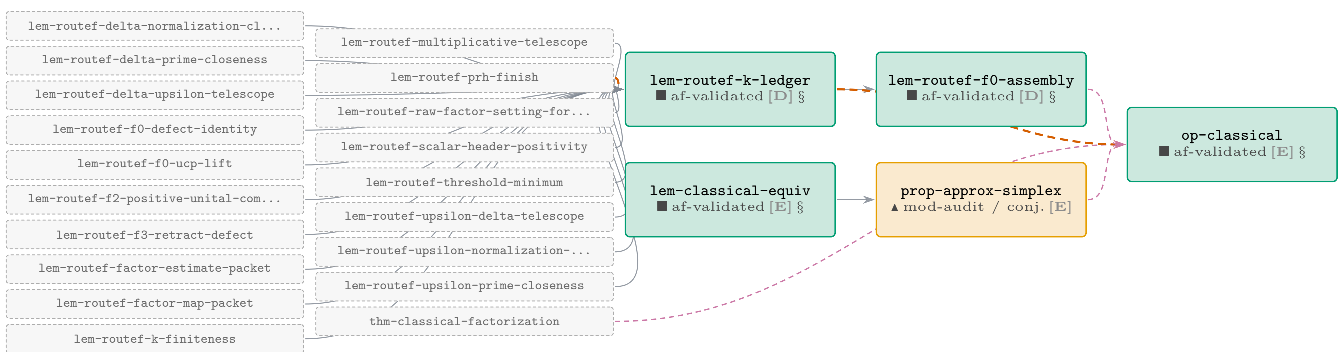


The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*



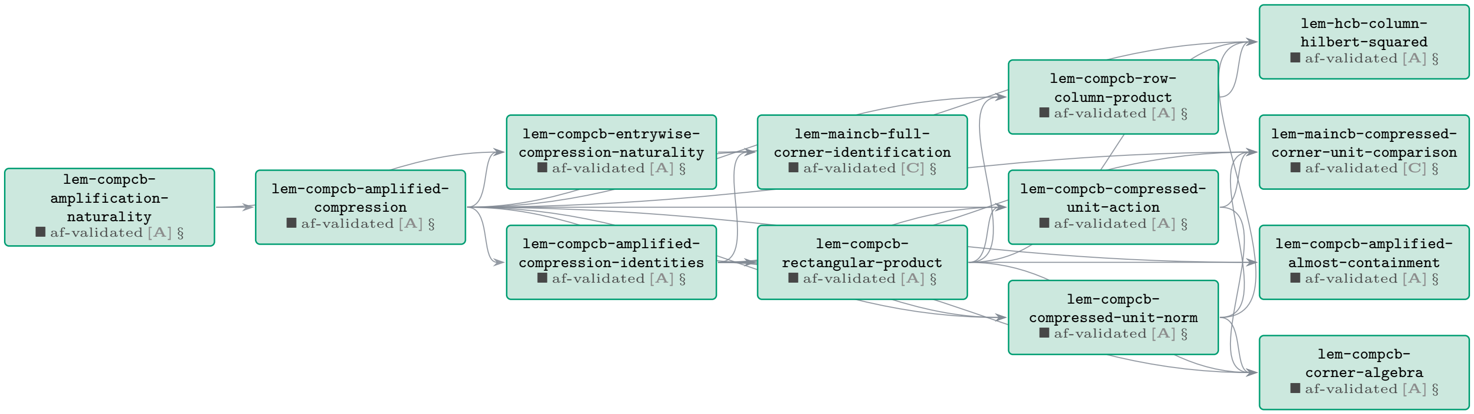
The Route-F proof chain for `op-classical` — complete sub-DAG *(cont.)*

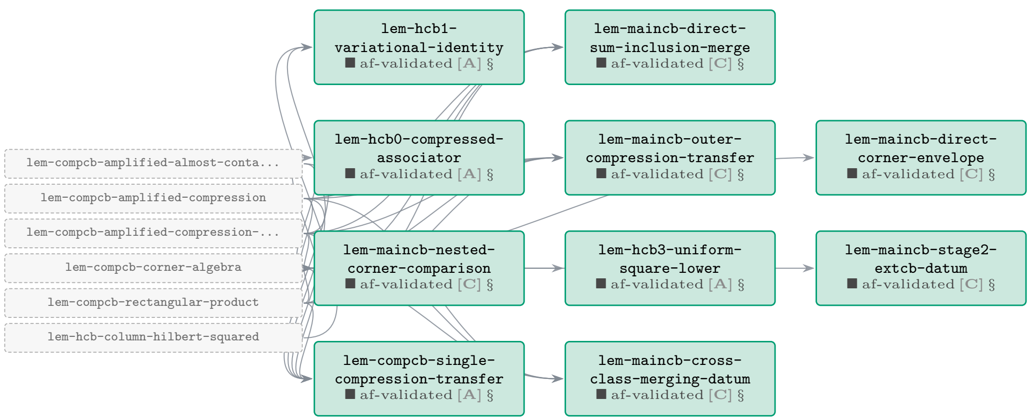




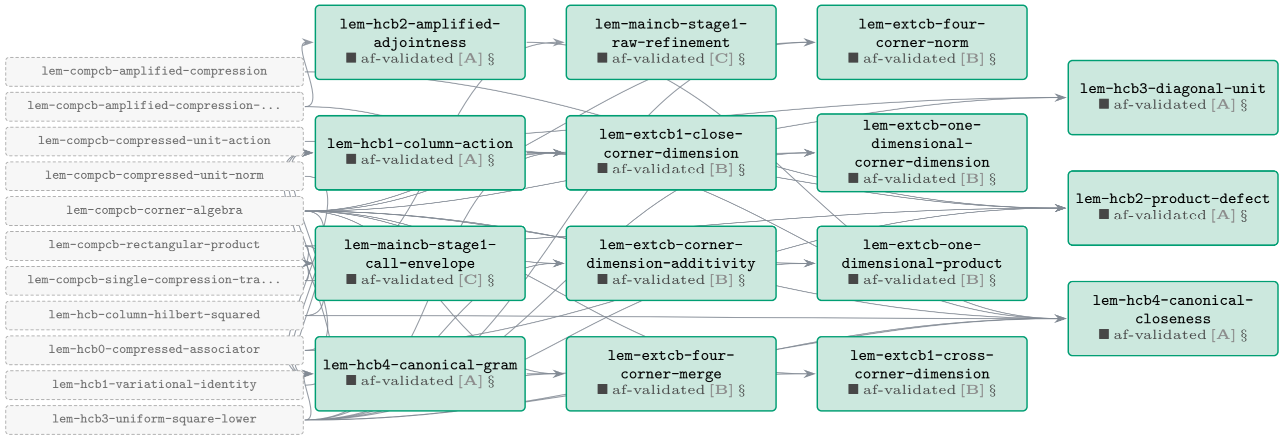
Phase A — COMP-CB / H-CB corner tower

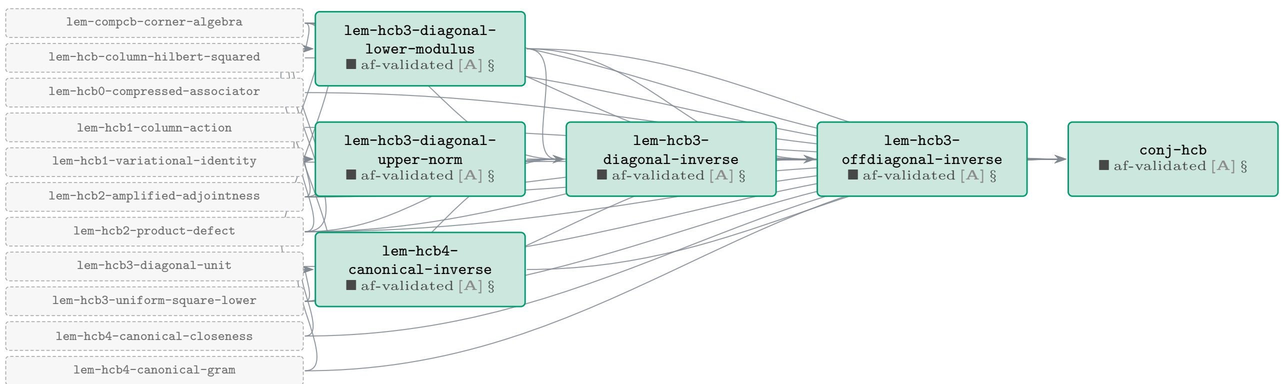
27 results (27 af-validated). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.



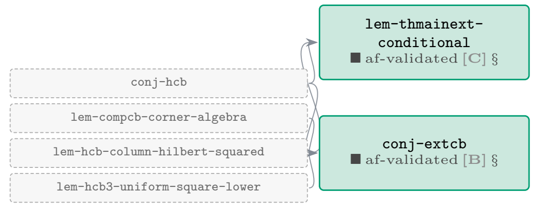


Phase A — COMP-CB / H-CB corner tower *(cont.)*



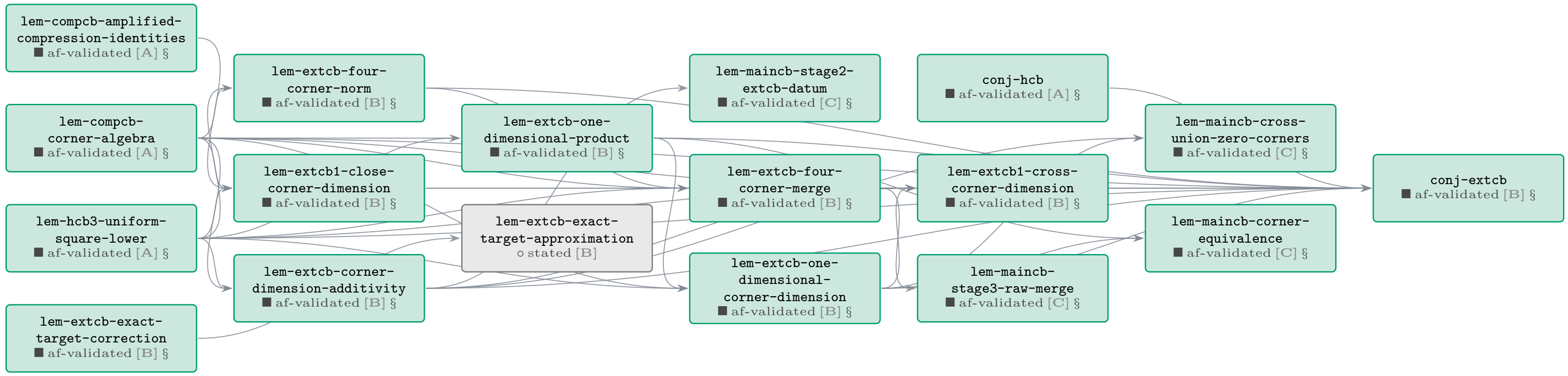


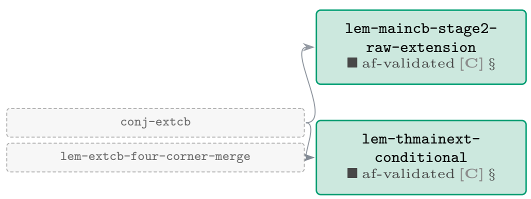
Phase A — COMP-CB / H-CB corner tower *(cont.)*



Phase B — EXT-CB corner extension

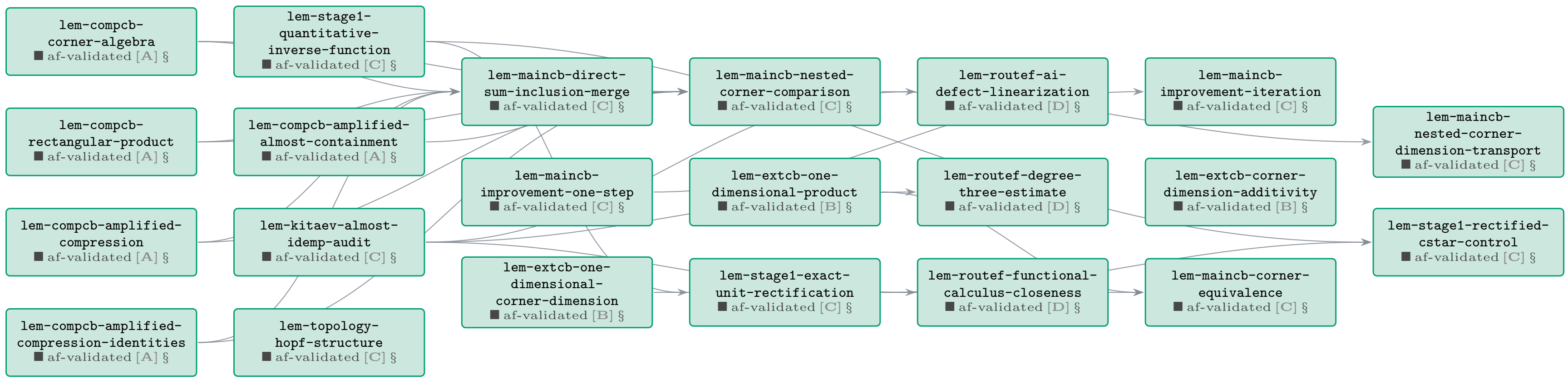
10 results (9 af-validated, 1 stated). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.

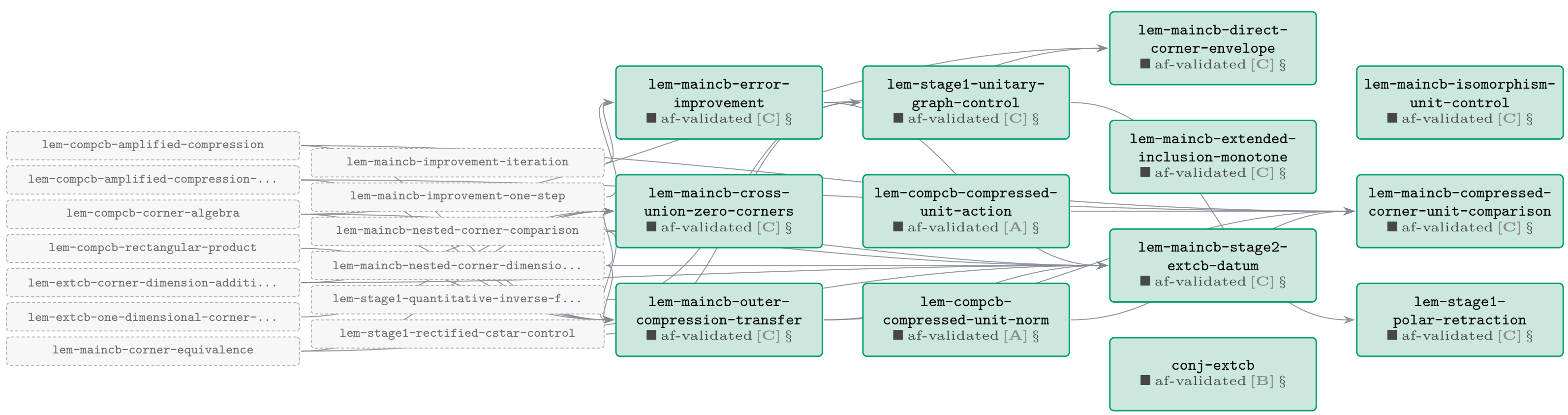




Phase C — Stage-1 reset, topology leaves & the Kitaev repair

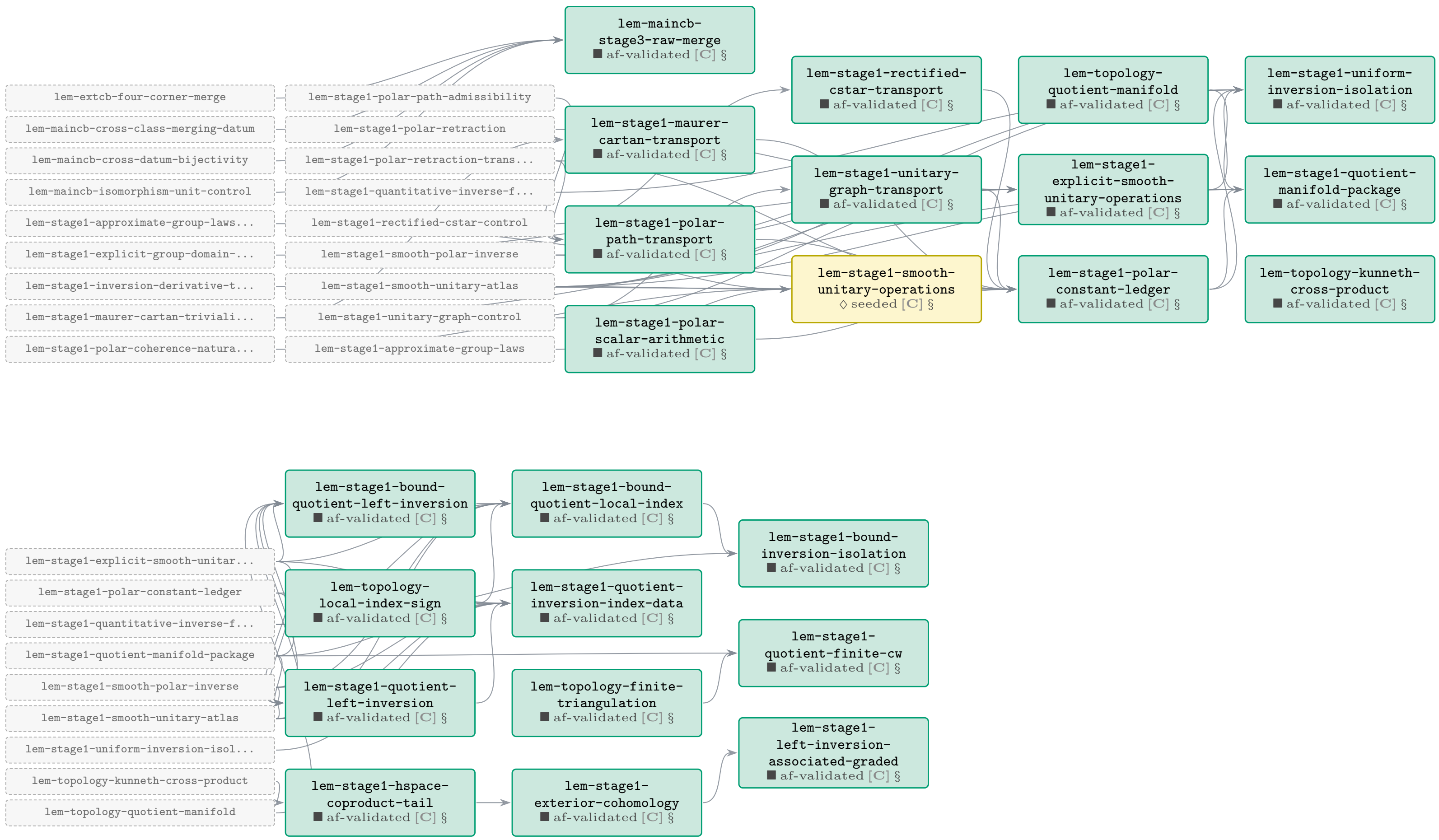
94 results (92 af-validated, 2 seeded). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.



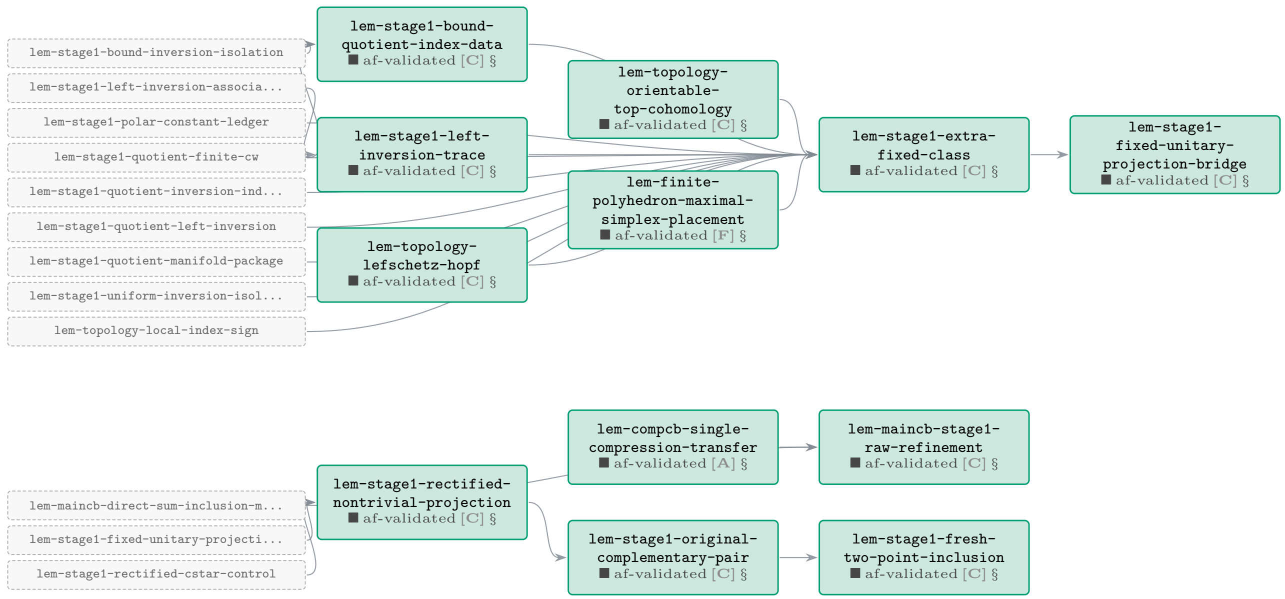


Phase C — Stage-1 reset, topology leaves & the Kitaev repair *(cont.)*

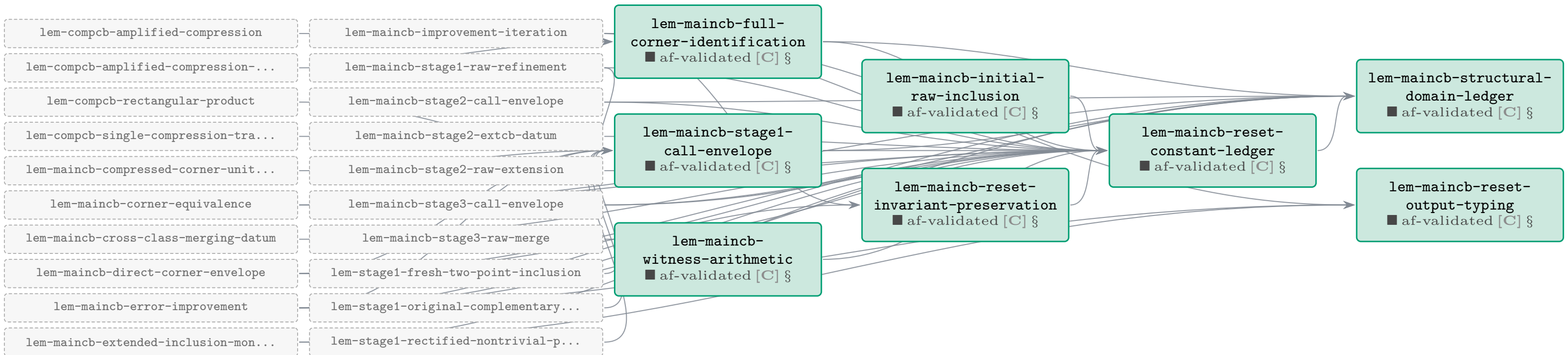
Phase C — Stage-1 reset, topology leaves & the Kitaev repair *(cont.)*

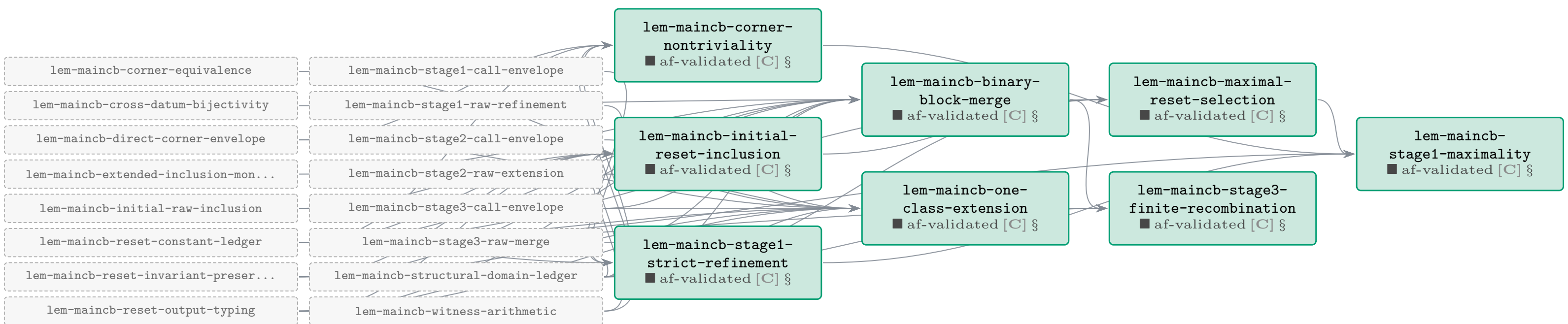


Phase C — Stage-1 reset, topology leaves & the Kitaev repair *(cont.)*

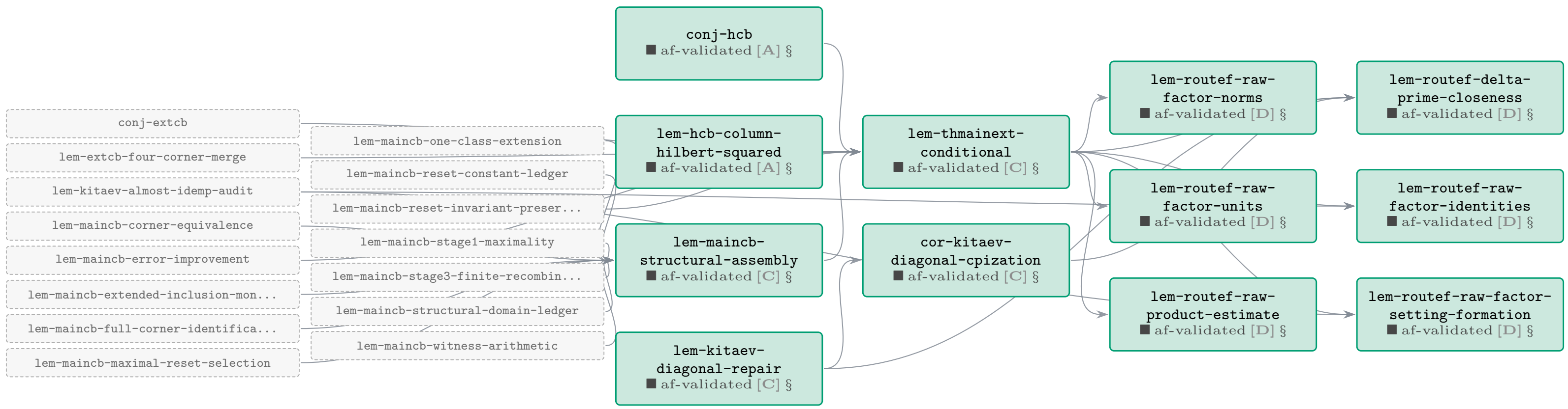


Phase C — Stage-1 reset, topology leaves & the Kitaev repair *(cont.)*



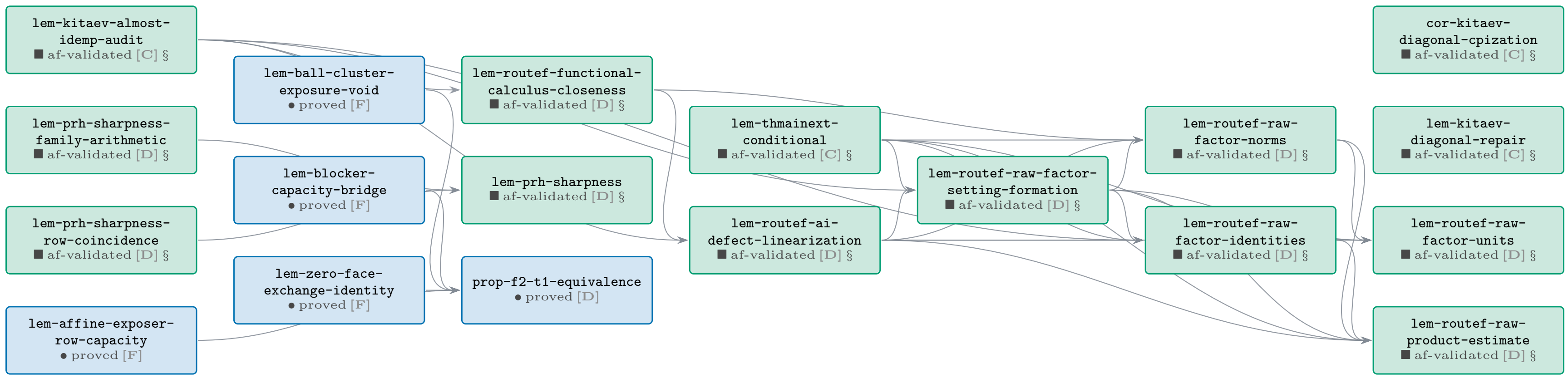


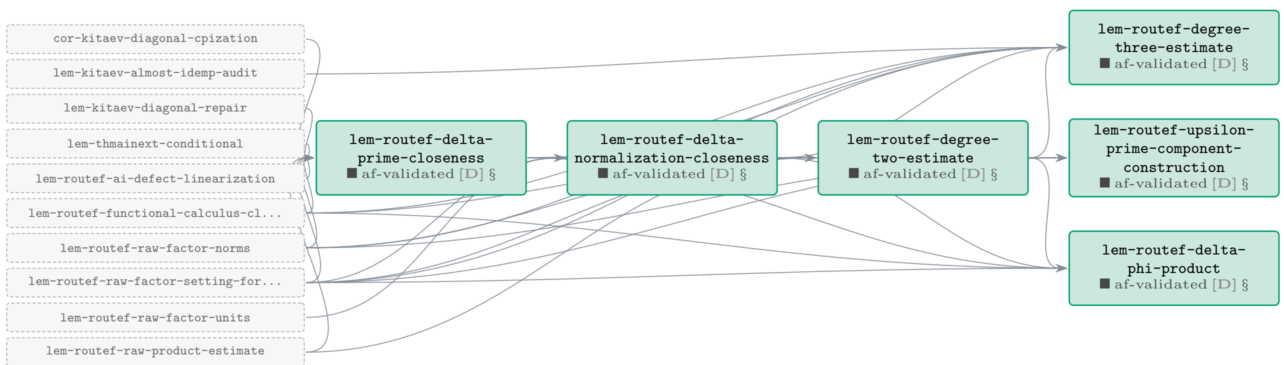
Phase C — Stage-1 reset, topology leaves & the Kitaev repair *(cont.)*



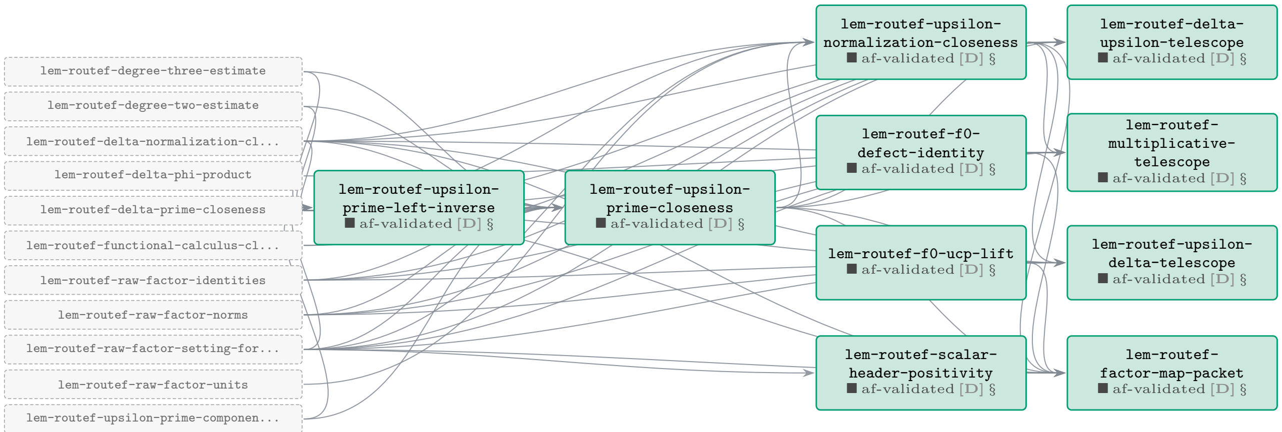
Phase D — Route F bridges, PRH & the K-ledger

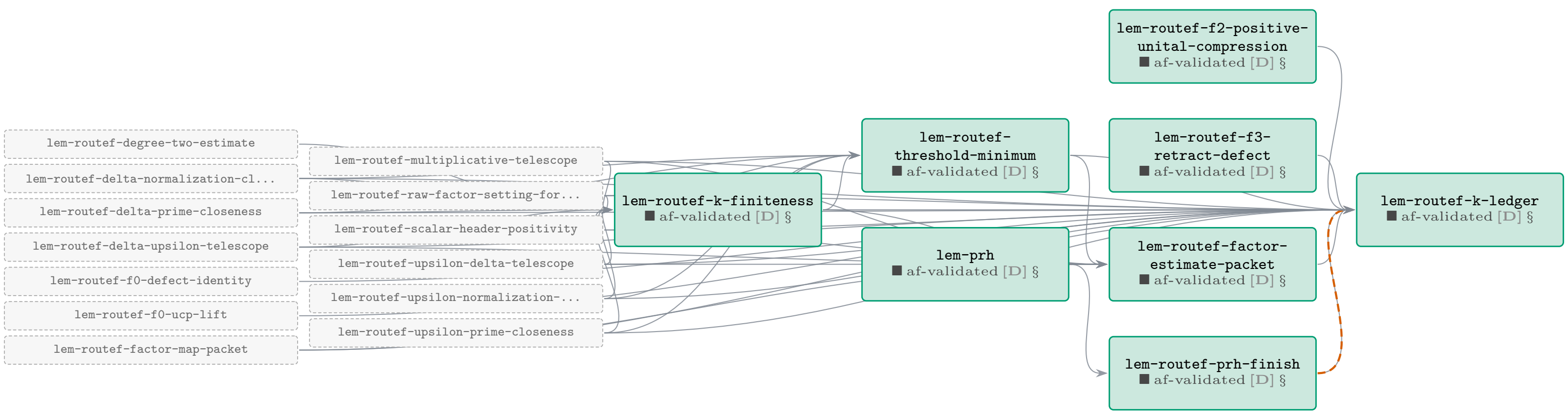
36 results (35 af-validated, 1 proved). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.



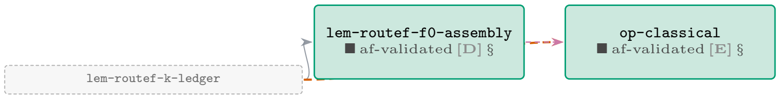


Phase D — Route F bridges, PRH & the K-ledger *(cont.)*



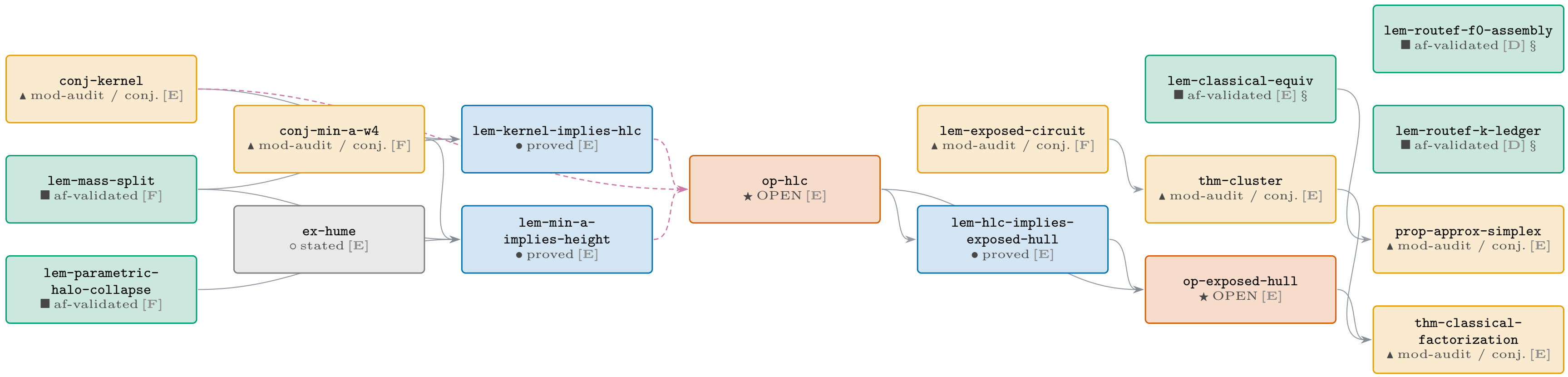


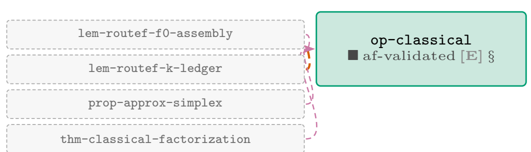
Phase D — Route F bridges, PRH & the K-ledger *(cont.)*



Phase E — root, classical reduction & the exposed-hull spine

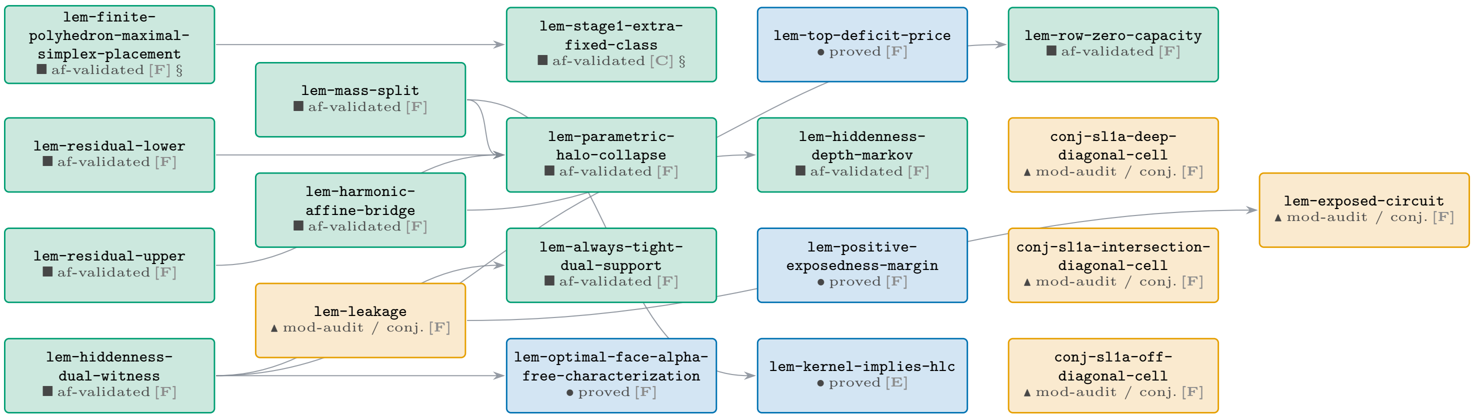
12 results (2 af-validated, 3 proved, 4 mod-audit / conj., 1 stated, 2 OPEN). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.

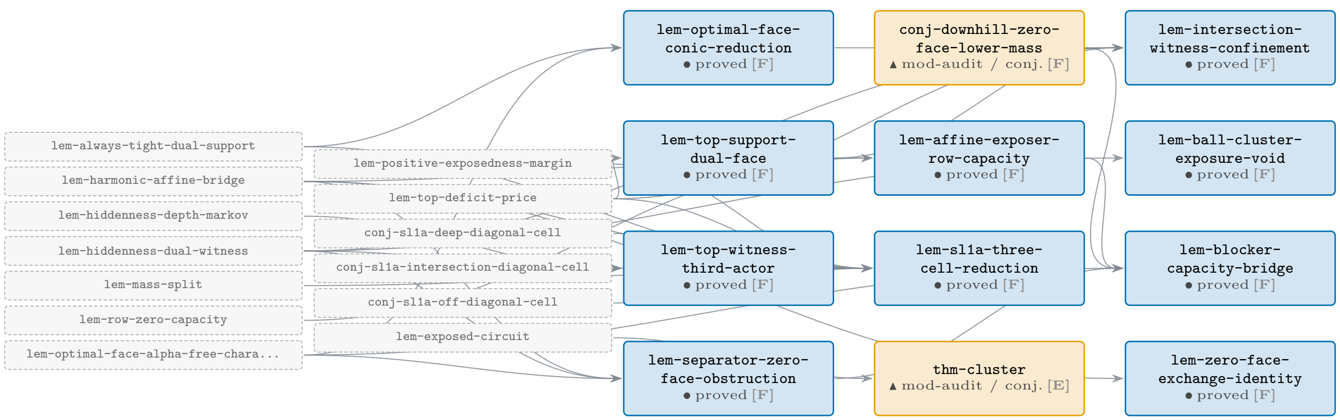




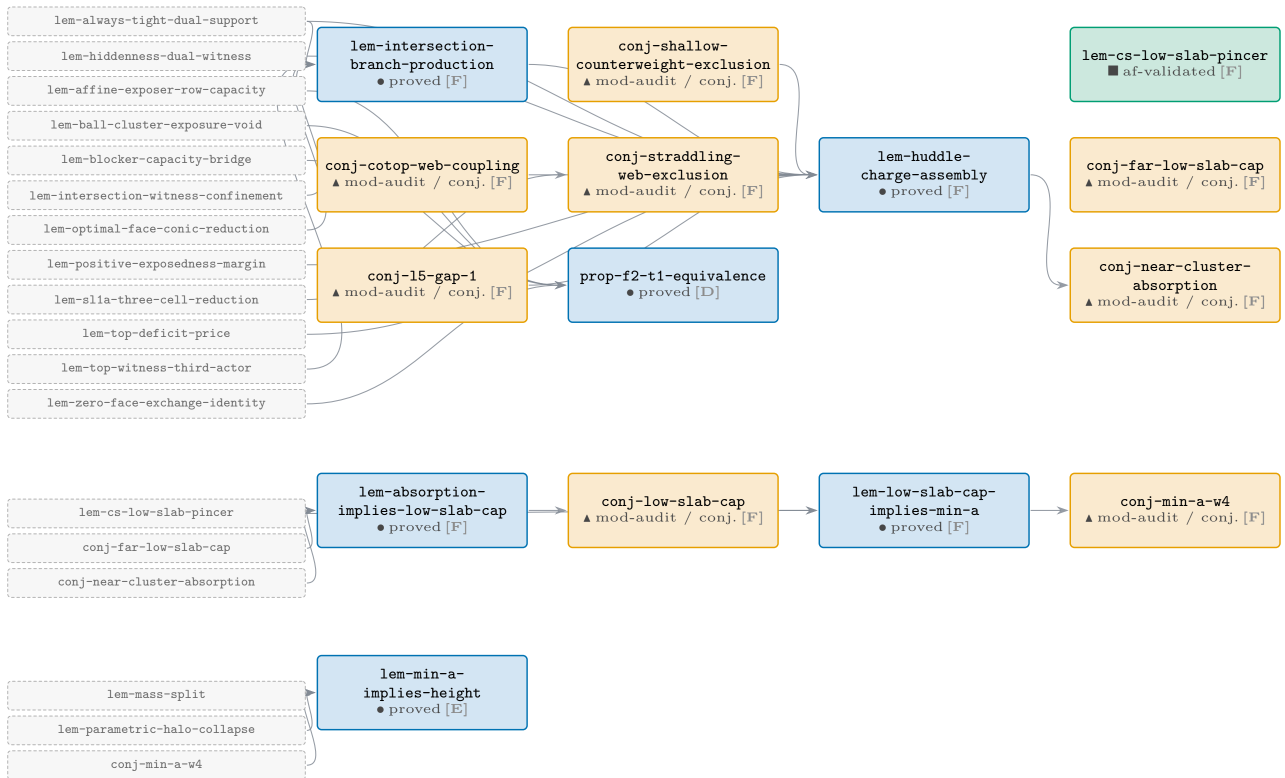
Phase F — signed-geometry trunk (ancestors of the exposed-hull route)

42 results (11 af-validated, 17 proved, 14 mod-audit / conj.). Boundary nodes of neighbouring phases are drawn as small dashed chips; click one to jump to its own page.





Phase F — signed-geometry trunk (ancestors of the exposed-hull route) *(cont.)*



Node index — every result in the chain

Anchors are `dag:<id>`. The **id** column links to the node in the atlas; **report** links to the typeset statement in this lab-book when one exists. Sorted by id (stable).

id	phase	kind	status	af	in	out	report
`conj-cotop-web-coupling`	trunk	lemma	conjecture	none	0	1	—
`conj-downhill-zero-face-lower-mass`	trunk	lemma	conjecture	none	0	1	—
`conj-extcb`	EXT-CB	lemma	proved	validated	10	2	§ conj:extcb
`conj-far-low-slab-cap`	trunk	lemma	conjecture	none	0	1	—
`conj-hcb`	COMP/H-CB	lemma	proved	validated	16	2	§ conj:hcb
`conj-kernel`	root	open-problem	conjecture	none	0	2	—
`conj-l5-gap-1`	trunk	lemma	conjecture	none	0	1	—
`conj-low-slab-cap`	trunk	lemma	conjecture	none	1	1	—
`conj-min-a-w4`	trunk	lemma	conjecture	none	1	1	—
`conj-near-cluster-absorption`	trunk	lemma	conjecture	none	1	1	—
`conj-shallow-counterweight-exclusion`	trunk	lemma	conjecture	none	0	1	—
`conj-sl1a-deep-diagonal-cell`	trunk	lemma	conjecture	none	0	1	—
`conj-sl1a-intersection-diagonal-cell`	trunk	lemma	conjecture	none	0	1	—
`conj-sl1a-off-diagonal-cell`	trunk	lemma	conjecture	none	0	1	—
`conj-straddling-web-exclusion`	trunk	lemma	conjecture	none	1	1	—
`cor-kitaev-diagonal-cpization`	Stage-1/topology	corollary	proved	validated	1	1	§ cor:kitaev-diagonal-cpization
`ex-hume`	root	obstruction	disproved	none	0	0	—
`lem-absorption-implies-low-slab-cap`	trunk	lemma	proved	none	3	1	—
`lem-affine-exposer-row-capacity`	trunk	lemma	proved	none	2	2	—
`lem-always-tight-dual-support`	trunk	lemma	proved	validated	1	4	—
`lem-ball-cluster-exposure-void`	trunk	lemma	proved	none	1	1	—
`lem-blocker-capacity-bridge`	trunk	lemma	proved	none	4	1	—
`lem-classical-equiv`	root	lemma	proved	validated	0	1	§ lem:classical-equiv
`lem-compcb-amplification-naturality`	COMP/H-CB	lemma	proved	validated	0	2	§ lem:compcb-amplification-naturality
`lem-compcb-amplified-almost-containment`	COMP/H-CB	lemma	proved	validated	2	2	§ lem:compcb-amplified-almost-containment
`lem-compcb-amplified-compression`	COMP/H-CB	lemma	proved	validated	1	15	§ lem:compcb-amplified-compression
`lem-compcb-amplified-compression-identities`	COMP/H-CB	lemma	proved	validated	1	15	§ lem:compcb-amplified-compression-identities
`lem-compcb-compressed-unit-action`	COMP/H-CB	lemma	proved	validated	3	3	§ lem:compcb-compressed-unit-action
`lem-compcb-compressed-unit-norm`	COMP/H-CB	lemma	proved	validated	2	4	§ lem:compcb-compressed-unit-norm
`lem-compcb-corner-algebra`	COMP/H-CB	lemma	proved	validated	4	24	§ lem:compcb-corner-algebra
`lem-compcb-entrywise-compression-naturality`	COMP/H-CB	lemma	proved	validated	2	1	§ lem:compcb-entrywise-compression-naturality
`lem-compcb-rectangular-product`	COMP/H-CB	lemma	proved	validated	2	12	§ lem:compcb-rectangular-product
`lem-compcb-row-column-product`	COMP/H-CB	lemma	proved	validated	2	1	§ lem:compcb-row-column-product
`lem-compcb-single-compression-transfer`	COMP/H-CB	lemma	proved	validated	5	2	§ lem:compcb-single-compression-transfer
`lem-cs-low-slab-pincer`	trunk	lemma	proved	validated	0	2	—
`lem-exposed-circuit`	trunk	lemma	proved-mod-audit	none	1	1	—
`lem-extcb-corner-dimension-additivity`	EXT-CB	lemma	proved	validated	2	4	§ lem:extcb-corner-dimension-additivity
`lem-extcb-exact-target-approximation`	EXT-CB	lemma	stated	none	1	0	—
`lem-extcb-exact-target-correction`	EXT-CB	lemma	proved	validated	0	1	§ lem:extcb-exact-target-correction
`lem-extcb-four-corner-merge`	EXT-CB	lemma	proved	validated	3	3	§ lem:extcb-four-corner-merge
`lem-extcb-four-corner-norm`	EXT-CB	lemma	proved	validated	2	2	§ lem:extcb-four-corner-norm

id	phase	kind	status	af	in	out	report
lem-extcb-one-dimensional-corner-dimension	EXT-CB	lemma	proved	validated	3	4	§ lem:extcb-one-dimensional-corner-dimension
lem-extcb-one-dimensional-product	EXT-CB	lemma	proved	validated	2	3	§ lem:extcb-one-dimensional-product
lem-extcb1-close-corner-dimension	EXT-CB	lemma	proved	validated	3	2	§ lem:extcb1-close-corner-dimension
lem-extcb1-cross-corner-dimension	EXT-CB	lemma	proved	validated	5	1	§ lem:extcb1-cross-corner-dimension
lem-finite-polyhedron-maximal-simplex-placement	trunk	lemma	proved	validated	0	1	§ lem:finite-polyhedron-maximal-simplex-placem
lem-harmonic-affine-bridge	trunk	lemma	proved	validated	0	5	—
lem-hcb-column-hilbert-squared	COMP/H-CB	lemma	proved	validated	4	5	§ lem:hcb-column-hilbert-squared
lem-hcb0-compressed-associator	COMP/H-CB	lemma	proved	validated	1	3	§ lem:hcb0-compressed-associator
lem-hcb1-column-action	COMP/H-CB	lemma	proved	validated	3	4	§ lem:hcb1-column-action
lem-hcb1-variational-identity	COMP/H-CB	lemma	proved	validated	2	2	§ lem:hcb1-variational-identity
lem-hcb2-amplified-adjointness	COMP/H-CB	lemma	proved	validated	1	6	§ lem:hcb2-amplified-adjointness
lem-hcb2-product-defect	COMP/H-CB	lemma	proved	validated	4	5	§ lem:hcb2-product-defect
lem-hcb3-diagonal-inverse	COMP/H-CB	lemma	proved	validated	5	2	§ lem:hcb3-diagonal-inverse
lem-hcb3-diagonal-lower-modulus	COMP/H-CB	lemma	proved	validated	4	3	§ lem:hcb3-diagonal-lower-modulus
lem-hcb3-diagonal-unit	COMP/H-CB	lemma	proved	validated	2	1	§ lem:hcb3-diagonal-unit
lem-hcb3-diagonal-upper-norm	COMP/H-CB	lemma	proved	validated	3	1	§ lem:hcb3-diagonal-upper-norm
lem-hcb3-offdiagonal-inverse	COMP/H-CB	lemma	proved	validated	6	1	§ lem:hcb3-offdiagonal-inverse
lem-hcb3-uniform-square-lower	COMP/H-CB	lemma	proved	validated	1	15	§ lem:hcb3-uniform-square-lower
lem-hcb4-canonical-closeness	COMP/H-CB	lemma	proved	validated	6	2	§ lem:hcb4-canonical-closeness
lem-hcb4-canonical-gram	COMP/H-CB	lemma	proved	validated	4	3	§ lem:hcb4-canonical-gram
lem-hcb4-canonical-inverse	COMP/H-CB	lemma	proved	validated	4	1	§ lem:hcb4-canonical-inverse
lem-hiddenness-depth-markov	trunk	lemma	proved	validated	1	1	—
lem-hiddenness-dual-witness	trunk	lemma	proved	validated	0	8	—
lem-hlc-implies-exposed-hull	root	lemma	proved	none	1	1	—
lem-huddle-charge-assembly	trunk	lemma	proved	none	9	1	—
lem-intersection-branch-production	trunk	lemma	proved	none	4	1	—
lem-intersection-witness-confinement	trunk	lemma	proved	none	4	1	—
lem-kernel-implies-hlc	root	lemma	proved	none	2	1	—
lem-kitaev-almost-idemp-audit	Stage-1/topology	lemma	proved	validated	0	5	§ lem:kitaev-almost-idemp-audit
lem-kitaev-diagonal-repair	Stage-1/topology	lemma	proved	validated	0	2	§ lem:kitaev-diagonal-repair
lem-leakage	trunk	lemma	proved-mod-audit	none	0	1	—
lem-low-slab-cap-implies-min-a	trunk	lemma	proved	none	2	1	—
lem-maincb-binary-block-merge	Stage-1/topology	lemma	proved	validated	8	1	§ lem:maincb-binary-block-merge
lem-maincb-compressed-corner-unit-comparison	Stage-1/topology	lemma	proved	validated	5	3	§ lem:maincb-compressed-corner-unit-compariso
lem-maincb-corner-equivalence	Stage-1/topology	lemma	proved	validated	2	6	§ lem:maincb-corner-equivalence
lem-maincb-corner-nontriviality	Stage-1/topology	lemma	proved	validated	4	1	§ lem:maincb-corner-nontriviality
lem-maincb-cross-class-merging-datum	Stage-1/topology	lemma	proved	validated	9	4	§ lem:maincb-cross-class-merging-datum
lem-maincb-cross-datum-bijectivity	Stage-1/topology	lemma	proved	validated	6	2	§ lem:maincb-cross-datum-bijectivity
lem-maincb-cross-union-zero-corners	Stage-1/topology	lemma	proved	validated	4	2	§ lem:maincb-cross-union-zero-corners
lem-maincb-direct-corner-envelope	Stage-1/topology	lemma	proved	validated	2	9	§ lem:maincb-direct-corner-envelope
lem-maincb-direct-sum-inclusion-merge	Stage-1/topology	lemma	proved	validated	4	1	§ lem:maincb-direct-sum-inclusion-merge
lem-maincb-error-improvement	Stage-1/topology	lemma	proved	validated	2	12	§ lem:maincb-error-improvement
lem-maincb-extended-inclusion-monotone	Stage-1/topology	lemma	proved	validated	0	6	§ lem:maincb-extended-inclusion-monotone
lem-maincb-full-corner-identification	Stage-1/topology	lemma	proved	validated	2	3	§ lem:maincb-full-corner-identification
lem-maincb-improvement-iteration	Stage-1/topology	lemma	proved	validated	1	4	§ lem:maincb-improvement-iteration
lem-maincb-improvement-one-step	Stage-1/topology	lemma	proved	validated	0	2	§ lem:maincb-improvement-one-step
lem-maincb-initial-raw-inclusion	Stage-1/topology	lemma	proved	validated	0	3	§ lem:maincb-initial-raw-inclusion

id	phase	kind	status	af	in	out	report
lem-maincb-initial-reset-inclusion	Stage-1/topology	lemma	proved	validated	6	1	§ lem:maincb-initial-reset-inclusion
lem-maincb-isomorphism-unit-control	Stage-1/topology	lemma	proved	validated	0	2	§ lem:maincb-isomorphism-unit-control
lem-maincb-maximal-reset-selection	Stage-1/topology	lemma	proved	validated	3	2	§ lem:maincb-maximal-reset-selection
lem-maincb-nested-corner-comparison	Stage-1/topology	lemma	proved	validated	5	4	§ lem:maincb-nested-corner-comparison
lem-maincb-nested-corner-dimension-transport	Stage-1/topology	lemma	proved	validated	1	3	§ lem:maincb-nested-corner-dimension-transport
lem-maincb-one-class-extension	Stage-1/topology	lemma	proved	validated	9	1	§ lem:maincb-one-class-extension
lem-maincb-outer-compression-transfer	Stage-1/topology	lemma	proved	validated	6	4	§ lem:maincb-outer-compression-transfer
lem-maincb-reset-constant-ledger	Stage-1/topology	lemma	proved	validated	17	10	§ lem:maincb-reset-constant-ledger
lem-maincb-reset-invariant-preservation	Stage-1/topology	lemma	proved	validated	2	6	§ lem:maincb-reset-invariant-preservation
lem-maincb-reset-output-typing	Stage-1/topology	lemma	proved	validated	3	4	§ lem:maincb-reset-output-typing
lem-maincb-stage1-call-envelope	Stage-1/topology	lemma	proved	validated	9	3	§ lem:maincb-stage1-call-envelope
lem-maincb-stage1-maximality	Stage-1/topology	lemma	proved	validated	4	1	§ lem:maincb-stage1-maximality
lem-maincb-stage1-raw-refinement	Stage-1/topology	lemma	proved	validated	2	3	§ lem:maincb-stage1-raw-refinement
lem-maincb-stage1-strict-refinement	Stage-1/topology	lemma	proved	validated	6	1	§ lem:maincb-stage1-strict-refinement
lem-maincb-stage2-call-envelope	Stage-1/topology	lemma	proved	validated	4	3	§ lem:maincb-stage2-call-envelope
lem-maincb-stage2-extcb-datum	Stage-1/topology	lemma	proved	validated	7	3	§ lem:maincb-stage2-extcb-datum
lem-maincb-stage2-raw-extension	Stage-1/topology	lemma	proved	validated	3	2	§ lem:maincb-stage2-raw-extension
lem-maincb-stage3-call-envelope	Stage-1/topology	lemma	proved	validated	4	3	§ lem:maincb-stage3-call-envelope
lem-maincb-stage3-finite-recombination	Stage-1/topology	lemma	proved	validated	3	1	§ lem:maincb-stage3-finite-recombination
lem-maincb-stage3-raw-merge	Stage-1/topology	lemma	proved	validated	4	2	§ lem:maincb-stage3-raw-merge
lem-maincb-structural-assembly	Stage-1/topology	lemma	proved	validated	10	1	§ lem:maincb-structural-assembly
lem-maincb-structural-domain-ledger	Stage-1/topology	lemma	proved	validated	8	8	§ lem:maincb-structural-domain-ledger
lem-maincb-witness-arithmetic	Stage-1/topology	lemma	proved	validated	0	4	§ lem:maincb-witness-arithmetic
lem-mass-split	trunk	lemma	proved	validated	0	4	—
lem-min-a-implies-height	root	lemma	proved	none	3	1	—
lem-optimal-face-alpha-free-characterization	trunk	lemma	proved	none	1	3	—
lem-optimal-face-conic-reduction	trunk	lemma	proved	none	2	2	—
lem-parametric-halo-collapse	trunk	lemma	proved	validated	3	1	—
lem-positive-exposedness-margin	trunk	lemma	proved	none	0	2	—
lem-prh	Route F	lemma	proved	validated	0	1	§ lem:prh
lem-prh-sharpness	Route F	lemma	proved	validated	2	0	§ lem:prh-sharpness
lem-prh-sharpness-family-arithmetic	Route F	lemma	proved	validated	0	1	§ lem:prh-sharpness-family-arithmetic
lem-prh-sharpness-row-coincidence	Route F	lemma	proved	validated	0	1	§ lem:prh-sharpness-row-coincidence
lem-residual-lower	trunk	lemma	proved	validated	0	1	—
lem-residual-upper	trunk	lemma	proved	validated	0	1	—
lem-routef-ai-defect-linearization	Route F	lemma	proved	validated	2	6	§ lem:routef-ai-defect-linearization
lem-routef-degree-three-estimate	Route F	lemma	proved	validated	7	2	§ lem:routef-degree-three-estimate
lem-routef-degree-two-estimate	Route F	lemma	proved	validated	6	6	§ lem:routef-degree-two-estimate
lem-routef-delta-normalization-closeness	Route F	lemma	proved	validated	3	14	§ lem:routef-delta-normalization-closeness
lem-routef-delta-phi-product	Route F	lemma	proved	validated	5	1	§ lem:routef-delta-phi-product
lem-routef-delta-prime-closeness	Route F	lemma	proved	validated	8	15	§ lem:routef-delta-prime-closeness
lem-routef-delta-upsilon-telescope	Route F	lemma	proved	validated	8	3	§ lem:routef-delta-upsilon-telescope
lem-routef-f0-assembly	Route F	lemma	proved	validated	1	1	§ lem:routef-f0-assembly
lem-routef-f0-defect-identity	Route F	lemma	proved	validated	0	2	§ lem:routef-f0-defect-identity
lem-routef-f0-ucp-lift	Route F	lemma	proved	validated	0	2	§ lem:routef-f0-ucp-lift
lem-routef-f2-positive-unital-compression	Route F	lemma	proved	validated	0	1	§ lem:routef-f2-positive-unital-compression
lem-routef-f3-retract-defect	Route F	lemma	proved	validated	0	1	§ lem:routef-f3-retract-defect

id	phase	kind	status	af	in	out	report
lem-routef-factor-estimate-packet	Route F	lemma	proved	validated	7	1	§ lem:routef-factor-estimate-packet
lem-routef-factor-map-packet	Route F	lemma	proved	validated	8	2	§ lem:routef-factor-map-packet
lem-routef-functional-calculus-closeness	Route F	lemma	proved	validated	1	8	§ lem:routef-functional-calculus-closeness
lem-routef-k-finiteness	Route F	lemma	proved	validated	9	3	§ lem:routef-k-finiteness
lem-routef-k-ledger	Route F	lemma	proved	validated	19	2	§ lem:routef-k-ledger
lem-routef-multiplicative-telescope	Route F	lemma	proved	validated	8	3	§ lem:routef-multiplicative-telescope
lem-routef-prh-finish	Route F	lemma	proved	validated	1	2	§ lem:routef-prh-finish
lem-routef-raw-factor-identities	Route F	lemma	proved	validated	4	5	§ lem:routef-raw-factor-identities
lem-routef-raw-factor-norms	Route F	lemma	proved	validated	4	10	§ lem:routef-raw-factor-norms
lem-routef-raw-factor-setting-formation	Route F	lemma	proved	validated	3	21	§ lem:routef-raw-factor-setting-formation
lem-routef-raw-factor-units	Route F	lemma	proved	validated	4	2	§ lem:routef-raw-factor-units
lem-routef-raw-product-estimate	Route F	lemma	proved	validated	5	2	§ lem:routef-raw-product-estimate
lem-routef-scalar-header-positivity	Route F	lemma	proved	validated	1	3	§ lem:routef-scalar-header-positivity
lem-routef-threshold-minimum	Route F	lemma	proved	validated	6	2	§ lem:routef-threshold-minimum
lem-routef-upsilon-delta-telescope	Route F	lemma	proved	validated	7	3	§ lem:routef-upsilon-delta-telescope
lem-routef-upsilon-normalization-closeness	Route F	lemma	proved	validated	5	7	§ lem:routef-upsilon-normalization-closeness
lem-routef-upsilon-prime-closeness	Route F	lemma	proved	validated	10	8	§ lem:routef-upsilon-prime-closeness
lem-routef-upsilon-prime-component-construction	Route F	lemma	proved	validated	5	2	§ lem:routef-upsilon-prime-component-construct
lem-routef-upsilon-prime-left-inverse	Route F	lemma	proved	validated	6	1	§ lem:routef-upsilon-prime-left-inverse
lem-row-zero-capacity	trunk	lemma	proved	validated	1	1	—
lem-separator-zero-face-obstruction	trunk	lemma	proved	none	3	1	—
lem-sl1a-three-cell-reduction	trunk	lemma	proved	none	8	1	—
lem-stage1-approximate-group-laws	Stage-1/topology	lemma	stated	seeded	4	1	§ lem:stage1-approximate-group-laws
lem-stage1-approximate-group-laws-transport	Stage-1/topology	lemma	proved	validated	3	1	§ lem:stage1-approximate-group-laws-transport
lem-stage1-bound-inversion-isolation	Stage-1/topology	lemma	proved	validated	2	1	§ lem:stage1-bound-inversion-isolation
lem-stage1-bound-quotient-index-data	Stage-1/topology	lemma	proved	validated	2	1	§ lem:stage1-bound-quotient-index-data
lem-stage1-bound-quotient-left-inversion	Stage-1/topology	lemma	proved	validated	5	1	§ lem:stage1-bound-quotient-left-inversion
lem-stage1-bound-quotient-local-index	Stage-1/topology	lemma	proved	validated	6	1	§ lem:stage1-bound-quotient-local-index
lem-stage1-exact-unit-rectification	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-exact-unit-rectification
lem-stage1-explicit-group-closeness	Stage-1/topology	lemma	proved	validated	1	3	§ lem:stage1-explicit-group-closeness
lem-stage1-explicit-group-domain-membership	Stage-1/topology	lemma	proved	validated	1	3	§ lem:stage1-explicit-group-domain-membership
lem-stage1-explicit-smooth-unitary-operations	Stage-1/topology	lemma	proved	validated	5	5	§ lem:stage1-explicit-smooth-unitary-operations
lem-stage1-exterior-cohomology	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-exterior-cohomology
lem-stage1-extra-fixed-class	Stage-1/topology	lemma	proved	validated	12	1	§ lem:stage1-extra-fixed-class
lem-stage1-fixed-unitary-projection-bridge	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-fixed-unitary-projection-bridge
lem-stage1-fresh-two-point-inclusion	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-fresh-two-point-inclusion
lem-stage1-group-closeness	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-group-closeness
lem-stage1-group-domain-membership	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-group-domain-membership
lem-stage1-hspace-coproduct-tail	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-hspace-coproduct-tail
lem-stage1-inversion-derivative-control	Stage-1/topology	lemma	proved	validated	3	0	§ lem:stage1-inversion-derivative-control
lem-stage1-inversion-derivative-transport	Stage-1/topology	lemma	proved	validated	4	1	§ lem:stage1-inversion-derivative-transport
lem-stage1-left-inversion-associated-graded	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-left-inversion-associated-graded
lem-stage1-left-inversion-trace	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-left-inversion-trace
lem-stage1-maurer-cartan-transport	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-maurer-cartan-transport
lem-stage1-maurer-cartan-trivialization	Stage-1/topology	lemma	proved	validated	1	2	§ lem:stage1-maurer-cartan-trivialization
lem-stage1-original-complementary-pair	Stage-1/topology	lemma	proved	validated	1	2	§ lem:stage1-original-complementary-pair
lem-stage1-polar-coherence-naturality	Stage-1/topology	lemma	proved	validated	1	3	§ lem:stage1-polar-coherence-naturality

id	phase	kind	status	af	in	out	report
`lem-stage1-polar-constant-ledger`	Stage-1/topology	lemma	proved	validated	8	5	§ lem:stage1-polar-constant-ledger
`lem-stage1-polar-path-admissibility`	Stage-1/topology	lemma	proved	validated	2	1	§ lem:stage1-polar-path-admissibility
`lem-stage1-polar-path-transport`	Stage-1/topology	lemma	proved	validated	2	1	§ lem:stage1-polar-path-transport
`lem-stage1-polar-retraction`	Stage-1/topology	lemma	proved	validated	1	13	§ lem:stage1-polar-retraction
`lem-stage1-polar-retraction-transport`	Stage-1/topology	lemma	proved	validated	1	2	§ lem:stage1-polar-retraction-transport
`lem-stage1-polar-scalar-arithmetic`	Stage-1/topology	lemma	proved	validated	0	1	§ lem:stage1-polar-scalar-arithmetic
`lem-stage1-quantitative-inverse-function`	Stage-1/topology	lemma	proved	validated	0	6	§ lem:stage1-quantitative-inverse-function
`lem-stage1-quotient-finite-cw`	Stage-1/topology	lemma	proved	validated	2	2	§ lem:stage1-quotient-finite-cw
`lem-stage1-quotient-inversion-index-data`	Stage-1/topology	lemma	proved	validated	8	1	§ lem:stage1-quotient-inversion-index-data
`lem-stage1-quotient-left-inversion`	Stage-1/topology	lemma	proved	validated	5	2	§ lem:stage1-quotient-left-inversion
`lem-stage1-quotient-manifold-package`	Stage-1/topology	lemma	proved	validated	4	5	§ lem:stage1-quotient-manifold-package
`lem-stage1-rectified-cstar-control`	Stage-1/topology	lemma	proved	validated	2	4	§ lem:stage1-rectified-cstar-control
`lem-stage1-rectified-cstar-transport`	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-rectified-cstar-transport
`lem-stage1-rectified-nontrivial-projection`	Stage-1/topology	lemma	proved	validated	2	2	§ lem:stage1-rectified-nontrivial-projection
`lem-stage1-smooth-polar-inverse`	Stage-1/topology	lemma	proved	validated	2	7	§ lem:stage1-smooth-polar-inverse
`lem-stage1-smooth-unitary-atlas`	Stage-1/topology	lemma	proved	validated	2	9	§ lem:stage1-smooth-unitary-atlas
`lem-stage1-smooth-unitary-operations`	Stage-1/topology	lemma	stated	seeded	4	0	§ lem:stage1-smooth-unitary-operations
`lem-stage1-uniform-inversion-isolation`	Stage-1/topology	lemma	proved	validated	5	2	§ lem:stage1-uniform-inversion-isolation
`lem-stage1-unitary-graph-control`	Stage-1/topology	lemma	proved	validated	2	7	§ lem:stage1-unitary-graph-control
`lem-stage1-unitary-graph-transport`	Stage-1/topology	lemma	proved	validated	1	1	§ lem:stage1-unitary-graph-transport
`lem-thmainext-conditional`	Stage-1/topology	lemma	proved	validated	7	6	§ lem:thmainext-conditional
`lem-top-deficit-price`	trunk	lemma	proved	none	0	4	—
`lem-top-support-dual-face`	trunk	lemma	proved	none	1	1	—
`lem-top-witness-third-actor`	trunk	lemma	proved	none	2	1	—
`lem-topology-finite-triangulation`	Stage-1/topology	lemma	proved	validated	0	1	§ lem:topology-finite-triangulation
`lem-topology-hopf-structure`	Stage-1/topology	lemma	proved	validated	0	0	§ lem:topology-hopf-structure
`lem-topology-kunneth-cross-product`	Stage-1/topology	lemma	proved	validated	0	1	§ lem:topology-kunneth-cross-product
`lem-topology-lefschetz-hopf`	Stage-1/topology	lemma	proved	validated	0	1	§ lem:topology-lefschetz-hopf
`lem-topology-local-index-sign`	Stage-1/topology	lemma	proved	validated	0	3	§ lem:topology-local-index-sign
`lem-topology-orientable-top-cohomology`	Stage-1/topology	lemma	proved	validated	0	1	§ lem:topology-orientable-top-cohomology
`lem-topology-quotient-manifold`	Stage-1/topology	lemma	proved	validated	0	2	§ lem:topology-quotient-manifold
`lem-zero-face-exchange-identity`	trunk	lemma	proved	none	1	1	—
`op-classical`	root	theorem	proved	validated	4	0	§ op:classical
`op-exposed-hull`	root	open-problem	open	none	2	1	—
`op-hlc`	root	open-problem	open	none	3	2	—
`prop-approx-simplex`	root	proposition	proved-mod-audit	seeded	1	1	—
`prop-f2-t1-equivalence`	Route F	proposition	proved	none	4	0	—
`thm-classical-factorization`	root	theorem	proved-mod-audit	seeded	2	1	—
`thm-cluster`	root	theorem	proved-mod-audit	seeded	1	1	—

Definition index — the vocabulary the chain imports

One row per `definitions/` shard imported by a drawn result (L2: a term is defined exactly once and only referenced elsewhere). Each row carries the anchor `dag:<def-id>`, so the definitions section can link here with `\dagdef{def-...}` and this table links back to the results that use it.

definition	status	uses	phases that import it
`def-actor-hull`	draft	7	trunk (7)
`def-almost-idempotent`	locked	9	Route F (7), root (2)
`def-approximate-unitary-space`	locked	33	Stage-1/topology (33)
`def-canonical-corner-identifications`	locked	4	COMP/H-CB (4)
`def-co-top`	draft	7	trunk (7)
`def-column-hilbert-corner`	locked	2	COMP/H-CB (2)
`def-compressed-associator`	locked	2	COMP/H-CB (2)
`def-compressed-corner`	locked	31	COMP/H-CB (11), EXT-CB (6), Stage-1/topology (14)
`def-delta-projection`	locked	33	COMP/H-CB (11), EXT-CB (4), Stage-1/topology (18)
`def-dual-witness`	draft	1	trunk (1)
`def-epsilon-cstar-algebra`	locked	40	EXT-CB (3), Stage-1/topology (37)
`def-exposed`	locked	38	Route F (1), root (4), trunk (33)
`def-extcb-datum`	locked	6	EXT-CB (3), Stage-1/topology (3)
`def-extended-delta-inclusion`	locked	37	COMP/H-CB (1), EXT-CB (3), Stage-1/topology (32), Route F (1)
`def-extended-epsilon-cstar-algebra`	locked	56	COMP/H-CB (13), EXT-CB (5), Stage-1/topology (36), Route F (2)
`def-fd-cstar-diagonal`	locked	7	EXT-CB (1), Stage-1/topology (5), Route F (1)
`def-four-corner-merging-datum`	locked	7	EXT-CB (2), Stage-1/topology (5)
`def-h-space-left-inversion`	locked	11	Stage-1/topology (11)
`def-ha-map`	locked	13	COMP/H-CB (12), EXT-CB (1)
`def-hcb-datum`	locked	15	COMP/H-CB (15)
`def-height`	locked	31	Route F (1), root (5), trunk (25)
`def-invisible-mass`	locked	7	root (3), trunk (4)
`def-lefschetz-fixed-point-data`	locked	7	Stage-1/topology (7)
`def-maincb-partition-state`	locked	18	Stage-1/topology (18)
`def-maincb-raw-call`	locked	17	Stage-1/topology (17)
`def-maincb-reset-state`	locked	18	Stage-1/topology (18)
`def-maincb-witness-ledger`	locked	21	Stage-1/topology (21)
`def-near-cluster`	draft	3	trunk (3)
`def-near-positive-projection`	locked	2	root (2)
`def-negative-mass`	locked	41	Route F (1), root (7), trunk (33)
`def-one-dimensional-delta-projection`	locked	8	COMP/H-CB (1), EXT-CB (2), Stage-1/topology (5)
`def-operator-space`	locked	16	Stage-1/topology (16)
`def-positive-approximate-retract`	locked	4	Route F (4)
`def-projection-basis`	locked	7	EXT-CB (2), Stage-1/topology (4), Route F (1)
`def-routef-raw-factor-setting`	locked	23	Route F (23)
`def-selected-corner`	draft	4	trunk (4)
`def-signed-idempotent`	locked	45	Route F (1), root (7), trunk (37)
`def-slab`	draft	2	trunk (2)
`def-stage1-polar-witness-data`	locked	15	Stage-1/topology (15)
`def-stochastic`	locked	20	Route F (12), root (7), trunk (1)
`def-theta-idempotent-approximation`	draft	2	COMP/H-CB (2)
`def-top-support-functional`	draft	6	trunk (6)
`def-ucp-map`	locked	14	Stage-1/topology (1), Route F (13)
`def-visible-set`	locked	37	Route F (1), root (5), trunk (31)
`def-zero-face`	draft	1	trunk (1)

Definitions in `definitions/` not imported by any result on this chain (2): `def-pivot`, `def-top-deficit`.

75 The campaign in numbers

75.1 The campaign at a glance

This section is machine-generated from the two repositories' own records: the `fr` controller log, the append-only `af` proof ledgers, the registry and definition shards, the run bundles, the work log, the issue tracker, and the git history. Table 5 places the progenitor campaign (`almost-idempotent-positive-maps`, where the classical portfolio was born) beside this one. Nothing below is a mathematical claim; nothing below promotes any result up the rigour ladder.

Table 5: The two campaigns side by side. The progenitor column is a snapshot of `../almost-idempotent-positive-maps`; that repository has no `fr` controller, so its exploration record lives only in its work log and git history.

Quantity	Progenitor	This repository
First commit	2026-06-02	2026-07-02
Last commit in snapshot	2026-06-13	2026-08-09
Commits	174	1 208
Days with a commit	11	28
Work-log entries	22	87
`fr` controller cycles	— (no controller)	1 281
Registry results	59	374
Definition shards	25	47
`af` workspaces	16	211
Workspaces ever recording a validated root	16	202
Current registry results at T0	16	200
Historically validated, not current T0	0	2
Adversarial proof nodes built	132	2 883
Ledger entries (append-only)	822	17 506
Numerical run bundles	–	38

"T0" is this project's current in-repository tag for a result whose `af` proof tree has a `validated` root under the fresh-prover/fresh-verifier protocol. It is not a Lean proof. The historical-root row is deliberately separate: append-only ledgers survive later demotions.

75.2 What is counted, and what cannot be

Counted, exactly. Every figure in this section is a count of an artifact that exists in one of the two repositories at the snapshot instant: a JSON record in `.frontier/log.jsonl`, a ledger entry under `proofs/*/ledger/`, a front-matter field in an `argument/lemmas/*.md` or `definitions/*.md` shard, a directory under `runs/` or `docs/`, a heading in `docs/worklog.md` or `FINDINGS.md`, a record in `.beads/issues.jsonl`, or a commit in `git log`. No figure is modelled, interpolated, or extrapolated.

Not recoverable: LLM token consumption. The single most-asked statistic about a campaign like this one — how many tokens, and at what cost — is *not* present in the record and is not reconstructed here. Neither repository ever logged token counts, context sizes, wall-clock durations, or billing data; the `af` ledger records who claimed which node and when, not how much text was produced to do it, and the `fr` log records a worker's `model:role` pair, not its usage. What *is* recoverable is the *claim-event* count — ledger node-claim records, a lower-bound proxy for prover/verifier invocations rather than a guaranteed-distinct API-call tally — and that is reported in Section 75.8. Multiplying a claim count by a guessed per-job token figure would produce a number with no evidentiary basis, so it is omitted rather than invented (CLAUDE.MD L0/L3: a number without a re-run command is not a finding).

Other honest gaps.

- The progenitor repository predates the `fr` controller, so it has no cycle/arm/outcome record at all. Its exploration history is reconstructible only from 22 work-log entries and 174 commits, at coarser granularity than this repository's 1 281 controller cycles.
- The progenitor repository's issue tracker is stored in an embedded Dolt database rather than an exported `issues.jsonl`, so its issue counts are not in this snapshot.
- `fr` evidence records carry a self-reported `verdict` field whose value is `claimed` on every one of the 564 records that has one. It is a protocol marker, not an adjudication, and is therefore not tabulated as an outcome.
- Wave numbering (`W1`–`W140`) is a human convention in the work log and wave documents; it is not a controller-enforced counter, and a few numbers were skipped or reused. Counts of wave *documents* are exact; the wave *index* is indicative.
- Cross-repository carry-over (what this campaign inherited from the progenitor) is described qualitatively in Section 75.10; it is a judgement about content, not a count.

75.3 The campaign calendar

The two campaigns ran as short, dense bursts rather than as steady background work. This repository put 1 208 commits into 28 calendar days with a commit (out of 39 days elapsed), and the progenitor put 174 into 11.

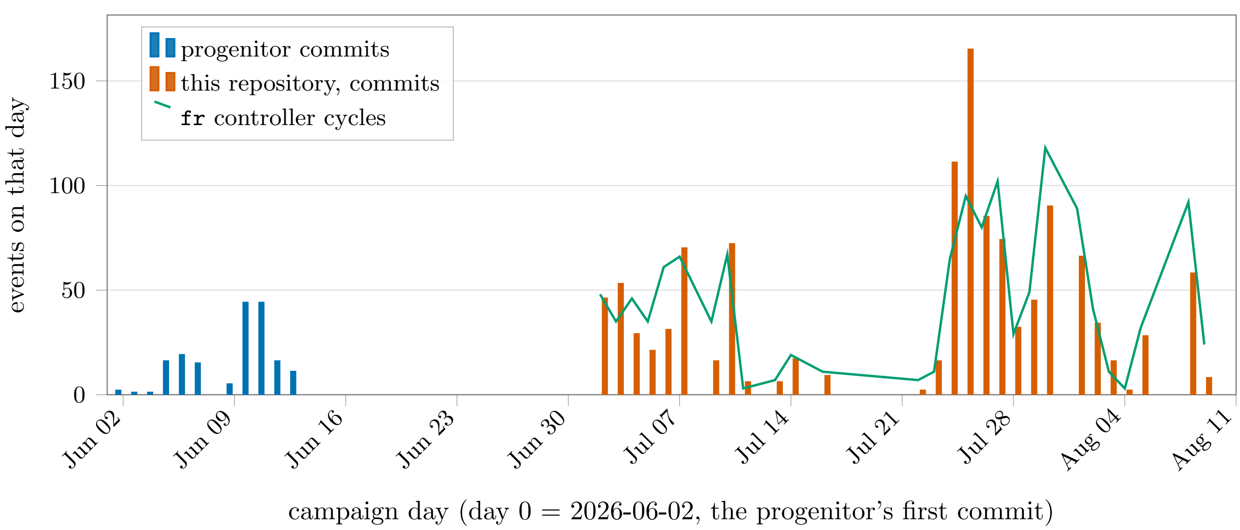


Figure 1: The campaign calendar. Bars are commits per calendar day in each repository; the line is `fr` controller cycles per day in this repository. The gap between the two bar families is the interregnum between the progenitor's last commit (2026-06-13) and this repository's stand-up (2026-07-02).

Table 6: The five heaviest days in this repository, by commit count.

Day	Commits	`fr` cycles	banked	`af` roots validated
2026-07-25	165	95	2	22
2026-07-24	111	65	0	14
2026-07-30	90	118	28	27
2026-07-26	85	80	6	13
2026-07-27	74	102	23	20

"banked" counts `fr` log records with outcome `banked`; the last column counts `af` proof trees whose root node was validated that day.

75.4 The rigour ladder

The project's whole point is that a claim's *status* is tracked honestly and separately from its plausibility. The registry therefore carries a status census, and the only rung that counts as rigorous in-repo is an `af`-validated tree (or a byte-matched citation). Of 374 registry results, 200 currently carry `af: validated` — 53.5% of the registry. The remainder are flagged non-rigorous in the open.

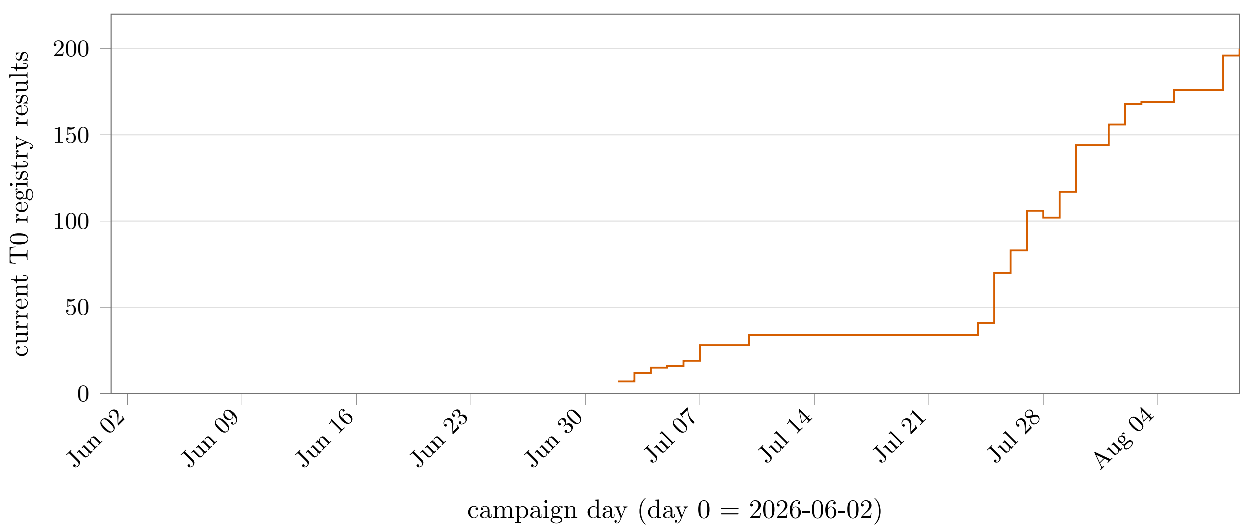


Figure 2: The rigour ladder over time: ledger validation dates supply the initial segment; from the first explicit `T0 before -> after` controller transition onward, the end-of-day current census follows those transitions. Downward steps are retained, so retractions and supersessions remain visible. The final level is cross-checked against current registry frontmatter.

Table 7: Current and discharge-era rigour-ladder census.

Census point	Count
Current T0 registry results	200
Registry results	374
Workspaces with a root validation ever recorded	202
Af workspaces	211
Historically validated but not current T0	2
Root discharge (2026-08-08, controller cycle 1 241)	196
Statistics-wave dispatch (374 registry results; cycle 1 264)	197

The root-discharge row is extracted from the controller record whose evidence artifact is `proofs/op-classical/export.md`; the current row is read independently from all registry frontmatters.

Table 8: Status census of the result registry.

Registry `status:`	Results	Share
`proved`	313	83.7%
`conjecture`	31	8.3%
`proved-mod-audit`	17	4.5%
`disproved`	3	0.8%
`stated`	3	0.8%
`heuristic`	2	0.5%
`numerical`	2	0.5%
`open`	2	0.5%
`obstruction`	1	0.3%

`proved` here means "the shard's proof has been carried out and reviewed"; it becomes *rigorous* only in combination with `af: validated` (next table). `proved-mod-audit` is the inherited-campaign tag: a paper proof that has not cleared an independent pass in this repository.

Table 9: Formalisation state of the registry: how many results have an `af` workspace, and how many reached a validated root.

`af:` field	Results	Share
`validated`	200	53.5%
`none`	163	43.6%
`seeded`	11	2.9%

`seeded` means a workspace exists with the registry contract as its root conjecture but the tree is not (yet) closed.

Table 10: The vocabulary layer: one definition per shard, provenance-gated.

Definition shards	Count
`original`	28
`cited`	13
`consensus`	6
`status: locked`	36
`status: draft`	11

`cited` definitions byte-match a local source under `refs/`; `consensus` and `original` are adopted/introduced here and lock only on recorded sign-off.

75.5 The exploration controller: arms and outcomes

Research directions are registered as `fr` *arms*; every harvested result is logged as one pull with an outcome, and a Stop-hook circuit-breaker forces a stalled arm to yield. Across 1 281 cycles, 894 carried an arm label, 384 were explicit no-wave orientation turns, and 387 records had no arm label in total. The controller recorded 576 explicit allocation decisions: `EXPLOIT` × 556, `EXPLORE` × 14, `PIVOT` × 6.

The Route-F hardening arm `FH` received 680 of the 894 arm-labelled pulls (76.1%). The live portfolio now parks 7 of its 10 arms at the tier-1 override priority; the table retains untried registered arms with a zero pull count.

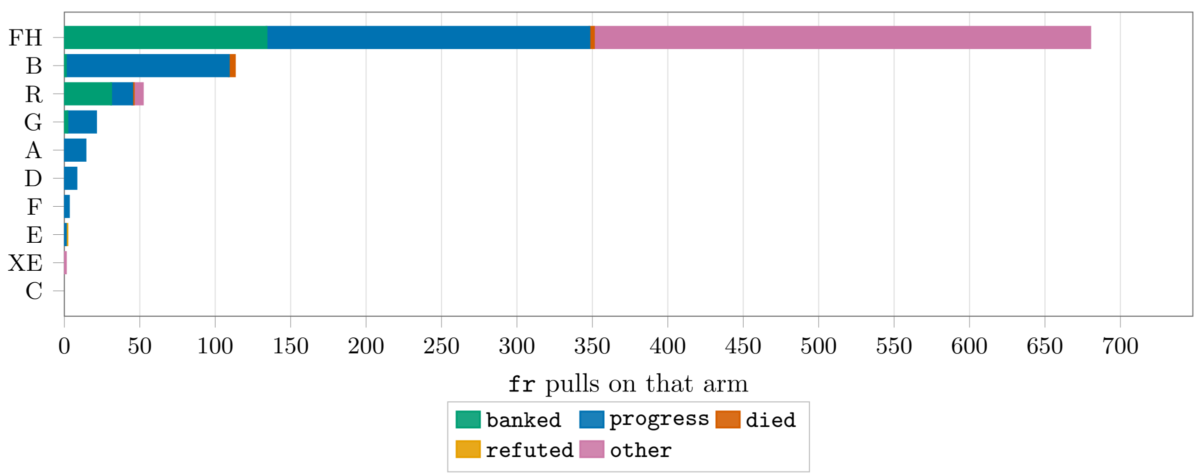


Figure 3: Attention allocation across research directions. Each bar is one `fr` arm; segments are the recorded outcomes of the pulls on it. Records with no arm (orientation and bookkeeping cycles) are excluded.

Table 11: Per-arm ledger: pulls, outcomes, activity window, and current portfolio priority.

Arm	Pulls	`banked`	`progress`	`died`	`refuted`	`other`	First	Last	Priority now
`FH`	680	134	214	3	0	329	2026-07-22	2026-08-09	primary
`B`	113	1	108	4	0	0	2026-07-02	2026-07-16	primary
`R`	52	31	14	1	0	6	2026-07-02	2026-07-26	parked-tier1-override

(continued)

(Table 11, continued)

Arm	Pulls	`banked`	`progress`	`died`	`refuted`	`other`	First	Last	Priority now
`G`	21	2	19	0	0	0	2026-07-03	2026-07-05	parked-tier1-override
`A`	14	0	14	0	0	0	2026-07-02	2026-07-05	parked-tier1-override
`D`	8	0	8	0	0	0	2026-07-03	2026-07-05	parked-tier1-override
`F`	3	0	3	0	0	0	2026-07-02	2026-07-04	parked-tier1-override
`E`	2	0	1	0	1	0	2026-07-05	2026-07-06	parked-tier1-override
`XE`	1	0	0	0	0	1	2026-07-26	2026-07-26	exploratory
`C`	0	0	0	0	0	0	–	–	parked-tier1-override

Outcome vocabulary: `banked` = a result harvested and recorded; `progress` = movement without a bankable result; `died` / `refuted` = the direction was killed, with a certificate; `other` combines dispatch, blocked, null, obstruction, discovery, and quarantined legacy free-text outcomes to keep the table portrait-safe.

Aggregated over all cycles the normalized outcome mix is 384 `orient`, 381 `progress`, 318 `dispatch`, 168 `banked`, 11 `null`, 8 `died`, 5 `legacy-free-text`, 3 `discovery`, 1 `blocked`, 1 `obstruction`, 1 `refuted`. Evidence records are tiered 219 `T0`, 175 `T1`, 170 `T2`. The normalizer quarantined 5 legacy records whose free-form harvest prose occupies the outcome field; they remain counted as records and pulls but no longer become spurious table columns.

75.6 Adversarial elevations

A result reaches the top in-repo rung only through an `af` workspace: the registry contract becomes the root conjecture, a fresh prover agent builds a proof tree, and every node is validated by a *separate* fresh verifier told that finding a gap is a success. The orchestrator never judges a step. 211 workspaces were opened; 202 have a root validation somewhere in their retained ledger, while 200 registry results are current T0 and 9 workspaces never recorded a validated root. Between them the workers built 2 883 proof nodes, raised 435 challenges (of which 428 were resolved), and forced 381 node amendments.

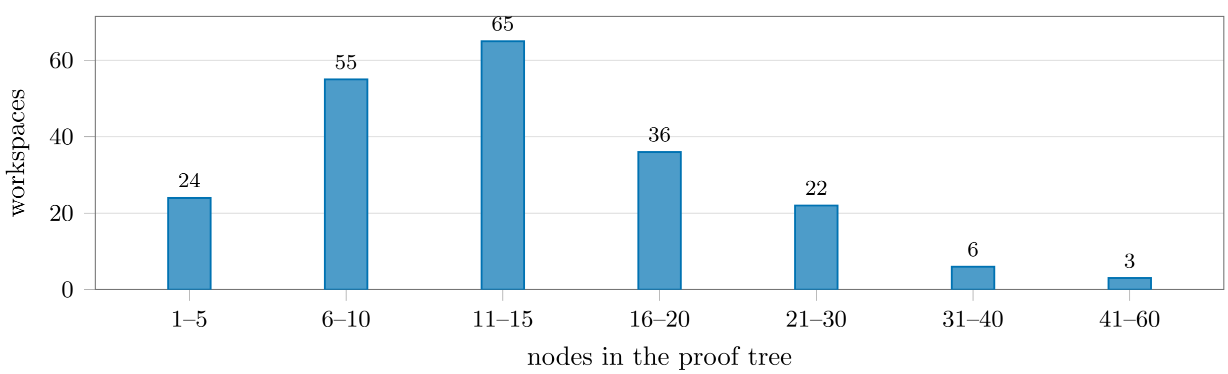


Figure 4: Size distribution of the 211 adversarial proof workspaces. The median workspace contains 12 nodes; the largest, `lem-residual-upper`, has 52.

Table 12: First-pass and correction metrics read from retained ledgers and controller records.

Mechanical pass metric	Trees/events	Rate
Initial build needed no prover-fix claim	78	38.6%
No verifier challenge recorded	75	37.1%
Root-validation events	203	–
Root-unvalidation events	1	–
Root re-validations after an unvalidation	1	–
Controller records explicitly marked re-validated or re-flipped	9	–
Controller records explicitly documenting a balloon abort/stop	25	–

A first pass here has a strict mechanical definition: a workspace recorded a validated root and no `pf-` (prover-fix) claim. This is not inferred from celebratory prose. Balloon rows count controller records matching explicit abort/stop language, not discarded trees that left no retained artifact.

The sharpness-family finding records 3 consecutive balloons, with live-tree sizes 27, 28, 27. The subsequent controller harvest records the prescribed `xhigh` remedy closing a 24-node tree under cap 26.

The most-contested tree was `conj-extcb` (24 challenges raised); the longest-running was `conj-extcb`, which needed 20 prover/verifier rounds before its root closed.

Table 13: The append-only `af` ledger, by entry type, summed over every workspace in this repository.

Ledger entry type	Count
`nodes_claimed`	4 633
`nodes_released`	4 633
`node_created`	2 883
`node_validated`	2 819
`def_added`	1 016
`challenge_raised`	435
`challenge_resolved`	428
`node_amended`	381
`proof_initialized`	211
`node_archived`	58
`node_unvalidated`	7
`approach_tried`	2

`nodes_claimed` is the operational unit of work: one claim event is one worker taking one node. It is the closest thing the record has to a job counter (Section 75.8).

Table 14: Every workspace in this repository that has ever recorded a validated root, in the order of its latest retained root-validation event.

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
1	`lem-classical-equiv`	29	21	6	4	4	7	37	2026-07-02
2	`obs-height-collapse`	19	19	2	1	5	5	29	2026-07-02
3	`lem-mass-split`	9	9	1	0	2	3	13	2026-07-02

(continued)

(Table 14, continued)

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
4	lem-residual-lower	36	32	9	2	0	11	56	2026-07-02
5	lem-residual-upper	52	49	11	1	0	5	79	2026-07-02
6	lem-halo-collapse	20	20	5	3	5	11	33	2026-07-02
7	lem-factorization	12	11	1	1	2	4	15	2026-07-02
8	lem-zerosum-triangle	10	10	0	0	0	2	13	2026-07-03
9	lem-weighted-min	8	8	1	1	0	4	11	2026-07-03
10	lem-fan-payment	15	15	1	0	0	6	21	2026-07-03
11	lem-negpart-subadditive	16	16	2	3	0	6	25	2026-07-03
12	lem-fan-payment-restricted	27	27	2	1	0	9	39	2026-07-03
13	lem-pivot-removing-move	9	9	0	2	1	2	14	2026-07-04
14	lem-collateral-import	32	31	2	0	1	5	41	2026-07-04
15	lem-cross-pivot-cancellation	23	23	3	2	1	5	33	2026-07-04
16	lem-import-reduction	10	10	0	0	1	2	13	2026-07-05
17	lem-parametric-halo-collapse	14	14	0	0	5	1	19	2026-07-06
18	lem-genuine-disintegration	19	19	3	1	6	8	30	2026-07-06
19	lem-top-concentration	19	18	4	6	6	8	34	2026-07-06
20	lem-hiddenness-dual-witness	11	11	0	0	4	2	15	2026-07-07
21	lem-cs-low-slab-pincer	14	14	2	1	2	5	22	2026-07-07
22	lem-harmonic-affine-bridge	18	18	0	0	1	1	23	2026-07-07
23	lem-row-far-dual-certificate	16	16	3	2	4	8	28	2026-07-07
24	lem-top-slab-companion	22	22	1	1	6	5	29	2026-07-07
25	lem-hiddenness-depth-markov	32	31	4	3	5	8	47	2026-07-07
26	lem-depth-d-halo-collapse	11	11	0	0	5	2	14	2026-07-07
27	lem-row-zero-capacity	18	18	0	0	2	2	24	2026-07-07
28	lem-always-tight-dual-support	18	18	3	0	4	2	27	2026-07-07
29	lem-starvation-completion-obstruction	7	7	0	0	2	2	9	2026-07-10
30	lem-hx-signed-variation-ledger	11	11	1	1	2	4	16	2026-07-10
31	lem-hx-financing-floor	14	12	5	3	2	8	32	2026-07-10
32	lem-hx-forced-exterior-coupling	12	12	0	0	2	3	17	2026-07-10
33	lem-hx-robust-scalar-starvation	12	12	0	2	2	2	18	2026-07-10
34	lem-hx-transverse-moment-identity	14	14	0	0	1	2	19	2026-07-10
35	lem-prh	14	14	2	0	2	6	24	2026-07-24
36	lem-compcb-amplification-naturality	12	12	0	0	1	3	16	2026-07-24
37	lem-compcb-amplified-compression	13	13	1	5	4	5	19	2026-07-24
38	lem-compcb-amplified-compression-identities	7	7	2	1	4	4	15	2026-07-24
39	lem-compcb-rectangular-product	12	12	1	0	4	5	19	2026-07-24
40	lem-compcb-compressed-unit-norm	15	12	6	4	6	9	33	2026-07-24
41	lem-compcb-amplified-almost-containment	13	13	1	1	6	5	20	2026-07-24
42	lem-compcb-compressed-unit-action	20	20	5	3	6	8	37	2026-07-24
43	lem-compcb-corner-algebra	14	14	2	10	6	4	22	2026-07-24
44	lem-compcb-single-compression-transfer	33	32	8	2	7	11	57	2026-07-24
45	lem-hcb0-compressed-associator	20	20	2	0	5	6	29	2026-07-24
46	lem-compcb-entrywise-compression-naturality	4	4	0	2	6	1	5	2026-07-24
47	lem-compcb-row-column-product	20	18	5	3	8	4	32	2026-07-24
48	lem-hcb-column-hilbert-squared	50	50	22	17	7	13	108	2026-07-24
49	lem-hcb1-variational-identity	8	8	0	0	5	1	9	2026-07-25
50	lem-hcb1-column-action	38	38	12	12	9	11	73	2026-07-25
51	lem-hcb2-amplified-adjointness	21	21	9	4	9	4	46	2026-07-25
52	lem-hcb2-product-defect	12	12	0	0	9	2	15	2026-07-25
53	lem-hcb3-diagonal-unit	18	18	3	9	9	8	31	2026-07-25
54	lem-hcb3-diagonal-upper-norm	30	30	11	7	9	13	61	2026-07-25
55	lem-hcb3-uniform-square-lower	18	18	6	4	11	15	37	2026-07-25
56	lem-hcb3-diagonal-lower-modulus	29	29	12	10	11	8	64	2026-07-25
57	lem-hcb3-diagonal-inverse	5	5	0	0	11	1	6	2026-07-25
58	lem-hcb3-offdiagonal-inverse	16	16	1	0	11	6	23	2026-07-25

(continued)

(Table 14, continued)

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
59	`lem-hcb4-canonical-gram`	9	9	0	4	12	4	13	2026-07-25
60	`lem-hcb4-canonical-closeness`	11	11	0	0	12	3	15	2026-07-25
61	`lem-hcb4-canonical-inverse`	19	19	3	3	13	6	30	2026-07-25
62	`conj-hcb`	11	11	0	0	15	1	12	2026-07-25
63	`lem-extcb-one-dimensional-product`	25	25	1	0	16	6	35	2026-07-25
64	`lem-extcb-corner-dimension-additivity`	39	39	9	3	18	9	76	2026-07-25
65	`lem-extcb-four-corner-norm`	18	18	6	2	19	11	35	2026-07-25
66	`lem-extcb-four-corner-merge`	26	22	5	9	21	10	44	2026-07-25
67	`lem-extcb-one-dimensional-corner-dimension`	8	8	0	0	20	2	11	2026-07-25
68	`lem-extcb1-close-corner-dimension`	16	16	2	0	21	5	24	2026-07-25
69	`lem-extcb1-cross-corner-dimension`	30	30	8	2	21	7	54	2026-07-25
70	`conj-extcb`	46	40	24	15	21	20	115	2026-07-25
71	`lem-stage1-quantitative-inverse-function`	14	14	3	2	0	4	26	2026-07-26
72	`lem-stage1-exact-unit-rectification`	6	6	1	0	1	4	9	2026-07-26
73	`lem-routef-prh-finish`	22	22	3	4	2	8	34	2026-07-26
74	`lem-kitaev-almost-idemp-audit`	24	24	5	0	1	12	39	2026-07-26
75	`lem-routef-functional-calculus-closeness`	11	11	0	0	1	3	17	2026-07-26
76	`lem-routef-ai-defect-linearization`	13	13	1	1	2	5	19	2026-07-26
77	`lem-topology-lefschetz-hopf`	2	2	0	1	1	1	5	2026-07-26
78	`lem-topology-kunneth-cross-product`	7	7	1	1	0	4	11	2026-07-26
79	`lem-topology-orientable-top-cohomology`	14	14	1	2	0	5	20	2026-07-26
80	`lem-topology-quotient-manifold`	4	4	3	0	0	5	13	2026-07-26
81	`lem-topology-hopf-structure`	13	13	1	2	1	5	20	2026-07-26
82	`lem-extcb-exact-target-correction`	6	6	0	0	4	3	9	2026-07-26
83	`lem-topology-local-index-sign`	23	23	9	10	1	7	55	2026-07-26
84	`lem-routef-f0-ucp-lift`	9	9	1	1	2	3	16	2026-07-27
85	`lem-routef-f0-defect-identity`	12	12	0	0	3	4	19	2026-07-27
86	`lem-routef-f2-positive-unital-compression`	22	22	2	1	3	9	32	2026-07-27
87	`lem-routef-f3-retract-defect`	11	11	0	0	1	4	15	2026-07-27
88	`lem-stage1-rectified-cstar-control`	17	17	5	0	1	9	33	2026-07-27
89	`lem-stage1-unitary-graph-control`	18	15	3	2	2	8	29	2026-07-27
90	`lem-stage1-maurer-cartan-trivialization`	15	15	2	0	2	9	24	2026-07-27
91	`lem-stage1-polar-retraction`	29	29	6	8	2	9	51	2026-07-27
92	`lem-stage1-polar-coherence-naturality`	10	10	0	0	1	4	13	2026-07-27
93	`lem-stage1-group-domain-membership`	10	10	3	1	4	8	17	2026-07-27
94	`lem-stage1-group-closeness`	12	12	0	0	2	3	15	2026-07-27
95	`lem-stage1-approximate-group-laws`	14	14	1	8	2	5	19	2026-07-27
96	`lem-stage1-polar-path-admissibility`	12	12	0	0	2	2	16	2026-07-27
97	`lem-stage1-smooth-unitary-atlas`	14	14	1	0	2	6	21	2026-07-27
98	`lem-stage1-smooth-polar-inverse`	21	21	0	1	2	3	28	2026-07-27
99	`lem-stage1-smooth-unitary-operations`	15	15	2	2	2	3	25	2026-07-27
100	`lem-stage1-polar-scalar-arithmetic`	15	15	0	0	1	2	20	2026-07-27
101	`lem-stage1-rectified-cstar-transport`	7	7	0	1	2	2	8	2026-07-27
102	`lem-stage1-unitary-graph-transport`	9	9	0	0	3	3	12	2026-07-27
103	`lem-stage1-polar-retraction-transport`	5	5	0	0	3	1	6	2026-07-27
104	`lem-stage1-explicit-group-domain-membership`	13	12	2	0	2	4	17	2026-07-28
105	`lem-stage1-explicit-group-closeness`	16	16	4	1	2	9	29	2026-07-29
106	`lem-stage1-explicit-smooth-unitary-operations`	12	12	1	2	2	5	16	2026-07-29
107	`lem-stage1-inversion-derivative-control`	10	10	0	0	2	1	11	2026-07-29
108	`lem-stage1-approximate-group-laws-transport`	16	16	3	3	3	5	24	2026-07-29
109	`lem-stage1-maurer-cartan-transport`	13	19	2	2	3	9	25	2026-07-29
110	`lem-stage1-polar-path-transport`	11	11	2	1	3	9	18	2026-07-29
111	`lem-stage1-inversion-derivative-transport`	25	25	3	3	3	6	41	2026-07-29
112	`lem-stage1-polar-constant-ledger`	11	11	0	0	3	2	13	2026-07-29
113	`lem-finite-polyhedron-maximal-simplex-placement`	3	3	0	0	0	1	4	2026-07-29

(continued)

(Table 14, continued)

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
114	`lem-stage1-uniform-inversion-isolation`	7	7	1	0	2	4	10	2026-07-29
115	`lem-stage1-quotient-manifold-package`	9	9	0	0	2	1	10	2026-07-29
116	`lem-stage1-quotient-left-inversion`	10	10	1	1	3	4	14	2026-07-29
117	`lem-stage1-quotient-inversion-index-data`	12	12	2	0	3	7	18	2026-07-29
118	`lem-topology-finite-triangulation`	6	6	0	2	0	2	8	2026-07-29
119	`lem-stage1-quotient-finite-cw`	4	4	0	0	2	2	6	2026-07-29
120	`lem-stage1-hspace-coproduct-tail`	14	14	1	0	1	3	20	2026-07-30
121	`lem-stage1-exterior-cohomology`	8	8	1	2	1	2	13	2026-07-30
122	`lem-stage1-left-inversion-associated-graded`	9	9	0	3	1	3	11	2026-07-30
123	`lem-stage1-left-inversion-trace`	6	6	0	5	1	2	7	2026-07-30
124	`lem-stage1-bound-quotient-left-inversion`	16	13	3	4	4	3	27	2026-07-30
125	`lem-stage1-bound-quotient-local-index`	19	19	5	1	5	4	39	2026-07-30
126	`lem-stage1-bound-inversion-isolation`	10	10	1	1	5	3	18	2026-07-30
127	`lem-stage1-bound-quotient-index-data`	7	7	0	0	5	2	9	2026-07-30
128	`lem-stage1-extra-fixed-class`	15	15	3	3	5	3	24	2026-07-30
129	`lem-stage1-fixed-unitary-projection-bridge`	10	10	3	2	4	4	20	2026-07-30
130	`lem-stage1-rectified-nontrivial-projection`	7	7	0	4	3	2	12	2026-07-30
131	`lem-stage1-original-complementary-pair`	9	9	1	3	2	5	17	2026-07-30
132	`lem-stage1-fresh-two-point-inclusion`	12	12	2	0	5	5	17	2026-07-30
133	`lem-maincb-initial-raw-inclusion`	16	7	3	4	4	3	22	2026-07-30
134	`lem-maincb-improvement-one-step`	12	12	7	11	5	2	33	2026-07-30
135	`lem-maincb-full-corner-identification`	10	8	1	2	5	3	16	2026-07-30
136	`lem-maincb-corner-equivalence`	8	8	1	0	6	3	14	2026-07-30
137	`lem-maincb-nested-corner-comparison`	18	18	4	2	5	4	35	2026-07-30
138	`lem-maincb-direct-sum-inclusion-merge`	14	14	5	5	5	3	32	2026-07-30
139	`lem-maincb-nested-corner-dimension-transport`	4	4	0	2	5	1	7	2026-07-30
140	`lem-maincb-stage1-raw-refinement`	5	5	1	1	6	3	9	2026-07-30
141	`lem-maincb-cross-union-zero-corners`	12	12	8	7	7	5	32	2026-07-30
142	`lem-maincb-outer-compression-transfer`	12	12	1	4	6	4	17	2026-07-30
143	`lem-maincb-improvement-iteration`	10	10	4	0	5	4	20	2026-07-30
144	`lem-maincb-error-improvement`	17	17	1	0	7	4	23	2026-07-30
145	`lem-maincb-direct-corner-envelope`	7	7	1	1	6	3	10	2026-07-30
146	`lem-maincb-stage2-extcb-datum`	12	12	0	0	10	2	15	2026-07-30
147	`lem-maincb-isomorphism-unit-control`	6	6	0	0	3	1	7	2026-08-01
148	`lem-maincb-witness-arithmetic`	6	6	0	0	2	1	7	2026-08-01
149	`lem-maincb-reset-invariant-preservation`	12	12	0	0	7	2	16	2026-08-01
150	`lem-maincb-stage2-raw-extension`	6	6	1	1	6	2	10	2026-08-01
151	`lem-maincb-compressed-corner-unit-comparison`	13	13	5	2	6	2	30	2026-08-01
152	`lem-maincb-stage1-call-envelope`	11	11	0	0	15	2	13	2026-08-01
153	`lem-maincb-cross-class-merging-datum`	11	11	1	0	21	2	14	2026-08-01
154	`lem-maincb-cross-datum-bijectivity`	12	12	1	4	10	2	17	2026-08-01
155	`lem-maincb-stage3-raw-merge`	6	6	0	0	6	2	7	2026-08-01
156	`lem-maincb-extended-inclusion-monotone`	3	3	1	1	4	1	8	2026-08-01
157	`lem-maincb-reset-constant-ledger`	15	15	1	1	4	2	20	2026-08-01
158	`lem-maincb-structural-domain-ledger`	12	12	4	0	5	2	24	2026-08-01
159	`lem-maincb-reset-output-typing`	8	8	0	0	7	2	9	2026-08-01
160	`lem-maincb-stage2-call-envelope`	10	10	0	9	8	2	20	2026-08-01
161	`lem-maincb-stage3-call-envelope`	19	19	6	1	8	7	35	2026-08-02
162	`lem-maincb-one-class-extension`	20	20	5	4	8	11	47	2026-08-02
163	`lem-maincb-initial-reset-inclusion`	6	6	1	2	7	5	11	2026-08-02
164	`lem-maincb-stage1-strict-refinement`	11	11	2	1	7	5	22	2026-08-02
165	`lem-maincb-maximal-reset-selection`	9	9	0	3	6	2	12	2026-08-02
166	`lem-maincb-binary-block-merge`	11	11	0	0	8	5	16	2026-08-02
167	`lem-maincb-stage3-finite-recombination`	7	7	0	0	6	2	9	2026-08-02
168	`lem-maincb-corner-nontriviality`	7	7	1	0	7	5	10	2026-08-02

(continued)

(Table 14, continued)

#	Result	Nodes	Val.	Chal.	Amd.	Defs	Rnds	Jobs	Closed
169	`lem-maincb-stage1-maximality`	5	5	0	0	6	1	6	2026-08-02
170	`lem-maincb-structural-assembly`	20	20	4	11	11	11	38	2026-08-02
171	`lem-thmainext-conditional`	4	4	0	0	6	2	5	2026-08-03
172	`lem-routef-raw-factor-setting-formation`	10	10	0	5	4	1	11	2026-08-05
173	`lem-routef-raw-factor-identities`	11	11	0	0	6	2	14	2026-08-05
174	`lem-routef-raw-factor-norms`	23	23	4	2	7	5	41	2026-08-05
175	`lem-routef-raw-product-estimate`	6	6	1	0	6	2	10	2026-08-05
176	`lem-routef-raw-factor-units`	8	8	4	5	6	2	21	2026-08-05
177	`lem-kitaev-diagonal-repair`	20	20	2	2	1	6	34	2026-08-05
178	`cor-kitaev-diagonal-cpization`	22	22	2	0	2	5	32	2026-08-05
179	`lem-routef-delta-prime-closeness`	15	15	1	1	7	5	21	2026-08-08
180	`lem-routef-delta-normalization-closeness`	6	6	0	5	6	1	7	2026-08-08
181	`lem-routef-degree-two-estimate`	7	7	0	2	6	1	8	2026-08-08
182	`lem-routef-delta-phi-product`	7	7	0	0	6	2	8	2026-08-08
183	`lem-routef-degree-three-estimate`	15	15	0	3	6	3	21	2026-08-08
184	`lem-routef-upsilon-prime-component-construction`	23	23	0	0	6	3	30	2026-08-08
185	`lem-routef-upsilon-prime-left-inverse`	14	14	2	3	6	3	22	2026-08-08
186	`lem-routef-upsilon-prime-closeness`	11	11	0	2	6	3	17	2026-08-08
187	`lem-routef-upsilon-normalization-closeness`	21	21	1	0	6	5	29	2026-08-08
188	`lem-routef-delta-upsilon-telescope`	4	4	0	1	6	1	5	2026-08-08
189	`lem-routef-multiplicative-telescope`	7	7	1	1	6	4	11	2026-08-08
190	`lem-routef-upsilon-delta-telescope`	13	13	1	0	6	5	18	2026-08-08
191	`lem-routef-k-finiteness`	18	18	1	1	6	4	28	2026-08-08
192	`lem-routef-threshold-minimum`	5	5	0	3	6	1	9	2026-08-08
193	`lem-routef-scalar-header-positivity`	16	14	2	2	5	3	22	2026-08-08
194	`lem-routef-factor-map-packet`	16	16	0	0	6	2	22	2026-08-08
195	`lem-routef-factor-estimate-packet`	16	16	0	0	6	3	22	2026-08-08
196	`lem-routef-k-ledger`	7	7	0	0	6	1	8	2026-08-08
197	`lem-routef-f0-assembly`	7	7	0	0	4	1	10	2026-08-08
198	`op-classical`	5	5	0	0	3	2	7	2026-08-08
199	`lem-prh-sharpness-family-arithmetic`	24	24	2	0	1	4	34	2026-08-09
200	`lem-prh-sharpness-row-coincidence`	19	19	2	0	1	4	27	2026-08-09
201	`lem-prh-sharpness`	12	12	0	0	2	2	16	2026-08-09
202	`cor-classical-sharpness`	5	5	0	0	3	1	6	2026-08-09

"Val." counts validated nodes; "Chal." counts challenges raised by verifiers against nodes of that tree; "Amd." counts node statements the prover had to change in response; "Defs" counts definitions imported into the workspace as byte-matched externals; "Rnds" is the highest prover/verifier round index reached; "Jobs" is the number of node-claim events, i.e. worker invocations against that tree; "Closed" is the day the root node was validated. Fewer validated nodes than created nodes means nodes were archived or superseded during the run.

The 9 workspaces that never recorded a validated root are `conj-degenerate-transport`, `conj-nsc`, `conj-rh`, `conj-sc`, `lem-hiddenness-alpha-slab-leakage`, `lem-negative-pivot-import`, `prop-approx-simplex`, `thm-classical-factorization`, `thm-cluster`. The registry currently has 11 `af: seeded` rows; the difference is explained by 2 retained workspaces with historical root validations that are no longer current T0: `lem-stage1-approximate-group-laws`, `lem-stage1-smooth-unitary-operations`.

75.7 Corrections, retractions, and report catch-up

`docs/LEARNINGS.md` contains 7 dated correction/retraction entries. Across the explicit downward T0 transitions written in that file, 11 result-slots were removed in 4 events. These are historical ledger counts, not a claim that the affected statements all remain demoted.

Table 15: Explicit downward T0 transitions recorded in the learning ledger.

Event	Before	After	Removed
1	107	105	2
2	105	101	4
3	159	156	3
4	156	154	2

Only literal `T0 before -> after` strings are counted. Re-seeded workspaces can erase the superseded tree, so this table complements rather than duplicates the append-only-ledger event counts.

The learning ledger has 1 entry headed by `ex-hume`; its current registry frontmatter is `status: disproved`, `af: none`. This is reported as a status transition only: the statistics layer does not restate the proposition.

The report-sync reconstruction joins current T0 frontmatter to actual `\label` anchors. The controller record supplies the sync section-number interval; removing that interval reconstructs the pre-wave surface. The table preserves the statistics-wave dispatch census separately from the live extraction census, because other elevation work continued concurrently. The sync is complete at the snapshot instant.

Table 16: T0 report-anchoring backlog before and after the W140 sync cohort.

Report coverage quantity	Count
Statistics-wave dispatch census (T0 / registry)	197 / 374
T0 results anchored before the sync cohort	75
T0 anchoring backlog at dispatch, before the sync cohort	122
Sync-cohort shards present	21 / 21
T0 results newly anchored by present sync shards	92
T0 results anchored after present sync shards	167
T0 anchoring backlog at dispatch, after present sync shards	30
Current live census (T0 / registry)	200 / 374
Current live T0 anchoring backlog	33

Backlog means current T0 results with no result anchor in a report section. It includes deliberately off-paper-track rows as well as recent paper-track debt; it is a coverage count, not a rigour judgement.

75.8 Worker and job census

Two independent records count delegated work. The `fr` log names the workers involved in each harvested cycle as `model:role` pairs; that is a curated, human-entered field and covers 565 worker mentions across 38 distinct model/role pairs. The `af` ledgers count *claim events* mechanically: every `nodes_claimed` entry is one claim event: one worker owner taking one node, and the owner string encodes whether it was the initial prover build, a prover fix round, or a verifier pass. That gives 4 633 claim events in this repository and 216 in the progenitor — 4 849 delegated adversarial claim events in total.

Table 17: Mechanically counted `af` node-claim events in this repository, by role. Provers and verifiers are always separate fresh agents; the orchestrator never appears here because it never claims a node.

`af` claim owner	Claim events
`verifier`	3 406
`prover (initial build)`	742
`prover (fix round)`	484
`other`	1

Table 18: Worker mix as recorded in the `fr` controller log (curated field; one cycle may name several workers).

Model	Role	Cycles
`codex`	`prover`	168
`codex`	`verifier`	138
`gpt-5.6-sol`	`verifier`	94
`gpt-5.6-sol`	`prover`	80
`opus`	`prover`	11
`codex`	`refuter`	7
`fable`	`author`	7
`codex`	`auditor`	6
`gpt-5.6-sol`	`designer`	6
`codex`	`applier`	5
`opus`	`author`	5
`codex`	`transcriber`	4
`codex-gpt-5.6-sol`	`prover`	3
`codex`	`adversary`	2
`codex`	`strategist`	2
`codex-gpt-5.6-sol`	`auditor`	2
`codex-gpt-5.6-sol`	`verifier`	2
`fable`	`prover`	2
20 further model/role pairs		21

`codex` entries are `gpt-5.6-sol` runs under the project's model/effort policy; a few rows carry the fuller `codex-gpt-5.6-sol` spelling from before the label settled. `opus`, `sonnet`, and `fable` are Claude models acting as orchestrator-side authors, strategists, and reviewers.

The progenitor's 216 claim events are counted but not decomposed: that campaign predates the `v-/pf-/prover-build` owner convention and used ad-hoc worker names (`agent-A`, `prover-direct-sum`, `vtb-2.1`, ...), so only its total is comparable with the column above.

What these numbers are not. A claim event is the closest retained job proxy, not a guaranteed count of distinct API invocations and not a token budget. Claim-event counts are exact and mechanically derived; token, cost, and wall-clock figures were never recorded by either repository and are not estimated here (Section 75.2).

75.9 Evidence, artifacts, and dead routes

Numerical work is quarantined by construction: every experiment is a run bundle with a re-run command, a schema, an index row, and a checkable invariant, and is never rigorous. 38 such bundles exist. The inherited campaign record (`docs/ingest/`) carries 60 experiment files. Its re-home CSV inventories at least 67 000 instances and 0 recorded counterexamples in the archive's own terminology; those fields remain L3 metadata and prove nothing.

Table 19: Document and tooling production in this repository.

Artifact class	Count
`docs/waves/` — wave records	107
`docs/plans/` — strategy and proof-sketch versions	96
`docs/audits/`, `lit-review/`, `recon/`, `tooling-feedback/`	5
`docs/ingest/` — the inherited classical portfolio	7
of which experiment files	60
`runs/` — numerical run bundles	38
producing scripts inside bundles	128
`report/sections/` — lab-book shards	76
lines of LaTeX in them	11 633
`refs/` — pinned ground-truth sources	10
`scripts/` — tooling files	31
lines of tooling code	10 612
of which validation gates	7
lines in the gates	1 999
gate unit-test files / lines	8 / 1 705

The negative record is kept as deliberately as the positive one. `FINDINGS.md` carries 53 dated entries with 16 explicit do-not-re-walk markers, indexed by mechanism; `docs/LEARNINGS.md` carries 7 dated correction/retraction entries.

Table 20: The dead-route census, read off the hand-built mechanism index in `FINDINGS.md`.

Mechanism tag in the findings index	Entries
`CERTIFICATE-OR-CENSUS`	14
`CONSTANT-OR-SCALE`	11
`DEAD-ROUTE`	10
`RETIRED-ROUTE`	1
`WALL`	10

DEAD-ROUTE is a permanent kill with a certificate; WALL is a structural limit found, not a route kill; RETIRED-ROUTE is superseded rather than refuted. The distinction is load-bearing: only the first forbids re-walking.

Table 21: The issue tracker at the snapshot instant (210 issues plus 9 persistent memory records).

Issue status	Count
`closed`	149
`open`	53
`in_progress`	6
`deferred`	2

75.10 Progenitor and successor

The classical portfolio was not born here. It began as a side-quest inside `almost-idempotent-positive-maps`, whose own target was the non-commutative problem; the classical case was its proving ground. That campaign ran 2026-06-02 to 2026-06-13 (174 commits, 22 work-log entries) and reached 59 registry results with 16 `af`-validated roots across 16 workspaces and 132 proof nodes. This repository was stood up 2026-07-02 — 19 days later — specifically to re-establish the classical result from scratch under a stricter ladder.

What carried over. The transferred artifacts were re-tagged *down* on arrival:

- **The result registry.** Imported paper-proof records re-entered as `stated` or `proved-mod-audit` — objects of study, not oracles — and 17 of the 374 registry results here still carry the latter tag.
- **The numerical record.** The re-home manifest inventories at least 67 000 instances; the archived files were copied into `docs/ingest/` (60 files) and re-homed as run bundles. Their status did not change: `numerical`, L3, never rigorous.
- **A named re-validation.** The `lem-classical-equiv` workspace was built again here from scratch: 29 nodes and 6 challenges are present in its retained ledger. This reports the local artifact only and makes no claim about unrecorded work.

75.11 Colophon: how this section stays true

Everything in Section 75 is generated by `scripts/gen-report-stats.py` into `report/generated/stats/`. The shard in `report/sections/` contains no numbers; it only `\input`s the generated body.

The staleness problem. Statistics about a repository depend on things that change with every commit — git history, the controller log, the proof ledgers. A byte-for-byte freshness gate over such a generator would be permanently red, because the very commit that lands the regenerated tables changes their inputs again. The generator therefore runs in two halves with a committed snapshot between them:

`python3 scripts/gen-report-stats.py --extract`

mines the live sources into `report/generated/stats/campaign-extract.json` *and* re-renders.

`python3 scripts/gen-report-stats.py`

renders the `.tex` files from that committed snapshot, deterministically.

`python3 scripts/gen-report-stats.py --check`

the gate: re-renders, byte-compares, and prints a non-fatal drift advisory.

Staleness semantics. Only the render step is gated, so the check is stable across ordinary commits and fails exactly when it should: a hand-edited generated file, or a renderer change without a re-render. The *numbers*, by contrast, are as of **2026-08-09T08:47:46Z** and go stale silently by design — refreshing them is a deliberate act. `--check` therefore also compares a few cheap live counts against the snapshot and prints how far behind it is, without blocking a commit.

Progenitor data. The `almost-idempotent-positive-maps` block is mined live when that repository sits beside this one and is otherwise carried forward from the committed snapshot, so the gate is green on machines that do not have the sibling. The current block was mined live at the last extraction.

Provenance of the figures. 1 281 controller records, 17 506 `af` ledger entries, 374 registry shards, 47 definition shards, 38 run bundles, 87 work-log entries, 210 issue records, and 1 382 commits were read to produce this section.

76 Status and outlook

What is validated here

The preceding sections reproduce forty-seven registry results, each `proved` with `af: validated`: the signed–stochastic bridge `lem-classical-equiv` (§3); positive-retract hardening `lem-prh` (§4); the COMP tier — `lem-compcb-amplification-naturality`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-entrywise-compression-naturality`, `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-u` `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`; the H-CB tier — `lem-hcb0-compressed-associator`, `lem-hcb-column-hilbert-squared`, `lem-hcb1-variational-identity`, `lem-hcb1-column-action`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-hcb3-diagonal-unit`, `lem-hcb3-diagonal` `lem-hcb3-uniform-square-lower`, `lem-hcb3-diagonal-lower-modulus`, `lem-hcb3-diagonal-inverse`, `lem-hcb3-offdiagonal-inverse` — its canonical layer — `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-clo` `lem-hcb4-canonical-inverse` — together with the tier's parent contract `conj-hcb` (§31); and the front end of the EXT tier — `lem-extcb-one-dimensional-product`, `lem-extcb-one-dimensional-corner-dime` `lem-extcb-corner-dimension-additivity`, `lem-extcb1-close-corner-dimension`, `lem-extcb1-cross-corne` `lem-extcb-four-corner-norm`, `lem-extcb-four-corner-merge`; and, closing that tier, its parent contract `conj-extcb` (§37); the Route F finish edge `lem-routef-prh-finish` (§38); and the Stage-1 rectification pair — `lem-stage1-quantitative-inverse-function` (§39) and `lem-stage1-exact-unit-rectifi` (§40); and the Route F upstream block — `lem-routef-f0-ucp-lift` and `lem-routef-f0-defect-identity` (§41), `lem-kitaev-almost-idemp-audit`, `lem-routef-functional-calculus-closeness` and `lem-routef-ai-d` (§42), `lem-routef-f2-positive-unital-compression` and `lem-routef-f3-retract-defect` (§43). The registry-index snapshot from which the table below is transcribed contains **73** results with that status pair. It predates the seven Route F upstream rows just listed, none of which appears in it, so the set difference is the thirty-three rows tabulated below.

The headline also needs qualifications. The inverse and lower-modulus results — `lem-hcb3-diagonal-lower-mo` `lem-hcb3-diagonal-inverse`, `lem-hcb3-offdiagonal-inverse` and the corresponding clauses of `conj-hcb` — are validated as *conditional* statements. Their level-one hypotheses are established in exactly one place in this document and for exactly one datum: §37 establishes them for the EXT-CB datum, in the order the tree fixes, before invoking the clauses. Nothing else here supplies them for any datum. Their conditional shape is itself a correction of record: the parent's inverse clause was unconditional until an exact $\mathbb{C}\oplus\mathbb{C}$ counterexample refuted it. `lem-extcb-four-corner-merge` (§36) carries a smallness hypothesis on the *total* defect $\rho+\varepsilon$; the transcribed version that constrained ρ alone is false, and the refuting example is reproduced there. Both corrections were produced by hostile verification, and both are reasons to distrust unconditional restatements of this material.

What is next, and what it is not

With §31 the H-CB tier is closed, and with §37 so is the EXT tier: `conj-extcb` is `proved` with `af: validated` and is reproduced above as Lemma 87. What it delivers is one extension step — from M_r to M_{r+1}, at one universal accuracy, with one unitary and four fixed corner maps serving every amplification level. It is not an induction, and nothing here iterates it.

Beyond the EXT tier lies the assembly, and the whole cone above it is now rigorous: the MAIN-CB assembly rows and `lem-thmainext-conditional` are `proved` with `af: validated` (not reproduced here), as are all fifteen original inputs of the K-ledger factorization — the formation backbone, ledger rows 1–14 with D2/D3, the two row-8 factoring sub-lemmas, the F0 seam pair, F2, F3, and the PRH finish — and, as of 2026-08-08, the top of the cone itself: the strengthened `lem-routef-k-ledger` replacement, its three helper rows (`lem-routef-scalar-header-positivity`, `lem-routef-factor-map-packet`, `lem-routef-factor-estimate-packet`), and `lem-routef-f0-assembly`, all elevated under the user-ratified, twice-hostile-audited KLEDGER-STRENGTHENED v2 package (fresh prover, separate fresh verifier per node, external oracle per bank), followed by the user-ratified root rewire (the audited OR-routes block) and the root's own af tree.

The honest summary is therefore: every edge of the live route, from the base tiers through the MAIN assembly, the ledger family, the strengthened K-ledger, and the F0 assembly, up to and including the root, has been made rigorous at this document's af-validated rung. `op-classical` is now **discharged**: `proved` with `af: validated`, with explicit dimension-free witnesses $\eta_0 = \eta_K$ and $C = K + 4\sqrt{2K}$. Sharpness of the exponent 1/2 is now carried at T0 by the af-validated `cor-classical-sharpness`; the former `ex-hume` contract is disproved historical record. What remains open is every rung above af-validation (no Lean/mathlib proof exists; that is the ladder's top rung and a separate campaign).

Validated registry results not reproduced here

The table is the exact set difference between the supplied index rows carrying `proved/validated` and the reproduced ids listed above. Its ordering follows the generated index.

Registry id	Status	Note
`lem-always-tight-dual-support`	`proved, af: validated`	off the live route; registry record retained
`lem-collateral-import`	`proved, af: validated`	off the live route; registry record retained
`lem-cross-pivot-cancellation`	`proved, af: validated`	off the live route; registry record retained
`lem-cs-low-slab-pincer`	`proved, af: validated`	off the live route; registry record retained
`lem-depth-d-halo-collapse`	`proved, af: validated`	off the live route; registry record retained
`lem-factorization`	`proved, af: validated`	off the live route; registry record retained
`lem-fan-payment`	`proved, af: validated`	off the live route; registry record retained
`lem-fan-payment-restricted`	`proved, af: validated`	off the live route; registry record retained
`lem-genuine-disintegration`	`proved, af: validated`	off the live route; registry record retained
`lem-halo-collapse`	`proved, af: validated`	off the live route; registry record retained
`lem-harmonic-affine-bridge`	`proved, af: validated`	off the live route; registry record retained
`lem-hiddenness-depth-markov`	`proved, af: validated`	off the live route; registry record retained
`lem-hiddenness-dual-witness`	`proved, af: validated`	off the live route; registry record retained
`lem-hx-financing-floor`	`proved, af: validated`	off the live route; registry record retained

Registry id	Status	Note
`lem-hx-forced-exterior-coupling`	`proved`, `af: validated`	off the live route; registry record retained
`lem-hx-robust-scalar-starvation`	`proved`, `af: validated`	off the live route; registry record retained
`lem-hx-signed-variation-ledger`	`proved`, `af: validated`	off the live route; registry record retained
`lem-hx-transverse-moment-identity`	`proved`, `af: validated`	off the live route; registry record retained
`lem-import-reduction`	`proved`, `af: validated`	off the live route; registry record retained
`lem-mass-split`	`proved`, `af: validated`	off the live route; registry record retained
`lem-negpart-subadditive`	`proved`, `af: validated`	off the live route; registry record retained
`lem-parametric-halo-collapse`	`proved`, `af: validated`	off the live route; registry record retained
`lem-pivot-removing-move`	`proved`, `af: validated`	off the live route; registry record retained
`lem-residual-lower`	`proved`, `af: validated`	off the live route; registry record retained
`lem-residual-upper`	`proved`, `af: validated`	off the live route; registry record retained
`lem-row-far-dual-certificate`	`proved`, `af: validated`	off the live route; registry record retained
`lem-row-zero-capacity`	`proved`, `af: validated`	off the live route; registry record retained
`lem-starvation-completion-obstruction`	`proved`, `af: validated`	off the live route; registry record retained
`lem-top-concentration`	`proved`, `af: validated`	off the live route; registry record retained
`lem-top-slab-companion`	`proved`, `af: validated`	off the live route; registry record retained
`lem-weighted-min`	`proved`, `af: validated`	off the live route; registry record retained
`lem-zerosum-triangle`	`proved`, `af: validated`	off the live route; registry record retained
`obs-height-collapse`	`proved`, `af: validated`	off the live route; registry record retained

Each id and status pair in the table is transcribed from `repo-inputs/argument-INDEX.md`. In particular, `conj-extcb` is absent from the table because it is *reproduced* above (§37), not because of its status: the index records it as `proved` with `af: validated`.